TYPE II REGULARITY FOR NAVIER–STOKES

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Abstract. We establish a local Type II exclusion criterion for the three-dimensional Navier–Stokes equations near a possible singular point. From a positive Caffarelli–Kohn–Nirenberg concentration sequence we select renormalized local windows around a retained core and analyze the resulting alternatives by compactness, pressure reconstruction, and local energy estimates. The argument combines a compact single-core exclusion criterion, a repaired-gauge representation for nondegenerate concentration cores, local Calderon–Zygmund pressure control, a Caccioppoli estimate on compact cylinders, multibubble and cascade reductions, and scale-collapse cost estimates. The retained compact scale-collapse branch is closed by local state-space routing into the scale-rigid discharge, so no additional stationary or self-similar classification theorem is used implicitly.

Contents

1. Introduction

The three-dimensional Navier–Stokes regularity problem is governed by the interaction between local energy control, scale-critical concentration, and the possible formation of singular structures at arbitrarily small parabolic scales. Since Leray's construction of global weak solutions [Ler34], the theory of suitable weak solutions and the partial regularity theorem of Scheffer and Caffarelli–Kohn–Nirenberg have provided the basic local language for this interaction [Sch76, CKN82, Lin98]. In that theory, singularity formation is detected by scale-invariant velocity-pressure quantities, and the critical L^3 scale plays a central role alongside the local energy inequality, pressure reconstruction, and compactness. The analytic background also includes the classical local and critical-space theory for Navier–Stokes [Lad69, Tem77, CF88, Kat84, LR02], the $L_{3,\infty}$ regularity theorem of Escauriaza–Seregin–Šverák [ESS03], and the rigidity theorem of Nečas–Růžička–Šverák for Leray self-similar profiles [NRŠ96].

This paper studies the local Type II alternative near a possible singular point. Starting from a positive Caffarelli–Kohn–Nirenberg concentration sequence, we select renormalized local windows around a retained core and analyze all branches that can arise from the scale-center dynamics. The result is a local Type II exclusion theorem in the class of alternatives covered by the analytic inputs stated in the body of the paper. It is deliberately formulated as a conditional PDE theorem: pressure reconstruction, repaired-gauge compactness, terminal profile decoupling, cost comparison, and local state-space closure are assumed or proved explicitly where they are used. No additional stationary, ancient, or self-similar classification theorem is invoked implicitly.

The proof begins with local concentration and the Type I/Type II separation. For a nondegenerate retained core, the solution is rewritten in local scale-translation variables. The repaired-gauge representation produces a renormalized Navier–Stokes equation with controlled modulation terms, local pressure reconstruction modulo functions of time, and compact-window suitable estimates. Within this representation, a compact single-core branch with positive CKN density and vanishing localized dissipation is impossible: local compactness gives a spatially constant limit, the zero constant contradicts pressure-stable persistence of the positive core, and a nonzero constant gives a bounded selected-window limit incompatible with the Type II condition.

The remaining sections remove the ways in which a Type II branch could avoid that compact single-core mechanism. Caccioppoli estimates convert finite outer velocity-pressure control into inner windowed H^1 control, so rough-core loss is returned to an explicit compact-cylinder alternative. Multibubble and cascade arguments separate active profiles, same-point cascades, and gauge-degenerate branches from a single retained core. Scale-collapse branches are treated by localized cost identities: finite scale-collapsing cost, and the corresponding finite absolute cost, are incompatible with a nonvanishing retained core, while thick autonomous windows lead only to the stated modulation, compactness, or retained compact scale-collapse state; the latter is routed into the scale-rigid discharge rather than into a new Liouville assumption. The terminal profile and corrected monotonicity sections then turn these local reductions into a branchwise classification, and the final assembly eliminates each covered Type II alternative.

The conclusion should therefore be read in its local and conditional sense. In the covered class, a positive local Type II concentration sequence cannot survive the repaired-gauge, compactness, multibubble, rough-core, and scale-collapse reductions. What remains outside the theorem is not hidden in the proof: it is precisely the failure of one of the stated analytic inputs or one of the explicitly named local state-space exits.

2. Further Local Reductions for Rough Cores, Scale Collapse, and Cascades

This section records the remaining local reductions used to separate the rough-core, scale-collapse, multibubble, and same-point cascade alternatives. All statements are conditional analytic statements for represented Type II branches.

2.1. **Vorticity-Controlled Rough-Core Alternative.** Let (V, P, a, b) be a repaired-gauge renormalized Navier–Stokes branch,

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu\Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0,$$

and set $\Omega = \nabla \times V$. Then, in distributions,

$$\partial_\tau \Omega + (V \cdot \nabla)\Omega - (\Omega \cdot \nabla)V = \nu\Delta\Omega + a(\tau)(2\Omega + y \cdot \nabla\Omega) + b(\tau) \cdot \nabla\Omega.$$

At the smooth approximation level, a localized enstrophy estimate contains

$$\int (\Omega \cdot \nabla V) \cdot \Omega\, \chi_R^2.$$

The standard bound is

$$\left| \int (\Omega \cdot \nabla V) \cdot \Omega\, \chi_R^2 \right| \le C\|\Omega\chi_R\|_{L^2}^{1/2} \|\nabla(\Omega\chi_R)\|_{L^2}^{3/2} \|\nabla V\|_{L^2(B_{2R})}.$$

After Young's inequality the differential inequality has a coefficient

$$C_\nu \|\nabla V\|_{L^2(B_{2R})}^4$$

in front of the localized enstrophy. Thus the vorticity equation does not by itself close a uniform enstrophy-window estimate without an independent gradient-window estimate. The vorticity argument below is therefore a conditional rough-core exclusion, while the primary rough-core exclusion remains the Caccioppoli estimate in Theorem 8.15.

Lemma 2.1 (Local div-curl estimate)**.** *For each $R > 0$ there is $C_R < \infty$ such that every $U \in L^2(B_{2R}; \mathbb{R}^3)$ satisfying*

$$\nabla \cdot U = 0 \quad \text{in } \mathcal{D}'(B_{2R})$$

and

$$\nabla \times U \in L^2(B_{2R}; \mathbb{R}^3)$$

belongs to $H^1(B_R; \mathbb{R}^3)$, with

$$\|\nabla U\|_{L^2(B_R)} \le C_R \left(\|\nabla \times U\|_{L^2(B_{2R})} + \|U\|_{L^2(B_{2R})} \right).$$

Proof. Choose $\chi \in C_c^\infty(B_{2R})$ with $\chi \equiv 1$ on B_R, and set

$$W = \chi U.$$

After extension by zero, $W \in L^2(\mathbb{R}^3)$ is compactly supported. Moreover,

$$\nabla \times W = \chi \nabla \times U + \nabla\chi \times U \in L^2(\mathbb{R}^3),$$

and, using $\nabla \cdot U = 0$,

$$\nabla \cdot W = \nabla\chi \cdot U \in L^2(\mathbb{R}^3).$$

For compactly supported distributions with W, $\nabla \times W$, and $\nabla \cdot W$ in L^2, the Fourier div-curl identity gives

$$\|\nabla W\|_{L^2(\mathbb{R}^3)}^2 = \|\nabla \times W\|_{L^2(\mathbb{R}^3)}^2 + \|\nabla \cdot W\|_{L^2(\mathbb{R}^3)}^2.$$

Indeed, Plancherel and

$$|\xi|^2 |\widehat{W}(\xi)|^2 = |\xi \times \widehat{W}(\xi)|^2 + |\xi \cdot \widehat{W}(\xi)|^2$$

yield the identity after integration in ξ. Hence $W \in H^1(\mathbb{R}^3)$. Since $W = U$ on B_R,

$$\|\nabla U\|_{L^2(B_R)} \le \|\nabla W\|_{L^2(\mathbb{R}^3)} \le C_R \left(\|\nabla \times U\|_{L^2(B_{2R})} + \|U\|_{L^2(B_{2R})}\right).$$

□

Lemma 2.2 (Critical L^3 control gives local L^2 control)**.** *If*

$$\sup_{\tau \ge \tau_0} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \le M,$$

then, for every $R > 0$*,*

$$\sup_{\tau \ge \tau_0} \|V(\tau)\|^2_{L^2(B_R)} \le CRM^2.$$

Proof. Hölder's inequality gives

$$\int_{B_R} |V(y,\tau)|^2 \, dy \le |B_R|^{1/3} \left(\int_{B_R} |V(y,\tau)|^3 \, dy\right)^{2/3} \le CRM^2.$$

Taking the supremum in τ proves the estimate. □

Proposition 2.3 (Vorticity-window control implies local windowed H^1)**.** *Assume V is a represented divergence-free branch,*

$$\sup_{\tau \ge \tau_0} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \le M,$$

$\Omega = \nabla \times V$ *in distributions, and*

$$\forall R > 0: \qquad \sup_{T \ge \tau_0 + 1} \int_T^{T+1} \|\Omega(\tau)\|^2_{L^2(B_{2R})} \, d\tau < \infty.$$

Then

$$\forall R > 0: \qquad \sup_{T \ge \tau_0 + 1} \int_T^{T+1} \|V(\tau)\|^2_{H^1(B_R)} \, d\tau < \infty.$$

Proof. Fix $R > 0$. By Theorem 2.1, for almost every τ,

$$\|\nabla V(\tau)\|^2_{L^2(B_R)} \le C_R \left(\|\Omega(\tau)\|^2_{L^2(B_{2R})} + \|V(\tau)\|^2_{L^2(B_{2R})}\right).$$

Integrating over $[T, T+1]$, the vorticity term is uniformly bounded by the assumed vorticity-window control, while the local L^2 term is uniformly bounded by Theorem 2.2:

$$\int_T^{T+1} \|V(\tau)\|^2_{L^2(B_{2R})} \, d\tau \le C_R M^2.$$

Therefore

$$\sup_{T \ge \tau_0 + 1} \int_T^{T+1} \|\nabla V(\tau)\|^2_{L^2(B_R)} \, d\tau < \infty.$$

The same local L^2 bound gives

$$\sup_{T \ge \tau_0 + 1} \int_T^{T+1} \|V(\tau)\|^2_{L^2(B_R)} \, d\tau < \infty,$$

and the asserted windowed H^1 bound follows. □

Corollary 2.4 (Vorticity-controlled rough-core exclusion)**.** *Let a represented Type II branch have positive finite critical mass and local vorticity-window control as in Theorem 2.3. Then the rough-core alternative, namely failure of local windowed H^1 control, cannot hold.*

Proof. The hypotheses give the local windowed H^1 bound by Theorem 2.3. This is the negation of the rough-core alternative. □

Consequently, a represented bounded-critical-norm rough-core branch must fail at least one of the following analytic hypotheses: local vorticity-window control, the representation or critical-mass hypotheses used before the vorticity argument, bounded modulation, Caccioppoli regularity, or one of their integrated refinements.

2.2. **Scale-Collapse Autonomous Reduction.** Let a represented repaired-gauge Type II branch be in the remaining scale-collapse class when it satisfies the representation hypotheses, positive finite critical mass, a nonzero localized L^2 core floor, bounded modulation, local windowed H^1 control, $\lambda(\tau) \to 0$, and the scale-collapse drift obstruction

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau = \infty, \qquad M_R(\tau) = \int |V(y,\tau)|^2 \phi_{R(\tau)}(y)\, dy.$$

The core floor supplies constants $m_0 > 0$, τ_1, and moving cutoffs such that

$$M_R(\tau) \geq m_0 \qquad \text{for a.e. } \tau \geq \tau_1.$$

This is the remaining scale-drift class; it is distinct from the radiative and rough-core alternatives.

Definition 2.5 (Thick negative-drift windows)**.** The branch has thick negative-drift windows if there exist $T > 0$, $c > 0$, and $\tau_n \to \infty$ such that

$$\frac{1}{T} \int_{\tau_n}^{\tau_n + T} a_-(\tau) M_R(\tau)\, d\tau \geq c \qquad \text{for every } n.$$

If $M_R \leq M_1$ on these windows, then

$$\frac{1}{T} \int_{\tau_n}^{\tau_n + T} a_-(\tau)\, d\tau \geq \frac{c}{M_1}.$$

The divergence of $a_- M_R$ alone does not imply the existence of Theorem 2.5; a bounded nonnegative function can have infinite integral while its unit-window averages tend to zero. Without thick windows, the branch remains in the thin-drift alternative rather than entering a self-similar regime.

Definition 2.6 (Autonomous modulation limit)**.** On fixed windows $[\tau_n, \tau_n + T]$, the branch has an autonomous modulation limit (a_∞, b_∞) if, after translation,

$$a(\tau_n + \cdot) \to a_\infty, \qquad b(\tau_n + \cdot) \to b_\infty$$

in a coefficient topology strong enough to pass the products

$$a(V + y \cdot \nabla V), \qquad b \cdot \nabla V$$

to the distributional limit. Strong $L^1(0,T)$ convergence is sufficient. The parameter a_∞ may be negative, zero, or any value allowed by the bounded modulation range. The case $a_\infty < 0$ is a generalized self-similar regime, while $a_\infty = 0$ gives a stationary Navier–Stokes regime, possibly with constant translation drift b_∞.

Lemma 2.7 (Thick autonomous drift gives a negative scale parameter)**.** *Assume Theorem 2.5 holds on windows $[\tau_n, \tau_n + T]$, $M_R \leq M_1$ on those windows, and*

$$a(\tau_n + \cdot) \to a_\infty \quad \text{in } L^1(0,T).$$

Then

$$a_\infty \leq -\frac{c}{M_1} < 0.$$

Proof. The upper mass bound and thick weighted drift give

$$\frac{1}{T}\int_{\tau_n}^{\tau_n+T} a_-(\tau)\,d\tau \geq \frac{c}{M_1}.$$

After translation and L^1 convergence of a to the constant a_∞, the left-hand side converges to $(-a_\infty)_+$. Hence

$$(-a_\infty)_+ \geq \frac{c}{M_1},$$

and $a_\infty \leq -c/M_1 < 0$. □

Definition 2.8 (Compactness on autonomous windows)**.** The branch has compactness on the windows (T, τ_n) if, for every ball B_R, the translated sequence

$$V_n(y,s) = V(y, \tau_n + s), \qquad 0 < s < T,$$

is precompact strongly in $L^2(0,T;L^q(B_R))$ for every $q < 6$, weakly in $L^2(0,T;H^1(B_R))$, and weak-* in

$$L^\infty(0,T;L^3(\mathbb{R}^3)).$$

The pressures are normalized modulo constants and converge in a topology sufficient to pass

$$-\Delta P = \partial_i\partial_j(V_iV_j)$$

to the local limit. The compactness, modulation convergence, and any thick drift condition are always imposed on the same window sequence τ_n.

Theorem 2.9 (Autonomous reduced-limit theorem)**.** *Assume the represented scale-collapse hypotheses above, compactness on autonomous windows, and an autonomous modulation limit (a_∞, b_∞). Then, after passing to a subsequence, the translated branch converges to a nonzero weak solution (U, Π) on $\mathbb{R}^3 \times (0,T)$ of*

$$\partial_s U + (U\cdot\nabla)U + \nabla\Pi = \nu\Delta U + a_\infty(U + y\cdot\nabla U) + b_\infty\cdot\nabla U, \qquad \nabla\cdot U = 0.$$

Moreover,

$$U \in L^\infty(0,T;L^3(\mathbb{R}^3)),$$

the pressure satisfies the Navier–Stokes pressure reconstruction, and the localized core floor passes to the limit on a smaller compact cutoff. In particular $U \not\equiv 0$.

Proof. The weak-* $L^\infty L^3$ bound follows from the positive finite critical annulus. The local L^2H^1 bound and time-derivative compactness are part of the compactness hypothesis, so Aubin–Lions gives strong local convergence in L^2L^q for $q < 6$. This strong convergence, together with weak L^2H^1 convergence, passes the nonlinear term to $(U\cdot\nabla)U$ in distributions. The pressure convergence passes the Poisson reconstruction to Π.

The modulation-limit hypothesis passes the coefficient terms to

$$a_\infty(U + y\cdot\nabla U), \qquad b_\infty\cdot\nabla U.$$

Therefore the limit solves the autonomous reduced equation. Finally, the localized L^2 core floor gives positive localized mass on the selected branch. The compactness hypothesis includes convergence strong enough for this localized pairing, or for a compact subcutoff below it, to pass to the limit. Hence $U \not\equiv 0$. □

Definition 2.10 (Stationary omega-limit hypothesis)**.** For a parameter pair (a_∞, b_∞), the autonomous limit of Theorem 2.9 has a stationary omega-limit if there is a nonzero stationary profile (W, Q) satisfying

$$(W\cdot\nabla)W + \nabla Q = \nu\Delta W + a_\infty(W + y\cdot\nabla W) + b_\infty\cdot\nabla W, \qquad \nabla\cdot W = 0,$$

with

$$W \in L^3(\mathbb{R}^3),$$

and with the admissibility hypotheses required by the rigidity theorem to be invoked.

The stationary omega-limit hypothesis is an additional analytic hypothesis: an autonomous bounded trajectory need not be stationary merely by time translation. A Lyapunov or asymptotic-compactness argument is required to produce W.

Definition 2.11 (Rigidity-covered stationary class)**.** For (a_∞, b_∞), the stationary class is rigidity-covered if every admissible stationary solution of

$$(W \cdot \nabla)W + \nabla Q = \nu \Delta W + a_\infty (W + y \cdot \nabla W) + b_\infty \cdot \nabla W, \qquad W \in L^3(\mathbb{R}^3),$$

is zero, or at least is not a terminal Type II blow-up branch in the class under consideration. In the classical backward self-similar normalization, after rescaling and constant translations, this is the role of the Nečas–Růžička–Šverák rigidity theorem. For forward self-similar classes, the required hypothesis is not nonexistence of profiles, but the narrower statement that the admissible forward profile is not a terminal Type II blow-up branch.

Theorem 2.12 (Conditional scale-collapse exclusion)**.** *Assume the represented scale-collapse hypotheses, compactness on autonomous windows, an autonomous modulation limit, a stationary omega-limit, and rigidity coverage for the corresponding parameter pair. Then the scale-collapse alternative is excluded for that branch.*

Proof. Theorem 2.9 gives a nonzero autonomous limit. The stationary omega-limit hypothesis yields a nonzero stationary profile (W, Q) in the admissible class for (a_∞, b_∞). Rigidity coverage says that no such nonzero profile can represent a terminal Type II branch in the class under consideration. This contradicts the assumption that the original branch remains in the remaining scale-collapse class. □

If, in addition, the branch is in the supercritical Type II scale alternative and the analytic implication from scale-collapse exclusion to that Type II alternative is assumed, then Theorem 2.12 excludes the corresponding Type II scale alternative.

The scale-collapse drift obstruction implies

$$\int_{\tau_0}^{\infty} a_-(\tau)\, d\tau = \infty$$

only when M_R is bounded above on the same moving cutoffs, because then

$$a_-(\tau) M_R(\tau) \le M_1 a_-(\tau).$$

The local mass floor is used for nontriviality of limits, not for this unweighted-drift implication. Thus the drift obstruction proves sustained weighted contraction, but by itself does not prove convergence of $a(\tau)$ to a negative constant, nor uniformly negative fixed-window averages.

The scale-collapse alternatives are therefore:

(i) thick autonomous drift, compactness, modulation convergence, and a stationary omega-limit give a rigidity-covered shortcut when such a stationary profile theorem is available;
(ii) thin or oscillatory drift is a named local drift alternative;
(iii) a thick autonomous limit which remains in the retained compact scale-collapse state is routed by Theorem 7.30 into the scale-rigid local discharge, rather than into a new global Liouville problem.

The remaining analytic alternatives are thin drift, compactness failure, failure of modulation convergence, and failure to remain in the retained compact scale-collapse state.

2.3. **Local L^3-Compactness from L^2-Compactness.** We record a standard interpolation argument that converts local compactness in L^2, under uniform local L^6 control, into local compactness in L^3 for translated renormalized Navier–Stokes trajectories. Let $R > 0$, $I = (0,1)$, $\overline{I} = [0,1]$, and let the shifted trajectories be defined on $\overline{I} \times B_{2R}$ by

$$W_n(s,y) := V(\tau_n + s, y), \qquad (s,y) \in \overline{I} \times B_{2R},$$

where $\tau_n \to \infty$. All spacetime norms below are taken over I; the notation $W_n(0,\cdot)$ denotes the endpoint trace whenever that trace is used.

Lemma 2.13 (Spacetime compactness in $L^2_t L^2_x$)**.** *Assume*

$$\sup_n \|W_n\|_{L^2(I;H^1(B_{2R}))} < \infty$$

and

$$\sup_n \|\partial_s W_n\|_{L^2(I;H^{-1}(B_R))} < \infty.$$

Then there are a subsequence, not relabeled, and a limit

$$W \in L^2(I; L^2(B_R))$$

such that

$$W_n \to W \qquad \textit{strongly in } L^2(I; L^2(B_R)).$$

Proof. Restrict W_n to $I \times B_R$. The first hypothesis gives (W_n) bounded in $L^2(I; H^1(B_R))$, and the second gives $(\partial_s W_n)$ bounded in $L^2(I; H^{-1}(B_R))$. The embeddings

$$H^1(B_R) \Subset L^2(B_R) \hookrightarrow H^{-1}(B_R)$$

are, respectively, compact and continuous. The Aubin–Lions–Simon compactness theorem therefore implies that (W_n) is relatively compact in

$$L^2(I; L^2(B_R)).$$

After passing to a subsequence, there is

$$W \in L^2(I; L^2(B_R))$$

such that $W_n \to W$ strongly in this space. □

Lemma 2.14 (Spacetime compactness in $L^2_t L^3_x$)**.** *Assume the hypotheses of Lemma 2.13. Assume also*

$$\sup_n \|W_n\|_{L^2(I;L^6(B_{2R}))} < \infty.$$

Then, after passing to a subsequence if necessary,

$$W_n \to W \qquad \textit{strongly in } L^2(I; L^3(B_R)).$$

Proof. By the $L^2(I; L^6(B_{2R}))$ bound, the restricted sequence is bounded in the reflexive space $L^2(I; L^6(B_R))$. Hence, after passing to a subsequence,

$$W_n \rightharpoonup \widetilde{W} \qquad \text{weakly in } L^2(I; L^6(B_R)).$$

The same subsequence converges strongly to W in $L^2(I; L^2(B_R))$ by Lemma 2.13. To identify the weak $L^2_t L^6_x$ limit, let

$$\varphi \in C_c^\infty(I \times B_R; \mathbb{R}^3).$$

Strong $L^2_t L^2_x$ convergence gives

$$\int_I \int_{B_R} W_n \cdot \varphi \, dy \, ds \to \int_I \int_{B_R} W \cdot \varphi \, dy \, ds,$$

while weak convergence in $L^2_tL^6_x$ gives the same limit with $\widetilde{W}$ in place of W. Hence $W = \widetilde{W}$ in $\mathcal{D}'(I \times B_R)$, and therefore

$$W \in L^2(I; L^6(B_R)),$$

with

$$\|W_n - W\|_{L^2(I;L^6(B_R))}$$

uniformly bounded in n. Indeed,

$$\|W_n - W\|_{L^2(I;L^6(B_R))} \leq \sup_k \|W_k\|_{L^2(I;L^6(B_R))} + \|W\|_{L^2(I;L^6(B_R))}.$$

The Gagliardo–Nirenberg interpolation inequality on the bounded domain B_R gives, for almost every $s \in I$,

$$\|W_n(s) - W(s)\|_{L^3(B_R)} \leq C_R \|W_n(s) - W(s)\|^{1/2}_{L^2(B_R)} \|W_n(s) - W(s)\|^{1/2}_{L^6(B_R)}$$

because $W_n(s) - W(s) \in L^2(B_R) \cap L^6(B_R)$ for a.e. s. Squaring and integrating in s gives

$$\int_0^1 \|W_n(s) - W(s)\|^2_{L^3(B_R)}\, ds \leq C_R \int_0^1 \|W_n(s) - W(s)\|_{L^2(B_R)} \|W_n(s) - W(s)\|_{L^6(B_R)}\, ds.$$

Hölder's inequality in time, applied to the product of the $L^2(B_R)$- and $L^6(B_R)$-norms, yields

$$\|W_n - W\|^2_{L^2(I;L^3(B_R))} \leq C_R \|W_n - W\|_{L^2(I;L^2(B_R))} \|W_n - W\|_{L^2(I;L^6(B_R))}.$$

The first factor tends to zero by Lemma 2.13, while the second factor is uniformly bounded. Therefore the left-hand side tends to zero. □

Lemma 2.15 (Fixed-time L^3 convergence from fixed-time L^2 convergence)**.** *Assume that, after passing to a subsequence, the time slices satisfy*

$$W_n(0, \cdot) \to f \qquad \textit{strongly in } L^2(B_R)$$

and

$$\sup_n \|W_n(0, \cdot)\|_{L^6(B_R)} < \infty.$$

Then

$$W_n(0, \cdot) \to f \qquad \textit{strongly in } L^3(B_R).$$

Proof. First we show that the strong L^2 limit belongs to $L^6(B_R)$. Since $L^6(B_R)$ is reflexive, the uniform $L^6(B_R)$ bound gives a subsequence and a function $g \in L^6(B_R)$ such that

$$W_n(0, \cdot) \rightharpoonup g \qquad \text{weakly in } L^6(B_R).$$

For every $\phi \in C_c^\infty(B_R)$, the strong $L^2(B_R)$ convergence also gives

$$\int_{B_R} W_n(0, y) \cdot \phi(y)\, dy \to \int_{B_R} f(y) \cdot \phi(y)\, dy.$$

The weak L^6 limit gives the same distributional limit with g in place of f, so $g = f$. Thus $f \in L^6(B_R)$. Returning to the original sequence, the triangle inequality gives

$$\sup_n \|W_n(0, \cdot) - f\|_{L^6(B_R)} \leq \sup_n \|W_n(0, \cdot)\|_{L^6(B_R)} + \|f\|_{L^6(B_R)} < \infty.$$

Interpolating between $L^2(B_R)$ and $L^6(B_R)$ at the fixed time $s = 0$,

$$\|W_n(0, \cdot) - f\|_{L^3(B_R)} \leq C_R \|W_n(0, \cdot) - f\|^{1/2}_{L^2(B_R)} \|W_n(0, \cdot) - f\|^{1/2}_{L^6(B_R)}.$$

The L^6 factor is uniformly bounded and the L^2 factor tends to zero by hypothesis. Hence the $L^3(B_R)$ norm tends to zero. □

Remark 2.16 (Limitation at a fixed time). The preceding two spacetime compactness lemmas do not by themselves imply convergence at the fixed time $s = 0$. The fixed-time conclusion in Lemma 2.15 requires a separate L^2 compactness hypothesis for the traces. One sufficient condition is

$$\sup_n \|W_n\|_{L^\infty(I;H^1(B_R))} < \infty$$

and, for some $q > 1$,

$$\sup_n \|\partial_s W_n\|_{L^q(I;H^{-1}(B_R))} < \infty.$$

Simon's compactness theorem then gives precompactness of (W_n) in

$$C([0,1]; L^2(B_R)),$$

and hence the fixed-time L^2 convergence after subsequence extraction. Without such an additional condition, fixed-time convergence at $s = 0$ is not a consequence of Aubin–Lions compactness in $L^2(I; L^2(B_R))$.

Lemma 2.17 (Time selection from averaged localized dissipation and local H^1 control)**.** *Let $J_n = [s_n, s_n + \ell]$ be time intervals with fixed length $\ell > 0$, and let $\delta_n \downarrow 0$. Assume that $\widetilde{\mathfrak{D}}_{R_0}$ is a nonnegative measurable function and, for a fixed radius R, that*

$$\int_{J_n} \widetilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau \le \delta_n$$

and

$$\int_{J_n} \|V(\tau)\|^2_{H^1(B_R)}\, d\tau \le C_R.$$

Then there exist times $\sigma_n \in J_n$ such that

$$\widetilde{\mathfrak{D}}_{R_0}(\sigma_n) \to 0, \qquad \|V(\sigma_n)\|_{H^1(B_R)} \le \left(\frac{2C_R}{\ell}\right)^{1/2}.$$

Consequently, after passing to a subsequence,

$$V(\sigma_n) \to V_* \qquad \textit{strongly in } L^2(B_R)$$

and

$$V(\sigma_n) \to V_* \qquad \textit{strongly in } L^3(B_R).$$

Proof. Define

$$E_n := \left\{\tau \in J_n : \widetilde{\mathfrak{D}}_{R_0}(\tau) \le \delta_n^{1/2}\right\}.$$

By Chebyshev's inequality,

$$|J_n \setminus E_n| \le \delta_n^{1/2}.$$

Indeed,

$$\delta_n^{1/2} |J_n \setminus E_n| \le \int_{J_n \setminus E_n} \widetilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau \le \delta_n.$$

For all sufficiently large n, $\delta_n^{1/2} \le \ell/2$, and hence $|E_n| \ge \ell/2$. Since

$$\int_{E_n} \|V(\tau)\|^2_{H^1(B_R)}\, d\tau \le C_R,$$

there exists $\sigma_n \in E_n$ such that

$$\|V(\sigma_n)\|^2_{H^1(B_R)} \le \frac{2C_R}{\ell}.$$

For otherwise the integral over E_n would be strictly larger than $(2C_R/\ell)|E_n| \geq C_R$, contradicting the displayed bound. Because $\sigma_n \in E_n$,

$$\widetilde{\mathfrak{D}}_{R_0}(\sigma_n) \leq \delta_n^{1/2} \to 0.$$

The estimate on $\|V(\sigma_n)\|_{H^1(B_R)}$ gives a bounded sequence in $H^1(B_R)$. By weak compactness in $H^1(B_R)$, after passing to a subsequence there is $V_* \in H^1(B_R)$ such that

$$V(\sigma_n) \rightharpoonup V_* \qquad \text{weakly in } H^1(B_R).$$

The Rellich compactness theorem, after passing to a further subsequence if necessary, gives the corresponding strong convergence

$$V(\sigma_n) \to V_* \qquad \text{strongly in } L^2(B_R).$$

The same $H^1(B_R)$ bound and the Sobolev embedding

$$H^1(B_R) \hookrightarrow L^6(B_R)$$

give a uniform $L^6(B_R)$ bound for $V(\sigma_n)$. Since $V_* \in H^1(B_R)$, the same embedding gives $V_* \in L^6(B_R)$, and $\|V(\sigma_n) - V_*\|_{L^6(B_R)}$ is uniformly bounded by the triangle inequality. Interpolation gives

$$\|V(\sigma_n) - V_*\|_{L^3(B_R)} \leq C_R \|V(\sigma_n) - V_*\|_{L^2(B_R)}^{1/2} \|V(\sigma_n) - V_*\|_{L^6(B_R)}^{1/2}.$$

The L^6 factor is uniformly bounded and the L^2 factor tends to zero. This proves strong convergence in $L^3(B_R)$. □

Remark 2.18 (Role of the time selection). Lemma 2.17 does not require endpoint trace compactness. Instead of first fixing times τ_n and then proving compactness of $V(\tau_n)$, one selects times of small localized dissipation inside intervals where the same intervals also carry local H^1 control. This supplies the fixed-time L^2-compactness hypothesis in Lemma 2.15 whenever the argument provides common intervals with these two properties.

Corollary 2.19 (Compactness at translated renormalized times)**.** *Assume that, for each fixed radius R, the shifted sequence satisfies the hypotheses of Lemma 2.13 and the time slices satisfy the two hypotheses of Lemma 2.15. Then, after passing to a subsequence,*

$$V(\tau_n) \to V_* \qquad \textit{strongly in } L^3_{\mathrm{loc}}(\mathbb{R}^3),$$

where V_* is the local L^2 limit of the time slices $V(\tau_n)$.

Proof. Apply Lemma 2.15 first with $R = 1$ and pass to a subsequence along which $V(\tau_n)$ converges strongly in $L^3(B_1)$. Apply the same argument to that subsequence with $R = 2$, and continue inductively. The diagonal subsequence converges strongly in $L^3(B_m)$ for every integer $m \geq 1$. The limits on nested balls agree because the convergence in $L^3(B_m)$ restricts to convergence in $L^3(B_k)$ whenever $k < m$. Hence these local limits define a single function $V_* \in L^3_{\mathrm{loc}}(\mathbb{R}^3)$, and

$$V(\tau_n) \to V_* \qquad \text{strongly in } L^3_{\mathrm{loc}}(\mathbb{R}^3)$$

along the diagonal subsequence. □

Remark 2.20 (Role in the compactness argument). The compactness argument uses the preceding statements as follows:

(i) a Caccioppoli-type estimate gives the $L^2(I; H^1)$ bound in Lemma 2.13;
(ii) a time-regularity estimate gives the $\partial_s W_n$ bound;
(iii) a local L^6-type estimate gives the hypothesis of Lemma 2.14 and fixed-time L^6 boundedness;
(iv) Lemmas 2.13 and 2.14 give spacetime L^3 compactness;
(v) an additional fixed-time L^2-compactness hypothesis gives Lemma 2.15;

(vi) Lemma 2.15 gives the fixed-time strong L^3-compactness used in vanishing-dissipation contradiction arguments.

This is the local compactness step needed to pass from renormalized L^2_{loc} compactness to the critical local L^3 topology at the selected times.

Corollary 2.21 (Local compactness from a common sequence of selected times)**.** *Assume there is a single sequence of selected times σ_n such that, for every fixed radius R,*

$$\sup_n \|V(\sigma_n)\|_{H^1(B_R)} < \infty$$

and

$$\widetilde{\mathfrak{D}}_{R_0}(\sigma_n) \to 0.$$

Then, after diagonal subsequence extraction,

$$V(\sigma_n) \to V_* \qquad \textit{strongly in } L^3_{\mathrm{loc}}(\mathbb{R}^3).$$

For each fixed radius, Lemma 2.17 supplies a radius-dependent sequence from intervals with local H^1 control. If the time selection is made on common intervals for an exhaustion of balls, then Lemma 2.17 replaces the separate fixed-time compactness hypothesis in Lemma 2.15.

Proof. Fix an integer $m \geq 1$. The uniform $H^1(B_m)$ bound gives a subsequence and a limit $V_*^{(m)} \in H^1(B_m)$ such that

$$V(\sigma_n) \rightharpoonup V_*^{(m)} \qquad \text{weakly in } H^1(B_m).$$

Rellich compactness gives, after passing to a further subsequence,

$$V(\sigma_n) \to V_*^{(m)} \qquad \text{strongly in } L^2(B_m),$$

after relabelling the subsequence. The Sobolev embedding $H^1(B_m) \hookrightarrow L^6(B_m)$ gives a uniform $L^6(B_m)$ bound, and the limit $V_*^{(m)}$ also belongs to $L^6(B_m)$. Therefore

$$\|V(\sigma_n) - V_*^{(m)}\|_{L^6(B_m)}$$

is uniformly bounded by the triangle inequality. The interpolation inequality

$$\|V(\sigma_n) - V_*^{(m)}\|_{L^3(B_m)} \leq C_m \|V(\sigma_n) - V_*^{(m)}\|_{L^2(B_m)}^{1/2} \|V(\sigma_n) - V_*^{(m)}\|_{L^6(B_m)}^{1/2}$$

then gives strong convergence in $L^3(B_m)$. Repeating this construction for $m = 1, 2, \ldots$ and taking a diagonal subsequence gives strong convergence on every B_m. The nested limits agree by restriction, so they define a single $V_* \in L^3_{\mathrm{loc}}(\mathbb{R}^3)$, and the diagonal subsequence converges to V_* strongly in $L^3_{\mathrm{loc}}(\mathbb{R}^3)$. ∎

2.4. **Reduction to Multibubble Alternatives.** Let the structural Type II hypotheses consist of exhaustion of the local Type II alternatives, repaired-gauge representation, positive finite critical mass, pressure reconstruction, bounded repaired-gauge modulation, Caccioppoli regularity, and the cost comparison used for the compact Type II criterion. If one of these hypotheses fails, the branch is classified by the corresponding analytic failure rather than by a multibubble alternative. If, in addition, the Navier–Stokes implication from compact scale exclusion to the relevant Type II scale alternative is included, the compact conclusion excludes that alternative.

Under these structural hypotheses, the following non-multibubble strata are excluded under separate analytic hypotheses.

(i) Far radiation consists of radiative critical mass whose physical radius stays away from the blow-up point:
$$\lambda(\tau_n)R_n \geq r_* > 0$$
along a radiative annular sequence $R_n \to \infty$. Under a single-point blow-up localization hypothesis, this mass lies in a physically regular exterior region. An exterior-regularity hypothesis asserts that such mass can be separated from the concentrating Type II core and does not count as a core-radiation alternative.

(ii) The rough-core stratum is excluded by the Caccioppoli implication
$$\text{representation}, \quad \text{bounded critical norm}, \quad \text{pressure reconstruction},$$
$$\text{bounded modulation}, \quad \text{Caccioppoli regularity}$$
which yields local windowed H^1 control.

(iii) The retained scale-collapse drift stratum is excluded by the local compact scale-collapse routing theorem Theorem 7.30: finite-cost scale collapse is removed by the localized cost exclusions, while a thick autonomous branch that keeps the retained compact-state inputs enters the scale-rigid local stratum. If one of those retained inputs fails, the branch is already assigned to a named local state-space exit.

Definition 2.22 (Same-point secondary concentration)**.** Same-point secondary concentration means that at least two concentration scales occur at the same physical blow-up point:
$$\lambda_1(t) \ll \lambda_2(t) \to 0.$$
In the primary renormalized coordinates, the secondary bubble appears as radiation at radius
$$R(t) = \frac{\lambda_2(t)}{\lambda_1(t)} \to \infty.$$
Regular strict cascades of this form are excluded by Theorem 2.33, and all same-point cascades are excluded under the nonlinear decoupling hypothesis of Theorem 2.36. Before that hypothesis is imposed, the remaining same-point alternative is precisely the failure of nonlinear no-splitting/profile compatibility for scale-separated profiles.

Definition 2.23 (Multi-point concentration)**.** Multi-point concentration means that at least two physical concentration centers are active. In a renormalization centered at one bubble, the other bubble appears as an escaping profile. This case is excluded by a separated-point decoupling hypothesis reducing each active physical point to its own single-center branch. Before that hypothesis is imposed, the remaining multi-point alternative is failure of separated-point decoupling.

Theorem 2.24 (Reduction to multibubble alternatives)**.** *Assume an admissible Navier–Stokes Type II branch satisfies the structural Type II hypotheses above, the single-point blow-up localization and exterior-regular removal hypotheses for far radiation, and the scale-collapse rigidity hypothesis for scale-collapse drift. If the branch is not a multibubble branch in either the same-point scale-separated form of Theorem 2.22 or the multi-point form of Theorem 2.23, then the compact Type II scale-exclusion conclusion holds. If the Navier–Stokes implication from this scale exclusion to the relevant Type II scale alternative is also included, then that Type II scale alternative is excluded.*

Proof. The structural hypotheses supply representation, critical-mass control, pressure reconstruction, bounded modulation, Caccioppoli regularity, and cost comparison. If the branch is globally L^3-tight, the Caccioppoli estimate gives local windowed H^1 control. Together with positive finite critical mass and tightness, the compact Type II criterion gives infinite localized renormalization cost, and the cost comparison gives the Type II scale-exclusion condition.

If uniform compact-cylinder tightness fails, the radiative branch is present. Far radiation is excluded by the single-point blow-up localization and exterior-regular removal hypotheses, because it lies in a physically smooth exterior region and is separated from the concentrating Type II core. Regular strict same-point cascades are excluded by Theorem 2.33, and all same-point cascades are excluded under the nonlinear same-point decoupling hypothesis. Separated-point multibubbles are excluded under the separated-point decoupling hypothesis. If the branch is assumed not to be in either multibubble form, no radiative or noncompact alternative remains.

If scale-collapse drift occurs, the localized cost exclusions first remove the finite scale-collapsing-cost cases. In the remaining thick autonomous case, failure of any retained compact-state input is an ordered local exit, not an unlabelled residual branch. If all retained inputs hold, the branch is a retained compact scale-collapse state and Theorem 7.30 routes it into the scale-rigid discharge, giving the required contradiction. The scale-collapse drift alternative is therefore excluded without a separate stationary classification theorem.

The remaining rough-core branch is excluded by the Caccioppoli implication, because the structural hypotheses include the pressure, modulation, and regularity hypotheses required for local windowed H^1 control. Thus all non-multibubble alternatives are exhausted, and the compact Type II scale-exclusion conclusion holds. The final Type II scale conclusion follows from the corresponding Navier–Stokes implication when that hypothesis is included. □

Thus, after the structural hypotheses, same-point regular cascade estimates, and separated-point reductions are supplied, the only remaining multibubble alternatives are failure of same-point nonlinear decoupling and failure of separated-point decoupling. Once nonlinear profile evolution and terminal active-frame compactness provide those decouplings, a multibubble failure is an earlier failure of profile completeness, representation, Caccioppoli regularity, retained compact scale-collapse routing, or scale-rigid discharge, not an additional singularity class.

2.5. **Same-Point Multiscale Cascade Exclusion.** For $\lambda > 0$, $x_0 \in \mathbb{R}^3$, and $\phi \in L^3(\mathbb{R}^3)$, define

$$(\Lambda_{\lambda,x_0}\phi)(x) = \lambda^{-1}\phi\left(\frac{x-x_0}{\lambda}\right).$$

Then

$$\|\Lambda_{\lambda,x_0}\phi\|_{L^3(\mathbb{R}^3)} = \|\phi\|_{L^3(\mathbb{R}^3)}.$$

Two parameter sequences (λ_i^n, x_i^n) and (λ_j^n, x_j^n) are asymptotically orthogonal if

$$\frac{\lambda_i^n}{\lambda_j^n} + \frac{\lambda_j^n}{\lambda_i^n} + \frac{|x_i^n - x_j^n|}{\max(\lambda_i^n, \lambda_j^n)} \longrightarrow \infty.$$

Lemma 2.25 (Finite L^3 decoupling)**.** *Let $K < \infty$, let $\phi^1, \dots, \phi^K \in L^3(\mathbb{R}^3)$, and set*

$$g_n^j = \Lambda_{\lambda_j^n, x_j^n}\phi^j.$$

If the parameter sequences are pairwise asymptotically orthogonal, then

$$\left\|\sum_{j=1}^K g_n^j\right\|_{L^3}^3 = \sum_{j=1}^K \|\phi^j\|_{L^3}^3 + o_n(1).$$

Proof. It suffices first to prove the statement for $\phi^j \in C_c^\infty(\mathbb{R}^3)$. The general L^3 case follows by density. For each $\varepsilon > 0$, choose $\psi^j \in C_c^\infty$ with

$$\|\phi^j - \psi^j\|_3 < \varepsilon.$$

Since each $\Lambda_{\lambda_j^n,x_j^n}$ is an L^3 isometry,

$$\left\| \sum_{j=1}^{K} \Lambda_{\lambda_j^n,x_j^n}(\phi^j - \psi^j) \right\|_3 \le K\varepsilon.$$

On bounded L^3 sets, $f \mapsto \|f\|_3^3$ is locally Lipschitz:

$$\big| \|f\|_3^3 - \|h\|_3^3 \big| \le C(\|f\|_3 + \|h\|_3)^2 \|f - h\|_3.$$

Thus the smooth compactly supported case implies the general case after $\varepsilon \downarrow 0$.

Assume now $\phi^j \in C_c^\infty$. The proof is by induction on K. The case $K = 1$ is the scaling invariance of the L^3 norm. Assume the result for $K - 1$, and write

$$G_n = \sum_{j=1}^{K-1} \Lambda_{\lambda_j^n,x_j^n}\phi^j, \qquad g_n^K = \Lambda_{\lambda_K^n,x_K^n}\phi^K.$$

Applying the inverse scaling of the K-th profile gives

$$(\Lambda^{-1}_{\lambda_K^n,x_K^n} G_n)(y) = \sum_{j=1}^{K-1} \frac{\lambda_K^n}{\lambda_j^n} \phi^j \left(\frac{\lambda_K^n y + x_K^n - x_j^n}{\lambda_j^n} \right).$$

For each $j < K$, asymptotic orthogonality gives, along a subsequence, one of the following alternatives:

$$\lambda_K^n/\lambda_j^n \to 0, \qquad \lambda_j^n/\lambda_K^n \to 0,$$

or the scale ratio stays bounded above and below while

$$|x_K^n - x_j^n|/\lambda_j^n \to \infty.$$

In the first case the displayed term is bounded pointwise by

$$\frac{\lambda_K^n}{\lambda_j^n} \|\phi^j\|_\infty \to 0.$$

In the second case, if $\operatorname{supp}\phi^j \subset B_A$, the set on which the term is nonzero is

$$E_n^j = \left\{ y : \left| \frac{\lambda_K^n y + x_K^n - x_j^n}{\lambda_j^n} \right| \le A \right\}.$$

Since $\lambda_j^n/\lambda_K^n \to 0$, the diameter of E_n^j is

$$O(\lambda_j^n/\lambda_K^n) \to 0.$$

Thus $\mathbf{1}_{E_n^j} \to 0$ pointwise outside at most one limiting point after passing to a subsequence, and the term converges to zero a.e. In the third case, the spatial translation tends to infinity in the rescaled variables; hence for each fixed compact B_R, the argument of ϕ^j lies outside $\operatorname{supp}\phi^j$ for all large n. Therefore

$$\Lambda^{-1}_{\lambda_K^n,x_K^n} G_n \to 0 \quad \text{a.e. on } \mathbb{R}^3.$$

The sequence is bounded in L^3, because the inverse scaling is an L^3 isometry and the sum is finite. The Brézis–Lieb lemma, applied to

$$\phi^K + \Lambda^{-1}_{\lambda_K^n,x_K^n} G_n,$$

gives

$$\left\| \phi^K + \Lambda^{-1}_{\lambda_K^n,x_K^n} G_n \right\|_3^3 = \|\phi^K\|_3^3 + \left\| \Lambda^{-1}_{\lambda_K^n,x_K^n} G_n \right\|_3^3 + o_n(1).$$

Using the L^3 isometry again,

$$\|G_n + g_n^K\|_3^3 = \|\phi^K\|_3^3 + \|G_n\|_3^3 + o_n(1).$$

The induction hypothesis gives

$$\|G_n\|_3^3 = \sum_{j=1}^{K-1} \|\phi^j\|_3^3 + o_n(1),$$

and the result follows for K. $\square$

Definition 2.26 (Admissible same-point critical cascade)**.** Let $u_n = u(\cdot, t_n)$, $t_n \uparrow T^*$. A same-point critical cascade decomposition at x_* consists of profiles $\{\phi^j\}_{j\in\mathcal{J}} \subset L^3(\mathbb{R}^3)$, scales $\lambda_j^n \downarrow 0$, centers $x_j^n \to x_*$, and remainders r_n^J such that, for every finite $J \subset \mathcal{J}$,

$$u_n = \sum_{j\in J} \Lambda_{\lambda_j^n, x_j^n}\phi^j + r_n^J.$$

The decomposition is admissible if:

(i) the parameters are pairwise asymptotically orthogonal:

$$\frac{\lambda_i^n}{\lambda_j^n} + \frac{\lambda_j^n}{\lambda_i^n} + \frac{|x_i^n - x_j^n|}{\max(\lambda_i^n, \lambda_j^n)} \to \infty \qquad (i \neq j);$$

(ii) critical L^3 decoupling with remainder holds:

$$\|u_n\|_3^3 = \sum_{j\in J} \|\phi^j\|_3^3 + \|r_n^J\|_3^3 + o_n(1);$$

(iii) there is $\eta > 0$ such that

$$\|\phi^j\|_3 \geq \eta \qquad \text{for every } j \in \mathcal{J}.$$

Definition 2.27 (Local active-cascade mass budget)**.** An admissible same-point critical cascade decomposition has local active mass budget M if, for every finite set $J \subset \mathcal{J}$,

$$\sum_{j\in J} \|\phi^j\|_{L^3(\mathbb{R}^3)}^3 \leq M^3.$$

This is a ledger condition on the extracted active profiles in the selected local state-space branch. It is not a global $L^3(\mathbb{R}^3)$ bound for the original solution.

Theorem 2.28 (Uniform bound on cascade length)**.** *Assume u_n admits an admissible same-point critical cascade decomposition with local active mass budget M. If every active profile satisfies*

$$\|\phi^j\|_3 \geq \eta > 0,$$

then the number of active profiles is finite and

$$\#\mathcal{J} \leq \left\lfloor \frac{M^3}{\eta^3} \right\rfloor.$$

Proof. Let $J \subset \mathcal{J}$ be finite. The local active mass budget gives

$$M^3 \geq \sum_{j\in J} \|\phi^j\|_3^3 \geq (\#J)\eta^3.$$

Thus

$$\#J \leq \frac{M^3}{\eta^3}$$

for every finite $J \subset \mathcal{J}$. If $\mathcal{J}$ had more than $\lfloor M^3/\eta^3 \rfloor$ elements, choosing

$$\#J = \left\lfloor \frac{M^3}{\eta^3} \right\rfloor + 1$$

would contradict the preceding inequality. $\square$

Definition 2.29 (Regular strict same-point cascade)**.** Assume there are $J \geq 2$ active profiles ordered by scale:

$$\lambda_1^n \ll \lambda_2^n \ll \cdots \ll \lambda_J^n, \qquad \lambda_j^n \to 0, \qquad x_j^n \to x_*.$$

Set

$$\mu_j^n = \frac{\lambda_1^n}{\lambda_j^n} \to 0 \qquad (j \geq 2).$$

The cascade is regular in the innermost coordinates if, after passing to a subsequence:

(i)

$$\xi_j^n = \frac{x_* - x_j^n}{\lambda_j^n} \longrightarrow \xi_j \in \mathbb{R}^3 \qquad (j \geq 2);$$

(ii) each outer profile is C^2 near ξ_j;

(iii) for every $R < \infty$,

$$\sup_{|z-\xi_j| \leq R\mu_j^n} \left(|\phi^j(z)| + |\nabla \phi^j(z)| + |\nabla^2 \phi^j(z)|\right) < \infty$$

uniformly for all large n;

(iv) the innermost repaired-gauge representation is available:

$$V_1^n(y,\tau) = \lambda_1(t) u(x_*(t) + \lambda_1(t) y, t)$$

and solves the repaired-gauge Navier–Stokes equation on compact (y,τ)-cylinders, modulo the outer-bubble perturbation.

If $|\xi_j^n| \to \infty$, the corresponding outer bubble is locally invisible in the innermost variables by Theorem 2.34, provided the pressure interaction satisfies the decoupling hypothesis stated below.

Lemma 2.30 (Inner expansion of regular outer bubbles)**.** *For $j \geq 2$, define*

$$W_j^n(y) = \lambda_1^n (\lambda_j^n)^{-1} \phi^j \left(\frac{\lambda_1^n y + x_* - x_j^n}{\lambda_j^n} \right) = \mu_j^n \phi^j(\xi_j^n + \mu_j^n y).$$

For every fixed $R < \infty$,

$$W_j^n(y) = \mu_j^n \phi^j(\xi_j) + (\mu_j^n)^2 D\phi^j(\xi_j) y + \mathcal{R}_j^n(y) \qquad \text{on } B_R,$$

where

$$\|\mathcal{R}_j^n\|_{L^\infty(B_R)} \leq C_{j,R} \left((\mu_j^n)^3 + \mu_j^n |\xi_j^n - \xi_j|^2 + (\mu_j^n)^2 |\xi_j^n - \xi_j|\right) = o(\mu_j^n) + O((\mu_j^n)^3).$$

In particular,

$$W_j^n = c_j^n + A_j^n y + o_{L^\infty(B_R)}(\mu_j^n),$$

with

$$c_j^n = \mu_j^n \phi^j(\xi_j), \qquad A_j^n = (\mu_j^n)^2 D\phi^j(\xi_j), \qquad \|A_j^n\| = O((\mu_j^n)^2).$$

Proof. Fix R. For $y \in B_R$, Taylor's theorem at ξ_j gives

$$\phi^j(\xi_j^n + \mu_j^n y) = \phi^j(\xi_j) + D\phi^j(\xi_j)(\xi_j^n - \xi_j + \mu_j^n y) + \frac{1}{2}(\xi_j^n - \xi_j + \mu_j^n y)^T D^2\phi^j(\zeta_{j,n,y})(\xi_j^n - \xi_j + \mu_j^n y),$$

where $\zeta_{j,n,y}$ lies on the segment between ξ_j and $\xi_j^n + \mu_j^n y$. Multiplying by μ_j^n yields

$$W_j^n(y) = \mu_j^n \phi^j(\xi_j) + \mu_j^n D\phi^j(\xi_j)(\xi_j^n - \xi_j) + (\mu_j^n)^2 D\phi^j(\xi_j) y + O\left(\mu_j^n |\xi_j^n - \xi_j + \mu_j^n y|^2\right).$$

The term $\mu_j^n D\phi^j(\xi_j)(\xi_j^n - \xi_j)$ is independent of y and may be absorbed into the constant part if one uses

$$c_j^n = \mu_j^n \phi^j(\xi_j^n).$$

With the displayed choice $c_j^n = \mu_j^n \phi^j(\xi_j)$, it is part of the remainder. Since $\xi_j^n \to \xi_j$ and $\mu_j^n \to 0$, the stated estimate follows. ∎

For the dynamical reduction it is useful to use the exact constant

$$c_j^n = \mu_j^n \phi^j(\xi_j^n),$$

so that

$$W_j^n(y) = c_j^n + (\mu_j^n)^2 D\phi^j(\xi_j^n) y + O_{L^\infty(B_R)}((\mu_j^n)^3 R^2).$$

Since Theorem 2.28 gives $J < \infty$,

$$c_{\text{out}}^n = \sum_{j=2}^{J} c_j^n = O\left(\sum_{j=2}^{J} \mu_j^n\right) \to 0$$

on every fixed cascade sequence, provided the profiles are locally bounded at their offsets.

Lemma 2.31 (Absorption of constant ambient velocity)**.** *Suppose on a compact cylinder*

$$V = V_{\text{in}} + c(\tau) + S,$$

where $c(\tau)$ is independent of y and S is a perturbation. Ignoring S, the equation for V_{in} is

$$\partial_\tau V_{\text{in}} + (V_{\text{in}} \cdot \nabla)V_{\text{in}} + \nabla P_{\text{in}} = \nu\Delta V_{\text{in}} + a(V_{\text{in}} + y \cdot \nabla V_{\text{in}}) + (b - c) \cdot \nabla V_{\text{in}},$$

modulo spatially constant forcing terms, which are absorbed by the moving translation gauge. Thus $b_{\text{eff}} = b - c$.

Proof. Since c is independent of y,

$$(V_{\text{in}} + c) \cdot \nabla(V_{\text{in}} + c) = (V_{\text{in}} \cdot \nabla)V_{\text{in}} + c \cdot \nabla V_{\text{in}}.$$

Terms depending only on τ, such as $\partial_\tau c - ac$, are spatially constant forces and are absorbed in the equivalent translation-gauge description. With the stated sign convention, the effective translation parameter is $b - c$. ∎

The leading constant part of every outer bubble in the inner coordinates is absorbed into b_{eff}. Since

$$|c_{\text{out}}^n| \le \sum_{j=2}^{J} \mu_j^n |\phi^j(\xi_j^n)|,$$

finite cascade length and local boundedness of the profiles imply

$$\sup_n |c_{\text{out}}^n| < \infty, \qquad c_{\text{out}}^n \to 0.$$

Thus bounded modulation is preserved whenever the unperturbed repaired-gauge b-parameter is bounded.

Definition 2.32 (Perturbative scale-collapse robustness)**.** Let E_n be the remaining inner-coordinate error after subtracting the innermost bubble and absorbing c^n_{out} into b_{eff}. It contains the linear shear terms

$$\sum_{j=2}^{J} (\mu^n_j)^2 D\phi^j(\xi^n_j)y,$$

Taylor remainders of size $O((\mu^n_j)^3)$ on fixed balls, the rescaled critical remainder, and pressure cross-terms generated by the interaction of the innermost profile with the outer perturbation. Perturbative scale-collapse robustness means that on every compact cylinder $B_R \times [\tau_0, \tau_0 + L]$:

(i) $E_n \to 0$ in $L^1_\tau H^{-1}_y(B_R)$, or in a stronger topology sufficient for the scale-collapse compactness passage;
(ii) the pressure generated by cross-terms converges to zero in the pressure topology used by the repaired-gauge and scale-collapse arguments;
(iii) local compactness, autonomous modulation, and recognition of the retained compact scale-collapse state are stable under adding $E_n \to 0$;
(iv) the perturbation creates no unrecorded terminal state: after passing to the scale-collapse limit, every failure of a retained compact-state input is a named local alternative, and otherwise the branch is eligible for Theorem 7.30.

Under Theorem 2.32, the innermost bubble gives a single-bubble scale-collapse branch with bounded modulation and positive critical mass.

Theorem 2.33 (Regular same-point cascade exclusion)**.** *Let u be a suitable Navier–Stokes solution on a terminal neighbourhood of (x_*, T^*). Let $t_n \uparrow T^*$ be a concentrating sequence at x_*. Assume:*

(i) *$u_n = u(\cdot, t_n)$ admits an admissible same-point critical cascade decomposition with local active mass budget M;*
(ii) *every active bubble satisfies $\|\phi^j\|_3 \geq \eta > 0$;*
(iii) *the active bubbles form a regular strict same-point cascade;*
(iv) *the innermost scale admits the repaired-gauge representation with bounded modulation, pressure reconstruction, Caccioppoli regularity, and positive finite critical mass;*
(v) *perturbative scale-collapse robustness holds;*
(vi) *the innermost scale-collapse branch is either already routed to a named local nonretained alternative or satisfies the retained compact scale-collapse state of Theorem 7.27.*

Then the cascade cannot occur.

Proof. By Theorem 2.28, the number of active bubbles is finite:

$$J \leq \left\lfloor \frac{M^3}{\eta^3} \right\rfloor.$$

Let λ^n_1 be the smallest scale, and for $j \geq 2$ set

$$\mu^n_j = \lambda^n_1 / \lambda^n_j \to 0.$$

Theorem 2.30 gives, on every fixed B_R,

$$\lambda^n_1 (\lambda^n_j)^{-1} \phi^j \left(\frac{\lambda^n_1 y + x_* - x^n_j}{\lambda^n_j} \right) = c^n_j + (\mu^n_j)^2 D\phi^j(\xi^n_j) y + O_{L^\infty(B_R)}((\mu^n_j)^3 R^2).$$

Summing over $2 \le j \le J$, the constant part

$$c^n_{\mathrm{out}} = \sum_{j=2}^{J} c^n_j$$

is bounded and tends to zero. By Theorem 2.31, it is absorbed into the repaired-gauge translation parameter:

$$b^n_{\mathrm{eff}} = b^n - c^n_{\mathrm{out}}.$$

The bounded-modulation hypothesis for b^n, together with boundedness of c^n_{out}, gives

$$\sup_n |b^n_{\mathrm{eff}}| < \infty.$$

The nonconstant outer contribution is

$$\sum_{j=2}^{J} (\mu^n_j)^2 D\phi^j(\xi^n_j) y + O_{L^\infty(B_R)}\left(\sum_{j=2}^{J} (\mu^n_j)^3 R^2 \right),$$

which converges to zero uniformly on each fixed B_R. The rescaled profile-decomposition remainder and pressure cross-terms are included in E_n, and perturbative scale-collapse robustness gives $E_n \to 0$ in the topology required by the retained compact scale-collapse passage.

Consequently, the innermost branch satisfies the repaired-gauge Navier–Stokes equation with bounded effective modulation, positive finite critical mass, pressure reconstruction, Caccioppoli regularity, and a perturbation that vanishes in the scale-collapse limit. The scale $\lambda_1(t) \to 0$ gives the scale-collapse hypothesis. If any retained compact-state input fails, Theorem 7.28 places the branch in a named local nonretained alternative, which is incompatible with being an unresolved regular strict retained cascade. Otherwise the branch is a retained compact scale-collapse state. Applying Theorem 7.30 excludes that state by routing it into the already discharged scale-rigid branch. This contradicts the active innermost mass floor $\|\phi^1\|_3 \ge \eta > 0$. Hence the regular strict same-point cascade cannot occur. □

The preceding theorem uses the following analytic hypotheses: strong L^3 profile decomposition with Brézis–Lieb decoupling and remainder decoupling; a uniform active-bubble mass floor $\|\phi^j\|_3 \ge \eta$; finite-offset C^2 nested-cascade regularity; repaired-gauge representation of the innermost branch with pressure, Caccioppoli regularity, bounded modulation, and positive finite critical mass; perturbative scale-collapse robustness; and retained compact scale-collapse routing for the innermost branch. Bounded critical mass alone proves only uniformly bounded cascade length. Full exclusion of the regular same-point cascade requires the inner-decoupling and local retained-state routing hypotheses.

Lemma 2.34 (Escaping offsets vanish locally)**.** *Let*

$$W^n_j(y) = \mu^n_j \phi^j(\xi^n_j + \mu^n_j y), \qquad \mu^n_j = \frac{\lambda^n_1}{\lambda^n_j} \to 0.$$

If $\phi^j \in L^3(\mathbb{R}^3)$ and $|\xi^n_j| \to \infty$, then, for every fixed $R < \infty$,

$$\|W^n_j\|_{L^3(B_R)} \to 0.$$

If additionally

$$\|\nabla W^n_j\|_{L^3(B_R)} + \|D^2 W^n_j\|_{L^3(B_R)} \to 0$$

for every fixed R, then $W^n_j \to 0$ in $W^{2,3}_{\mathrm{loc}}$.

Proof. With $z = \xi_j^n + \mu_j^n y$ and $dy = (\mu_j^n)^{-3}\,dz$,

$$\|W_j^n\|_{L^3(B_R)}^3 = \int_{B_R} (\mu_j^n)^3 |\phi^j(\xi_j^n + \mu_j^n y)|^3\,dy = \int_{B(\xi_j^n, \mu_j^n R)} |\phi^j(z)|^3\,dz.$$

Since $|\xi_j^n| \to \infty$ and $\mu_j^n R \to 0$, the balls

$$B(\xi_j^n, \mu_j^n R)$$

eventually lie in $\{|z| > A\}$ for every fixed $A < \infty$. Since $\phi^j \in L^3(\mathbb{R}^3)$,

$$\int_{|z|>A} |\phi^j(z)|^3\,dz \to 0 \quad \text{as } A \to \infty.$$

Thus, for all sufficiently large n,

$$\|W_j^n\|_{L^3(B_R)}^3 \le \int_{|z|>A} |\phi^j(z)|^3\,dz,$$

and the right-hand side is arbitrarily small for large A. This proves local L^3 vanishing. The $W^{2,3}_{\mathrm{loc}}$ conclusion is precisely the additional derivative-tail hypothesis. □

The pressure contribution of an escaping-offset bubble is not automatic from local velocity vanishing, because pressure is nonlocal. The needed escaping-offset pressure-decoupling hypothesis is that, for every compact B_R and compact time window, the pressure generated by the cross source

$$U_n \odot W_j^n + W_j^n \odot U_n + W_j^n \otimes W_j^n$$

converges to zero modulo constants in the pressure topology used for the repaired-gauge and scale-collapse passages. Under this hypothesis, escaping-offset outer profiles are perturbative for the innermost scale-collapse branch.

Proposition 2.35 (Perturbative estimates for the same-point cascade)**.** *Let U_n be the innermost renormalized branch after absorbing c^n_{out} into b^n, and let S_n be the remaining outer perturbation on $B_R \times I$. Assume*

$$\|S_n\|_{L^\infty_\tau W^{2,\infty}_y(B_R)} + \|\partial_\tau S_n\|_{L^1_\tau H^{-1}_y(B_R)} \longrightarrow 0,$$
$$\|U_n\|_{L^\infty_\tau L^3_y(B_{2R})} + \|U_n\|_{L^2_\tau H^1_y(B_{2R})} \le C_R,$$

and assume the pressure reconstruction satisfies the local Calderon–Zygmund estimate for cross terms. Then every perturbative term produced by replacing $U_n + S_n$ with U_n vanishes in

$$L^1_\tau H^{-1}_y(B_R).$$

Proof. The time and Laplacian perturbations satisfy

$$\|\partial_\tau S_n\|_{L^1 H^{-1}} \to 0, \qquad \|\Delta S_n\|_{L^1 H^{-1}} \le C_R \|S_n\|_{L^1 W^{2,\infty}} \to 0.$$

The linear transport terms obey

$$\|(U_n \cdot \nabla) S_n\|_{L^1 H^{-1}(B_R)} \le C_R \|U_n\|_{L^2 L^2(B_R)} \|\nabla S_n\|_{L^2 L^\infty(B_R)} \to 0,$$

and

$$\|(S_n \cdot \nabla) U_n\|_{L^1 H^{-1}(B_R)} \le C_R \|S_n\|_{L^\infty L^\infty(B_R)} \|\nabla U_n\|_{L^1 L^2(B_R)} \to 0.$$

The quadratic perturbation satisfies

$$\|(S_n \cdot \nabla) S_n\|_{L^1 H^{-1}(B_R)} \le C_R \|S_n\|_{L^\infty L^\infty} \|\nabla S_n\|_{L^1 L^\infty} \to 0.$$

The modulation terms satisfy

$$\|a_n(S_n + y \cdot \nabla S_n) + b_n \cdot \nabla S_n\|_{L^1 H^{-1}(B_R)} \le C_R(M_{ab}) \|S_n\|_{L^1 W^{1,\infty}(B_R)} \to 0.$$

For the pressure, the cross source is

$$2U_n \odot S_n + S_n \otimes S_n.$$

On B_{2R},

$$\|U_n \odot S_n\|_{L^1_\tau L^{3/2}_y} \le \|S_n\|_{L^\infty_{\tau,y}} \|U_n\|_{L^1_\tau L^3_y} \to 0,$$

and

$$\|S_n \otimes S_n\|_{L^1_\tau L^{3/2}_y} \to 0.$$

The local pressure estimate then gives convergence of the cross pressure modulo constants in the required pressure topology, hence its gradient vanishes in $L^1_\tau H^{-1}_y(B_R)$. □

For a regular finite strict cascade, Theorem 2.30 and bounded relative modulation give the required $S_n \to 0$ estimates. Therefore regular nested-cascade structure, bounded modulation, pressure reconstruction, and local windowed H^1 control imply perturbative scale-collapse robustness.

Definition 2.36 (Same-point nonlinear decoupling)**.** Same-point nonlinear decoupling means that every finite same-point strict scale-separated profile family satisfies:

(i) strong critical decomposition,

$$u_n = \sum_{j=1}^{J} \Lambda_{\lambda_j^n, x_j^n} \phi^j + r_n, \qquad \|u_n\|_3^3 = \sum_{j=1}^{J} \|\phi^j\|_3^3 + \|r_n\|_3^3 + o_n(1);$$

(ii) nonlinear profile stability: profiles below the critical small-data threshold are perturbative, and every active singular profile has L^3-mass at least $\varepsilon_{\mathrm{sd}}$;

(iii) innermost velocity decoupling: after rescaling by the smallest scale and absorbing all constant ambient velocities into the translation gauge,

$$V_n = U_n + S_n, \qquad S_n \to 0$$

in the local velocity topology required by perturbative scale-collapse robustness;

(iv) pressure decoupling: all pressure cross-terms between the innermost profile, outer profiles, and the remainder vanish modulo constants in the repaired-gauge scale-collapse pressure topology;

(v) modulation compatibility: the effective repaired-gauge parameters of the innermost branch remain bounded after ambient constants are absorbed;

(vi) scale-collapse topology compatibility: the full interaction error vanishes in the retained compact scale-collapse passage.

Theorem 2.37 (Conditional no same-point multibubble theorem)**.** *Assume an admissible Navier–Stokes Type II branch has the structural hypotheses needed to enter the repaired-gauge Type II class. Assume same-point nonlinear decoupling, a finite local active-cascade mass budget M, the innermost repaired-gauge representation, and the local retained compact scale-collapse routing statement for the innermost branch: either a retained compact-state input fails and the branch is assigned to a named local exit, or the retained compact scale-collapse theorem Theorem 7.30 applies. Then no same-point multibubble cascade of active singular profiles can occur.*

Proof. Let a same-point strict scale-separated cascade be given. By Theorem 2.36, the decomposition is strongly decoupled in critical L^3, and every active singular bubble carries at least $\varepsilon_{\mathrm{sd}}$ critical mass. The selected active-profile ledger has a finite local budget M, so Theorem 2.28 gives

$$J \le \left\lfloor \frac{M^3}{\varepsilon_{\mathrm{sd}}^3} \right\rfloor.$$

Choose the innermost scale λ_1^n. The nonlinear decoupling hypothesis says that, in innermost variables and after translation-gauge absorption of constant ambient velocities,

$$V_n = U_n + S_n, \qquad S_n \to 0$$

in the perturbative topology required for scale-collapse compactness, including pressure and modulation compatibility. Hence the innermost branch U_n satisfies the repaired-gauge Navier–Stokes equation with bounded effective modulation, positive finite critical mass, pressure reconstruction, and Caccioppoli regularity, up to an error that vanishes in the scale-collapse limiting passage.

The scale $\lambda_1^n \to 0$ gives the scale-collapse hypothesis. If a retained compact-state input fails, Theorem 7.28 places the branch in a named local exit rather than in an unresolved same-point cascade. Otherwise Theorem 7.30 applies to the retained innermost branch and excludes it by the scale-rigid discharge. The nonzero branch input is the active bubble mass floor:

$$\|\phi^1\|_3 \geq \varepsilon_{\mathrm{sd}} > 0.$$

Thus the routed scale-collapse branch contradicts the existence of the active innermost bubble. □

Thus same-point cascade exclusion leaves only the failure of same-point nonlinear decoupling, namely the no-splitting/profile-compatibility multibubble alternative. If a regular strict same-point cascade of active singular profiles is not excluded, then one of the following has failed: profile decomposition or remainder decoupling, active-bubble small-data separation, escaping-offset pressure decoupling, local profile regularity or innermost representation, or the retained compact scale-collapse routing statement for the innermost branch. These are analytic profile, representation, interaction, or local state-space routing failures; a below-threshold profile is regular and is not a singular cascade bubble.

3. Local Single-Core Criterion

3.1. **Setup and definitions.** For $R > 0$, write

$$Q_R = B_R \times (-R^2, 0).$$

For a suitable pair (v, q) on Q_R, define

$$A_v(R) = R^{-1} \operatorname*{ess\,sup}_{-R^2<s<0} \int_{B_R} |v(y,s)|^2 \, dy,$$

$$E_v(R) = R^{-1} \int_{Q_R} |\nabla v|^2 \, dy \, ds,$$

$$C_v(R) = R^{-2} \int_{Q_R} |v|^3 \, dy \, ds,$$

$$D_q(R) = R^{-2} \int_{-R^2}^{0} \int_{B_R} |q(y,s) - (q)_{B_R}(s)|^{3/2} \, dy \, ds.$$

All pressure quantities are understood modulo functions of time.

Definition 3.1 (Bounded selected-window limit)**.** Let (V_n, P_n) be a sequence of represented local rescalings on a compact cylinder Q_r. A selected-window subsequence has a bounded selected-window limit on Q_r if, after passing to a subsequence,

$$V_n \to V_* \qquad \text{strongly in } L^3(Q_r),$$

where $V_* \in L^\infty(Q_r)$. The limit is nonzero if $V_* \not\equiv 0$ on Q_r.

Definition 3.2 (Local Type II sequence)**.** A represented sequence is called a local Type II sequence if it has positive local Caffarelli–Kohn–Nirenberg concentration on some fixed compact cylinder and no selected-window subsequence has a nonzero bounded selected-window limit on any compact cylinder in the sense of Theorem 3.1.

Definition 3.3 (Compact single-core vanishing-dissipation branch)**.** Fix radii $0 < r_* < R_*$, constants $M < \infty$, $\eta_0 > 0$, and a sequence of represented local rescalings (V_n, P_n, a_n, b_n) on Q_{R_*}. The sequence is a compact single-core vanishing-dissipation branch if:

(i) each (V_n, P_n) is suitable on Q_{R_*} and solves the represented Navier–Stokes equation

$$\partial_s V_n + (V_n \cdot \nabla) V_n + \nabla P_n = \nu \Delta V_n + a_n(s)(V_n + y \cdot \nabla V_n) + b_n(s) \cdot \nabla V_n, \qquad \nabla \cdot V_n = 0; \tag{1}$$

(ii) the compact-cylinder suitable estimates are uniformly bounded:

$$A_{V_n}(R_*) + E_{V_n}(R_*) + C_{V_n}(R_*) + D_{P_n}(R_*) \le M;$$

(iii) the selected core has positive CKN concentration on Q_{r_*}:

$$C_{V_n}(r_*) + D_{P_n}(r_*) \ge \eta_0 \qquad \text{for all } n;$$

(iv) the localized scale and center are nondegenerate for the selected core, so the representation theorem gives the variables in (1);

(v) the modulation coefficients are uniformly bounded on the selected cylinder:

$$\|a_n\|_{L^\infty(-R_*^2,0)} + \|b_n\|_{L^\infty(-R_*^2,0)} \le M;$$

(vi) the branch has vanishing localized dissipation in the sense of Theorem 3.4 below.

Definition 3.4 (Vanishing localized dissipation)**.** Let $r_* < \rho < R_*$. Selected windows are first translated and rescaled to the fixed cylinders $Q_{r_*} \Subset Q_\rho \Subset Q_{R_*}$. The localized dissipation on Q_ρ is

$$\mathcal{D}_n(\rho) = \int_{Q_\rho} |\nabla V_n|^2 \, dy \, ds.$$

The branch has vanishing localized dissipation if, after passing to a selected-window subsequence, there is $\rho \in (r_*, R_*)$ such that

$$\mathcal{D}_n(\rho) \to 0.$$

3.2. **Local analytic inputs.** The compact single-core argument is based on the following local analytic statements.

Theorem 3.5 (Local concentration dichotomy)**.** *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T-\delta, T)$, and suppose $\Sigma(T) \neq \emptyset$. Then, for every $x_0 \in \Sigma(T)$, using the pressure-normalized quantities C, D, there are $\eta > 0$ and $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \qquad \textit{for all } n.$$

Moreover, at each singular point exactly one of the following alternatives holds: a local Type I scale-invariant bound holds, or the point lies in the local Type II alternative, meaning that no such local Type I bound holds.

Proof. By the Caffarelli–Kohn–Nirenberg epsilon-regularity criterion, there is a universal $\varepsilon_{\mathrm{CKN}} > 0$ such that if

$$C((x_0, T), r) + D((x_0, T), r) < \varepsilon_{\mathrm{CKN}}$$

for all sufficiently small r, then u is locally bounded in a smaller backward cylinder ending at (x_0, T). Since $x_0 \in \Sigma(T)$, this regularity conclusion is impossible. Hence for every k there exists $0 < r_k < 1/k$ with

$$C((x_0, T), r_k) + D((x_0, T), r_k) \geq \varepsilon_{\mathrm{CKN}}.$$

After passing to a decreasing subsequence and setting $\eta = \varepsilon_{\mathrm{CKN}}$, the displayed concentration sequence follows. The Type I/Type II split is the dichotomy between existence and failure of the local Type I scale-invariant bound at the same point. □

Proposition 3.6 (Passage to local Type II sequences)**.** *Let (x_0, T) be a concentration point in the local Type II alternative of Theorem 3.5. After choosing the positive concentration scales given there and applying the local scale-translation selection used in the present section to a nondegenerate compact single core, the represented selected windows form a sequence with positive local CKN concentration on a fixed compact cylinder. In the selected-window Type II branch, no subsequence has a nonzero bounded selected-window limit on any compact cylinder. Hence the represented selected windows are a local Type II sequence in the sense of Theorem 3.2.*

Proof. The local concentration dichotomy supplies the positive CKN concentration sequence and the exclusion of the local Type I alternative at the chosen concentration point. The scale-translation selection rewrites those shrinking cylinders as fixed selected windows. The last condition is the selected-window formulation of the Type II branch considered here: any branch with a nonzero bounded selected-window limit belongs to the bounded-window alternative, not to Theorem 3.2. □

Proposition 3.7 (Local representation input)**.** *For a nondegenerate selected single core, the local scale-translation construction gives represented variables (V_n, P_n, a_n, b_n) satisfying (1) on compact cylinders. Failure of the localized nondegeneracy condition is a multibubble, cascade, or gauge-degenerate alternative, not a compact single-core branch.*

Proposition 3.8 (Local pressure decomposition and stability). *For every $B_r \Subset B_R$, the represented pressure decomposes as*

$$P_n = P_{\mathrm{loc},n} + H_n, \qquad -\Delta P_{\mathrm{loc},n} = \partial_i\partial_j(\zeta V_{n,i}V_{n,j}),$$

where $\zeta \in C_c^\infty(B_R)$ equals one on B_r, and H_n is harmonic in B_r. The local part satisfies the Calderon–Zygmund bound [CZ52, Ste70]

$$\|P_{\mathrm{loc},n}\|_{L^{3/2}(B_R)} \leq C\|V_n\|^2_{L^3(B_R)}$$

for a.e. time, and the harmonic part is controlled modulo spatial means by the local pressure norm.

Moreover, if $V_n \to 0$ strongly in $L^3(Q_R)$, then for every $r < R$,

$$P_n - (P_n)_{B_r} \to 0 \quad \textit{strongly in } L^{3/2}(Q_r).$$

Lemma 3.9 (Localization of pressure oscillation bounds). *Let $0 < r < R$, and let $q \in L^{3/2}(Q_R)$. Then*

$$D_q(r) \leq C\left(\frac{R}{r}\right)^2 D_q(R),$$

where C is universal.

Proof. For a.e. $s \in (-r^2, 0)$, Jensen's inequality gives

$$|(q)_{B_r}(s) - (q)_{B_R}(s)|^{3/2} \leq \frac{1}{|B_r|}\int_{B_r} |q(y,s) - (q)_{B_R}(s)|^{3/2}\,dy.$$

Hence

$$\int_{B_r} |q - (q)_{B_r}(s)|^{3/2}\,dy \leq C\int_{B_r} |q - (q)_{B_R}(s)|^{3/2}\,dy.$$

Integrating over $(-r^2, 0)$ and using $Q_r \subset Q_R$,

$$\begin{aligned} D_q(r) &\leq Cr^{-2}\int_{-r^2}^0\int_{B_r} |q - (q)_{B_R}(s)|^{3/2}\,dy\,ds \\ &\leq Cr^{-2}\int_{-R^2}^0\int_{B_R} |q - (q)_{B_R}(s)|^{3/2}\,dy\,ds = C\left(\frac{R}{r}\right)^2 D_q(R). \end{aligned}$$

□

3.3. **Compactness and pressure stability.**

Proposition 3.10 (Compactness on selected windows). *Let (V_n, P_n) be suitable on Q_{R_*}, solve* (1), *and satisfy*

$$A_{V_n}(R_*) + E_{V_n}(R_*) + C_{V_n}(R_*) + D_{P_n}(R_*) \leq M,$$

with

$$\|a_n\|_{L^\infty(-R_*^2,0)} + \|b_n\|_{L^\infty(-R_*^2,0)} \leq M.$$

Then for every $r < R_*$, after passing to a subsequence there are (V, P) such that

$$\begin{aligned} V_n &\overset{*}{\rightharpoonup} V \quad \textit{in } L^\infty(-r^2, 0; L^2(B_r)), \\ V_n &\rightharpoonup V \quad \textit{in } L^2(-r^2, 0; H^1(B_r)), \\ V_n &\to V \quad \textit{strongly in } L^3(Q_r), \end{aligned}$$

and

$$P_n - (P_n)_{B_r} \rightharpoonup P \quad \textit{in } L^{3/2}(Q_r).$$

After passing to a further subsequence, a_n and b_n converge weak- in L^∞ to bounded coefficients, and the limit satisfies the corresponding represented equation distributionally on smaller cylinders.*

Proof. The bounds on A and E give weak-* compactness in $L^\infty L^2$ and weak compactness in $L^2 H^1$ on smaller cylinders. Fix $r < r_1 < R_*$, and choose m so large that

$$H_0^m(B_{r_1}) \hookrightarrow W_0^{1,\infty}(B_{r_1}).$$

We claim that $\partial_s V_n$ is uniformly bounded in

$$L^1(-r_1^2, 0; H^{-m}(B_{r_1})).$$

Let $\varphi \in H_0^m(B_{r_1})$. Pairing (1) with φ, integrating by parts in space where needed, and using the embedding above gives

$$|\langle (V_n\cdot\nabla)V_n, \varphi\rangle| = \left|\int_{B_{r_1}} V_{n,i}V_{n,j}\partial_j\varphi_i\,dy\right| \le C\|V_n\|_{L^3(B_{r_1})}^2\|\varphi\|_{H^m},$$

$$|\langle \nu\Delta V_n, \varphi\rangle| \le C\|\nabla V_n\|_{L^2(B_{r_1})}\|\varphi\|_{H^m},$$

$$|\langle \nabla P_n, \varphi\rangle| = \left|\int_{B_{r_1}} (P_n - (P_n)_{B_{r_1}})\nabla\cdot\varphi\,dy\right| \le C\|P_n - (P_n)_{B_{r_1}}\|_{L^{3/2}(B_{r_1})}\|\varphi\|_{H^m}.$$

The modulation terms obey

$$|\langle a_n V_n, \varphi\rangle| \le C\|V_n\|_{L^2(B_{r_1})}\|\varphi\|_{H^m},$$

$$|\langle a_n\, y\cdot\nabla V_n, \varphi\rangle| = \left|\int_{B_{r_1}} a_n V_{n,i}\partial_j(y_j\varphi_i)\,dy\right| \le C\|V_n\|_{L^2(B_{r_1})}\|\varphi\|_{H^m},$$

and

$$|\langle b_n\cdot\nabla V_n, \varphi\rangle| = \left|\int_{B_{r_1}} V_{n,i}b_{n,j}\partial_j\varphi_i\,dy\right| \le C\|V_n\|_{L^2(B_{r_1})}\|\varphi\|_{H^m},$$

where the constant uses r_1 and the uniform L^∞ bounds for a_n, b_n. Integrating these estimates in time gives a uniform bound because

$$\int_{-r_1^2}^0 \|V_n(s)\|_{L^3(B_{r_1})}^2\,ds \le r_1^{2/3}\|V_n\|_{L^3(Q_{r_1})}^2,$$

$$\int_{-r_1^2}^0 \|\nabla V_n(s)\|_{L^2(B_{r_1})}\,ds \le r_1\|\nabla V_n\|_{L^2(Q_{r_1})},$$

$$\int_{-r_1^2}^0 \|P_n(s) - (P_n)_{B_{r_1}}(s)\|_{L^{3/2}(B_{r_1})}\,ds \le r_1^{2/3}\|P_n - (P_n)_{B_{r_1}}\|_{L^{3/2}(Q_{r_1})},$$

and

$$\int_{-r_1^2}^0 \|V_n(s)\|_{L^2(B_{r_1})}\,ds \le r_1^2\|V_n\|_{L^\infty(-r_1^2,0;L^2(B_{r_1}))}.$$

The uniform C, E, D, and A bounds therefore prove the claimed time-derivative bound; the D-bound on Q_{r_1} follows from the bound on Q_{R_*} by Theorem 3.9.

The compact embedding $H^1(B_{r_1}) \Subset L^2(B_r)$, together with Aubin–Lions compactness [Aub63, LM72, Sim87], gives

$$V_n \to V \qquad \text{strongly in } L^2(Q_r)$$

after passing to a subsequence. The same $L^\infty L^2 \cap L^2 H^1$ bounds give the parabolic interpolation estimate

$$\|V_n\|_{L^{10/3}(Q_r)} \le C(r, M).$$

Indeed, for a.e. s, the Sobolev and Gagliardo–Nirenberg inequalities give

$$\|V_n(s)\|_{L^{10/3}(B_r)} \le C\|V_n(s)\|_{L^2(B_r)}^{2/5}\|\nabla V_n(s)\|_{L^2(B_r)}^{3/5}.$$

Raising to the power 10/3, integrating in time, and using the A and E bounds yields

$$\int_{-r^2}^{0} \|V_n(s)\|_{L^{10/3}(B_r)}^{10/3}\, ds \le C \left(\operatorname*{ess\,sup}_{-r^2<s<0} \|V_n(s)\|_{L^2(B_r)}^2 \right)^{2/3} \int_{Q_r} |\nabla V_n|^2\, dy\, ds \le C(r, M).$$

The same estimate applies to V by weak lower semicontinuity, and therefore to $V_n - V$ by the triangle inequality. Interpolating between L^2 and $L^{10/3}$, one obtains

$$\|V_n - V\|_{L^3(Q_r)} \le \|V_n - V\|_{L^2(Q_r)}^{1/6} \|V_n - V\|_{L^{10/3}(Q_r)}^{5/6}.$$

The second factor is uniformly bounded and the first tends to zero, so

$$V_n \to V \qquad \text{strongly in } L^3(Q_r).$$

The pressure compactness follows from the uniform D-bound: after subtracting spatial means,

$$P_n - (P_n)_{B_r} \rightharpoonup P \qquad \text{in } L^{3/2}(Q_r).$$

We pass to the limit distributionally using the preceding convergences. It remains to justify the modulation terms. For every compactly supported test field φ,

$$\int a_n(s)\, y \cdot \nabla V_n \cdot \varphi = -\int a_n(s)\, V_{n,i}\, \partial_j(y_j \varphi_i),$$

where summation over repeated spatial indices is understood. Equivalently,

$$\partial_j(y_j\varphi_i) = 3\varphi_i + y_j \partial_j \varphi_i.$$

Hence this term passes to the limit by strong local L^2 convergence of V_n and weak-* convergence of a_n: if $F_n \to F$ in L^1 and $a_n \rightharpoonup^* a$ in L^∞, then

$$\int a_n F_n - \int aF = \int a_n(F_n - F) + \int (a_n - a)F \to 0.$$

The term $a_n V_n$ is immediate from the same argument, while

$$\int b_n(s) \cdot \nabla V_n \cdot \varphi = -\int V_{n,i}\, b_{n,j}(s)\, \partial_j \varphi_i$$

passes to the limit by strong local L^2 convergence of V_n and weak-* convergence of b_n. These passages give the limiting represented equation in the sense of distributions. □

Remark 3.11 (Use of bounded modulation). The uniform L^∞ bounds on a_n and b_n enter the compactness argument in exactly two places. First, they give the negative Sobolev time-derivative bound for the three lower-order modulation terms in (1). Second, after subsequential weak-* convergence of a_n, b_n, they allow the modulation terms to pass to the distributional limit once $V_n \to V$ strongly in L^2_{loc}.

Lemma 3.12 (Stability of local pressure decomposition)**.** *Let $V_n \to V$ strongly in $L^3(Q_R)$, and use the pressure decomposition of Theorem 3.8. Then for every $r < R$,*

$$P_{\mathrm{loc},n} \to P_{\mathrm{loc}} \quad \textit{strongly in } L^{3/2}(Q_r),$$

where $-\Delta P_{\mathrm{loc}} = \partial_i\partial_j(\zeta V_i V_j)$. If $V \equiv 0$ on Q_R, then

$$P_n - (P_n)_{B_r} \to 0 \quad \textit{strongly in } L^{3/2}(Q_r).$$

More locally in time, if $V_n \to 0$ strongly in

$$L^3(B_R \times (-r^2, 0)),$$

then

$$P_n - (P_n)_{B_r} \to 0 \quad \textit{strongly in } L^{3/2}(Q_r).$$

Proof. Strong convergence in L^3 gives

$$V_{n,i}V_{n,j} \to V_iV_j \quad \text{strongly in } L^{3/2}(Q_R).$$

Indeed,

$$\|V_{n,i}V_{n,j} - V_iV_j\|_{L^{3/2}(Q_R)} \le \big(\|V_n\|_{L^3(Q_R)} + \|V\|_{L^3(Q_R)}\big)\|V_n - V\|_{L^3(Q_R)}.$$

Calderon–Zygmund estimates for the compactly supported localized source yield strong convergence of the localized pressures in $L^{3/2}$ on smaller balls:

$$\|P_{\mathrm{loc},n} - P_{\mathrm{loc}}\|_{L^{3/2}(Q_r)} \le C_{r,R}\|V_{n,i}V_{n,j} - V_iV_j\|_{L^{3/2}(Q_R)}.$$

Assume now that $V \equiv 0$ on Q_R. Then $P_{\mathrm{loc}} = 0$. By the zero-limit stability part of Theorem 3.8,

$$P_n - (P_n)_{B_r} \to 0 \quad \text{strongly in } L^{3/2}(Q_r).$$

The same proof is local in time. If $V_n \to 0$ strongly in $L^3(B_R \times (-r^2, 0))$, the Calderon–Zygmund estimate and harmonic pressure control are applied only for times $s \in (-r^2, 0)$, giving the strong convergence on Q_r. □

Lemma 3.13 (Zero-dissipation limit)**.** *If*

$$V_n \rightharpoonup V \quad \text{in } L^2(I; H^1(B_\rho))$$

and

$$\int_I \int_{B_\rho} |\nabla V_n|^2 \, dy \, ds \to 0,$$

then $\nabla V = 0$ a.e. on $B_\rho \times I$. Hence for a.e. time, $V(\cdot, s)$ is spatially constant on B_ρ.

Proof. Weak lower semicontinuity gives

$$\int_I \int_{B_\rho} |\nabla V|^2 \, dy \, ds \le \liminf_{n\to\infty} \int_I \int_{B_\rho} |\nabla V_n|^2 \, dy \, ds = 0.$$

Thus $\nabla V = 0$ a.e. on $B_\rho \times I$. By Fubini, for a.e. $s \in I$ one has $\nabla V(\cdot, s) = 0$ in $L^2(B_\rho)$. Since B_ρ is connected, every $H^1(B_\rho)$ function with zero weak gradient is a.e. equal to a constant. Hence $V(\cdot, s)$ is spatially constant for a.e. s. □

Lemma 3.14 (Spatially constant limits)**.** *Let V be a limit on Q_ρ and suppose $\nabla V = 0$ a.e. there. Then $V(y, s) = c(s)$ on Q_ρ, with $c \in L^\infty(-\rho^2, 0)$. If $c \not\equiv 0$ on Q_ρ, then the approximating selected-window subsequence has a nonzero bounded selected-window limit.*

Proof. The spatial constancy follows from Theorem 3.13. Since $V(\cdot, s) = c(s)$ on B_ρ, the local $L^\infty_s L^2_y$ bound gives

$$|c(s)|^2 |B_\rho| \le \operatorname*{ess\,sup}_{-\rho^2<s<0} \int_{B_\rho} |V(y, s)|^2 \, dy,$$

so $c \in L^\infty(-\rho^2, 0)$. If $c \not\equiv 0$ on Q_ρ, then the strong L^3 convergence of Theorem 3.10 gives a subsequential limit to a nonzero bounded function on Q_ρ, which is precisely a nonzero bounded selected-window limit in the sense of Theorem 3.1. □

3.4. **Contradiction lemmas.**

Lemma 3.15 (Persistence of positive core concentration)**.** *For a compact single-core branch in the sense of Theorem 3.3, there are fixed $r_* > 0$ and $\eta_0 > 0$ such that along the selected-window subsequence,*

$$C_{V_n}(r_*) + D_{P_n}(r_*) \geq \eta_0 \qquad \textit{for all } n.$$

Proof. The lower bound is part of the definition of the selected compact core and is preserved by passing to subsequences of the selected windows. $\square$

Lemma 3.16 (Zero constant contradicts positive CKN concentration)**.** *Let $r_* < \rho$. Suppose the compactness limit in Theorem 3.10 is spatially constant on Q_ρ and is identically zero on Q_{r_*}. Then*

$$C_{V_n}(r_*) + D_{P_n}(r_*) \to 0.$$

Consequently this case contradicts Theorem 3.15.

Proof. Strong $L^3(Q_{r_*})$ convergence to zero gives

$$C_{V_n}(r_*) = r_*^{-2} \int_{Q_{r_*}} |V_n|^3 \, dy \, ds = r_*^{-2} \|V_n\|^3_{L^3(Q_{r_*})} \to 0.$$

It remains to control the pressure term. Since V is spatially constant on Q_ρ, the identity $V = 0$ on Q_{r_*} implies

$$V = 0 \qquad \text{on } B_\rho \times (-r_*^2, 0).$$

Indeed, for a.e. $s \in (-r_*^2, 0)$, the spatially constant value on B_ρ vanishes on the nonempty subball B_{r_*}, hence vanishes on all of B_ρ. The strong L^3_{loc} convergence therefore gives

$$V_n \to 0 \qquad \text{strongly in } L^3(B_\rho \times (-r_*^2, 0)).$$

Applying Theorem 3.12 with $R = \rho$ and $r = r_*$, in its local-in-time form, gives

$$P_n - (P_n)_{B_{r_*}} \to 0 \qquad \text{strongly in } L^{3/2}(Q_{r_*}).$$

Therefore

$$D_{P_n}(r_*) = r_*^{-2} \|P_n - (P_n)_{B_{r_*}}\|^{3/2}_{L^{3/2}(Q_{r_*})} \to 0.$$

Thus

$$C_{V_n}(r_*) + D_{P_n}(r_*) \to 0,$$

contradicting the uniform lower bound

$$C_{V_n}(r_*) + D_{P_n}(r_*) \geq \eta_0.$$

$\square$

Lemma 3.17 (Nonzero constant contradicts the Type II regime)**.** *If the compactness limit is spatially constant and nonzero on a compact subcylinder, then the original sequence is not a local Type II sequence.*

Proof. By Theorem 3.14, the selected-window subsequence has a nonzero bounded selected-window limit. This contradicts Theorem 3.2. $\square$

3.5. **Main theorem.**

Theorem 3.18 (Local single-core Type II criterion)**.** *Under the analytic inputs Theorems 3.6 to 3.8, a compact single-core vanishing-dissipation branch in the sense of Theorem 3.3 cannot be a local Type II sequence.*

Proof. By the definition of vanishing localized dissipation, after passing to a selected-window subsequence there is a fixed radius $\rho \in (r_*, R_*)$ such that

$$\int_{Q_\rho} |\nabla V_n|^2 \, dy \, ds \to 0$$

and the positive core bound on Q_{r_*} remains valid along this subsequence. Apply compactness on the fixed compact cylinder Q_ρ and pass to a subsequence. The compactness proposition gives

$$V_n \rightharpoonup V \quad \text{in } L^2(-\rho^2, 0; H^1(B_\rho))$$

and strong convergence in L^3 on every smaller cylinder. Since the localized dissipation tends to zero, the zero-dissipation lemma gives

$$\nabla V = 0 \quad \text{a.e. on } Q_\rho.$$

Thus $V(y, s) = c(s)$ on Q_ρ, and in particular on the selected core cylinder Q_{r_*}.

If this spatially constant limit is zero on the core cylinder, the zero-limit contradiction lemma gives

$$C_{V_n}(r_*) + D_{P_n}(r_*) \to 0,$$

contradicting persistence of positive core concentration.

If the spatially constant limit is not identically zero on Q_{r_*}, the nonzero-limit lemma produces a nonzero bounded selected-window limit, contradicting the definition of a local Type II sequence. The two cases exhaust all spatially constant limits, so the branch cannot occur. ☐

Remark 3.19 (Minimal local inputs)*.* The proof uses two local analytic inputs from the Type II argument: the localized representation for a nondegenerate selected core, the compact-ball pressure reconstruction with stability on smaller balls. The compact-cylinder A, E, C, D bounds and vanishing localized dissipation are part of the branch hypothesis, not separate analytic inputs. Multibubble, cascade, gauge-degenerate, and compactness-failure alternatives are outside this single-core theorem; they are treated by the local multibubble and cascade exclusion theorem.

4. Repaired-Gauge Construction

4.1. **Assumptions and notation.** Let $u : \mathbb{R}^3 \times (T-\delta, T) \to \mathbb{R}^3$, $p : \mathbb{R}^3 \times (T-\delta, T) \to \mathbb{R}$ be a suitable weak Navier–Stokes solution with viscosity $\nu > 0$. For $z = (x_0, t_0)$ and $R > 0$,

$$Q_R(z) := B_R(x_0) \times (t_0 - R^2, t_0), \qquad Q_R^* := B_R \times I_*.$$

Fix a reference physical center-time $(x_\bullet, t_\star)$ around which $B_{R_*}(x_\bullet)$ is taken. For a fixed spatial center x_0 we also use

$$Q_R(t) := Q_R(x_0, t).$$

Fix a compact window $I_* := [\tau_0 - \ell, \tau_0 + \ell]$, $\ell > 0$, and write $I_{\mathrm{ph}} := [t_\star - \Lambda_{\max}^2 \ell,\ t_\star + \Lambda_{\max}^2 \ell]$.

For a cylinder $Q_R(x_0, t_0) = B_R(x_0) \times (t_0 - R^2, t_0)$, set

$$\begin{aligned}
A(u; Q_R(x_0,t_0)) &:= R^{-1} \sup_{s \in (t_0 - R^2, t_0)} \int_{B_R(x_0)} |u(x,s)|^2\, dx, \\
E(u; Q_R(x_0,t_0)) &:= R^{-1} \int_{Q_R(x_0,t_0)} |\nabla u|^2\, dx\, ds, \\
C(u; Q_R(x_0,t_0)) &:= R^{-2} \int_{Q_R(x_0,t_0)} |u|^3\, dx\, ds, \\
D(p; Q_R(x_0,t_0)) &:= R^{-2} \int_{t_0 - R^2}^{t_0} \int_{B_R(x_0)} \left| p(x,s) - \langle p(s) \rangle_{B_R(x_0)} \right|^{3/2} dx\, ds.
\end{aligned}$$

Here $\langle p(s) \rangle_{B_R(x_0)} := |B_R|^{-1} \int_{B_R(x_0)} p(x,s)\, dx$; pressure is always understood modulo time-dependent constants.

Definition 4.1 (Single-core hypotheses)**.** Let $(R_*, r_*, M, \eta, \kappa, \Lambda_{\min}, \Lambda_{\max})$ satisfy $\Lambda_{\min} \le \Lambda_{\max} < \infty$. We say (u,p) satisfies $\mathrm{SC}(R_*, r_*, M, \eta, \kappa)$ on $Q_{R_*}(x_\bullet, t_\star)$ if the following hold:

(SC1) $A(u; Q_{R_*}(x_\bullet, t)) + E(u; Q_{R_*}(x_\bullet, t)) + C(u; Q_{R_*}(x_\bullet, t)) + D(p; Q_{R_*}(x_\bullet, t)) \le M$ for every $t \in [t_\star - R_*^2, t_\star)$.

(SC2) There exists a physically selected core map $(x_{\mathrm{c}}(t), \rho_{\mathrm{c}}(t))$ on I_{ph} such that $\rho_{\mathrm{c}}(t) \in (0, R_*]$, $x_{\mathrm{c}}(t) \in B_{R_*/2}(x_\bullet)$, and for some $\Theta_0 > 0$,

$$\int_{B_{r_*}(0)} |\rho_{\mathrm{c}}(t)^{-1} u(x_{\mathrm{c}}(t) + \rho_{\mathrm{c}}(t) y, t)|^3 \mathrm{d}y \ge \eta \quad \text{for a.e. } t \in I_{\mathrm{ph}}.$$

(SC3) There exists $0 < r_{\mathrm{sep}} \le r_*/2$ such that for every $0 < r \le r_{\mathrm{sep}}$, every competitor pair

$$\mathcal{C}_{\mathrm{core}}(t) := \left\{ (\bar{x}, \bar{\rho}) : |\bar{x} - x_\bullet| \le \frac{R_*}{2},\ 0 < \bar{\rho} \le R_* \right\},$$

$(\bar{x}, \bar{\rho}) \in \mathcal{C}_{\mathrm{core}}(t) \setminus \{(x_{\mathrm{c}}(t), \rho_{\mathrm{c}}(t))\}$ with

$$\rho_{\mathrm{c}}(t)^{-1} |\bar{x} - x_{\mathrm{c}}(t)| + \left| \frac{\bar{\rho}}{\rho_{\mathrm{c}}(t)} - 1 \right| \ge r_{\mathrm{sep}}$$

has separated local weighted mass in the same core-scale variables:

$$\left| \int_{B_r} |\rho_{\mathrm{c}}(t)^{-1} u(x_{\mathrm{c}}(t) + \rho_{\mathrm{c}}(t) y, t)|^3\, \mathrm{d}y - \int_{B_r} |\bar{\rho}^{-1} u(\bar{x} + \bar{\rho} y, t)|^3\, \mathrm{d}y \right| \ge \eta.$$

This quantitative non-competing core condition is imposed on the selected physical core directly; it is not a consequence of the repaired-gauge construction.

(SC4) For the selected core,

$$\sup_{t\in I_{\rm ph}}\int_{B_{2R_*}}|y|^{-p}|\rho_{\rm c}(t)^{-1}u(x_{\rm c}(t)+\rho_{\rm c}(t)y,t)|^3\,dy\le M,$$

for some $0<p<3$.

(SC5) The preliminary frame associated with the selected core is the absolutely continuous representative

$$0<\Lambda_{\min}\le\Lambda_{\rm pre}(\tau)\le\Lambda_{\max},\quad \Lambda_{\rm pre}\in AC(I_*),\quad X_{\rm pre}\in AC(I_*;\mathbb{R}^3).$$

It is tied to the selected physical core by the time change $dt_{\rm c}/d\tau=\Lambda_{\rm pre}(\tau)^2$, $t_{\rm c}(\tau_0)=t_\star$: one has

$$X_{\rm pre}(\tau)=x_{\rm c}(t_{\rm c}(\tau)),\qquad \Lambda_{\rm pre}(\tau)=\rho_{\rm c}(t_{\rm c}(\tau))$$

for a.e. $\tau\in I_*$.

The parameter κ denotes the quantitative constants obtained in the local gauge estimates and is not used in items (SC1)–(SC5).

Remark 4.2. The selected core $(x_{\rm c},\rho_{\rm c})$ and its separation/nondegeneracy conditions are specified in physical variables; no repaired-gauge constraint is used in their definition. This avoids the circularity of defining the core via the chart to be constructed.

4.2. **Preliminary frame map from the selected core.** The transport map is constructed from the selected core; the core selection itself is part of the hypotheses.

Define

$$\frac{dt_{\rm c}}{d\tau}=\Lambda_{\rm pre}(\tau)^2,\qquad t_{\rm c}(\tau_0)=t_\star,$$

and use the compatibility in (SC5) to identify $\Lambda_{\rm pre}(\tau)=\rho_{\rm c}(t_{\rm c}(\tau))$ and $X_{\rm pre}(\tau)=x_{\rm c}(t_{\rm c}(\tau))$ a.e.

Proposition 4.3 (Preliminary frame map regularity). *Under* $\mathrm{SC}(R_*,r_*,M,\eta,\kappa)$*, the map*

$$\Phi_{\rm pre}(\tau,Y):=(x,t)=(X_{\rm pre}(\tau)+\Lambda_{\rm pre}(\tau)Y,\ t_{\rm c}(\tau))$$

is well-defined and absolutely continuous in τ*. Moreover,* $1/\Lambda_{\rm pre}$*,* $\Lambda_{\rm pre}\in L^\infty(I_*)$*,* $X_{\rm pre}\in W^{1,1}_{\rm loc}(I_*;\mathbb{R}^3)$*, and for every* $R\le R_*$*,*

$$\Phi_{\rm pre}:Q^*_R\to\Phi_{\rm pre}(Q^*_R)\subset Q_{2R_*}(x_\bullet,t_\star)$$

is bi-Lipschitz with $\det\nabla_Y\Phi_{\rm pre}=\Lambda^3_{\rm pre}$*, hence Jacobian and inverse-Jacobian bounds depend only on* $\Lambda_{\min},\Lambda_{\max}$*.*

Proof. From (SC5), $\Lambda_{\rm pre}\in AC(I_*)$, $0<\Lambda_{\min}\le\Lambda_{\rm pre}\le\Lambda_{\max}<\infty$, and $X_{\rm pre}\in AC(I_*;\mathbb{R}^3)$. The scalar ODE

$$\frac{dt_{\rm c}}{d\tau}=\Lambda_{\rm pre}(\tau)^2,\qquad t_{\rm c}(\tau_0)=t_\star$$

therefore has an absolutely continuous strictly increasing solution on I_*, and

$$\frac{dt_{\rm c}}{d\tau}=\Lambda_{\rm pre}(\tau)^2\quad\text{a.e. on }I_*.$$

The compatibility condition (SC5) identifies this absolutely continuous frame with the selected physical core a.e.; no differentiability of $x_{\rm c}$ or $\rho_{\rm c}$ beyond the frame regularity in (SC5) is used. For fixed τ, $\nabla_Y\Phi_{\rm pre}(\tau,\cdot)=\Lambda_{\rm pre}(\tau)I$, so

$$\det\nabla_Y\Phi_{\rm pre}(\tau,\cdot)=\Lambda_{\rm pre}(\tau)^3,\qquad \Lambda^3_{\min}\le\det\nabla_Y\Phi_{\rm pre}\le\Lambda^3_{\max}.$$

Hence, if $Y_1,Y_2\in B_R$,

$$\Lambda_{\min}|Y_1-Y_2|\le|\Phi_{\rm pre}(\tau,Y_1)-\Phi_{\rm pre}(\tau,Y_2)|\le\Lambda_{\max}|Y_1-Y_2|,$$

and therefore $\Phi_{\rm pre}(\tau,\cdot)$ is bi-Lipschitz on B_R. Its inverse has Lipschitz constant $\Lambda_{\min}^{-1}$, so Jacobian and inverse-Jacobian bounds are controlled by $\Lambda_{\min},\Lambda_{\max}$.

For image inclusions, if $x = X_{\rm pre}(\tau)+\Lambda_{\rm pre}(\tau)Y$ with $|Y|\le R$, then

$$|x - X_{\rm pre}(\tau)| \le \Lambda_{\max}R, \qquad X_{\rm pre}(\tau)\in B_{R_*/2}(x_\bullet),$$

so $x\in B_{R_*/2+\Lambda_{\max}R}(x_\bullet)$. As $\tau\in I_*$, we also have $t=t_{\rm c}(\tau)\in I_{\rm ph}$. Therefore

$$\Phi_{\rm pre}(Q_R^*)\subset B_{R_*/2+\Lambda_{\max}R}(x_\bullet)\times I_{\rm ph}.$$

□

Definition 4.4 (Preliminary profile)**.** For $(x,t)=\Phi_{\rm pre}(\tau,Y)$, define

$$V_{\rm pre}(Y,\tau) := \Lambda_{\rm pre}(\tau)\,u(x,t), \qquad P_{\rm pre}(Y,\tau) := \Lambda_{\rm pre}(\tau)^2\,p(x,t).$$

Hence $u(x,t)=\Lambda_{\rm pre}(\tau)^{-1}V_{\rm pre}(Y,\tau)$ and $p(x,t)=\Lambda_{\rm pre}(\tau)^{-2}P_{\rm pre}(Y,\tau)$ under $\Phi_{\rm pre}$.

4.3. **Exact renormalized change of variables.** Define renormalized fields on Q_{2R}^* by

$$Y=\frac{x-X(\tau)}{\Lambda(\tau)}, \qquad u(x,t)=\Lambda(\tau)^{-1}V(Y,\tau), \qquad p(x,t)=\Lambda(\tau)^{-2}P(Y,\tau),$$

where $dt/d\tau=\Lambda(\tau)^2$, $X: I_*\to\mathbb{R}^3$, $\Lambda: I_*\to(0,\infty)$, and $\Lambda, X\in AC(I_*)$. Write

$$a(\tau) := -\Lambda'(\tau), \qquad b(\tau) := -X'(\tau).$$

Lemma 4.5 (Exact differential identities)**.** *For smooth data and fixed (x,t), the chain rule gives*

$$\partial_t u = \Lambda^{-3}\left(\partial_\tau V - (X'+\Lambda' Y)\cdot\nabla_Y V - \Lambda' V\right),$$

$$\nabla_x u = \Lambda^{-2}\nabla_Y V, \qquad \Delta_x u = \Lambda^{-3}\Delta_Y V, \qquad \nabla_x p = \Lambda^{-3}\nabla_Y P.$$

In particular

$$\nabla_x\cdot u = \Lambda^{-2}\nabla_Y\cdot V.$$

Proof. For fixed (x,t), $\tau=\tau(t)$ and $Y=(x-X(\tau))/\Lambda(\tau)$. Differentiate

$$u(x,t)=\Lambda(\tau)^{-1}V\big(Y,\tau\big)$$

in t:

$$\partial_t u = -\frac{\Lambda'}{\Lambda^2}V + \frac{1}{\Lambda}\partial_\tau V\,\frac{d\tau}{dt} + \frac{1}{\Lambda}\nabla_Y V\cdot\frac{dY}{dt}.$$

Since $d\tau/dt=\Lambda^{-2}$ and

$$Y_\tau = -\Lambda^{-1}X' - \frac{\Lambda'}{\Lambda}Y, \qquad \frac{dY}{dt} = Y_\tau\,\frac{d\tau}{dt},$$

therefore

$$\partial_t u = \Lambda^{-3}\left(\partial_\tau V - (X'+\Lambda' Y)\cdot\nabla_Y V - \Lambda' V\right).$$

For spatial derivatives, $Y=(x-X)/\Lambda$ gives $\nabla_x Y=\Lambda^{-1}I$, hence

$$\nabla_x u = \Lambda^{-2}\nabla_Y V, \qquad \Delta_x u = \Lambda^{-3}\Delta_Y V.$$

Similarly $p(x,t)=\Lambda^{-2}P(Y,\tau)$ yields

$$\nabla_x p = \Lambda^{-3}\nabla_Y P.$$

Taking divergence of $u=\Lambda^{-1}V$ gives

$$\nabla_x\cdot u = \Lambda^{-2}\nabla_Y\cdot V.$$

□

Theorem 4.6 (Renormalized Navier–Stokes equation)**.** *Assume u, p solve*

$$\partial_t u + (u \cdot \nabla_x)u + \nabla_x p - \nu\Delta_x u = 0, \qquad \nabla_x \cdot u = 0$$

in distributions. Under the transformation above, V, P satisfy, on the image renormalized cylinder, the exact equation

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P - \nu\Delta V + a(\tau)\big(V + Y \cdot \nabla V\big) + b(\tau) \cdot \nabla V = 0, \qquad \nabla \cdot V = 0,$$

with the sign convention fixed by $a = -\Lambda'$, $b = -X'$. Conversely, any V, P satisfying this equation pull back to a distributional solution u, p of Navier–Stokes on the physical image.

Proof. To avoid hidden regularity loss, first write the physical equation tested against $\Phi \in C_c^\infty$ in the physical cylinder and then use the smooth pull-back formula (an approximation argument gives distributions). By Lemma 4.5,

$$\partial_t u + \big(u \cdot \nabla_x\big)u + \nabla_x p - \nu\Delta_x u = 0$$

becomes

$$\Lambda^{-3}\big(\partial_\tau V - (X' + \Lambda' Y) \cdot \nabla V - \Lambda' V\big) + \Lambda^{-3}(V \cdot \nabla)V + \Lambda^{-3}\nabla P - \Lambda^{-3}\nu\Delta V = 0.$$

Multiplying by Λ^3 and using $a = -\Lambda'$, $b = -X'$ gives

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P - \nu\Delta V + a\big(V + Y \cdot \nabla V\big) + b \cdot \nabla V = 0.$$

The divergence condition is

$$\nabla_x \cdot u = \Lambda^{-2}\nabla \cdot V = 0 \quad \Longrightarrow \quad \nabla \cdot V = 0.$$

Conversely, assume (V, P) solve this equation on the renormalized cylinder. Set u, p by the same change-of-variables map. Then the preceding identities run in reverse and give

$$\partial_t u + (u \cdot \nabla_x)u + \nabla_x p - \nu\Delta_x u = 0, \qquad \nabla_x \cdot u = 0$$

in $\mathcal{D}'$ on the physical image. □

4.4. **Domain localization and support.** Let $R > 0$ be fixed and $0 < \ell \le \ell_0$. Set $Q_R^* = B_R \times I_*$. With Φ as above define physical support maps

$$\mathcal{Q}_R^{\mathrm{ph}} := \Phi(Q_R^*), \qquad \mathcal{Q}_{2R}^{\mathrm{ph}} := \Phi(Q_{2R}^*).$$

Proposition 4.7 (Compact support transport)**.** *Assume $\Lambda_{\min} \le \Lambda \le \Lambda_{\max}$ on I_*. Then*

$$X(\tau) \in B_{R_*/2}(x_\bullet), \qquad |x - X(\tau)| \le \Lambda_{\max} R,$$
$$\mathcal{Q}_R^{\mathrm{ph}} \subset B_{R_*/2 + \Lambda_{\max} R}(x_\bullet) \times I_{\mathrm{ph}},$$
$$\mathcal{Q}_{2R}^{\mathrm{ph}} \subset B_{R_*/2 + 2\Lambda_{\max} R}(x_\bullet) \times I_{\mathrm{ph}},$$

for every $\tau \in I_*$ and $x = X(\tau) + \Lambda(\tau)Y$ with $Y \in B_{2R}$. In particular, if $|c(\tau)| \le \delta$ from Proposition 4.20, then $X(\tau) \in B_{R_*/2 + \Lambda_{\max}\delta}(x_\bullet)$ on I_*. Hence, $\chi_R \in C_c^\infty(B_{2R})$ pulls back to $\chi_R^\tau := \chi_R \circ \Phi^{-1} \in C_c^\infty(\mathcal{Q}_{2R}^{\mathrm{ph}})$ with support depending only on $R, \Lambda_{\min}, \Lambda_{\max}$.

Proof. Fix $\tau \in I_*$ and $Y \in B_R$. Since $x = X(\tau) + \Lambda(\tau)Y$,

$$|x - X(\tau)| \le |Y|\,\Lambda_{\max} \le \Lambda_{\max} R, \qquad \Lambda_{\max}^{-1}|x - X(\tau)| \le |Y|.$$

Because $X(\tau) \in B_{R_*/2}(x_\bullet)$, this gives

$$x \in B_{R_*/2 + \Lambda_{\max} R}(x_\bullet).$$

Similarly, if $Y \in B_{2R}$, then $x \in B_{R_*/2 + 2\Lambda_{\max} R}(x_\bullet)$. Together with $t = t_c(\tau) \in I_{\mathrm{ph}}$, these give the claimed inclusions for $\mathcal{Q}_R^{\mathrm{ph}}$ and $\mathcal{Q}_{2R}^{\mathrm{ph}}$.

For pull-backs of cutoffs, Φ is bi-Lipschitz at every τ, so composition

$$\chi_R^\tau(x,t) := \chi_R\left(\frac{x - X(\tau)}{\Lambda(\tau)}\right)$$

is smooth on $\mathcal{Q}_{2R}^{\mathrm{ph}}$, supported where $Y \in B_{2R}$, hence supported where $(x,t) \in \mathcal{Q}_{2R}^{\mathrm{ph}}$. The support size depends only on $R, \Lambda_{\min}, \Lambda_{\max}$ and the fixed C_c^∞ norm of χ_R, not on the particular branch. $\square$

4.5. **Transport of local quantities under the chart.** Let $t = t(\tau) = t_{\mathrm{c}}(\tau)$ and, for $R > 0$, define

$$J_\tau^-(R) := \big(t(\tau) - \Lambda_{\max}^2 R^2,\ t(\tau)\big].$$

Define renormalized local quantities on $Q_R^*(\tau)$ by

$$A_V(\tau,R) := \sup_{s\in(\tau-R^2,\tau)} \int_{B_R} |V(\cdot,s)|^2\, dY, \qquad E_V(\tau,R) := \int_{\tau-R^2}^{\tau}\int_{B_R} |\nabla V|^2\, dY\, ds,$$

$$C_V(\tau,R) := \int_{\tau-R^2}^{\tau}\int_{B_R} |V|^3\, dY\, ds, \qquad D_V(\tau,R) := \int_{\tau-R^2}^{\tau}\int_{B_R} \Big|P - \langle P\rangle_{B_R}\Big|^{3/2} dY\, ds.$$

For interval notation and a center X, define

$$A_u(J,R;X) := \sup_{s\in J} \int_{B_R(X)} |u(\cdot,s)|^2\, dx, \quad E_u(J,R;X) := \int_J \int_{B_R(X)} |\nabla u|^2\, dx\, ds,$$

$$C_u(J,R;X) := \int_J \int_{B_R(X)} |u|^3\, dx\, ds, \quad D_u(J,R;X) := \int_J \int_{B_R(X)} |p - \langle p\rangle_{B_R(X)}|^{3/2}\, dx\, ds.$$

In the inequalities below we abbreviate

$$A_u(J,R) := A_u(J,R;X(\tau)), \quad E_u(J,R) := E_u(J,R;X(\tau)),$$
$$C_u(J,R) := C_u(J,R;X(\tau)), \quad D_u(J,R) := D_u(J,R;X(\tau)).$$

Proposition 4.8 (Cylinder comparison on compact windows)**.** *Assume $\Lambda_{\min} \le \Lambda \le \Lambda_{\max}$ and formulas of Lemma 4.5. Then for every $\tau \in I_*$, $R > 0$,*

$$A_V(\tau,R) \le \Lambda_{\min}^{-1} A_u(J_\tau^-(R), \Lambda_{\max} R),$$

$$E_V(\tau,R) \le \Lambda_{\min}^{-3} E_u(J_\tau^-(R), \Lambda_{\max} R),$$

$$C_V(\tau,R) \le \Lambda_{\min}^{-2} C_u(J_\tau^-(R), \Lambda_{\max} R),$$

$$D_V(\tau,R) \le \Lambda_{\min}^{-2} D_u(J_\tau^-(R), \Lambda_{\max} R).$$

In particular, all four renormalized quantities are bounded whenever the physical quantities are bounded on the corresponding compact physical cylinders.

Proof. These are direct changes of variables under Φ_{pre}:

$$V(\cdot,\tau) = \Lambda(\tau) u(\cdot, t(\tau)), \qquad P(\cdot,\tau) = \Lambda(\tau)^2 p(\cdot, t(\tau)),$$

$$\nabla_Y V = \Lambda \nabla_x u, \qquad dY\, d\tau = \Lambda^{-5}\, dx\, dt, \qquad d\tau = \Lambda^2\, dt.$$

Insert these formulas into each definition and use $\Lambda_{\min} \le \Lambda \le \Lambda_{\max}$, plus

$$\tau - s \in (0, R^2) \implies t(\tau) - \Lambda_{\max}^2 R^2 \le t(s) \le t(\tau) - \Lambda_{\min}^2 R^2.$$

$\square$

4.6. **Local repaired gauge map.** Fix $R>0$, $0<p<3$, and $\chi_R\in C_c^\infty(B_{2R})$, $\chi_R\equiv 1$ on B_R. Let the local energy profile space be

$$\mathcal{V}_R := \Big\{V\in L^3(B_{2R})\cap W^{1,3/2}(B_{2R};\mathbb{R}^3) : |Y|^{-p/3}V\in L^3(B_{2R})\Big\},$$

with norm

$$\|V\|_{\mathcal{V}_R} := \|V\|_{L^3(B_{2R})} + \||Y|^{-p/3}V\|_{L^3(B_{2R})} + \|\nabla V\|_{L^{3/2}(B_{2R})}.$$

For differentiating the repaired scale and centering gauges we use the admissible gauge class

$$\mathcal{A}_{p,R} := \left\{ V : \begin{array}{l} V\in L^3(B_{4R})\cap W^{1,3}_{\mathrm{loc}}(B_{4R};\mathbb{R}^3), \\ \displaystyle\int_{B_{4R}} |Y|^{-p}|V|^3\,dY<\infty, \quad \int_{B_{4R}} |Y|^{-p}|V|\,|Y\cdot\nabla V|\,dY<\infty, \\ \displaystyle\int_{B_{4R}} |Y|^{-p-1}|V|^3\,dY<\infty \end{array} \right\}.$$

The larger ball is only a domain buffer: all gauge integrals are supported in B_{2R}, while (λ,c) is restricted near $(1,0)$ so that $\lambda^{-1}(B_{2R}-c)\subset B_{4R}$.

Definition 4.9 (Gauge actions)**.** For $(\lambda,c)\in(0,\infty)\times\mathbb{R}^3$, define

$$V_{\lambda,c}(Y) := \lambda^{-1}V(\lambda^{-1}(Y-c)).$$

For pressure profiles use

$$P_{\lambda,c}(Y) := \lambda^{-2}P(\lambda^{-1}(Y-c)).$$

We also write $V^{\lambda,c}:=V_{\lambda,c}$ and $P^{\lambda,c}:=P_{\lambda,c}$.

Definition 4.10 (Finite-dimensional gauge map)**.** For $V\in\mathcal{V}_R$ and $(\lambda,c)\in(0,\infty)\times\mathbb{R}^3$, set

$$\mathcal{F}(\lambda,c;V) := \big(\mathcal{F}_0,\mathcal{F}_1,\mathcal{F}_2,\mathcal{F}_3\big),$$

with

$$\mathcal{F}_0(\lambda,c;V) := \int_{\mathbb{R}^3}\chi_R(Y)\,|Y|^{-p}|V_{\lambda,c}(Y)|^3\mathrm{d}Y-\Theta_0, \qquad \mathcal{F}_j(\lambda,c;V) := \int_{\mathbb{R}^3} Y_j\,\chi_R(Y)\,|V_{\lambda,c}(Y)|^2\mathrm{d}Y$$

for $j=1,2,3$. The gauge conditions are $\mathcal{F}(\lambda,c;V)=0$. Codomain is $\mathbb{R}^4$, domain $(0,\infty)\times\mathbb{R}^3\times\mathcal{V}_R$.

Definition 4.11 (Localized modulation matrix)**.** For $J\subset I_*$ and a profile path $W:J\to\mathcal{V}_R$, define

$$J_W(\tau) := D_{(\lambda,c)}\mathcal{F}\big(1,0;W(\tau)\big)\in\mathbb{R}^{4\times4}.$$

The rows are ordered as the scale constraint followed by the three centering constraints.

Lemma 4.12 (Weighted integrability and differentiation under integrals)**.** *Assume $V,W\in\mathcal{A}_{p,R}$. Then $G_0(V):=\int\chi_R|Y|^{-p}|V|^3$ is finite and*

$$\frac{\mathrm{d}}{\mathrm{d}\epsilon}\Big|_{\epsilon=0}\int\chi_R|Y|^{-p}|V+\epsilon W|^3\,\mathrm{d}Y = \int 3\chi_R|Y|^{-p}|V|\,V\cdot W\,\mathrm{d}Y.$$

Proof. For $W\in\mathcal{A}_{p,R}$, define

$$F_\varepsilon(Y) := \chi_R(Y)|Y|^{-p}|V(Y)+\varepsilon W(Y)|^3.$$

For each Y, $\varepsilon\mapsto F_\varepsilon(Y)$ is C^1 and

$$\partial_\varepsilon F_\varepsilon = 3\chi_R|Y|^{-p}|V+\varepsilon W|(V+\varepsilon W)\cdot W.$$

For $|\varepsilon|\le 1$,

$$|F_\varepsilon(Y)|+|\partial_\varepsilon F_\varepsilon|\le C\chi_R|Y|^{-p}\big(|V|^3+|W|^3\big)\in L^1(B_{2R})$$

by the defining weighted L^3 condition of $\mathcal{A}_{p,R}$. Hence dominated convergence allows differentiation under the integral, and

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\Big|_{\varepsilon=0}\int_{B_{2R}}\chi_R|Y|^{-p}|V+\epsilon W|^3\,\mathrm{d}Y=\int_{B_{2R}}3\chi_R|Y|^{-p}|V|\,V\cdot W\,\mathrm{d}Y.$$

□

Lemma 4.13 (Gauge covariance)**.** *For $V\in\mathcal{V}_R$, $(\lambda,c)\in(0,\infty)\times\mathbb{R}^3$, and $\alpha=0,\dots,3$,*

$$\mathcal{F}_\alpha(1,0;V_{\lambda,c})=\mathcal{F}_\alpha(\lambda,c;V).$$

Proof. For each component this is a direct substitution:

$$\mathcal{F}_0(1,0;V_{\lambda,c})=\int\chi_R(Y)|Y|^{-p}|V_{\lambda,c}(Y)|^3-\Theta_0\,\mathrm{d}Y=\mathcal{F}_0(\lambda,c;V),$$

$$\mathcal{F}_j(1,0;V_{\lambda,c})=\int Y_j\chi_R(Y)|V_{\lambda,c}(Y)|^2\,\mathrm{d}Y=\mathcal{F}_j(\lambda,c;V),\quad j=1,2,3.$$

□

Proposition 4.14 (Differentiability of $\mathcal{F}$)**.** *For each fixed $V\in\mathcal{A}_{p,R}$, the map $(\lambda,c)\mapsto\mathcal{F}(\lambda,c;V)$ is C^1 in a neighborhood of $(1,0)$. In each component $\mathcal{F}_\alpha$ all derivatives under the integral sign are legitimate.*

Proof. Fix (λ,c) near $(1,0)$, and abbreviate

$$V_{\lambda,c}(Y):=\lambda^{-1}V\left(\lambda^{-1}(Y-c)\right).$$

By direct differentiation,

$$\partial_\lambda V_{\lambda,c}=-\lambda^{-1}V_{\lambda,c}-\lambda^{-1}(Y-c)\cdot\nabla V_{\lambda,c},\qquad\partial_{c_k}V_{\lambda,c}=-\lambda^{-2}\partial_kV(\lambda^{-1}(Y-c))=-\partial_kV_{\lambda,c}.$$

The neighborhood of $(1,0)$ is chosen so that the arguments of V remain inside B_{4R} for $Y\in\operatorname{supp}\chi_R$. The scale derivative is controlled by the two weighted integrability conditions

$$\int|Y|^{-p}|V|^3<\infty,\qquad\int|Y|^{-p}|V|\,|Y\cdot\nabla V|<\infty,$$

transported by the smooth change of variables. The translation derivative in the scale component is treated by the source identity

$$3\int|Y|^{-p}|V|\,V\cdot\partial_kV=p\int Y_k|Y|^{-p-2}|V|^3,$$

obtained first for smooth compactly supported profiles and then by approximation in $\mathcal{A}_{p,R}$; the right-hand side is finite because $\int|Y|^{-p-1}|V|^3<\infty$. The translation components are supported in B_{2R} and are controlled by $V\in L^2_{\mathrm{loc}}$ and $\nabla V\in L^3_{\mathrm{loc}}$. These bounds give an integrable majorant for the difference quotients, so dominated convergence applies.

$$\mathcal{F}_0(\lambda,c;V)=\int_{\mathbb{R}^3}\chi_R(Y)\,|Y|^{-p}|V_{\lambda,c}|^3\,\mathrm{d}Y-\Theta_0,$$

for which dominated convergence gives

$$\partial_\lambda\mathcal{F}_0=\int_{\mathbb{R}^3}\chi_R|Y|^{-p}\,3|V_{\lambda,c}|\,V_{\lambda,c}\cdot\partial_\lambda V_{\lambda,c}\,dY,$$

$$\partial_{c_k}\mathcal{F}_0=\int_{\mathbb{R}^3}\chi_R|Y|^{-p}\,3|V_{\lambda,c}|\,V_{\lambda,c}\cdot\partial_{c_k}V_{\lambda,c}\,dY.$$

Similarly

$$\mathcal{F}_j(\lambda,c;V)=\int_{\mathbb{R}^3}Y_j\chi_R(Y)\,|V_{\lambda,c}|^2\,\mathrm{d}Y,\quad j=1,2,3,$$

hence

$$\partial_\lambda \mathcal{F}_j = \int_{\mathbb{R}^3} 2Y_j \chi_R \, V_{\lambda,c} \cdot \partial_\lambda V_{\lambda,c} \, dY,$$

$$\partial_{c_k} \mathcal{F}_j = \int_{\mathbb{R}^3} 2Y_j \chi_R \, V_{\lambda,c} \cdot \partial_{c_k} V_{\lambda,c} \, dY.$$

Thus each component admits pointwise C^1 formulas and these are continuous in (λ, c) near $(1,0)$. Therefore $(\lambda, c) \mapsto \mathcal{F}(\lambda, c; V) \in C^1$ near $(1,0)$. □

Proposition 4.15 (Full local Jacobian at $(1,0)$)**.** *Let $V \in \mathcal{A}_{p,R}$. Set $V_0(Y) = V(Y)$. At $(\lambda, c) = (1,0)$, the matrix*

$$J(V) := D_{(\lambda,c)} \mathcal{F}(1,0;V) \in \mathbb{R}^{4\times 4}$$

has rows

$$J_\alpha = \big(\partial_\lambda \mathcal{F}_\alpha,\ \partial_{c_1} \mathcal{F}_\alpha,\ \partial_{c_2} \mathcal{F}_\alpha,\ \partial_{c_3} \mathcal{F}_\alpha\big), \quad \alpha = 0, \ldots, 3.$$

Writing $W_{\mathrm{scl}} = V_0 + Y \cdot \nabla V_0$, one has

$$\partial_\lambda \mathcal{F}_0(1,0;V) = -3 \int \chi_R |Y|^{-p} |V_0| \, V_0 \cdot W_{\mathrm{scl}} \, \mathrm{d}Y, \tag{2}$$

$$\partial_{c_k} \mathcal{F}_0(1,0;V) = -3 \int_{\mathbb{R}^3} \chi_R |Y|^{-p} |V_0| \, V_0 \cdot \partial_k V_0 \, \mathrm{d}Y, \tag{3}$$

$$\partial_\lambda \mathcal{F}_j(1,0;V) = -2 \int_{\mathbb{R}^3} Y_j \chi_R \, V_0 \cdot W_{\mathrm{scl}} \, \mathrm{d}Y, \tag{4}$$

$$\partial_{c_k} \mathcal{F}_j(1,0;V) = -2 \int_{\mathbb{R}^3} \chi_R Y_j V_0 \cdot \partial_k V_0 \, \mathrm{d}Y, \tag{5}$$

for $j, k = 1, 2, 3$.

Proof. Differentiate the formulas in Definition 4.10 at $(\lambda, c) = (1,0)$.

Since $V_{\lambda,c}(Y) = \lambda^{-1} V(\lambda^{-1}(Y - c))$,

$$\partial_\lambda V_{\lambda,c} = -\lambda^{-1} V_{\lambda,c} - \lambda^{-1} Y \cdot \nabla V_{\lambda,c}, \qquad \partial_{c_k} V_{\lambda,c} = -\lambda^{-2} \partial_k V(\lambda^{-1}(Y - c)).$$

At $(1,0)$, $\partial_\lambda V_{\lambda,c}|_{(1,0)} = -(V + Y \cdot \nabla V)$, $\partial_{c_k} V_{\lambda,c}|_{(1,0)} = -\partial_k V$.

For $\mathcal{F}_0$, write

$$\mathcal{F}_0(\lambda, c; V) = \int \chi_R(Y) |Y|^{-p} |V_{\lambda,c}(Y)|^3 \, dY - \Theta_0.$$

Derivative in λ:

$$\partial_\lambda \mathcal{F}_0 = \int \chi_R |Y|^{-p} 3 |V_{\lambda,c}| \, V_{\lambda,c} \cdot \partial_\lambda V_{\lambda,c} \, dY$$

Evaluating at $(1,0)$ yields (2).

For $k = 1, 2, 3$, c_k-derivative of $\mathcal{F}_0$ has no cutoff term:

$$\partial_{c_k} \mathcal{F}_0(1,0;V) = \int \chi_R |Y|^{-p} \, 3|V| \, V \cdot (-\partial_k V) \, dY,$$

which is (3).

For $\mathcal{F}_j$,

$$\mathcal{F}_j(\lambda, c; V) = \int Y_j \chi_R(Y) \, |V_{\lambda,c}|^2 \, dY.$$

Hence

$$\partial_\lambda \mathcal{F}_j = \int 2Y_j \chi_R \, V_{\lambda,c} \cdot \partial_\lambda V_{\lambda,c} \, dY,$$

and at $(1,0)$:

$$\partial_\lambda \mathcal{F}_j(1,0;V) = -2\int Y_j\chi_R\, V\cdot W_{\mathrm{scl}}\, dY,$$

which is (4). Similarly,

$$\partial_{c_k}\mathcal{F}_j(1,0;V) = \int 2Y_j\chi_R\, V\cdot \partial_{c_k}V_{\lambda,c}\big|_{(1,0)}\, dY = -2\int \chi_R Y_j V\cdot\partial_k V\, dY,$$

giving (5). $\square$

Proposition 4.16 (Translation block and Schur-complement invertibility)**.** *Let $V : J\to\mathcal{A}_{p,R}$ satisfy the repaired gauge constraints*

$$\mathcal{F}(1,0;V(\tau)) = 0 \qquad \textit{for a.e. } \tau\in J.$$

Write the modulation matrix in block form

$$J(V(\tau)) = \begin{pmatrix}\alpha(\tau) & u(\tau)^T\\ v(\tau) & A(\tau)\end{pmatrix},$$

where $\alpha = \partial_\lambda\mathcal{F}_0$, $u_\ell = \partial_{c_\ell}\mathcal{F}_0$, $v_j = \partial_\lambda\mathcal{F}_j$, and $A_{j\ell} = \partial_{c_\ell}\mathcal{F}_j$. Assume that for a.e. $\tau\in J$

$$m_\chi(\tau) := \int |V(Y,\tau)|^2\chi_R(Y)\, dY \ge m_0 > 0,$$

$$\int_{A_R}|V(Y,\tau)|^2\, dY \le \varepsilon_0, \qquad 3C_\chi\varepsilon_0 \le \frac{m_0}{2},$$

where $A_R = \{R\le |Y|\le 2R\}$ and

$$C_\chi := \max_{j,\ell}\|Y_j\partial_\ell\chi_R\|_{L^\infty}.$$

Assume also

$$C_u := \operatorname*{ess\,sup}_{\tau\in J}\int |Y|^{-p-1}|V(Y,\tau)|^3\, dY < \infty$$

and, with $C_\nabla := \||Y|\nabla\chi_R\|_{L^\infty}$,

$$6p\left(\frac{2}{m_0}\right)C_u\, RC_\nabla\varepsilon_0 \le \frac{p\Theta_0}{2}.$$

Then $J(V(\tau))$ is invertible for a.e. $\tau\in J$, and

$$\operatorname*{ess\,sup}_{\tau\in J}\|J(V(\tau))^{-1}\|_{\infty\to\infty} \le K(R,p,\Theta_0,m_0,C_u,\varepsilon_0,\chi_R).$$

Proof. The entries are finite by $V(\tau)\in\mathcal{A}_{p,R}$ and the compact support of χ_R. With the present parameter convention, $\partial_\lambda V_{\lambda,c}|_{(1,0)} = -(V + Y\cdot\nabla V)$. Hence the scale entry satisfies the signed transversality identity

$$\alpha(\tau) = -p\int |Y|^{-p}|V(Y,\tau)|^3\, dY = -p\Theta_0$$

on the gauge surface. Thus $|\alpha|\ge p\Theta_0$.

For the centering block, integration by parts gives

$$A_{j\ell} = -\delta_{j\ell}\int |V|^2\chi_R - \int |V|^2 Y_j\partial_\ell\chi_R.$$

Writing $A = -m_\chi I_3 + E$, the annular support of $\nabla\chi_R$ gives

$$|E_{j\ell}| \le C_\chi\int_{A_R}|V|^2 \le C_\chi\varepsilon_0, \qquad \|E\|_{\infty\to\infty}\le 3C_\chi\varepsilon_0 \le m_0/2.$$

Thus $A = -m_\chi(I_3 - m_\chi^{-1}E)$ and $\|m_\chi^{-1}E\|_{\infty\to\infty}\le 1/2$, so A^{-1} exists and $\|A^{-1}\|_{\infty\to\infty}\le 2/m_0$.

For the scale-to-translation block, the scale-constraint formula gives

$$|u_\ell| \le p\int |Y|^{-p-1}|V|^3\,dY \le p\int |Y|^{-p-1}|V|^3\,dY,$$

so $\|u\|_\infty \le pC_u$. For the translation-to-scale block,

$$v_j = -\int |V|^2 Y_j (Y\cdot\nabla\chi_R)\,dY$$

because the centering constraint cancels the interior term. Therefore

$$\|v\|_\infty \le 2R\,C_\nabla\varepsilon_0.$$

The Schur complement $S = \alpha - u^T A^{-1} v$ satisfies

$$|u^T A^{-1} v| \le 3\|u\|_\infty \|A^{-1}\|_{\infty\to\infty}\|v\|_\infty \le 6p\left(\frac{2}{m_0}\right) C_u R C_\nabla \varepsilon_0 \le \frac{p\Theta_0}{2}.$$

Since $|\alpha| = p\Theta_0$, this gives $|S| \ge p\Theta_0/2$. The block inverse formula

$$J^{-1} = \begin{pmatrix} S^{-1} & -S^{-1}u^T A^{-1} \\ -A^{-1}vS^{-1} & A^{-1} + A^{-1}vS^{-1}u^T A^{-1}\end{pmatrix}$$

then gives the displayed bound. □

4.7. **Repaired gauge by implicit inversion in time.**

Lemma 4.17 (Time chain rule for preliminary profile)**.** *For each $\alpha = 0,\dots,3$, set $G_\alpha(\tau) := \mathcal{F}_\alpha(1,0;V_{\mathrm{pre}}(\tau))$. Then $G_\alpha \in W^{1,1}_{\mathrm{loc}}(I_*)$ and for a.e. τ,*

$$G'_\alpha(\tau) = D_V\mathcal{F}_\alpha(1,0;V_{\mathrm{pre}}(\tau))[\partial_\tau V_{\mathrm{pre}}(\tau)].$$

The right-hand side is obtained from the profile formulas of Proposition 4.14 and is in $L^1_{\mathrm{loc}}(I_*)$.

Proof. Let $V^\varepsilon_{\mathrm{pre}}$ be a spatial and temporal Steklov mollification inside a slightly larger compact cylinder. For the smoothed profiles the ordinary chain rule gives

$$\frac{d}{d\tau}\mathcal{F}_\alpha(1,0;V^\varepsilon_{\mathrm{pre}}(\tau)) = D_V\mathcal{F}_\alpha(1,0;V^\varepsilon_{\mathrm{pre}}(\tau))[\partial_\tau V^\varepsilon_{\mathrm{pre}}(\tau)].$$

The transformed equation expresses $\partial_\tau V^\varepsilon_{\mathrm{pre}}$ as the mollification of $\nu\Delta V_{\mathrm{pre}} - (V_{\mathrm{pre}}\cdot\nabla)V_{\mathrm{pre}} - \nabla P_{\mathrm{pre}}$ plus the chart transport terms. Pairing this distribution with the compactly supported Gateaux gradients below is legitimate by the local suitable bounds, the pressure bound, and the weighted L^3 control in (SC4). Passing $\varepsilon \downarrow 0$ gives the asserted $W^{1,1}_{\mathrm{loc}}$ identity. For $W \in \mathcal{V}_R$, these derivative formulas hold:

$$D_V\mathcal{F}_0(1,0;V)[W] = \int_{\mathbb{R}^3} 3\chi_R|Y|^{-p}|V|\,V\cdot W\,\mathrm{d}Y,$$
$$D_V\mathcal{F}_j(1,0;V)[W] = \int_{\mathbb{R}^3} 2Y_j\chi_R\,V\cdot W\,\mathrm{d}Y, \qquad j=1,2,3,$$

where both right-hand sides are finite on $\mathcal{V}_R$ by compact support, $p<3$, and $V,W \in L^3\cap W^{1,3/2}_{\mathrm{loc}}$. The right-hand side is finite by the explicit formulas above and by the distributional equation paired with compactly supported test gradients; the local energy, pressure, and weighted L^3 bounds give an $L^1_{\mathrm{loc}}(I_*)$ majorant. □

Lemma 4.18 (AC implicit inversion with parameters)**.** *Let $U\subset\mathbb{R}^4$ be open and $G : I_*\times U \to \mathbb{R}^4$. Assume:*

(i) for each $s\in U$, $\tau\mapsto G(\tau,s)\in W^{1,1}_{\mathrm{loc}}(I_*;\mathbb{R}^4)$, and $D_\tau G \in L^1_{\mathrm{loc}}(I_*;L^\infty(U;\mathbb{R}^4))$;

(ii) for a.e. $\tau\in I_*$, $s\mapsto G(\tau,s)$ is C^1 on U, with D_sG continuous in (τ,s);

(iii) for some $(\tau_0, s_0) \in I_* \times U$, $G(\tau_0, s_0) = 0$, *and there is* $\rho, C > 0$ *such that*

$$\sup_{(\tau,s)\in I_*\times B_\rho(s_0)} \|D_sG(\tau,s)^{-1}\|_{\mathcal{L}(\mathbb{R}^4,\mathbb{R}^4)} \le C.$$

Then after possibly shrinking I_* *to a subinterval* $J \ni \tau_0$, *there is a unique map* $s \in AC(J;U)$, $s(\tau_0) = s_0$, *with*

$$G(\tau, s(\tau)) = 0 \quad \text{a.e. in } J.$$

Moreover

$$s'(\tau) = -D_sG(\tau, s(\tau))^{-1}\, D_\tau G(\tau, s(\tau)) \quad \text{a.e. on } J.$$

Proof. By (ii), the matrix field D_sG is continuous in s, so there is $r_0 \in (0,\rho)$ such that $\inf_{(\tau,s)\in I_*\times B_{r_0}(s_0)} |\det D_sG(\tau,s)| > 0$ after possibly shrinking I_*; this keeps D_sG^{-1} bounded and continuous on $B_{r_0}(s_0)$.

Set

$$F(\tau, s) := -D_sG(\tau, s)^{-1}D_\tau G(\tau, s).$$

Then $F(\tau,\cdot)$ is continuous on $B_{r_0}(s_0)$ for a.e. τ and $F(\cdot, s) \in L^1_{\mathrm{loc}}(I_*;\mathbb{R}^4)$ by (i), hence F is a Carathéodory right-hand side. Since the gauge map is C^1 in the finite variables, after shrinking r_0 there is $L \in L^1(J)$ such that

$$|F(\tau, s_1) - F(\tau, s_2)| \le L(\tau)|s_1 - s_2| \quad \text{for } s_1, s_2 \in B_{r_0}(s_0).$$

Choose $\delta > 0$ so small that

$$\int_{\tau_0-\delta}^{\tau_0+\delta} \sup_{s\in B_{r_0}(s_0)} |F(\sigma, s)|\, d\sigma < r_0.$$

On $J = [\tau_0 - \delta, \tau_0 + \delta] \cap I_*$ define the Picard operator

$$(\mathcal{T}s)(\tau) := s_0 + \int_{\tau_0}^{\tau} F(\sigma, s(\sigma))\, d\sigma$$

on $C(J; B_{r_0}(s_0))$. The preceding bound keeps $\mathcal{T}$ in the same closed ball, and the Lipschitz estimate gives

$$\|\mathcal{T}s_1 - \mathcal{T}s_2\|_{C(J)} \le \left(\int_J L(\sigma)\, d\sigma\right) \|s_1 - s_2\|_{C(J)}.$$

After decreasing δ once more, the coefficient is < 1, so Banach's fixed-point theorem gives a unique solution

$$s(\tau) = s_0 + \int_{\tau_0}^{\tau} F(\sigma, s(\sigma))\, d\sigma, \qquad \tau \in J,$$

which stays in $B_{r_0}(s_0) \subset U$. This solution is absolutely continuous.

Define $H(\tau) := G(\tau, s(\tau))$. Since $G(\cdot, s(\cdot)) \in W^{1,1}_{\mathrm{loc}}$ from (i), the chain rule gives for a.e. $\tau \in J$

$$H'(\tau) = D_\tau G(\tau, s(\tau)) + D_sG(\tau, s(\tau))s'(\tau) = D_\tau G(\tau, s(\tau)) + D_sG(\tau, s(\tau))F(\tau, s(\tau)) = 0.$$

Hence $H \equiv G(\tau_0, s_0) = 0$ on J, which is the claimed constraint. Uniqueness follows from Grönwall on the difference of two AC solutions on $B_{r_0}(s_0)$, using Lipschitz continuity of $F(\tau,\cdot)$ on that ball. □

Definition 4.19 (Retained nondegenerate repaired-gauge regime)**.** Let $J \subset I_*$ be compact. The retained nondegenerate repaired-gauge regime over the preliminary profile V_{pre} consists of branches for which there are

$$\lambda \in AC(J; (0,\infty)), \qquad c \in AC(J;\mathbb{R}^3),$$

such that, for a.e. $\tau \in J$,

$$V(\tau) := V_{\mathrm{pre}}^{\lambda(\tau),c(\tau)} \in \mathcal{A}_{p,R}, \qquad \mathcal{F}(1,0;V(\tau)) = 0.$$

The path is quantitatively nondegenerate if the translation-block, annular, weighted first-moment, and Schur-complement bounds in Proposition 4.16 hold on J. If no such local AC selection exists, or if the quantitative nondegeneracy hypotheses fail persistently, the branch lies in the gauge-degenerate, multibubble, or cascade alternatives rather than in the retained regime.

Proposition 4.20 (Gauge regularity on the retained regime)**.** *Let (λ, c) be the repaired-gauge selection of the retained nondegenerate regime on J. Then*

$$\mathcal{F}\big(\lambda(\tau), c(\tau); V_{\mathrm{pre}}(\tau)\big) = 0 \quad \textit{for a.e. } \tau \in J,$$

and $(\lambda, c) \in AC(J;\mathbb{R}^4)$ by definition. If the path is quantitatively nondegenerate, then

$$\operatorname*{ess\,sup}_{\tau\in J} \|J(V(\tau))^{-1}\| < \infty.$$

Proof. The constraint follows from Lemma 4.13 applied to $V(\tau) = V_{\mathrm{pre}}^{\lambda(\tau),c(\tau)}$. The absolute continuity is part of Definition 4.19. The uniform inverse bound is Proposition 4.16. ☐

Definition 4.21 (Repaired chart)**.** Define

$$\Lambda(\tau) := \Lambda_{\mathrm{pre}}(\tau)\lambda(\tau), \qquad X(\tau) := X_{\mathrm{pre}}(\tau) + \Lambda_{\mathrm{pre}}(\tau)c(\tau),$$

$$V(\tau) := V_{\mathrm{pre}}^{\lambda(\tau),c(\tau)}, \qquad P(\tau) := P_{\mathrm{pre}}^{\lambda(\tau),c(\tau)} \text{ (same physical pressure pull-back under } \Phi).$$

Theorem 4.22 (AC repaired parameters)**.** *Assume* $\mathrm{SC}(R_*, r_*, M, \eta, \kappa)$*, and suppose the branch lies in the retained nondegenerate repaired-gauge regime on J, with selection (λ, c). Then the corrected chart (Λ, X) is absolutely continuous on J, satisfies*

$$a(\tau) = -\Lambda'(\tau) \in L^1_{\mathrm{loc}}(J), \qquad b(\tau) = -X'(\tau) \in L^1_{\mathrm{loc}}(J),$$

and the gauge constraints $\mathcal{F}(1,0;V(\tau)) = 0$ hold a.e. Equivalently,

$$\mathcal{F}(\lambda(\tau), c(\tau); V_{\mathrm{pre}}(\tau)) = 0$$

for the repaired-gauge selection (λ, c).

Proof. By Definition 4.19, $(\lambda, c) \in AC(J)$ and the repaired gauge constraints hold for $V(\tau)$ a.e. By definition $V(\tau) = V_{\mathrm{pre}}^{\lambda(\tau),c(\tau)}$, so Lemma 4.13 applied at $(\lambda, c) = (\lambda(\tau), c(\tau))$ gives

$$\mathcal{F}(1,0;V_{\mathrm{pre}}^{\lambda,c}) = \mathcal{F}(\lambda, c; V_{\mathrm{pre}})$$

implies $\mathcal{F}(1,0;V(\tau)) = 0$ a.e.

Since $\Lambda = \Lambda_{\mathrm{pre}}\lambda$ and $X = X_{\mathrm{pre}} + \Lambda_{\mathrm{pre}}c$, and $\Lambda_{\mathrm{pre}}, X_{\mathrm{pre}}$ are AC by (SC5), the products and sums preserve AC, hence $(\Lambda, X) \in AC(I_*)$. Differentiating,

$$a = -\Lambda' = -(\Lambda'_{\mathrm{pre}}\lambda + \Lambda_{\mathrm{pre}}\lambda') \in L^1_{\mathrm{loc}}, \qquad b = -X' = -(X'_{\mathrm{pre}} + \Lambda'_{\mathrm{pre}}c + \Lambda_{\mathrm{pre}}c') \in L^1_{\mathrm{loc}}.$$

This gives the claimed regularity. ☐

Remark 4.23 (Gauge degeneracy). Failure of invertibility of J is the gauge-degenerate alternative to the retained single-core regime for this fixed renormalization mechanism. In the Type II decomposition of alternatives this case belongs to the multibubble/cascade and scale-rigidity alternatives, rather than to the representation theorem.

4.8. **Pressure transport and decomposition.**

Lemma 4.24 (Renormalized pressure equation)**.** *If V, P satisfy the renormalized Navier–Stokes equation in Theorem 4.6, then*

$$-\Delta P = \partial_i \partial_j (V_i V_j) \qquad \text{in } \mathcal{D}'(Q^*_{2R}).$$

Proof. Take test $\varphi \in C_c^\infty(Q^*_{2R})$. Since $V, P \in L^2_{\rm loc} \times L^{3/2}_{\rm loc}$ on Q^*_{2R}, all pairings below are legitimate. Expand

$$\nabla \cdot \big((V \cdot \nabla)V\big) = \sum_{i,j} \partial_i \partial_j (V_i V_j) - \sum_{i,j} \partial_i V_i \, \partial_j V_j.$$

Because $\nabla \cdot V = 0$, this reduces to $\partial_i \partial_j (V_i V_j)$. Taking divergence of

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P - \nu \Delta V + a(V + Y \cdot \nabla V) + b \cdot \nabla V = 0$$

and using $\nabla \cdot V = 0$, $\nabla \cdot (V + Y \cdot \nabla V) = 0$, $\nabla \cdot (b \cdot \nabla V) = b \cdot \nabla(\nabla \cdot V) = 0$, and $\nabla \cdot \Delta V = \Delta(\nabla \cdot V) = 0$, the following identity holds in $\mathcal{D}'$:

$$\Delta P + \partial_i \partial_j (V_i V_j) = 0.$$

Hence $-\Delta P = \partial_i \partial_j (V_i V_j)$ on Q^*_{2R}. □

Theorem 4.25 (Localized pressure decomposition)**.** *Let $\zeta \in C_c^\infty(B_{2R})$, $\zeta \equiv 1$ on B_R, and assume $V \in L^1_{\rm loc}(I_*; L^3(B_{2R}))$, $P \in L^1_{\rm loc}(I_*; L^{3/2}(B_{2R}))$. For a.e. $\tau \in I_*$ there exists*

$$P(\cdot, \tau) = P_{\rm loc}(\cdot, \tau) + H(\cdot, \tau)$$

on B_R such that

$$\begin{cases} -\Delta P_{\rm loc}(\cdot, \tau) = \partial_i \partial_j \big(\zeta V_i V_j\big), \\ -\Delta H(\cdot, \tau) = 0 \ \text{ on } B_R, \quad ⨍_{B_R} H(\cdot, \tau)\, dY = 0. \end{cases}$$

Moreover

$$\|P_{\rm loc}(\tau)\|_{L^{3/2}(B_R)} \lesssim \|V(\tau)\|^2_{L^3(B_{2R})}, \qquad \|H(\tau)\|_{L^q(B_r)} \lesssim_{q,R,r} \|P(\tau)\|_{L^{3/2}(B_{2R})} + \|V(\tau)\|^2_{L^3(B_{2R})}$$

for each $1 \le q < \infty$ and each $0 < r < R$, with constants depending only on R, r, and q.

Proof. For fixed τ, set

$$P_{\rm loc} = \mathcal{R}_i \mathcal{R}_j (\zeta V_i V_j), \qquad H := P - P_{\rm loc}.$$

$\zeta V_i V_j \in L^1_{\rm loc}(I_*; L^{3/2}(B_{2R}))$, so the Calderón–Zygmund operator $f \mapsto \mathcal{R}_i \mathcal{R}_j f$ gives

$$P_{\rm loc}(\cdot, \tau) = \mathcal{R}_i \mathcal{R}_j (\zeta V_i V_j)(\cdot, \tau), \qquad \|P_{\rm loc}(\tau)\|_{L^{3/2}(B_R)} \lesssim \|V(\tau)\|^2_{L^3(B_{2R})}.$$

Because $\mathcal{R}_i \mathcal{R}_j$ is linear bounded on $L^{3/2}$,

$$P_{\rm loc} \in L^1_{\rm loc}(I_*; L^{3/2}(B_R)).$$

Since $-\Delta P_{\rm loc} = \partial_i \partial_j (\zeta V_i V_j)$ and $\zeta \equiv 1$ on B_R,

$$-\Delta H(\cdot, \tau) = \partial_i \partial_j \big((1 - \zeta) V_i V_j\big) = 0 \quad \text{in } B_R,$$

so $H(\cdot, \tau)$ is harmonic on B_R. Imposing $⨍_{B_R} H = 0$ fixes the time-dependent additive constant uniquely in each time slice. For $q \in [1, \infty)$ and $0 < r < R$, harmonicity and local L^p-to-L^q interior estimates imply

$$\|H(\tau)\|_{L^q(B_r)} \lesssim_{q,R,r} \|H(\tau)\|_{L^{3/2}(B_R)}.$$

Finally, $\|H(\tau)\|_{L^{3/2}(B_R)} \le \|P(\tau)\|_{L^{3/2}(B_R)} + \|P_{\rm loc}(\tau)\|_{L^{3/2}(B_R)}$, giving the announced bound in terms of P and V. □

Remark 4.26 (Measurability and normalization). If V, P are measurable in τ, then $\tau \mapsto P_{\mathrm{loc}}(\cdot, \tau)$ is measurable in $L^{3/2}(B_R)$ by boundedness of $\mathcal{R}_i\mathcal{R}_j$, hence $\tau \mapsto H(\cdot, \tau)$ is measurable in each $L^q(B_r)$, $0 < r < R$. The normalization $\fint_{B_R} H(\cdot, \tau)\, dY = 0$ fixes the time-dependent additive constant in H uniquely for each τ, and therefore the decomposition is a legitimate renormalized pressure splitting.

4.9. **Modulation from differentiated constraints.**

Lemma 4.27 (Differentiation of constraints)**.** *For the repaired profile V constructed above, with $|Y|^{-p}|V|^3 \in L^1_{\mathrm{loc}}(I_*; L^1)$, Then $G_\alpha(\tau) := \mathcal{F}_\alpha(1, 0; V(\tau))$ are absolutely continuous and*

$$\frac{\mathrm{d}}{\mathrm{d}\tau} G_\alpha(\tau) = \int_{\mathbb{R}^3} \mathbf{D}_\alpha(V, \nabla V, \nabla\chi_R)\, \partial_\tau V \,\mathrm{d}Y$$

for a.e. τ. Here $\mathbf{D}_\alpha$ is the L^1-integrable Gateaux gradient associated to $\mathcal{F}_\alpha$.

Proof.

$$G_\alpha(\tau) = \mathcal{F}_\alpha(1, 0; V(\tau))$$

is first evaluated on spatial-temporal mollifications V^ε. For those smooth profiles the ordinary chain rule gives

$$\frac{d}{d\tau} G^\varepsilon_\alpha(\tau) = D_V \mathcal{F}_\alpha(1, 0; V^\varepsilon(\tau))\, [\partial_\tau V^\varepsilon(\tau)].$$

The local suitable bounds, pressure estimates, and compact support of the Gateaux gradients below allow passage to the limit $\varepsilon \downarrow 0$, which gives the displayed derivative formula for V. For $\mathcal{F}_0$,

$$\mathbf{D}_0(V, \nabla V, \nabla\chi_R) = 3\chi_R |Y|^{-p}|V|\, V;$$

for $\alpha = 1, 2, 3$,

$$\mathbf{D}_\alpha(V, \nabla V, \nabla\chi_R) = 2\chi_R\, Y_\alpha V.$$

Each coefficient is $L^1(B_{2R})$: $3\chi_R|Y|^{-p}|V|\, V \in L^{3/2}(B_{2R}) \subset L^1(B_{2R})$ from $p < 3$ and $|Y|^{-p}|V|^3 \in L^1$, and $2\chi_R Y_\alpha V \in L^2(B_{2R}) \subset L^1(B_{2R})$ by compact support.

$$\frac{d}{d\tau} G_\alpha(\tau) = \int_{\mathbb{R}^3} \mathbf{D}_\alpha(V, \nabla V, \nabla\chi_R) \cdot \partial_\tau V \,\mathrm{d}Y,$$

Since $\mathbf{D}_\alpha(V, \nabla V, \nabla\chi_R) \in L^1(B_{2R})$ and $\partial_\tau V \in L^1_{\mathrm{loc}}(I_*; L^2(B_{2R}))$, the right-hand side is in $L^1_{\mathrm{loc}}(I_*)$, so $G_\alpha \in W^{1,1}_{\mathrm{loc}}(I_*) \subset AC_{\mathrm{loc}}(I_*)$. ☐

Lemma 4.28 (Directional derivatives in the profile variable)**.** *For $\mathcal{A}_\tau = V + Y \cdot \nabla V$ and $\alpha = 0, 1, 2, 3$,*

$$D_V\mathcal{F}_\alpha(1, 0; V)[\mathcal{A}_\tau] = -\partial_\lambda \mathcal{F}_\alpha(1, 0; V), \qquad D_V\mathcal{F}_\alpha(1, 0; V)[\partial_k V] = -\partial_{c_k}\mathcal{F}_\alpha(1, 0; V).$$

Indeed, from $V_{\lambda,c}(Y) = \lambda^{-1}V(\lambda^{-1}(Y - c))$,

$$\partial_\lambda V_{\lambda,c}\big|_{(1,0)} = -(V + Y \cdot \nabla V) = -\mathcal{A}_\tau, \qquad \partial_{c_k} V_{\lambda,c}\big|_{(1,0)} = -\partial_k V.$$

Substituting these into the Gateaux formulas from Proposition 4.14 gives the two identities.

Theorem 4.29 (Modulation system)**.** *Let (V, P) be the repaired pair on the compact window under consideration. Writing $\theta := (a, b)$, define*

$$\mathcal{A}_\tau(Y) := V(\tau, Y) + Y \cdot \nabla V(\tau, Y).$$

For $\alpha = 0, 1, 2, 3$, define

$$\mathcal{R}_\alpha(\tau) := -\int_{\mathbb{R}^3} \mathbf{D}_\alpha\big(V, \nabla V, \nabla\chi_R\big) \cdot \big((V \cdot \nabla)V + \nabla P - \nu\Delta V\big)\,\mathrm{d}Y.$$

Then for a.e. $\tau \in I_*$*,*

$$\partial_\lambda \mathcal{F}_\alpha(1,0;V)\, a(\tau) + \sum_{k=1}^{3} \partial_{c_k} \mathcal{F}_\alpha(1,0;V)\, b_k(\tau) = \mathcal{R}_\alpha(\tau),$$

Equivalently,

$$J(\tau)\theta(\tau) = \mathcal{R}(\tau).$$

Hence

$$\theta(\tau) = J(\tau)^{-1}\mathcal{R}(\tau) \in L^1_{\mathrm{loc}}(I_*)^4, \qquad |\theta(\tau)| \le \kappa\, |\mathcal{R}(\tau)|.$$

Proof. Set $G_\alpha(\tau) := \mathcal{F}_\alpha(1,0;V(\tau))$. By Theorem 4.22, $G_\alpha(\tau) = 0$ a.e.; by Lemma 4.27, $G_\alpha \in W^{1,1}_{\mathrm{loc}}$. Hence for a.e. τ,

$$0 = \frac{d}{d\tau} G_\alpha(\tau) = D_V \mathcal{F}_\alpha(1,0;V(\tau))\, [\partial_\tau V(\tau)].$$

Insert the equation from Theorem 4.6,

$$\partial_\tau V = -(V\cdot\nabla)V - \nabla P + \nu\Delta V - a\mathcal{A}_\tau - b\cdot\nabla V.$$

This gives

$$0 = -D_V\mathcal{F}_\alpha\big[(V\cdot\nabla)V + \nabla P - \nu\Delta V\big] - a\, D_V\mathcal{F}_\alpha[\mathcal{A}_\tau] - \sum_{k=1}^{3} b_k\, D_V\mathcal{F}_\alpha[\partial_k V].$$

By Lemma 4.27, for any profile direction W,

$$D_V\mathcal{F}_\alpha(1,0;V)[W] = \int_{\mathbb{R}^3} \mathbf{D}_\alpha(V,\nabla V,\nabla\chi_R)\cdot W\,\mathrm{d}Y.$$

By Lemma 4.28,

$$D_V\mathcal{F}_\alpha[\mathcal{A}_\tau] = -\partial_\lambda\mathcal{F}_\alpha(1,0;V), \qquad D_V\mathcal{F}_\alpha[\partial_k V] = -\partial_{c_k}\mathcal{F}_\alpha(1,0;V).$$

Substituting into the previous line yields the linear system

$$\partial_\lambda \mathcal{F}_\alpha(1,0;V)\, a(\tau) + \sum_{k=1}^{3} \partial_{c_k} \mathcal{F}_\alpha(1,0;V)\, b_k(\tau) = \mathcal{R}_\alpha(\tau),$$

which is equivalent to $J(\tau)\theta(\tau) = \mathcal{R}(\tau)$. By definition of $\mathbf{D}_\alpha$, each $\mathcal{R}_\alpha \in L^1_{\mathrm{loc}}(I_*)$, and Proposition 4.16 gives $|J^{-1}(\tau)| \le K$ a.e., hence

$$\theta(\tau) = J^{-1}(\tau)\mathcal{R}(\tau), \quad \theta \in L^1_{\mathrm{loc}}(I_*), \quad |\theta| \le K|\mathcal{R}| \text{ a.e.}.$$

□

Corollary 4.30 (Regularity of a, b)**.** *For the repaired pair of Theorem 4.29,* $a, b \in L^1_{\mathrm{loc}}(I_*)$*. If in addition* $\mathcal{R} \in L^\infty_{\mathrm{loc}}(I_*)$*, then* $a, b \in L^\infty_{\mathrm{loc}}(I_*)$*.*

Proof. By Theorem 4.29, $\theta = (a,b) = J^{-1}\mathcal{R}$ a.e., and $J^{-1} \in L^\infty_{\mathrm{loc}}(I_*)$ by Theorem 4.16. The first claim is therefore immediate from $\mathcal{R} \in L^1_{\mathrm{loc}}$, while the second follows if $\mathcal{R} \in L^\infty_{\mathrm{loc}}$. □

4.10. **Repaired suitable structure in renormalized variables.**

Proposition 4.31 (Transport of suitable local energy inequality)**.** *Suppose (u,p) is suitable on $\mathcal{Q}^{\mathrm{ph}}_{2R}$, and (V,P) is obtained by the repaired map Φ. Set*

$$\tilde a(\tau) := -\Lambda(\tau)^{-1}\Lambda'(\tau), \qquad \tilde b(\tau) := -\Lambda(\tau)^{-1}X'(\tau).$$

Then for every nonnegative $\phi \in C_c^\infty(Q_R^*)$ and $\tau_1 < \tau_2$,

$$\begin{aligned}\mathscr{E}_\phi(V,P) := {}& \frac12|V|^2\partial_\tau(\phi^2) + \frac{\nu}{2}|V|^2\Delta_Y\phi^2 + \frac12|V|^2\, V\cdot\nabla(\phi^2)\\ &+ (P_{\mathrm{loc}}+H)\, V\cdot\nabla(\phi^2) + \frac{\tilde a}{2}|V|^2\phi^2 + \frac{\tilde a}{2}|V|^2\, Y\cdot\nabla(\phi^2) + \frac12|V|^2\,\tilde b\cdot\nabla(\phi^2).\end{aligned}$$

$$\begin{aligned}\frac12\int_{B_R}|V(\tau_2)|^2\phi(\cdot,\tau_2)^2 + \nu\int_{\tau_1}^{\tau_2}\!\!\int_{B_R}|\nabla V|^2\phi^2\,dY\,d\tau \le \frac12\int_{B_R}|V(\tau_1)|^2\phi(\cdot,\tau_1)^2\\ + \int_{\tau_1}^{\tau_2}\!\!\int_{B_R}\mathscr{E}_\phi(V,P)\,dY\,d\tau.\end{aligned}$$

The right-hand side is controlled on compact windows by $\|\tilde a\|_{L^1}, \|\tilde b\|_{L^1}, M$, and the local bounds from Theorem 4.25, and support bounds from Section 4.4. Hence (V,P) is locally suitable on Q_R^*.

Proof. Apply the physical local energy inequality on $\mathcal{Q}^{\mathrm{ph}}_{2R}$: for nonnegative $\psi \in C_c^\infty(\mathcal{Q}^{\mathrm{ph}}_{2R})$,

$$\frac12\int|u|^2\psi\Big|_{t_1}^{t_2} + \nu\int_{t_1}^{t_2}\!\!\int|\nabla_x u|^2\psi\,dxdt \le \int_{t_1}^{t_2}\!\!\int\Big(\frac12|u|^2(\partial_t\psi+\nu\Delta_x\psi) + \Big(\frac{|u|^2}{2}u + p\,u\Big)\cdot\nabla_x\psi\Big)\,dxdt.$$

Choose

$$\psi(x,t) := \Lambda(t)^{-1}\phi\Big(\frac{x-X(t)}{\Lambda(t)},\tau(t)\Big)^2, \qquad \tau=\tau(t),\ d\tau/dt = \Lambda^{-2},$$

let $t_i = t(\tau_i)$, $i=1,2$. By Proposition 4.3 and Section 4.4, $\psi \in C_c^\infty(\mathcal{Q}^{\mathrm{ph}}_{2R})$.

$\nabla_x\psi = \Lambda^{-2}\nabla_Y\phi^2$, $\quad \Delta_x\psi = \Lambda^{-3}\Delta_Y\phi^2$, $\quad \partial_t\psi = -\Lambda^{-4}\Lambda'\phi^2 + \Lambda^{-3}\partial_\tau(\phi^2) - \Lambda^{-4}(X'+\Lambda'Y)\cdot\nabla_Y\phi^2$.

Using $u = \Lambda^{-1}V$, $p = \Lambda^{-2}P$, $\nabla_x u = \Lambda^{-2}\nabla_Y V$, $dx\,dt = \Lambda^5\,dY\,d\tau$, each term in the physical inequality becomes

$$\frac12\int|u|^2\psi\Big|_{t_1}^{t_2} = \frac12\int|V|^2\phi^2\Big|_{\tau_1}^{\tau_2},$$

$$\nu\int|\nabla_x u|^2\psi\,dxdt = \nu\int|\nabla V|^2\phi^2\,dY\,d\tau,$$

$$\frac12\int|u|^2\partial_t\psi\,dxdt = \frac12\int|V|^2\left(\partial_\tau(\phi^2) + \tilde a\phi^2 + \tilde a\,Y\cdot\nabla_Y\phi^2 + \tilde b\cdot\nabla_Y\phi^2\right)\,dY\,d\tau,$$

$$\frac{\nu}{2}\int|u|^2\Delta_x\psi\,dxdt = \frac{\nu}{2}\int|V|^2\Delta_Y\phi^2\,dY\,d\tau,$$

$$\int\left(\frac{|u|^2}{2}u + p\,u\right)\cdot\nabla_x\psi\,dxdt = \int\left(\frac12|V|^2V + (P_{\mathrm{loc}}+H)V\right)\cdot\nabla_Y\phi^2\,dY\,d\tau.$$

Substituting these identities and grouping terms gives the proposition's right-hand side and coefficients $\tilde a, \tilde b$. ∎

4.11. **Stability under subsequences.**

Proposition 4.32. *Let $(u^{(n)}, p^{(n)})$ satisfy $\mathrm{SC}(R_*, r_*, M, \eta, \kappa)$ with the same data parameters and let $(\Lambda^{(n)}, X^{(n)}, V^{(n)}, P^{(n)}, a^{(n)}, b^{(n)})$ be obtained from retained quantitatively nondegenerate repaired-gauge selections on the same I_*, with the constants in Proposition 4.16 uniform in n. After extraction $n_j \to \infty$, there exist subsequences and limits (Λ, X, V, P, a, b) such that*

$$\Lambda^{(n_j)} \to \Lambda \text{ in } L^1_{\mathrm{loc}}, \quad X^{(n_j)} \to X \text{ in } L^1_{\mathrm{loc}},$$
$$\Lambda^{(n_j)\prime} \rightharpoonup \Lambda' \text{ in } L^1_{\mathrm{loc}}, \quad X^{(n_j)\prime} \rightharpoonup X' \text{ in } L^1_{\mathrm{loc}},$$
$$V^{(n_j)} \to V \text{ in } L^2_{\mathrm{loc}}(I_*; L^2_{\mathrm{loc}}), \quad V^{(n_j)} \rightharpoonup V \text{ in } L^2_{\mathrm{loc}}(I_*; H^1_{\mathrm{loc}}),$$
$$P^{(n_j)} \rightharpoonup P \text{ in } L^{3/2}_{\mathrm{loc}}(Q^*_R), \quad (a^{(n_j)}, b^{(n_j)}) \rightharpoonup (a, b) \text{ in } L^1_{\mathrm{loc}}(I_*)^4.$$

Moreover $a = -\Lambda'$, $b = -X'$ a.e. on I_*, and the limit obeys all conclusions below.

Proof. From SC and the retained repaired-gauge selections, $(V^{(n)}, P^{(n)})$ are uniformly bounded in $L^\infty_\tau L^2(B_{2R}) \cap L^2_\tau H^1(B_{2R})$ and $L^{3/2}_{\mathrm{loc}}(Q^*_R)$. Theorem 4.16 gives uniform invertibility bounds for $J^{(n)}$, and Theorem 4.29 gives $(a^{(n)}, b^{(n)})$ uniformly in L^1_{loc} (since $\mathcal{R}^{(n)} \in L^1_{\mathrm{loc}}$). Hence $\Lambda^{(n)}$ and $X^{(n)}$ are uniformly bounded in $W^{1,1}_{\mathrm{loc}}$, so by BV compactness and 1D subsequences,

$$\Lambda^{(n_j)} \to \Lambda \text{ in } L^1_{\mathrm{loc}}, \qquad X^{(n_j)} \to X \text{ in } L^1_{\mathrm{loc}},$$

and $a^{(n_j)} \rightharpoonup -\Lambda'$, $b^{(n_j)} \rightharpoonup -X'$ in L^1_{loc}. From the renormalized equation,

$$\partial_\tau V^{(n)} \in L^1_\tau(H^{-1}_{\mathrm{loc}}) + L^{4/3}_\tau(L^{12/11}_{\mathrm{loc}}),$$

uniformly on compact windows. Aubin–Lions compactness on bounded balls gives

$$V^{(n_j)} \to V \text{ in } L^2_{\mathrm{loc}}(I_*; L^2_{\mathrm{loc}}), \quad V^{(n_j)} \rightharpoonup V \text{ in } L^2_{\mathrm{loc}}(I_*; H^1_{\mathrm{loc}}).$$

By Calderón–Zygmund and Theorem 4.25, $P^{(n_j)} \rightharpoonup P$ in $L^{3/2}_{\mathrm{loc}}(Q^*_R)$, and with normalized pressure split $P = P_{\mathrm{loc}} + H$, the same for P_{loc} and H. Passing to limits in $\mathcal{F}(1, 0; V^{(n)}(\tau)) = 0$, Theorem 4.6, and the linear modulation system yields the corresponding limit constraints and equations. □

4.12. **Final representation theorem.**

Theorem 4.33 (Chart and repaired-gauge representation)**.** *Assume $\mathrm{SC}(R_*, r_*, M, \eta, \kappa)$, and suppose the branch lies in the retained repaired-gauge regime with selection (λ, c) on $J = [\tau_0 - \ell, \tau_0 + \ell] \subset I_*$. Then there are fields (V, P) on Q^J_{2R}, with $(\Lambda, X) \in AC(J)$, $a = -\Lambda'$, $b = -X' \in L^1_{\mathrm{loc}}(J)$, such that*

$$u(x,t) = \Lambda(\tau)^{-1} V\left(\frac{x - X(\tau)}{\Lambda(\tau)}, \tau\right), \quad \frac{dt}{d\tau} = \Lambda(\tau)^2,$$

and $\mathcal{F}(1, 0; V(\tau)) = 0$ for $\tau \in J$.

Proof. From (SC5) and Proposition 4.3, $(\Lambda_{\mathrm{pre}}, X_{\mathrm{pre}})$ are AC on I_*, so V_{pre} is defined on Q^*_{2R}. The retained repaired-gauge regime supplies $(\lambda, c) \in AC(J; (0, \infty) \times \mathbb{R}^3)$ and $\mathcal{F}(1, 0; V^{\lambda,c}_{\mathrm{pre}}) = 0$ a.e. By Definition 4.21 and Theorem 4.22, setting

$$\Lambda(\tau) = \Lambda_{\mathrm{pre}}(\tau)\lambda(\tau), \qquad X(\tau) = X_{\mathrm{pre}}(\tau) + \Lambda_{\mathrm{pre}}(\tau)c(\tau), \qquad V(\tau) = V^{\lambda(\tau), c(\tau)}_{\mathrm{pre}},$$

and $P(\tau) = P^{\lambda(\tau), c(\tau)}_{\mathrm{pre}}$, gives $\Lambda, X \in AC(J)$ and $a = -\Lambda'$, $b = -X' \in L^1_{\mathrm{loc}}(J)$, and

$$\mathcal{F}(1, 0; V(\tau)) = 0 \text{ for a.e. } \tau \in J.$$

Finally $u = \Lambda^{-1} V(\cdot, \tau) \circ \Phi^{-1}$ by definition of the repaired map, hence the physical identity with $\frac{dt}{d\tau} = \Lambda^2$ follows. □

Theorem 4.34 (Pressure and modulation)**.** *Assume, in addition to Theorem 4.33, that the retained repaired-gauge selection is quantitatively nondegenerate in the sense of Proposition 4.16. Then (V,P) satisfies the renormalized Navier–Stokes equation with the sign convention of Theorem 4.6; there is a decomposition $P = P_{\mathrm{loc}} + H$ on Q_R^* as in Theorem 4.25; and (a,b) is the unique solution of*

$$J(\tau)\begin{pmatrix} a(\tau) \\ b(\tau) \end{pmatrix} = \mathcal{R}(\tau), \qquad \begin{pmatrix} a \\ b \end{pmatrix} \in L^1_{\mathrm{loc}}(J)^4,$$

with bounds $|(a,b)| \le K|\mathcal{R}|$, where K depends on $R, p, \Theta_0, m_0, C_u, \varepsilon_0, \chi_R$.

Proof. Equation and sign convention are inherited directly from Theorem 4.6 applied to (Λ, X) from Theorem 4.33 and the definition of V. By Theorem 4.25 applied at each time slice, the decomposition on Q_R^J is defined and unique once $\fint_{B_R} H = 0$ is imposed. The modulation identity is Theorem 4.29; uniqueness follows from invertibility of J (Theorem 4.16) and the pointwise identity $(a,b) = J^{-1}\mathcal{R}$. Uniform bounds on (a,b) are obtained by the bound on J^{-1} there and the stated bound on $\mathcal{R}$. □

Theorem 4.35 (Compact-window representation theorem)**.** *Under the assumptions of Theorems 4.33 and 4.34, for every compact window $J \subset I_*$ and each fixed $R > 0$, setting $Q_R^J := B_R \times J$, one has:*

(1) the exact renormalized equation on Q_{2R}^J,
(2) the four repaired-gauge constraints $\mathcal{F}(1, 0; V(\tau)) = 0$,
(3) bounds on $a, b \in L^1_{\mathrm{loc}}(J)$ (and L^∞_{loc} under $\mathcal{R} \in L^\infty_{\mathrm{loc}}$),
(4) local pressure decomposition $P = P_{\mathrm{loc}} + H$ with $L^{3/2}$ local bounds and harmonic remainder controls,
(5) renormalized local suitable energy inequality on Q_R^J,
(6) stability under subsequences in the topologies above.

The later reductions use only these conclusions.

Proof. From Theorem 4.33 and the choice of $J \subset I_*$, all quantities are defined on Q_{2R}^J. Items (1) and (2) are Theorems 4.6 and 4.22; item (3) is Theorem 4.29 together with the inverse bound from Proposition 4.16; item (4) is Theorem 4.25; item (5) is Proposition 4.31; item (6) is Proposition 4.32. Combining these results gives the stated compact-window representation. □

Remark 4.36. The preceding results separate two alternatives:

(1) the nondegenerate single-core repaired-gauge regime, where the representation theorems apply;
(2) gauge degeneracy ($\det J = 0$) or compact-structure failure, which belongs to the multi-bubble, cascade, or scale-rigidity alternatives.

5. Multibubble and Cascade Exclusion

Analytic inputs.

The local reduction theorem uses the following analytic inputs from the preceding sections, with profile decompositions and decoupling estimates in their standard critical-space sense [Gal01, BL83]. The remaining work is the multibubble, cascade, and terminal-frame decoupling.

(i) From Section 3, the compact single-core criterion Theorem 3.18: a compact single-core finite-cost Type II branch with positive local CKN concentration, nondegenerate scale-center selection, pressure reconstruction, bounded compact-cylinder suitable estimates, bounded modulation, and vanishing localized dissipation is excluded.

(ii) From Section 4, the renormalized equation Theorem 4.6, pressure decomposition Theorem 4.25, modulation system Theorem 4.29, and repaired-gauge representation Theorems 4.33 to 4.35: the renormalized equation, pressure decomposition modulo functions of time, bounded modulation coefficients, renormalized suitable structure, and stability of these structures under subsequences.

(iii) From Section 6, the compact-cylinder Caccioppoli estimate Theorem 6.4 and rough-core exclusion Theorem 6.7.

(iv) From Section 7, the collapse cost exclusion Theorems 7.6 and 7.7 and autonomous reduced-limit theorem Theorem 7.19 for scale-rigid or scale-collapsing branches.

5.1. **Local setting and active frames.** *Normalized local Navier–Stokes setting.* All quantities are written in rescaled local variables $(x,t) \in Q_R = B_R \times (-R^2, 0)$ with viscosity $\nu = 1$ (critical scaling has absorbed ν). A local pair (u_n, p_n) on such a cylinder satisfies

$$\partial_t u_n - \Delta u_n + (u_n \cdot \nabla) u_n + \nabla p_n = 0, \qquad \nabla \cdot u_n = 0,$$

and pressure is normalised through spatial averaging

$$(p_n)_{B_\rho}(t) := |B_\rho|^{-1} \int_{B_\rho} p_n(x,t)\, dx$$

to factor out time-dependent constants.

For $r > 0$, $x_0 \in \mathbb{R}^3$, and $t_0 \in \mathbb{R}$, let

$$Q_r(x_0,t_0) := B_r(x_0) \times (t_0 - r^2, t_0), \qquad Q_r := Q_r(0,0) = B_r \times (-r^2, 0).$$

Let (u_n, p_n) be parabolic rescalings around a potential singular point with positive local Caffarelli–Kohn–Nirenberg concentration on compact cylinders. Concretely, for a base triple (x_n, t_n, λ_n),

$$u_n(y,s) = \lambda_n u(x_n + \lambda_n y,\ t_n + \lambda_n^2 s), \qquad p_n(y,s) = \lambda_n^2 p(x_n + \lambda_n y,\ t_n + \lambda_n^2 s).$$

For each n and $R > 0$, define the scaling-invariant local quantities

$$A_n(R) := R^{-1} \operatorname*{ess\,sup}_{-R^2<t<0} \int_{B_R} |u_n(x,t)|^2\, dx,$$

$$E_n(R) := R^{-1} \int_{Q_R} |\nabla u_n(x,t)|^2\, dx\, dt,$$

$$C_n(R) := R^{-2} \int_{Q_R} |u_n(x,t)|^3\, dx\, dt, \qquad D_n(R) := R^{-2} \int_{-R^2}^0 \int_{B_R} |p_n - (p_n)_{B_R}(t)|^{3/2}\, dx\, dt,$$

where

$$(p_n)_{B_R}(t) := |B_R|^{-1} \int_{B_R} p_n(x,t)\, dx.$$

For an active frame (x_n, λ_n), $\lambda_n > 0$, define the critical rescaling

$$\mathcal{T}_{x_n,\lambda_n} f(y) := \lambda_n f(x_n + \lambda_n y).$$

If $\phi \in L^3_\sigma(\mathbb{R}^3)$ is observed in this frame, its contribution is

$$\phi_n(y) = \mathcal{T}_{x_n,\lambda_n}\phi(y) = \lambda_n \phi(x_n + \lambda_n y).$$

The critical Jacobian identity is exact:

$$\|\mathcal{T}_{x_n,\lambda_n} f\|^3_{L^3(E)} = \int_{x_n+\lambda_n E} |f(x)|^3\, dx$$

for every measurable $E \subset \mathbb{R}^3$. In particular

$$\|\mathcal{T}_{x_n,\lambda_n} f\|^3_{L^3(\mathbb{R}^3)} = \|f\|^3_{L^3(\mathbb{R}^3)},$$

so $\mathcal{T}_{x_n,\lambda_n}$ is an isometry on $L^3(\mathbb{R}^3)$.

Definition 5.1 (Bounded selected-window limit)**.** A selected-window subsequence has a bounded selected-window limit on a compact cylinder Q_R if there exists $u_\infty \in L^\infty(Q_R)$ such that, after extracting a subsequence,

$$\forall \varepsilon > 0\ \exists N :\ \forall n \geq N,\ \|u_n - u_\infty\|_{L^3(Q_R)} < \varepsilon.$$

The limit is nonzero if $u_\infty \not\equiv 0$ on Q_R.

Definition 5.2 (Local Type II sequence)**.** A positive local concentration sequence is called a local Type II sequence if every selected-window rescaling has zero bounded selected-window limit on every compact cylinder Q_R:

$$\forall R > 0, \qquad \lim_{n\to\infty} \|u_n\|_{L^3(Q_R)} = 0.$$

Definition 5.3 (Active frame)**.** Fix a concentrating time sequence $t_n \uparrow T$ and a compact time interval $I \Subset (-\infty, 0]$. An active frame is a tuple

$$\mathfrak{F}^j = \big(x^j_n, \lambda^j_n, I, U^j_n, P^j_n\big), \qquad \lambda^j_n \downarrow 0,$$

where $x^j_n \in \mathbb{R}^3$ and

$$U^j_n(y,s) := \lambda^j_n u(x^j_n + \lambda^j_n y, t_n + (\lambda^j_n)^2 s), \qquad P^j_n(y,s) := (\lambda^j_n)^2 p(x^j_n + \lambda^j_n y, t_n + (\lambda^j_n)^2 s)$$

on $B_{2R} \times I$ for every fixed compact cylinder under consideration. The frame is active at threshold $\eta_* > 0$ if for some fixed $R_0 < \infty$

$$\liminf_{n\to\infty} \int_{Q_{R_0}} |U^j_n|^3\, dy\, ds \geq \eta_*^3.$$

Definition 5.4 (Relations between active frames)**.** Let $\mathfrak{F}^i$ and $\mathfrak{F}^j$ be two active frames. After passing to a subsequence exactly one of the following parameter regimes holds.

(i) They are *same-point comparable-scale* if

$$0 < c \leq \frac{\lambda^i_n}{\lambda^j_n} \leq C < \infty, \qquad \frac{|x^i_n - x^j_n|}{\max(\lambda^i_n, \lambda^j_n)} \leq C$$

for all large n.

(ii) They form a *same-point cascade* if, after relabeling,

$$\frac{\lambda^i_n}{\lambda^j_n} \to 0, \qquad \frac{|x^i_n - x^j_n|}{\lambda^j_n} = O(1).$$

The frame i is then the inner frame and j is an outer frame.

(iii) They are *separated-point* if

$$\frac{|x_n^i - x_n^j|}{\lambda_n^i} \to \infty \quad \text{and} \quad \frac{|x_n^i - x_n^j|}{\lambda_n^j} \to \infty.$$

Definition 5.5 (Terminal frame and perturbative remainder)**.** Given a finite active family, a terminal frame is chosen by selecting an active scale with minimal order:

$$\lambda_n^{j_*} \le C_j \lambda_n^j \qquad \text{for every active } j$$

after passing to a subsequence. Comparable frames at the same physical point are grouped before this choice; if several grouped frames remain tied, one is chosen by a fixed lexicographic ordering of their labels. A remainder r_n in a terminal frame is perturbative on $B_{2R} \times I$ if

$$\|r_n\|_{L_I^\infty L^3(B_{2R})} \to 0$$

and its equation residual and pressure satisfy

$$\|f_n\|_{L_I^1 H^{-1}(B_R)} + \|\nabla \pi_n^r\|_{L_I^1 H^{-1}(B_R)} \to 0$$

for every compact $B_R \times I$.

Definition 5.6 (Local multibubble/cascade branch)**.** A local multibubble/cascade branch is a positive-concentration local Type II branch for which at least one of the following alternatives occurs:

(i) the compact-cylinder suitable estimates $A_n + E_n + C_n + D_n$ are unbounded on some fixed rescaled cylinder;
(ii) positive CKN mass splits into two or more comparable active cores;
(iii) a strict same-point scale cascade appears inside bounded rescaled cylinders;
(iv) separated physical points remain active in different local rescaled frames;
(v) the localized scale or centering selection degenerates (no unique core selected).

Definition 5.7 (Active profile family)**.** An active profile family is a family

$$\{(\phi^j, x_n^j, \lambda_n^j)\}_{j\in J}, \qquad \phi^j \in L^3_\sigma(\mathbb{R}^3),$$

with finite or countable index set J, extracted along a positive local concentration sequence. In physical variables write

$$\Lambda_{\lambda,x_0}\phi(x) := \lambda^{-1}\phi\left(\frac{x - x_0}{\lambda}\right).$$

For every finite $J_0 \subset J$ the family gives an expansion

$$u(\cdot, t_n) = \sum_{j\in J_0} \Lambda_{\lambda_n^j, x_n^j}\phi^j + r_n^{J_0}$$

on the compact terminal windows under consideration. After the finite active subfamily has been selected, the corresponding remainder satisfies the perturbative bounds of Theorem 5.5. The parameters are considered only after grouping same-point comparable-scale profiles into compound profiles as in Theorem 5.8. A profile is active if

$$\|\phi^j\|_{L^3(\mathbb{R}^3)} \ge \eta_* > 0.$$

The family has finite critical mass when

$$\sum_{j\in J} \|\phi^j\|^3_{L^3(\mathbb{R}^3)} \le M_* < \infty.$$

It captures positive local concentration if every compact terminal-window sequence with

$$\liminf_{n\to\infty}\int_{Q_R}|u_n|^3>0$$

either contains an active profile observed in that window or is absorbed into the selected terminal frame; equivalently, no positive terminal-window concentration remains in the perturbative remainder.

Definition 5.8 (Compound profile)**.** Let $J_k\subset J$ be a same-point comparable-scale class. Choose one representative $(x_n^{j_k},\lambda_n^{j_k})$. After passing to a subsequence,

$$\rho_j:=\lim_{n\to\infty}\frac{\lambda_n^j}{\lambda_n^{j_k}}\in(0,\infty),\qquad z_j:=\lim_{n\to\infty}\frac{x_n^j-x_n^{j_k}}{\lambda_n^{j_k}}\in\mathbb{R}^3.$$

The compound profile in the representative frame is the L^3_σ-sum

$$\Phi^k(y):=\sum_{j\in J_k}\rho_j^{-1}\phi^j\left(\frac{y-z_j}{\rho_j}\right),$$

whenever J_k is finite; finiteness follows for active classes from Theorem 5.9. If $\|\Phi^k\|_{L^3}<\varepsilon_\star$, where $\varepsilon_\star$ is the critical small-data threshold, the class is perturbative. Otherwise Φ^k is treated as one active object in its representative frame.

Lemma 5.9 (Finite active profile count)**.** *Every active profile family with finite critical mass contains only finitely many active profiles. More precisely,*

$$\#J\le M_*\eta_*^{-3}.$$

Proof. For each active profile $j\in J$,

$$\|\phi^j\|^3_{L^3(\mathbb{R}^3)}\ge\eta_*^3.$$

Let $J_0\subset J$ be finite. Summing over J_0 gives

$$\#J_0\,\eta_*^3\le\sum_{j\in J_0}\|\phi^j\|^3_{L^3(\mathbb{R}^3)}\le\sum_{j\in J}\|\phi^j\|^3_{L^3(\mathbb{R}^3)}\le M_*.$$

Hence $\#J_0\le M_*\eta_*^{-3}$. Taking the supremum over all finite subsets J_0 yields the same bound for the cardinality of J, so J is finite and

$$\#J\le M_*\eta_*^{-3}.$$

□

Lemma 5.10 (Exhaustive parameter partition)**.** *Let $\mathfrak{F}^i,\mathfrak{F}^j$ be two active frames. After passing to a subsequence, either the pair is same-point comparable-scale, same-point cascade, or separated-point in the sense of Theorem 5.4; if none of these relations holds in a selected compact frame, then one of the two contributions is locally invisible there in L^3.*

Proof. Pass to a subsequence so that

$$\frac{\lambda_n^i}{\lambda_n^j}\to\ell\in[0,\infty]$$

in the extended half-line. If $0<\ell<\infty$, then either

$$\frac{|x_n^i-x_n^j|}{\max(\lambda_n^i,\lambda_n^j)}$$

is bounded along a further subsequence, giving the same-point comparable-scale case, or it tends to infinity, giving separated points because the two scales are comparable.

If $\ell = 0$, then either

$$\frac{|x_n^i - x_n^j|}{\lambda_n^j}$$

is bounded, giving a same-point cascade with i inner and j outer, or it tends to infinity. In the latter case the j-profile is locally invisible in the i-frame by Theorem 5.13. The case $\ell = \infty$ is the same after interchanging i and j. These alternatives exhaust all subsequential limits of the scale ratio and normalized center distance. □

Theorem 5.11 (Minimal bad-configuration extraction). *Let a positive local Type II branch admit an active profile family with finite critical mass, perturbative remainder in the sense of Theorem 5.5, and concentration capture in the sense of Theorem 5.7. If the branch is not already a single active terminal frame satisfying the single-core hypotheses, then after passing to a subsequence one of the following occurs:*

(i) *compact-cylinder suitable control fails, and Theorem 5.15 produces a bounded-cylinder concentration core;*
(ii) *there are at least two active frames in a same-point comparable-scale class, hence a compound profile in the sense of Theorem 5.8;*
(iii) *there is a same-point cascade;*
(iv) *there are separated active physical points;*
(v) *the scale-center selection of the selected compound or terminal frame is degenerate.*

In every case the positive local concentration is carried by an active frame or compound profile, not by the perturbative remainder.

Proof. If compact-cylinder suitable control fails, item (i) follows from Theorem 5.15. Assume therefore that the compact-cylinder family is bounded on the windows under consideration.

By concentration capture, positive local concentration cannot remain solely in the perturbative remainder. Hence either one selected active frame carries the terminal-window concentration, or another active frame remains non-negligible. If there is one nondegenerate selected active frame and no competing active object, the branch is in the single-core regime. Since this case is excluded by hypothesis, either the selected scale-center map is degenerate, giving item (v), or at least two active frames remain.

For two active frames, apply Theorem 5.10. Same-point comparable-scale frames give item (ii), same-point scale separation gives item (iii), and separated points give item (iv). The remaining alternative in Theorem 5.10 is local invisibility of one contribution in the selected compact frame; such a contribution is absorbed into the perturbative remainder and cannot carry the retained positive local concentration by the capture property. This gives the extraction statement. □

5.2. **Elementary local decoupling estimates.**

Lemma 5.12 (Critical rescalings that escape are locally invisible). *Let $\phi \in L^3(\mathbb{R}^3)$, let $R < \infty$, and define*

$$W_n(y) = \rho_n \phi(z_n + \rho_n y),$$

where $\rho_n > 0$. Assume that the sets

$$z_n + \rho_n B_R$$

either have measure tending to zero, or escape to spatial infinity in the sense that for every compact set $K \subset \mathbb{R}^3$,

$$(z_n + \rho_n B_R) \cap K = \varnothing$$

for all sufficiently large n. Then

$$\|W_n\|_{L^3(B_R)} \to 0.$$

Proof. By the critical change of variables,

$$\|W_n\|^3_{L^3(B_R)} = \int_{B_R} \rho_n^3 |\phi(z_n + \rho_n y)|^3 \, dy = \int_{z_n + \rho_n B_R} |\phi(x)|^3 \, dx.$$

Set

$$m_n := \int_{z_n + \rho_n B_R} |\phi(x)|^3 \, dx.$$

Since $\|W_n\|^3_{L^3(B_R)} = m_n$, it is enough to show $m_n \to 0$.

Fix $\varepsilon > 0$. We split into the two structural alternatives.

Case 1: shrinking measure. If $|z_n + \rho_n B_R| \to 0$, then by absolute continuity of the integral $|\phi|^3 \in L^1(\mathbb{R}^3)$ there exists $\delta > 0$ such that

$$|E| < \delta \Longrightarrow \int_E |\phi|^3 \, dx < \varepsilon.$$

Choose N with $|z_n + \rho_n B_R| < \delta$ for all $n \geq N$. Then

$$m_n = \int_{z_n + \rho_n B_R} |\phi|^3 \, dx < \varepsilon, \qquad n \geq N.$$

Case 2: escape to infinity. Assume now that for every compact set $K \subset \mathbb{R}^3$, $(z_n + \rho_n B_R) \cap K = \varnothing$ for all sufficiently large n. By absolute continuity at infinity, choose compact $K \subset \mathbb{R}^3$ such that

$$\int_{\mathbb{R}^3 \setminus K} |\phi|^3 < \varepsilon.$$

For every such compact K, there is $N(K)$ with $(z_n + \rho_n B_R) \cap K = \varnothing$ for all $n \geq N(K)$. For these indices,

$$z_n + \rho_n B_R \subset \mathbb{R}^3 \setminus K,$$

and therefore, for $n \geq N(K)$,

$$m_n = \int_{z_n + \rho_n B_R} |\phi|^3 \, dx \leq \int_{\mathbb{R}^3 \setminus K} |\phi|^3 < \varepsilon.$$

In either case, for every $\varepsilon > 0$, $m_n < \varepsilon$ for n sufficiently large. Hence $m_n \to 0$, and therefore

$$\|W_n\|_{L^3(B_R)} \to 0.$$

$\square$

Lemma 5.13 (Separated active profiles vanish in the selected frame)**.** *Let (x_n^p, λ_n^p) and (x_n^q, λ_n^q) be two active frames with separated physical centers in the sense that*

$$\frac{|x_n^q - x_n^p|}{\lambda_n^p} \to \infty.$$

If $\phi^q \in L^3(\mathbb{R}^3)$, then the q-profile observed in the p-frame converges to zero locally in L^3: for every $R < \infty$,

$$\left\| \frac{\lambda_n^p}{\lambda_n^q} \phi^q \left(\frac{x_n^p - x_n^q}{\lambda_n^q} + \frac{\lambda_n^p}{\lambda_n^q} y \right) \right\|_{L^3(B_R)} \to 0.$$

Proof. The displayed function has the form in Theorem 5.12 with

$$z_n = \frac{x_n^p - x_n^q}{\lambda_n^q}, \qquad \rho_n = \frac{\lambda_n^p}{\lambda_n^q}.$$

The corresponding physical set in the q-profile variables is

$$z_n + \rho_n B_R = \frac{x_n^p - x_n^q}{\lambda_n^q} + \frac{\lambda_n^p}{\lambda_n^q} B_R.$$

For every $y \in B_R$,

$$|z_n + \rho_n y| \geq |z_n| - \rho_n R.$$

Set $E_n := z_n + \rho_n B_R$. Then

$$\operatorname{dist}(E_n, 0) = |z_n| - \rho_n R = \frac{\lambda_n^p}{\lambda_n^q} \left(\frac{|x_n^q - x_n^p|}{\lambda_n^p} - R \right).$$

Since $\frac{|x_n^q - x_n^p|}{\lambda_n^p} \to \infty$, there exists $N_0(R)$ such that

$$\frac{|x_n^q - x_n^p|}{\lambda_n^p} - R \geq 1, \quad n \geq N_0(R),$$

and hence for such n, $\operatorname{dist}(E_n, 0) \geq \rho_n$.

There are two cases.

(a) If $\rho_n \to 0$, then

$$|z_n + \rho_n B_R| = \rho_n^3 |B_R| \to 0.$$

Then the first alternative in Theorem 5.12 applies.

(b) Otherwise, after passing to a subsequence, $\rho_n \geq \rho_* > 0$. Then

$$\operatorname{dist}(z_n + \rho_n B_R, 0) \geq \rho_* \left(\frac{|x_n^q - x_n^p|}{\lambda_n^p} - R \right) \to \infty,$$

so the sets escape every fixed compact subset of $\mathbb{R}^3$. The second alternative in Theorem 5.12 applies.

In both subcases the $L^3(B_R)$-norm of the displayed expression converges to zero. □

Lemma 5.14 (Strict outer scales vanish in the innermost frame)**.** *Let several active profiles have the same physical center and scales $\lambda_n^1, \dots, \lambda_n^J$. Suppose λ_n^1 is an innermost scale and*

$$\frac{\lambda_n^1}{\lambda_n^j} \to 0 \qquad (j \neq 1).$$

Then every outer profile $\phi^j \in L^3(\mathbb{R}^3)$, viewed in the λ_n^1-frame, converges to zero in $L^3(B_R)$ for each fixed $R < \infty$. Any constant ambient velocity arising from the regular part of the outer flow is absorbed into the translation modulation.

Proof. In the innermost variables, the j-th outer contribution has the form

$$W_n^j(y) = \frac{\lambda_n^1}{\lambda_n^j} \phi^j \left(\frac{x_n^1 - x_n^j}{\lambda_n^j} + \frac{\lambda_n^1}{\lambda_n^j} y \right).$$

Set

$$\rho_n := \frac{\lambda_n^1}{\lambda_n^j} \to 0, \qquad z_n^j := \frac{x_n^1 - x_n^j}{\lambda_n^j}.$$

Then

$$W_n^j(y) = \rho_n \phi^j \left(z_n^j + \rho_n y \right),$$

and by the critical change of variables,

$$\|W_n^j\|_{L^3(B_R)}^3 = \int_{z_n^j+\rho_n B_R} |\phi^j(x)|^3\,dx.$$

Since $\rho_n \to 0$, the translated domains have

$$|z_n^j + \rho_n B_R| = \rho_n^3 |B_R| \to 0.$$

By absolute continuity of the integral for $\phi^j \in L^3(\mathbb{R}^3)$,

$$\|W_n^j\|_{L^3(B_R)} \to 0.$$

To treat a general ϕ^j, let $\phi_\varepsilon^j \in C_c^\infty(\mathbb{R}^3)$ with $\|\phi^j - \phi_\varepsilon^j\|_{L^3} < \varepsilon$. By critical isometry,

$$\|W_n^j - (W_n^j)_\varepsilon\|_{L^3(B_R)} \le \|\phi^j - \phi_\varepsilon^j\|_{L^3} < \varepsilon,$$

where $(W_n^j)_\varepsilon$ is the same scaling of ϕ_ε^j. The compactly supported term has vanishing local norm by the shrinking-domain argument above, and then letting first $n \to \infty$, then $\varepsilon \downarrow 0$ implies the claimed convergence. A constant vector field from the regular part of an outer flow contributes only a Galilean drift in the selected frame; it is therefore incorporated into the translation modulation and does not remain as an exterior perturbation. □

Lemma 5.15 (Compactness failure produces another core)**.** *If the compact-cylinder suitable estimates are unbounded on a fixed rescaled cylinder, then after passing to a subsequence the local $C_n + D_n$ quantity is unbounded on a comparable cylinder. Consequently the branch contains an additional local CKN concentration core.*

Proof. Fix $r > 0$. Apply the local Caccioppoli inequality with outer radius $2r$. For each index n, suitable weak solutions satisfy, with C_r independent of n,

$$A_n(r) + E_n(r) \le C_r\big(1 + C_n(2r) + C_n(2r)^{2/3} + D_n(2r)\big),$$

where the pressure is normalized by subtracting spatial means $(p_n)_{B_{2r}}(t)$.

Suppose that $A_n(r) + E_n(r) + C_n(r) + D_n(r)$ is unbounded along a subsequence. If $C_n(2r) + D_n(2r)$ were bounded on this subsequence, the preceding estimate would give uniform boundedness of $A_n(r)+E_n(r)$. Monotonicity of the local integrals also gives $C_n(r)+D_n(r) \le C_n(2r)+D_n(2r)$, contradicting unboundedness of the full family on Q_r. Therefore, after extracting a subsequence,

$$C_n(2r) + D_n(2r) \to \infty.$$

Define the localized CKN concentration at variable center and radius

$$G_n(x,t,\rho) := \rho^{-2}\int_{Q_\rho(x,t)} |u_n|^3\,dx\,ds + \rho^{-2}\int_{Q_\rho(x,t)} |p_n - (p_n)_{B_\rho}(s)|^{3/2}\,dx\,ds.$$

Set

$$\mathbf{M}_n := \sup\{G_n(x,t,\rho) : 0 < \rho \le 2r,\ Q_\rho(x,t) \subset Q_{4r}\}.$$

The choice $(x,t,\rho) = (0,0,2r)$ is admissible because $Q_{2r}(0,0) \subset Q_{4r}$, so

$$\mathbf{M}_n \ge C_n(2r) + D_n(2r) \to \infty.$$

After extracting a further subsequence, we may assume $\mathbf{M}_n \ge 1$ for all n. Choose (x_n, t_n, ρ_n) with $0 < \rho_n \le 2r$, $Q_{\rho_n}(x_n,t_n) \subset Q_{4r}$, such that

$$G_n(x_n,t_n,\rho_n) \ge \tfrac{1}{2}\,\mathbf{M}_n \ge \tfrac{1}{2}.$$

Rescale around (x_n, t_n, ρ_n):

$$v_n(y,s) := \rho_n u_n(x_n + \rho_n y, t_n + \rho_n^2 s), \qquad q_n(y,s) := \rho_n^2 p_n(x_n + \rho_n y, t_n + \rho_n^2 s).$$

Using the critical Jacobian identity,

$$\int_{Q_1} |v_n|^3 \, dy \, ds + \int_{Q_1} |q_n - (q_n)_{B_1}(s)|^{3/2} \, dy \, ds = G_n(x_n, t_n, \rho_n) \geq \tfrac{1}{2}.$$

Hence this new frame carries positive local critical mass, i.e. a new local CKN concentration core, as required. □

5.3. **Terminal active-frame decoupling.**

Definition 5.16 (Admissible terminal active family)**.** Fix a compact cylinder $B_{2R} \times I$ and a selected active frame. A terminal active family is admissible on this cylinder if the following hold.

(i) The active profiles satisfy the finite-mass and active-mass conditions of Theorem 5.7.

(ii) Every nonselected active profile is either separated from the selected physical point in the sense of Theorem 5.13, or has the same physical point and a strictly outer scale in the sense of Theorem 5.14, after comparable same-point scales have been grouped.

(iii) The rescaled remainder r_n satisfies

$$\|r_n\|_{L^\infty_I L^3(B_{2R})} \to 0,$$

and there exist fields (π^r_n, f_n) such that on $B_{2R} \times I$,

$$\partial_t r_n - \Delta r_n + a_n\big(r_n + y \cdot \nabla r_n\big) + b_n \cdot \nabla r_n + \nabla \pi^r_n = f_n, \qquad \nabla \cdot r_n = 0,$$

with

$$\|f_n\|_{L^1_I H^{-1}(B_R)} \to 0,$$

$$\|\nabla \pi^r_n\|_{L^1_I H^{-1}(B_R)} \to 0.$$

(iv) The selected velocity satisfies

$$\sup_n \|U_n\|_{L^\infty_I L^3(B_{2R})} < \infty.$$

(v) The selected modulation coefficients satisfy

$$\sup_n \big(\|a_n\|_{L^\infty(I)} + \|b_n\|_{L^\infty(I)}\big) < \infty.$$

(vi) Local pressure decomposition is stable under the splitting: if tensor coefficients tend to zero in $L^1_I L^{3/2}(B_R)$, then the corresponding pressure gradients tend to zero in $L^1_I H^{-1}(B_R)$, after subtracting spatial means.

(vii) Cutoff commutators generated by localizing to B_R are controlled by the $L^\infty_I L^3$, $L^1_I L^{3/2}$, and pressure-gradient norms appearing above. In particular, for any compact $\chi_R \in C^\infty_c(B_{2R})$ with $\chi_R \equiv 1$ on B_R, the commutators satisfy

$$\|[\Delta, \chi_R]u\|_{H^{-1}(B_R)} + \|[\nabla \chi_R, \nabla \cdot](u \otimes u)\|_{H^{-1}(B_R)} \leq C_R \|u\|^2_{L^3(B_{2R})}.$$

uniformly for $u \in L^3(B_{2R})$ and the same type of estimate for S_n-quadratic commutator terms.

Theorem 5.17 (Terminal profile-evolution decoupling)**.** *Consider an admissible terminal active family on $B_{2R} \times I$. Let U_n be the selected-frame velocity and let S_n be the sum, in that frame, of all nonselected active profiles together with the rescaled remainder. Then on $B_R \times I$:*

(i) $\nabla \cdot S_n = 0$ *distributionally;*

(ii) $S_n \to 0$ *in* $L^\infty_I L^3(B_{2R})$*;*

(iii) $U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n \to 0$ *in* $L^1_I L^{3/2}(B_R)$*;*

(iv)

$$\exists\,\pi_n,\ F_n:\quad \partial_t S_n - \Delta S_n + a_n\big(y\cdot\nabla S_n + S_n\big) + b_n\cdot\nabla S_n + \nabla\pi_n = F_n,$$
$$\|F_n\|_{L^1_I H^{-1}(B_R)} \to 0, \qquad \nabla\cdot S_n = 0 \quad \textit{in } \mathcal{D}'(B_R\times I).$$
$$\|\nabla\pi_n\|_{L^1_I H^{-1}(B_R)} \to 0.$$

(v) *every pressure source containing at least one factor of S_n has vanishing pressure gradient in $L^1_I H^{-1}(B_R)$;*

(vi) *localization errors introduced by compact cutoffs vanish in $L^1_I H^{-1}(B_R)$;*

(vii) *the effective modulation parameters of the selected frame remain bounded;*

(viii) *after subtracting S_n, the selected frame remains a positive finite-mass local Type II branch with pressure reconstruction and the compact-cylinder bounds required by the single-core analysis.*

Proof. Each profile in the family is divergence-free, and the rescaled remainder is divergence-free. Since divergence commutes with the Navier–Stokes critical rescalings, finite sums remain divergence-free in distributions. This establishes (i).

We prove (ii). By admissibility, every nonselected active profile is either separated from the selected physical point or is a same-point strictly outer scale. The separated profiles vanish locally by Theorem 5.13, and the same-point outer profiles vanish locally by Theorem 5.14. The remainder tends to zero in $L^\infty_I L^3(B_{2R})$ by Theorem 5.16. The number of active profiles is finite by Theorem 5.9. Since every summand in S_n tends to zero in $L^\infty_I L^3(B_{2R})$, the triangle inequality gives

$$\|S_n\|_{L^\infty_I L^3(B_{2R})} \le \sum_{j\neq j_*} \|S_n^j\|_{L^\infty_I L^3(B_{2R})} + \|r_n\|_{L^\infty_I L^3(B_{2R})} \to 0.$$

This establishes (ii).

For (iii), admissibility gives

$$\sup_n \|U_n\|_{L^\infty_I L^3(B_{2R})} < \infty.$$

Hence, by Hölder's inequality,

$$\|U_n\otimes S_n\|_{L^1_I L^{3/2}(B_R)} \le |I|\|U_n\|_{L^\infty_I L^3(B_R)}\|S_n\|_{L^\infty_I L^3(B_R)} \to 0,$$

and the same estimate applies to $S_n\otimes U_n$. Finally,

$$\|S_n\otimes S_n\|_{L^1_I L^{3/2}(B_R)} \le |I|\|S_n\|^2_{L^\infty_I L^3(B_R)} \to 0.$$

For (iv), write the represented Navier–Stokes equation for the full profile sum $U_n + S_n$ and subtract the represented equation for the selected component U_n. The remaining equation for S_n has the displayed linear left-hand side

$$\partial_t S_n - \Delta S_n + a_n\big(y\cdot\nabla S_n + S_n\big) + b_n\cdot\nabla S_n.$$

Collect all right-hand terms into

$$F_n := -\nabla\cdot(U_n\otimes S_n + S_n\otimes U_n + S_n\otimes S_n) + f_n + E_n^{\mathrm{cut}} + E_n^{\mathrm{mod}},$$

where f_n is the remainder forcing from item (iii) of admissibility, and $E_n^{\mathrm{cut}}, E_n^{\mathrm{mod}}$ are localization and modulation errors. The nonlinear stresses tend to zero in $L^1_I L^{3/2}(B_R)$, hence in $L^1_I H^{-1}(B_R)$, because $L^{3/2}(B_R)\hookrightarrow H^{-1}(B_R)$ on a bounded three-dimensional ball. The residual term vanishes by Theorem 5.16. The cutoff terms vanish by the localization part of admissibility, recorded explicitly in (vi). This gives the displayed convergence in $L^1_I H^{-1}(B_R)$.

For (v), decompose the pressure source of $U_n + S_n$ into the pure selected source, the pure exterior source, and the two mixed sources. The mixed sources are

$$\partial_i\partial_j(U_{n,i}S_{n,j} + S_{n,i}U_{n,j}) \quad \text{and} \quad \partial_i\partial_j(S_{n,i}S_{n,j}).$$

By (iii), their tensor coefficients tend to zero in $L^1_I L^{3/2}(B_R)$. The pressure-stability part of admissibility, equivalently local Calderon–Zygmund estimates plus harmonic pressure control modulo spatial means, gives convergence of the corresponding pressure gradients to zero in $L^1_I H^{-1}(B_R)$. Pure exterior sources are generated by profiles that are locally invisible in the selected frame; applying the same pressure stability after Theorems 5.13 and 5.14 gives their vanishing pressure gradients.

For (vi), let χ_R be a cutoff equal to one on B_R and supported in B_{2R}. The commutators are finite sums of terms of the following types:

$$(\nabla\chi_R)\cdot\nabla S_n, \qquad (\Delta\chi_R)S_n, \qquad (\nabla\chi_R)\cdot(U_n\otimes S_n + S_n\otimes U_n + S_n\otimes S_n),$$

plus analogous pressure cutoff terms. Their supports lie in the fixed annulus $B_{2R}\setminus B_R$. By the localization part of admissibility, these commutators are controlled by the norms already shown to vanish in (ii), (iii), and (v). Hence all localization errors vanish in $L^1_I H^{-1}(B_R)$.

For (vii), the only exterior contribution that can remain nonzero at first order in a compact selected frame is a constant ambient velocity. This has already been absorbed into the translation modulation. The remaining exterior profiles vanish locally, so their contribution to the modulation equations is small in the same local L^3 topology used above. The selected modulation coefficients are bounded by admissibility; hence the effective parameters remain bounded.

For $(viii)$, the positive CKN mass of the selected profile is unchanged by subtracting a term that tends to zero in local L^3 and whose mixed pressure sources vanish by (v). The pressure reconstruction is stable under the same decomposition by the pressure-stability condition in Theorem 5.16. The compact-cylinder bounds pass to the selected component because the full family is bounded and the nonretained terms vanish on compact cylinders. Thus the selected frame remains a positive finite-mass local Type II branch with the analytic inputs needed for the single-core analysis. ☐

5.4. **Same-point and separated-point reductions.**

Theorem 5.18 (Same-point cascade reduction)**.** *Every same-point multibubble or strict same-point cascade reduces, in the selected innermost or compound frame, to one of the following: the single-core criterion, scale-rigidity, or a perturbative terminal branch covered by Theorem 5.17, provided the terminal family generated in the selected innermost frame is admissible in the sense of Theorem 5.16.*

Proof. Step 1: scale-comparability classes. Split active same-point profiles into equivalence classes by

$$i\sim j \iff \exists c, C\in(0,\infty) : c\le \frac{\lambda_n^i}{\lambda_n^j}\le C \quad \text{for all } n$$

after passing to a subsequence. Thus within each class there are constants c_{ij}, C_{ij} with $c_{ij}\le \lambda_n^i/\lambda_n^j\le C_{ij}$ for all n, so one may define a single common rescaling sequence (choose $\lambda_n^{(k)}=\lambda_n^{j_k}$) and write all class-$k$ profiles in the same coordinates. Within each class, finite sums of divergence-free profiles remain divergence-free.

Step 2: compound profile reduction inside each class. For each class set

$$\Phi_n^k := \sum_{j\in J_k}\lambda_n^j\phi^j(x_n^j+\lambda_n^j y).$$

If the L^3-size of this rescaled sum satisfies

$$\left\|\sum_{j\in J_k}\phi^j\right\|_{L^3(\mathbb{R}^3)} < \varepsilon_\star,$$

with $\varepsilon_\star$ the perturbative small-data threshold, then standard local small-data theory for Navier–Stokes in the critical space eliminates this class from the active alternatives.

If the class is above threshold, it is retained as a single compound active core; at most one active core from that class is needed for local selection.

Step 3: local alternatives. If the retained compound core is unique and its scale-centering Jacobian is nondegenerate, we obtain the single selected core alternative. If this Jacobian is degenerate, the localized choice of active center or scale is not unique, which is the gauge-degenerate alternative in Theorem 5.6.

Step 4: strict same-point scale separation. Assume now the branch is not reduced to comparable scales and choose the innermost active scale λ_n^1 in that physical point. By Theorem 5.14, every profile with larger scale vanishes in the λ_n^1-frame in local L^3, up to constant drift terms. These constant drifts are absorbed into translation modulation by admissibility. Let S_n be the sum of all such outer profiles plus the rescaled remainder r_n. Then, by Step 2 and Theorem 5.14,

$$S_n \to 0 \quad \text{in } L_I^\infty L^3(B_{2R}).$$

Subtracting the equation for the innermost selected profile from the full-frame equation gives a perturbation equation for the selected profile whose forcing belongs to $L_I^1 H^{-1}(B_R)$ and vanishes. Thus Theorem 5.17 applies, and one of three outcomes holds:

(a) **Single-core branch:** after subtracting S_n, the branch is governed by one active component, so it reduces to the compact single-core branch of Theorem 5.20;

(b) **Scale-rigid branch:** the relative Jacobian between comparable scales becomes rigid in the sense of Theorem 5.21, so Theorem 5.21 excludes it;

(c) **Purely perturbative branch:** $S_n \to 0$ and the selected profile satisfies a vanishing forcing perturbation in $L_I^1 H^{-1}(B_R)$. No independent same-point cascade mechanism remains; after subtracting S_n, the selected component is reduced to the single-core or scale-rigidity analysis.

These are the strict same-point scale-separation alternatives, so the same-point reduction is exhausted. □

Theorem 5.19 (Separated-point decoupling)**.** *Separated active physical points are locally invisible in each other's active rescaled frames. Consequently every separated-point multibubble reduces in each selected active frame to a single-core, scale-rigid, or perturbative terminal branch, provided the terminal family in that selected active frame is admissible in the sense of Theorem 5.16.*

Proof. Fix an active point p with frame (x_n^p, λ_n^p). Let $q \neq p$ be another active point. By separation,

$$\frac{|x_n^q - x_n^p|}{\lambda_n^p} \to \infty.$$

Thus Theorem 5.13 gives local L^3 convergence to zero of the q-profile in the p-frame on every compact ball. The same conclusion holds for each profile centered at any physical point different from p. Since only finitely many active profiles remain after the critical mass threshold is imposed, the sum of all profiles outside the p-frame tends to zero locally in L^3.

The pure pressure generated by a separated profile is locally invisible by the same change of variables whenever its profile pressure belongs to $L^{3/2}_{\mathrm{loc}}$ with the standard Calderon–Zygmund control. Mixed pressure terms contain at least one exterior velocity factor, hence vanish in $L^1 H^{-1}_{\mathrm{loc}}$ by the pressure part of Theorem 5.17. Localization and modulation commutators are localized to $B_{2R} \setminus B_R$, so their H^{-1}-norm is controlled by the L^3-smallness already proved in Theorems 5.13 and 5.14 and by admissibility item (vii).

Therefore, in the selected p-frame, all other physical points contribute only a perturbative forcing tending to zero in $L^1 H^{-1}$ on compact cylinders. The selected frame remains a positive finite-mass branch with the same compact-cylinder and pressure reconstruction properties. Hence the branch reduces to the single-core case, the scale-rigid case, or the perturbative terminal branch. Since p was arbitrary, the same reduction holds in every active frame. □

5.5. **Multibubble and cascade exclusion.** The following statements isolate the analytic inputs needed for the multibubble reduction: the (already proved) single-core exclusion, the scale-rigidity reduction (a remaining hypothesis), and terminal active-frame admissibility (deduced here from the terminal nonlinear decoupling theorem Theorem 9.27). They are stated explicitly so that the following theorem uses no unstated compactness, pressure, or profile completeness conclusion.

Proposition 5.20 (Local single-core exclusion)**.** *Every compact single-core branch with positive local CKN concentration, nondegenerate scale-center selection, pressure reconstruction, bounded compact-cylinder suitable estimates, bounded modulation, and vanishing localized dissipation is excluded from the local Type II regime.*

Proof. This is precisely the conclusion of the local single-core Type II criterion, Theorem 3.18, applied to the represented compact single-core branch in the renormalized chart. The hypotheses listed here are the input data of that theorem: positive local CKN concentration provides the nondegenerate selected core, the nondegenerate scale-center selection together with the pressure reconstruction and bounded compact-cylinder suitable estimates are the inputs of the compactness statement Theorem 3.10, bounded modulation is Theorem 3.11, and vanishing localized dissipation is Theorem 3.4. The conclusion of Theorem 3.18 is exactly the stated exclusion. □

Assumption 5.21 (Local scale-rigidity exclusion)**.** Every scale-rigid local branch produced by the same-point or separated-point reductions, including a branch with degenerate relative scale-center Jacobian, either has a nonzero bounded selected-window limit or reduces to the single-core branch in Theorem 5.20. Consequently no such branch is compatible with the local Type II condition.

Remark 5.22 (Status of Theorem 5.21)*.* The remainder of this subsection proves Theorem 5.21 by an explicit local compactness stratification. The proof consists of the local lemmas Theorems 5.25 to 5.37 and the main decomposition Theorem 5.38. The final Theorem 5.40 then proves the conclusion of Theorem 5.21 from these results. All proofs use only the local Type II definition Theorem 3.2, the represented variables of Theorem 4.35, the local pressure decomposition of Theorem 4.25, the modulation L^∞ control of Theorem 4.34, the same-point and separated-point reductions Theorems 5.18 and 5.19, the minimal bad-configuration extraction Theorem 5.11, the single-core exclusion Theorem 3.18, the terminal CKN regularity criterion Theorem 9.34, and Banach–Alaoglu plus Calderón–Zygmund on fixed cylinders [CZ52, Ste70]. No global PDE bound, no Liouville theorem, and no profile-decomposition classification are invoked.

5.5.1. *Local stratification of scale-rigid branches.* We now identify the named scale-rigid local stratum and prove that every branch placed there by Theorems 5.18 and 5.19 either exits the local Type II class through Theorem 3.2 or relaxes back to one of the already excluded local strata.

Definition 5.23 (Scale-rigid local branch)**.** A scale-rigid local branch is a represented branch (V_n, P_n, a_n, b_n) on a fixed compact selected cylinder Q_{R_*}, produced by Step 3 or Step 4 of Theorem 5.18 or by Theorem 5.19, such that:

(R1) the compact-cylinder suitable estimates are uniformly bounded,

$$A_{V_n}(R_*) + E_{V_n}(R_*) + C_{V_n}(R_*) + D_{P_n}(R_*) \le M;$$

(R2) the branch carries positive local CKN concentration on a fixed inner cylinder $Q_{r_*} \Subset Q_{R_*}$,

$$C_{V_n}(r_*) + D_{P_n}(r_*) \geq \eta_0;$$

(R3) the modulation coefficients are bounded on the selected cylinder,

$$\|a_n\|_{L^\infty(-R_*^2,0)} + \|b_n\|_{L^\infty(-R_*^2,0)} \leq M;$$

(R4) the relative scale-center selection between the retained core and any previously selected core is rigid in one of the senses produced by Theorem 5.18: either the relative scale-center Jacobian degenerates along the subsequence (the gauge-degenerate alternative of Theorem 5.6 (v)), or the inner core is the innermost member of a same-point cascade in which all outer profiles have been absorbed into the perturbative remainder by Theorem 5.14.

Remark 5.24 (Inputs (R1)–(R3) are automatic on the selected cylinder)*.* The bounds (R1)–(R3) are not extra hypotheses but are inherited from the preceding reductions. Bound (R1) is the compact-cylinder hypothesis carried through Theorem 5.11; if it failed, Theorem 5.15 would assign the branch to the multicore alternative, not to the scale-rigid stratum. Bound (R2) is the positive concentration capture in Theorem 5.7. Bound (R3) is the selected-cylinder modulation control of Theorem 4.34, which is part of the represented-variable output of Theorem 4.35. No additional analytic input is needed to enter the scale-rigid stratum.

Lemma 5.25 (Bounded selected-window limits exit Type II)**.** *Let (V_n, P_n, a_n, b_n) be a represented sequence on a compact selected cylinder Q_r, and assume that, after passing to a selected-window subsequence, there is $V_* \in L^\infty(Q_r)$ with $V_* \not\equiv 0$ such that $V_n \to V_*$ strongly in $L^3(Q_r)$. Then the original sequence is not a local Type II sequence in the sense of Theorem 3.2.*

Proof. The hypothesis is exactly the statement that the subsequence has a nonzero bounded selected-window limit on Q_r in the sense of Theorem 3.1. By Theorem 3.2, a local Type II sequence has no selected-window subsequence with a nonzero bounded selected-window limit on any compact cylinder. Hence the sequence fails to be a local Type II sequence. ☐

Lemma 5.26 (Pressure and modulation stability under rigid-frame limits)**.** *Let (V_n, P_n, a_n, b_n) be a scale-rigid local branch on Q_{R_*} in the sense of Theorem 5.23. Fix $\rho \in (r_*, R_*)$. After passing to a subsequence:*

(i) *$V_n \rightharpoonup V_*$ weakly in $L^3(Q_\rho) \cap L_t^2 H_x^1(Q_\rho)$, and $V_n \to V_*$ strongly in $L^3(Q'_\rho)$ for every $\rho' \in (r_*, \rho)$;*

(ii) *the mean-normalized pressure $\widetilde{P}_n := P_n - \overline{P_n}_{B_\rho}$ converges weakly in $L^{3/2}(Q_\rho)$ to a limit P_* such that $-\Delta P_* = \partial_i \partial_j (V_{*,i} V_{*,j})$ on Q_ρ in the sense of distributions modulo a function of time;*

(iii) *$a_n \rightharpoonup^* a_*$ and $b_n \rightharpoonup^* b_*$ weakly-$*$ in $L^\infty(-R_*^2, 0)$, with $\|a_*\|_\infty + \|b_*\|_\infty \leq M$;*

(iv) *the limit (V_*, P_*, a_*, b_*) is a suitable weak solution of the represented Navier–Stokes equation (1) on Q_ρ, with $V_* \in L^3(Q_\rho) \cap L_t^2 H_x^1(Q_\rho)$.*

This lemma is only a compactness and stability statement. It does not assert boundedness of V_*. Bounded selected-window limits are obtained below from the separate no-subconcentration/CKN argument.

Proof. (i) Compactness of V_n. By (R1), the family (V_n) is bounded in $L^3(Q_\rho)$ (from $C_{V_n}(R_*)$), in $L_t^2 H_x^1(Q_\rho)$ (from $A_{V_n}(R_*) + E_{V_n}(R_*)$), and in $L_t^\infty L_x^2(Q_\rho)$ (from $A_{V_n}(R_*)$). Banach–Alaoglu yields the weak limits in $L^3(Q_\rho)$ and $L_t^2 H_x^1(Q_\rho)$ along a subsequence. The represented Navier–Stokes equation (1) bounds $\partial_s V_n$ in $L_t^{4/3} H_x^{-1}(Q_\rho)$ by Sobolev product estimates on the bounded cylinder, since each term on the right-hand side is controlled by $A_{V_n}(R_*)$, $E_{V_n}(R_*)$, $C_{V_n}(R_*)$, $D_{P_n}(R_*)$, and $\|a_n\|_\infty + \|b_n\|_\infty \le M$; the modulation correction $a_n(s)(V_n + y\cdot\nabla V_n) + b_n(s)\cdot\nabla V_n$ is bounded in $L^2_t L^{6/5}_x(Q_\rho) \subset L^{4/3}_t H^{-1}_x(Q_\rho)$ on the bounded cylinder. Hence Aubin–Lions [Aub63, Sim87] applied on the fixed cylinder Q_ρ yields strong convergence $V_n \to V_*$ in $L^3(Q_{\rho'})$ for every $\rho' \in (r_*, \rho)$.

(ii) Pressure stability. By (R1) the local pressure quantity $D_{P_n}(R_*)$ is bounded, equivalently $\widetilde{P}_n$ is bounded in $L^{3/2}(Q_\rho)$ by the local Calderón–Zygmund mean-normalization estimate [CZ52, Ste70] applied to the local pressure identity in Theorem 4.25. Banach–Alaoglu yields $\widetilde{P}_n \rightharpoonup P_*$ weakly in $L^{3/2}(Q_\rho)$. Pass to the limit in the local pressure identity

$$-\Delta\widetilde{P}_n = \partial_i\partial_j(V_{n,i}V_{n,j}) - \partial_i\partial_j(\overline{V_{n,i}V_{n,j}}_{B_\rho})$$

on Q_ρ. The right-hand side converges in $\mathcal{D}'(Q_\rho)$ by the strong L^3 convergence of (i) (which implies weak $L^{3/2}$ convergence of $V_{n,i}V_{n,j}$ by Hölder). This proves (ii).

(iii) Modulation limits. By (R3), a_n, b_n are bounded in $L^\infty(-R_*^2, 0)$; Banach–Alaoglu applied to the dual of L^1 yields the weak-$*$ limits with the same bound M.

(iv) Suitable weak solution status. Each (V_n, P_n) is suitable on Q_ρ (by (R1) and the represented-equation output of Theorem 4.35). The local energy inequality and the distributional momentum equation pass to the limit using (i)–(iii) and Hölder on the fixed cylinder Q_ρ: the dissipation term is lower semicontinuous under weak $L^2_t H^1_x$ convergence, the convection term converges by strong $L^3(Q_{\rho'})$ convergence, and the pressure-velocity pairing $\int \widetilde{P}_n V_n\cdot\nabla\chi$ converges by weak $L^{3/2}$ times strong L^3 on the fixed cylinder. The modulation correction converges by weak-$*$ L^∞ for (a_n, b_n) against the strong L^1_t convergence of $(V_n, y\cdot\nabla V_n, \nabla V_n)$ tested against fixed test functions on Q_ρ. This proves the represented suitable limit in the displayed local energy class. No L^∞ bound is inferred from the compact-cylinder estimates. $\square$

Lemma 5.27 (Compact chart comparability in scale-rigid gauges)**.** *Let (V, P, a, b) be a represented branch on $Q_{2\rho}(z_0)$ obtained from the repaired-gauge chart of Theorem 4.35. Assume that the associated chart (Λ, X) satisfies*

$$a = -\Lambda', \qquad b = -X', \qquad \|a\|_{L^\infty(I_{2\rho})} + \|b\|_{L^\infty(I_{2\rho})} \le M$$

on the time interval $I_{2\rho}$ of $Q_{2\rho}(z_0)$, and set $\Lambda_0 := \Lambda(s_0) > 0$ for $z_0 = (y_0, s_0)$. There exists $\rho_c = \rho_c(M, \Lambda_0, \rho) > 0$ such that, for every $0 < r \le \rho_c$, the chart on $Q_{2r}(z_0)$ obeys

$$\frac{\Lambda_0}{2} \le \Lambda(s) \le \frac{3\Lambda_0}{2}, \qquad |X(s) - X(s_0)| \le 4Mr^2, \qquad s \in (s_0 - 4r^2, s_0].$$

After a harmless concentric shrink, if needed, the physical image remains inside the support buffer of Theorem 4.7. Consequently Theorems 4.7and 4.8 apply on the retained subcylinder with

$$\Lambda_{\min} = \Lambda_0/2, \qquad \Lambda_{\max} = 3\Lambda_0/2,$$

and represented and physical CKN quantities on comparable cylinders are equivalent with constants depending only on M, Λ_0, r, ρ and the fixed buffer.

Proof. The identities $a = -\Lambda'$ and $b = -X'$ are part of the exact repaired change of variables in Theorems 4.5, 4.33 and 4.35. Hence, for $|s - s_0| \le 4r^2$,

$$|\Lambda(s) - \Lambda_0| \le \int_s^{s_0} |a(\sigma)|\, d\sigma \le 4Mr^2, \qquad |X(s) - X(s_0)| \le \int_s^{s_0} |b(\sigma)|\, d\sigma \le 4Mr^2.$$

Choose

$$\rho_c \le \min\{\rho/4, (\Lambda_0/(16M))^{1/2}\}$$

when $M > 0$, and choose $\rho_c = \rho/4$ when $M = 0$. Then $|\Lambda(s) - \Lambda_0| \le \Lambda_0/4$, which implies the displayed lower-upper scale bounds after enlarging the upper constant to $3\Lambda_0/2$. The center estimate is the preceding integral bound. A final concentric shrink, depending only on the fixed distance to the boundary of the selected compact window, keeps the image inside the domain buffer. Theorems 4.7 and 4.8 then give the stated comparison of the represented and physical quantities. No solution boundedness, and no vanishing dissipation, is used. □

Lemma 5.28 (Two-sided local CKN transfer in comparable charts)**.** *Let (V, P) be related to a physical suitable pair (u, p) by the repaired scale-translation chart on $Q_{2r}(z_0)$, and assume the chart comparability conclusion of Theorem 5.27 there:*

$$0 < \Lambda_{\min} \le \Lambda(s) \le \Lambda_{\max} < \infty, \qquad |X(s) - X(s_0)| \le Mr^2.$$

Set

$$\mathcal{C}_V(z_0, r) := r^{-2} \int_{Q_r(z_0)} |V|^3 + r^{-2} \int_{Q_r(z_0)} |P - (P)_{B_r}|^{3/2},$$

and define the physical CKN quantity $\mathcal{C}_u(x, t, \rho)$ analogously, with pressure normalized by spatial averages. There are constants

$$N_{\mathrm{tr}} < \infty, \qquad c_{\mathrm{tr}} \in (0, 1), \qquad C_{\mathrm{tr}} < \infty,$$

depending only on $M, \Lambda_{\min}, \Lambda_{\max}$ and the fixed support buffer, and physical cylinders

$$Q_\ell^{\mathrm{ph}} := B_{\rho_\ell}(x_\ell) \times (t_\ell - \rho_\ell^2, t_\ell), \qquad c_{\mathrm{tr}}\Lambda_{\min} r \le \rho_\ell \le C_{\mathrm{tr}}\Lambda_{\max} r, \qquad 1 \le \ell \le N_{\mathrm{tr}},$$

such that the physical image of $Q_{c_{\mathrm{tr}}r}(z_0)$ is contained in $\cup_\ell Q_\ell^{\mathrm{ph}}$, each Q_ℓ^{ph} is contained in the physical image of $Q_r(z_0)$, and

$$\mathcal{C}_u(x_\ell, t_\ell, \rho_\ell) \le C_{\mathrm{tr}} \mathcal{C}_V(z_0, r) \qquad (1 \le \ell \le N_{\mathrm{tr}}).$$

Conversely, if a finite physical cylinder cover of the image of $Q_r(z_0)$ with comparable radii satisfies $\mathcal{C}_u(x_\ell, t_\ell, \rho_\ell) \le \varepsilon$ for every member of the cover, then

$$\mathcal{C}_V(z_0, c_{\mathrm{tr}} r) \le C_{\mathrm{tr}} N_{\mathrm{tr}} \varepsilon.$$

Proof. The chart is

$$x = X(s) + \Lambda(s) y, \qquad dt = \Lambda(s)^2\, ds, \qquad V(y, s) = \Lambda(s) u(x, t), \qquad P(y, s) = \Lambda(s)^2 p(x, t).$$

The bounds on Λ and X imply that the image of a represented parabolic cylinder is trapped between parabolic cylinders whose radii are comparable to $\Lambda(s_0) r$. A standard Besicovitch covering with the bounded eccentricity supplied by $\Lambda_{\min}, \Lambda_{\max}$ gives the finite cover and the bounded overlap constant N_{tr}.

For the velocity term the change of variables gives, on every comparison cylinder,

$$\rho_\ell^{-2} \int_{Q_\ell^{\mathrm{ph}}} |u|^3 \, dx\, dt \le C_{\mathrm{tr}} r^{-2} \int_{Q_r(z_0)} |V|^3 \, dy\, ds.$$

For the pressure term we use the invariant seminorm $\inf_{c(s)} \int |P - c(s)|^{3/2}$. Spatial averages are equivalent to this seminorm up to the universal factor $2^{3/2}$, because for every ball B and $q \ge 1$,

$$\int_B |F - F_B|^q \le 2^q \inf_c \int_B |F - c|^q.$$

The same Jacobian identity therefore gives the displayed physical estimate for the pressure oscillation after subtracting spatial means. Applying the same argument to the inverse chart, whose scale is bounded between $\Lambda_{\max}^{-1}$ and $\Lambda_{\min}^{-1}$ on the comparison cover, gives the converse estimate. No direction of Theorem 4.8 is used as a black box here; both estimates come from the exact change of variables and the pressure oscillation seminorm. □

Lemma 5.29 (CKN regularity in comparable repaired gauges)**.** *Let (V, P, a, b) be a represented suitable weak solution on $Q_{2r}(z_0)$ obtained from the repaired-gauge chart of Theorem 4.35. Assume that the chart satisfies the comparability conclusion of Theorem 5.27 on this cylinder, with constants $\Lambda_{\min}, \Lambda_{\max}$, and that*

$$\|a\|_{L^\infty} + \|b\|_{L^\infty} \le M$$

there. There are constants

$$\varepsilon_{\rm rg} = \varepsilon_{\rm rg}(M, \Lambda_{\min}, \Lambda_{\max}) > 0, \qquad c_{\rm rg} = c_{\rm rg}(M, \Lambda_{\min}, \Lambda_{\max}) \in (0,1),$$

such that if

$$r^{-2} \int_{Q_r(z_0)} |V|^3 + r^{-2} \int_{Q_r(z_0)} |P - (P)_{B_r}|^{3/2} \le \varepsilon_{\rm rg},$$

then V is bounded on $Q_{c_{\rm rg} r}(z_0)$, with a bound depending only on $M, \Lambda_{\min}, \Lambda_{\max}, r$, and the preceding displayed quantity.

Proof. Apply the forward direction of Theorem 5.28 to $Q_r(z_0)$. Choose $\varepsilon_{\rm rg}$ so small that every physical cylinder in the resulting finite comparison cover has CKN quantity below the threshold $\varepsilon_{\rm CKN}$ from Theorem 9.34. That theorem gives boundedness of the physical velocity on smaller physical cylinders. Pulling those estimates back through the same comparable repaired chart gives the asserted L^∞ bound for V on $Q_{c_{\rm rg} r}(z_0)$. Thus boundedness is obtained only from epsilon regularity after two-sided chart transfer and small CKN quantity, not from compactness or gauge degeneracy. ∎

Lemma 5.30 (Scaled local pressure decay)**.** *Let (V, P) be a represented suitable pair on $Q_{2r}(z_0)$, and assume the localized pressure decomposition of Theorem 4.25 is available with a cutoff supported in $B_{2r}(y_0)$ and equal to 1 on $B_r(y_0)$. For*

$$C_V(\rho; z_0) := \rho^{-2} \int_{Q_\rho(z_0)} |V|^3, \qquad D_P(\rho; z_0) := \rho^{-2} \int_{Q_\rho(z_0)} |P - (P)_{B_\rho}|^{3/2},$$

there is a universal constant $C_{\rm dec}$ such that, whenever $0 < \theta \le 1/4$,

$$D_P(\theta r; z_0) \le C_{\rm dec} \theta^{5/2} D_P(r; z_0) + C_{\rm dec} \theta^{-2} C_V(2r; z_0).$$

Proof. Write $z_0 = (y_0, s_0)$. For a.e. time $s \in (s_0 - r^2, s_0]$, apply Theorem 4.25 on $B_r(y_0)$ with cutoff supported in $B_{2r}(y_0)$:

$$P(\cdot, s) = P_{\rm loc}(\cdot, s) + H(\cdot, s), \qquad -\Delta H(\cdot, s) = 0 \quad \text{in } B_r(y_0).$$

The Calderón–Zygmund part satisfies

$$\int_{B_{\theta r}(y_0)} |P_{\rm loc}|^{3/2} \le C \int_{B_{2r}(y_0)} |V|^3.$$

After integrating over the shorter time interval and multiplying by $(\theta r)^{-2}$, this contributes at most $C\theta^{-2} C_V(2r; z_0)$.

For the harmonic part, the interior oscillation estimate for harmonic functions gives, for $q = 3/2$,

$$\int_{B_{\theta r}(y_0)} |H - (H)_{B_{\theta r}}|^q \le C\theta^{3+q} \int_{B_r(y_0)} |H - (H)_{B_r}|^q.$$

Since $3 + q = 9/2$, scaling the cylinder by $(\theta r)^{-2}$ leaves the factor $\theta^{5/2}$. Moreover

$$\int_{B_r} |H - (H)_{B_r}|^{3/2} \le C \int_{B_r} |P - (P)_{B_r}|^{3/2} + C \int_{B_r} |P_{\rm loc}|^{3/2},$$

and the last term is again bounded by $C\int_{B_{2r}}|V|^3$. Absorbing the resulting $C\theta^{5/2}C_V(2r;z_0)$ into $C\theta^{-2}C_V(2r;z_0)$ gives the displayed estimate. Spatial averages may be replaced by the minimizing constants at the cost of a universal factor, as in Theorem 5.28. □

Lemma 5.31 (Dyadic persistence of CKN concentration)**.** *Let (V,P) be suitable on a compact cylinder Q_ρ, and let*

$$\mathcal{C}_{V,P}(z,r) := r^{-2}\int_{Q_r(z)}|V|^3 + r^{-2}\int_{Q_r(z)}|P-(P)_{B_r}|^{3/2}.$$

If $z_* \in Q_\rho$ is a Caffarelli–Kohn–Nirenberg concentration point in the sense that

$$\limsup_{r\downarrow 0}\mathcal{C}_{V,P}(z_*,r) \geq \kappa_* > 0,$$

then for every $0<\theta<1$ and every sufficiently small outer radius r_0 with $Q_{2r_0}(z_*) \Subset Q_\rho$, there exist $z_m \to z_*$, integers $L_m \to \infty$, and dyadic radii $\rho_m = \theta^{L_m}r_0$ such that

$$Q_{\rho_m}(z_m) \Subset Q_{r_0}(z_*), \qquad \mathcal{C}_{V,P}(z_m,\rho_m) \geq c_\theta\kappa_*,$$

where $c_\theta > 0$ depends only on θ.

Proof. Choose radii $\sigma_m \downarrow 0$ and points $z_m \to z_*$ with $Q_{\sigma_m}(z_m) \Subset Q_{r_0}(z_*)$ and $\mathcal{C}_{V,P}(z_m,\sigma_m) \geq \kappa_*/2$. Pick L_m so that

$$\theta^{L_m+1}r_0 < \sigma_m \leq \theta^{L_m}r_0 =: \rho_m.$$

Then $L_m \to \infty$ and $\rho_m \leq \theta^{-1}\sigma_m$. The velocity term on $Q_{\rho_m}(z_m)$ controls the velocity term on $Q_{\sigma_m}(z_m)$ with loss at most θ^2, because the normalization changes by $(\sigma_m/\rho_m)^2$. For the pressure term, spatial averages are compared through the minimizing constant:

$$\int_{B_{\sigma_m}}|P-(P)_{B_{\sigma_m}}|^{3/2} \leq 2^{3/2}\int_{B_{\sigma_m}}|P-c|^{3/2} \leq 2^{3/2}\int_{B_{\rho_m}}|P-c|^{3/2}.$$

Taking $c=(P)_{B_{\rho_m}}$, integrating in time, and accounting for the same normalization loss gives

$$\mathcal{C}_{V,P}(z_m,\rho_m) \geq 2^{-5/2}\theta^2\mathcal{C}_{V,P}(z_m,\sigma_m).$$

Thus the conclusion holds with $c_\theta = 2^{-7/2}\theta^2$. □

Lemma 5.32 (Exterior pressure persistence is a structural exit)**.** *Let a scale-rigid branch have a pressure lower bound on shrinking selected cylinders that is not accounted for by the local source $\partial_i\partial_j(V_iV_j)$ in any compact selected frame after the decomposition of Theorem 4.25. Then the branch enters the separated/exterior structural alternative of Theorem 5.6 (iv), and it does not remain in the residual scale-rigid stratum.*

Proof. The unaccounted part of the pressure is harmonic in the selected compact frame and is generated by sources at positive distance in the corresponding physical variables. If the exterior velocity source has a terminal CKN concentration on one of the annuli separating it from the selected core, then Theorem 9.38 gives an exterior concentration branch. In the local variables this is a separated active physical point, hence the separated-point alternative Theorem 5.6 (iv) and the reduction Theorem 5.19 apply.

If no such exterior concentration exists on the separating annuli, then Theorem 9.41 gives regular exterior control and Theorem 9.44 gives convergence of the exterior pressure contribution modulo functions of time on the selected compact frame. This contradicts the assumed persistent pressure lower bound there. Thus the only non-vanishing exterior-pressure case is already a separated/exterior structural exit. □

Lemma 5.33 (Persistent CKN mass activates an L^3 core or exits structurally)**.** *Let (V_n, P_n) be a compact-cylinder scale-rigid sequence on Q_{2R} with*

$$\sup_n \big(C_{V_n}(2R) + D_{P_n}(2R)\big) \le M_0,$$

and assume the pressure decomposition of Theorem 4.25 is available on Q_{2R}. Fix $\kappa_0 > 0$ and choose $\theta \in (0, 1/8)$ so that

$$C_{\mathrm{dec}}\theta^{5/2} \le \frac{1}{4}.$$

Then there are constants $\eta_{\mathrm{act}} > 0$ and $L_0 \in \mathbb{N}$, depending only on M_0, R, κ_0, θ, the fixed-scale pressure bound below, and the pressure-decomposition constants, with the following property. Suppose $0 < R_0 < R$, $z_n \to z$, $Q_{2R_0}(z_n) \Subset Q_{2R}$ for all large n, and suppose the fixed outer scale used for the iteration has a uniform local pressure bound

$$D_{P_n}(R_0; z_n) \le D_0 < \infty.$$

Let $L_n \to \infty$, $\rho_n := \theta^{L_n} R_0$, and assume

$$\mathcal{C}_n(\rho_n; z_n) := \rho_n^{-2} \int_{Q_{\rho_n}(z_n)} |V_n|^3 + \rho_n^{-2} \int_{Q_{\rho_n}(z_n)} |P_n - (P_n)_{B_{\rho_n}}|^{3/2} \ge \kappa_0$$

for all large n. Then, after passing to a subsequence, either:

(a) *for some integers $0 \le j_n < L_n$, with $s_n := 2\theta^{j_n} R_0$,*

$$s_n^{-2} \int_{Q_{s_n}(z_n)} |V_n|^3 \ge \eta_{\mathrm{act}}^3,$$

so the rescaled branch at scale s_n contains an active L^3 core;

(b) *the retained pressure lower bound is exterior pressure persistence, hence the branch is assigned to the separated/exterior structural alternative by Theorem 5.32.*

Proof. Choose L_0 so large that

$$(1/4)^{L_0} D_0 \le \kappa_0/16,$$

and choose $\eta_{\mathrm{act}} > 0$ so small that

$$C_{\mathrm{dec}}\theta^{-2} \sum_{m=0}^{\infty} (1/4)^m \eta_{\mathrm{act}}^3 \le \kappa_0/16, \qquad 4\theta^{-2}\eta_{\mathrm{act}}^3 \le \kappa_0/4.$$

For large n, $L_n \ge L_0$. Suppose item (a) fails. Then for each $0 \le j < L_n$,

$$C_{V_n}(2\theta^j R_0; z_n) < \eta_{\mathrm{act}}^3.$$

Iterating Theorem 5.30 from the fixed outer scale R_0 down to $\theta^{L_n} R_0 = \rho_n$ gives

$$D_{P_n}(\rho_n; z_n) \le (1/4)^{L_n} D_{P_n}(R_0; z_n) + C_{\mathrm{dec}}\theta^{-2} \sum_{j=0}^{L_n - 1} (1/4)^{L_n - 1 - j} C_{V_n}(2\theta^j R_0; z_n) \le \kappa_0/8.$$

Here the initial pressure is bounded by D_0 because the iteration starts from this fixed compact scale, not from the shrinking target scale. Moreover $Q_{\rho_n}(z_n) \subset Q_{2\theta^{L_n - 1} R_0}(z_n)$, so failure of item (a) at $j = L_n - 1$ gives

$$\rho_n^{-2} \int_{Q_{\rho_n}(z_n)} |V_n|^3 \le 4\theta^{-2}\eta_{\mathrm{act}}^3 \le \kappa_0/4.$$

Hence $\mathcal{C}_n(\rho_n; z_n) < \kappa_0$, contradicting the assumed lower bound, unless the pressure term used in the lower bound is not the local pressure governed by the compact-frame decomposition. The latter possibility is exactly exterior pressure persistence and is item (b) by Theorem 5.32. ∎

Lemma 5.34 (Activated cores are captured by the finite active ledger)**.** *Let a scale-rigid branch inherit an active profile family in the sense of Theorem 5.7, with finite critical mass M_* and concentration capture. Suppose Theorem 5.33 produces represented scales $s_n > 0$ and centers z_n such that*

$$s_n^{-2} \int_{Q_{s_n}(z_n)} |V_n|^3 \geq \eta_{\mathrm{act}}^3.$$

After lowering the active threshold once, if necessary, to $\eta_*^{\mathrm{new}} := \min\{\eta_*, \eta_{\mathrm{act}}/2\}$, exactly one of the following occurs after passing to a subsequence:

(a) *the activated core is already represented by an existing active profile or compound profile in the inherited family;*
(b) *it is absorbed into the selected terminal frame, so it does not create an independent residual obstruction;*
(c) *it is assigned to the profile-completeness obligation rather than to the scale-rigid residual branch.*

In particular, under the stated concentration-capture hypothesis, pressure-core activation cannot create an unregistered active object inside the residual scale-rigid branch.

Proof. Rescale the branch around (z_n, s_n) inside the represented window; in physical variables the corresponding scale is the selected physical scale multiplied by s_n, hence still tends to zero along the Type II sequence. The displayed lower bound is exactly the active-frame condition in Theorem 5.3 on Q_1, with threshold η_{act}. The concentration-capture clause in Theorem 5.7 says that a compact terminal-window sequence with positive L^3 mass is either seen by the active family or is absorbed into the selected terminal frame; it cannot remain in the perturbative remainder. Hence alternatives (a) and (b) hold unless the frame is genuinely absent from the captured profile family. That last possibility is not a scale-rigid residual mechanism; it is precisely failure of the terminal active-family completeness/capture input and is assigned to that separate profile-completeness obligation. Therefore the scale-rigid refinement below only uses active objects that already belong to the finite captured family, together with objects absorbed into the selected terminal frame. □

Lemma 5.35 (No sub-concentration gives a bounded selected-window limit)**.** *Let (V_n, P_n, a_n, b_n) be a scale-rigid local branch and let (V_*, P_*) be a compact limit supplied by Theorem 5.26 on Q_ρ. Then, after shrinking once to a compact subcylinder $Q_{\rho'} \Subset Q_\rho$, exactly one of the following alternatives holds:*

(a) *(V_*, P_*) has no Caffarelli–Kohn–Nirenberg concentration point in $Q_{\rho'}$. Then $V_* \in L^\infty(Q_{\rho''})$ for every $Q_{\rho''} \Subset Q_{\rho'}$, and the selected subsequence has a nonzero bounded selected-window limit on some such $Q_{\rho''}$.*
(b) *(V_*, P_*) has a Caffarelli–Kohn–Nirenberg concentration point in $Q_{\rho'}$. Then the original branch either contains a further captured active L^3 concentration core after rescaling around that point, the pressure mass is exterior and the branch is already in the separated/exterior structural alternative, or the core is assigned to the separate profile-completeness obligation. According to its relative scale-center position, every captured core enters one of the already named multicore, same-point cascade, or separated/exterior structural alternatives, rather than remaining a residual scale-rigid branch.*

Proof. Let $\varepsilon_{\mathrm{rg}}$ be the represented-gauge regularity threshold from Theorem 5.29. Before using that threshold, apply Theorem 5.27 on each retained compact subcylinder. The chart stability in Theorem 4.32, together with the L^∞ modulation bound (R3), gives a limiting chart (Λ_*, X_*) with $a_* = -\Lambda_*'$ and $b_* = -X_*'$ on the retained subwindow. Thus the compactness of the subcylinder and (R3) give a uniform lower-upper scale comparison after one shrink, so the constants in

Theorem 5.29 depend only on the retained chart geometry and not on any unproved boundedness of V_n or V_*.

Assume first that there is no CKN concentration point of (V_*, P_*) in $Q_{\rho'}$. For every point $z \in Q_{\rho'}$ there is a parabolic cylinder $Q_{r_z}(z) \Subset Q_\rho$ such that the scale-invariant velocity-pressure quantity of (V_*, P_*) on $Q_{r_z}(z)$ is below $\varepsilon_{\mathrm{rg}}$. A finite subcover of each $Q_{\rho''} \Subset Q_{\rho'}$, followed by Theorem 5.29 on the cover, gives $V_* \in L^\infty(Q_{\rho''})$.

It remains to check that the bounded limit is nonzero. If $V_* \equiv 0$ on the retained inner cylinder, then the strong L^3 convergence from Theorem 5.26 and the pressure stability argument of Theorem 3.16 imply

$$C_{V_n}(r_*) + D_{P_n}(r_*) \to 0,$$

contradicting the retained lower bound (R2) in Theorem 5.23. Hence $V_* \not\equiv 0$ on some smaller compact cylinder. Since $V_n \to V_*$ strongly in L^3 there, the selected subsequence has a nonzero bounded selected-window limit.

Now suppose a CKN concentration point z_* remains in $Q_{\rho'}$. Then there is $\kappa_* > 0$ such that

$$\limsup_{r \downarrow 0} \mathcal{C}_{V_*,P_*}(z_*, r) \geq \kappa_*.$$

Choose $R_0 > 0$ with $Q_{2R_0}(z_*) \Subset Q_\rho$, and choose $\theta \in (0, 1/8)$ satisfying the pressure-decay smallness condition in Theorem 5.33. By Theorem 5.31, there are $z_m \to z_*$, $L_m \to \infty$, and $\rho_m = \theta^{L_m} R_0$ such that

$$\mathcal{C}_{V_*,P_*}(z_m, \rho_m) \geq c_\theta \kappa_*.$$

For each fixed m, the strong L^3 convergence of V_n and weak lower semicontinuity of the pressure oscillation seminorm in $L^{3/2}/\{\text{spatial constants}\}$ imply, along a diagonal subsequence,

$$\mathcal{C}_{V_n,P_n}(z_m, \rho_m) \geq \frac{1}{2} c_\theta \kappa_*.$$

Relabeling the diagonal sequence, apply Theorem 5.33 with $\kappa_0 = \frac{1}{2} c_\theta \kappa_*$ on the compact cylinder where (R1) holds. The additional fixed-scale pressure bound required by that lemma holds because R_0 is fixed, $Q_{2R_0}(z_m) \Subset Q_\rho$ for all large m, and the mean-normalized pressures are uniformly bounded in $L^{3/2}(Q_\rho)$ by Theorem 5.26; comparison of spatial averages on the fixed nested balls gives $D_{P_n}(R_0; z_m) \leq D_0(R_0, \rho, M)$. If the persistent CKN mass is produced by an interior velocity core, then Theorem 5.34 either identifies that core with the finite captured active ledger or assigns it to the separate profile-completeness obligation, after the one-time lowering of the active threshold. If the mass is carried by harmonic pressure generated outside the selected compact frame, Theorem 5.32 assigns the branch to the separated/exterior structural alternative, so it exits the residual scale-rigid stratum.

The relative position of this new core with respect to the retained scale is exhausted by Theorem 5.10: it is comparable at the same point, strictly separated in scale at the same point, separated in physical center, or locally invisible in the retained frame. The invisible case cannot carry the retained positive concentration by the capture clause in Theorem 5.7. The remaining cases are precisely the multicore, same-point cascade, and separated/exterior structural alternatives already named in the manuscript. Thus the branch does not remain in the residual scale-rigid stratum in this case. □

Theorem 5.36 (Single-core reduction compatibility)**.** *Let (V_n, P_n, a_n, b_n) be a scale-rigid local branch in the sense of Theorem 5.23, and assume the branch reduces, after the rigid-frame limit of Theorem 5.26, to a single retained nondegenerate core on a fixed inner cylinder $Q_{r_*} \Subset Q_\rho \Subset Q_{R_*}$. Then, after passing to a subsequence, one of the following alternatives holds:*

(i) *the branch has a nonzero bounded selected-window limit;*

(ii) *the original selected sequence has vanishing localized dissipation on a retained compact cylinder, hence is a compact single-core vanishing-dissipation branch in the sense of Theorem 3.3 and is excluded by Theorem 3.18;*

(iii) *a further active concentration core is produced, so the branch leaves the single-core reduction and enters a multicore, cascade, or separated/exterior structural alternative.*

Proof. If the original selected sequence has vanishing localized dissipation on some $Q_{\rho'}$, $Q_{r_*} \Subset Q_{\rho'} \Subset Q_{R_*}$, then we first verify Theorem 3.3. Suitability and the represented equation are supplied by Theorem 4.35. Compact-cylinder bounds, positive selected-core CKN concentration, and bounded modulation are exactly (R1), (R2), and (R3) of Theorem 5.23. The nondegenerate scale-center selection is the hypothesis of this single-core case. The vanishing localized dissipation is the present hypothesis on the original selected sequence, not on a residual after subtracting a limit. Thus the branch satisfies every clause of Theorem 3.3, and Theorem 3.18 excludes it. This is item (ii).

Apply Theorem 5.35 to the compact limit on a slightly smaller cylinder. If the no-subconcentration alternative holds, item (i) follows. If a further active concentration core is produced, item (iii) follows. □

Lemma 5.37 (Degenerate Jacobian resolution)**.** *Let (V_n, P_n, a_n, b_n) be a represented sequence reaching the gauge-degenerate alternative Theorem 5.6 (v), i.e. along the subsequence the relative scale-center Jacobian of the selected compound or terminal frame degenerates. Then, after passing to a further subsequence and reselecting the active center in the sense of Theorem 5.7 on the same physical point, exactly one of the following four resolutions occurs:*

(D1) *the reselection produces an additional active core or active compound carrying positive L^3 mass at the fixed active threshold; hence the branch enters the multicore, same-point cascade, or separated/exterior alternative Theorem 5.6 (ii)–(iv) and is treated by Theorem 5.11;*

(D2) *the reselection collapses the compound to a single nondegenerate core and the branch enters the single-core compatibility dichotomy governed by Theorem 5.36;*

(D3) *the reselection produces a perturbative compound that is below the critical small-data threshold of Step 2 of Theorem 5.18, hence is absorbed into the perturbative remainder and treated by Theorem 5.17;*

(D4) *the reselection still has degenerate scale-center Jacobian on every further subsequence and has no further active subconcentration; in this case the no-subconcentration alternative of Theorem 5.35 gives a nontrivial bounded selected-window limit on a compact cylinder.*

Proof. By the gauge-degeneracy alternative Theorem 5.6 (v), the localized scale or centering selection of the selected compound or terminal frame is non-unique along the subsequence. Apply the parameter partition lemma Theorem 5.10 to the selected compound together with any candidate alternate selection produced by the gauge degeneracy. Three mutually exclusive structural outcomes are possible.

Case A: alternate selection is comparable to the original at the same physical point. Then both selections lie in the same scale comparability class of Step 1 of Theorem 5.18. Step 2 of that theorem groups them into a single compound profile. If the compound is below the small-data threshold, we are in case (D3); otherwise the compound is above threshold and Step 3 of the theorem applies. If the compound is then nondegenerate, we are in case (D2); if it is still degenerate, the gauge degeneracy persists on the reselected compound, and we proceed to Case D below.

Case B: alternate selection is at a strictly different scale at the same physical point. By Theorem 5.14, the strictly outer profile vanishes locally in the innermost frame; absorbing it into the perturbative remainder lands us in case (D3). If instead the reselection produces a strictly inner core not previously detected, then either compact-cylinder suitable control fails, in which case Theorem 5.15 produces the new CKN core, or compact control persists and Theorem 5.33 converts the retained CKN mass into an active L^3 core unless the branch has already exited to the separated/exterior pressure alternative. These outcomes are case (D1).

Case C: alternate selection is at a separated physical point. By Theorem 5.19, the new selection is locally invisible in the original frame, hence absorbed into the perturbative remainder; this is case (D3). If the new selection retains positive CKN concentration in its own frame, then Theorem 5.33 makes it an additional separated active core, or else assigns it to the separated/exterior pressure alternative. Both outcomes are case (D1) via Theorem 5.6 (iv).

Case D: persistent degeneracy. Suppose every reselection on every further subsequence still has degenerate scale-center Jacobian. Apply Theorem 5.26 and then Theorem 5.35. If the compact limit has a CKN subconcentration point, that lemma produces an additional active core; this is case (D1). If no such subconcentration remains, the same lemma gives a nonzero bounded selected-window limit; this is case (D4). No L^∞ bound is inferred from the Jacobian degeneracy itself. □

Theorem 5.38 (Scale-rigid stratum decomposition)**.** *Every scale-rigid local branch in the sense of Theorem 5.23 enters one of the following named local strata after passing to a subsequence:*

(i) *the multicore alternative Theorem 5.6 (ii)–(iv), governed by Theorem 5.11;*
(ii) *the compact single-core stratum, governed by Theorem 5.36;*
(iii) *the perturbative remainder of Theorem 5.7, governed by Theorem 5.17;*
(iv) *the bounded selected-window limit alternative Theorem 3.1, which by Theorem 5.25 is incompatible with Theorem 3.2.*

Proof. By Theorem 5.23 (R4), the branch enters the scale-rigid stratum either via a degenerate relative scale-center Jacobian or as an innermost cascade core after outer profiles have been absorbed perturbatively.

Degenerate Jacobian case. Apply Theorem 5.37. Cases (D1)–(D4) of that lemma give items (i)–(iv) respectively.

Innermost cascade case. By Step 4 of Theorem 5.18, after subtracting the outer-profile sum $S_n \to 0$ in $L^\infty_I L^3(B_{2R})$ and absorbing constant drifts into the translation modulation, the residual is governed by Theorem 5.17. That theorem produces three outcomes already enumerated as (a) single-core, (b) scale-rigid (degenerate Jacobian), or (c) purely perturbative. Case (a) is item (ii) here via Theorem 5.36; case (c) is item (iii); case (b) loops back into the degenerate Jacobian case treated above and is therefore reduced by Theorem 5.37 to one of (i)–(iv). □

Lemma 5.39 (Finite refinement of scale-rigid structural exits)**.** *Let ω be a scale-rigid local branch produced by Theorem 5.18 or Theorem 5.19, with the inherited active family of finite critical mass and with concentration capture in the sense of Theorem 5.7. Resolve every structural exit of Theorem 5.38 by applying only Theorem 5.11, Theorem 5.18, Theorem 5.19, and Theorem 5.17, never Theorem 5.42. Then the refinement terminates after finitely many ledger-decreasing refinements. At the terminal residual branch one has either a nonzero bounded selected-window limit or a compact single-core vanishing-dissipation branch excluded by Theorem 3.18.*

Proof. After the active threshold has been lowered once, every core produced by Theorem 5.33 is handled by Theorem 5.34: it is already in the captured inherited family, is absorbed into the terminal frame, or exits to the separate profile-completeness obligation. The residual scale-rigid ledger therefore uses only the captured active family. Write η_*^{new} for the lowered active threshold. By Theorem 5.9, the resulting family contains at most

$$N_* := \lfloor M_* (\eta_*^{\text{new}})^{-3} \rfloor$$

active profiles. Group same-point comparable-scale profiles into compound profiles as in Theorem 5.8. There are therefore only finitely many active objects and only finitely many unordered pair relations among them.

Define the refinement ledger to consist of:

(L1) every active object not yet selected, grouped, absorbed, or declared terminal;
(L2) every pair relation not yet classified as comparable same-point, strict same-point cascade, separated point, or locally invisible;
(L3) every degenerate scale-center selection not yet resolved by Theorem 5.37;
(L4) every perturbative remainder not yet removed by Theorem 5.17.

The ledger is finite because it is built from a finite active family and finitely many compounds. Order ledgers lexicographically by the number of unresolved entries in (L1), (L2), (L3), and (L4).

At each nonterminal step, no multibubble exclusion theorem is used. If Theorem 5.38 gives a multicore, same-point cascade, or separated/exterior structural exit, apply Theorem 5.11 to select the corresponding active object. Comparable same-point objects are grouped into a compound profile; below-threshold compounds are removed by the small-data step in Theorem 5.18, while above-threshold compounds count as active objects. Strict same-point cascades are reduced by Theorem 5.18; separated physical centers are reduced by Theorem 5.19. Perturbative exterior and remainder terms are discarded only through Theorem 5.17, whose output preserves the selected positive concentration and produces no new active object.

Each move strictly decreases the ledger. Adding a previously unseen active core consumes one entry of (L1), and this can happen at most N_* times. Grouping comparable same-point objects, ordering a same-point cascade, separating two physical centers, or declaring one object locally invisible resolves a previously unresolved pair and consumes one entry of (L2). Applying Theorem 5.37 resolves the selected degenerate frame or produces an active structural exit already counted in (L1)–(L2), so it consumes one entry of (L3). Applying Theorem 5.17 to a perturbative remainder consumes one entry of (L4). Once an entry is resolved it is not reintroduced: later passes use the grouped compound, the selected innermost representative, the separated-frame assignment, or the discarded perturbative remainder. Hence no cycle is possible.

After the ledger is exhausted, item (i) of Theorem 5.38 cannot occur again, because it would create either a new active object or an unresolved active relation, contradicting exhaustion of (L1)–(L2). Item (iii) cannot carry the retained positive concentration by the capture clause in Theorem 5.7; a purely perturbative remainder has already been absorbed through (L4). Item (iv) is exactly the nonzero bounded selected-window-limit alternative.

It remains only to handle item (ii). Apply Theorem 5.36. Its bounded-limit alternative gives the first terminal outcome. Its further-active-core alternative would add a new ledger entry, impossible after ledger exhaustion. Its vanishing-localized-dissipation alternative verifies Theorem 3.3 for the original selected sequence and is excluded by Theorem 3.18. These are precisely the two terminal residual outcomes claimed. □

Proposition 5.40 (Scale-rigidity exclusion)**.** *Every scale-rigid local branch produced by Theorem 5.18 or Theorem 5.19, including a branch with degenerate relative scale-center Jacobian, either has a nonzero bounded selected-window limit on a compact cylinder (and so fails Theorem 3.2 by Theorem 5.25) or reduces to the compact single-core branch governed by Theorem 3.18 (equivalently by the local proposition Theorem 5.20). Consequently no such branch is compatible with the local Type II condition. This is the conclusion of Theorem 5.21, proved here as a theorem in the stated class.*

Proof. Let ω be a scale-rigid local branch. Theorem 5.38 places ω into one of items (i)–(iv).

If item (i) occurs, the branch enters a named multicore, same-point cascade, or separated/exterior structural exit. Resolve that exit by the finite refinement procedure of Theorem 5.39.

This procedure uses only Theorem 5.11, Theorem 5.18, Theorem 5.19, and Theorem 5.17; it does not invoke Theorem 5.42, so there is no circular use of Theorem 5.21. The refinement terminates at a bounded selected-window limit or at the compact single-core vanishing-dissipation branch.

In item (ii), apply Theorem 5.36. Its bounded-limit case is the first alternative in the proposition. Its vanishing-localized-dissipation case verifies Theorem 3.3 for the original selected sequence and is excluded by Theorem 3.18; this is the "reduces to the single-core branch" alternative. Its further-active-core case is resolved by the same finite refinement lemma used for item (i).

In item (iii), the branch is the perturbative remainder; by Theorem 5.17 it produces no independent multibubble obstruction. The retained positive concentration is carried by the selected component, not by the perturbative remainder, so the residual selected component is reduced to item (ii) or item (iv) by Theorem 5.39.

In item (iv), by Theorem 5.25 the branch is not a local Type II sequence in the sense of Theorem 3.2. This is the "nonzero bounded selected-window limit" alternative.

In every case the branch either has a nonzero bounded selected-window limit or reduces to the single-core branch. This is the conclusion of Theorem 5.21. □

Proposition 5.41 (Terminal admissibility in active frames)**.** *Assume finite local critical norm, terminal windowed state-space completeness in the sense of Theorem 9.16, local perturbative residual stability, the exterior annulus alternative Theorem 9.47, the repaired-gauge representation of Theorem 4.35, and local Caccioppoli regularity. Then every terminal family produced by the same-point and separated-point reductions is admissible on each compact selected cylinder in the sense of Theorem 5.16.*

Proof. The terminal nonlinear local-state decoupling theorem, Theorem 9.27, applies under the stated hypotheses and produces, on every compact terminal cylinder, the conclusions of Theorem 9.13. We verify each clause of Theorem 5.16.

Items (i) and (ii) are the active-family enumeration of Theorem 5.7 together with the same-point and separated-point grouping of Theorems 5.13 and 5.14; they are properties of the active family that survive the terminal-frame construction by definition of the terminal active class.

Item (iii) is the velocity decoupling Theorem 9.18, the linear residual decoupling Theorem 9.24, and the pressure decoupling Theorem 9.23; together they give the $L^\infty_I L^3$ vanishing of the rescaled remainder r_n and the existence of (π^r_n, f_n) with the stated $L^1_I H^{-1}(B_R)$ decay.

Item (iv) is the retained $L^\infty_I L^3(B_{2R})$ bound established in the proof of Theorem 9.26, which uses the finite local critical norm and the retained local-state bounds.

Item (v) is the bounded-modulation hypothesis, supplied by the repaired-gauge representation Theorem 4.35 and Theorem 4.34.

Item (vi) is the local pressure-decomposition stability of Theorem 9.23, whose proof relies only on the Calderon–Zygmund pressure-tail decoupling Theorem 9.12 together with the kernel boundedness on $L^{3/2}$ provided by the classical Calderon–Zygmund theorem [CZ52, Ste70].

Item (vii) follows from the Leibniz formula $[\Delta, \chi_R]u = (\Delta\chi_R)u + 2\nabla\chi_R \cdot \nabla u$ and the identity $[\nabla\chi_R, \nabla\cdot](u\otimes u) = (\nabla\chi_R)(u\otimes u)$, combined with the duality $\|fu\|_{H^{-1}(B_R)} \le C_R\|f\|_{C^1(\overline{B}_R)}\|u\|_{L^{6/5}(B_R)}$ and the embedding $L^3(B_R) \hookrightarrow L^{6/5}(B_R)$ on bounded balls. Indeed, for $\varphi \in H^1_0(B_R)$,

$$\langle \nabla\chi_R \cdot \nabla u, \varphi\rangle = -\langle u, (\Delta\chi_R)\varphi + \nabla\chi_R \cdot \nabla\varphi\rangle,$$

giving $\|[\Delta, \chi_R]u\|_{H^{-1}(B_R)} \le C_R\|u\|_{L^3(B_{2R})}$. The pointwise bound $|(\nabla\chi_R)(u \otimes u)| \le \|\nabla\chi_R\|_\infty |u|^2$ gives $\|(\nabla\chi_R)(u \otimes u)\|_{L^{3/2}(B_R)} \le C_R\|u\|^2_{L^3(B_{2R})}$, and $L^{3/2} \hookrightarrow H^{-1}$ on B_R supplies the displayed bound. The combined estimate

$$\|[\Delta, \chi_R]u\|_{H^{-1}(B_R)} + \|[\nabla\chi_R, \nabla\cdot](u\otimes u)\|_{H^{-1}(B_R)} \le C_R\big(\|u\|_{L^3(B_{2R})} + \|u\|^2_{L^3(B_{2R})}\big) \le C'_R\|u\|^2_{L^3(B_{2R})}$$

holds whenever $\|u\|_{L^3(B_{2R})}$ is bounded below by a fixed constant, which is the case in the active selected frame. The same estimate applies to the S_n-quadratic commutator terms thanks to the uniform $L^\infty_I L^3(B_{2R})$ control of U_n and the $L^1_I L^{3/2}(B_R)$ control of the cross stresses produced by Theorem 9.20.

All items in Theorem 5.16 are therefore satisfied, which is the asserted admissibility. □

Theorem 5.42 (Local multibubble and cascade exclusion)**.** *Use the local single-core exclusion Theorem 5.20, the scale-rigidity exclusion Theorem 5.21, and the terminal admissibility in active frames Theorem 5.41. Let a positive local Type II branch admit an active family satisfying the hypotheses of Theorem 5.11. Then no local multibubble, cascade, or gauge-degenerate Type II case can occur in this class.*

Proof. Let a local Type II branch be in the multibubble/cascade class of Theorem 5.6. Apply Theorem 5.11. The positive local concentration is carried by an active frame or compound local state, and one of the alternatives listed there occurs.

Case 1: compactness-failure reduction. Compactness failure yields, after changing radius by a fixed factor and rescaling as in Theorem 5.15, a bounded-cylinder concentration branch. Relabel this branch as an active core. After passing to a subsequence, the new core and every previously retained active core have one of three relations: comparable scale at the same physical point, strictly separated scale at the same physical point, or separated physical center. Comparable same-point cores are grouped into compound local states. A compound state with no detecting compact cylinder is removed by local perturbative residual stability; an active compound state is treated as one selected core in its active frame. If the scale-center selection inside such a compound core is degenerate, the branch is gauge-degenerate and belongs to the non-uniqueness alternative in Theorem 5.6, rather than to a separate core dynamics.

Case 1a: gauge degeneracy. If the scale-center selection of the selected compound or terminal frame is degenerate, then the branch is in the gauge-degenerate alternative of Theorem 5.6. By the repaired-gauge representation from Section 4, nondegeneracy is the hypothesis needed to enter the single-core branch. Persistent degeneracy therefore belongs to the scale-rigidity alternative in Theorem 5.21; if the degeneracy disappears after passing to a subsequence, the branch enters the single-core or terminal-frame alternatives below.

Case 1b: same-point comparable scales. Same-point comparable-scale active frames are grouped into a compound local state by the local state-space selection rule. If the compound state has no detecting compact cylinder, it is part of the perturbative residual. If it has the local detection threshold and its scale-center selection is nondegenerate, it is one selected active core and enters the single-core branch. If the corresponding scale-center selection is degenerate, it is covered by Case 1a. Thus comparable same-point states do not produce an additional multibubble mechanism.

Case 2: same-point cascading. Strict same-point scale cascades are treated by Theorem 5.18. In the innermost frame, all outer scales are locally invisible after constant drifts are absorbed. The branch therefore reduces to one of the three terminal alternatives

(A) single-core, (B) scale-rigid, (C) terminal perturbative.

Case 3: separated-point interactions. For separated physical points, Theorem 5.19 gives the same three terminal alternatives as in Case 2:

(A) single-core, (B) scale-rigid, (C) terminal perturbative.

The single-core alternative is excluded by Theorem 5.20. The scale-rigid alternative is excluded by Theorem 5.21, which includes the scale-collapse and autonomous-limit exclusions in Section 7 in this setting. The terminal perturbative alternative gives no independent multibubble obstruction:

Theorem 5.17 removes all exterior profiles and remainders from the selected active equation in $L^1 H^{-1}_{\mathrm{loc}}$, preserves the positive finite-mass selected branch, and leaves only the single-core or scale-rigid possibilities already excluded. Consequently every alternative in Theorem 5.6 is incompatible with the local Type II regime. □

Corollary 5.43 (Removal of the multibubble alternative)**.** *Under the hypotheses of Theorem 5.42, every failure of the single nondegenerate compact core alternative is excluded.*

Proof. By Theorem 5.6, failure of the single nondegenerate compact core alternative means that at least one of the following occurs: the compact-cylinder suitable estimates are unbounded; positive local CKN mass splits between active cores; a strict same-point scale cascade occurs; separated physical points remain active in distinct local frames; or the localized scale or center selection is degenerate. The compactness-failure case produces an additional core by Theorem 5.15. Same-point and separated cases are reduced by Theorems 5.18 and 5.19. Theorem 5.42 excludes the remaining cases. Hence no failure of the single nondegenerate compact core alternative remains. □

6. Local Caccioppoli and Rough-Core Exclusion

6.1. **Analytic inputs and notation.** The renormalized local energy inequality from the preceding renormalized-chart analysis is used as an analytic input. The renormalized suitable structure and pressure reconstruction are not rederived. All estimates are conditional on the following local hypotheses on the compact renormalized cylinders under consideration.

Proposition 6.1 (Renormalized local energy inequality)**.** *Let (V, P, a, b) be defined on a compact renormalized cylinder $Q_{2R,J} := B_{2R} \times J$, where $J \subset \mathbb{R}$ is a bounded interval. Assume*

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P - \nu\Delta V = a(\tau)V + a(\tau)(y \cdot \nabla V) + b(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0, \tag{6}$$

with $\nu > 0$, $\nabla \cdot V = 0$, and $a, b \in L^\infty(J)$. For every nonnegative $\phi \in C_c^\infty(Q_{2R,J})$ and every $\tau_1 < \tau_2$ in J,

$$\begin{aligned}
&\int_{B_{2R}} \frac{|V(\cdot,\tau_2)|^2}{2}\phi(\cdot,\tau_2) - \int_{B_{2R}} \frac{|V(\cdot,\tau_1)|^2}{2}\phi(\cdot,\tau_1) \\
&\quad + \nu\int_{\tau_1}^{\tau_2}\int_{B_{2R}} \phi\,|\nabla V|^2 \le \int_{\tau_1}^{\tau_2}\int_{B_{2R}} \frac{|V|^2}{2}(\partial_\tau\phi + \nu\Delta\phi) \\
&\quad + \int_{\tau_1}^{\tau_2}\int_{B_{2R}} \frac{|V|^2}{2} V \cdot \nabla\phi + \int_{\tau_1}^{\tau_2}\int_{B_{2R}} PV \cdot \nabla\phi \\
&\quad + \int_{\tau_1}^{\tau_2}\int_{B_{2R}} a\frac{|V|^2}{2}(-\phi - y \cdot \nabla\phi) \\
&\quad - \int_{\tau_1}^{\tau_2}\int_{B_{2R}} \frac{|V|^2}{2} b \cdot \nabla\phi.
\end{aligned} \tag{7}$$

The pressure is understood modulo functions of time, and the mean-normalized pressure on compact balls belongs to the local class used below.

Proof. Theorem 4.31 transports the physical local energy inequality of Caffarelli–Kohn–Nirenberg [CKN82] into the renormalized variables (y, τ) and yields the inequality stated above with test function of the form ϕ^2, where $\phi \in C_c^\infty(Q_R^*)$ is nonnegative. Setting $\Phi := \phi^2$ and noting that every nonnegative $\Phi \in C_c^\infty(Q_{2R,J})$ is the limit, in $C_c^k(Q_{2R,J})$ for every k, of squares of nonnegative smooth compactly supported functions $\phi_\varepsilon := (\Phi + \varepsilon^2)^{1/2} - \varepsilon$, the inequality extends by approximation to all such Φ; this is the standard reformulation used in the theory of suitable weak solutions [CKN82, Lin98]. The compact-window representation theorem Theorem 4.35 provides the cylinder $Q_{2R,J} = B_{2R} \times J$ and the divergence-free renormalized equation (6), while Theorem 4.25 supplies the localized pressure decomposition with the mean-normalized pressure on compact balls. Identifying the modulation coefficients $a = \tilde{a}$ and $b = \tilde{b}$ of Theorem 4.31 with those of (6) and grouping the renormalized error terms exactly as in (7) yields the stated inequality. □

Throughout, for a space–time set $E \subset \mathbb{R}^3 \times \mathbb{R}$, write

$$\int_E f := \int_E f(y,\tau)\,dy\,d\tau.$$

For $r > 0$ and a bounded interval $J \subset \mathbb{R}$, define

$$Q_{r,J} := B_r \times J, \qquad |Q_{r,J}| := |B_r|\,|J|.$$

For a ball $B \subset \mathbb{R}^3$, denote the spatial mean of pressure by

$$(P)_B(\tau) := \frac{1}{|B|}\int_B P(y,\tau)\,dy.$$

The local quantities are

$$\mathcal{A}_J(r) := \operatorname*{ess\,sup}_{\tau\in J} \int_{B_r} |V(y,\tau)|^2\,dy,$$

$$\mathcal{H}_J(r) := \int_J\int_{B_r} |\nabla V|^2\,dy\,d\tau,$$

$$\mathcal{C}_J(r) := \int_J\int_{B_r} |V|^3\,dy\,d\tau, \qquad \mathcal{D}_J(r) := \int_J\int_{B_r} |P-(P)_{B_r}(\tau)|^{3/2}\,dy\,d\tau.$$

Lemma 6.2 (Finite-measure conversion)**.** *For every bounded interval J and every $r>0$,*

$$\int_J\int_{B_r} |V|^2 \le |Q_{r,J}|^{1/3}\mathcal{C}_J(r)^{2/3} \le |Q_{r,J}|^{1/3}\big(1+\mathcal{C}_J(r)\big).$$

Proof. Hölder's inequality on $Q_{r,J}$ gives

$$\int_J\int_{B_r} |V|^2 \le |Q_{r,J}|^{1/3}\left(\int_J\int_{B_r} |V|^3\right)^{2/3}.$$

The second inequality follows from $s^{2/3}\le 1+s$ for $s\ge 0$. □

Lemma 6.3 (Comparison of pressure normalizations)**.** *Let $1\le p<\infty$ and $0<r\le R$. For almost every τ such that $P(\tau)\in L^p(B_R)$,*

$$\|P(\tau)-(P)_{B_r}(\tau)\|_{L^p(B_r)} \le 2\|P(\tau)-(P)_{B_R}(\tau)\|_{L^p(B_R)}.$$

Consequently,

$$\mathcal{D}_J(r)\le 2^{3/2}\mathcal{D}_J(R)$$

whenever the right-hand side is finite.

Proof. Set $c=(P)_{B_R}(\tau)$. Then

$$\|P-(P)_{B_r}\|_{L^p(B_r)} \le \|P-c\|_{L^p(B_r)} + |B_r|^{1/p}|(P)_{B_r}-c|.$$

Since

$$|(P)_{B_r}-c| \le |B_r|^{-1}\int_{B_r}|P-c| \le |B_r|^{-1/p}\|P-c\|_{L^p(B_r)},$$

the displayed pointwise estimate follows. Raising to the power $3/2$ and integrating in time gives the last assertion. □

6.2. **Compact-cylinder Caccioppoli estimate.**

Theorem 6.4 (Compact-cylinder Caccioppoli estimate)**.** *Let $R>0$, let $J\subset\mathbb{R}$ be a bounded open interval, and let $J'\Subset J$ be a compact subinterval. Assume Theorem 6.1 on $Q_{2R,J}$ and assume*

$$\mathcal{C}_J(2R)+\mathcal{D}_J(2R)<\infty.$$

Then there is a constant

$$C = C\big(R,|J|,\nu,\|a\|_{L^\infty(J)},\|b\|_{L^\infty(J)},\operatorname{dist}(J',\partial J)\big)$$

such that

$$\mathcal{A}_{J'}(R)+\mathcal{H}_{J'}(R) \le C\Big(1+\mathcal{C}_J(2R)+\mathcal{D}_J(2R)\Big).$$

The estimate is outer-to-inner: all right-hand quantities are measured on $B_{2R}\times J$, while the conclusion is on $B_R\times J'$.

Proof. **Step 1: localized cut-off and the two energy inequalities.** Set

$$\delta := \operatorname{dist}(J', \partial J) > 0.$$

Choose $\eta \in C_c^\infty(J)$ and $\zeta \in C_c^\infty(B_{2R})$ with

$$0 \le \eta \le 1, \qquad 0 \le \zeta \le 1, \qquad \eta \equiv 1 \text{ on } J', \qquad \zeta \equiv 1 \text{ on } B_R,$$

and

$$\|\eta'\|_{L^\infty(J)} \le C_\eta\, \delta^{-1}, \qquad \|\nabla\zeta\|_{L^\infty(B_{2R})} \le C_\zeta R^{-1}, \qquad \|\Delta\zeta\|_{L^\infty(B_{2R})} \le C_\zeta R^{-2}.$$

Set

$$\phi(y,\tau) := \eta(\tau)\, \zeta(y)^2.$$

Then $\phi \in C_c^\infty(Q_{2R,J})$, $0 \le \phi \le 1$, and

$$\phi \equiv 1 \text{ on } B_R \times J'.$$

Choose $\tau_- < \inf J'$ and $\tau_+ > \sup J'$ in J with $\eta(\tau_-) = \eta(\tau_+) = 0$. Applying (7) on (τ_-, τ_+) gives

$$\nu \int_{\tau_-}^{\tau_+} \int_{B_{2R}} \phi\, |\nabla V|^2 \le |R_\tau| + |R_\Delta| + |R_{\mathrm{conv}}| + |R_{\mathrm{pres}}| + |R_{\mathrm{mod}}|,$$

where

$$R_\tau := \int_J \int_{B_{2R}} \frac{|V|^2}{2}\, \partial_\tau \phi, \qquad R_\Delta := \nu \int_J \int_{B_{2R}} \frac{|V|^2}{2}\, \Delta\phi,$$

$$R_{\mathrm{conv}} := \int_J \int_{B_{2R}} \big(\tfrac{|V|^2}{2} V\big) \cdot \nabla\phi, \qquad R_{\mathrm{pres}} := \int_J \int_{B_{2R}} \big(P - (P)_{B_{2R}}(\tau)\big) V \cdot \nabla\phi,$$

$$R_{\mathrm{mod}} := \int_J \int_{B_{2R}} a \frac{|V|^2}{2} (-\phi - y \cdot \nabla\phi) - \int_J \int_{B_{2R}} \frac{|V|^2}{2} b \cdot \nabla\phi.$$

For almost every $\sigma \in J'$, applying (7) on (τ_-, σ) gives the same right-hand estimates and leaves the terminal term $\frac{1}{2}\int_{B_R} |V(\sigma)|^2$ on the left. Therefore the bounds below control both $\mathcal{H}_{J'}(R)$ and $\mathcal{A}_{J'}(R)$. Define

$$L := \int_J \int_{B_{2R}} |V|^2, \qquad C_3 := \mathcal{C}_J(2R), \qquad D_{3/2} := \mathcal{D}_J(2R).$$

By Theorem 6.2,

$$L \le |Q_{2R,J}|^{1/3} C_3^{2/3} \le C(R, |J|)(1 + C_3).$$

Since $\phi(y,\tau) = \eta(\tau)\zeta(y)^2$,

$$\partial_\tau \phi = \eta'(\tau)\zeta(y)^2, \qquad \nabla\phi = 2\eta(\tau)\zeta(y)\nabla\zeta(y), \qquad \Delta\phi = \eta(\tau)\Delta(\zeta(y)^2),$$

hence

$$\|\partial_\tau\phi\|_{L^\infty(Q_{2R,J})} \le C_\eta \delta^{-1}, \qquad \|\nabla\phi\|_{L^\infty(Q_{2R,J})} \le 2C_\zeta R^{-1}, \qquad \|\Delta\phi\|_{L^\infty(Q_{2R,J})} \le C_\zeta R^{-2}.$$

Step 2: derivative terms from the cut-off.

$$|R_\tau| \le \frac{1}{2} \|\partial_\tau \phi\|_{L^\infty(Q_{2R,J})} L \le C\delta^{-1}(1 + C_3),$$

$$|R_\Delta| \le \frac{\nu}{2} \|\Delta\phi\|_{L^\infty(Q_{2R,J})} L \le C\nu R^{-2}(1 + C_3).$$

Step 3: convection term. Using $|\nabla\phi| \le CR^{-1}$,

$$|R_{\mathrm{conv}}| \le \frac{1}{2} \|\nabla\phi\|_{L^\infty(Q_{2R,J})} \int_J \int_{B_{2R}} |V|^3 \le CR^{-1} C_3.$$

Step 4: pressure term. For each time τ,

$$\int_{B_{2R}} (P - (P)_{B_{2R}}(\tau))V \cdot \nabla\phi = \int_{B_{2R}} PV \cdot \nabla\phi,$$

with

$$\int_{B_{2R}} (P)_{B_{2R}}(\tau)V \cdot \nabla\phi = (P)_{B_{2R}}(\tau) \int_{B_{2R}} V \cdot \nabla\phi = -(P)_{B_{2R}}(\tau) \int_{B_{2R}} \phi \nabla \cdot V = 0$$

(using $\phi \in C_c^\infty(B_{2R})$ and $\nabla \cdot V = 0$). Thus subtracting any time-dependent pressure constant is legitimate in the pressure flux term. Hence

$$|R_{\mathrm{pres}}| \le \|\nabla\phi\|_{L^\infty(Q_{2R,J})} \int_J \|P - (P)_{B_{2R}}(\tau)\|_{L_x^{3/2}(B_{2R})} \|V(\tau,\cdot)\|_{L_x^3(B_{2R})}\, d\tau \le CR^{-1} D_{3/2}^{2/3} C_3^{1/3}.$$

Young's inequality gives

$$|R_{\mathrm{pres}}| \le C(R)(C_3 + D_{3/2}).$$

Step 5: modulation terms. The local energy inequality already contains the modulation terms after the spatial integrations by parts from the repaired-gauge equation:

$$\int_{B_{2R}} (y \cdot \nabla V) \cdot V\, \phi = \frac{1}{2} \int_{B_{2R}} y \cdot \nabla(|V|^2)\phi = -\frac{1}{2} \int_{B_{2R}} |V|^2 \nabla \cdot (y\phi),$$

and

$$\int_{B_{2R}} (b(\tau) \cdot \nabla V) \cdot V\, \phi = \frac{1}{2} \int_{B_{2R}} b(\tau) \cdot \nabla(|V|^2)\phi = -\frac{1}{2} \int_{B_{2R}} |V|^2 b(\tau) \cdot \nabla\phi.$$

Thus we estimate the additional terms directly. Split

$$R_{\mathrm{mod}} = R_a + R_{ay} + R_b,$$

where

$$R_a := -\int_J\int_{B_{2R}} a\,\frac{|V|^2}{2}\phi, \quad R_{ay} := -\int_J\int_{B_{2R}} a\,\frac{|V|^2}{2}\, y \cdot \nabla\phi, \quad R_b := -\int_J\int_{B_{2R}} \frac{|V|^2}{2} b \cdot \nabla\phi.$$

$$M_a := \|a\|_{L^\infty(J)}, \qquad M_b := \|b\|_{L^\infty(J)}.$$

For R_a,

$$|R_a| \le \frac{1}{2} M_a L \le CM_a(1 + C_3).$$

For the scale-gradient part, $|y| \le 2R$ on B_{2R} and $\|\nabla\phi\|_{L^\infty(Q_{2R,J})} \le CR^{-1}$, hence

$$|y \cdot \nabla\phi| \le C.$$

Therefore

$$|R_{ay}| \le CM_a L \le CM_a(1 + C_3).$$

For R_b,

$$|R_b| \le CM_b R^{-1} L \le CM_b R^{-1}(1 + C_3).$$

Step 6: absorption and conclusion. Collecting estimates from Steps 2–5 yields

$$\nu \int_{\tau_-}^{\tau_+} \int_{B_{2R}} \phi |\nabla V|^2 \le C\big(1 + C_3 + D_{3/2}\big),$$

where C depends only on

$$R, |J|, \nu, \|a\|_{L^\infty(J)}, \|b\|_{L^\infty(J)}, \delta.$$

Since $\phi \equiv 1$ on $B_R \times J'$,

$$\mathcal{H}_{J'}(R) \le C\big(1 + \mathcal{C}_J(2R) + \mathcal{D}_J(2R)\big).$$

Repeating the same estimates with terminal time $\sigma \in J'$ gives

$$\frac{1}{2}\int_{B_R} |V(y,\sigma)|^2\,dy \le C\big(1 + \mathcal{C}_J(2R) + \mathcal{D}_J(2R)\big)$$

for almost every $\sigma \in J'$. Taking the essential supremum over $\sigma \in J'$ and renaming C proves the theorem. □

Definition 6.5 (Local compact-cylinder control)**.** Local Caffarelli–Kohn–Nirenberg control on $Q_{r,J}$ means

$$\mathcal{C}_J(r) + \mathcal{D}_J(r) < \infty.$$

Local windowed H^1 control on $Q_{r,J}$ means

$$\mathcal{H}_J(r) < \infty.$$

Corollary 6.6 (Outer control implies inner suitable control)**.** *Let $R > 0$, let $J \subset \mathbb{R}$ be a bounded open interval, and let $J' \Subset J$ be a compact subinterval. Assume Theorem 6.1 on $Q_{2R,J}$ and $\mathcal{C}_J(2R) + \mathcal{D}_J(2R) < \infty$. Then*

$$\mathcal{A}_{J'}(R) + \mathcal{H}_{J'}(R) < \infty.$$

More quantitatively, the bound is the one in Theorem 6.4. If an argument uses pressure means on a smaller ball, Theorem 6.3 converts those normalizations to the outer-ball normalization above.

Proof. The estimate follows from Theorem 6.4. The comparison of pressure normalizations follows from Theorem 6.3. □

Corollary 6.7 (Rough-core exclusion on compact windows)**.** *Let $R > 0$, let $J \subset \mathbb{R}$ be a bounded open interval, and let $J' \Subset J$ be a compact subinterval. Assume Theorem 6.1 on $Q_{2R,J}$. If*

$$\mathcal{C}_J(2R) + \mathcal{D}_J(2R) < \infty,$$

then

$$\mathcal{H}_{J'}(R) < \infty.$$

Equivalently, if $\mathcal{H}_{J'}(R) = +\infty$ for an inner cylinder $Q_{R,J'}$, then for every enclosing cylinder $Q_{2R,J}$ with $J' \Subset J$ on which the local energy inequality and bounded modulation hypotheses hold,

$$\mathcal{C}_J(2R) = +\infty \qquad \text{or} \qquad \mathcal{D}_J(2R) = +\infty.$$

Proof. The direct implication is the gradient part of Theorem 6.4. The last assertion is its contrapositive with the same outer-to-inner geometry: finite outer $\mathcal{C} + \mathcal{D}$ on $Q_{2R,J}$ forces finite inner $\mathcal{H}$ on $Q_{R,J'}$. □

Remark 6.8 (Meaning of rough core)*.* Here a rough core means failure of local windowed H^1 control on compact inner renormalized cylinders while the modulation coefficients are bounded. Theorem 6.7 shows that such a failure is not independent of compact-cylinder Caffarelli–Kohn–Nirenberg control: it can occur only if the velocity-pressure bounds $\mathcal{C} + \mathcal{D}$ fail on every relevant larger enclosing compact cylinder.

6.3. **Consequences for the assembly.** We shall use the following consequences in the final assembly.

1. On every compact renormalized window satisfying the local energy inequality and bounded modulation, finite $\mathcal{C} + \mathcal{D}$ on $Q_{2R,J}$ implies finite $\mathcal{A} + \mathcal{H}$ on $Q_{R,J'}$ whenever $J' \Subset J$.
2. The pressure normalization may be taken on any comparable ball, after using Theorem 6.3.
3. Failure of local windowed H^1 control on an inner compact cylinder forces failure of the compact-cylinder $\mathcal{C} + \mathcal{D}$ bounds on each larger enclosing cylinder to which Theorem 6.4 applies.

7. Scale-Collapse Cost and Autonomous Limits

Notation and functional conventions. For a bounded spatial region $K \subset \mathbb{R}^3$, we use

$$(f,g)_{L^2(K)} := \int_K f \cdot g \, dx, \qquad \langle F, \phi \rangle_{H^{-1}(K),H_0^1(K)} := \int_0^T \langle F(\cdot,s), \phi(\cdot,s) \rangle_{H^{-1}(K),H_0^1(K)} \, ds$$

whenever $F \in L^1(0,T;H^{-1}(K))$ and $\phi \in L^\infty(0,T;H_0^1(K))$. Weak limits and weak convergences are taken in the distributional sense on C_c^∞-test functions.

$$\|f\|_{L_t^p L_x^q} := \left(\int_0^T \|f(\cdot,s)\|_{L^q(\mathbb{R}^3)}^p \, ds \right)^{1/p},$$

for any fixed $T > 0$, with the usual modification $p = \infty$.

7.1. Renormalized scale-collapse setting.

Position of the result. The scale-collapse analysis has two components:

(1) Sections 7.2–7.4 contain the cost exclusion arguments, which do not use compactness.
(2) Section 7.5 contains the autonomous compactness statement under explicitly stated compactness hypotheses.

Throughout the scale-collapse analysis, the renormalized representation established earlier is assumed: on the time interval under consideration, the fields (V,P) satisfy (8) and its distributional weak formulation. In particular, the solenoidal formulation is obtained by restricting the test fields below to divergence-free fields.

Let (u,p) be a divergence-free suitable weak solution of

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \qquad \nabla \cdot u = 0,$$

in $\mathbb{R}^3$, with viscosity $\nu > 0$. Let $\lambda(\tau) > 0$, $X(\tau) \in \mathbb{R}^3$, and a measurable gauge function $c(t)$ be given, and define the renormalized fields by

$$u(x,t) = \lambda(\tau)^{-1} V\left(\frac{x - X(\tau)}{\lambda(\tau)}, \tau \right), \qquad p(x,t) = \lambda(\tau)^{-2} P\left(\frac{x - X(\tau)}{\lambda(\tau)}, \tau \right) + c(t).$$

Pressure is defined only up to an arbitrary function of time; $c(t)$ has no effect on ∇p, so it is irrelevant for all subsequent estimates. Assume that (V,P) satisfies on $(0,\infty) \times \mathbb{R}^3$

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0. \tag{8}$$

For every $\psi \in C_c^\infty((0,\infty) \times \mathbb{R}^3; \mathbb{R}^3)$ with $\nabla \cdot \psi = 0$, the pair (V,P) satisfies the weak formulation

$$\begin{aligned} 0 = &\int_0^\infty \int_{\mathbb{R}^3} \big(-V \cdot \partial_\tau \psi - (V \otimes V) : \nabla \psi + \nu \nabla V : \nabla \psi \big) \, dy \, d\tau \\ &+ \int_0^\infty \int_{\mathbb{R}^3} \big(-a(\tau)(V + y \cdot \nabla V) \cdot \psi - (b(\tau) \cdot \nabla V) \cdot \psi + P \, \nabla \cdot \psi \big) \, dy \, d\tau. \end{aligned}$$

For every $0 < T < \infty$, restricting the τ-integration to $(0,T)$ gives the same identity on $(0,T) \times \mathbb{R}^3$. Equivalently, for every $\psi \in C_c^\infty((0,\infty) \times \mathbb{R}^3; \mathbb{R}^3)$, not necessarily divergence free,

$$\begin{aligned} 0 = &\int_0^\infty \int_{\mathbb{R}^3} \big(-V \cdot \partial_\tau \psi - (V \otimes V) : \nabla \psi \nu \nabla V : \nabla \psi - P \, \nabla \cdot \psi \big) \, dy \, d\tau \\ &+ \int_0^\infty \int_{\mathbb{R}^3} \big(-a(\tau)(V + y \cdot \nabla V) \cdot \psi - (b(\tau) \cdot \nabla V) \cdot \psi \big) \, dy \, d\tau. \end{aligned}$$

Define positive and negative parts of the scalar field a by

$$a_+(\tau) := \max\{a(\tau), 0\}, \qquad a_-(\tau) := \max\{-a(\tau), 0\}.$$

Then for a.e. τ, $a = a_+ - a_-$, $a_+, a_- \geq 0$, and $a_+ a_- = 0$. Assume

$$a \in L^1_{\mathrm{loc}}([\tau_0, \infty)), \qquad b \in L^1_{\mathrm{loc}}([\tau_0, \infty); \mathbb{R}^3),$$

and the scale convention

$$\frac{d}{d\tau} \log \lambda(\tau) = a(\tau) \tag{9}$$

for a.e. $\tau \geq \tau_0$, with $\log \lambda$ absolutely continuous on compact intervals.

All integrals below are interpreted in $[0, \infty]$. When we write $\int_{\tau_0}^{\infty}$, partial integrals are taken over $[\tau_0, T]$ and monotone convergence is used as $T \to \infty$.

Fix $R > 0$. Choose $\Phi \in C_c^\infty(\mathbb{R}^3; [0,1])$ satisfying

$$\Phi(y) = 1 \text{ if } |y| \leq R, \quad \operatorname{supp}_y \Phi \subset B_{2R}. \tag{10}$$

Define the localized kinetic mass

$$M_R(\tau) := \int_{\mathbb{R}^3} |V(y,\tau)|^2 \Phi(y)\, dy.$$

Since Φ is fixed in τ, M_R is measurable in τ, and only this measurability is used below.

Definition 7.1 (Nonvanishing localized core)**.** A renormalized branch has nonvanishing localized core if there exist constants $m_0 > 0$ and $\tau_1 \geq \tau_0$ such that

$$M_R(\tau) \geq m_0 \quad \text{for a.e. } \tau \in [\tau_1, \infty).$$

In applications, this lower bound is supplied by the selected-core construction; here it is stated as an explicit hypothesis.

Definition 7.2 (Scale-collapse and absolute scale-drift densities)**.** Define

$$\mathcal{C}_{\mathrm{sc}}(\tau) := a_-(\tau) M_R(\tau), \qquad \mathcal{C}_{\mathrm{abs}}(\tau) := \nu \int_{\mathbb{R}^3} |\nabla V(y,\tau)|^2 \Phi(y)\, dy + |a(\tau)| M_R(\tau).$$

$\mathcal{C}_{\mathrm{sc}}$ is the negative drift weighted by localized core mass, while $\mathcal{C}_{\mathrm{abs}}$ controls both collapse drift and local dissipation. Their total costs are extended real-valued by

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{sc}}(\tau)\, d\tau, \qquad \int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{abs}}(\tau)\, d\tau.$$

Definition 7.3 (Finite-cost branch)**.** A branch has finite scale-collapsing cost if

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{sc}}(\tau)\, d\tau < \infty.$$

It has finite absolute cost if

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{abs}}(\tau)\, d\tau < \infty.$$

7.2. **Elementary alternative.**

Lemma 7.4 (Finite-or-infinite alternative)**.** *Let $f \geq 0$ be measurable on $[\tau_0, \infty)$. Then exactly one of the following holds:*

$$\int_{\tau_0}^{\infty} f(\tau)\, d\tau < \infty \quad \textit{or} \quad \int_{\tau_0}^{\infty} f(\tau)\, d\tau = \infty.$$

Proof. (1) For $T > \tau_0$, define partial costs

$$F(T) := \int_{\tau_0}^{T} f(\tau)\, d\tau \in [0, \infty].$$

(2) If $T_2 > T_1$, then by nonnegativity of f,

$$F(T_2) - F(T_1) = \int_{T_1}^{T_2} f(\tau)\, d\tau \geq 0,$$

so F is monotone nondecreasing.

(3) A monotone function on $[\tau_0, \infty)$ has an extended limit $L \in [0, \infty]$,

$$L := \lim_{T \to \infty} F(T).$$

By definition of improper integral, this limit equals $\int_{\tau_0}^{\infty} f(\tau)\, d\tau$.

(4) Therefore L is either finite or equal to $+\infty$, and these cases are mutually exclusive.

Hence exactly one item in the lemma holds. □

Lemma 7.5 (Logarithmic scale identity)**.** *Under* (9)*, for all* τ_1, τ_2 *with* $\tau_0 \leq \tau_1 < \tau_2 < \infty$*,*

$$\int_{\tau_1}^{\tau_2} a(\tau)\, d\tau = \log \lambda(\tau_2) - \log \lambda(\tau_1).$$

Proof. (1) Since $a \in L^1_{\mathrm{loc}}$, the map $\log \lambda$ is absolutely continuous on $[\tau_1, \tau_2]$, hence has a representative with a.e. derivative equal to a.

(2) By the fundamental theorem of calculus for absolutely continuous functions, for every $\tau \in [\tau_1, \tau_2]$,

$$\log \lambda(\tau) = \log \lambda(\tau_1) + \int_{\tau_1}^{\tau} a(s)\, ds.$$

(3) Setting $\tau = \tau_2$ yields the identity.

□

7.3. **Scale collapse versus localized core lower bounds.**

Theorem 7.6 (Collapse with nonvanishing localized core forces infinite scale-collapsing cost)**.** *Assume:*

(1) $\lambda(\tau) \to 0$ *as* $\tau \to \infty$*;*

(2) the branch has nonvanishing localized core.

Then

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{sc}}(\tau)\, d\tau = \int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau = \infty.$$

Proof. (1) Choose $m_0 > 0$ and $\tau_1 \geq \tau_0$ from the nonvanishing core assumption so that

$$M_R(\tau) \geq m_0 \quad \text{for a.e. } \tau \in [\tau_1, \infty).$$

(2) By Lemma 7.5, for any $\tau_2 > \tau_1$,

$$\int_{\tau_1}^{\tau_2} a(\tau)\, d\tau = \log \lambda(\tau_2) - \log \lambda(\tau_1).$$

Taking $\tau_2 \to \infty$ and using $\lambda(\tau_2) \to 0$,

$$\lim_{\tau_2 \to \infty} \int_{\tau_1}^{\tau_2} a(\tau)\, d\tau = -\infty.$$

(3) Decompose $a = a_+ - a_-$. Then for each $\tau_2 > \tau_1$,

$$\int_{\tau_1}^{\tau_2} a(\tau)\, d\tau = \int_{\tau_1}^{\tau_2} a_+(\tau)\, d\tau - \int_{\tau_1}^{\tau_2} a_-(\tau)\, d\tau \geq - \int_{\tau_1}^{\tau_2} a_-(\tau)\, d\tau.$$

Hence

$$\int_{\tau_1}^{\tau_2} a_-(\tau)\,d\tau \ge -\int_{\tau_1}^{\tau_2} a(\tau)\,d\tau.$$

(4) Taking limits and using Step 2 yields

$$\int_{\tau_1}^{\infty} a_-(\tau)\,d\tau = \infty.$$

Indeed, by Step 2,

$$-\int_{\tau_1}^{\tau_2} a(\tau)\,d\tau \to +\infty.$$

Since the partial integrals of a_- dominate this quantity, they are unbounded. Because $a_- \ge 0$, monotone convergence gives $\int_{\tau_1}^{\infty} a_-(\tau)\,d\tau = \infty$.

(5) Split the cost into early and late times:

$$\int_{\tau_0}^{\infty} a_- M_R\,d\tau = \int_{\tau_0}^{\tau_1} a_- M_R\,d\tau + \int_{\tau_1}^{\infty} a_- M_R\,d\tau.$$

The first term is finite or infinite independently of the argument; the second term satisfies

$$\int_{\tau_1}^{\infty} a_-(\tau) M_R(\tau)\,d\tau \ge m_0 \int_{\tau_1}^{\infty} a_-(\tau)\,d\tau = \infty$$

by the bound in (1) and Step 4. Hence the total integral is infinite.

□

7.4. **Finite-cost exclusions.**

Theorem 7.7 (Scale-collapse cost exclusion)**.** *No branch with finite scale-collapsing cost (Definition 7.3) can satisfy simultaneously $\lambda(\tau) \to 0$ and nonvanishing localized core.*

Proof. Assume, for contradiction, that such a branch exists. The two hypotheses are exactly those of Theorem 7.6; therefore

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{sc}}(\tau)\,d\tau = \infty.$$

This contradicts finite scale-collapsing cost. Therefore no such branch exists. □

Lemma 7.8 (Absolute-cost domination)**.** *For a.e. $\tau \in [\tau_0, \infty)$,*

$$\mathcal{C}_{\mathrm{abs}}(\tau) \ge \mathcal{C}_{\mathrm{sc}}(\tau).$$

Hence, pointwise and after integrating in τ,

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{sc}}(\tau)\,d\tau \le \int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{abs}}(\tau)\,d\tau,$$

as extended real numbers. In particular, finite absolute cost implies finite scale-collapsing cost, and equivalently

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{sc}}(\tau)\,d\tau = \infty \quad\Longrightarrow\quad \int_{\tau_0}^{\infty} \mathcal{C}_{\mathrm{abs}}(\tau)\,d\tau = \infty.$$

Proof. At each time τ,

$$\mathcal{C}_{\mathrm{abs}}(\tau) = \nu \int |\nabla V|^2 \Phi + (a_+ + a_-) M_R \ge (a_+ + a_-) M_R \ge a_- M_R = \mathcal{C}_{\mathrm{sc}}(\tau),$$

since $a_+, a_- \ge 0$ and $\nu \int |\nabla V|^2 \Phi \ge 0$. Integrating over $[\tau_0, \infty)$ yields the claimed implication. □

Corollary 7.9 (Absolute scale-cost exclusion)**.** *No branch with finite absolute cost in the sense of Definition 7.3 can satisfy simultaneously $\lambda(\tau) \to 0$ and nonvanishing localized core.*

Proof. Assume instead such a branch exists. By Theorem 7.6, $\int \mathcal{C}_{\rm sc} = \infty$. By Lemma 7.8, the absolute cost is pointwise above $\mathcal{C}_{\rm sc}$, so its integral must also be infinite, contradicting finite absolute cost. ☐

Remark 7.10. The exclusions above are purely integral and use neither compactness nor any autonomous-limit argument.

7.5. **Autonomous reduced-limit theorem from the selected-window estimates.** The preceding exclusions are integral statements and do not rely on compactness. On a fixed finite autonomous window, compactness requires not only the cost bound but also the selected-window estimates supplied by the preceding renormalized representation, pressure reconstruction, and local Caccioppoli estimates.

Analytic inputs. The preceding sections supply the following ingredients on selected compact renormalized windows:

(1) the repaired-gauge renormalized equation and weak formulation for (V, P);
(2) pressure reconstruction modulo spatial means, with local $L^{3/2}$ bounds on compact cylinders;
(3) local windowed bounds $V \in L^\infty_t L^2_x \cap L^2_t H^1_x$ on compact cylinders;
(4) a retained compact core lower bound preventing the selected profile from vanishing.

Together with the thick-window convergence of the modulation coefficients, these estimates imply the compactness and limit passage below.

Definition 7.11 (Thick autonomous windows)**.** A branch has *thick autonomous windows* if there exist constants $T > 0$, $c > 0$, $M_1 \in (0, \infty)$, a sequence (τ_n) with $\tau_n \to \infty$, and limits $a_\infty \in \mathbb{R}$, $b_\infty \in \mathbb{R}^3$, such that for every n:

(i)
$$\frac{1}{T}\int_{\tau_n}^{\tau_n+T} a_-(\tau) M_R(\tau)\, d\tau \geq c;$$

(ii)
$$0 \leq M_R(\tau) \leq M_1 \quad \text{for a.e. } \tau \in [\tau_n, \tau_n + T];$$

(iii)
$$a(\tau_n + \cdot) \to a_\infty \text{ in } L^1(0,T), \qquad b(\tau_n + \cdot) \to b_\infty \text{ in } L^1(0,T;\mathbb{R}^3).$$

Remark 7.12. This notion isolates a limiting drift coefficient $a_\infty < 0$. The compactness to extract a profile limit is obtained from the selected-window compactness estimates, not from the scalar cost inequalities of Sections 7.2–7.4.

Lemma 7.13 (Thick drift implies negative autonomous limit parameter)**.** *On thick autonomous windows, the limiting autonomous coefficient is negative:*

$$a_\infty \leq -\frac{c}{M_1} < 0.$$

Proof. (1) From (ii), for a.e. $\tau \in [\tau_n, \tau_n + T]$, $0 \leq M_R(\tau) \leq M_1$. Therefore

$$\frac{a_-(\tau) M_R(\tau)}{M_1} \leq a_-(\tau).$$

Therefore

$$\frac{1}{T}\int_{\tau_n}^{\tau_n+T} a_-(\tau)\, d\tau \geq \frac{1}{M_1 T}\int_{\tau_n}^{\tau_n+T} a_-(\tau) M_R(\tau)\, d\tau \geq \frac{c}{M_1}.$$

(2) Set $a_n(s) := a(\tau_n + s)$. By (iii), $a_n \to a_\infty$ in $L^1(0,T)$. The map $r \mapsto r^-$ is 1-Lipschitz, so $a_n^- \to a_\infty^-$ in $L^1(0,T)$:

$$\|a_n^- - a_\infty^-\|_{L^1(0,T)} \le \|a_n - a_\infty\|_{L^1(0,T)} \to 0.$$

Thus

$$\lim_{n\to\infty} \frac{1}{T}\int_{\tau_n}^{\tau_n+T} a_-(\tau)\,d\tau = \frac{1}{T}\int_0^T (a_\infty)^-\,ds = (a_\infty)^-.$$

(3) Combine Step 1 and Step 2: $(a_\infty)^- \ge c/M_1$, i.e. $a_\infty \le -c/M_1 < 0$.

□

Proposition 7.14 (Selected-window suitable estimates)**.** *Let (τ_n) be a thick autonomous-window sequence and set*

$$V_n(y,s) := V(y,\tau_n + s), \qquad P_n(y,s) := P(y,\tau_n+s),$$
$$\alpha_n(s) := a(\tau_n+s), \qquad \beta_n(s) := b(\tau_n+s), s \in (0,T).$$

For every bounded Lipschitz domain $K \Subset \mathbb{R}^3$, there is a constant $C_K < \infty$, independent of n, such that

$$\|V_n\|_{L^\infty(0,T;L^2(K))} + \|V_n\|_{L^2(0,T;H^1(K))} + \|P_n - (P_n)_K\|_{L^{3/2}((0,T)\times K)} \le C_K,$$

where

$$(P_n)_K(s) := \frac{1}{|K|}\int_K P_n(y,s)\,dy.$$

Each V_n is divergence free and there exist $\chi \in C_c^\infty(B_R)$, $\chi \ge 0$, $\eta > 0$, and $N \in \mathbb{N}$ such that for all $n \ge N$,

$$\int_{\mathbb{R}^3} |V_n(y,s)|^2\chi(y)\,dy \ge \eta \quad \textit{for a.e. } s \in (0,T).$$

Proof. The localized lower bound is the conclusion of the selected core retention theorem Theorem 7.21 below, which is purely a property of the selected-core construction and is independent of the autonomous-window selection. With the lower bound in hand, Theorem 7.15 below derives the local $L_t^\infty L_x^2$, $L_t^2 H_x^1$, and pressure bounds from the repaired-gauge renormalized representation Theorem 4.35, the localized pressure decomposition Theorem 4.25, and the compact-cylinder Caccioppoli estimate Theorem 6.4. These two inputs together yield all items in the statement. □

Proposition 7.15 (Derivation of the selected-window estimates from the preceding sections)**.** *Let a repaired-gauge branch satisfy the renormalized representation and pressure reconstruction of the compact-window representation theorem, Theorem 4.35, on the windows $[\tau_n, \tau_n + T]$. Assume that on each compact cylinder $(0,T)\times K$ the local compact-cylinder Caffarelli–Kohn–Nirenberg quantities are finite uniformly in n, and that the selected core has the localized lower bound stated in Assumption 7.14. Then Assumption 7.14 holds.*

Proof. The renormalized representation theorem supplies the shifted equation, divergence-free condition, and pressure reconstruction modulo functions of time. The pressure reconstruction gives, after subtracting spatial means,

$$\sup_n \|P_n - (P_n)_K\|_{L^{3/2}((0,T)\times K)} < \infty$$

on every compact K. The local Caccioppoli estimate applied on a slightly larger compact cylinder gives the inner bounds

$$\sup_n \|V_n\|_{L^\infty(0,T;L^2(K))} + \sup_n \|V_n\|_{L^2(0,T;H^1(K))} < \infty.$$

These are precisely the local velocity and pressure estimates in Assumption 7.14. The retained selected-core lower bound is the last condition in that assumption. Hence all items in Assumption 7.14 follow from the representation, pressure, and local Caccioppoli results. $\square$

Lemma 7.16 (Local time regularity on autonomous windows)**.** *Assume thick autonomous windows and Assumption 7.14. Let $K_0 \Subset K_1 \Subset \mathbb{R}^3$ be bounded Lipschitz domains, and choose an integer m so large that $H_0^m(K_1) \hookrightarrow W_0^{1,\infty}(K_1)$. Then*

$$\{\partial_s V_n\}_{n\geq 1} \quad \textit{is bounded in } L^1(0,T;H^{-m}(K_1)).$$

Proof. Fix $\zeta \in C_c^\infty(K_1)$ with $\zeta \equiv 1$ on a neighborhood of K_0. Let $\phi \in H_0^m(K_1;\mathbb{R}^3)$. Since $H_0^m(K_1) \hookrightarrow W_0^{1,\infty}(K_1)$, there is C_m such that

$$\|\phi\|_{L^\infty(K_1)} + \|\nabla\phi\|_{L^\infty(K_1)} \leq C_m\|\phi\|_{H^m(K_1)}.$$

The shifted equation is

$$\partial_s V_n = -(V_n\cdot\nabla)V_n - \nabla P_n + \nu\Delta V_n + \alpha_n(V_n + y\cdot\nabla V_n) + \beta_n\cdot\nabla V_n$$

in distributions. We estimate each term against ϕ.

For the convection term, integration by parts gives

$$|\langle (V_n\cdot\nabla)V_n,\phi\rangle| = \left|\int_{K_1} V_{n,i}V_{n,j}\partial_j\phi_i\,dy\right| \leq C_m\|V_n(s)\|^2_{L^2(K_1)}\|\phi\|_{H^m(K_1)}.$$

For the viscous term,

$$|\langle \nu\Delta V_n,\phi\rangle| = \nu\left|\int_{K_1}\nabla V_n : \nabla\phi\,dy\right| \leq \nu C_m\|\nabla V_n(s)\|_{L^2(K_1)}|K_1|^{1/2}\|\phi\|_{H^m(K_1)}.$$

For the pressure term, subtracting the spatial mean does not change the gradient:

$$|\langle\nabla P_n,\phi\rangle| = \left|\int_{K_1}(P_n - (P_n)_{K_1})\,\nabla\cdot\phi\,dy\right| \leq C_m|K_1|^{1/3}\|P_n - (P_n)_{K_1}\|_{L^{3/2}(K_1)}\|\phi\|_{H^m(K_1)}.$$

For the V_n-part of the scaling drift,

$$|\alpha_n(s)|\left|\int_{K_1} V_n\cdot\phi\,dy\right| \leq C_m|\alpha_n(s)|\,|K_1|^{1/2}\|V_n(s)\|_{L^2(K_1)}\|\phi\|_{H^m(K_1)}.$$

For the $y\cdot\nabla V_n$-part, integrate by parts:

$$\int_{K_1}(y\cdot\nabla V_n)\cdot\phi\,dy = -\int_{K_1} V_n\cdot(y\cdot\nabla\phi + 3\phi)\,dy,$$

because ϕ has zero trace. Hence

$$|\alpha_n(s)|\left|\int_{K_1}(y\cdot\nabla V_n)\cdot\phi\,dy\right| \leq C_{K_1,m}|\alpha_n(s)|\|V_n(s)\|_{L^2(K_1)}\|\phi\|_{H^m(K_1)}.$$

Similarly,

$$\left|\int_{K_1}(\beta_n\cdot\nabla V_n)\cdot\phi\,dy\right| = \left|\int_{K_1} V_n\cdot(\beta_n\cdot\nabla\phi)\,dy\right| \leq C_m|\beta_n(s)|\,|K_1|^{1/2}\|V_n(s)\|_{L^2(K_1)}\|\phi\|_{H^m(K_1)}.$$

Combining the estimates and taking the supremum over $\|\phi\|_{H^m(K_1)} \leq 1$, we obtain

$$\begin{aligned}\|\partial_s V_n(s)\|_{H^{-m}(K_1)} \leq C_{K_1,m}\big(&\|V_n(s)\|^2_{L^2(K_1)} + \|\nabla V_n(s)\|_{L^2(K_1)}\\ &+ \|P_n(s) - (P_n)_{K_1}(s)\|_{L^{3/2}(K_1)} + (|\alpha_n(s)| + |\beta_n(s)|)\|V_n(s)\|_{L^2(K_1)}\big).\end{aligned}$$

Integrating in $s \in (0,T)$, using Assumption 7.14, and using $\alpha_n \to a_\infty$, $\beta_n \to b_\infty$ in L^1, hence uniform L^1-bounds for α_n, β_n, proves the claimed $L^1(0,T;H^{-m})$ bound. $\square$

Lemma 7.17 (Local compactness on autonomous windows)**.** *Assume thick autonomous windows and Assumption 7.14. Then after passing to a subsequence there exist*

$$U \in L^\infty(0,T;L^2_{\mathrm{loc}}(\mathbb{R}^3)) \cap L^2(0,T;H^1_{\mathrm{loc}}(\mathbb{R}^3)), \qquad \Pi \in L^{3/2}_{\mathrm{loc}}((0,T)\times\mathbb{R}^3),$$

such that, for every compact $K \Subset \mathbb{R}^3$,

(a)
$$V_n \to U \quad \textit{strongly in } L^2(0,T;L^2(K));$$

(b)
$$V_n \rightharpoonup U \quad \textit{weakly in } L^2(0,T;H^1(K));$$

(c)
$$V_n \rightharpoonup^* U \quad \textit{weakly-* in } L^\infty(0,T;L^2(K));$$

(d)
$$P_n - (P_n)_K \rightharpoonup \Pi_K \quad \textit{weakly in } L^{3/2}((0,T)\times K),$$

where the local distributions $\nabla\Pi_K$ agree on overlaps and define a pressure gradient $\nabla\Pi$ modulo functions of time.

Proof. Fix $K_0 \Subset K_1 \Subset \mathbb{R}^3$. Assumption 7.14 gives boundedness of V_n in $L^2(0,T;H^1(K_1))$, and Lemma 7.16 gives boundedness of $\partial_s V_n$ in $L^1(0,T;H^{-m}(K_1))$ for m large. Since

$$H^1(K_1) \Subset L^2(K_0) \hookrightarrow H^{-m}(K_1),$$

the Aubin–Lions–Simon compactness theorem implies, after extracting a subsequence,

$$V_n \to U \quad \text{strongly in } L^2(0,T;L^2(K_0)).$$

The local $L^\infty L^2$ and L^2H^1 bounds also give, after extraction,

$$V_n \rightharpoonup^* U \quad \text{in } L^\infty(0,T;L^2(K_0)), \qquad V_n \rightharpoonup U \quad \text{in } L^2(0,T;H^1(K_0)).$$

The pressure bound in Assumption 7.14 gives weak compactness of $P_n-(P_n)_{K_0}$ in $L^{3/2}((0,T)\times K_0)$. A diagonal extraction over balls B_j gives a single subsequence for every compact K. On overlaps, two pressure limits can differ only by a function of s, because both represent the same distributional pressure gradient in the limit equation. Thus the gradients assemble into $\nabla\Pi$, with Π defined modulo functions of time. □

Lemma 7.18 (Compatibility of local pressure limits)**.** *Let $K_1, K_2 \Subset \mathbb{R}^3$ be bounded Lipschitz domains with nonempty connected overlap $K_{12} := K_1 \cap K_2$. Suppose that, after passing to a common subsequence,*

$$P_n - (P_n)_{K_i} \rightharpoonup \Pi_i \quad \textit{in } L^{3/2}((0,T)\times K_i), \qquad i = 1,2.$$

Then there exists $c_{12} \in L^{3/2}(0,T)$ such that

$$\Pi_1(y,s) - \Pi_2(y,s) = c_{12}(s) \quad \textit{for a.e. } (y,s) \in K_{12}\times(0,T).$$

Consequently the pressure gradient $\nabla\Pi$ obtained from the local limits is globally well-defined on $(0,T)\times\mathbb{R}^3$, with the pressure itself defined modulo functions of time.

Proof. For each n,

$$\big(P_n - (P_n)_{K_1}\big) - \big(P_n - (P_n)_{K_2}\big) = (P_n)_{K_2}(s) - (P_n)_{K_1}(s)$$

on K_{12}. Hence its spatial gradient is zero in $\mathcal{D}'((0,T)\times K_{12})$. Passing to the weak limit gives

$$\nabla_y(\Pi_1 - \Pi_2) = 0 \quad \text{in } \mathcal{D}'((0,T)\times K_{12}).$$

Since K_{12} is connected, the standard distributional characterization of functions with zero spatial gradient implies that $\Pi_1 - \Pi_2 = c_{12}(s)$ for some $c_{12} \in L^{3/2}(0,T)$. The last assertion follows because adding a time-dependent scalar to pressure does not change $\nabla\Pi$. □

Theorem 7.19 (Autonomous reduced-limit theorem)**.** *Assume that a branch has thick autonomous windows and satisfies Assumption 7.14 on the same window sequence. Let (U, Π) be the limit extracted in Lemma 7.17. Then*

$$\nabla \cdot U = 0,$$
$$U \not\equiv 0$$

and (U, Π) satisfies

$$\partial_s U + (U \cdot \nabla)U + \nabla \Pi = \nu \Delta U + a_\infty(U + y \cdot \nabla U) + b_\infty \cdot \nabla U, \qquad \nabla \cdot U = 0, \tag{11}$$

in $\mathcal{D}'((0,T) \times \mathbb{R}^3)$. Equivalently, for every $\varphi \in C_c^\infty((0,T) \times \mathbb{R}^3; \mathbb{R}^3)$,

$$\int_0^T \int \big(-U \cdot \partial_s \varphi - (U \otimes U) : \nabla \varphi + \nu \nabla U : \nabla \varphi - \Pi\, \nabla \cdot \varphi - a_\infty (U + y \cdot \nabla U) \cdot \varphi - (b_\infty \cdot \nabla U) \cdot \varphi\big)\, dy\, ds = 0.$$

Remark 7.20. The theorem asserts only the compact finite-window autonomous limit from the selected-window estimates. A rigidity contradiction for an arbitrary autonomous limit would require a separate Liouville theorem for the class containing (U, Π). The retained compact scale-collapse branch used in the terminal proof is handled differently: failures of retained inputs are named local exits, and the remaining retained autonomous case is routed by Theorem 7.30 into the scale-rigid discharge.

Proof. Step 1: divergence-free property of the limit.

Each V_n is divergence free by Assumption 7.14, so for any compactly supported scalar $\psi \in C_c^\infty((0,T) \times \mathbb{R}^3)$,

$$\int_{(0,T)\times\mathbb{R}^3} V_n \cdot \nabla \psi\, dy\, ds = 0.$$

Hence, for every such ψ, since $\nabla \psi \in C_c^\infty((0,T) \times \mathbb{R}^3)$ and $V_n \to U$ in $L^2(0,T; L^2_{\mathrm{loc}})$,

$$\int_{(0,T)\times\mathbb{R}^3} (V_n - U) \cdot \nabla \psi\, dy\, ds \to 0,$$

we deduce

$$\int_{(0,T)\times\mathbb{R}^3} U \cdot \nabla \psi\, dy\, ds = 0.$$

Hence $\nabla \cdot U = 0$ in $\mathcal{D}'((0,T) \times \mathbb{R}^3)$.

Step 2: nontriviality of the limit profile. Define for $s \in (0,T)$

$$f_n(s) := \int_{\mathbb{R}^3} |V_n(y,s)|^2 \chi(y)\, dy, \qquad f(s) := \int_{\mathbb{R}^3} |U(y,s)|^2 \chi(y)\, dy.$$

Here $K_\chi := \operatorname{supp} \chi$, so $\chi^{1/2} V_n, \chi^{1/2} U \in L^2(0,T; L^2(K_\chi))$. Then

$$f_n - f = \int_{\mathbb{R}^3} (V_n - U) \cdot (V_n + U) \chi\, dy.$$

Applying Cauchy–Schwarz in space,

$$|f_n(s) - f(s)| \le \|\chi^{1/2}(V_n(s) - U(s))\|_{L^2(\mathbb{R}^3)} \|\chi^{1/2}(V_n(s) + U(s))\|_{L^2(\mathbb{R}^3)}.$$

Integrating in s and applying Cauchy–Schwarz in time gives

$$\|f_n - f\|_{L^1(0,T)} \le \|\chi^{1/2}(V_n - U)\|_{L^2(0,T;L^2)} \|\chi^{1/2}(V_n + U)\|_{L^2(0,T;L^2)}.$$

Since K_χ is compact, Assumption 7.14 and the membership $U \in L^2(0,T; L^2(K_\chi))$ give

$$\sup_n \|\chi^{1/2}(V_n + U)\|_{L^2(0,T;L^2)} \le C_\chi \left(\sup_n \|V_n\|_{L^2(0,T;L^2(K_\chi))} + \|U\|_{L^2(0,T;L^2(K_\chi))} \right) < \infty.$$

Moreover

$$\|\chi^{1/2}(V_n - U)\|_{L^2(0,T;L^2)} \le \|\chi\|_{L^\infty}^{1/2}\|V_n - U\|_{L^2(0,T;L^2(K_\chi))} \to 0.$$

Therefore

$$\|f_n - f\|_{L^1(0,T)} \to 0.$$

Hence $f_n \to f$ in $L^1(0,T)$.

Assumption 7.14 gives for large n, $f_n(s) \ge \eta$ for a.e.s. Therefore

$$\int_0^T f(s)\,ds = \lim_{n\to\infty}\int_0^T f_n(s)\,ds \ge \eta T > 0.$$

Hence $f \not\equiv 0$, so $U \not\equiv 0$.

Step 3: shifted weak formulation for V_n. Take any test field $\varphi \in C_c^\infty((0,T)\times\mathbb{R}^3;\mathbb{R}^3)$, and set

$$K_\varphi := \overline{\{x\in\mathbb{R}^3 : \exists s\in(0,T),\ \varphi(x,s)\neq 0\}} \subset \mathbb{R}^3, \qquad K_\varphi \text{ bounded}.$$

Since (V,P) satisfies (8) in distributions, the change of variables $\tau = \tau_n + s$ gives the shifted weak formulation

$$\begin{aligned} 0 = &\int_0^T\int_{\mathbb{R}^3}\Big(-V_n\cdot\partial_s\varphi - (V_n\otimes V_n):\nabla\varphi + \nu\nabla V_n:\nabla\varphi - P_n\nabla\cdot\varphi\Big)\,dy\,ds \\ &+ \int_0^T\int_{\mathbb{R}^3}\Big(-a(\tau_n+s)(V_n + y\cdot\nabla V_n)\cdot\varphi - (b(\tau_n+s)\cdot\nabla V_n)\cdot\varphi\Big)\,dy\,ds. \end{aligned} \tag{12}$$

Step 4: term-by-term passage to the limit. Write $\alpha_n(s) := a(\tau_n+s)$, $\beta_n(s) := b(\tau_n+s)$.

(i) Time derivative.

By strong convergence in $L^2(0,T;L^2(K_\varphi))$,

$$\left|\int_0^T\int (V_n - U)\cdot\partial_s\varphi\,dy\,ds\right| \le \|V_n - U\|_{L^2(0,T;L^2(K_\varphi))}\|\partial_s\varphi\|_{L^2(0,T;L^2(K_\varphi))} \to 0.$$

Hence

$$\int V_n\cdot\partial_s\varphi \to \int U\cdot\partial_s\varphi.$$

(ii) Convection term. Decompose

$$V_n\otimes V_n - U\otimes U = (V_n - U)\otimes V_n + U\otimes(V_n - U).$$

Then

$$\begin{aligned} &\left|\int_0^T\int ((V_n\otimes V_n - U\otimes U):\nabla\varphi)\,dy\,ds\right| \\ &\quad\le \|\nabla\varphi\|_{L^\infty_{x,t}}\|V_n - U\|_{L^2(0,T;L^2(K_\varphi))}\big(\|V_n\|_{L^2(0,T;L^2(K_\varphi))} + \|U\|_{L^2(0,T;L^2(K_\varphi))}\big) \to 0, \end{aligned}$$

using boundedness of V_n in $L^2(0,T;L^2(K_\varphi))$. Therefore $\int(V_n\otimes V_n):\nabla\varphi \to \int(U\otimes U):\nabla\varphi$.

(iii) Viscous term. From weak convergence of V_n in $L^2(0,T;H^1_{\mathrm{loc}})$,

$$\int_0^T\int \nabla V_n:\nabla\varphi\,dy\,ds \to \int_0^T\int \nabla U:\nabla\varphi\,dy\,ds.$$

(iv) Pressure term. Let K be a bounded Lipschitz domain with $K_\varphi \Subset K$. Since φ is compactly supported in K,

$$\int_K (P_n)_K(s)\,\nabla\cdot\varphi(y,s)\,dy = (P_n)_K(s)\int_K \nabla\cdot\varphi(y,s)\,dy = 0.$$

Therefore

$$\int_0^T \int_{\mathbb{R}^3} P_n \nabla \cdot \varphi \, dy \, ds = \int_0^T \int_K (P_n - (P_n)_K) \nabla \cdot \varphi \, dy \, ds.$$

By Lemma 7.17,

$$P_n - (P_n)_K \rightharpoonup \Pi_K \quad \text{in } L^{3/2}((0,T) \times K),$$

and $\nabla \cdot \varphi \in L^3((0,T) \times K)$. The compatibility Lemma 7.18 permits us to write the resulting pressure distribution as Π, modulo functions of time. Hence

$$\int_0^T \int_{\mathbb{R}^3} P_n \nabla \cdot \varphi \, dy \, ds \to \int_0^T \int_K \Pi_K \nabla \cdot \varphi \, dy \, ds.$$

The value of the last integral is independent of the chosen K, because replacing Π_K by $\Pi_K + c(s)$ changes the integral by $\int_0^T c(s) \int_K \nabla \cdot \varphi \, dy \, ds = 0$. The limit is therefore the pressure term $\int \Pi \nabla \cdot \varphi$.

(v) Drift term with a: part involving V_n. Define

$$Y_n^0(s) := \int_{\mathbb{R}^3} V_n(y,s) \cdot \varphi(y,s) \, dy, \qquad Y^0(s) := \int_{\mathbb{R}^3} U(y,s) \cdot \varphi(y,s) \, dy.$$

For each s, Cauchy–Schwarz gives

$$|Y_n^0(s)| \le \|V_n(s)\|_{L^2(K_\varphi)} \|\varphi(s)\|_{L^2(K_\varphi)}.$$

By Assumption 7.14, there is $C_0 < \infty$ such that $\sup_n \|Y_n^0\|_{L^\infty(0,T)} \le C_0$. Moreover,

$$Y_n^0 \to Y^0 \quad \text{in } L^2(0,T),$$

because $V_n \to U$ in $L^2(0,T; L^2(K_\varphi))$. Now

$$\int_0^T \alpha_n(s) Y_n^0(s) \, ds = \int_0^T (\alpha_n(s) - a_\infty) Y_n^0(s) \, ds + a_\infty \int_0^T Y_n^0(s) \, ds.$$

For the first term,

$$\left| \int_0^T (\alpha_n - a_\infty) Y_n^0 \, ds \right| \le \sup_{s \in (0,T)} |Y_n^0(s)| \, \|\alpha_n - a_\infty\|_{L^1(0,T)} \to 0.$$

For the second term, strong convergence in $L^2(0,T)$ implies $\int_0^T Y_n^0 \to \int_0^T Y^0$, so

$$\int_0^T \alpha_n(s) \int_{\mathbb{R}^3} V_n \cdot \varphi \, dy \, ds \to a_\infty \int_0^T \int_{\mathbb{R}^3} U \cdot \varphi \, dy \, ds.$$

Thus the signed contribution in (12) satisfies

$$-\int_0^T \alpha_n(s) \int_{\mathbb{R}^3} V_n \cdot \varphi \, dy \, ds \to -a_\infty \int_0^T \int_{\mathbb{R}^3} U \cdot \varphi \, dy \, ds.$$

(vi) Drift term with a: part involving $y \cdot \nabla V_n$. Since $\varphi \in C_c^\infty$, integrating by parts in y gives

$$\int_{\mathbb{R}^3} (y \cdot \nabla V_n) \cdot \varphi = -\int_{\mathbb{R}^3} V_n \cdot (y \cdot \nabla \varphi + 3\varphi) \, dy.$$

Define

$$Y_n^1(s) := \int_{\mathbb{R}^3} V_n \cdot (y \cdot \nabla \varphi + 3\varphi) \, dy, \qquad Y^1(s) := \int_{\mathbb{R}^3} U \cdot (y \cdot \nabla \varphi + 3\varphi) \, dy.$$

For each s, Cauchy–Schwarz gives

$$|Y_n^1(s)| \le \|V_n(s)\|_{L^2(K_\varphi)} \|y \cdot \nabla \varphi(s) + 3\varphi(s)\|_{L^2(K_\varphi)}.$$

The $L^\infty(0,T;L^2(K_\varphi))$ bound in Assumption 7.14 gives

$$\sup_n \|Y_n^1\|_{L^\infty(0,T)} \le C_1 < \infty.$$

Also,

$$\|Y_n^1 - Y^1\|_{L^2(0,T)} \le \|V_n - U\|_{L^2(0,T;L^2(K_\varphi))} \|y\cdot\nabla\varphi + 3\varphi\|_{L^\infty(0,T;L^2(K_\varphi))} \to 0.$$

Therefore

$$-\int_0^T \alpha_n(s)\int_{\mathbb{R}^3}(y\cdot\nabla V_n)\cdot\varphi\,dy\,ds \to a_\infty\int_0^T\int_{\mathbb{R}^3} U\cdot(y\cdot\nabla\varphi+3\varphi)\,dy\,ds.$$

Integrating by parts once more in the limit gives

$$a_\infty\int_0^T\int_{\mathbb{R}^3} U\cdot(y\cdot\nabla\varphi+3\varphi)\,dy\,ds = -a_\infty\int_0^T\int_{\mathbb{R}^3}(y\cdot\nabla U)\cdot\varphi\,dy\,ds.$$

(vii) Term with b. Since $\beta_n(s)$ is independent of y,

$$\int_{\mathbb{R}^3}(\beta_n\cdot\nabla V_n)\cdot\varphi = -\int_{\mathbb{R}^3} V_n\cdot(\beta_n\cdot\nabla\varphi)\,dy,$$

so

$$-\int_0^T\int(\beta_n\cdot\nabla V_n)\cdot\varphi\,dy\,ds = \int_0^T\int V_n\cdot(\beta_n\cdot\nabla\varphi)\,dy\,ds.$$

Let

$$Z_n(s) := \left(\int_{\mathbb{R}^3} V_n\cdot\partial_1\varphi\,dy,\ \int_{\mathbb{R}^3} V_n\cdot\partial_2\varphi\,dy,\ \int_{\mathbb{R}^3} V_n\cdot\partial_3\varphi\,dy\right),$$
$$Z(s) := \left(\int_{\mathbb{R}^3} U\cdot\partial_1\varphi\,dy,\ \int_{\mathbb{R}^3} U\cdot\partial_2\varphi\,dy,\ \int_{\mathbb{R}^3} U\cdot\partial_3\varphi\,dy\right).$$

Then

$$\int_{\mathbb{R}^3} V_n\cdot(\beta_n\cdot\nabla\varphi)\,dy = \beta_n(s)\cdot Z_n(s), \qquad \int_{\mathbb{R}^3} U\cdot(b_\infty\cdot\nabla\varphi)\,dy = b_\infty\cdot Z(s).$$

Consequently

$$\left|\int_0^T(\beta_n - b_\infty)\cdot Z_n\,ds\right| \le \|\beta_n - b_\infty\|_{L^1(0,T)}\sup_{s\in(0,T)}|Z_n(s)| \to 0,$$

because

$$|Z_n(s)| \le \|V_n(s)\|_{L^2(K_\varphi)}\left(\sum_{j=1}^3\|\partial_j\varphi(s)\|^2_{L^2(K_\varphi)}\right)^{1/2}$$

and Assumption 7.14 bounds the right-hand side uniformly in n and s. Furthermore,

$$\|Z_n - Z\|_{L^2(0,T;\mathbb{R}^3)} \le \|V_n - U\|_{L^2(0,T;L^2(K_\varphi))}\left(\sum_{j=1}^3\|\partial_j\varphi\|^2_{L^\infty(0,T;L^2(K_\varphi))}\right)^{1/2} \to 0.$$

Therefore

$$\int_0^T b_\infty\cdot Z_n(s)\,ds \to \int_0^T b_\infty\cdot Z(s)\,ds.$$

Hence

$$-\int_0^T\int(\beta_n\cdot\nabla V_n)\cdot\varphi\,dy\,ds \to \int_0^T\int U\cdot(b_\infty\cdot\nabla\varphi)\,dy\,ds.$$

Since b_∞ is constant in y, integration by parts yields

$$\int_0^T \int U \cdot (b_\infty \cdot \nabla\varphi)\, dy\, ds = -\int_0^T \int (b_\infty \cdot \nabla U) \cdot \varphi\, dy\, ds.$$

Step 5: assemble the limit. Passing to the limit in (12) in each term (Steps 4(i)–4(vii)) gives

$$\int_0^T \int \big(-U \cdot \partial_s \varphi - (U \otimes U) : \nabla\varphi + \nu \nabla U : \nabla\varphi - \Pi \nabla \cdot \varphi - a_\infty (U + y \cdot \nabla U) \cdot \varphi - (b_\infty \cdot \nabla U) \cdot \varphi\big)\, dy\, ds = 0.$$

Since this holds for every compactly supported vector field φ, equation (11) holds in distributions on $(0, T) \times \mathbb{R}^3$. ☐

7.6. **Decomposition of alternatives of the scale-collapse branch.** The cost argument in Sections 7.2 and 7.4 excludes the finite-cost scale-collapse branch. The remaining scale-collapse obstruction is treated by a decomposition of alternatives, following the local estimate and autonomous-reduction argument of the preceding sections. The conclusion separates the excluded branches from the residual alternatives; it does not assert a Liouville theorem for every autonomous negative-drift equation.

Theorem 7.21 (Selected core retention)**.** *Let a represented Type II branch be obtained from the selected-core construction of the preceding sections. Then there exist $R > 0$, a cutoff $\chi \in C_c^\infty(B_R)$, $\chi \geq 0$, a constant $\eta > 0$, and a terminal time τ_1 such that, along every selected window used below,*

$$\int_{\mathbb{R}^3} |V(y, \tau)|^2 \chi(y)\, dy \geq \eta \quad \text{for a.e. } \tau \geq \tau_1.$$

In particular, every translated sequence $V_n(y, s) = V(y, \tau_n + s)$ satisfies the localized nontriviality condition in Assumption 7.14, after passing to a tail of the sequence.

Proof. The selected-core construction chooses a compact renormalized core carrying a fixed positive amount of localized kinetic mass. Since the cutoff χ is supported strictly inside that core and equals a positive spatial weight there, the selected-core lower bound gives the stated estimate. Translating $\tau = \tau_n + s$ gives the final assertion on every finite selected window. ☐

Theorem 7.22 (Autonomous modulation selection)**.** *Let a represented scale-collapse branch satisfy the modulation bounds obtained in the repaired gauge construction. Suppose that a sequence of windows $[\tau_n, \tau_n + T]$ is selected in the autonomous regime. Then, after passing to a subsequence, there exist constants $a_\infty \in \mathbb{R}$ and $b_\infty \in \mathbb{R}^3$ such that*

$$a(\tau_n + \cdot) \to a_\infty \quad \text{in } L^1(0, T), \qquad b(\tau_n + \cdot) \to b_\infty \quad \text{in } L^1(0, T; \mathbb{R}^3).$$

Proof. The autonomous-window selection in the repaired-gauge modulation analysis gives convergence of the modulation functions to constant autonomous parameters on the fixed window length T. Passing to the corresponding subsequence gives the displayed L^1-convergences. ☐

Theorem 7.23 (Scale-collapse decomposition of alternatives)**.** *Let a represented Type II branch have genuine scale collapse and nonvanishing selected core. Then exactly one of the following alternatives occurs after passing to selected terminal windows:*

(i) the branch has finite scale-collapsing cost, which is excluded by Theorem 7.7;
(ii) the branch has finite absolute cost, which is excluded by Corollary 7.9;
(iii) the branch admits thick autonomous windows in the sense of Definition 7.11, satisfies the selected-window estimates of Assumption 7.14, and has autonomous modulation convergence;
(iv) the branch has a thin-drift alternative: no selected fixed-length window carries a uniform positive lower bound for the averaged weighted negative drift;

(v) the branch has an autonomous-modulation alternative: thick windows exist, but no subsequence has a constant-coefficient L^1 modulation limit in the sense needed for Theorem 7.19;
(vi) the branch has a compactness alternative: the local estimates needed for Assumption 7.14 fail on the selected windows.

Proof. The scale-collapse selection theorem decomposes terminal scale-collapse windows according to the averaged negative-drift mass

$$\frac{1}{T}\int_{\tau_n}^{\tau_n+T} a_-(\tau)M_R(\tau)\,d\tau.$$

If the accumulated cost is finite, alternatives (i) or (ii) apply according to the cost functional under consideration. If the averaged negative-drift mass stays bounded below on a subsequence, one obtains thick windows; otherwise the branch lies in the thin-drift alternative. On thick windows, either the local estimates and core retention hold, in which case Proposition 7.15 and Theorem 7.21 give Assumption 7.14, or the branch lies in the compactness alternative. If the estimates hold but no subsequence has the required constant-coefficient modulation limit, the branch lies in the autonomous-modulation alternative. The remaining case is the thick autonomous alternative. □

Definition 7.24 (Stationary omega-limit class)**.** Let (U,Π) be a nonzero autonomous limit produced by Theorem 7.19. We say that it has a stationary omega-limit in the admissible class if there exists a nonzero pair

$$(W,Q), \qquad W \in L^3(\mathbb{R}^3),$$

with the local energy and pressure reconstruction properties inherited from the selected windows, such that

$$(W\cdot\nabla)W + \nabla Q = \nu\Delta W + a_\infty(W + y\cdot\nabla W) + b_\infty\cdot\nabla W, \qquad \nabla\cdot W = 0,$$

in $\mathcal{D}'(\mathbb{R}^3)$.

Definition 7.25 (Rigidity-covered stationary class)**.** For a parameter pair (a_∞, b_∞), the stationary class is rigidity-covered if every admissible stationary solution in Definition 7.24 is either zero or is not a terminal Type II branch. In the classical backward self-similar normalization this is the role of the Nečas–Růžička–Šverák rigidity theorem, after the constant translation drift is removed [NRŠ96]. Other parameter regimes require their own stated Liouville or classification theorem.

Theorem 7.26 (Stationary-rigidity exclusion)**.** *Assume a thick autonomous branch produces a nonzero autonomous limit by Theorem 7.19. If that autonomous limit has a stationary omega-limit in the admissible class and the corresponding stationary class is rigidity-covered, then the branch cannot occur as a terminal Type II scale-collapse branch.*

Proof. The stationary omega-limit gives a nonzero stationary profile (W,Q) in the admissible class. The rigidity-covered hypothesis says that no such nonzero profile can represent a terminal Type II branch. This contradicts membership in the terminal scale-collapse Type II class. □

Definition 7.27 (Retained compact scale-collapse state)**.** A represented scale-collapse branch is in the retained compact scale-collapse state if, on the selected terminal windows, it has:

(i) the repaired-gauge representation and local pressure reconstruction of Theorems 4.25 and 4.35;
(ii) the compact-cylinder suitable estimates used in Theorem 7.14;
(iii) thick autonomous selected windows in the sense of Theorem 7.11, together with a subsequential constant-coefficient L^1 modulation limit in the sense used by Theorem 7.19;

(iv) bounded modulation on the selected compact cylinders;
(v) a retained active Caffarelli–Kohn–Nirenberg mass floor on some fixed inner selected cylinder;
(vi) the finite scale-collapsing-cost and finite absolute-cost alternatives have already been removed by Theorems 7.7 and 7.9;
(vii) no unresolved thin-drift, multicore, same-point cascade, separated/exterior, rough-core, autonomous-modulation-failure, or compactness-failure label after the ordered local state-space tests have been applied.

Lemma 7.28 (Failure of retained compact scale-collapse is a named exit)**.** *Let a represented genuine scale-collapse branch with nonvanishing selected core be tested against Theorem 7.27 after the ordered scale-collapse stratification has been applied. If the branch is not in the retained compact scale-collapse state, then its first failed retained-state clause is one of the named local exits: representation/pressure failure, thin-drift, finite scale-cost or finite absolute-cost exclusion, compactness failure, autonomous-modulation failure, no retained active core, multicore/same-point cascade, separated/exterior concentration, or rough-core loss.*

Proof. The clauses of Theorem 7.27 are ordered in the same way as the local tests. Failure of clause (i) is exactly the representation or pressure-reconstruction exit. Failure of (ii) is the compactness alternative in Theorem 7.23. Failure of (iii) is either the thin-drift alternative or the autonomous-modulation alternative in that theorem. Failure of (iv) is a modulation-control exit in the repaired-gauge state-space tests. Failure of (v) means the selected scale-collapse branch carries no retained active Caffarelli–Kohn–Nirenberg core and is therefore not the retained terminal branch. Failure of (vi) is precisely one of the two cost alternatives already excluded by Theorems 7.7 and 7.9. Failure of (vii) is, by definition, one of the listed ordered local state-space exits. Thus no failed retained-state input remains unlabelled. □

Lemma 7.29 (Retained autonomous scale collapse enters the scale-rigid stratum)**.** *Let a represented Type II branch be in the retained compact scale-collapse state of Theorem 7.27. Suppose it has thick autonomous windows and satisfies Theorem 7.14 on the same window sequence. Then the shifted selected-window sequence*

$$V_n(y,s) = V(y,\tau_n+s), \qquad P_n(y,s) = P(y,\tau_n+s),$$

restricted to a fixed compact cylinder carrying the retained active mass, is a scale-rigid local branch in the sense of Theorem 5.23, or it is the compact single-core branch excluded by Theorem 3.18, unless the branch has already entered one of the named local alternatives excluded from the retained compact state.

Proof. The compact-cylinder suitable bounds in Theorem 7.27(ii) give Theorem 5.23(R1) on a fixed compact selected cylinder after shrinking once. The retained active CKN mass floor in (v) gives (R2) on an inner cylinder. Bounded modulation in (iv) gives (R3).

It remains to verify the scale-rigid structural clause (R4). Apply the minimal active-family extraction Theorem 5.11 to the selected retained core and to the nearest previously selected active scale-center, if such a scale-center is present. If no such active relation is present, the branch is the compact single-core state rather than a genuine scale-collapse residual; this is the single-core branch excluded by Theorem 3.18, not a remaining case in this lemma.

Otherwise the pair is an input to the same-point or separated-point reduction, Theorems 5.18 and 5.19. Those reductions are an ordered partition. Their non-rigid outputs are exactly the named exits: another active core at comparable scale, a strict same-point cascade not yet absorbed, a separated physical point, an exterior pressure/concentration label, a rough-core label, or failure of the compact estimates/modulation convergence needed for the autonomous scale-collapse passage. The comparable, same-point, separated, and exterior outcomes are the multicore, same-point cascade, or separated/exterior alternatives of Theorem 5.6; the rough-core outcome is the local windowed-H^1 failure alternative; and the last two outcomes are the compactness or autonomous-modulation alternatives of Theorem 7.23. All are excluded from the retained compact state by Theorem 7.27(vii).

After those exits are removed, the same ordered reductions leave only the structural outputs named in Theorem 5.23(R4): either the relative scale-center Jacobian degenerates, or the selected core is the innermost member after all strict outer same-point states have been absorbed into the perturbative remainder by Theorem 5.14. Thus the shifted sequence is produced by the scale-rigid reduction and satisfies (R4) on the selected compact cylinder. □

Theorem 7.30 (Compact scale-collapse rigidity from local routing)**.** *Let a represented Type II branch have genuine scale collapse, nonvanishing selected core, and belong to the retained compact scale-collapse state of Theorem 7.27. Then no compact scale-collapse branch remains. In particular, the compact scale-collapse stationary-rigidity input is not an independent assumption on the retained local state-space branch.*

Proof. Apply the scale-collapse stratification Theorem 7.23. The finite scale-collapsing cost and finite absolute cost cases are excluded by Theorems 7.7 and 7.9.

The thin-drift, autonomous-modulation, and compactness alternatives are not retained compact scale-collapse states. They are exactly named local alternatives in the ordered stratification and are excluded from the present hypothesis by Theorem 7.27(vii).

It remains to consider the thick autonomous case. By Theorem 7.29, the shifted selected-window sequence is either a scale-rigid local branch or the compact single-core branch, unless the branch has already left the retained compact state. Since the present branch is retained, the exit case is unavailable. If the compact single-core branch occurs, it is excluded by Theorem 3.18. If the scale-rigid branch occurs, Theorem 5.40 applies: the branch either has a nonzero bounded selected-window limit, which is incompatible with the local Type II definition by Theorem 5.25, or it reduces to the compact single-core branch, again excluded by Theorem 3.18. Hence the thick autonomous retained case is impossible.

The stationary omega-limit and stationary-rigidity theorem remain valid shortcuts when available, by Theorem 7.26, but they are not needed for the retained local state-space branch: a nonstationary autonomous obstruction is already visible as a scale-rigid local branch and is removed by the scale-rigidity discharge. □

Theorem 7.31 (Scale-collapse alternatives)**.** *Let a represented Type II branch have genuine scale collapse and a nonvanishing selected core. Then one of the following mutually exclusive outcomes occurs:*

(i) the finite scale-collapsing cost exclusion applies;
(ii) the finite absolute cost exclusion applies;
(iii) the branch has a thin-drift alternative;
(iv) the branch has an autonomous-modulation alternative;
(v) the branch has a compactness alternative;
(vi) the branch enters the retained compact scale-collapse state, in which case Theorem 7.30 excludes it.

Proof. Apply Theorem 7.23. The finite-cost alternatives are items (i) and (ii). The thin-drift, modulation, and compactness alternatives are items (iv)–(vi). It remains only to treat item (iii), the thick autonomous branch with selected-window estimates and autonomous modulation convergence. If one of the retained compact-state inputs in Theorem 7.27 fails, then Theorem 7.28 assigns the branch to a named local alternative, hence to one of the preceding nonretained outcomes. Otherwise the branch is in the retained compact scale-collapse state and Theorem 7.30 excludes it. □

8. Critical Tightness and Windowed-Control Estimates

This section records the local estimates used to exclude failure of uniform critical tightness and failure of local windowed H^1 control. The statements are phrased as analytic alternatives for a renormalized Navier–Stokes solution $V(\tau)$.

8.1. **Uniform critical tightness.**

Definition 8.1 (Uniform critical tightness)**.** A renormalized orbit $V(\tau)$, $\tau \geq \tau_0$, is uniformly critically tight if

$$\forall \varepsilon > 0\ \exists R_\varepsilon < \infty : \sup_{\tau \geq \tau_0} \int_{|y|>R_\varepsilon} |V(y,\tau)|^3 \, dy < \varepsilon.$$

The non-tight alternative is the failure of this property.

Lemma 8.2 (Critical precompactness implies uniform tightness)**.** *If*

$$\mathcal{O} := \{V(\tau) : \tau \geq \tau_0\}$$

is contained in a compact subset of $L^3(\mathbb{R}^3)$, then V is uniformly critically tight.

Proof. Compact subsets of $L^3(\mathbb{R}^3)$ are uniformly tight. Let $K \subset L^3$ be compact and fix $\varepsilon > 0$. Set $\delta = \varepsilon^{1/3}/2$. Choose a finite δ-net $f_1, \dots, f_N$ in L^3. For each i, choose R_i so that

$$\|f_i\|_{L^3(|y|>R_i)} < \delta.$$

Let $R = \max_i R_i$. For any $f \in K$, choose i with $\|f - f_i\|_3 < \delta$. Then

$$\|f\|_{L^3(|y|>R)} \leq \|f - f_i\|_3 + \|f_i\|_{L^3(|y|>R)} < \varepsilon^{1/3}.$$

Taking the cube gives

$$\int_{|y|>R} |f(y)|^3 \, dy < \varepsilon.$$

Applying this to the compact set containing $\mathcal{O}$ gives uniform critical tightness. □

Definition 8.3 (Compact core with vanishing critical remainder)**.** The orbit has a compact critical core with vanishing remainder if there exist a compact set $\mathcal{K}^{\mathrm{c}} \subset L^3(\mathbb{R}^3)$, a map $Q : [\tau_0, \infty) \to \mathcal{K}^{\mathrm{c}}$, and a remainder $r(\tau) \in L^3(\mathbb{R}^3)$, such that

$$V(\tau) = Q(\tau) + r(\tau), \qquad \lim_{\tau\to\infty} \|r(\tau)\|_{L^3} = 0,$$

and, for every $T > \tau_0$, the set

$$\{V(\tau) : \tau \in [\tau_0, T]\}$$

is uniformly L^3-tight.

Lemma 8.4 (Compact core plus vanishing remainder implies tightness)**.** *If Theorem 8.3 holds, then V is uniformly critically tight.*

Proof. Fix $\varepsilon > 0$ and set $\delta = \varepsilon^{1/3}$. By compactness of $\mathcal{K}^{\mathrm{c}}$, apply Theorem 8.2 to $\mathcal{K}^{\mathrm{c}}$ and choose R_1 such that

$$\sup_{q\in\mathcal{K}^{\mathrm{c}}} \|q\|_{L^3(|y|>R_1)} < \delta/3.$$

Since $\|r(\tau)\|_3 \to 0$, choose T so that

$$\|r(\tau)\|_3 < \delta/3 \qquad \text{for all } \tau \geq T.$$

Then for $\tau \geq T$,

$$\|V(\tau)\|_{L^3(|y|>R_1)} \leq \|Q(\tau)\|_{L^3(|y|>R_1)} + \|r(\tau)\|_3 < 2\delta/3.$$

By finite-initial-window tightness on $[\tau_0, T]$, choose R_2 such that

$$\sup_{\tau\in[\tau_0,T]} \int_{|y|>R_2} |V(y,\tau)|^3 \, dy < \varepsilon.$$

With $R = \max(R_1, R_2)$, the L^3-norm of the tail of $V(\tau)$ is $< \delta$ for all $\tau \geq T$, while the tail integral is $< \varepsilon$ on $[\tau_0, T]$. Cubing on the tail $\tau \geq T$ gives the required L^3-tail integral $< \varepsilon$ for all $\tau \geq \tau_0$. □

Definition 8.5 (No-splitting profile condition)**.** A critical profile decomposition has no critical-mass splitting if it has:

(i) a primary compact core family $\mathcal{K}^{\mathrm{c}} \subset L^3$;
(ii) asymptotic L^3-decoupling of core, secondary profiles, and remainder;
(iii) single-profile saturation, so every non-primary profile has zero critical L^3-mass;
(iv) a remainder whose L^3-norm vanishes on the renormalized tail;
(v) finite-initial-window tightness.

Lemma 8.6 (No splitting gives compact core with vanishing remainder)**.** *If Theorem 8.5 holds, then Theorem 8.3 holds.*

Proof. By single-profile saturation and nonnegative L^3-mass decoupling, every secondary profile has zero critical L^3-mass and therefore vanishes in L^3. The profile decomposition then reduces to a primary compact core plus a remainder whose L^3-norm tends to zero on the renormalized tail. These provide the core $Q(\tau)$, compact set $\mathcal{K}^{\mathrm{c}}$, and remainder $r(\tau)$ required by Theorem 8.3, with finite-initial-window tightness supplied as part of Theorem 8.5. □

Corollary 8.7 (No splitting gives uniform critical tightness)**.** *If Theorem 8.5 holds, then V is uniformly critically tight.*

Proof. Theorem 8.6 gives a compact core with vanishing critical remainder, and Theorem 8.4 gives uniform critical tightness. □

Definition 8.8 (Finite critical profile approximation)**.** The orbit admits a finite critical profile approximation if there are finitely many profiles

$$\mathcal{B}_{NS}^{II} := \{B_1, \dots, B_N\} \subset L^3(\mathbb{R}^3)$$

such that, for every $\delta > 0$, there is τ_δ with

$$\forall \tau \geq \tau_\delta \; \exists i(\tau) \in \{1, \dots, N\} : \|V(\tau) - B_{i(\tau)}\|_{L^3} < \delta,$$

and the finite initial window

$$\{V(\tau) : \tau \in [\tau_0, \tau_\delta]\}$$

is uniformly L^3-tight for each $\delta > 0$.

Lemma 8.9 (Finite critical profile approximation implies tightness)**.** *If Theorem 8.8 holds, then V is uniformly critically tight.*

Proof. Fix $\varepsilon > 0$ and set $\delta = \varepsilon^{1/3}$. Since the profile family is finite and each $B_i \in L^3(\mathbb{R}^3)$, there is R_1 such that

$$\max_{1\leq i\leq N} \|B_i\|_{L^3(|y|>R_1)} < \delta/3.$$

Choose the approximation tolerance $\delta/3$. For $\tau \geq \tau_{\delta/3}$, pick $i(\tau)$ with $\|V(\tau) - B_{i(\tau)}\|_3 < \delta/3$. Then

$$\|V(\tau)\|_{L^3(|y|>R_1)} \leq \|V(\tau) - B_{i(\tau)}\|_3 + \|B_{i(\tau)}\|_{L^3(|y|>R_1)} < 2\delta/3.$$

By finite-initial-window tightness, choose R_2 so that the L^3 tail is $< \varepsilon$ on $[\tau_0, \tau_{\delta/3}]$. With $R = \max(R_1, R_2)$, the tail integral is $< \varepsilon$ for every $\tau \geq \tau_0$, so the orbit is uniformly L^3-tight. □

Definition 8.10 (Compact parameter realization)**.** The orbit has a compact parameter realization if there exist a compact parameter set Θ_{II}, a continuous map

$$\Theta_{II} \ni \theta \mapsto V_\theta \in L^3(\mathbb{R}^3),$$

a curve $\theta(\tau) \in \Theta_{II}$, and a remainder $r(\tau) \to 0$ in L^3, such that

$$V(\tau) = V_{\theta(\tau)} + r(\tau),$$

and, for every $T > \tau_0$, the set

$$\{V(\tau) : \tau \in [\tau_0, T]\}$$

is uniformly L^3-tight.

Lemma 8.11 (Compact realization implies tightness)**.** *If Theorem 8.10 holds, then V is uniformly critically tight.*

Proof. The continuous image of compact Θ_{II} in L^3 is compact. Thus

$$\mathcal{K}^c := \{V_\theta : \theta \in \Theta_{II}\}$$

is compact in L^3. The representation $V(\tau) = V_{\theta(\tau)} + r(\tau)$ with $r(\tau) \to 0$ in L^3 gives the compact-core vanishing-remainder situation, because finite-initial-window tightness is included in Theorem 8.10. Theorem 8.4 gives uniform critical tightness. □

Theorem 8.12 (Exclusion of the non-tight alternative)**.** *For a renormalized Type II Navier–Stokes solution, any one of the following conditions gives uniform critical tightness:*

(i) $\mathcal{O} \Subset L^3(\mathbb{R}^3)$*;*
(ii) *compact core with vanishing critical remainder;*
(iii) *no-splitting profile condition;*
(iv) *finite critical profile approximation;*
(v) *compact parameter realization.*

Proof. The five alternatives are handled respectively by Theorems 8.2, 8.4, 8.7, 8.9 and 8.11. In every case V is uniformly critically tight, so the failure of uniform critical tightness is unavailable. □

Corollary 8.13 (After exclusion of the non-tight alternative)**.** *Assume a finite-cost renormalized Type II solution not excluded by the compact cost-divergence criterion has a two-alternative classification whose only remaining alternatives are failure of uniform critical tightness and failure of tail windowed H^1 control. If Theorem 8.1 holds, then the only remaining alternative is failure of tail windowed H^1 control.*

Proof. The two-alternative classification gives exactly one of the two alternatives. Uniform critical tightness rules out failure of tightness, so only the windowed H^1-failure alternative remains. □

8.2. **Tail windowed H^1 control.**

Lemma 8.14 (Uniform local pressure control)**.** *Assume*

$$M_{3,R} := \sup_{\tau\geq\tau_0} \|V(\tau)\|_{L^3(B_{4R})} < \infty$$

and

$$-\Delta P = \partial_i\partial_j(V_iV_j) \quad \text{in } \mathbb{R}^3$$

for each $\tau \geq \tau_0$. Assume also the local tail estimate

$$\mathcal{T}_R(\tau) := \sup_{x\in B_R} \int_{|z|>2R} \frac{|V(z,\tau)|^2}{|x-z|^5}\, dz, \qquad \sup_{\tau\geq\tau_0} \mathcal{T}_R(\tau) \leq CR^{-4}M_{3,R}^2.$$

Then there is a measurable choice of constants $c_R(\tau) \in \mathbb{R}$ such that

$$\sup_{\tau \geq \tau_0} \|P(\tau) - c_R(\tau)\|_{L^{3/2}(B_R)} \leq C_R M_{3,R}^2.$$

Proof. The local pressure decomposition gives

$$\|P(\tau) - c_R(\tau)\|_{L^{3/2}(B_R)} \leq C \left(\|V(\tau)\|_{L^3(B_{4R})}^2 + R^2 \mathcal{T}_R(\tau) \right).$$

The tail hypothesis gives

$$\sup_{\tau \geq \tau_0} \mathcal{T}_R(\tau) \leq C R^{-4} M_{3,R}^2,$$

and the local critical bound gives

$$\|V(\tau)\|_{L^3(B_{4R})} \leq M_{3,R}.$$

Combining the two estimates yields

$$\sup_{\tau \geq \tau_0} \|P(\tau) - c_R(\tau)\|_{L^{3/2}(B_R)} \leq C_R M_{3,R}^2.$$

The constants $c_R(\tau)$ can be chosen by fixing the displayed pressure decomposition, hence measurably. □

Proposition 8.15 (Renormalized Caccioppoli gives uniform windowed gradients)**.** *Assume V, P are smooth on compact renormalized cylinders and solve*

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0.$$

Assume local compact-cylinder critical control on the outer cylinders used in the estimate, bounded modulation

$$M_{ab} := \sup_{\tau \geq \tau_0} (|a(\tau)| + |b(\tau)|) < \infty,$$

and pressure normalized as in Theorem 8.14. Then for every $R > 0$ there exists

$$C_R = C(R, \nu, M_{3,R}, M_{ab})$$

such that

$$\sup_{T \geq \tau_0 + 1} \int_T^{T+1} \int_{B_R} |\nabla V(y, \tau)|^2 \, dy \, d\tau \leq C_R.$$

Proof. Fix $R > 0$ and $T \geq \tau_0 + 1$. Use the renormalized Caccioppoli estimate with outer radius $2R$, inner radius R, and time intervals

$$I_{2R} := (T-1, T+2), \qquad I_R := [T, T+1].$$

Thus $R_{\text{out}} - R_{\text{in}} = R$ and the inner time length is 1. Keeping the pressure term before taking absolute values gives

$$\nu \int_T^{T+1} \int_{B_R} |\nabla V|^2 \leq C_R \iint_{(T-1,T+2) \times B_{2R}} |V|^2 + C_R \iint_{(T-1,T+2) \times B_{2R}} |V|^3 + C_R \mathcal{P}_{T,R} + C_R M_{ab} \iint_{(T-1,T+2) \times B_{2R}} |V|^2,$$

where C_R absorbs the powers of R^{-1}, R^{-2}, and ν, and

$$\mathcal{P}_{T,R} := \left| \iint_{(T-1,T+2) \times B_{2R}} P \, V \cdot \nabla \zeta \right|.$$

The L^2-term is controlled by the local critical bound and boundedness of B_{2R}:

$$\int_{B_{2R}} |V(\tau)|^2\,dy \le |B_{2R}|^{1/3}\|V(\tau)\|^2_{L^3(B_{2R})} \le C_R M_{3,R}^2.$$

Since the outer time interval has length 3,

$$\iint_{(T-1,T+2)\times B_{2R}} |V|^2 \le C_R M_{3,R}^2,$$

and similarly

$$\iint_{(T-1,T+2)\times B_{2R}} |V|^3 \le 3M_{3,R}^3.$$

For any time-dependent constant $c(\tau)$,

$$\int_{B_{2R}} c(\tau)V\cdot\nabla\zeta\,dy = -\int_{B_{2R}} c(\tau)\zeta\,\nabla\cdot V\,dy = 0.$$

Thus P may be replaced by $P - c_{2R}(\tau)$. Holder's inequality gives

$$\mathcal{P}_{T,R} \le C_R \iint_{(T-1,T+2)\times B_{2R}} |P - c_{2R}(\tau)|\,|V|\,dy\,d\tau.$$

For almost every τ,

$$\int_{B_{2R}} |P - c_{2R}(\tau)|\,|V(\tau)|\,dy \le \|P(\tau) - c_{2R}(\tau)\|_{L^{3/2}(B_{2R})}\|V(\tau)\|_{L^3(B_{2R})}.$$

The pressure estimate with radius $2R$ gives

$$\|P(\tau) - c_{2R}(\tau)\|_{L^{3/2}(B_{2R})} \le C_R M_{3,R}^2,$$

and the local critical bound gives

$$\|V(\tau)\|_{L^3(B_{2R})} \le M_{3,R}.$$

Hence $\mathcal{P}_{T,R} \le C_R M_{3,R}^3$. Substituting these estimates into Caccioppoli yields

$$\nu\int_T^{T+1}\int_{B_R} |\nabla V|^2 \le C_R\left(M_{3,R}^2 + M_{3,R}^3 + M_{ab}M_{3,R}^2\right),$$

uniformly in $T \ge \tau_0 + 1$. Dividing by ν and renaming the constant proves the claim. □

Corollary 8.16 (Tail windowed local H^1 bound)**.** *Under the hypotheses of Theorem 8.15, for every $R > 0$,*

$$\sup_{T\ge\tau_0+1}\int_T^{T+1} \|V(\tau)\|^2_{H^1(B_R)}\,d\tau < \infty.$$

Proof. The gradient part is Theorem 8.15. The L^2-part follows from the same bounded-set Holder estimate:

$$\int_T^{T+1} \|V(\tau)\|^2_{L^2(B_R)}\,d\tau \le C_R\int_T^{T+1} \|V(\tau)\|^2_{L^3(B_{2R})}\,d\tau \le C_R M_{3,R}^2.$$

Adding the two bounds gives the asserted $H^1(B_R)$ estimate. □

Corollary 8.17 (Exclusion of failure of local windowed H^1 control)**.** *Assume the renormalized Type II solution satisfies the repaired-gauge renormalized Navier–Stokes equation on compact cylinders, local compact-cylinder critical control*

$$\sup_{\tau\geq\tau_0}\|V(\tau)\|_{L^3(B_{2R})}<\infty$$

for each relevant R, bounded modulation

$$\sup_{\tau\geq\tau_0}(|a(\tau)|+|b(\tau)|)<\infty,$$

and pressure reconstruction

$$-\Delta P=\partial_i\partial_j(V_iV_j).$$

Then failure of local windowed H^1 control,

$$\exists m\geq 1:\sup_{T\geq\tau_0+1}\int_T^{T+1}\|V(\tau)\|^2_{H^1(B_m)}\,d\tau=\infty$$

is impossible.

Proof. Apply Theorem 8.16 with $R=m$. The resulting finite bound contradicts the displayed failure of local windowed H^1 control. □

8.3. **Modulation and weighted forcing inputs.**

Lemma 8.18 (Positive finite critical annulus implies bounded critical norm)**.** *If there exist $0<\eta\leq M<\infty$ such that*

$$\eta\leq\|V(\tau)\|_{L^3(\mathbb{R}^3)}\leq M\qquad\forall\tau\geq\tau_0,$$

then

$$\sup_{\tau\geq\tau_0}\|V(\tau)\|_{L^3(\mathbb{R}^3)}\leq M<\infty.$$

Proof. This is obtained by taking the supremum in τ in the displayed upper bound. □

Definition 8.19 (Modulation matrix Schur data)**.** Write the repaired modulation matrix in block form

$$M(V)=\begin{pmatrix}\alpha(V) & u(V)^T\\ v(V) & A(V)\end{pmatrix}.$$

Assume

$$\alpha(V(\tau))\geq\alpha_0>0,\qquad\|A(V(\tau))^{-1}\|_{\infty\to\infty}\leq C_A,$$
$$\|u(V(\tau))\|_\infty\leq C_u',\qquad\|v(V(\tau))\|_\infty\leq C_v',$$

and

$$S(V(\tau)):=\alpha(V(\tau))-u(V(\tau))^TA(V(\tau))^{-1}v(V(\tau))\geq\sigma_0>0.$$

Lemma 8.20 (Uniform inverse from Schur data)**.** *Under Theorem 8.19,*

$$\sup_{\tau\geq\tau_0}\|M(V(\tau))^{-1}\|_{\infty\to\infty}\leq\sigma_0^{-1}+3\sigma_0^{-1}C_u'C_A+\sigma_0^{-1}C_AC_v'+C_A+3C_A^2C_v'C_u'\sigma_0^{-1}.$$

Proof. For each τ, write

$$M(V(\tau))=\begin{pmatrix}\alpha & u^T\\ v & A\end{pmatrix},\qquad S=\alpha-u^TA^{-1}v.$$

By the translation-block inverse bound, A^{-1} exists and is uniformly bounded. By the Schur gap, S^{-1} exists and $|S^{-1}|\leq\sigma_0^{-1}$. The block inverse formula gives

$$M^{-1}=\begin{pmatrix}S^{-1} & -S^{-1}u^TA^{-1}\\ -A^{-1}vS^{-1} & A^{-1}+A^{-1}vS^{-1}u^TA^{-1}\end{pmatrix}.$$

Using the ℓ^∞ matrix norm and the finite dimension of the mixed blocks,

$$\|S^{-1}u^T A^{-1}\|_{\infty\to\infty} \le 3\sigma_0^{-1} C_u' C_A,$$

$$\|A^{-1}vS^{-1}\|_{\infty\to\infty} \le \sigma_0^{-1} C_A C_v',$$

and

$$\|A^{-1}vS^{-1}u^T A^{-1}\|_{\infty\to\infty} \le 3C_A^2 C_v' C_u' \sigma_0^{-1}.$$

Adding the four block estimates gives the displayed uniform bound. □

Lemma 8.21 (Scale-normalization nondegeneracy gives Schur data)**.** *Assume the scale normalization constraint is satisfied, the scale derivative obeys*

$$\alpha(V) = DG_{\rm sc}(V)[Z_{\rm sc}(V)] = p\Theta_0 > 0,$$

the translation block has a uniform inverse bound, and the mixed blocks obey the weighted first-moment bound and the Schur-compatible estimate

$$\alpha(V(\tau)) - u(V(\tau))^T A(V(\tau))^{-1} v(V(\tau)) \ge \frac{\alpha_0}{2}.$$

Then the hypotheses of Theorem 8.19 hold with $\sigma_0 = \alpha_0/2$.

Proof. The scale-normalization identity gives $\alpha(V) = DG_{\rm sc}(V)[Z_{\rm sc}(V)] = p\Theta_0$, hence the scale transversality bound. The translation-block inverse hypothesis gives the uniform bound on A^{-1}. The mixed-block hypotheses give the bounds on u and v, and the displayed estimate is the Schur gap with $\sigma_0 = \alpha_0/2$. □

Definition 8.22 (Weighted scale-force components)**.** Let $0 < p < 3$ and set

$$H(V,P) = \nu\Delta V - (V\cdot\nabla)V - \nabla P.$$

Define

$$I_\Delta(\tau) := \int_{\mathbb{R}^3} |y|^{-p}|V|V\cdot\Delta V\,dy,$$

$$I_{\rm nl}(\tau) := \int_{\mathbb{R}^3} |y|^{-p}|V|V\cdot((V\cdot\nabla)V)\,dy,$$

and

$$I_P(\tau) := \int_{\mathbb{R}^3} |y|^{-p}|V|V\cdot\nabla P\,dy,$$

whenever these integrals are absolutely convergent or are defined by the distributional pairing used for the renormalized solution.

Lemma 8.23 (Scale-force decomposition)**.** *If*

$$\sup_\tau |I_\Delta(\tau)| \le C_\Delta, \qquad \sup_\tau |I_{\rm nl}(\tau)| \le C_{\rm nl}, \qquad \sup_\tau |I_P(\tau)| \le C_P,$$

then

$$\sup_{\tau\ge\tau_0} \left| \int |y|^{-p}|V|V\cdot H(V,P)\,dy \right| \le \nu C_\Delta + C_{\rm nl} + C_P.$$

Proof. By the definition of H,

$$\int |y|^{-p}|V|V\cdot H(V,P)\,dy = \nu I_\Delta - I_{\rm nl} - I_P.$$

The triangle inequality gives the displayed uniform bound. □

Lemma 8.24 (Weighted fourth moment controls the nonlinear scale component)**.** *Assume $V(\tau)$ is divergence-free and belongs to the approximation class needed to justify the weighted integration by parts. If*

$$\sup_{\tau\geq\tau_0}\int_{\mathbb{R}^3}|y|^{-p-1}|V(y,\tau)|^4\,dy<\infty,$$

then

$$|I_{\mathrm{nl}}(\tau)|\leq\frac{p}{3}\int_{\mathbb{R}^3}|y|^{-p-1}|V(y,\tau)|^4\,dy.$$

Proof. For smooth compactly supported approximants,

$$|V|V\cdot((V\cdot\nabla)V)=\frac{1}{3}V\cdot\nabla(|V|^3).$$

Since $\nabla\cdot V=0$, integration by parts gives

$$I_{\mathrm{nl}}=\frac{1}{3}\int|y|^{-p}V\cdot\nabla(|V|^3)\,dy=-\frac{1}{3}\int|V|^3V\cdot\nabla(|y|^{-p})\,dy.$$

Because

$$\nabla(|y|^{-p})=-p|y|^{-p-2}y,$$

we obtain

$$I_{\mathrm{nl}}=\frac{p}{3}\int|y|^{-p-2}(V\cdot y)|V|^3\,dy.$$

Therefore

$$|I_{\mathrm{nl}}|\leq\frac{p}{3}\int|y|^{-p-1}|V|^4\,dy.$$

Taking the supremum in τ proves the assertion. The general admissible case follows by the same approximation convention used for the weighted scale normalization. ☐

Corollary 8.25 (Pointwise scale-force bound)**.** *If*

$$\sup_{\tau}|I_\Delta(\tau)|<\infty,\qquad\sup_{\tau\geq\tau_0}\int|y|^{-p-1}|V(y,\tau)|^4\,dy<\infty,\qquad\sup_{\tau}|I_P(\tau)|<\infty,$$

then

$$\sup_{\tau\geq\tau_0}\left|\int|y|^{-p}|V|V\cdot H(V,P)\,dy\right|<\infty.$$

Proof. Theorem 8.24 bounds I_{nl}. Together with the Laplacian and pressure bounds, this gives the hypotheses of Theorem 8.23. ☐

Lemma 8.26 (Bounded modulation from the modulation system)**.** *Assume*

$$M(V(\tau))\begin{pmatrix}a(\tau)\\ b(\tau)\end{pmatrix}=-F(V(\tau)),$$

$$\sup_{\tau\geq\tau_0}\|M(V(\tau))^{-1}\|_{\infty\to\infty}<\infty,\qquad\sup_{\tau\geq\tau_0}\|F(V(\tau))\|_\infty<\infty.$$

Then

$$\sup_{\tau\geq\tau_0}(|a(\tau)|+\|b(\tau)\|_\infty)<\infty.$$

Proof. The system gives

$$\begin{pmatrix}a(\tau)\\ b(\tau)\end{pmatrix}=-M(V(\tau))^{-1}F(V(\tau)),$$

so

$$\max\{|a(\tau)|,\|b(\tau)\|_\infty\}\leq\|M(V(\tau))^{-1}\|_{\infty\to\infty}\|F(V(\tau))\|_\infty.$$

Taking the supremum in τ gives bounded modulation. ☐

Proposition 8.27 (Windowed H^1 exclusion from modulation bounds)**.** *Assume a renormalized Type II solution has the repaired-gauge orbit and pressure reconstruction, bounded critical L^3 norm, uniformly invertible modulation matrix, uniformly bounded modulation forcing vector, and the compact-cylinder regularity needed for Caccioppoli. Then*

$$\forall R > 0 : \qquad \sup_{T \geq \tau_0+1} \int_T^{T+1} \|V(\tau)\|^2_{H^1(B_R)} \, d\tau < \infty,$$

and failure of local windowed H^1 control is excluded for this solution.

Proof. The inverse matrix and forcing bounds imply bounded modulation by Theorem 8.26. The renormalized solution supplies the renormalized equation and pressure reconstruction. Bounded critical norm gives the local compact-cylinder critical bounds, and the regularity hypothesis allows the Caccioppoli estimate. Therefore Theorem 8.16 applies. Failure of local windowed H^1 control is the negation of the displayed tail windowed H^1 estimate on some bounded ball. □

Corollary 8.28 (Two-alternative exclusion after windowed control)**.** *Assume a finite-cost renormalized Type II solution not excluded by the compact cost-divergence criterion has reached the two-alternative classification. If the hypotheses of Theorem 8.27 are supplied for that solution, then failure of local windowed H^1 control is excluded. If, in addition, uniform critical tightness is available, then no finite-cost alternative remains outside the compact cost-divergence criterion.*

Proof. Theorem 8.27 gives the positive tail windowed H^1 bound, contradicting failure of local windowed H^1 control. If uniform critical tightness is also available, then failure of tightness is excluded as well, and the two-alternative classification has no remaining finite-cost alternative outside the compact cost-divergence criterion. □

Theorem 8.29 (Exclusion of local windowed H^1 failure)**.** *If every renormalized Type II solution in the class satisfies the hypotheses of Theorem 8.27, then no such solution can satisfy*

$$\exists m \geq 1 : \sup_{T \geq \tau_0+1} \int_T^{T+1} \|V(\tau)\|^2_{H^1(B_m)} \, d\tau = \infty.$$

Proof. Apply Theorem 8.27 to each renormalized solution. The displayed condition is the negation of the resulting windowed H^1-bound for $R = m$. □

8.4. **Window-Integrated Modulation Forcing and Weighted Scale Components.** This subsection records sufficient analytic estimates for the modulation forcing. The estimates are stated directly for the differentiated modulation conditions and for the weighted scale component; no pointwise modulation bound is used unless it is explicitly assumed.

Let the modulation forcing operator be

$$H(V, P) := \nu \Delta V - (V \cdot \nabla)V - \nabla P.$$

For gauge functionals G_k, $0 \leq k \leq 3$, define

$$F_k(V) := DG_k(V)[H(V, P)].$$

Thus the modulation system has the form

$$M(V(\tau)) \begin{pmatrix} a(\tau) \\ b(\tau) \end{pmatrix} = -F(V(\tau)).$$

Lemma 8.30 (Forcing bound for the translation constraints)**.** *Let*

$$G_j(V) = \int_{\mathbb{R}^3} y_j |V(y)|^2 \psi_R(y) \, dy, \qquad 1 \leq j \leq 3,$$

where $\psi_R \in C_c^\infty(B_{2R})$, and set

$$\Phi_j(y) := y_j \psi_R(y).$$

For

$$F_j(V) := DG_j(V)[H(V,P)] = 2\int_{\mathbb{R}^3} \Phi_j\, V \cdot H(V,P)\, dy,$$

assume, uniformly for $\tau \geq \tau_0$,

$$\|V(\tau)\|_{L^3(B_{2R})} \leq M_3, \qquad \|\nabla V(\tau)\|_{L^2(B_{2R})} \leq M_\nabla,$$

and

$$\inf_{c\in\mathbb{R}} \|P(\tau) - c\|_{L^{3/2}(B_{2R})} \leq M_P.$$

Then

$$\sup_{\tau\geq\tau_0} \max_{1\leq j\leq 3} |F_j(V(\tau))| \leq C(R,\nu,\psi_R)\left(M_\nabla^2 + M_3^2 + M_3^3 + M_P M_3\right).$$

Proof. Fix j. Since Φ_j is supported in B_{2R},

$$F_j = 2\nu I_\Delta - 2I_{\mathrm{nl}} - 2I_P,$$

where

$$I_\Delta := \int \Phi_j V \cdot \Delta V, \qquad I_{\mathrm{nl}} := \int \Phi_j V \cdot (V\cdot\nabla V), \qquad I_P := \int \Phi_j V \cdot \nabla P.$$

For the Laplacian term, using compact support of Φ_j and summing over the component index α,

$$\begin{aligned} I_\Delta &= \sum_{\alpha=1}^{3} \int \Phi_j V_\alpha \Delta V_\alpha\, dy \\ &= -\sum_{\alpha,k} \int \partial_k(\Phi_j V_\alpha)\partial_k V_\alpha\, dy \\ &= -\int \Phi_j |\nabla V|^2\, dy - \sum_{\alpha,k}\int (\partial_k \Phi_j) V_\alpha \partial_k V_\alpha\, dy. \end{aligned}$$

Equivalently,

$$I_\Delta = -\int \Phi_j |\nabla V|^2 - \int (\nabla\Phi_j \cdot \nabla V)\cdot V.$$

Consequently, by the Cauchy–Schwarz inequality,

$$|I_\Delta| \leq \|\Phi_j\|_{L^\infty} \|\nabla V\|^2_{L^2(B_{2R})} + \|\nabla \Phi_j\|_{L^\infty} \|V\|_{L^2(B_{2R})} \|\nabla V\|_{L^2(B_{2R})}.$$

Hölder's inequality on the bounded set B_{2R} gives

$$\|V\|_{L^2(B_{2R})} \leq |B_{2R}|^{1/6} \|V\|_{L^3(B_{2R})} \leq C_R M_3.$$

Therefore

$$|I_\Delta| \leq C_R(M_\nabla^2 + M_3 M_\nabla) \leq C_R(M_\nabla^2 + M_3^2).$$

The last inequality is Young's inequality, $ab \leq (a^2+b^2)/2$.

For the nonlinear term,

$$V\cdot(V\cdot\nabla V) = V_i V_\alpha \partial_i V_\alpha = \frac{1}{2} V_i \partial_i(|V|^2) = \frac{1}{2} V\cdot\nabla(|V|^2).$$

Since $\nabla\cdot V = 0$ and Φ_j is compactly supported,

$$I_{\mathrm{nl}} = \frac{1}{2}\int \Phi_j V\cdot\nabla(|V|^2) = -\frac{1}{2}\int |V|^2 V\cdot\nabla\Phi_j.$$

Thus

$$|I_{\mathrm{nl}}| \le \frac{1}{2}\|\nabla\Phi_j\|_{L^\infty}\|V\|^3_{L^3(B_{2R})} \le C_R M_3^3.$$

For the pressure term, replace P by $P - c(\tau)$. Since $\nabla \cdot V = 0$, the identity

$$\nabla \cdot (\Phi_j V) = V \cdot \nabla \Phi_j$$

holds distributionally. Therefore

$$I_P = \int \Phi_j V \cdot \nabla(P - c) = -\int (P-c) V \cdot \nabla\Phi_j.$$

Consequently

$$|I_P| \le \|\nabla\Phi_j\|_{L^\infty}\|P - c\|_{L^{3/2}(B_{2R})}\|V\|_{L^3(B_{2R})} \le C_R M_P M_3.$$

Taking the infimum over c, summing the three estimates in F_j, and then taking the supremum in τ proves the bound. □

Remark 8.31 (Independence of the local gradient hypothesis). The pointwise local gradient bound in Theorem 8.30 is an additional analytic input. When the translation estimate is used to prove local windowed H^1-control, this input is required to come from an independent Caccioppoli, regularity, or approximation estimate, not from the conclusion being proved.

Definition 8.32 (Weighted scale forcing)**.** Let

$$G_{\mathrm{sc}}(V) = \int_{\mathbb{R}^3} |y|^{-p}|V|^3\,dy - \Theta_0, \qquad 0 < p < 3.$$

The weighted scale component is

$$F_0(V) := DG_{\mathrm{sc}}(V)[H(V,P)] = 3\int_{\mathbb{R}^3} |y|^{-p}|V|V \cdot H(V,P)\,dy.$$

All weighted pairings in this subsection are assumed to be defined either as absolutely convergent integrals or as specified distributional pairings represented by $L^1_{\mathrm{loc}}(d\tau)$ functions.

Lemma 8.33 (Weighted scale forcing bound)**.** *Assume*

$$\sup_{\tau\ge\tau_0}\left|\int_{\mathbb{R}^3} |y|^{-p}|V|V\cdot H(V,P)\,dy\right| < \infty.$$

Then

$$\sup_{\tau\ge\tau_0}|F_0(V(\tau))| < \infty.$$

Proof. By the derivative formula in Theorem 8.32,

$$|F_0(V(\tau))| \le 3\left|\int_{\mathbb{R}^3} |y|^{-p}|V|V\cdot H(V,P)\,dy\right|.$$

Taking the supremum in τ gives the result. □

Corollary 8.34 (Pointwise modulation forcing bound)**.** *Assume the hypotheses of Theorem 8.30 for the three translation components and the weighted scale forcing hypothesis of Theorem 8.33. Then*

$$\sup_{\tau\ge\tau_0}\|F(V(\tau))\|_\infty < \infty.$$

Proof. The vector $F(V) \in \mathbb{R}^4$ has the scale component F_0 and the three translation components F_j, $1 \le j \le 3$. The scale component is bounded by Theorem 8.33, and the translation components are bounded by Theorem 8.30. Taking the maximum over the four components proves the ℓ^∞-bound. □

Corollary 8.35 (Analytic inputs for pointwise modulation forcing)**.** *Pointwise boundedness of the modulation forcing follows from the following analytic estimates:*

(i) *local L^3, local H^1, and local pressure-oscillation control on the support of the compactly supported centering constraints;*

(ii) *weighted control of the Laplacian, pressure, and fourth-moment terms in the scale component, sufficient to bound the weighted scale forcing.*

The scale-component estimate is independent of the algebraic scale transversality identity

$$DG_{\mathrm{sc}}(V)[Z_{\mathrm{sc}}(V)] = p\Theta_0.$$

Proof. The first item is the set of hypotheses used in Theorem 8.30. The second item bounds the weighted scale component through Theorem 8.33, or through the explicit decomposition in Theorem 8.52 below in the corresponding integrated setting. Theorem 8.34 then gives the modulation forcing bound. The displayed transversality identity controls the modulation matrix, not the forcing vector, so it is a separate algebraic input. □

Lemma 8.36 (Window-integrated forcing gives window-integrated modulation)**.** *Assume*

$$\sup_{\tau \geq \tau_0} \|M(V(\tau))^{-1}\|_{\infty \to \infty} \leq C_M < \infty$$

and

$$\sup_{T \geq \tau_0} \int_T^{T+1} \|F(V(\tau))\|_\infty \, d\tau \leq C_F < \infty.$$

Then

$$\sup_{T \geq \tau_0} \int_T^{T+1} \big(|a(\tau)| + \|b(\tau)\|_\infty\big) \, d\tau \leq 2C_M C_F.$$

Proof. For almost every τ,

$$\begin{pmatrix} a(\tau) \\ b(\tau) \end{pmatrix} = -M(V(\tau))^{-1} F(V(\tau)).$$

Hence

$$\max\{|a(\tau)|, \|b(\tau)\|_\infty\} \leq C_M \|F(V(\tau))\|_\infty.$$

Since

$$|a(\tau)| + \|b(\tau)\|_\infty \leq 2 \max\{|a(\tau)|, \|b(\tau)\|_\infty\},$$

we have, for every $T \geq \tau_0$,

$$\begin{aligned} \int_T^{T+1} \big(|a(\tau)| + \|b(\tau)\|_\infty\big) \, d\tau &\leq 2C_M \int_T^{T+1} \|F(V(\tau))\|_\infty \, d\tau \\ &\leq 2C_M C_F. \end{aligned}$$

Taking the supremum over T proves the claim. □

Lemma 8.37 (Selection from an L^1 modulation bound)**.** *Let $I_n = [T_n, T_n + 1]$, where $T_n \to \infty$. Let*

$$d_n : I_n \to [0, \infty), \qquad q_n : I_n \to [0, \infty)$$

be measurable functions satisfying

$$\int_{I_n} d_n(\tau) \, d\tau \to 0, \qquad \sup_n \int_{I_n} q_n(\tau) \, d\tau \leq Q < \infty.$$

Then, after choosing measurable representatives, there are times $\tau_n \in I_n$ such that

$$d_n(\tau_n) \to 0$$

and, if $Q > 0$,

$$q_n(\tau_n) \le 2Q \qquad \text{for every } n.$$

If $Q = 0$, the times may be chosen so that

$$q_n(\tau_n) = 0, \qquad d_n(\tau_n) \to 0.$$

Proof. Assume first that $Q > 0$, and set

$$E_n := \{\tau \in I_n : q_n(\tau) \le 2Q\}.$$

By Chebyshev's inequality,

$$|I_n \setminus E_n| \le \frac{1}{2Q} \int_{I_n} q_n(\tau)\, d\tau \le \frac{1}{2}.$$

Thus $|E_n| \ge 1/2$. Since $d_n \ge 0$,

$$\inf_{\tau \in E_n} d_n(\tau) \le \frac{1}{|E_n|} \int_{E_n} d_n(\tau)\, d\tau \le 2 \int_{I_n} d_n(\tau)\, d\tau.$$

Choose $\tau_n \in E_n$, outside the null set on which the representatives are undefined, such that

$$d_n(\tau_n) \le 2 \int_{I_n} d_n(\tau)\, d\tau + \frac{1}{n}.$$

The right-hand side tends to zero, hence $d_n(\tau_n) \to 0$, and the definition of E_n gives $q_n(\tau_n) \le 2Q$.

If $Q = 0$, then $q_n = 0$ almost everywhere on I_n. Applying the same averaging argument to d_n on the full-measure set where $q_n = 0$ gives the asserted times. □

Corollary 8.38 (Selection on unit windows)**.** *Assume*

$$q(\tau) := |a(\tau)| + \|b(\tau)\|_\infty$$

satisfies

$$\sup_{T \ge \tau_0} \int_T^{T+1} q(\tau)\, d\tau < \infty.$$

For every sequence of unit windows $I_n = [T_n, T_n + 1]$ and every nonnegative measurable density d_n satisfying

$$\int_{I_n} d_n(\tau)\, d\tau \to 0,$$

there are times $\tau_n \in I_n$ such that

$$d_n(\tau_n) \to 0, \qquad \sup_n \big(|a(\tau_n)| + \|b(\tau_n)\|_\infty\big) < \infty.$$

Proof. Apply Theorem 8.37 with $q_n = q|_{I_n}$ and

$$Q = \sup_n \int_{I_n} q(\tau)\, d\tau.$$

This Q is finite because of the uniform window L^1-bound. If $Q > 0$, the selected times satisfy

$$|a(\tau_n)| + \|b(\tau_n)\|_\infty = q(\tau_n) \le 2Q.$$

If $Q = 0$, the same lemma gives $q(\tau_n) = 0$. In both cases $d_n(\tau_n) \to 0$ and the modulation is uniformly bounded along the selected times. □

Lemma 8.39 (Product-form windowed modulation estimate)**.** *Assume*

$$\sup_{T\geq\tau_0}\int_T^{T+1}\|M(V(\tau))^{-1}\|_{\infty\to\infty}\|F(V(\tau))\|_\infty\,d\tau<\infty.$$

Then

$$\sup_{T\geq\tau_0}\int_T^{T+1}\big(|a(\tau)|+\|b(\tau)\|_\infty\big)\,d\tau<\infty.$$

Proof. The modulation system gives

$$\begin{pmatrix}a(\tau)\\ b(\tau)\end{pmatrix}=-M(V(\tau))^{-1}F(V(\tau)).$$

Therefore

$$|a(\tau)|+\|b(\tau)\|_\infty\leq 2\|M(V(\tau))^{-1}\|_{\infty\to\infty}\|F(V(\tau))\|_\infty.$$

Integrating this pointwise estimate on $[T,T+1]$ and taking the supremum in T gives the result. □

Lemma 8.40 (Integrated translation forcing bound)**.** *Let G_j, Φ_j, $H(V,P)$, and $F_j(V)$ be as in Theorem 8.30. Assume that, for some $R>0$,*

$$\sup_{\tau\geq\tau_0}\|V(\tau)\|_{L^3(B_{2R})}\leq M_3,$$

$$\sup_{T\geq\tau_0}\int_T^{T+1}\|\nabla V(\tau)\|^2_{L^2(B_{2R})}\,d\tau\leq A_R,$$

and

$$\sup_{\tau\geq\tau_0}\inf_{c\in\mathbb{R}}\|P(\tau)-c\|_{L^{3/2}(B_{2R})}\leq M_P.$$

Then

$$\sup_{T\geq\tau_0}\int_T^{T+1}\max_{1\leq j\leq 3}|F_j(V(\tau))|\,d\tau\leq C(R,\nu,\psi_R)(A_R+M_3^2+M_3^3+M_PM_3).$$

Proof. Fix j. The proof of Theorem 8.30 gives, for almost every τ,

$$|F_j(V(\tau))|\leq C(R,\nu,\psi_R)\left(\|\nabla V(\tau)\|^2_{L^2(B_{2R})}+M_3\|\nabla V(\tau)\|_{L^2(B_{2R})}+M_3^3+M_PM_3\right).$$

Integrate this estimate over $I_T=[T,T+1]$. The first term contributes at most A_R. For the mixed term, Cauchy's inequality on the unit interval gives

$$\int_{I_T}M_3\|\nabla V(\tau)\|_{L^2(B_{2R})}\,d\tau\leq M_3\left(\int_{I_T}\|\nabla V(\tau)\|^2_{L^2(B_{2R})}\,d\tau\right)^{1/2}\leq M_3A_R^{1/2}.$$

Since

$$M_3A_R^{1/2}\leq\frac{1}{2}(M_3^2+A_R),$$

the mixed term is bounded by $C(A_R+M_3^2)$. The terms M_3^3 and M_PM_3 are constant on the unit window. Hence

$$\int_T^{T+1}|F_j(V(\tau))|\,d\tau\leq C(R,\nu,\psi_R)(A_R+M_3^2+M_3^3+M_PM_3).$$

Since $\max_{1\leq j\leq 3}|F_j|\leq\sum_{j=1}^3|F_j|$, taking the maximum over the three centering constraints changes only the constant. □

Remark 8.41 (Independence of the windowed gradient hypothesis). The windowed gradient input in Theorem 8.40 is an independent hypothesis whenever the integrated translation estimate is used to prove windowed H^1-control. It may come, for example, from a preliminary Caccioppoli estimate depending only on already available data, from a direct approximation or regularity estimate on the support of the centering functions, or from an independent verification obtained without using the conclusion of the windowed H^1 estimate under consideration.

Lemma 8.42 (Window-integrated scale forcing). *Assume the window-integrated weighted scale-forcing bound*

$$\sup_{T\geq\tau_0}\int_T^{T+1}\left|\int_{\mathbb{R}^3}|y|^{-p}|V|V\cdot H(V,P)\,dy\right|d\tau<\infty,$$

where the weighted pairing is interpreted as in Theorem 8.32. Then

$$\sup_{T\geq\tau_0}\int_T^{T+1}|F_0(V(\tau))|\,d\tau<\infty.$$

Proof. By the identity

$$F_0(V)=3\int_{\mathbb{R}^3}|y|^{-p}|V|V\cdot H(V,P)\,dy$$

we have, for each $T\geq\tau_0$,

$$\int_T^{T+1}|F_0(V(\tau))|\,d\tau=3\int_T^{T+1}\left|\int_{\mathbb{R}^3}|y|^{-p}|V|V\cdot H(V,P)\,dy\right|d\tau.$$

Taking the supremum over T gives the asserted L^1-bound. □

Lemma 8.43 (Full integrated modulation forcing). *Assume the integrated translation forcing estimate*

$$\sup_{T\geq\tau_0}\int_T^{T+1}\max_{1\leq j\leq 3}|F_j(V(\tau))|\,d\tau<\infty$$

and the window-integrated weighted scale-forcing bound in Theorem 8.42. Then

$$\sup_{T\geq\tau_0}\int_T^{T+1}\|F(V(\tau))\|_\infty\,d\tau<\infty.$$

Proof. The vector $F(V)\in\mathbb{R}^4$ consists of F_0 and the three translation components F_j, $1\leq j\leq 3$. Since

$$\|F(V(\tau))\|_\infty\leq|F_0(V(\tau))|+\max_{1\leq j\leq 3}|F_j(V(\tau))|,$$

for almost every τ. Integrating this inequality on $[T,T+1]$ gives

$$\int_T^{T+1}\|F(V(\tau))\|_\infty\,d\tau\leq\int_T^{T+1}|F_0(V(\tau))|\,d\tau+\int_T^{T+1}\max_{1\leq j\leq 3}|F_j(V(\tau))|\,d\tau.$$

The first term is uniformly bounded in T by Theorem 8.42, and the second term is uniformly bounded by the assumed integrated translation forcing estimate. Taking the supremum over T proves the claim. □

Corollary 8.44 (Integrated modulation criterion on selected windows). *Assume the uniform inverse bound for $M(V(\tau))$, the integrated translation forcing estimate, and the window-integrated weighted scale-forcing bound. Then*

$$\sup_{T\geq\tau_0}\int_T^{T+1}\big(|a(\tau)|+\|b(\tau)\|_\infty\big)\,d\tau<\infty.$$

Consequently every sequence of unit windows with vanishing nonnegative measurable density contains selected times where this density vanishes asymptotically and the modulation remains uniformly bounded.

Proof. Theorem 8.43 gives the window-integrated bound on $\|F(V)\|_\infty$. Combining this with the uniform inverse bound in Theorem 8.36 gives the window-integrated modulation bound. The selection conclusion is Theorem 8.38. □

Definition 8.45 (Weighted scale-force components)**.** Let $w(y) = |y|^{-p}$, $0 < p < 3$. Define

$$F_0^\Delta(V) := 3\nu \int_{\mathbb{R}^3} w|V|V \cdot \Delta V \, dy,$$

$$F_0^{\mathrm{nl}}(V) := -3 \int_{\mathbb{R}^3} w|V|V \cdot (V \cdot \nabla V) \, dy,$$

and

$$F_0^P(V, P) := -3 \int_{\mathbb{R}^3} w|V|V \cdot \nabla P \, dy.$$

The three pairings are required to be compatible with the decomposition of $H(V, P)$ into diffusion, transport, and pressure.

Lemma 8.46 (Integrated scale-force decomposition)**.** *For every time at which the three pairings in Theorem 8.45 are defined,*

$$F_0(V(\tau)) = F_0^\Delta(V(\tau)) + F_0^{\mathrm{nl}}(V(\tau)) + F_0^P(V(\tau), P(\tau)).$$

Proof. Substitute

$$H(V, P) = \nu\Delta V - (V \cdot \nabla)V - \nabla P$$

into

$$F_0(V) = 3 \int w|V|V \cdot H(V, P) \, dy.$$

Then

$$F_0(V) = 3\nu \int w|V|V \cdot \Delta V \, dy - 3 \int w|V|V \cdot (V \cdot \nabla V) \, dy$$
$$- 3 \int w|V|V \cdot \nabla P \, dy.$$

The three terms coincide with $F_0^\Delta(V)$, $F_0^{\mathrm{nl}}(V)$, and $F_0^P(V, P)$, respectively. Thus linearity of the weighted pairing gives the stated decomposition. □

Lemma 8.47 (Component bounds imply integrated scale-force control)**.** *Assume*

$$\sup_{T \geq \tau_0} \int_T^{T+1} |F_0^\Delta(V(\tau))| \, d\tau < \infty,$$

$$\sup_{T \geq \tau_0} \int_T^{T+1} |F_0^{\mathrm{nl}}(V(\tau))| \, d\tau < \infty,$$

and

$$\sup_{T \geq \tau_0} \int_T^{T+1} |F_0^P(V(\tau), P(\tau))| \, d\tau < \infty.$$

Then the window-integrated weighted scale-forcing bound in Theorem 8.42 holds.

Proof. By Theorem 8.46,

$$|F_0(V(\tau))| \le |F_0^{\Delta}(V(\tau))| + |F_0^{\mathrm{nl}}(V(\tau))| + |F_0^{P}(V(\tau), P(\tau))|$$

for almost every τ. Integrating on $[T, T+1]$, taking the supremum in T, and applying the three component bounds gives

$$\sup_{T \ge \tau_0} \int_T^{T+1} |F_0(V(\tau))|\, d\tau < \infty.$$

Since F_0 is three times the weighted pairing in Theorem 8.32, this is equivalent to the window-integrated weighted scale-forcing bound. □

Definition 8.48 (Singular-weight convective integration by parts)**.** The singular-weight convective integration-by-parts condition means that, for almost every τ,

$$-\int_{\mathbb{R}^3} w\, V \cdot \nabla(|V|^3)\, dy = \int_{\mathbb{R}^3} |V|^3 V \cdot \nabla w\, dy,$$

with no boundary contribution from $y = 0$ or from spatial infinity, either classically or by convergence of smooth compactly supported divergence-free approximants.

Lemma 8.49 (Convective scale-force bound)**.** *Assume the condition of Theorem 8.48 and*

$$\sup_{T \ge \tau_0} \int_T^{T+1} \int_{\mathbb{R}^3} |y|^{-p-1} |V(y,\tau)|^4\, dy\, d\tau < \infty.$$

Then

$$\sup_{T \ge \tau_0} \int_T^{T+1} |F_0^{\mathrm{nl}}(V(\tau))|\, d\tau < \infty.$$

More precisely,

$$|F_0^{\mathrm{nl}}(V(\tau))| \le p \int_{\mathbb{R}^3} |y|^{-p-1} |V(y,\tau)|^4\, dy$$

for almost every τ.

Proof. Fix such a time τ and suppress it from the notation. Since

$$\partial_i(|V|^3) = 3|V| V_j \partial_i V_j$$

for each i, and therefore

$$V \cdot \nabla(|V|^3) = 3|V| V_i V_j \partial_i V_j = 3|V|\, V \cdot ((V \cdot \nabla)V),$$

we have

$$w|V| V \cdot (V \cdot \nabla V) = \frac{1}{3} w\, V \cdot \nabla(|V|^3).$$

Therefore

$$F_0^{\mathrm{nl}}(V) = -3 \int w|V| V \cdot (V \cdot \nabla V) = -\int w\, V \cdot \nabla(|V|^3).$$

Using $\nabla \cdot V = 0$ and Theorem 8.48,

$$-\int w\, V \cdot \nabla(|V|^3) = \int |V|^3 V \cdot \nabla w.$$

Since

$$\nabla w(y) = -p|y|^{-p-2} y, \qquad y \ne 0,$$

we have

$$|V \cdot \nabla w| = p|y|^{-p-2} |V \cdot y| \le p|y|^{-p-1} |V|.$$

Thus

$$|F_0^{\mathrm{nl}}(V)| \le p \int |y|^{-p-1} |V|^4\, dy.$$

Integrating over $[T, T+1]$ and taking the supremum in T proves the window-integrated convective bound. $\square$

Definition 8.50 (Annular convective regularity)**.** Annular convective regularity means that, for almost every τ, $V(\tau)$ is divergence free and, on every compact annulus $0 < r < |y| < R < \infty$, the chain-rule identities

$$\partial_i(|V|^3) = 3|V|V_j\partial_i V_j, \qquad V \cdot \nabla(|V|^3) = 3|V|V_iV_j\partial_i V_j$$

hold in the sense of distributions, either classically or by passage to the limit from smooth divergence-free approximants on annuli.

Lemma 8.51 (Annular regularity gives singular-weight integration by parts)**.** *Assume annular convective regularity and the weighted L^4-window bound*

$$\sup_{T\geq\tau_0}\int_T^{T+1}\int_{\mathbb{R}^3}|y|^{-p-1}|V(y,\tau)|^4\,dy\,d\tau < \infty.$$

Then the singular-weight convective integration-by-parts condition of Theorem 8.48 holds for almost every τ.

Proof. Fix a time τ for which

$$\int_{\mathbb{R}^3}|y|^{-p-1}|V(y,\tau)|^4\,dy < \infty$$

and annular convective regularity holds. Such times form a full-measure set on every unit window.

Let $\eta \in C^\infty([0,\infty))$ satisfy $0 \leq \eta \leq 1$, $\eta(s) = 0$ for $s \leq 1$, and $\eta(s) = 1$ for $s \geq 2$. Define

$$\chi_{\varepsilon,R}(y) := \eta(|y|/\varepsilon)\eta(R/|y|).$$

Then $\chi_{\varepsilon,R}$ is supported in $\{\varepsilon < |y| < R\}$, equals one on $\{2\varepsilon < |y| < R/2\}$, and satisfies

$$|\nabla\chi_{\varepsilon,R}(y)| \leq C\varepsilon^{-1}\mathbf{1}_{\{\varepsilon<|y|<2\varepsilon\}} + CR^{-1}\mathbf{1}_{\{R/2<|y|<R\}}.$$

By annular convective regularity, the following integration by parts is valid on the annulus $\{\varepsilon < |y| < R\}$. Since $\nabla \cdot V = 0$ and $w\chi_{\varepsilon,R}|V|^3V$ is compactly supported in this annulus,

$$0 = \int \nabla \cdot \left(w\chi_{\varepsilon,R}|V|^3V\right)dy.$$

Expanding the divergence gives

$$0 = \int |V|^3V \cdot \nabla(w\chi_{\varepsilon,R})\,dy + \int w\chi_{\varepsilon,R}V \cdot \nabla(|V|^3)\,dy,$$

and hence

$$-\int w\chi_{\varepsilon,R}V \cdot \nabla(|V|^3) = \int |V|^3V \cdot \nabla(w\chi_{\varepsilon,R}).$$

Expanding the derivative,

$$-\int w\chi_{\varepsilon,R}V \cdot \nabla(|V|^3) = \int \chi_{\varepsilon,R}|V|^3V \cdot \nabla w + \int w|V|^3V \cdot \nabla\chi_{\varepsilon,R}.$$

On the inner cutoff region $\{\varepsilon < |y| < 2\varepsilon\}$, one has $|y| \simeq \varepsilon$, and on the outer cutoff region $\{R/2 < |y| < R\}$, one has $|y| \simeq R$. Therefore the derivative bound for $\chi_{\varepsilon,R}$ implies

$$\left|\int w|V|^3V \cdot \nabla\chi_{\varepsilon,R}\right| \leq C\int_{\{\varepsilon<|y|<2\varepsilon\}}|y|^{-p-1}|V|^4\,dy + C\int_{\{R/2<|y|<R\}}|y|^{-p-1}|V|^4\,dy.$$

The integrand $|y|^{-p-1}|V|^4$ belongs to $L^1(\mathbb{R}^3)$ at the fixed time. Hence absolute continuity of the Lebesgue integral gives

$$\int_{\{\varepsilon<|y|<2\varepsilon\}} |y|^{-p-1}|V|^4\,dy \to 0 \qquad \text{as } \varepsilon \downarrow 0,$$

and integrability at infinity gives

$$\int_{\{R/2<|y|<R\}} |y|^{-p-1}|V|^4\,dy \to 0 \qquad \text{as } R \uparrow \infty.$$

Moreover

$$|\chi_{\varepsilon,R}|V|^3V\cdot\nabla w| \le p|y|^{-p-1}|V|^4 \in L^1(\mathbb{R}^3),$$

so dominated convergence gives

$$\int \chi_{\varepsilon,R}|V|^3V\cdot\nabla w \to \int |V|^3V\cdot\nabla w.$$

Passing first $\varepsilon \downarrow 0$ and then $R \uparrow \infty$ yields

$$-\int_{\mathbb{R}^3} w\,V\cdot\nabla(|V|^3)\,dy = \int_{\mathbb{R}^3} |V|^3V\cdot\nabla w\,dy,$$

with no boundary contribution at $y=0$ or at infinity. □

Corollary 8.52 (Reduced integrated scale-force criterion)**.** *Assume annular convective regularity, the weighted L^4-window bound, and the two component estimates*

$$\sup_{T\ge\tau_0}\int_T^{T+1} |F_0^{\Delta}(V(\tau))|\,d\tau < \infty,$$

$$\sup_{T\ge\tau_0}\int_T^{T+1} |F_0^{P}(V(\tau),P(\tau))|\,d\tau < \infty.$$

Then the window-integrated weighted scale-forcing bound holds.

Proof. Theorem 8.51 gives the singular-weight integration-by-parts condition. Theorem 8.49 then gives the integrated convective scale-force bound. Combining this with the diffusive and pressure component estimates, and applying Theorem 8.47, gives the window-integrated weighted scale-forcing bound. □

Corollary 8.53 (Remaining analytic alternatives for the scale component)**.** *Before the singular-weight integration by parts is justified, failure of the window-integrated weighted scale-forcing bound is reduced to failure of at least one of the following four analytic statements:*

(i) *the integrated diffusive weighted scale-force bound;*
(ii) *the singular-weight convective integration-by-parts identity;*
(iii) *the weighted L^4-window bound;*
(iv) *the integrated weighted pressure scale-force bound.*

If annular convective regularity is available, then the second item is replaced by failure of annular convective regularity itself, together with the same weighted L^4-window input.

Proof. Theorem 8.47 shows that the diffusive, convective, and pressure component bounds imply the integrated scale-force bound. Theorem 8.49 reduces the convective component to the singular-weight integration-by-parts identity and the weighted L^4-window bound. Finally, Theorem 8.51 shows that annular convective regularity and the weighted L^4-window bound imply the singular-weight integration-by-parts identity. Therefore the listed failures exhaust the remaining ways in which this scale-component estimate can fail. □

8.5. **Pointwise Type II Exclusion.**

Definition 8.54 (Pointwise exclusion estimates)**.** For a renormalized Type II solution ω, the non-tight alternative is excluded when V_ω satisfies Theorem 8.1. The windowed-control alternative is excluded when

$$\forall R > 0: \qquad \sup_{T \geq \tau_0+1} \int_T^{T+1} \|V_\omega(\tau)\|^2_{H^1(B_R)}\, d\tau < \infty.$$

The universal versions require the corresponding pointwise statement for every renormalized solution in the class.

Definition 8.55 (Remaining Type II case after local estimates)**.** A renormalized Type II solution remains after the local estimates if it reaches the local two-alternative classification, is not already excluded by the compact cost-divergence criterion, and is not excluded by the local Navier–Stokes cost-to-exclusion theorem.

Theorem 8.56 (Type II exclusion after tightness and windowed control)**.** *Assume the renormalized Type II class satisfies:*

(i) *the local classification has exactly the following alternatives: the compact cost-divergence regime, to be further stratified by carrier validation, failure of uniform critical tightness, and failure of local windowed H^1 control;*
(ii) *every renormalized solution is uniformly critically tight;*
(iii) *every renormalized solution satisfies the tail windowed H^1 bound;*
(iv) *the compact cost-divergence regime is evaluated as the retained compact cost-divergence stratum after the carrier-validation alternatives of Theorem 8.87 have been tested;*
(v) *every carrier-validation adjacent stratum arising from this regime is treated by its corresponding local stratum argument.*

Then every renormalized Type II solution is either excluded in a carrier-validation adjacent stratum or excluded on the retained compact cost-divergence stratum by the local Navier–Stokes cost-to-exclusion theorem. Consequently no remaining Type II case exists.

Proof. Fix a renormalized Type II solution ω. By the local classification, ω reaches exactly one of the following alternatives:

(1) the compact cost-divergence regime, with its carrier-validation subalternatives;
(2) failure of uniform critical tightness;
(3) failure of windowed H^1 control, after uniform critical tightness has been established.

By the universal tightness hypothesis and Theorem 8.54, the second alternative is unavailable. By the universal windowed H^1 hypothesis and Theorem 8.54, the third alternative is unavailable. The only remaining alternative is the compact cost-divergence regime. By hypothesis this regime has already been passed through the ordered carrier-validation alternatives of Theorem 8.87. If a carrier-validation test fails, ω leaves the retained compact stratum and belongs to the corresponding adjacent stratum, where it is handled by the assumed local stratum argument. On the retained compact stratum, the same theorem gives local monotonicity-carrier data for the validated compact cost. These data give the local cost-to-exclusion data by Theorem 8.73; then Theorem 8.77 excludes this case. Since ω was arbitrary, no remaining Type II case exists. □

Corollary 8.57 (Finite-cost exclusion after pointwise tests)**.** *Under the hypotheses of Theorem 8.56, there is no finite-cost renormalized Type II solution outside the compact cost-divergence criterion.*

Proof. Theorem 8.56 shows that every renormalized Type II solution is excluded. Hence no renormalized solution remains outside the compact cost-divergence criterion, regardless of whether its Type II cost is finite. □

8.6. **Physical energy versus renormalized Type II cost.**

Definition 8.58 (Navier–Stokes cost-to-exclusion data)**.** A renormalized Navier–Stokes Type II branch satisfying the compact cost-divergence criterion has Navier–Stokes cost-to-exclusion data if the validated compact cost admits local monotonicity-carrier data in the sense of Theorem 8.72. In particular, this may be verified either by the fixed-cutoff criterion of Theorem 8.74 or by the moving-cutoff criterion of Theorem 8.75. This is a data condition, not a primitive cost-exclusion assumption. On the retained compact cost-divergence stratum it is supplied by Theorem 8.87. Failure of this data condition is therefore not an unclassified obstruction: it is the carrier validation exit recorded in Theorem 8.98 and is treated as an adjacent compactness stratum.

Lemma 8.59 (Navier–Stokes cost-divergence exclusion from carrier data)**.** *Assume a renormalized Type II solution has the active supercritical scaling condition, satisfies the compact cost-divergence criterion, and has the Navier–Stokes cost-to-exclusion data of Theorem 8.58. Then the solution is excluded from the Type II class under consideration.*

Proof. Theorem 8.58 supplies local monotonicity-carrier data for the validated compact cost. By Theorem 8.73, this gives the local cost-to-exclusion data. Therefore Theorem 8.77 applies to the active supercritical scaling condition and compact cost-divergence criterion, giving exclusion from the stated Type II class. □

Lemma 8.60 (Physical dissipation in renormalized variables)**.** *Let*

$$u(x,t) = \lambda(t)^{-1} V\left(\frac{x - x_c(t)}{\lambda(t)}, \tau(t)\right), \qquad \frac{d\tau}{dt} = \lambda(t)^{-2}.$$

Then

$$\|\nabla_x u(t)\|^2_{L^2_x} = \lambda(t)^{-1} \|\nabla_y V(\tau(t))\|^2_{L^2_y}.$$

Consequently,

$$\int_{t_1}^{t_2} \|\nabla_x u(t)\|^2_{L^2_x}\, dt = \int_{\tau(t_1)}^{\tau(t_2)} \lambda(\tau) \|\nabla_y V(\tau)\|^2_{L^2_y}\, d\tau.$$

Proof. Since $u = \lambda^{-1} V$ and $y = (x - x_c)/\lambda$, $\nabla_x u = \lambda^{-2} \nabla_y V$. With $dx = \lambda^3 dy$,

$$\|\nabla_x u\|^2_{L^2_x} = \int \lambda^{-4} |\nabla_y V|^2 \lambda^3\, dy = \lambda^{-1} \|\nabla_y V\|^2_{L^2_y}.$$

Since $d\tau/dt = \lambda^{-2}$, $dt = \lambda^2 d\tau$. Multiplying gives the second identity. □

Definition 8.61 (Physical-renormalized dissipation cost)**.** Define

$$\mathfrak{D}_{\mathrm{phys-ren}}(\tau) := \nu \lambda(\tau) \|\nabla_y V(\tau)\|^2_{L^2_y}$$

and the localized version

$$\mathfrak{D}_{\mathrm{phys-ren},R_0}(\tau) := \nu \lambda(\tau) \int |\nabla_y V|^2 \phi_{R_0}.$$

Theorem 8.62 (Physical energy controls scale-weighted dissipation)**.** *Assume finite physical energy and the Navier–Stokes energy inequality*

$$E(t) + \nu \int_0^t \|\nabla_x u(s)\|^2_{L^2_x}\, ds \le E(0), \qquad E(t) = \frac{1}{2} \|u(t)\|^2_{L^2_x}.$$

Then, for every $t_1 < T^*$,

$$\int_{\tau(t_1)}^{\infty} \mathfrak{D}_{\mathrm{phys-ren}}(\tau)\, d\tau \le E(t_1) \le E(0).$$

In particular,

$$\int_{\tau(t_1)}^{\infty} \mathfrak{D}_{\mathrm{phys-ren},R_0}(\tau)\, d\tau < \infty.$$

Proof. The physical energy inequality on $[t_1, t]$ gives

$$\nu \int_{t_1}^{t} \|\nabla_x u(s)\|_2^2\, ds \le E(t_1).$$

Use Theorem 8.60 and let $t \uparrow T^*$. Monotone convergence gives the full weighted bound. The localized estimate follows from $0 \le \phi_{R_0} \le 1$. □

Definition 8.63 (Weighted positive scale-drift cost control)**.** Let

$$a_+(\tau) = \max\{a(\tau), 0\}.$$

Weighted positive scale-drift cost control means

$$\int_{\tau_1}^{\infty} \lambda(\tau) a_+(\tau) \int |V(y,\tau)|^2 \phi_{R_0}(y)\, dy\, d\tau < \infty.$$

Definition 8.64 (Scale-weighted Type II cost)**.** For

$$\widetilde{\mathfrak{D}}_{R_0}(\tau) = \nu \int |\nabla_y V|^2 \phi_{R_0} + a_+(\tau) \int |V|^2 \phi_{R_0},$$

define

$$\mathfrak{C}^{NS}_{\mathrm{II,phys}}(\tau) := \lambda(\tau) \widetilde{\mathfrak{D}}_{R_0}(\tau).$$

Theorem 8.65 (Physical energy controls the scale-weighted Type II cost)**.** *Assume finite physical energy, the Navier–Stokes energy inequality, and Theorem 8.63. Then*

$$\int_{\tau_1}^{\infty} \mathfrak{C}^{NS}_{\mathrm{II,phys}}(\tau)\, d\tau < \infty.$$

Proof. By definition,

$$\mathfrak{C}^{NS}_{\mathrm{II,phys}} = \lambda\nu \int |\nabla V|^2 \phi_{R_0} + \lambda a_+ \int |V|^2 \phi_{R_0}.$$

The first term is integrable by Theorem 8.62; the second is integrable by Theorem 8.63. □

Definition 8.66 (Scale-decoupled infinite Type II cost)**.** A renormalized Type II solution has scale-decoupled infinite Type II cost if

$$\int_{\tau_0}^{\infty} \widetilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty, \qquad \int_{\tau_0}^{\infty} \lambda(\tau) \widetilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau < \infty.$$

The physically forbidden weighted-infinite-cost alternative is

$$\int_{\tau_0}^{\infty} \mathfrak{C}^{NS}_{\mathrm{II,phys}}(\tau)\, d\tau = \infty,$$

which is impossible under Theorem 8.65.

Theorem 8.67 (Infinite-cost trichotomy)**.** *Assume finite physical energy, the Navier–Stokes energy inequality, and Theorem 8.63. Let a renormalized Type II solution satisfy*

$$\int_{\tau_0}^{\infty} \widetilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty.$$

Then:

(i) *the compact Type II cost criterion gives a scale-collapse cost-divergence conclusion;*

(ii) *if the solution remains in the retained compact cost-divergence stratum after carrier validation, then it is excluded from the Type II class under consideration;*

(iii) *relative to physical energy, exactly one of the following occurs: the physically forbidden weighted-infinite-cost alternative or the scale-decoupled infinite-cost alternative of Theorem 8.66.*

Proof. The unweighted infinite-cost conclusion is precisely the compact Type II cost divergence, so the compact criterion gives a scale-collapse cost-divergence conclusion. If the solution remains in the retained compact cost-divergence stratum after carrier validation, Theorem 8.87 gives local monotonicity-carrier data, and Theorem 8.95 gives exclusion from the Type II class under consideration.

For the physical-energy classification, evaluate $\int \mathfrak{C}^{NS}_{\mathrm{II,phys}}$. If it is infinite, then the physically forbidden weighted-infinite-cost alternative occurs, contradicting Theorem 8.65. If it is finite, then the solution is scale-decoupled by Theorem 8.66. The two physical-energy outcomes are disjoint and exhaustive. □

Remark 8.68. Finite physical energy controls the scale-weighted renormalized cost, not the unweighted compact Type II cost. Thus infinite scale-weighted renormalized cost forces infinite physical energy dissipation, while infinite unweighted renormalized cost does not by itself force infinite physical energy without an additional scale-weight estimate. Likewise, a compact Type II cost-divergence conclusion does not automatically imply exclusion from the Type II class unless the local corrected-monotonicity data of Theorem 8.87 are present, equivalently unless the branch remains in the retained compact stratum after the adjacent carrier-failure alternatives have been evaluated.

8.7. **Navier–Stokes Cost-Divergence Exclusion.**

Definition 8.69 (Type II exclusion conclusion)**.** For a renormalized Type II solution, the exclusion conclusion means that after the active supercritical scaling condition and the compact cost-divergence criterion have both been established, the solution is excluded from the Type II class under consideration. This is stronger than cost divergence alone, which only states that the solution satisfies the compact Type II cost-divergence criterion.

Definition 8.70 (Local cost-to-exclusion data)**.** Let ω be a repaired-gauge Navier–Stokes Type II branch on a terminal renormalized tail $[\tau_0,\infty)$. Local cost-to-exclusion data for the validated compact cost consist of a nonnegative locally integrable cost $\mathcal{D}_\omega$, a locally absolutely continuous corrected localized energy $\mathcal{H}_\omega$, a constant $c_\omega>0$, and a nonnegative locally integrable remainder $\mathcal{R}_\omega$ such that:

(i) the validated compact cost-divergence conclusion is

$$\int_{\tau_0}^{\infty} \mathcal{D}_\omega(\tau)\,d\tau=\infty;$$

(ii) $\mathcal{H}_\omega(\tau_0)<\infty$ and

$$\inf_{\tau\ge\tau_0} \mathcal{H}_\omega(\tau)>-\infty;$$

(iii) the corrected localized energy inequality

$$\frac{d}{d\tau}\mathcal{H}_\omega(\tau)+c_\omega\mathcal{D}_\omega(\tau)+\mathcal{R}_\omega(\tau)\le 0$$

holds in distributions on (τ_0,∞).

For the identity cost one takes $\mathcal{D}_\omega=\tilde{\mathfrak{D}}_{R_0}$. For an explicitly comparable localized cost, $\mathcal{D}_\omega$ is the localized cost whose divergence is validated by the compact criterion, provided that the same cost is controlled by the corrected localized energy inequality on the terminal tail.

Definition 8.71 (Energy-controlled validated cost)**.** Let (V, P, a, b) be a repaired-gauge branch and let

$$\tilde{\mathfrak{D}}_{R_0}(\tau) = \nu \int |\nabla V|^2 \phi_{R_0} + a_+(\tau) \int |V|^2 \phi_{R_0}$$

be the identity localized Navier–Stokes cost. A validated compact cost $\mathfrak{C}_{\mathrm{II}}^{NS}$ is energy-controlled by the identity cost on a terminal tail if there are constants $C_1 < \infty$ and $\tau_1 \geq \tau_0$ such that

$$0 \leq \mathfrak{C}_{\mathrm{II}}^{NS}(\tau) \leq C_1 \tilde{\mathfrak{D}}_{R_0}(\tau) \qquad \text{for a.e. } \tau \geq \tau_1.$$

The identity cost is energy-controlled with $C_1 = 1$ and $\tau_1 = \tau_0$. For a non-identity cost, this is the upper comparison needed for a corrected localized energy inequality controlling $\tilde{\mathfrak{D}}_{R_0}$ to also control the validated cost.

Definition 8.72 (Local monotonicity carrier for a validated cost)**.** Let ω be a repaired-gauge Navier–Stokes Type II branch on a terminal renormalized tail $[\tau_0, \infty)$, and let $\mathfrak{C}_{\mathrm{II}}^{NS}$ be the validated compact cost for that branch. A local monotonicity carrier for $\mathfrak{C}_{\mathrm{II}}^{NS}$ consists of a nonnegative locally integrable carrier cost $\mathcal{A}_\omega$, a locally absolutely continuous corrected localized energy $\mathcal{H}_\omega$, a nonnegative locally integrable remainder $\mathcal{R}_\omega$, and constants $c_A > 0$, $0 < C_A < \infty$ such that:

(i) $\mathcal{H}_\omega(\tau_0) < \infty$ and
$$\inf_{\tau \geq \tau_0} \mathcal{H}_\omega(\tau) > -\infty;$$
(ii) the carrier monotonicity inequality
$$\frac{d}{d\tau}\mathcal{H}_\omega(\tau) + c_A \mathcal{A}_\omega(\tau) + \mathcal{R}_\omega(\tau) \leq 0$$
holds in distributions on (τ_0, ∞);
(iii) the validated compact cost is controlled by the carrier on the terminal tail:
$$0 \leq \mathfrak{C}_{\mathrm{II}}^{NS}(\tau) \leq C_A \mathcal{A}_\omega(\tau) \qquad \text{for a.e. } \tau \geq \tau_0.$$

The carrier is local: $\mathcal{A}_\omega$ may be the fixed cutoff identity cost, a moving-cutoff identity cost, or another localized cost with an explicit terminal comparison to the validated compact cost.

Lemma 8.73 (Monotonicity carrier gives local cost-to-exclusion data)**.** *Let ω be a repaired-gauge Navier–Stokes Type II branch. Assume that the validated compact cost $\mathfrak{C}_{\mathrm{II}}^{NS}$ is nonnegative, locally integrable on finite renormalized intervals, and satisfies*

$$\int_{\tau_0}^{\infty} \mathfrak{C}_{\mathrm{II}}^{NS}(\tau)\, d\tau = \infty.$$

If $\mathfrak{C}_{\mathrm{II}}^{NS}$ admits a local monotonicity carrier in the sense of Theorem 8.72, then ω carries the local cost-to-exclusion data of Theorem 8.70 with $\mathcal{D}_\omega = \mathfrak{C}_{\mathrm{II}}^{NS}$.

Proof. Let $(\mathcal{A}_\omega, \mathcal{H}_\omega, \mathcal{R}_\omega^{\mathrm{car}}, c_A, C_A)$ be the carrier data. The lower bound and finite initial value of $\mathcal{H}_\omega$ are exactly the energy bounds required in Theorem 8.70. The terminal comparison gives

$$\mathcal{A}_\omega(\tau) \geq C_A^{-1} \mathfrak{C}_{\mathrm{II}}^{NS}(\tau) \qquad \text{for a.e. } \tau \geq \tau_0.$$

Substituting this into the carrier monotonicity inequality yields

$$\frac{d}{d\tau}\mathcal{H}_\omega(\tau) + \frac{c_A}{C_A} \mathfrak{C}_{\mathrm{II}}^{NS}(\tau) + \mathcal{R}_\omega^{\mathrm{car}}(\tau) \leq 0$$

in distributions. Thus Theorem 8.70 holds with

$$\mathcal{D}_\omega = \mathfrak{C}_{\mathrm{II}}^{NS}, \qquad c_\omega = c_A / C_A, \qquad \mathcal{R}_\omega = \mathcal{R}_\omega^{\mathrm{car}}.$$

The divergent-cost item is the validated compact cost-divergence conclusion. $\square$

Lemma 8.74 (Fixed-cutoff monotonicity gives local cost-to-exclusion data)**.** *Let ω be represented on a terminal tail by a repaired-gauge Navier–Stokes branch (V, P, a, b). Let ϕ_{R_0} be the cutoff used in the validated compact cost, let $\mathfrak{C}_{\mathrm{II}}^{NS}$ be a nonnegative locally integrable validated compact cost, and set*

$$\mathcal{E}_\phi(\tau) = \frac{1}{2}\int |V(y,\tau)|^2 \phi_{R_0}(y)\, dy.$$

Assume, after increasing τ_0 if necessary, that:

(i) *$\mathcal{E}_\phi(\tau_0) < \infty$;*
(ii) *the fixed-cutoff localized energy identity (13) holds on the terminal tail;*
(iii) *the fixed-cutoff error density B_{R_0} of Theorem 10.9 satisfies*

$$\int_{\tau_0}^{\infty} B_{R_0}(\tau)\, d\tau < \infty;$$

(iv) *the validated compact cost $\mathfrak{C}_{\mathrm{II}}^{NS}$ is energy-controlled by $\tilde{\mathfrak{D}}_{R_0}$ in the sense of Theorem 8.71;*
(v) *the validated compact cost-divergence conclusion is*

$$\int_{\tau_0}^{\infty} \mathfrak{C}_{\mathrm{II}}^{NS}(\tau)\, d\tau = \infty.$$

Then ω carries the local cost-to-exclusion data of Theorem 8.70 with $\mathcal{D}_\omega = \mathfrak{C}_{\mathrm{II}}^{NS}$.

Proof. Let C_1, τ_1 be the constants in the energy-control comparison. All statements are terminal. Since $\mathfrak{C}_{\mathrm{II}}^{NS}$ is locally integrable on finite intervals, removing any finite initial subinterval does not change divergence of its tail integral. After the allowed increase and relabeling of τ_0, we may therefore assume that $\tau_0 \geq \tau_1$, that $\mathcal{E}_\phi(\tau_0) < \infty$, and that the upper comparison holds on the whole tail.

Define the corrected localized energy

$$\mathcal{H}_\omega(\tau) = \mathcal{E}_\phi(\tau) + \int_\tau^\infty B_{R_0}(s)\, ds.$$

The finite-error hypothesis makes the correction tail finite for every $\tau \geq \tau_0$ and locally absolutely continuous. Since $\mathcal{E}_\phi \geq 0$ and $B_{R_0} \geq 0$, one has

$$\inf_{\tau \geq \tau_0} \mathcal{H}_\omega(\tau) \geq 0, \qquad \mathcal{H}_\omega(\tau_0) < \infty.$$

Theorem 10.11 gives, in distributions,

$$\frac{d}{d\tau}\mathcal{H}_\omega(\tau) + \frac{1}{2}\tilde{\mathfrak{D}}_{R_0}(\tau) \leq 0.$$

By energy-control of the validated cost,

$$\tilde{\mathfrak{D}}_{R_0}(\tau) \geq C_1^{-1}\mathfrak{C}_{\mathrm{II}}^{NS}(\tau) \quad \text{for a.e. } \tau \geq \tau_0.$$

Therefore

$$\frac{d}{d\tau}\mathcal{H}_\omega(\tau) + \frac{1}{2C_1}\mathfrak{C}_{\mathrm{II}}^{NS}(\tau) \leq 0.$$

This is the corrected localized energy inequality in Theorem 8.70, with

$$\mathcal{D}_\omega = \mathfrak{C}_{\mathrm{II}}^{NS}, \qquad c_\omega = (2C_1)^{-1}, \qquad \mathcal{R}_\omega = 0.$$

The divergent-cost item is exactly the validated compact cost-divergence conclusion. Hence all local cost-to-exclusion data hold. $\square$

Lemma 8.75 (Moving-cutoff monotonicity gives local cost-to-exclusion data). *Let ω be represented on a terminal tail by a repaired-gauge Navier–Stokes branch (V, P, a, b), and let $\mathfrak{C}_{\mathrm{II}}^{NS}$ be a nonnegative locally integrable validated compact cost. Let*

$$\Phi(\tau, y) = \phi(y/R(\tau))$$

be a moving cutoff as in Theorem 10.18, with $R(\tau) \geq R_0$ locally absolutely continuous and nondecreasing, and set

$$\mathcal{E}_R(\tau) = \frac{1}{2}\int |V(y,\tau)|^2\Phi(\tau, y)\,dy, \qquad \tilde{\mathfrak{D}}_{R(\tau)}(\tau) = \nu\int |\nabla V|^2\Phi + a_+(\tau)\int |V|^2\Phi.$$

Assume, after increasing τ_0 if necessary, that:

(i) $\mathcal{E}_R(\tau_0) < \infty$;
(ii) *the localized energy identity is valid with the time-dependent cutoff Φ;*
(iii) *the moving finite-error condition holds, for instance through summable moving-annulus errors:*
$$\int_{\tau_0}^{\infty} B_{\mathrm{mov}}(\tau)\,d\tau < \infty;$$
(iv) *the validated compact cost is controlled by the moving identity cost on the terminal tail: for some $C_m < \infty$,*
$$0 \leq \mathfrak{C}_{\mathrm{II}}^{NS}(\tau) \leq C_m\tilde{\mathfrak{D}}_{R(\tau)}(\tau) \qquad \text{for a.e. } \tau \geq \tau_0;$$
(v) *the validated compact cost-divergence conclusion is*
$$\int_{\tau_0}^{\infty} \mathfrak{C}_{\mathrm{II}}^{NS}(\tau)\,d\tau = \infty.$$

Then ω carries the local cost-to-exclusion data of Theorem 8.70 with $\mathcal{D}_\omega = \mathfrak{C}_{\mathrm{II}}^{NS}$.

Proof. The moving finite-error condition gives $B_{\mathrm{mov}} \in L^1(\tau_0, \infty)$; if it is verified through Theorem 10.20, this is precisely Theorem 10.21. Define

$$\mathcal{H}_\omega(\tau) = \mathcal{E}_R(\tau) + \int_{\tau}^{\infty} B_{\mathrm{mov}}(s)\,ds.$$

The finite moving-error condition makes the correction tail finite and locally absolutely continuous. Since $\mathcal{E}_R \geq 0$ and $B_{\mathrm{mov}} \geq 0$,

$$\inf_{\tau\geq\tau_0} \mathcal{H}_\omega(\tau) \geq 0, \qquad \mathcal{H}_\omega(\tau_0) < \infty.$$

Theorem 10.19 gives

$$\frac{d}{d\tau}\mathcal{H}_\omega(\tau) + \frac{1}{2}\tilde{\mathfrak{D}}_{R(\tau)}(\tau) \leq 0$$

in distributions. Thus $\tilde{\mathfrak{D}}_{R(\tau)}$ is a local monotonicity carrier for the validated compact cost, with $c_A = 1/2$, $C_A = C_m$, and $\mathcal{R}_\omega = 0$. Applying Theorem 8.73 gives the claimed local cost-to-exclusion data. □

Lemma 8.76 (Corrected monotonicity bounds the validated cost). *Let ω carry local cost-to-exclusion data in the sense of Theorem 8.70, except do not assume the divergence condition in item* (i). *Then*

$$\int_{\tau_0}^{\infty} \mathcal{D}_\omega(\tau)\,d\tau < \infty.$$

Proof. Since $\mathcal{H}_\omega$ is locally absolutely continuous and the corrected localized energy inequality holds in distributions, integration over $[\tau_0, T]$ gives, for every $T > \tau_0$,

$$\mathcal{H}_\omega(T) - \mathcal{H}_\omega(\tau_0) + c_\omega \int_{\tau_0}^{T} \mathcal{D}_\omega(\tau)\, d\tau + \int_{\tau_0}^{T} \mathcal{R}_\omega(\tau)\, d\tau \le 0.$$

The remainder is nonnegative, hence

$$c_\omega \int_{\tau_0}^{T} \mathcal{D}_\omega(\tau)\, d\tau \le \mathcal{H}_\omega(\tau_0) - \mathcal{H}_\omega(T) \le \mathcal{H}_\omega(\tau_0) - \inf_{\sigma \ge \tau_0} \mathcal{H}_\omega(\sigma).$$

The right-hand side is finite and independent of T. Letting $T \to \infty$ and using monotone convergence for the nonnegative cost proves the claim. $\square$

Theorem 8.77 (Local Navier–Stokes cost-to-exclusion theorem)**.** *Let ω be a renormalized Navier–Stokes Type II branch on a retained terminal compact stratum. If ω has active supercritical scaling, the validated compact Type II cost-divergence conclusion, and local cost-to-exclusion data of Theorem 8.70, then ω is excluded from the Type II class under consideration. Equivalently, on this local stratum,*

$$\Pi_{\mathrm{NS,II}}(\omega) : \mathcal{S}_\omega \wedge \mathcal{B}_\omega \Longrightarrow \mathcal{E}_\omega$$

is a theorem.

Proof. Assume, for contradiction, that the branch remains in the Type II class on the retained compact stratum. The local cost-to-exclusion data give a corrected localized energy inequality for the same nonnegative cost $\mathcal{D}_\omega$ whose divergent tail integral is the validated compact cost-divergence conclusion. By Theorem 8.76,

$$\int_{\tau_0}^{\infty} \mathcal{D}_\omega(\tau)\, d\tau < \infty.$$

This contradicts the validated compact cost-divergence conclusion. Hence the branch cannot remain in the Type II class under consideration. $\square$

Definition 8.78 (Pointwise cost-divergence exclusion data)**.** For a renormalized Type II solution ω, the pointwise cost-divergence exclusion data are available when the following local analytic statements hold:

(i) class membership: ω lies in the renormalized Navier–Stokes Type II class and satisfies the Type II analytic data before the Type II cost criterion is evaluated;
(ii) cost compatibility: the active compact Type II cost criterion uses either the Navier–Stokes identity-cost criterion
$$\mathfrak{C}_{\mathrm{II}}^{NS} = \tilde{\mathfrak{D}}_{R_0},$$
or a localized cost explicitly comparable to $\tilde{\mathfrak{D}}_{R_0}$ on a terminal tail in the sense of Theorem 12.26; in either case its conclusion is the validated compact Type II cost-divergence conclusion for ω;
(iii) stability of the active scaling condition: the active supercritical scaling condition remains in force after the repaired-gauge representation, cost comparison, and criterion evaluation are attached;
(iv) stability of the cost-divergence conclusion: every renormalized continuation preserving the active scaling condition and the validated compact Type II cost-divergence conclusion remains in the same compact Type II cost-divergence regime, unless it first enters one of the local alternatives already isolated above;

(v) local monotonicity-carrier data: on the retained compact stratum, the validated compact cost admits a carrier in the sense of Theorem 8.72. This may be supplied by the fixed or moving carrier subidentity of Theorem 8.101, the corresponding finite-error condition, and the corresponding cost comparison. Exact fixed or moving localized energy identities are stronger sufficient inputs through Theorems 8.74 and 8.75.

Lemma 8.79 (Canonical cost compatibility)**.** *Let ω be represented on a terminal tail by a repaired-gauge Navier–Stokes branch (V, P, a, b). Assume that the localized identity cost*

$$\tilde{\mathfrak{D}}_{R_0}(\tau) = \nu \int_{\mathbb{R}^3} |\nabla V(y,\tau)|^2 \phi_{R_0}(y)\, dy + a_+(\tau) \int_{\mathbb{R}^3} |V(y,\tau)|^2 \phi_{R_0}(y)\, dy$$

is measurable, nonnegative, and locally integrable on finite τ-intervals. If the compact Type II criterion is evaluated with the canonical Navier–Stokes identity cost, then

$$\mathfrak{C}_{\mathrm{II}}^{NS} = \tilde{\mathfrak{D}}_{R_0}$$

is a valid Type II scale-exclusion cost, the comparison holds with

$$c_0 = 1, \qquad \tau_1 = \tau_0,$$

and the validated compact cost-divergence conclusion for ω is the identity-cost divergence statement

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty.$$

If the compact criterion is not evaluated with this identity cost, the branch is not being treated by the canonical compatibility statement and must instead come with an explicit comparison as in Theorem 8.80.

Proof. The first assertion is the identity-cost case of the localized cost definitions. Since $\tilde{\mathfrak{D}}_{R_0}$ is assumed measurable, nonnegative, and locally integrable on finite intervals, the canonical choice

$$\mathfrak{C}_{\mathrm{II}}^{NS} := \tilde{\mathfrak{D}}_{R_0}$$

satisfies the defining properties of a Type II scale-exclusion cost. The comparison inequality is the identity

$$\mathfrak{C}_{\mathrm{II}}^{NS}(\tau) = \tilde{\mathfrak{D}}_{R_0}(\tau) \geq 1 \cdot \tilde{\mathfrak{D}}_{R_0}(\tau)$$

for a.e. $\tau \geq \tau_0$, so $c_0 = 1$ and $\tau_1 = \tau_0$. Therefore divergence of the active compact cost is precisely divergence of $\tilde{\mathfrak{D}}_{R_0}$ on the terminal tail. This is the canonical case of Theorems 12.27 and 12.28. ∎

Lemma 8.80 (Comparable cost compatibility)**.** *Let ω be represented on a terminal tail by a repaired-gauge Navier–Stokes branch (V, P, a, b). Let $\mathfrak{C}_{\mathrm{II}}^{NS}$ be a nonnegative, measurable, locally integrable localized cost on finite τ-intervals. Assume that, on a later terminal tail $[\tau_1, \infty)$, there is a constant $c_0 > 0$ such that*

$$\mathfrak{C}_{\mathrm{II}}^{NS}(\tau) \geq c_0\, \tilde{\mathfrak{D}}_{R_0}(\tau) \quad \textit{for a.e. } \tau \geq \tau_1.$$

Then every identity-cost divergence conclusion

$$\int_{\tau_1}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty$$

implies the corresponding compact Type II cost-divergence conclusion

$$\int_{\tau_1}^{\infty} \mathfrak{C}_{\mathrm{II}}^{NS}(\tau)\, d\tau = \infty.$$

Consequently the use of $\mathfrak{C}_{\mathrm{II}}^{NS}$ introduces no additional exit from the compact-cost stratum beyond the explicit failure of this comparison.

Proof. For every $T > \tau_1$, the assumed comparison gives

$$\int_{\tau_1}^{T} \mathfrak{C}_{\mathrm{II}}^{NS}(\tau)\, d\tau \geq c_0 \int_{\tau_1}^{T} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau.$$

Letting $T \to \infty$ proves the implication from identity-cost divergence to divergence of the selected compact Type II cost. This is precisely the one-sided comparison needed by Theorem 12.26 and used in Theorem 12.28. If the inequality is unavailable, the selected diagnostic cost is not connected to the Navier–Stokes identity dissipation on this terminal tail; this is a cost-comparison failure, not a new unlabelled Type II branch. □

Lemma 8.81 (Local cost well-posedness)**.** *Let (V, P, a, b) be a repaired-gauge branch on a finite renormalized interval $I = [\alpha, \beta]$, and let $\phi_{R_0} \in C_c^\infty(B_{R_0})$, $\phi_{R_0} \geq 0$. Assume:*

(i) *$V \in L^2(I; H^1(B_{R_0}))$;*
(ii) *$a \in L^1(I)$;*
(iii) *either $V \in L^\infty(I; L^3(\mathbb{R}^3))$, or more locally*

$$\operatorname*{ess\,sup}_{\tau\in I} \|V(\tau)\|_{L^2(B_{R_0})} < \infty.$$

Then the localized identity cost

$$\tilde{\mathfrak{D}}_{R_0}(\tau) = \nu \int |\nabla V(y,\tau)|^2 \phi_{R_0}(y)\, dy + a_+(\tau) \int |V(y,\tau)|^2 \phi_{R_0}(y)\, dy$$

is measurable, nonnegative, and integrable on I. Consequently, on every compact renormalized window satisfying these local bounds, $\tilde{\mathfrak{D}}_{R_0}$ is a well-defined finite-interval cost.

Proof. The two summands are measurable because V is strongly measurable in the local Sobolev spaces on the finite interval and a_+ is measurable. They are nonnegative since $\nu > 0$, $\phi_{R_0} \geq 0$, and $a_+ \geq 0$.

For the gradient term, boundedness of ϕ_{R_0} and the support condition give

$$\int_I \int |\nabla V|^2 \phi_{R_0} \leq \|\phi_{R_0}\|_{L^\infty} \int_I \|\nabla V(\tau)\|_{L^2(B_{R_0})}^2\, d\tau < \infty.$$

For the drift term, first suppose

$$\operatorname*{ess\,sup}_{\tau\in I} \|V(\tau)\|_{L^2(B_{R_0})} < \infty.$$

Then

$$\int_I a_+(\tau) \int |V|^2 \phi_{R_0}\, dy\, d\tau \leq \|\phi_{R_0}\|_{L^\infty} \left(\operatorname*{ess\,sup}_{\tau\in I} \|V(\tau)\|_{L^2(B_{R_0})}^2 \right) \|a_+\|_{L^1(I)} < \infty.$$

If instead $V \in L^\infty(I; L^3(\mathbb{R}^3))$, then Hölder on B_{R_0} gives the local L^2 bound

$$\|V(\tau)\|_{L^2(B_{R_0})}^2 \leq |B_{R_0}|^{1/3} \|V(\tau)\|_{L^3(\mathbb{R}^3)}^2$$

for a.e. $\tau \in I$, reducing to the previous case. Thus both summands are integrable on I. □

Lemma 8.82 (Local modulation integrability from absolutely continuous repaired gauges)**.** *Let u, p be a Navier–Stokes solution on a terminal physical time interval, and let (V, P) be its repaired-gauge renormalization on a terminal renormalized tail. Assume that, on every compact renormalized subinterval, the physical concentration chart and the repaired-gauge path satisfy Theorems 10.45 and 10.47. Let a, b be the final modulation coefficients in the repaired-gauge renormalized equation, as in Theorem 10.49. Then*

$$a, b \in L^1_{\mathrm{loc}}([\tau_0, \infty)).$$

In particular, for every unit terminal window

$$I_n = [\tau_0 + n, \tau_0 + n + 1],$$

one has $a \in L^1(I_n)$*. If the absolutely continuous chart, the absolutely continuous repaired gauge, or the resulting local integrability of the modulation coefficients is unavailable on such a compact window, then the branch does not lie in the retained compact-cost stratum and falls into the corresponding representation, repaired-gauge, or modulation-integrability alternative.*

Proof. On each compact renormalized window, the assumed absolutely continuous chart and repaired-gauge path place the branch under the hypotheses of Theorem 10.54. In particular, Theorem 10.53 gives

$$a, b \in L^1_{\mathrm{loc}},$$

and Theorem 10.68 identifies these locally integrable functions as the final coefficients in the repaired-gauge equation. Restricting to I_n gives $a \in L^1(I_n)$. If the absolutely continuous local chart, repaired-gauge path, or resulting coefficient integrability is missing, then the ordered stratification records the first missing input as a representation, repaired-gauge, or modulation-integrability failure. $\square$

Lemma 8.83 (Local state data give local L^2 windows)**.** *Let* (V, P, a, b) *be a repaired-gauge Type II branch on a terminal tail. Fix* $R_0 < \infty$ *and a unit terminal window* I_n*. Suppose either*

$$\operatorname*{ess\,sup}_{\tau \in I_n} \|V(\tau)\|_{L^2(B_{R_0})} < \infty,$$

as supplied by the local suitable-energy class on compact repaired-gauge cylinders, or the positive finite critical annulus of Theorem 12.21 holds on the terminal tail:

$$\|V(\tau)\|_{L^3(\mathbb{R}^3)} \le M, \qquad \tau \ge \tau_0.$$

Then

$$\operatorname*{ess\,sup}_{\tau \in I_n} \|V(\tau)\|_{L^2(B_{R_0})}$$

is finite. Thus the local L^2*-input required for the finite-window identity cost is available on every retained compact cylinder either from the local energy class or, as a sufficient bookkeeping route, from the global critical-annulus test. Failure of both local* L^2 *control on the selected compact cylinder and the finite critical-annulus route is recorded by the ordered local validation as a local-energy, rough-core, or critical-mass exit.*

Proof. If the displayed local $L^\infty_\tau L^2_y(B_{R_0})$ bound is part of the retained local state data, there is nothing to prove. This is the case used in the physical local no-Type-II route: the repaired-gauge pullback of a suitable weak solution belongs to the local energy class on compact cylinders, and the ordered validation records absence of that bound as a local-energy or rough-core failure rather than repairing it by a global estimate.

In the optional critical-annulus route, for a.e. $\tau \in I_n$, Holder's inequality on the finite-measure ball gives

$$\|V(\tau)\|_{L^2(B_{R_0})} \le |B_{R_0}|^{1/6} \|V(\tau)\|_{L^3(B_{R_0})} \le |B_{R_0}|^{1/6} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \le |B_{R_0}|^{1/6} M.$$

Taking the essential supremum over I_n proves finiteness in this second case. The final assertion is exactly the first-failure convention in Theorem 8.86: a missing compact-cylinder local energy bound is not kept inside the retained compact-cost branch. $\square$

Corollary 8.84 (Compact-window data give cost well-posedness)**.** *Let* (V, P, a, b) *be a repaired-gauge branch on a terminal tail. Fix* $R_0 < \infty$*. Suppose that, for every unit terminal window*

$$I_n = [\tau_0 + n, \tau_0 + n + 1], \qquad n \in \mathbb{N},$$

the branch has the local compact-cylinder $L^\infty_\tau L^2_y(B_{R_0})$ *input of* Theorem 8.83 *and local windowed* $H^1(B_{R_0})$ *control. Suppose also that the local modulation-integrability conclusion of* Theorem 8.82 *holds. Then* $\tilde{\mathfrak{D}}_{R_0}$ *is a measurable nonnegative cost, integrable on each* I_n*. If the branch also has the identity cost or an explicitly comparable localized cost, then the compact Type II cost criterion can be evaluated on every retained terminal window.*

The same conclusion holds by the older sufficient route when the positive finite critical annulus supplies the compact-cylinder L^2 *input through* Theorem 8.83*. Failure of the conclusion is exactly one of the local failures in the hypotheses: local compact-cylinder* L^2 *control, windowed* H^1 *control, modulation-integrability, or cost-comparison failure.*

Proof. Theorem 8.83 supplies the $L^\infty(I_n; L^2(B_{R_0}))$ input on every unit window, and Theorem 8.82 supplies $a \in L^1(I_n)$. The assumed windowed $H^1(B_{R_0})$ control gives

$$V \in L^2(I_n; H^1(B_{R_0})).$$

Thus the hypotheses of Theorem 8.81 hold on each I_n. The identity-cost case is then covered by Theorem 8.79; the explicitly comparable case is covered by Theorem 8.80. Hence the criterion is locally meaningful on every retained terminal window. Conversely, if the conclusion fails before criterion evaluation, at least one of the listed local inputs is absent; by the ordered stratification, that absence is recorded as the corresponding local alternative. □

Proposition 8.85 (Active scaling stability criterion)**.** *Let* ω *be an admissible local Type II branch with active supercritical scaling on a terminal tail. A passage from a raw Type II chart to a repaired-gauge chart is called scale-admissible on a later tail if its raw and repaired physical scales satisfy*

$$0 < m \le \frac{\lambda_{\mathrm{rep}}(t)}{\lambda_{\mathrm{raw}}(t)} \le M < \infty$$

on that tail. On the compact-cost stratum, the active supercritical scaling condition is unchanged by the following operations:

(i) *restriction to a later terminal tail;*
(ii) *passage from an admissible raw Type II chart to a scale-admissible repaired-gauge representation;*
(iii) *replacement of the canonical cost by an explicitly comparable localized cost;*
(iv) *evaluation of the compact Type II cost criterion.*

If the repaired chart is not scale-admissible on any later tail, the branch is not retained in the compact-cost stratum; it is assigned instead to the corresponding representation, repaired-gauge, or scale-degeneracy alternative.

Proof. The active supercritical scaling condition is a terminal property of the scale-center chart and is invariant under removal of a finite initial renormalized interval. Thus restriction from $[\tau_0, \infty)$ to $[\tau_1, \infty)$, $\tau_1 \ge \tau_0$, cannot change the terminal scale alternative.

Suppose next that a scale-admissible repaired-gauge representation is available. Two-sided comparability of λ_{rep} and λ_{raw} implies that any terminal statement depending only on the scale class, such as supercriticality relative to the Type I rate, holds for one scale if and only if it holds for the other after passing to the same tail. The repaired gauge changes coordinates and modulation variables inside this comparable scale class; it does not replace the branch by a different terminal scaling alternative. If the representation cannot be constructed, or if every repaired chart loses scale comparability, then the first failure is a representation, repaired-gauge, or scale-degeneracy alternative.

Cost comparison is scale-passive. It changes only the nonnegative diagnostic functional used to record cost divergence and does not alter the center, physical scale, renormalized time, or active scaling hypothesis. If the comparison is unavailable, the failure is cost-comparison failure. Finally, criterion evaluation is a logical operation applied to the already constructed branch and cost. It does not modify the branch or its scale. Hence active scaling is stable through the listed operations exactly on the local stratum on which those operations are admissible. □

Lemma 8.86 (No unrecorded exit from compact cost divergence)**.** *Let ω be an admissible local Type II branch with active supercritical scaling on a terminal tail. Before declaring that a continuation lies in the compact cost-divergence stratum, run the ordered local validation tests:*

(i) *Type II analytic data and exhaustion data;*
(ii) *raw-chart selection, repaired-gauge representation, and scale admissibility;*
(iii) *local compactness decomposition into the alternatives of Theorem 11.13;*
(iv) *retained finite critical-mass window, read either as the compact positive finite critical L^3 annulus or, in the physical local route, as the compact-window finite critical-mass input of Theorem 10.82;*
(v) *the critical-tightness/exterior-leakage test at the retained scale;*
(vi) *local windowed H^1 control;*
(vii) *local modulation integrability, cost well-posedness, and compatibility of the chosen compact Type II cost;*
(viii) *local monotonicity-carrier data for the validated compact cost, supplied by the fixed-cutoff criterion, the moving-cutoff criterion, or another explicit localized carrier;*
(ix) *the compact Type II cost-divergence conclusion for the validated cost.*

Then exactly one of the following occurs:

(a) *all tests pass, and ω remains in the compact Type II cost-divergence regime on the retained terminal branch;*
(b) *the first failed test gives one of the already named local alternatives: ordered exhaustion failure, representation failure, repaired-gauge or scale-degeneracy failure, multibubble/multicore splitting, cascade, exterior tightness failure, loss of local windowed H^1 control, finite-cost scale-collapse alternative, scale-rigid residual alternative, critical-mass alternative, modulation-integrability failure, cost-compatibility failure, carrier-subidentity failure, finite fixed-cutoff error failure, summable moving-annulus error failure, carrier-comparison failure for the validated cost, or failure of compact cost divergence for the validated cost.*

Thus a renormalized continuation preserving active scaling and validated compact cost divergence cannot leave the compact cost-divergence regime without being recorded by the local compactness stratification.

Proof. The proof is the ordered-alternative argument. First test whether the Type II analytic and exhaustion data hold. If not, the first failure is one of the ordered exhaustion failures of Theorem 12.3. If they hold, test for raw-chart selection, repaired-gauge representation, and scale admissibility. Failure at this stage is one of the ordered representation failures of Theorem 12.12, possibly with the specific scale-degeneracy recorded by Theorem 8.85.

Once a scale-admissible repaired branch is available, apply the local compactness decomposition Theorem 11.13. If the branch falls into the multibubble, multicore, cascade, gauge-degenerate, finite-cost scale-collapse, local H^1-loss, or scale-rigid residual alternatives, it has left the compact single-core cost-divergence stratum through a named local case. In particular, an additional exterior concentration mechanism is not discarded; if it retains critical mass it appears as a multibubble or multicore branch, and if it escapes the retained compact region it appears as failure of critical tightness.

The first alternative in Theorem 11.13 is retained only as a compact single-core branch; it is not yet identified with compact cost divergence. On that retained branch the remaining tests below decide whether the branch belongs to the compact cost-divergence stratum, has finite validated cost, or exits through one of the remaining compact tests.

On the retained compact single-core branch, test the finite critical-mass row. For the compact-cost bookkeeping criterion this is the ordered critical L^3 annulus test of Theorem 12.22. For the physical local route it is the compact-window version of Theorem 10.82: lower active mass failure means non-retention, and failure of a finite local upper bound produces a comparable retained label or a named adjacent active label. Thus failure of the finite critical-mass row is recorded as a critical-mass or adjacent local alternative, not as a hidden global estimate. Once the row passes, the remaining compact tests are critical tightness and local windowed H^1 control. Failure of either is precisely one of the alternatives in Theorem 12.30.

If all compact tests pass, local cost well-posedness and compatibility are evaluated. Local modulation integrability is supplied by Theorem 8.82; the local cost well-posedness is supplied by Theorem 8.84. The compatibility is the identity case of Theorem 8.79 or the explicit comparison case of Theorem 8.80; failure of both is a cost-compatibility failure. Next test whether the validated compact cost has a local monotonicity carrier. In the fixed-cutoff case this means the fixed carrier subidentity, finite fixed-cutoff error, and energy-control comparison. In the moving-cutoff case this means the moving carrier subidentity, summable moving-annulus errors, and moving-cost comparison. Exact localized energy identities are sufficient for these subidentities, but the one-sided suitable local energy inequality is the actual input needed. Failure of these inputs is recorded as a carrier-subidentity failure, a finite fixed-cutoff error failure, a summable moving-annulus error failure, or a carrier-comparison failure for the validated cost. Finally, evaluate the compact cost-divergence conclusion for the validated cost. If it fails, the branch is a finite-cost or non-divergent compact branch and is not counted as a compact cost-divergence branch. The tests are ordered, so the first failure is unique. If no failure occurs, the branch has the analytic data, scale-admissible repaired-gauge representation, compact single-core local compactness position, the retained finite critical-mass window, critical tightness, local windowed H^1 control, local modulation integrability, a compatible compact cost, local monotonicity-carrier data, and the validated compact cost-divergence conclusion. Together with the assumed active scaling, these hypotheses place the branch in the compact Type II cost-divergence regime. Therefore there is no unrecorded exit. □

Theorem 8.87 (Carrier validation alternative on the retained compact stratum)**.** *Let ω be an admissible local Type II branch with active supercritical scaling. Suppose that the ordered validation tests in Theorem 8.86 have reached the retained compact single-core branch, that cost well-posedness and cost compatibility have passed, and that the validated compact cost-divergence conclusion holds. Then exactly one of the following alternatives occurs:*

(i) *the validated compact cost admits local monotonicity-carrier data in the sense of Theorem 8.72;*
(ii) *the branch leaves the retained compact cost-divergence stratum through the first failed carrier-validation test: carrier-subidentity failure, finite fixed-cutoff error failure, summable moving-annulus error failure, or carrier-comparison failure for the validated cost.*

Consequently, on the retained compact cost-divergence stratum, where the adjacent carrier-failure alternatives are not part of the stratum, the validated compact cost admits local monotonicity-carrier data.

Proof. After the branch has passed the Type II analytic, representation, compactness, retained finite critical mass, critical-tightness, local windowed H^1, modulation, cost well-posedness, and cost-compatibility tests, the next ordered test in Theorem 8.86 is precisely the local monotonicity-carrier test for the validated compact cost. If this test passes, alternative (i) holds. If it fails, the first failed input is recorded as one of the named adjacent carrier failures listed in Theorem 8.86: failure of the carrier subidentity, failure of the fixed finite-error condition, failure of the moving finite-error condition through the summable moving-annulus test, or failure of the carrier comparison for the validated cost. The ordered first-failure convention makes the two alternatives disjoint and exhaustive.

By definition, the retained compact cost-divergence stratum keeps only the branch for which all preceding compact tests and the carrier-validation test have passed before the validated compact cost-divergence conclusion is attached. Therefore a branch that remains in this stratum cannot realize one of the carrier-failure alternatives, and the local monotonicity carrier exists. □

Proposition 8.88 (Local verification of the cost-exclusion data)**.** *On a retained terminal branch in the compact Type II cost-divergence stratum, the pointwise data of Theorem 8.78 are local analytic statements, not additional global assumptions. More precisely, assume that the branch:*

- (i) *has the local Type II analytic data and has not entered an exhaustion, representation, modulation-integrability, multibubble/multicore, cascade, exterior-tightness, scale-collapse, scale-rigid, critical-mass, or local H^1-loss alternative;*
- (ii) *admits a scale-admissible repaired-gauge representation on retained windows from absolutely continuous local scale-center and repaired-gauge data;*
- (iii) *has the retained finite critical-mass window, critical tightness at the retained scale, local windowed H^1 control, active supercritical scaling, and the compact cost-divergence conclusion for the validated cost;*
- (iv) *uses either the canonical identity cost $\widetilde{\mathfrak{D}}_{R_0}$ or an explicitly comparable localized cost;*
- (v) *remains in the retained compact cost-divergence stratum after the carrier-validation alternatives of Theorem 8.87 have been evaluated.*

Then class membership, cost compatibility, stability of the active scaling condition, stability of the validated cost-divergence conclusion, and the cost-to-exclusion implication all hold for this branch. Consequently, on this local stratum, all clauses of Theorem 8.78 are verified; no estimate outside the retained local stratum and no exclusion of secondary exterior concentration are being assumed.

Proof. Class membership is the retained local Type II analytic-data alternative after the ordered exhaustion and representation tests have passed. The local inputs needed to evaluate the compact cost are available by Theorems 8.82 and 8.83 and Theorem 8.84. Cost compatibility then follows from Theorem 8.79 in the identity-cost case and from Theorem 8.80 in the explicitly comparable-cost case. Stability of the active scaling condition is Theorem 8.85. Stability of the validated cost-divergence conclusion, with all exits assigned to named local alternatives, is Theorem 8.86. These prove clauses (i)–(iv) of Theorem 8.78. Since the branch remains in the retained compact cost-divergence stratum after carrier validation, Theorem 8.87 gives local monotonicity-carrier data for the validated compact cost. These data provide the local cost-to-exclusion data through Theorem 8.73. Then Theorem 8.77 gives $\Pi_{\mathrm{NS,II}}(\omega)$, which is the analytic content of clause (v). □

Definition 8.89 (Classwise retained compact coverage)**.** The renormalized Type II class has classwise retained compact coverage for the validated compact cost if every branch with active supercritical scaling and validated compact cost divergence either remains in the retained compact cost-divergence stratum after the ordered validation tests of Theorem 8.86, or has already entered one of the named adjacent alternatives in that ordered stratification.

Theorem 8.90 (Classwise retained compact coverage from ordered validation)**.** *Every admissible local Type II branch with active supercritical scaling and validated compact Type II cost divergence has classwise retained compact coverage in the sense of Theorem 8.89.*

Proof. Apply the ordered validation tests of Theorem 8.86 to the branch. The lemma gives two possibilities. If all tests pass, the branch remains in the compact Type II cost-divergence regime on the retained terminal branch, which is precisely the retained compact cost-divergence stratum.

If a test fails, the first failed test is one of the named local alternatives in Theorem 8.86. Since the present theorem assumes the validated compact cost-divergence conclusion, the last listed failure, failure of compact cost divergence for the validated cost, cannot occur. Every other first failure is an adjacent compactness stratum: exhaustion, representation, repaired-gauge or scale-degeneracy failure, multibubble/multicore splitting, cascade, exterior tightness failure, local H^1-loss, finite-cost scale-collapse, scale-rigid residual behavior, critical-mass failure, modulation-integrability failure, cost-compatibility failure, carrier-subidentity failure, fixed-cutoff finite-error failure, moving-annulus summability failure, or carrier-comparison failure. Therefore every branch satisfying the hypotheses either remains retained or has already entered a named adjacent alternative. This is exactly classwise retained compact coverage. □

Remark 8.91 (Scope of classwise retained compact coverage)*.* Theorem 8.88 is deliberately local. It proves the interface hypotheses for a branch only after the branch has been placed in the validated compact cost-divergence stratum by the compactness stratification. Branches with multibubble, cascade, exterior leakage, scale-collapse, scale-rigid, or rough-core behavior are not forced into this stratum; they are assigned to their own local alternatives. Thus the cost-divergence exclusion step is unconditional on the retained compact stratum, while a classwise statement is obtained only after the adjacent strata have been treated by their own local arguments. The coverage part itself is no longer an assumption; it is supplied by Theorem 8.90.

Definition 8.92 (Abstract cost-divergence exclusion theorem applicability)**.** An abstract Type II cost-divergence exclusion theorem is applicable to a renormalized Type II solution if the following analytic identifications are available:

(i) an abstract Type II exclusion theorem based on localized renormalization-cost divergence;
(ii) hypotheses supplying supercritical scaling, a Type II concentration scenario with bounded physical energy, and a localized energy monotonicity formula at the active scale $\lambda(t)$;
(iii) an identification of the theorem's renormalization-cost integral with a validated compact Navier–Stokes cost $\mathfrak{C}_{\mathrm{II}}^{NS}$, where either

$$\mathfrak{C}_{\mathrm{II}}^{NS} = \tilde{\mathfrak{D}}_{R_0}$$

or $\mathfrak{C}_{\mathrm{II}}^{NS}$ is a localized cost satisfying the terminal one-sided comparison with $\tilde{\mathfrak{D}}_{R_0}$ in Theorem 12.26;
(iv) a domain embedding placing the repaired-gauge Navier–Stokes orbit and its scale variables into the parabolic domain expected by the theorem without changing the active scaling condition, validated cost-divergence conclusion, or exclusion conclusion;
(v) identification of the theorem's conclusion with Theorem 8.69.

Lemma 8.93 (Applicability of the abstract cost-divergence theorem)**.** *Assume a renormalized Type II solution has the active supercritical scaling condition, the validated compact Type II cost-divergence conclusion, and the abstract applicability data of Theorem 8.92. Then the solution is excluded from the Type II class under consideration.*

Proof. The abstract Type II theorem is applied with supercritical scaling, bounded-energy Type II concentration, and the localized energy monotonicity formula at the active scale. The cost identification places the validated compact Navier–Stokes cost in the theorem's renormalization-cost integral. In the identity case this cost is $\tilde{\mathfrak{D}}_{R_0}$. In the comparable-cost case, the terminal one-sided comparison in Theorem 12.26 and Theorem 8.80 show that the validated Navier–Stokes cost-divergence conclusion implies divergence of the localized cost used by the abstract theorem. The domain embedding places the repaired-gauge Navier–Stokes solution in the theorem's parabolic domain, and the conclusion is identified with Theorem 8.69. Therefore the theorem gives exclusion from the Type II class. □

Theorem 8.94 (Pointwise Navier–Stokes cost-divergence exclusion)**.** *Let ω be a renormalized Type II solution. If ω has the active supercritical scaling condition, the validated compact Type II cost-divergence conclusion, and the pointwise data of Theorem 8.78, then ω is excluded from the Type II class under consideration.*

Proof. Class membership and stability of the active scaling condition ensure that ω is evaluated in the renormalized Type II class and that the active supercritical scaling condition is still the relevant one. Cost compatibility ensures that the validated compact cost-divergence conclusion is the Navier–Stokes compact cost conclusion for ω, not an external or incomparable cost statement. Stability of the cost-divergence conclusion rules out an exit from the compact Type II cost-divergence regime to a Type II case outside the compact cost-divergence criterion inside the same class.

It remains only to justify the transition from cost divergence to exclusion. The local monotonicity-carrier data in Theorem 8.78 give the local cost-to-exclusion data by Theorem 8.73. Therefore Theorem 8.77 applies to the active supercritical scaling condition and the validated compact cost-divergence conclusion, giving the claimed exclusion. In an abstract theorem setting, the same implication is also licensed by Theorem 8.93. □

Corollary 8.95 (Cost-divergence exclusion on the retained compact stratum)**.** *Let ω be a renormalized Type II solution with active supercritical scaling and validated compact Type II cost divergence. If ω remains in the retained compact cost-divergence stratum after the ordered validation tests of Theorem 8.86, then ω is excluded from the Type II class under consideration.*

Proof. By Theorem 8.88, the pointwise cost-divergence exclusion data of Theorem 8.78 are verified on this retained stratum. Applying Theorem 8.94 gives the exclusion conclusion. □

Definition 8.96 (Adjacent-stratum closure for the compact-cost regime)**.** The compact-cost regime has adjacent-stratum closure if every named adjacent alternative produced by Theorem 8.86 for a branch with active supercritical scaling and validated compact Type II cost divergence has its own stratum-local argument giving the same Type II exclusion conclusion, either directly or by placing the branch in a previously listed non-compact-cost alternative whose own closure theorem already gives the Type II exclusion conclusion.

Corollary 8.97 (Uniform Navier–Stokes cost-divergence exclusion)**.** *Assume adjacent-stratum closure for the compact-cost regime in the sense of Theorem 8.96. Then every renormalized Type II solution with active supercritical scaling condition and validated compact Type II cost-divergence conclusion is either excluded in its adjacent stratum or excluded by Theorem 8.95.*

Proof. By Theorem 8.90, a branch with active supercritical scaling and validated compact cost divergence either lies in one of the named adjacent strata or remains in the retained compact cost-divergence stratum. The adjacent cases are handled by the adjacent-stratum closure hypothesis. In the retained compact case, Theorem 8.95 applies. □

Definition 8.98 (Adjacent exits from cost-divergence exclusion)**.** If the active supercritical scaling condition and validated compact cost-divergence conclusion are present but the branch is not retained in the compact cost-divergence stratum, then the first exit from the cost-exclusion argument lies in the ordered list:

class membership, cost compatibility, stability of the active scaling condition,
stability of the validated cost-divergence conclusion, local monotonicity-carrier validation.

On the retained compact stratum none of these exits occurs. The final item is expanded by Theorem 8.87 into the carrier-subidentity, fixed-finite-error, moving-annulus, and carrier-comparison adjacent strata.

Lemma 8.99 (Carrier exits are adjacent strata, not retained compact branches)**.** *Assume a renormalized Type II solution already has the active supercritical scaling condition and the validated compact Type II cost-divergence conclusion. If one of the exits in Theorem 8.98 occurs, then the branch is not in the retained compact cost-divergence stratum. If no such exit occurs, the local cost-to-exclusion data are available.*

Proof. The retained compact cost-divergence stratum is reached only after the ordered tests of Theorem 8.86 have passed. Hence a failure of class membership, cost compatibility, active-scaling stability, cost-divergence stability, or carrier validation is by definition an exit to an adjacent stratum. Conversely, if no such exit occurs, the carrier-validation test has passed. By Theorem 8.87, the validated compact cost admits local monotonicity-carrier data, and then Theorem 8.73 gives the local cost-to-exclusion data. □

Definition 8.100 (Pre-carrier validated compact-cost regime)**.** A renormalized Type II branch lies in the pre-carrier validated compact-cost regime if, on a terminal tail, it has active supercritical scaling and the validated compact Type II cost-divergence conclusion, and the ordered validation tests of Theorem 8.86 have passed through the following stages:

(i) the Type II analytic data and ordered exhaustion tests;
(ii) raw-chart selection, repaired-gauge representation, pressure reconstruction, modulation data, and scale-admissibility of the repaired chart;
(iii) the local compactness decomposition without entering a multibubble, multicore, cascade, exterior-tightness, local H^1-loss, finite-cost scale-collapse, or scale-rigid residual alternative;
(iv) the retained finite critical-mass window: in the compact-cost bookkeeping route this may be the positive finite critical L^3 annulus of Theorem 12.21, while in the physical local route it is the compact-window finite critical-mass input supplied by Theorem 10.82;
(v) critical tightness at the retained scale and local windowed H^1 control;
(vi) local modulation integrability, local cost well-posedness, and compatibility of the chosen compact Type II cost with the validated Navier–Stokes cost.

This regime deliberately stops immediately before local monotonicity-carrier validation. Thus it contains all compact-cost interface data needed to ask for the fixed-cutoff or moving-cutoff carrier estimates, but it does not assert that those estimates have already been proved. The finite critical-mass row is read locally in the final no-Type-II assembly: failure of the whole-space critical annulus is not repaired by assuming a global L^3 bound; it is routed by the local terminal critical-mass and exterior/state-space alternatives.

Definition 8.101 (Fixed and moving carrier subidentities)**.** For the fixed cutoff ϕ_{R_0}, the fixed carrier subidentity is the distributional estimate

$$\frac{d}{d\tau}\mathcal{E}_\phi(\tau) + \frac{1}{2}\tilde{\mathfrak{D}}_{R_0}(\tau) \le B_{R_0}(\tau), \qquad \mathcal{E}_\phi(\tau) = \frac{1}{2}\int |V|^2\phi_{R_0}.$$

For a moving cutoff $\Phi(\tau, y) = \phi(y/R(\tau))$, the moving carrier subidentity is the distributional estimate

$$\frac{d}{d\tau}\mathcal{E}_R(\tau) + \frac{1}{2}\tilde{\mathfrak{D}}_{R(\tau)}(\tau) \le B_{\mathrm{mov}}(\tau), \qquad \mathcal{E}_R(\tau) = \frac{1}{2}\int |V|^2\Phi.$$

Here B_{R_0} and B_{mov} are the absolute error densities of Theorems 10.9 and 10.18. The subidentity is the one-sided estimate needed for corrected monotonicity; an exact localized energy identity is sufficient but not necessary.

Lemma 8.102 (Suitable local energy gives carrier subidentities)**.** *Assume the repaired-gauge representation of a pre-carrier branch includes the local suitable energy inequality on compact renormalized cylinders. Then the fixed carrier subidentity holds for every nonnegative compactly supported fixed cutoff. The moving carrier subidentity holds for every nonnegative finite-valued moving cutoff $\Phi(\tau, y) = \phi(y/R(\tau))$ with $R \in AC_{\mathrm{loc}}$, after the usual approximation of R by smooth positive radii on compact τ-intervals.*

Proof. Apply the local suitable energy inequality in the physical variables to the pullback of the nonnegative renormalized test function. The repaired-gauge change of variables is locally absolutely continuous in time and smooth in space on compact cylinders, so the distributional chain rule gives the renormalized local energy inequality with the drift terms $a(V + y\cdot\nabla V)$ and $b\cdot\nabla V$. For a fixed cutoff this is the fixed localized energy identity with equality replaced by $\le$. Adding $\frac{1}{2}\tilde{\mathfrak{D}}_{R_0}$ to both sides and bounding the cutoff, pressure, translation, and scale terms by their absolute values gives the fixed carrier subidentity.

For a moving cutoff, use smooth positive approximations $R_k \to R$ in AC_{loc} on each compact time interval. The local energy inequality applies to $\Phi_k(\tau, y) = \phi(y/R_k(\tau))$, and the preceding argument gives the moving subidentity with $B_{\mathrm{mov},k}$. Passing to the limit in distributions uses the local finite-energy and local H^1 bounds already present on the pre-carrier regime. The term $\partial_\tau\Phi$ is interpreted as the L^1_{loc} derivative coming from $R \in AC_{\mathrm{loc}}$. The limit is exactly the moving carrier subidentity. ☐

Definition 8.103 (Fixed-cutoff localized-energy closure criterion)**.** A branch in the pre-carrier validated compact-cost regime satisfies the fixed-cutoff localized-energy closure criterion if, for the cutoff ϕ_{R_0} used to evaluate the validated compact cost and for

$$\mathcal{E}_\phi(\tau) = \frac{1}{2}\int |V(y,\tau)|^2\phi_{R_0}(y)\,dy,$$

the following local analytic statements hold on a terminal tail:

(i) $\mathcal{E}_\phi(\tau_0) < \infty$;
(ii) the fixed carrier subidentity of Theorem 8.101 is valid for the repaired-gauge branch;
(iii) the fixed-cutoff error density B_{R_0} of Theorem 10.9 has finite tail integral,

$$\int_{\tau_0}^{\infty} B_{R_0}(\tau)\,d\tau < \infty;$$

(iv) the validated compact cost is energy-controlled by $\tilde{\mathfrak{D}}_{R_0}$ in the sense of Theorem 8.71.

Definition 8.104 (Moving-cutoff localized-energy closure criterion)**.** A branch in the pre-carrier validated compact-cost regime satisfies the moving-cutoff localized-energy closure criterion if there is a finite-valued nondecreasing locally absolutely continuous radius $R(\tau) \ge R_0$, with $R(\tau) \to \infty$ and moving cutoff $\Phi(\tau, y) = \phi(y/R(\tau))$, such that, on a terminal tail:

(i)

$$\mathcal{E}_R(\tau_0) = \frac{1}{2}\int |V(y,\tau_0)|^2\Phi(\tau_0, y)\,dy < \infty;$$

(ii) the moving carrier subidentity of Theorem 8.101 is valid with the time-dependent cutoff Φ;

(iii) the moving-annulus errors are summable in the sense of Theorem 10.20, which implies

$$\int_{\tau_0}^{\infty} B_{\mathrm{mov}}(\tau)\, d\tau < \infty$$

by Theorem 10.21;

(iv) the validated compact cost is controlled by the moving identity cost on the terminal tail: for some $C_m < \infty$,

$$0 \le \mathfrak{C}_{\mathrm{II}}^{NS}(\tau) \le C_m \tilde{\mathfrak{D}}_{R(\tau)}(\tau) \qquad \text{for a.e. } \tau \ge \tau_0.$$

Theorem 8.105 (Non-carrier adjacent exits are identified before carrier validation)**.** *Let ω be an admissible local Type II branch with active supercritical scaling and validated compact Type II cost divergence. Apply the ordered tests of Theorem 8.86. If a first failed test exists and is not one of the carrier-validation failures*

carrier subidentity, fixed finite error, moving-annulus summability, carrier comparison,

then the branch has already failed to enter the pre-carrier validated compact-cost regime of Theorem 8.100. More precisely, its first failure is one of the following named non-carrier strata:

ordered exhaustion failure, representation failure, repaired-gauge or scale-degeneracy failure,

multibubble or multicore splitting, cascade, exterior-tightness failure, local H^1-loss,

finite-cost scale-collapse, scale-rigid residual behavior, critical-mass failure,

modulation-integrability failure, cost-well-posedness failure, or cost-compatibility failure.

In any ambient class in which the corresponding local closure theorems for these named strata are imposed, such a branch is not a remaining compact-cost branch.

Proof. The validation procedure in Theorem 8.86 is ordered. The pre-carrier regime is the set of branches for which all tests before local monotonicity-carrier validation have passed. Therefore any first failure before the carrier-validation row prevents membership in Theorem 8.100.

The same lemma lists every possible first failure before carrier validation: the exhaustion failures are those of Theorem 12.3; representation and repaired-gauge failures are those of Theorems 12.12 and 12.19 together with scale-degeneracy from Theorem 8.85; the compactness failures are the alternatives in Theorem 11.13; critical-mass failures are the alternatives of Theorem 12.22; compact tightness and local H^1-loss are the tests of Theorem 12.30; and the modulation, cost-well-posedness, and compatibility failures are exactly the failures recorded in Theorems 8.79, 8.80, 8.82 and 8.84. Thus the displayed list is exhaustive for non-carrier first failures.

If the surrounding Type II class includes the local closure theorem assigned to the named stratum, then the branch is handled there. It is therefore not a branch remaining to be treated by the retained compact-cost localized-energy argument. □

Theorem 8.106 (Pre-carrier reduction to carrier validation)**.** *Let ω be in the pre-carrier validated compact-cost regime. Then exactly one of the following alternatives occurs:*

(i) *the validated compact cost admits local monotonicity-carrier data in the sense of Theorem 8.72;*

(ii) *the branch exits the retained compact cost-divergence stratum through the first failed carrier-validation test: carrier-subidentity failure, fixed-cutoff finite-error failure, moving-annulus summability failure, or carrier-comparison failure for the validated cost.*

If the first alternative occurs, ω is excluded by the retained compact-cost argument.

Proof. By definition of the pre-carrier regime, every ordered validation test before carrier validation has passed, and the validated compact cost-divergence conclusion is already present. Thus the next unresolved row in Theorem 8.86 is precisely the local monotonicity-carrier row. Applying Theorem 8.87 gives the two alternatives listed in the statement, and the first-failure convention makes them disjoint.

If the carrier exists, then Theorem 8.73 gives the local cost-to-exclusion data, and Theorem 8.77 excludes the branch. Equivalently, the same conclusion is Theorem 8.95. □

Theorem 8.107 (Fixed-cutoff carrier closure)**.** *Let ω be in the pre-carrier validated compact-cost regime. If ω satisfies the fixed-cutoff localized-energy closure criterion of Theorem 8.103, then ω is excluded from the Type II class under consideration.*

Proof. The pre-carrier regime supplies active supercritical scaling, local well-posedness and compatibility of the validated compact cost, and the validated compact cost-divergence conclusion. The fixed-cutoff localized-energy criterion supplies finite initial localized energy, the fixed carrier subidentity, finite fixed-cutoff error, and energy-control comparison of the validated cost with $\tilde{\mathfrak{D}}_{R_0}$. Let C_1, τ_1 be the constants in that comparison and increase the tail initial time so that the comparison holds from τ_0 onward. Define

$$\mathcal{H}_\omega(\tau) = \mathcal{E}_\phi(\tau) + \int_\tau^\infty B_{R_0}(s)\, ds.$$

The finite-error condition makes $\mathcal{H}_\omega$ locally absolutely continuous and bounded below. The fixed carrier subidentity gives

$$\frac{d}{d\tau}\mathcal{H}_\omega(\tau) + \frac{1}{2}\tilde{\mathfrak{D}}_{R_0}(\tau) \le 0$$

in distributions. Since $\mathfrak{C}_{\mathrm{II}}^{NS} \le C_1 \tilde{\mathfrak{D}}_{R_0}$, this implies

$$\frac{d}{d\tau}\mathcal{H}_\omega(\tau) + \frac{1}{2C_1}\mathfrak{C}_{\mathrm{II}}^{NS}(\tau) \le 0.$$

Together with the validated compact cost-divergence conclusion, these are the local cost-to-exclusion data with $\mathcal{D}_\omega = \mathfrak{C}_{\mathrm{II}}^{NS}$. Equivalently, the fixed criterion is a successful carrier-validation row. Since all pre-carrier tests and the validated compact cost-divergence conclusion are already present, ω is on the retained compact cost-divergence stratum after carrier validation. Applying Theorem 8.77 gives the Type II exclusion conclusion. □

Theorem 8.108 (Moving-cutoff carrier closure)**.** *Let ω be in the pre-carrier validated compact-cost regime. If ω satisfies the moving-cutoff localized-energy closure criterion of Theorem 8.104, then ω is excluded from the Type II class under consideration.*

Proof. The pre-carrier regime supplies active supercritical scaling, local well-posedness and compatibility of the validated compact cost, and the validated compact cost-divergence conclusion. The moving-cutoff carrier criterion supplies the finite initial moving localized energy, the moving carrier subidentity, summable moving-annulus errors, and the terminal comparison

$$0 \le \mathfrak{C}_{\mathrm{II}}^{NS}(\tau) \le C_m \tilde{\mathfrak{D}}_{R(\tau)}(\tau).$$

By Theorem 10.21, the summability condition gives $B_{\mathrm{mov}} \in L^1(\tau_0, \infty)$. Define

$$\mathcal{H}_\omega(\tau) = \mathcal{E}_R(\tau) + \int_\tau^\infty B_{\mathrm{mov}}(s)\, ds.$$

The finite moving-error condition makes $\mathcal{H}_\omega$ locally absolutely continuous and bounded below. The moving carrier subidentity gives

$$\frac{d}{d\tau}\mathcal{H}_\omega(\tau) + \frac{1}{2}\tilde{\mathfrak{D}}_{R(\tau)}(\tau) \le 0$$

in distributions. The moving comparison gives

$$\frac{d}{d\tau}\mathcal{H}_\omega(\tau) + \frac{1}{2C_m}\mathfrak{C}_{\mathrm{II}}^{NS}(\tau) \le 0.$$

Together with the validated compact cost-divergence conclusion, these are the local cost-to-exclusion data with $\mathcal{D}_\omega = \mathfrak{C}_{\mathrm{II}}^{NS}$. Equivalently, the moving criterion is a successful carrier-validation row. Since all pre-carrier tests and the validated compact cost-divergence conclusion are already present, ω is on the retained compact cost-divergence stratum after carrier validation. Applying Theorem 8.77 excludes the branch. □

Lemma 8.109 (Pre-carrier local energy gives finite initial localized energy)**.** *Let ω be in the pre-carrier validated compact-cost regime, represented by V on a terminal tail. Fix any later time $T \ge \tau_0$. After possibly replacing the tail by one whose initial time $\tau_1 \in [T, T+1]$ is a compact-cylinder local-energy Lebesgue time, every finite $R < \infty$ satisfies*

$$\int_{B_R} |V(y,\tau_1)|^2\, dy < \infty.$$

Consequently the initial localized-energy clauses in the fixed and moving localized-energy closure criteria are available after a harmless terminal restriction for every compactly supported fixed cutoff and every moving cutoff with finite initial radius. This is a local compact-cylinder statement; it does not require $V(\tau) \in L^3(\mathbb{R}^3)$ or a whole-space critical bound.

Proof. The physical branch is suitable and the repaired-gauge chart is locally absolutely continuous on compact renormalized cylinders in the pre-carrier regime. Pulling back the local energy class gives

$$V \in L^\infty_{\mathrm{loc}}([\tau_0,\infty); L^2(B_R)) \quad \text{for every } R < \infty$$

after choosing the usual representative, or at minimum

$$V \in L^2_{\mathrm{loc}}([\tau_0,\infty); L^2(B_R))$$

with local-energy Lebesgue times on every compact interval. If the stronger representative is unavailable at a prescribed endpoint, choose $\tau_1 \in [T, T+1]$ which is a Lebesgue time simultaneously for the finitely many compact supports used by the fixed or moving cutoff under consideration. This is possible because the local L^2-in-time norms are finite on $[T, T+1]$.

At such a τ_1, every compactly supported fixed cutoff has finite localized energy. If $\Phi(\tau, y) = \phi(y/R(\tau))$ and $R(\tau_1) < \infty$, then $\Phi(\tau_1, \cdot)$ is compactly supported, so the same local L^2 finiteness gives $\mathcal{E}_R(\tau_1) < \infty$. Passing from τ_0 to τ_1 removes only a finite initial renormalized interval and therefore does not change any terminal divergence or tail-integrability alternative. □

Lemma 8.110 (Nested moving cutoffs give carrier comparison)**.** *Let ω be in the pre-carrier validated compact-cost regime, and assume the validated compact cost is energy-controlled by the fixed identity cost:*

$$0 \le \mathfrak{C}_{\mathrm{II}}^{NS}(\tau) \le C_1 \tilde{\mathfrak{D}}_{R_0}(\tau) \qquad \text{for a.e. } \tau \ge \tau_1.$$

Let $\Phi(\tau, y)$ be a moving cutoff such that, after increasing the initial time if necessary,

$$\Phi(\tau, y) \ge \phi_{R_0}(y) \qquad \text{for all } y \in \mathbb{R}^3 \text{ and a.e. } \tau \ge \tau_1.$$

Then the validated compact cost is controlled by the moving identity cost:

$$0 \le \mathfrak{C}_{\mathrm{II}}^{NS}(\tau) \le C_1 \tilde{\mathfrak{D}}_{R(\tau)}(\tau) \qquad \text{for a.e. } \tau \ge \tau_1.$$

In particular, for the canonical identity cost the carrier comparison is automatic with $C_1 = 1$.

Proof. Because $\Phi(\tau,\cdot) \ge \phi_{R_0}$ and both cutoffs are nonnegative,

$$\nu \int |\nabla V|^2 \phi_{R_0} \le \nu \int |\nabla V|^2 \Phi(\tau,\cdot),$$

and, since $a_+(\tau) \ge 0$,

$$a_+(\tau) \int |V|^2 \phi_{R_0} \le a_+(\tau) \int |V|^2 \Phi(\tau,\cdot).$$

Adding these two inequalities gives

$$\tilde{\mathfrak{D}}_{R_0}(\tau) \le \tilde{\mathfrak{D}}_{R(\tau)}(\tau) \qquad \text{for a.e. } \tau \ge \tau_1.$$

Combining this with the fixed energy-control comparison yields the displayed moving comparison. The canonical identity cost is energy-controlled by Theorem 8.71 with $C_1 = 1$. □

Definition 8.111 (Finite-error carrier selection)**.** The pre-carrier validated compact-cost regime has finite-error carrier selection if every branch in that regime whose validated compact cost is energy-controlled by the fixed identity cost satisfies, after passing to a later terminal tail if necessary, at least one of the following two local alternatives:

(i) the fixed carrier subidentity of Theorem 8.101 is valid and

$$\int_{\tau_0}^{\infty} B_{R_0}(\tau)\, d\tau < \infty;$$

(ii) there exists a finite-valued nondecreasing locally absolutely continuous radius $R(\tau) \ge R_0$, with $R(\tau) \to \infty$ and moving cutoff $\Phi(\tau,y) = \phi(y/R(\tau))$, such that

$$\Phi(\tau,y) \ge \phi_{R_0}(y) \qquad \text{for all } y \text{ and a.e. } \tau,$$

the moving carrier subidentity of Theorem 8.101 is valid with the time-dependent cutoff Φ, and the moving-annulus errors are summable in the sense of Theorem 10.20.

This condition contains only the local one-sided energy estimate and finite-error selection statements. It does not include finite initial localized energy or carrier comparison, because those are supplied by Theorems 8.109 and 8.110.

Definition 8.112 (Annular finite-error selection)**.** The pre-carrier validated compact-cost regime has annular finite-error selection if every branch in that regime whose validated compact cost is energy-controlled by the fixed identity cost satisfies, after passing to a later terminal tail if necessary, at least one of the following two alternatives:

(i) the fixed error density has finite tail integral,

$$\int_{\tau_0}^{\infty} B_{R_0}(\tau)\, d\tau < \infty;$$

(ii) there exists a finite-valued nondecreasing locally absolutely continuous nested radius $R(\tau) \ge R_0$, with $R(\tau) \to \infty$ and $\Phi(\tau,y) = \phi(y/R(\tau)) \ge \phi_{R_0}(y)$, such that the moving-annulus errors are summable in the sense of Theorem 10.20.

This is the purely finite-error part of the carrier problem. It does not include local energy subidentities, finite initial energy, or carrier comparison.

Definition 8.113 (Componentwise fixed-or-moving finite-error estimates)**.** For a branch in the pre-carrier validated compact-cost regime, the componentwise annular finite-error estimates hold if, after passing to a later terminal tail if necessary, either the six fixed-cutoff components

$$I_1 = \left| \int |V|^2 \Delta \phi_{R_0} \right|, \qquad I_2 = \left| \int |V|^2 V \cdot \nabla \phi_{R_0} \right|, \qquad I_3 = \left| \int P\, V \cdot \nabla \phi_{R_0} \right|,$$
$$I_4 = |b| \left| \int |V|^2 \nabla \phi_{R_0} \right|, \qquad I_5 = a_- M_\phi, \qquad I_6 = a_+ \int |V|^2 \big(-y \cdot \nabla \phi_{R_0}\big)$$

belong to $L^1(\tau_0, \infty)$, or there is a finite-valued nondecreasing locally absolutely continuous nested radius $R(\tau) \geq R_0$, with $R(\tau) \to \infty$ and $\Phi(\tau, y) = \phi(y/R(\tau)) \geq \phi_{R_0}(y)$, such that each of the seven moving annular components

$$J_1 = \left| \int |V|^2 \partial_\tau \Phi \right|, \qquad J_2 = \left| \int |V|^2 \Delta \Phi \right|, \qquad J_3 = \left| \int |V|^2 V \cdot \nabla \Phi \right|,$$
$$J_4 = \left| \int P\, V \cdot \nabla \Phi \right|, \qquad J_5 = |b| \left| \int |V|^2 \nabla \Phi \right|,$$
$$J_6 = a_- \int |V|^2 \Phi, \qquad J_7 = a_+ \int |V|^2 \big(-y \cdot \nabla \Phi\big)$$

belongs to $L^1(\tau_0, \infty)$.

Remark 8.114 (What the componentwise estimates isolate). The quantities in Theorem 8.113 are exactly the fixed and moving carrier-error components: in the fixed-cutoff case, viscous cutoff, convection, pressure-tail, center drift, negative scale drift, and the positive scale-shell drift; in the moving-cutoff case, cutoff transition, viscous cutoff, convection, pressure-tail, center drift, negative scale drift, and the positive scale-shell drift. These diagnostics need not be closed by a single global tail estimate. In the stratified use of Section 7, persistent failure of the cutoff, viscous, convection, or pressure diagnostics is read as shell mass that cannot be evacuated from selected windows and is therefore assigned to exterior leakage, exterior concentration, loss of critical tightness, or local windowed H^1-loss. Persistent failure of the translation diagnostic is read as a modulation-driven adjacent branch, while persistent failure of the positive scale-shell term is read as a scale-dynamics branch. Thus the component list records the local diagnostics seen by the carrier test; it is not intended to force the proof into a single classical PDE closure argument. For radial nonincreasing cutoffs, the scale-shell term

$$-\frac{a}{2} \int |V|^2 y \cdot \nabla \Phi$$

has only an a_+-contribution in the residual carrier error. The a_- part is favorable and is already absorbed into the negative-drift alternative. The negative scale-drift component J_6 is not to be replaced by an unproved equivalence with compact-cost divergence: if J_6 is not summable, it must be handled through the explicit scale-drift alternatives of Theorems 10.30 and 10.38 or treated by a direct summability estimate.

Lemma 8.115 (Componentwise estimates give annular finite-error selection)**.** *If every branch in the pre-carrier validated compact-cost regime whose validated compact cost is energy-controlled by the fixed identity cost satisfies the componentwise annular finite-error estimates of Theorem 8.113, then the regime has annular finite-error selection in the sense of Theorem 8.112.*

Proof. Let ω be such a branch. If the fixed alternative in Theorem 8.113 holds, with $I_1, \ldots, I_6$ denoting the fixed components appearing in that definition, then Theorem 10.9 gives

$$B_{R_0} = \frac{\nu}{2} I_1 + \frac{1}{2} I_2 + I_3 + \frac{1}{2} I_4 + \frac{1}{2} I_5 + \frac{1}{2} I_6.$$

Since B_{R_0} is a finite positive linear combination of integrable fixed components, $B_{R_0} \in L^1(\tau_0, \infty)$. This is the first alternative in Theorem 8.112. Otherwise the definition supplies a finite-valued nested moving radius $R(\tau) \geq R_0$, $R(\tau) \to \infty$, for which the seven moving annular components $J_1, \dots, J_7$ are integrable. These seven components are exactly the summability requirements in Theorem 10.20, with the same moving cutoff Φ. Hence the moving-annulus errors are summable, which is the second alternative in Theorem 8.112. □

Definition 8.116 (Fixed annular PDE estimates)**.** Let ϕ_{R_0} be the fixed cutoff used in the validated compact cost and set

$$A_\phi := \operatorname{supp}(\nabla \phi_{R_0}) \cup \operatorname{supp}(\Delta \phi_{R_0}).$$

A branch in the pre-carrier validated compact-cost regime satisfies the fixed annular PDE estimates if, on a terminal tail,

(i)
$$\int_{\tau_0}^{\infty} \int_{A_\phi} |V(y,\tau)|^2 \, dy \, d\tau < \infty;$$

(ii)
$$\int_{\tau_0}^{\infty} \int_{A_\phi} |V(y,\tau)|^3 \, dy \, d\tau < \infty;$$

(iii) the fixed pressure flux is integrable:
$$\int_{\tau_0}^{\infty} \left| \int P(y,\tau) V(y,\tau) \cdot \nabla \phi_{R_0}(y) \, dy \right| d\tau < \infty;$$

(iv) the fixed translation-annulus term is integrable:
$$\int_{\tau_0}^{\infty} |b(\tau)| \int_{A_\phi} |V(y,\tau)|^2 \, dy \, d\tau < \infty;$$

(v) the fixed negative-scale term is integrable:
$$\int_{\tau_0}^{\infty} a_-(\tau) M_\phi(\tau) \, d\tau < \infty, \qquad M_\phi(\tau) = \int |V(y,\tau)|^2 \phi_{R_0}(y) \, dy.$$

(vi) the fixed positive scale-shell term is integrable:
$$\int_{\tau_0}^{\infty} a_+(\tau) \int |V(y,\tau)|^2 \big(-y \cdot \nabla \phi_{R_0}(y)\big) \, dy \, d\tau < \infty.$$

Lemma 8.117 (Fixed PDE estimates gives fixed finite error)**.** *If a pre-carrier branch satisfies the fixed annular PDE estimates of Theorem 8.116, then the fixed alternative in Theorem 8.113 holds.*

Proof. Since $\phi_{R_0} \in C_c^\infty$, there are finite constants

$$C_{\Delta,\phi} = \|\Delta \phi_{R_0}\|_{L^\infty}, \qquad C_{\nabla,\phi} = \|\nabla \phi_{R_0}\|_{L^\infty}, \qquad C_{y\nabla,\phi} = \|y \cdot \nabla \phi_{R_0}\|_{L^\infty}.$$

Because $\nabla \phi_{R_0}$ and $\Delta \phi_{R_0}$ vanish outside A_ϕ,

$$I_1 = \left| \int |V|^2 \Delta \phi_{R_0} \right| \leq C_{\Delta,\phi} \int_{A_\phi} |V|^2,$$

$$I_2 = \left| \int |V|^2 V \cdot \nabla \phi_{R_0} \right| \leq C_{\nabla,\phi} \int_{A_\phi} |V|^3,$$

$$I_4 = |b| \left| \int |V|^2 \nabla \phi_{R_0} \right| \leq C_{\nabla,\phi} |b| \int_{A_\phi} |V|^2,$$

and

$$I_6 = a_+ \int |V|^2 \big(-y \cdot \nabla \phi_{R_0}\big) \le C_{y\nabla,\phi} a_+ \int_{A_\phi} |V|^2.$$

The pressure term I_3 is integrable by item (iii) of the fixed criterion, $I_5 = a_- M_\phi$ is integrable by item (v), and I_6 is integrable by item (vi). Therefore all six fixed components in Theorem 8.113 belong to $L^1(\tau_0, \infty)$, which is exactly the fixed alternative there. □

Definition 8.118 (Moving annular PDE estimates)**.** A branch in the pre-carrier validated compact-cost regime satisfies the moving annular PDE estimates if there is a finite-valued nondecreasing locally absolutely continuous nested radius $R(\tau) \ge R_0$, with $R(\tau) \to \infty$ and $\Phi(\tau, y) = \phi(y/R(\tau)) \ge \phi_{R_0}(y)$, such that on a terminal tail:

(i) the moving derivative/convection annular packing holds:

$$\int_{\tau_0}^{\infty} \left(\frac{|\dot R(\tau)|}{R(\tau)} \int_{A_{R(\tau)}} |V|^2 + \frac{1}{R(\tau)^2} \int_{A_{R(\tau)}} |V|^2 + \frac{1}{R(\tau)} \int_{A_{R(\tau)}} |V|^3 \right) d\tau < \infty;$$

(ii) the moving pressure flux is integrable:

$$\int_{\tau_0}^{\infty} \left| \int P(y,\tau) V(y,\tau) \cdot \nabla \Phi(\tau, y)\, dy \right| d\tau < \infty;$$

(iii) the moving translation-annulus packing holds:

$$\int_{\tau_0}^{\infty} \frac{|b(\tau)|}{R(\tau)} \int_{A_{R(\tau)}} |V|^2\, d\tau < \infty;$$

(iv) the moving positive scale-shell term is integrable:

$$\int_{\tau_0}^{\infty} a_+(\tau) \int |V(y,\tau)|^2 \big(-y \cdot \nabla \Phi(\tau, y)\big)\, dy\, d\tau < \infty;$$

(v) either the moving negative-scale term is integrable,

$$\int_{\tau_0}^{\infty} a_-(\tau) \int |V(y,\tau)|^2 \Phi(\tau, y)\, dy\, d\tau < \infty,$$

or the branch lies in the scale-collapse drift obstruction of Theorem 10.25.

Remark 8.119 (Pressure item in the moving PDE estimates)*.* The pressure-flux clause in Theorem 8.118 is not a separate global pressure hypothesis. By Theorem 8.122, it follows from enlarged- shell L^3 packing together with a summable $R(\tau)^{-1}$ schedule. Thus the genuinely new moving-cutoff burden is concentrated in the annular velocity packing, the translation term, the positive scale-shell term, and the negative-drift alternative.

Lemma 8.120 (Moving PDE estimates gives moving finite error or scale-drift alternative)**.** *If a pre-carrier branch satisfies the moving annular PDE estimates of Theorem 8.118, then either the moving alternative in Theorem 8.113 holds or the branch lies in the scale-collapse drift obstruction of Theorem 10.25.*

Proof. Let

$$C_{\phi,1} = \|\nabla \phi\|_{L^\infty}, \qquad C_{\phi,2} = \|\Delta \phi\|_{L^\infty}, \qquad C_{\phi,3} = \sup_{1 \le |z| \le 2} |z| |\nabla \phi(z)|.$$

Because ϕ is supported in B_2 and constant on B_1, the support of $\nabla\Phi$, $\Delta\Phi$, and $\partial_\tau \Phi$ is contained in $A_{R(\tau)}$. Moreover

$$\partial_\tau \Phi(\tau, y) = -\frac{\dot R(\tau)}{R(\tau)} \frac{y}{R(\tau)} \cdot \nabla \phi(y/R(\tau)),$$

so on $A_{R(\tau)}$,

$$|\partial_\tau \Phi(\tau, y)| \leq C_{\phi,3} \frac{|\dot{R}(\tau)|}{R(\tau)}.$$

Likewise,

$$|\nabla\Phi(\tau, y)| \leq C_{\phi,1} R(\tau)^{-1} \mathbf{1}_{A_{R(\tau)}}(y), \qquad |\Delta\Phi(\tau, y)| \leq C_{\phi,2} R(\tau)^{-2} \mathbf{1}_{A_{R(\tau)}}(y),$$

and

$$|y \cdot \nabla\Phi(\tau, y)| \leq C_{\phi,3} \mathbf{1}_{A_{R(\tau)}}(y).$$

Therefore

$$J_1 = \left| \int |V|^2 \partial_\tau \Phi \right| \leq C_{\phi,3} \frac{|\dot{R}|}{R} \int_{A_{R(\tau)}} |V|^2,$$

$$J_2 = \left| \int |V|^2 \Delta\Phi \right| \leq C_{\phi,2} R(\tau)^{-2} \int_{A_{R(\tau)}} |V|^2,$$

$$J_3 = \left| \int |V|^2 V \cdot \nabla\Phi \right| \leq C_{\phi,1} R(\tau)^{-1} \int_{A_{R(\tau)}} |V|^3,$$

$$J_5 = |b| \left| \int |V|^2 \nabla\Phi \right| \leq C_{\phi,1} \frac{|b|}{R(\tau)} \int_{A_{R(\tau)}} |V|^2,$$

and

$$J_7 = a_+ \int |V|^2 \big(-y \cdot \nabla\Phi\big) \leq C_{\phi,3} a_+ \int_{A_{R(\tau)}} |V|^2.$$

Hence items (i) and (iii) of the moving criterion make J_1, J_2, J_3, J_5 integrable, and item (iv) makes J_7 integrable. Item (ii) makes J_4 integrable. If the first branch of item (v) holds, then J_6 is integrable as well, so all seven moving components in Theorem 8.113 belong to $L^1(\tau_0, \infty)$, which is exactly the moving alternative there. Otherwise the second branch of item (v) places the branch in the scale-collapse drift obstruction. □

Lemma 8.121 (Annular pressure-flux estimate from local shell mass)**.** *Let $\phi \in C_c^\infty(\mathbb{R}^3)$ be radial and nonincreasing, with $\phi = 1$ on B_1 and $\phi = 0$ outside B_2. For $R \geq 1$, set*

$$\Phi_R(y) = \phi(y/R), \qquad A_R = \{R \leq |y| \leq 2R\}, \qquad \widetilde{A}_R = \{R/4 < |y| < 8R\}.$$

Assume $P = \mathcal{R}_i\mathcal{R}_j(V_iV_j)$ modulo functions of time and

$$\|V(\tau)\|_{L^3(\mathbb{R}^3)} \leq M \qquad \textit{for a.e. } \tau \geq \tau_0.$$

Then for a.e. $\tau \geq \tau_0$,

$$\left| \int P(y,\tau) V(y,\tau) \cdot \nabla\Phi_R(y)\, dy \right| \leq C_\phi R^{-1} \int_{\widetilde{A}_R} |V(y,\tau)|^3\, dy + C_\phi M^3 R^{-1}.$$

Proof. Choose $\chi_R \in C_c^\infty(\widetilde{A}_R)$, radial, $0 \leq \chi_R \leq 1$, with $\chi_R \equiv 1$ on A_R. Decompose

$$P = P_R^{\mathrm{ann}} + H_R, \qquad P_R^{\mathrm{ann}} := \mathcal{R}_i\mathcal{R}_j(\chi_R V_i V_j),$$

so that $H_R := P - P_R^{\mathrm{ann}}$ is harmonic on the neighborhood $\{R/2 < |y| < 4R\}$ of A_R.

By Calderon–Zygmund boundedness on $L^{3/2}$,

$$\|P_R^{\mathrm{ann}}(\tau)\|_{L^{3/2}(A_R)} \leq C\|V(\tau)\|^2_{L^3(\widetilde{A}_R)}.$$

Since $\nabla\Phi_R$ is supported in A_R and $\|\nabla\Phi_R\|_{L^\infty} \le C_\phi R^{-1}$,

$$\left|\int P_R^{\mathrm{ann}} V\cdot\nabla\Phi_R\right| \le C_\phi R^{-1}\|P_R^{\mathrm{ann}}\|_{L^{3/2}(A_R)}\|V\|_{L^3(A_R)}$$
$$\le C_\phi R^{-1}\|V\|^3_{L^3(\widetilde{A}_R)} \le C_\phi R^{-1}\int_{\widetilde{A}_R}|V|^3.$$

For the harmonic part, let $c_R(\tau)$ be the average of $H_R(\cdot,\tau)$ over A_R. Since constants disappear against $V\cdot\nabla\Phi_R$, one has

$$\int H_R V\cdot\nabla\Phi_R = \int (H_R - c_R)V\cdot\nabla\Phi_R.$$

It therefore suffices to bound the oscillation of H_R on A_R.

Fix $y \in \{R/2 < |y| < 4R\}$. Because $1-\chi_R$ is supported in $\{|z| \le R/4\}\cup\{|z| \ge 8R\}$, every such z satisfies $|y-z| \ge cR$. Differentiating the Calderon–Zygmund kernel gives

$$|\nabla H_R(y,\tau)| \le C\int_{|z|\le R/4}|y-z|^{-4}|V(z,\tau)|^2\,dz + C\int_{|z|\ge 8R}|y-z|^{-4}|V(z,\tau)|^2\,dz.$$

For the inner part, Hölder on $B_{R/4}$ yields

$$\int_{|z|\le R/4}|V(z,\tau)|^2\,dz \le |B_{R/4}|^{1/3}\|V(\tau)\|^2_{L^3(\mathbb{R}^3)} \le CRM^2,$$

hence

$$\int_{|z|\le R/4}|y-z|^{-4}|V(z,\tau)|^2\,dz \le CR^{-4}(CRM^2) = CR^{-3}M^2.$$

For the outer part, $|y-z|\simeq|z|$ and Hölder give

$$\int_{|z|\ge 8R}|y-z|^{-4}|V(z,\tau)|^2\,dz \le C\int_{|z|\ge 8R}|z|^{-4}|V(z,\tau)|^2\,dz$$
$$\le C\|V(\tau)\|^2_{L^3(\mathbb{R}^3)}\left(\int_{|z|\ge 8R}|z|^{-12}\,dz\right)^{1/3}$$
$$\le CR^{-3}M^2.$$

Therefore

$$\|\nabla H_R(\tau)\|_{L^\infty(\{R/2<|y|<4R\})} \le CR^{-3}M^2.$$

Since A_R has diameter $O(R)$, the mean-value theorem gives

$$\|H_R(\tau) - c_R(\tau)\|_{L^\infty(A_R)} \le CR^{-2}M^2.$$

Using again that $\nabla\Phi_R$ is supported in A_R,

$$\left|\int (H_R - c_R)V\cdot\nabla\Phi_R\right| \le C_\phi R^{-1}\|H_R - c_R\|_{L^\infty(A_R)}\|V\|_{L^1(A_R)}$$
$$\le C_\phi R^{-1}(CR^{-2}M^2)|A_R|^{2/3}\|V\|_{L^3(A_R)}$$
$$\le C_\phi M^2R^{-1}\|V\|_{L^3(\mathbb{R}^3)} \le C_\phi M^3R^{-1}.$$

Adding the annular-local and harmonic-tail bounds proves the claim. □

Corollary 8.122 (Moving pressure flux from enlarged-shell L^3 packing)**.** *Let $R(\tau)\ge 1$ be measurable and set*

$$\widetilde{A}_{R(\tau)} = \{R(\tau)/4 < |y| < 8R(\tau)\},\qquad \Phi(\tau,y) = \phi(y/R(\tau)).$$

Assume

$$\|V(\tau)\|_{L^3(\mathbb{R}^3)} \le M \qquad \textit{for a.e. } \tau \ge \tau_0,$$

and

$$\int_{\tau_0}^{\infty} \frac{1}{R(\tau)} \int_{\widetilde{A}_{R(\tau)}} |V(y,\tau)|^3 \, dy \, d\tau < \infty,$$

$$\int_{\tau_0}^{\infty} \frac{d\tau}{R(\tau)} < \infty.$$

Then

$$\int_{\tau_0}^{\infty} \left| \int P(y,\tau) V(y,\tau) \cdot \nabla \Phi(\tau, y) \, dy \right| d\tau < \infty.$$

Proof. Apply Theorem 8.121 at almost every τ with radius $R(\tau)$. Integrating the resulting pointwise bound gives

$$\begin{aligned} &\int_{\tau_0}^{\infty} \left| \int P \, V \cdot \nabla \Phi \right| d\tau \\ &\le C_\phi \int_{\tau_0}^{\infty} \frac{1}{R(\tau)} \int_{\widetilde{A}_{R(\tau)}} |V|^3 \, dy \, d\tau + C_\phi M^3 \int_{\tau_0}^{\infty} \frac{d\tau}{R(\tau)}, \end{aligned}$$

and both terms on the right are finite by hypothesis. □

Theorem 8.123 (Carrier closure reduced to explicit PDE estimates and scale-drift alternative)**.** *Assume repaired-gauge suitable local energy gives the fixed and moving carrier subidentities in the sense of Theorem 8.102. Let ω be a branch in the pre-carrier validated compact-cost regime whose validated compact cost is energy-controlled by the fixed identity cost. Assume that ω satisfies either the fixed annular PDE estimates of Theorem 8.116 or the moving annular PDE estimates of Theorem 8.118. Then exactly one of the following occurs:*

(i) *ω is excluded by the fixed-cutoff carrier closure theorem Theorem 8.107;*
(ii) *ω is excluded by the moving-cutoff carrier closure theorem Theorem 8.108;*
(iii) *ω enters the scale-collapse drift obstruction of Theorem 10.25.*

Proof. If the fixed annular PDE estimates hold, then Theorem 8.117 gives the fixed alternative in Theorem 8.113. Writing $I_1, \dots, I_6$ for those six fixed components, the identity

$$B_{R_0} = \frac{\nu}{2} I_1 + \frac{1}{2} I_2 + I_3 + \frac{1}{2} I_4 + \frac{1}{2} I_5 + \frac{1}{2} I_6$$

from Theorem 10.9 yields $B_{R_0} \in L^1(\tau_0, \infty)$. The suitable local energy lemma gives the fixed carrier subidentity. Finite initial localized energy and carrier comparison are automatic on the pre-carrier regime by Theorem 8.109 and the assumed energy-control of the validated cost by the fixed identity cost. Thus the fixed-cutoff localized-energy closure criterion holds, and Theorem 8.107 excludes ω.

Suppose instead that the moving annular PDE estimates hold. By Theorem 8.120, either the branch lies in the scale-collapse drift obstruction, giving alternative (iii), or the moving alternative in Theorem 8.113 holds. In the latter case, the seven moving components $J_1, \dots, J_7$ are integrable, which is exactly the summable moving-annulus condition of Theorem 10.20. The suitable local energy lemma gives the moving carrier subidentity. Finite initial moving energy and moving carrier comparison are automatic on the pre-carrier regime by Theorems 8.109 and 8.110. Therefore the moving-cutoff localized-energy closure criterion holds, and Theorem 8.108 excludes ω. The three alternatives are disjoint by construction. □

Corollary 8.124 (Explicit PDE estimates leave no unresolved compact-cost branch)**.** *Assume the non-carrier adjacent strata listed in Theorem 8.105 are closed by their corresponding local compactness or interface arguments in the ambient Type II class. Assume also that any branch entering the scale-collapse drift obstruction is closed by its assigned Section 6 local argument, for example through Theorem 10.38. If every branch in the pre-carrier validated compact-cost regime whose validated compact cost is energy-controlled by the fixed identity cost satisfies either the fixed annular PDE estimates of Theorem 8.116 or the moving annular PDE estimates of Theorem 8.118, then every branch with active supercritical scaling and validated compact Type II cost divergence is either excluded on the retained compact stratum, excluded by fixed or moving carrier closure, or assigned to a non-carrier or scale-drift branch that is already closed in the ambient class. In particular, the compact-cost regime leaves no unresolved branch.*

Proof. Let ω be a branch with active supercritical scaling and validated compact Type II cost divergence. By Theorem 8.90, either ω lies in the retained compact stratum or it enters one of the named adjacent strata. The retained compact stratum is excluded by Theorem 8.95. If ω enters a non-carrier adjacent stratum, the first assumption closes it. The only remaining case is that ω lies in the pre-carrier validated compact-cost regime. Then the explicit PDE-estimate hypothesis and Theorem 8.123 show that ω is either excluded by fixed carrier closure, excluded by moving carrier closure, or assigned to the scale-collapse drift obstruction, which is closed by the second assumption. Hence every branch is either excluded on the retained stratum or handled in a closed non-carrier or scale-drift branch. This is exactly the claimed no-unresolved-branch conclusion. □

Remark 8.125 (The explicit PDE estimates are not the remaining target). The explicit PDE estimates of Theorems 8.116 and 8.118 give a sufficient analytic realization of carrier closure, but they are not the remaining analytic target. The proof does not require one global tail estimate that simultaneously absorbs every carrier error. Instead, persistent failure of a moving-carrier ingredient is supposed to force entry into a neighboring compactness stratum. The remainder of this section therefore isolates only the small alternative statements needed for that local compactness reduction.

Theorem 8.126 (Finite-error selection reduced to annular finite-error)**.** *Assume that repaired-gauge suitable local energy gives the fixed and moving carrier subidentities in the sense of Theorem 8.102. If the pre-carrier regime has annular finite-error selection in the sense of Theorem 8.112, then it has finite-error carrier selection in the sense of Theorem 8.111.*

Proof. Let ω be a branch in the pre-carrier regime whose validated compact cost is energy-controlled by the fixed identity cost. By annular finite-error selection, after restricting to a later tail either the fixed error density is integrable or there is a nested moving radius for which the moving-annulus errors are summable. The repaired-gauge suitable local energy lemma supplies the fixed carrier subidentity in the first case and the moving carrier subidentity in the second. These are exactly the two alternatives in Theorem 8.111. □

Theorem 8.127 (Carrier completion reduced to finite-error selection)**.** *Assume the pre-carrier validated compact-cost regime has finite-error carrier selection in the sense of Theorem 8.111. Then every branch in the pre-carrier regime whose validated compact cost is energy-controlled by the fixed identity cost satisfies the fixed-cutoff or moving-cutoff carrier closure criterion. Consequently that branch is excluded from the Type II class under consideration.*

Proof. Let ω be such a branch. The positive finite critical L^3 annulus is not used here as a hidden global estimate. The finite initial localized energy required by either localized-energy criterion is supplied locally, after a harmless terminal restriction, by Theorem 8.109. The assumed energy-control comparison gives the fixed carrier-comparison clause.

If the fixed alternative in Theorem 8.111 holds, then the fixed carrier subidentity and the finite B_{R_0}-tail are precisely the two remaining clauses of the fixed-cutoff localized-energy closure criterion. Hence Theorem 8.107 excludes the branch.

If the moving alternative holds, the summable moving-annulus condition gives

$$\int_{\tau_0}^{\infty} B_{\mathrm{mov}}(\tau)\, d\tau < \infty$$

by Theorem 10.21. The moving carrier subidentity is part of the moving alternative. The nested cutoff condition and fixed energy-control comparison give the moving carrier comparison by Theorem 8.110. Together with finite initial moving energy from Theorem 8.109, these are exactly the moving localized-energy closure criterion. Therefore Theorem 8.108 excludes the branch.

The two alternatives in finite-error carrier selection exhaust the branch, so the claimed fixed-or-moving carrier completion and exclusion follow. □

Theorem 8.128 (Localized-energy criterion closes the pre-carrier regime)**.** *Let ω be in the pre-carrier validated compact-cost regime. If ω satisfies either the fixed-cutoff localized-energy closure criterion of Theorem 8.103 or the moving-cutoff localized-energy closure criterion of Theorem 8.104, then ω is excluded from the Type II class under consideration.*

Proof. In the fixed-cutoff case apply Theorem 8.107. In the moving-cutoff case apply Theorem 8.108. These are the only cases in the stated disjunction. □

Definition 8.129 (Carrier-estimate completion for the compact-cost regime)**.** The compact-cost regime has carrier-estimate completion if every branch in the pre-carrier validated compact-cost regime satisfies, after passing to a later terminal tail if necessary, either the fixed-cutoff localized-energy closure criterion of Theorem 8.103 or the moving-cutoff localized-energy closure criterion of Theorem 8.104.

Corollary 8.130 (Finite-error selection gives carrier-estimate completion)**.** *Assume every branch in the pre-carrier validated compact-cost regime uses a validated compact cost that is energy-controlled by the fixed identity cost; in particular this holds for the canonical identity cost. If finite-error carrier selection holds in the sense of Theorem 8.111, then the compact-cost regime has carrier-estimate completion in the sense of Theorem 8.129.*

Proof. Let ω be any branch in the pre-carrier validated compact-cost regime. By the energy-control hypothesis, ω is one of the branches covered by Theorem 8.127. That theorem gives either the fixed-cutoff localized-energy closure criterion or the moving-cutoff localized-energy closure criterion on a terminal tail. This is exactly Theorem 8.129. The parenthetical canonical case follows from Theorem 8.71. □

Corollary 8.131 (Annular finite-error selection gives carrier-estimate completion)**.** *Assume every branch in the pre-carrier validated compact-cost regime uses a validated compact cost that is energy-controlled by the fixed identity cost. Assume also that repaired-gauge suitable local energy gives the fixed and moving carrier subidentities in the sense of Theorem 8.102. If annular finite-error selection holds in the sense of Theorem 8.112, then the compact-cost regime has carrier-estimate completion.*

Proof. Theorem 8.126 converts annular finite-error selection into finite-error carrier selection. Then Theorem 8.130 gives carrier-estimate completion. □

Theorem 8.132 (Adjacent closure reduced to carrier-estimate completion)**.** *Assume the non-carrier adjacent strata listed in Theorem 8.105 are closed by their corresponding local compactness or interface arguments in the ambient Type II class. If the compact-cost regime has carrier-estimate* *completion in the sense of Theorem 8.129, then the compact-cost regime has adjacent-stratum closure in the sense of Theorem 8.96.*

Proof. Let ω be a branch with active supercritical scaling and validated compact Type II cost divergence that does not remain in the retained compact cost-divergence stratum. By Theorem 8.90, ω enters one of the named adjacent strata produced by Theorem 8.86.

If the adjacent stratum is non-carrier, then Theorem 8.105 identifies it with one of the previously named exhaustion, representation, compactness, critical-mass, compact-test, modulation, cost-well-posedness, or cost-compatibility strata. By the first hypothesis of the present theorem, that stratum is closed by its own local argument, so the Type II exclusion conclusion holds there, or the branch is assigned to a previously closed non-compact-cost alternative.

It remains to consider a branch that reaches the pre-carrier regime. By carrier-estimate completion, after restriction to a later tail if needed, it satisfies either the fixed-cutoff or moving-cutoff localized-energy closure criterion. In the fixed or moving case, Theorem 8.128 gives the Type II exclusion conclusion. Thus every named adjacent stratum arising from the compact-cost regime has a local exclusion or an assignment to a previously closed alternative, which is exactly Theorem 8.96. □

Definition 8.133 (Stratified moving-carrier alternative)**.** Assume the validated compact cost is energy-controlled by the fixed identity cost. The pre-carrier validated compact-cost regime has stratified moving-cutoff alternative if every branch ω in that regime, after passing to a later terminal tail if necessary, satisfies at least one of the following local alternatives:

(i) ω satisfies the fixed-cutoff localized-energy closure criterion of Theorem 8.103;

(ii) ω satisfies the moving-cutoff localized-energy closure criterion of Theorem 8.104;

(iii) ω satisfies the exterior/leakage alternative: either there are unit windows $I_n = [T_n, T_n+1]$, selected times $\tau_n \in I_n$, radii $R_n \to \infty$, and $\eta > 0$ such that

$$\sup_n \big(|a(\tau_n)| + \|b(\tau_n)\|_\infty\big) < \infty,$$

$$\frac{1}{R_n^2}\int_{A_{R_n}} |V(y,\tau_n)|^2\,dy + \frac{1}{R_n}\int_{A_{R_n}} |V(y,\tau_n)|^3\,dy + \frac{1}{R_n}\int_{\widetilde{A}_{R_n}} |V(y,\tau_n)|^3\,dy \ge \eta$$

for every n, and this selected-window shell persistence is assigned to one of the already named critical-tightness failure, exterior concentration, or local windowed H^1-loss branches; or the normalized transition-carrier blocks of the moving cutoff have a local occupation-state limit with the transition-leakage carrier label of Theorem 8.147, which is identified there with the same exterior/leakage cluster.

(iv) ω satisfies the translation alternative: there are unit windows $I_n = [T_n, T_n+1]$, selected times $\tau_n \in I_n$, radii $R_n \to \infty$, cutoffs $\Phi_n(y) = \phi(y/R_n)$, and $\eta > 0$ such that

$$\sup_n \big(|a(\tau_n)| + \|b(\tau_n)\|_\infty\big) < \infty,$$

$$\frac{|b(\tau_n)|}{R_n}\int_{A_{R_n}} |V(y,\tau_n)|^2\,dy \ge \eta$$

for every n, and this persistent translation transport is assigned to the already named modulation-integrability failure or to a multicore/exterior branch produced after selected-window recentering;

(v) ω satisfies the positive-scale alternative: either there are unit windows $I_n = [T_n, T_n+1]$, selected times $\tau_n \in I_n$, radii $R_n \to \infty$, cutoffs $\Phi_n(y) = \phi(y/R_n)$, and $\eta > 0$ such that

$$\sup_n \big(|a(\tau_n)| + \|b(\tau_n)\|_\infty\big) < \infty,$$

$$a_+(\tau_n)\int |V(y,\tau_n)|^2\big(-y\cdot\nabla\Phi_n(y)\big)\,dy \geq \eta$$

for every n, and this persistent positive scale forcing is assigned to the already named scale-rigid residual alternative or to a bounded selected-window limit refinement of that alternative; or the normalized positive-scale-work carrier blocks have a retained compact occupation-state limit whose compact-core realization satisfies the scale-rigid branch clauses of Theorem 5.23. If the occupation state instead has an earlier exterior/leakage, modulation-driven recentering, or negative-drift label, the branch is read in that earlier alternative rather than in the positive-scale alternative.

(vi) ω enters the negative scale-drift obstruction of Theorem 10.25.

The fixed-cutoff case is auxiliary here. The canonical local alternative is the moving one: either a nested moving cutoff closes the carrier, or a persistent failure produces one of the displayed selected-window diagnostics and is assigned to the corresponding adjacent stratum.

Remark 8.134 (How the moving diagnostics match the carrier components)*.* The exterior/leakage alternative in Theorem 8.133 groups the moving cutoff, viscous, convection, and pressure diagnostics. The point is that Theorem 8.121 already converts pressure failure into enlarged-shell L^3 persistence plus an R^{-1} tail, and the R^{-1}-tail can be made harmless on discrete selected radii. Thus the pressure obstruction is not an independent global theorem; it is another way of detecting shell mass that refuses to leave the selected windows. The translation alternative contains the J_5 term and is meant to be combined with the bounded selected-window modulation provided by Theorems 8.38 and 8.44. The positive-scale alternative contains the J_7 term and isolates the only genuinely positive scale-dynamics obstruction left after the favorable a_--part has been separated. The negative-drift term J_6 is already assigned to Section 6 and is not to be replaced by an unproved compact-cost equivalence. The occupation-state clauses are included so that a diffuse nonintegrable error is not incorrectly converted into a pointwise lower bound; it is instead read as an occupation state in the local ordered stratification.

Definition 8.135 (Moving test family)**.** For a branch in the pre-carrier validated compact-cost regime, a moving test family is a finite-valued nondecreasing locally absolutely continuous radius $R(\tau)\geq R_0$, with $R(\tau)\to\infty$, together with the nested moving cutoff

$$\Phi(\tau,y)=\phi(y/R(\tau))\geq \phi_{R_0}(y).$$

For such a family, let $J_1,\dots,J_7$ denote the seven moving carrier components of Theorem 8.113.

Lemma 8.136 (A successful moving test family gives moving carrier closure)**.** *Assume repaired-gauge suitable local energy gives the moving carrier subidentity in the sense of Theorem 8.102. Let ω be a branch in the pre-carrier validated compact-cost regime whose validated compact cost is energy-controlled by the fixed identity cost. If there exists a moving test family in the sense of Theorem 8.135 such that, for some $\tau_1\geq\tau_0$,*

$$J_k\in L^1(\tau_1,\infty)\qquad \textit{for each } 1\leq k\leq 7,$$

then ω satisfies the moving-cutoff localized-energy closure criterion of Theorem 8.104 after restriction to the later tail $[\tau_1,\infty)$. In particular, ω is excluded by Theorem 8.108.

Proof. Restrict the branch and the moving test family to the tail $[\tau_1,\infty)$. Let $R(\tau)$ and $\Phi(\tau,y)$ be the restricted moving test family, with components $J_1,\dots,J_7$. By Theorem 10.20, the integrability of these seven components is exactly the summable moving-annulus condition for Φ. Hence $\int_{\tau_1}^{\infty}B_{\mathrm{mov}}(\tau)\,d\tau<\infty$ by Theorem 10.21. The suitable local energy lemma gives the moving carrier subidentity for the same cutoff. Finite initial localized energy on the pre-carrier regime is automatic by Theorem 8.109, and the moving carrier comparison is automatic by Theorem 8.110. These are exactly the clauses of Theorem 8.104. Therefore Theorem 8.108 excludes ω. $\square$

Lemma 8.137 (Retained pressure tails vanish on windowwise annuli)**.** *Let ω be a branch in the pre-carrier validated compact-cost regime, represented by (V, P, a, b) on a terminal tail, and let $I = [A, B]$ be a compact renormalized time interval. Assume that no pressure-reconstruction, exterior-leakage, separated-core, or critical-tightness failure label occurs on the retained branch over I. Then, for every radial cutoff $\phi \in C_c^\infty(B_2)$ with $\phi = 1$ on B_1,*

$$\lim_{S\to\infty} \sup_{\substack{R\in AC(I),\ R(\tau)\geq S\\ R(\tau)<\infty}} \int_I \left| \int P(y,\tau)V(y,\tau)\cdot\nabla\phi(y/R(\tau))\,dy \right| d\tau = 0.$$

Consequently, on the retained branch one may choose any prescribed summable sequence $\varepsilon_n > 0$ and radii $S_n \to \infty$ so that every finite-valued moving radius $R(\tau) \geq S_n$ on the n-th unit window has pressure flux at most ε_n on that window.

Proof. Argue by contradiction on a compact interval I. If the displayed limit fails, then there are $\eta > 0$, radii $S_m \to \infty$, and finite-valued moving radii $R_m \in AC(I)$, $R_m \geq S_m$, such that the pressure flux through the annulus of $\phi(\cdot/R_m(\tau))$ has L^1_τ-mass at least η.

Localize each annulus by a finite smooth partition after rescaling it to a fixed reference annulus. The Calderon–Zygmund part generated by the velocity stress inside the recorded annulus is controlled by the local annular L^3-mass. Since the retained branch has not failed critical tightness or exterior compactness, those annular L^3-masses vanish along $S_m \to \infty$ on the selected compact interval. Hence the local Calderon–Zygmund pressure part cannot carry the lower bound η.

The remaining pressure contribution is harmonic on the retained compact subannulus after subtracting functions of time. The quantitative nested-ball harmonic-tail lemma Theorem 9.21, applied after the same annular rescaling, says that any non-vanishing retained harmonic pressure must be generated by a non-vanishing recorded exterior annular source. If that source is present, the ordered pressure/exterior routing Theorems 9.22 and 9.38 assigns the branch to a pressure-reconstruction, exterior-leakage, or separated-core label. If no such source is present, the exterior pressure vanishes modulo functions of time by the retained pressure lemma used in Theorem 9.22. Both possibilities contradict the hypotheses that the branch is retained on I and that the pressure flux carries mass η.

Therefore the displayed uniform tail limit holds. Applying it on each unit window with ε_n gives the final scheduling statement. □

Lemma 8.138 (Critical tightness gives a tailored windowwise radius schedule)**.** *Let ω be a branch in the pre-carrier validated compact-cost regime, represented by V on a terminal tail. For the unit windows*

$$I_n = [\tau_0 + n, \tau_0 + n + 1], \qquad n \in \mathbb{N},$$

set

$$A_n := 1 + \int_{I_n} \big(|a(\tau)| + \|b(\tau)\|_\infty\big)\,d\tau.$$

Then there exists a nondecreasing sequence $R_n \to \infty$, with

$$R_n \geq 4\cdot 2^n,$$

such that

$$\sup_{\tau\in I_n} \int_{|y|>R_n/4} |V(y,\tau)|^3\,dy \leq 2^{-3n}A_n^{-3/2} \qquad \textit{for every } n.$$

The sequence may also be chosen so that every finite-valued moving radius $\mathcal{R}(\tau) \geq R_n$ on I_n satisfies

$$\int_{I_n} \left| \int P(y,\tau)V(y,\tau)\cdot\nabla\phi(y/\mathcal{R}(\tau))\,dy \right| d\tau \leq 2^{-n}.$$

Proof. Since ω lies in the pre-carrier validated compact-cost regime, the critical-tightness test and the modulation-integrability test in Theorem 8.86 have already passed. Thus Theorem 8.1 holds and Theorem 8.82 gives

$$a, b \in L^1_{\mathrm{loc}}([\tau_0, \infty)).$$

In particular each A_n is finite.

For every n, set

$$\varepsilon_n := 2^{-3n} A_n^{-3/2} > 0.$$

By uniform critical tightness, for each ε_n there exists $S_n < \infty$ such that

$$\sup_{\tau \geq \tau_0} \int_{|y|>S_n} |V(y,\tau)|^3 \, dy \leq \varepsilon_n.$$

Since the branch is retained in the pre-carrier regime, pressure reconstruction and the exterior/critical-tightness rows have not failed on I_n. Applying Theorem 8.137 on I_n with tolerance 2^{-n}, choose $P_n < \infty$ such that every finite-valued moving radius $R(\tau) \geq P_n$ on I_n has pressure flux at most 2^{-n} on that window. Choose R_n recursively so that

$$R_n \geq \max\{4S_n, \ P_n, \ 4 \cdot 2^n, \ R_{n-1}\}$$

for $n \geq 1$, and $R_0 \geq \max\{4S_0, P_0, 4\}$. Then R_n is nondecreasing, $R_n \to \infty$, $R_n \geq 4 \cdot 2^n$, and

$$\sup_{\tau \in I_n} \int_{|y|>R_n/4} |V(y,\tau)|^3 \, dy \leq \sup_{\tau \geq \tau_0} \int_{|y|>S_n} |V(y,\tau)|^3 \, dy \leq \varepsilon_n,$$

which is the desired estimate. □

Lemma 8.139 (Canonical windowwise moving family controls shell, pressure, and translation terms)**.** *Let ω be a branch in the pre-carrier validated compact-cost regime. Let R_n be the sequence supplied by Theorem 8.138. On the unit windows $I_n = [\tau_0 + n, \tau_0 + n + 1]$, define the canonical interpolated radius $R(\tau)$ by*

$$R(\tau) := R_n + (\tau - (\tau_0 + n))(R_{n+1} - R_n), \qquad \tau \in I_n,$$

and the associated moving cutoff

$$\Phi(\tau, y) := \phi(y/R(\tau)).$$

Write

$$J_2(\tau) = \left| \int |V|^2 \Delta\Phi(\tau, \cdot) \right|, \qquad J_3(\tau) = \left| \int |V|^2 V \cdot \nabla\Phi(\tau, \cdot) \right|,$$

$$J_4(\tau) = \left| \int P \, V \cdot \nabla\Phi(\tau, \cdot) \right|, \qquad J_5(\tau) = |b(\tau)| \left| \int |V|^2 \nabla\Phi(\tau, \cdot) \right|.$$

Then

$$\int_{\tau_0}^{\infty} \big(J_2(\tau) + J_3(\tau) + J_4(\tau) + J_5(\tau)\big) \, d\tau < \infty.$$

Proof. The sequence R_n is nondecreasing and tends to $+\infty$. Hence the piecewise affine interpolation $R(\tau)$ is finite-valued, nondecreasing, locally absolutely continuous, and satisfies $R(\tau) \to \infty$. Thus it is an admissible moving test family in the sense of Theorem 8.135.

By Theorem 8.138,

$$\sup_{\tau \in I_n} \int_{|y|>R_n/4} |V(y,\tau)|^3 \, dy \leq \varepsilon_n := 2^{-3n} A_n^{-3/2},$$

where

$$A_n = 1 + \int_{I_n} \big(|a(\tau)| + \|b(\tau)\|_\infty\big) \, d\tau.$$

For $\tau \in I_n$, one has $R(\tau) \geq R_n$, hence

$$A_{R(\tau)} \cup \widetilde{A}_{R(\tau)} \subset \{|y| > R(\tau)/4\} \subset \{|y| > R_n/4\}.$$

Therefore, for every $\tau \in I_n$,

$$\int_{A_{R(\tau)}} |V(y,\tau)|^3\, dy + \int_{\widetilde{A}_{R(\tau)}} |V(y,\tau)|^3\, dy \leq \varepsilon_n.$$

Using Hölder on the annulus $A_{R(\tau)}$,

$$\int_{A_{R(\tau)}} |V(y,\tau)|^2\, dy \leq |A_{R(\tau)}|^{1/3} \left(\int_{A_{R(\tau)}} |V(y,\tau)|^3\, dy \right)^{2/3} \leq C_\phi R(\tau) \varepsilon_n^{2/3}.$$

Since $|\Delta\Phi(\tau,\cdot)| \leq C_\phi R(\tau)^{-2} \mathbf{1}_{A_{R(\tau)}}$,

$$J_2(\tau) \leq C_\phi R(\tau)^{-2} \int_{A_{R(\tau)}} |V|^2 \leq C_\phi R(\tau)^{-1} \varepsilon_n^{2/3} \leq C_\phi R_n^{-1} \varepsilon_n^{2/3}.$$

Integrating on I_n and using $R_n \geq 4 \cdot 2^n$,

$$\int_{I_n} J_2(\tau)\, d\tau \leq C_\phi R_n^{-1} \varepsilon_n^{2/3} \leq C_\phi 2^{-3n} 2^{-2n} A_n^{-1}.$$

Since $|\nabla\Phi(\tau,\cdot)| \leq C_\phi R(\tau)^{-1} \mathbf{1}_{A_{R(\tau)}}$,

$$J_3(\tau) \leq C_\phi R(\tau)^{-1} \int_{A_{R(\tau)}} |V|^3 \leq C_\phi R_n^{-1} \varepsilon_n,$$

and therefore

$$\int_{I_n} J_3(\tau)\, d\tau \leq C_\phi R_n^{-1} \varepsilon_n = C_\phi R_n^{-1} 2^{-3n} A_n^{-3/2}.$$

The pressure component is controlled by the retained pressure-tail scheduling included in Theorem 8.138. Since $R(\tau) \geq R_n$ on I_n, the choice of R_n gives

$$\int_{I_n} J_4(\tau)\, d\tau = \int_{I_n} \left| \int P(y,\tau) V(y,\tau) \cdot \nabla\Phi(\tau,y)\, dy \right| d\tau \leq 2^{-n}.$$

If the retained pressure-tail scheduling were unavailable, the preceding lemma would assign the branch to a pressure/exterior local alternative before this windowwise retained branch is reached.

Finally,

$$J_5(\tau) \leq C_\phi \|b(\tau)\|_\infty R(\tau)^{-1} \int_{A_{R(\tau)}} |V|^2 \leq C_\phi \|b(\tau)\|_\infty \varepsilon_n^{2/3}.$$

Hence

$$\int_{I_n} J_5(\tau)\, d\tau \leq C_\phi \varepsilon_n^{2/3} \int_{I_n} \|b(\tau)\|_\infty\, d\tau \leq C_\phi 2^{-2n} A_n^{-1}(A_n - 1) \leq C_\phi 2^{-2n}.$$

The four resulting series are summable in n, proving the claim. ☐

Corollary 8.140 (Only transition and scale terms remain on the windowwise schedule)**.** *Under the hypotheses of Theorem 8.139, the canonical windowwise moving family*

$$R(\tau) = R_n + (\tau - (\tau_0 + n))(R_{n+1} - R_n), \qquad \tau \in I_n,$$

has summable moving carrier components J_2, J_3, J_4, J_5. Thus the only unresolved moving-carrier terms on that genuine moving family are the transition term J_1, the positive scale-shell term J_7, and the negative scale-drift term J_6.

Proof. This is exactly the conclusion of Theorem 8.139, together with the fact that the moving carrier has exactly seven components, of which Theorem 8.139 already controls J_2, J_3, J_4, J_5. □

Definition 8.141 (Windowwise transition/leakage alternative statement)**.** The pre-carrier validated compact-cost regime has windowwise transition/leakage alternative if, for every branch ω in that regime, the canonical windowwise moving family of Theorem 8.139 has the following property: if

$$J_1 \notin L^1(\tau_1, \infty)$$

on every later tail $[\tau_1, \infty)$, then ω enters the exterior/leakage alternative of Theorem 8.133.

Definition 8.142 (Windowwise positive-scale alternative statement)**.** The pre-carrier validated compact-cost regime has windowwise positive-scale alternative if, for every branch ω in that regime, the canonical windowwise moving family of Theorem 8.139 has the following property: if

$$J_7 \notin L^1(\tau_1, \infty)$$

on every later tail $[\tau_1, \infty)$, then ω enters the positive-scale alternative of Theorem 8.133, unless it has already entered the exterior/leakage, modulation-driven recentering, or negative scale-drift alternatives of the same stratified moving-localized-energy criterion.

8.7.1. *Proof of the windowwise alternative statements.* We now prove that the two alternative properties of Theorems 8.141 and 8.142 hold for the canonical windowwise moving family of Theorem 8.139. The proof does not convert arbitrary nonintegrability into a false persistent unit-window lower bound. Instead it uses the local compactness interpretation of a failed carrier component: the nonintegrable component defines normalized carrier blocks; each block is then labelled by the ordered first-failure stratification already used in Theorem 8.86.

Definition 8.143 (Normalized carrier blocks for a failed component)**.** Let $g \in L^1_{\mathrm{loc}}([\tau_0, \infty))$ be nonnegative and assume

$$\int_T^\infty g(\tau)\, d\tau = \infty \qquad \text{for every } T \geq \tau_0.$$

A normalized g-carrier block sequence on a tail $[\tau_1, \infty)$ is a sequence of intervals $[s_m, t_m]$, with $\tau_1 \leq s_m < t_m$, $s_m \to \infty$, $t_m \leq s_{m+1}$, and

$$\int_{s_m}^{t_m} g(\tau)\, d\tau = 1.$$

The associated carrier probability measures are

$$d\mu_m(\tau) := g(\tau)\mathbf{1}_{[s_m,t_m]}(\tau)\, d\tau.$$

For $g = J_i$, $i \in \{1, 7\}$, a J_i-occupation state means the terminal local occupation state obtained as follows. Fix any compact retained renormalized cylinder and record on it the repaired-gauge variables (V, P, a, b), the moving radius R, the cutoff Φ, and the component density J_i, using the weak local topology supplied by Theorem 3.10 and the pressure reconstruction for (V, P), the compactified repaired-gauge topology for (a, b, R, Φ), and the one-point compactification of $[0, \infty)$ for the scalar component density. Push μ_m forward by this local-data map and pass, on every compact cylinder and then diagonally, to a weak-$*$ limit of probability measures on this metrizable local state space. Any such limit is a J_i-occupation state. Its label row is the first closed stratum met by its support in the ordered validation procedure of Theorem 8.86, with the carrier row refined by the active component J_i. If several named diagnostics are met at the same row, they are read as the declared label cluster for that row.

Lemma 8.144 (Closed local carrier diagnostics)**.** *On the local state space of Theorem 8.143, each row of the ordered validation procedure in Theorem 8.86 is represented by a closed diagnostic set, after grouping diagnostics that are declared to belong to the same row. The component-refined carrier diagnostic for J_i, $i \in \{1, 7\}$, is also closed on normalized J_i-carrier states.*

Proof. Fix a compact retained cylinder. The solution variables are recorded in the weak local compactness topology of Theorem 3.10; pressure is recorded modulo spatial constants in the corresponding weak $L^{3/2}_{\mathrm{loc}}$ topology; and the auxiliary scale, center, radius, and cutoff data are recorded in the compactified repaired-gauge topology, with the boundary faces labelled by loss of scale admissibility, modulation integrability, exterior tightness, multicore escape, or local H^1 control. Thus leaving a retained compact row is convergence to one of the prescribed boundary faces, hence a closed condition.

The quantitative rows are closed for the standard lower-semicontinuity reasons used throughout the retained compactness arguments: L^3, pressure $L^{3/2}$, $L^2_t H^1_x$, and local energy quantities are lower semicontinuous under the recorded weak/strong local convergence, while failure of a required finite bound is the closed face at infinity in the compactified coordinate. Critical-annulus, local H^1-loss, modulation-loss, cost-compatibility, and carrier-comparison failures are therefore closed diagnostic sets in this topology. Equal-scale, separated, same-point cascade, gauge-degenerate, and scale-drift alternatives are closed because they are defined by limits of the recorded scale-center parameters in the finite parameter partition.

Finally, the component-refined carrier row is the closed face consisting of nonzero $J_i\,d\tau$-occupation states. Each approximating block has total occupation mass

$$\mu_m(\mathfrak{X}) = \int_{s_m}^{t_m} J_i(\tau)\,d\tau = 1,$$

where $\mathfrak{X}$ denotes the local occupation-state space. Weak-$*$ limits of probability measures have the same total mass. Hence the row "J_i is not negligible/summable on the carrier tail" is closed on the normalized occupation-measure compactification. This is a statement about the occupation measure; it does not choose pointwise times and does not assert a unit-window lower bound. □

Lemma 8.145 (Compactness and labels for normalized occupation states)**.** *Let $i \in \{1, 7\}$. Every normalized J_i-carrier block sequence for a pre-carrier branch has a subsequential J_i-occupation state in the sense of Theorem 8.143. The state has a unique first label row in the finite ordered list of Theorem 8.86. If no non-carrier label occurs before carrier validation, then the first carrier label is the component-refined J_i-moving-annulus summability failure.*

Proof. On every compact retained cylinder, the pre-carrier hypotheses give the represented suitable estimates, pressure reconstruction, local L^3 bounds, and local $L^2_t H^1_x$ bounds used in Theorem 3.10. The repaired-gauge modulation data are locally integrable by Theorem 8.82, and the moving radius is locally absolutely continuous. Hence the local-data maps take values in a metrizable product of weakly compact bounded sets after restricting to a subsequence and a compact cylinder. Banach–Alaoglu, the pressure compactness part of Theorem 3.10, and a diagonal extraction over compact cylinders give weak-$*$ convergence of the pushed-forward probability measures. This is precisely the occupation state in Theorem 8.143.

The ordered validation list is finite and its strata are evaluated in order. By Theorem 8.144, each row is a closed diagnostic set on the local occupation-measure compactification. The first row whose closed diagnostic meets the occupation-state support is therefore unique; ties inside that row are exactly the declared label cluster. If a non-carrier diagnostic appears before carrier validation, that row is the label row. If all non-carrier diagnostics pass on the support, the carrier row is reached. Because the measures are normalized by $\int J_i\,d\tau = 1$, the limiting occupation state has total carrier mass one. Thus the active carrier row cannot be the statement that J_i is summable and negligible on the carrier tail. The component-refined failure of J_i-moving-annulus summability is then the first carrier label. This is a compactness label, not a pointwise lower bound for J_i. $\square$

Lemma 8.146 (Occupation-state extraction from nonintegrability)**.** *Let $g \in L^1_{\mathrm{loc}}([\tau_0, \infty))$ be nonnegative and assume*

$$\int_T^\infty g(\tau)\, d\tau = \infty \qquad \textit{for every } T \geq \tau_0.$$

Then every tail $[\tau_1, \infty)$ contains a normalized g-carrier block sequence in the sense of Theorem 8.143. After passing to a subsequence, exactly one of the following two density regimes occurs:

(C1) Persistent unit-window carrier mass. *There are $\gamma > 0$ and unit intervals $U_m = [T_m, T_m+1]$, $T_m \to \infty$, such that*

$$\int_{U_m} g(\tau)\, d\tau \geq \gamma.$$

(C2) Diffuse carrier mass. *For the normalized block subsequence retained in this case,*

$$\sup_{T \in [s_m, t_m]} \int_{[T, T+1] \cap [s_m, t_m]} g(\tau)\, d\tau \longrightarrow 0.$$

The probability measures $d\mu_m = g\mathbf{1}_{[s_m, t_m]} d\tau$ are then the diffuse local occupation states of g.

If, in addition, $q(\tau) := |a(\tau)| + \|b(\tau)\|_\infty$ satisfies the uniform unit-window bound

$$Q := \sup_{T \geq \tau_0} \int_T^{T+1} q(\tau)\, d\tau < \infty.$$

then in case (C1) *either there are selected times $\tau_m \in U_m$ such that*

$$g(\tau_m) \geq \gamma/2, \qquad q(\tau_m) \leq 4Q/\gamma.$$

after relabelling, or the normalized g-carrier mass on U_m is carried by the high-modulation sets

$$H_m := \{\tau \in U_m : q(\tau) > 4Q/\gamma\}, \qquad \int_{H_m} g(\tau)\, d\tau \geq \gamma/2.$$

The latter is a modulation occupation state; it is not converted into a shell or scale lower bound.

Proof. Block construction. Fix a tail $[\tau_1, \infty)$. Choose $s_1 \geq \tau_1$. Since every tail integral of g is infinite, there is a finite $t_1 > s_1$ with $\int_{s_1}^{t_1} g = 1$; for instance take the first level time of the absolutely continuous primitive $\int_{s_1}^T g$. Having chosen t_m, choose $s_{m+1} \geq t_m + 1$ and then $t_{m+1} > s_{m+1}$ with $\int_{s_{m+1}}^{t_{m+1}} g = 1$. This gives normalized carrier blocks and $s_m \to \infty$.

Persistent versus diffuse. If case (C1) fails, then for each j there is $m(j)$ such that every interval of the form $[T, T+1] \cap [s_m, t_m]$ inside every retained block with index $m \geq m(j)$ has g-mass at most $1/j$. A diagonal subsequence gives the displayed diffuse condition. Conversely, if (C1) holds then the block sequence has a persistent unit-window carrier by definition. The two alternatives are therefore exhaustive after subsequence extraction.

Bounded modulation on persistent windows. In case (C1), set

$$Q := \sup_{T \geq \tau_0} \int_T^{T+1} q(\tau)\, d\tau.$$

If $Q = 0$, then $q = 0$ for a.e. time in every U_m. Since $\int_{U_m} g \geq \gamma$ and $|U_m| = 1$, the set $\{\tau \in U_m : q(\tau) = 0,\ g(\tau) \geq \gamma/2\}$ has positive measure after modifying g, q only on null sets;

otherwise $g < \gamma/2$ a.e. on $\{q = 0\} \cap U_m$ and $\int_{U_m} g \le \gamma/2 < \gamma$. Choose τ_m to be a Lebesgue time in that set. This gives the low-modulation alternative. Assume now that $Q > 0$. Set

$$H_m := \{\tau \in U_m : q(\tau) > 4Q/\gamma\}, \qquad E_m := U_m \setminus H_m.$$

Chebyshev gives $|H_m| \le \gamma/4$. Since $\int_{U_m} g \ge \gamma$, either $\int_{E_m} g \ge \gamma/2$ or $\int_{H_m} g \ge \gamma/2$ along a subsequence. In the first case $|E_m| \le 1$ gives a Lebesgue time $\tau_m \in E_m$ with $g(\tau_m) \ge \gamma/2$ and $q(\tau_m) \le 4Q/\gamma$. In the second case the displayed high-modulation carrier mass holds. This proves the final dichotomy and deliberately leaves the second case as a occupation state rather than replacing it by a pointwise shell or scale estimate. □

Lemma 8.147 (Transition carrier label is exterior/leakage)**.** *Let ω be a branch in the pre-carrier validated compact-cost regime, and let $R(\tau)$ be the canonical windowwise moving family of Theorem 8.139. If*

$$J_1 \notin L^1(\tau_1, \infty)$$

on every later tail, then every J_1-occupation state of Theorem 8.143 satisfies the exterior/leakage alternative of Theorem 8.133. More precisely, either an earlier non-carrier exterior/leakage label occurs, or the component-refined J_1-moving-annulus summability failure is the transition-leakage carrier label, which belongs to the same exterior/leakage cluster.

Proof. By Theorem 8.145, every normalized J_1-block has a subsequential labelled occupation state. Since ω is in the pre-carrier regime, the analytic, representation, retained finite critical-mass, compact-tightness, local H^1, modulation, and cost-compatibility rows in Theorem 8.100 have already passed on the retained branch. By Theorem 8.144, these rows are closed diagnostic sets in the occupation-measure compactification. Hence if one fails in the carrier limit, the failure is exactly a loss of retained compactness, exterior concentration/multicore escape, or local windowed H^1-loss by the ordered classification in Theorem 8.86. Those are exterior/leakage labels.

It remains to identify the first carrier label when all earlier rows remain retained. The component

$$J_1(\tau) = \left| \int |V|^2 \partial_\tau \Phi(\tau, \cdot) \right|$$

is the time-transition part of the moving cutoff. It contains no pressure, convection, translation, or scale-drift factor. The exact identity

$$\partial_\tau \Phi(\tau, y) = -\frac{R'(\tau)}{R(\tau)^2}\, y \cdot \nabla\phi(y/R(\tau))$$

shows that J_1 measures only occupation of the moving boundary of the retained compact window. Therefore the component-refined carrier row supplied by Theorem 8.145 is the *transition-leakage carrier label*: the retained moving boundary carries non-summable transition occupation. This is not a new compact-cost stratum. In the finite compactness list it is assigned to the same exterior/leakage cluster as critical-tightness failure, exterior concentration, and local windowed H^1-loss, because the only local interpretation of non-summable boundary transition in a retained compact single-core state is that mass is crossing the retained boundary or the compact regularity needed to ignore that crossing has failed. Thus every possible first label satisfies the exterior/leakage alternative. □

Theorem 8.148 (Windowwise transition/leakage alternative by occupation states)**.** *Let ω be a branch in the pre-carrier validated compact-cost regime, and let $R(\tau)$ be the canonical windowwise moving family of Theorem 8.139. If*

$$J_1 \notin L^1(\tau_1, \infty)$$

on every later tail, then ω enters the exterior/leakage alternative of Theorem 8.133.

Proof. Apply Theorem 8.146 with $g = J_1$. It gives normalized J_1-carrier blocks on every terminal tail, with either persistent unit-window mass or diffuse carrier mass. In the persistent case, the supplementary modulation selection gives bounded selected-time representatives when the low-modulation subcase occurs; in the high-modulation subcase and in the diffuse case one keeps the normalized occupation state instead of imposing an artificial pointwise lower bound.

All three outputs are J_1-occupation states in the sense of Theorem 8.143. By Theorem 8.147, each such state either has an earlier exterior/leakage label or has the transition-leakage carrier label identified there with the exterior/leakage cluster. This is exactly the exterior/leakage alternative allowed in Theorem 8.133. □

Corollary 8.149 (Windowwise transition/leakage alternative holds)**.** *The pre-carrier validated compact-cost regime has the windowwise transition/leakage alternative property in the sense of Theorem 8.141.*

Proof. This is the definition of Theorem 8.141 applied with Theorem 8.148. □

Lemma 8.150 (Retained positive scale-work realizes a scale-rigid core)**.** *Let ω be a branch in the pre-carrier validated compact-cost regime, and let $R(\tau)$ be the canonical windowwise moving family. Let μ_m be any normalized J_7-carrier block sequence and let $\mathfrak{s}$ be a subsequential J_7-occupation state. Assume that $\mathfrak{s}$ has no earlier exterior/leakage label, no modulation-driven recentering label, and no negative scale-drift label. Then, after admissible scale-center reselection, the represented sequence carried by $\mathfrak{s}$ is produced by Step 3 or Step 4 of Theorem 5.18 or by Theorem 5.19, and satisfies the four clauses (R1)–(R4) of Theorem 5.23.*

Proof. Step 1: what the J_7-carrier records. By definition,

$$J_7(\tau) = a_+(\tau) \int |V(y,\tau)|^2 \big(-y \cdot \nabla\Phi(\tau, y)\big)\, dy.$$

Thus each normalized block satisfies

$$\int_{s_m}^{t_m} a_+(\tau) \int |V|^2 (-y \cdot \nabla\Phi)\, dy\, d\tau = 1.$$

This is positive scale-generator work carried by the retained moving window. It is a scale-center compactness datum, not a consequence of an annular L^2-to-CKN conversion.

Step 2: compact-core realization. By Theorem 8.145, $\mathfrak{s}$ is a labelled local occupation state. The absence of an exterior/leakage label means that the state remains in the retained compact single-core window: critical tightness has not failed, no exterior/multicore loss has occurred, and local windowed H^1 control is retained. Consequently the compactness, pressure-reconstruction, and suitable-energy estimates inherited from the pre-carrier regime give a fixed compact selected cylinder Q_{R_*} on which the represented sequence has the uniform $A + E + C + D$ bound required in (R1). This is only the usual compact-cylinder suitable bound; it is not an L^∞ or regularity assertion.

The positive inner concentration needed for (R2) is inherited from the retained finite critical-mass row itself, not inferred from the J_7-density. In the physical local route this is the compact-window active mass supplied by Theorem 10.82; in the older compact-cost bookkeeping route it may be represented by the positive finite critical L^3-annulus test. Since the present occupation state has no exterior/leakage label, the corresponding critical mass remains inside the retained compact single-core region. Cover a slightly smaller retained cylinder by finitely many CKN cylinders. If each member of this finite cover had arbitrarily small velocity-pressure CKN quantity along the represented sequence, then the retained positive local critical mass would either vanish on the compact core or exit through the exterior face; the first contradicts the retained mass row and the second is exactly the exterior/leakage label excluded above. Thus, after passing to a subsequence and reselecting the scale-center within this finite cover, there are $Q_{r_*} \Subset Q_{R_*}$ and $\eta_0 > 0$ with positive CKN mass. This is (R2).

If the modulation were unbounded on the selected compact cylinder, the component-refined carrier label would be the modulation-driven recentering branch. That label is excluded by hypothesis, so the repaired-gauge modulation remains bounded on the selected cylinder; this is (R3).

Step 3: scale-rigid relative selection. The normalized J_7-mass is nonzero positive scale-generator work. If this work could be absorbed by a summable scale correction, the J_7-moving-annulus row would pass on a terminal carrier tail, contradicting the normalized carrier label. If the work instead caused scale collapse or negative scale drift, the first label would be the negative-drift branch, excluded by hypothesis.

Apply the finite parameter partition Theorem 5.10 to the retained compact core selected in Step 2 and to the inherited active ledger. A separated physical-center relation is exactly the input of Theorem 5.19. A same-point comparable-scale relation is reduced by Step 3 of Theorem 5.18, and a strict same-point scale separation is reduced by Step 4 of that theorem. The nonretained outputs of this partition are exterior/leakage, modulation-driven recentering, negative drift, or perturbative absorption; the first three are excluded by hypothesis and perturbative absorption would make the J_7-row summable. The only retained compact scale-center label left is therefore the scale-rigid residual alternative produced by those reductions: the relative scale-center selection is rigid in the sense of (R4), either through the gauge-degenerate selection or through the innermost same-point cascade after outer profiles have been absorbed. Thus the production requirement in Theorem 5.23 and clauses (R1)–(R4) all hold. □

Theorem 8.151 (Positive scale-work occupation states enter the scale-rigid branch)**.** *Let ω be a branch in the pre-carrier validated compact-cost regime, and let $R(\tau)$ be the canonical windowwise moving family. Let μ_m be any normalized J_7-carrier block sequence, and let $\mathfrak{s}$ be any subsequential J_7-occupation state generated by μ_m. If $\mathfrak{s}$ is not labelled by exterior/leakage, by modulation-driven recentering, or by negative scale drift, then the represented branch carried by $\mathfrak{s}$ is excluded from the local Type II class by Theorem 5.40.*

Proof. Theorem 8.150 realizes the J_7-occupation state $\mathfrak{s}$ as a scale-rigid local branch satisfying Theorem 5.23. Applying Theorem 5.40 excludes that branch. □

Lemma 8.152 (Positive-scale occupation-state stratification)**.** *Let ω be a branch in the pre-carrier validated compact-cost regime, and let $R(\tau)$ be the canonical windowwise moving family of Theorem 8.139, with cutoff $\Phi(\tau, y) = \phi(y/R(\tau))$ and J_7 defined as in Theorem 8.113. If*

$$J_7 \notin L^1(\tau_1, \infty)$$

on every later tail, then every J_7-occupation state is labelled either by exterior/leakage, by modulation-driven recentering, by the Section 6 negative scale-drift branch, or by the scale-rigid residual branch excluded in Theorem 8.151.

Proof. Occupation-state extraction is Theorem 8.146 with $g = J_7$. The ordered first-failure map of Theorem 8.86 first removes exterior loss, critical-tightness failure, local H^1-loss, and modulation-driven recentering. The scale decomposition then separates the favorable negative scale-drift obstruction, already assigned to Section 6, from retained compact states carrying positive scale work. The retained compact states are realized as scale-rigid states by Theorem 8.150 and excluded by Theorem 8.151. These alternatives exhaust the J_7-carrier because no other moving component is being tested here and J_2, J_3, J_4, J_5 are already summable on the canonical windowwise family by Theorem 8.139. □

Theorem 8.153 (Windowwise positive-scale alternative by occupation states)**.** *Let ω be a branch in the pre-carrier validated compact-cost regime, and let $R(\tau)$ be the canonical windowwise moving family of Theorem 8.139, with cutoff $\Phi(\tau, y) = \phi(y/R(\tau))$ and J_7 defined as in Theorem 8.113. If*

$$J_7 \notin L^1(\tau_1, \infty)$$

on every later tail, then ω enters the positive-scale alternative of Theorem 8.133, unless it has already entered the exterior/leakage, modulation-driven recentering, or negative-drift alternatives.

Proof. Apply Theorem 8.152. Exterior/leakage carrier labels already satisfy alternative (iii) of Theorem 8.133, modulation-driven recentering satisfies alternative (iv), and negative drift satisfies alternative (vi). In the retained positive-scale case, Theorem 8.151 places the carrier in the scale-rigid residual branch and then applies Theorem 5.40. The persistent unit-window subcase gives the pointwise selected-time form of alternative (v) whenever bounded-modulation points are selected; the high-modulation and diffuse subcases give the occupation-state form of the same alternative. Hence there is no unhandled diffuse J_7 tail. □

Corollary 8.154 (Windowwise positive-scale alternative holds)**.** *The pre-carrier validated compact-cost regime has the windowwise positive-scale alternative property in the sense of Theorem 8.142.*

Proof. This is exactly the alternatives allowed by Theorem 8.142, supplied by Theorem 8.153. □

Remark 8.155 (Status of the windowwise alternative definitions). Theorems 8.149 and 8.154 together prove unconditionally that the pre-carrier validated compact-cost regime has the alternative properties of Theorems 8.141 and 8.142. The two definitions are therefore not independent assumptions in the present manuscript; they are properties whose proof has been reduced to the occupation-measure compactness and extraction argument Theorems 8.144 to 8.146, the transition carrier label in Theorem 8.147, the positive scale-work compact-core realization Theorem 8.150, the scale-rigid exclusion argument Theorem 8.151, and the canonical windowwise schedule of Theorem 8.138. The proof does not use the invalid implication from nonintegrability to persistent unit-window lower bounds; diffuse tails are retained as normalized occupation states and classified by the local ordered stratification.

Theorem 8.156 (Canonical windowwise family reduces the alternatives to transition, positive scale, and negative drift)**.** *Assume every branch in the pre-carrier validated compact-cost regime uses a validated compact cost that is energy-controlled by the fixed identity cost. Assume repaired-gauge suitable local energy gives the moving carrier subidentity in the sense of Theorem 8.102. Assume also that the regime has the windowwise transition/leakage and windowwise positive-scale alternative statements of Theorems 8.141 and 8.142, and that failure of $J_6 \in L^1(\tau_1, \infty)$ on every later tail for the canonical windowwise moving family forces entry into the negative scale-drift obstruction of Theorem 10.25. Then the regime has stratified moving-cutoff alternative in the sense of Theorem 8.133.*

Proof. Let ω be a branch in the pre-carrier validated compact-cost regime, and let $R(\tau)$ be the canonical windowwise moving family of Theorem 8.139. Write $J_1, \dots, J_7$ for its moving carrier components.

By Theorem 8.139,

$$J_2, J_3, J_4, J_5 \in L^1(\tau_0, \infty).$$

If there exists $\tau_1 \geq \tau_0$ such that

$$J_1, J_6, J_7 \in L^1(\tau_1, \infty),$$

then all seven components are integrable on $[\tau_1, \infty)$. Therefore Theorem 8.136 gives the moving-cutoff localized-energy closure criterion, so ω satisfies alternative (ii) of Theorem 8.133.

Assume now that no later tail has J_1, J_6, J_7 all integrable. Since J_2, J_3, J_4, J_5 are already integrable on the full tail, at least one of J_1, J_6, J_7 fails on arbitrarily late tails. Fix such a component. As in the proof of Theorem 8.167, failure on arbitrarily late tails implies failure on every later tail: otherwise the component would be integrable on some tail and hence on every subsequent tail. If J_1 fails on every later tail, the windowwise transition/leakage alternative statement gives alternative (iii) of Theorem 8.133. If J_6 fails on every later tail, the negative scale-drift hypothesis gives alternative (vi). If J_7 fails on every later tail, the windowwise positive-scale alternative statement gives alternative (v), unless the branch has already entered alternative (iii), (iv), or (vi). Therefore one of the alternatives in Theorem 8.133 always holds. □

Corollary 8.157 (Adjacent closure reduced to windowwise transition and positive-scale alternative)**.** *Assume the non-carrier adjacent strata listed in Theorem 8.105 are closed by their corresponding local compactness or interface arguments in the ambient Type II class. Assume also that the exterior concentration, local windowed H^1-loss, modulation-driven recentering branches, scale-rigid residual branches, and the negative scale-drift obstruction are closed by their assigned local arguments in the ambient class. Assume every branch in the pre-carrier validated compact-cost regime uses a validated compact cost that is energy-controlled by the fixed identity cost, that repaired-gauge suitable local energy gives the moving carrier subidentity, and that the windowwise transition/leakage and windowwise positive-scale alternative statements of Theorems 8.141 and 8.142 hold, together with the negative-drift alternative statement used in Theorem 8.156. Then the compact-cost regime has adjacent-stratum closure in the sense of Theorem 8.96.*

Proof. Theorem 8.156 gives the stratified moving-cutoff alternative. Then Theorem 8.169 gives the adjacent-stratum closure conclusion. □

Lemma 8.158 (Ambient non-carrier exits close by ordered local routing)**.** *Let ω be an admissible physical local Type II branch which has passed the covered-entry theorem and then reaches the compact-cost validation layer. If ω leaves the retained compact-cost stratum through a non-carrier first failure listed in Theorem 8.105, then the branch is either class-incompatible or is assigned to a named local state-space alternative which is already eliminated by the local packages of the paper.*

Proof. The ordered validation in Theorems 8.86 and 8.105 lists the possible non-carrier first failures. We check them in that order. Exhaustion failures are routed by Theorem 10.101; the only alternatives not retained there are class-incompatible alternatives or named local exits. Representation, pressure, and modulation failures are the ordered failures in Theorems 12.13 and 12.20 and the absolutely continuous repaired-gauge consequences of Theorem 10.54; when the repaired chart is available, pressure and modulation are consequences, and when it is not available the failure is the representation alternative. On a physical branch which has already passed covered entry, representation failures cannot remain as unclassified analytic gaps: if a nondegenerate represented chart is not available, the local state-decomposition places the branch in a multibubble/cascade, gauge-degenerate, scale-degenerate, or scale-rigid alternative, all eliminated by Theorem 11.14.

Critical-mass failures are used here only as local terminal-window diagnostics. Failure of lower retained mass means that the candidate was not a retained active core, while failure of a finite local upper L^3 bound on a compact selected cylinder produces an additional compact active label; this is routed by Theorem 10.82. The whole-space critical $L^3(\mathbb{R}^3)$ alternatives in Theorem 12.24 are therefore pre-compact-data exits for the compact-cost criterion, not global assumptions in the local no-Type-II proof. Once the retained local finite critical-mass window is reached, the compact tests are closed by Theorems 10.96 and 10.97. Exterior tightness and leakage are routed by Theorem 9.38. Local windowed H^1-loss is the rough-core row and is eliminated by Theorems 11.7 and 11.14. Multibubble, multicore, cascade, and gauge-degenerate rows are alternatives in Theorem 11.13 and are eliminated by Theorem 11.14. Finite-cost scale collapse is eliminated by Theorem 11.8. Scale-rigid residual and retained compact scale-collapse rows are eliminated by Theorems 5.40 and 7.30. Finally, cost well-posedness and cost-comparison failures are precisely the ordered cost rows of Theorems 12.25 to 12.28; for the canonical identity cost the comparison is a theorem, and any non-identity failure is recorded as its own cost-compatibility exit. The thin-drift, autonomous-modulation, and compactness failures which can arise inside the scale-collapse stratification are the named first failures of Theorem 7.28; they are non-retained scale-collapse exits, while the retained compact scale-collapse state is excluded by Theorem 7.30. Thus every non-carrier first failure is already class-incompatible or assigned to a closed named local branch. No step uses a global L^3 bound, a global profile theorem, or vanishing localized dissipation outside the selected local branch. □

Lemma 8.159 (Carrier applicability on physical compact-cost branches)**.** *Every physical local Type II branch in the pre-carrier validated compact-cost regime either exits through the ordered cost-compatibility row, or else satisfies the carrier applicability hypotheses used in Theorem 8.156 on the retained canonical channel: the validated compact cost is the fixed identity cost and is energy-controlled with constant* 1*, and suitable repaired-gauge local energy supplies the fixed and moving carrier subidentities.*

Proof. The pre-carrier regime already includes the repaired-gauge representation, local pressure reconstruction, local cost well-posedness, and compatibility of the validated compact cost. The ordered cost row is deliberately read before carrier validation. If the selected non-identity diagnostic has only the lower comparison used for cost-divergence transfer, but lacks the upper energy-control comparison needed by the carrier inequality, the branch is not silently promoted to an energy-controlled carrier branch; it is recorded as the cost-compatibility or carrier-comparison exit handled by Theorem 8.158. On the retained carrier branch the canonical cost channel is selected:

$$\mathfrak{C}_{\mathrm{II}}^{NS} = \tilde{\mathfrak{D}}_{R_0}.$$

By Theorem 12.27 this is a valid localized cost, and by Theorem 8.71 it is energy-controlled with constant 1. This uses the identity equality only; no unproved upper comparison for a non-identity cost is invoked.

The branch comes from a suitable physical solution and has an absolutely continuous repaired-gauge chart on each compact renormalized window. Hence Theorem 10.67 gives the renormalized local energy inequality, and Theorem 8.102 converts it into the fixed and moving carrier subidentities. These are local compact-window statements; no whole-space energy or global compactness estimate is used. □

Lemma 8.160 (Negative drift is an assigned local branch)**.** *On every physical local Type II branch in the canonical windowwise pre-carrier regime, failure of $J_6 \in L^1(\tau_1, \infty)$ on every later tail is exactly the negative scale-drift branch of Theorem 10.25. This branch is routed by the local scale-drift alternatives and the local scale-collapse packages; it is not an additional ambient hypothesis.*

Proof. The J_6 component is the negative-scale contribution in the moving localized-energy identity. By Theorems 10.25 and 10.30, its nonintegrability on every later tail is precisely the scale-collapse drift obstruction. From this point the branch is not closed by feeding the same adjacent-closure statement back into itself; it is passed to the local scale-collapse stratification.

Before applying scale-collapse rigidity, verify the local prerequisites by the same first-failure order. The pre-carrier retained branch supplies the repaired-gauge representation, active supercritical scaling, retained compact-window local critical mass, critical tightness, local H^1 control, and suitable local energy on compact cylinders. If any of these inputs fails, or if the selected core is not nonvanishing in the retained local window, then the first failure is a critical-mass, exterior/leakage, rough-core, representation, or compactness exit already closed by Theorem 8.158. If the scale path is not a genuine retained collapse path, the branch is a finite-cost, scale-degenerate, or bounded-window/scale-rigid exit, again a named local alternative. Thus the only J_6-failure case remaining after ordered routing is a represented branch with a nonvanishing selected core and genuine retained scale collapse.

Apply Theorems 7.31 and 11.9 to that remaining branch. The finite scale-cost and finite absolute-cost alternatives are excluded by Theorem 11.8. Thin-drift, autonomous-modulation failure, compactness failure, exterior/separated-core loss, rough-core loss, and multicore or same-point cascade are exactly the non-retained first failures listed in Theorem 7.28; by Theorem 8.158 those labels are already routed to closed local state-space branches. The only retained scale-collapse state left after those exits is the retained compact scale-collapse state, and it is excluded by Theorem 7.30; if the retained state becomes scale-rigid, it is excluded by Theorem 5.40. Thus J_6-failure is a routed local scale-collapse branch and not a hidden assumption. □

Theorem 8.161 (Ambient adjacent closure for physical Type II branches)**.** *Every physical local Type II branch which reaches the compact-cost validation layer satisfies adjacent-stratum closure for the compact-cost regime in the sense of Theorem 8.96. Equivalently, the hypotheses of Theorem 8.157 are theorems on the physical local Type II state-space branch.*

Proof. Non-carrier adjacent exits are closed by Theorem 8.158. The exterior, local H^1-loss, modulation-driven recentering, scale-rigid residual, and negative scale-drift branches appearing in the stratified moving-carrier alternative are the named local branches routed in Theorems 5.40, 8.160, 9.38, 10.54 and 11.7. Carrier applicability and energy-control of the canonical cost are supplied by Theorem 8.159. The windowwise transition and positive-scale alternative statements are no longer assumptions: Theorems 8.149 and 8.154 prove them on the canonical windowwise moving family. The negative-drift alternative required by Theorem 8.156 is Theorem 8.160.

All hypotheses of Theorem 8.157 are therefore verified branchwise. Applying that corollary gives compact-cost adjacent-stratum closure. □

Corollary 8.162 (Ambient adjacent wrapper discharged)**.** *The ambient adjacent-stratum closure and carrier-applicability obligation for the unconditional no-Type-II theorem is discharged by local state-space stratification and the canonical windowwise carrier package.*

Proof. This is Theorem 8.161. □

Definition 8.163 (Transition/leakage alternative statement)**.** The pre-carrier validated compact-cost regime has transition/leakage alternative if, for every branch ω in that regime and every moving test family in the sense of Theorem 8.135, failure of at least one of

$$J_1, \qquad J_2, \qquad J_3$$

to belong to $L^1(\tau_1, \infty)$ on every later tail $[\tau_1, \infty)$ forces ω into the exterior/leakage alternative of Theorem 8.133.

Definition 8.164 (Pressure alternative statement)**.** The pre-carrier validated compact-cost regime has pressure alternative if, for every branch ω in that regime and every moving test family in the sense of Theorem 8.135, failure of

$$J_4 \in L^1(\tau_1, \infty)$$

on every later tail $[\tau_1, \infty)$ forces ω into the exterior/leakage alternative of Theorem 8.133.

Definition 8.165 (Translation alternative statement)**.** The pre-carrier validated compact-cost regime has the translation alternative if, for every branch ω in that regime and every moving test family in the sense of Theorem 8.135, failure of

$$J_5 \in L^1(\tau_1, \infty)$$

on every later tail $[\tau_1, \infty)$ forces ω into the translation alternative of Theorem 8.133.

Definition 8.166 (Positive-scale alternative statement)**.** The pre-carrier validated compact-cost regime has positive-scale alternative if, for every branch ω in that regime and every moving test family in the sense of Theorem 8.135, failure of

$$J_7 \in L^1(\tau_1, \infty)$$

on every later tail $[\tau_1, \infty)$ forces ω into the positive-scale alternative of Theorem 8.133.

Theorem 8.167 (Small alternative statements imply stratified moving-cutoff alternative)**.** *Assume every branch in the pre-carrier validated compact-cost regime uses a validated compact cost that is energy-controlled by the fixed identity cost. Assume repaired-gauge suitable local energy gives the moving carrier subidentity in the sense of Theorem 8.102. Assume also that the regime has the transition/leakage, pressure, translation, and positive-scale alternative statements of Theorems 8.163 to 8.166, and that failure of $J_6 \in L^1(\tau_1, \infty)$ on every later tail forces entry into the negative scale-drift obstruction of Theorem 10.25. Then the regime has stratified moving-cutoff alternative in the sense of Theorem 8.133.*

Proof. Let ω be a branch in the pre-carrier validated compact-cost regime, and choose the affine moving test family

$$R(\tau) = R_0 + (\tau - \tau_0), \qquad \Phi(\tau, y) = \phi(y/R(\tau)).$$

This is admissible by Theorem 8.135. Let $J_1, \dots, J_7$ be the associated moving components.

If there exists $\tau_1 \geq \tau_0$ such that

$$J_k \in L^1(\tau_1, \infty) \qquad \text{for all } 1 \leq k \leq 7,$$

then Theorem 8.136 gives the moving-cutoff localized-energy closure criterion. Hence ω satisfies alternative (ii) in Theorem 8.133.

Assume now that no later tail has all seven components integrable. Since there are only seven components, one may choose an index $k \in \{1, \dots, 7\}$ that fails on arbitrarily late tails. Such a component then fails on every later tail: otherwise $J_k \in L^1(\tau_1, \infty)$ for some τ_1, hence also on every later tail beyond τ_1, contrary to the choice of k. If $k \in \{1, 2, 3\}$, the transition/leakage alternative statement places ω in alternative (iii) of Theorem 8.133. If $k = 4$, the pressure alternative statement gives the same conclusion. If $k = 5$, the translation alternative statement gives alternative (iv). If $k = 6$, the negative scale-drift hypothesis gives alternative (vi). If $k = 7$, the positive-scale alternative statement gives alternative (v). Therefore one of the alternatives in Theorem 8.133 always holds. ☐

Theorem 8.168 (Stratified moving-cutoff alternative leaves only named branches)**.** *Assume every branch in the pre-carrier validated compact-cost regime uses a validated compact cost that is energy-controlled by the fixed identity cost. Assume also that the regime has stratified moving-cutoff alternative in the sense of Theorem 8.133. Then every branch in the pre-carrier validated compact-cost regime, after restriction to a later terminal tail if necessary, satisfies at least one of the following conclusions:*

(i) *it is excluded by the fixed-cutoff carrier closure theorem Theorem 8.107;*
(ii) *it is excluded by the moving-cutoff carrier closure theorem Theorem 8.108;*
(iii) *it is assigned to one of the already named critical-tightness failure, exterior concentration, or local windowed H^1-loss branches;*

(iv) *it is assigned to the already named modulation-integrability failure or to a multicore/exterior branch produced after recentering;*
(v) *it is assigned to the already named scale-rigid residual alternative or to a bounded selected-window limit refinement of that alternative;*
(vi) *it enters the negative scale-drift obstruction of Theorem 10.25.*

Proof. Let ω be a branch in the pre-carrier validated compact-cost regime. By the energy-control hypothesis, ω is one of the branches covered by Theorem 8.133. If alternative (i) in that definition holds, then Theorem 8.107 excludes ω. If alternative (ii) holds, then Theorem 8.108 excludes ω. In the remaining alternatives, the branch assignment is part of the definition of the stratified alternative itself. This yields the displayed list. □

Corollary 8.169 (Adjacent closure reduced to stratified moving-cutoff alternative)**.** *Assume the non-carrier adjacent strata listed in Theorem 8.105 are closed by their corresponding local compactness or interface arguments in the ambient Type II class. Assume also that the exterior concentration, local windowed H^1-loss, modulation-driven recentering branches, scale-rigid residual branches, and the negative scale-drift obstruction appearing in Theorem 8.133 are closed by their assigned local arguments in the ambient class. If the pre-carrier validated compact-cost regime has stratified moving-cutoff alternative, then the compact-cost regime has adjacent-stratum closure in the sense of Theorem 8.96.*

Proof. Let ω be a branch with active supercritical scaling and validated compact Type II cost divergence. By Theorem 8.90, either ω remains in the retained compact stratum or it enters one of the named adjacent strata. The retained compact stratum is excluded by Theorem 8.95. If ω enters a non-carrier adjacent stratum before the pre-carrier regime, then the first hypothesis closes it by Theorem 8.105. It remains to consider a branch in the pre-carrier validated compact-cost regime. Theorem 8.168 then shows that ω is either excluded by fixed or moving carrier closure, or is assigned to one of the branches closed by the second hypothesis. Therefore every compact-cost branch is either excluded on the retained stratum or assigned to a previously closed adjacent branch. This is exactly Theorem 8.96. □

Corollary 8.170 (Adjacent closure reduced to the small alternative statements)**.** *Assume the non-carrier adjacent strata listed in Theorem 8.105 are closed by their corresponding local compactness or interface arguments in the ambient Type II class. Assume also that the exterior concentration, local windowed H^1-loss, modulation-driven recentering branches, scale-rigid residual branches, and the negative scale-drift obstruction are closed by their assigned local arguments in the ambient class. Assume every branch in the pre-carrier validated compact-cost regime uses a validated compact cost that is energy-controlled by the fixed identity cost, that repaired-gauge suitable local energy gives the moving carrier subidentity, and that the transition/leakage, pressure, translation, and positive-scale alternative statements of Theorems 8.163 to 8.166 hold, together with the negative-drift alternative statement used in Theorem 8.167. Then the compact-cost regime has adjacent-stratum closure in the sense of Theorem 8.96.*

Proof. Theorem 8.167 gives the stratified moving-cutoff alternative. Then Theorem 8.169 gives the adjacent-stratum closure conclusion. □

Remark 8.171 (Cost divergence on the retained compact stratum). The compact Type II cost argument gives infinite localized renormalization cost under the compact PDE hypotheses. Passing from that cost divergence to exclusion from the Type II class uses the corrected localized energy inequality constructed in Theorem 8.73. On the retained compact stratum, the carrier exists by Theorem 8.87; failure of the carrier test is not a retained compact branch but an adjacent stratum.

9. Terminal Profile Decoupling and Exterior Regularity

This section records the terminal local estimates used to remove multibubble configurations, small residual components, and components carried at a positive physical distance from the selected Type II core. The statements are conditional on the analytic inputs explicitly stated below: terminal state-space completeness on compact windows, repaired-gauge representation, local Caccioppoli estimates, exterior regularity, and the local compact scale-collapse routing theorem Theorem 7.30.

9.1. **Renormalized frames and active local states.** A renormalized frame consists of a center $x_c(t)$, a scale $\lambda(t) > 0$, and renormalized time

$$\frac{d\tau}{dt} = \lambda(t)^{-2}.$$

It represents the physical solution by

$$V(y,\tau) = \lambda(t)u(x_c(t) + \lambda(t)y, t).$$

When the repaired gauge is imposed, the represented velocity satisfies

$$\partial_\tau V + (V\cdot\nabla)V + \nabla P = \nu\Delta V + a(\tau)(V + y\cdot\nabla V) + b(\tau)\cdot\nabla V, \qquad \nabla\cdot V = 0,$$

where

$$a(\tau) = \partial_\tau \log\lambda(\tau),$$

and b is the translation modulation coefficient determined by the centering condition. In a fixed frame, every other local state either remains at bounded relative position and comparable scale, escapes to spatial infinity in the frame variables, appears at a much larger same-point scale as a perturbative ambient field in the innermost frame, or lies at a smaller scale and therefore requires selecting that smaller active frame.

Definition 9.1 (Active local terminal state data)**.** Let $t_n \uparrow T^*$. Active local terminal state data consist of local state-space classes $\mathcal{A}_j$, scales $\lambda_j^n \to 0$, centers $x_j^n \to x_j^*$, compact terminal windows I_j, and representatives W_j^n in the corresponding frames. Each active class has a detecting compact cylinder $B_{R_j} \times I_j$ and a local mass floor

$$\limsup_{n\to\infty} \|W_j^n\|_{L^\infty(I_j;L^3(B_{R_j}))} \geq \varepsilon_{\mathrm{loc}} > 0.$$

After grouping comparable parameters based at the same physical point, distinct classes are separated in the local state-space sense: they have different physical points, strictly separated scales, or have already been assigned to an adjacent local alternative. On every compact terminal cylinder and for every finite retained set J, the local field is written

$$V_n = U_n^J + S_n^J,$$

where U_n^J is the grouped retained local state and S_n^J is the nonretained residual. Positive local mass in S_n^J is not a new unlabelled object: by terminal state-space completeness it is assigned either to a named adjacent local alternative or to the retained comparable class, in which case it must be included in U_n^J. No global profile decomposition, Brézis–Lieb identity, or heat-flow remainder smallness is part of this data. The corresponding local active mass floor is stated and proved in Theorem 9.32.

Lemma 9.2 (Finite active local state count on a terminal slice)**.** *Fix a compact terminal cylinder $B_R \times I$, a selected time sequence $\tau_n \in I$, and assume*

$$\sup_n \|V_n\|_{L^\infty(I;L^3(B_R))} \leq M_R < \infty.$$

Assume every active local state detected on the slice τ_n inside B_R has local mass at least $\varepsilon_{\mathrm{loc}}$, and that, after grouping comparable retained classes, the detecting balls on that slice have overlap multiplicity at most N_R. Then the number of active local states detected on that terminal slice is finite and satisfies

$$\#\mathcal{J}_{\mathrm{act}}(R, I) \leq \left\lfloor \frac{N_R M_R^3}{\varepsilon_{\mathrm{loc}}^3} \right\rfloor .$$

Proof. For each active state choose a detecting ball on the selected time slice, shrunk if necessary so that the family has overlap multiplicity at most N_R. At that common detecting time, the local mass floor gives

$$\int_{B_j} |V_n(y, \tau_n)|^3 \, dy \geq \varepsilon_{\mathrm{loc}}^3 + o_n(1)$$

for the ball B_j associated with the j-th state. Summing over any finite active subfamily and using the overlap bound,

$$(\#J)\varepsilon_{\mathrm{loc}}^3 \leq N_R \sup_n \sup_{\tau \in I} \int_{B_R} |V_n(y, \tau)|^3 \, dy \leq N_R M_R^3.$$

Thus every finite active subfamily has cardinality at most $N_R M_R^3/\varepsilon_{\mathrm{loc}}^3$. If there were more active states than the displayed integer bound, a finite subfamily with one additional element would contradict this estimate. □

9.2. Same-point and separated-point reductions.

Lemma 9.3 (Comparable same-point states form a compound state)**.** *Let a finite set $\mathcal{C}$ of active local states have the same physical center $x_j^* = x_*$. Suppose that for a representative center and scale (x_*^n, λ_*^n),*

$$\frac{\lambda_j^n}{\lambda_*^n} \to \rho_j \in (0, \infty), \qquad \frac{x_j^n - x_*^n}{\lambda_*^n} \to z_j \in \mathbb{R}^3.$$

Let $W_{j,n}$ denote the local representative of the j-th state in its own frame on compact cylinders, and assume

$$W_{j,n} \to W_j \quad \textit{strongly in } L^3_{\mathrm{loc}}$$

on the relevant terminal slice, or in $L^\infty_{\mathrm{loc}} L^3_{\mathrm{loc}}$ on a compact terminal window. Write

$$\rho_j^n := \frac{\lambda_j^n}{\lambda_*^n}, \qquad z_j^n := \frac{x_j^n - x_*^n}{\lambda_*^n}.$$

Then, in the frame (x_*^n, λ_*^n),

$$(\rho_j^n)^{-1} W_{j,n}\left(\frac{y - z_j^n}{\rho_j^n}\right) \to \rho_j^{-1} W_j \left(\frac{y - z_j}{\rho_j}\right)$$

strongly in the same local topology on every compact subcylinder on which the right-hand side is defined. Hence the comparable class has compound local representative

$$\Phi_{\mathcal{C}}(y) := \sum_{j \in \mathcal{C}} \rho_j^{-1} W_j \left(\frac{y - z_j}{\rho_j}\right).$$

If the compound representative carries the local mass floor

$$\|\Phi_{\mathcal{C}}\|_{L^3(B_R)} \geq \varepsilon_{\mathrm{loc}}$$

on a detecting compact cylinder, it is one active local state in the selected frame. If no compact cylinder carries the local mass floor, it belongs to the perturbative residual class by Theorem 9.31.

Proof. Set

$$\rho_j^n = \frac{\lambda_j^n}{\lambda_*^n}, \qquad z_j^n = \frac{x_j^n - x_*^n}{\lambda_*^n}.$$

The transformed local representative is

$$T_n W_{j,n}(y) = (\rho_j^n)^{-1} W_{j,n}\left(\frac{y - z_j^n}{\rho_j^n}\right),$$

and the proposed limit is

$$TW_j(y) = \rho_j^{-1} W_j\left(\frac{y - z_j}{\rho_j}\right).$$

Here $T_{\rho,z}F(y) := \rho^{-1}F((y-z)/\rho)$. Thus $T_n W_{j,n} = T_{\rho_j^n, z_j^n} W_{j,n}$ and $TW_j = T_{\rho_j, z_j} W_j$. Fix a compact ball B_R in the representative frame. For all large n, the preimage of B_R under $y \mapsto (y - z_j^n)/\rho_j^n$ is contained in a fixed compact ball $B_{R'}$ in the j-th frame. The local strong convergence gives

$$\|T_{\rho_j^n, z_j^n}(W_{j,n} - W_j)\|_{L^3(B_R)} \le C\|W_{j,n} - W_j\|_{L^3(B_{R'})} \to 0.$$

It remains to move the fixed local limit W_j. Translation and dilation are strongly continuous on $L^3(B_{R'})$ after restriction to B_R: prove this first for $C_c^\infty(B_{R'})$ by uniform convergence on compact sets, then approximate W_j in $L^3(B_{R'})$. This proves the local convergence. The same argument applied uniformly in τ gives the $L^\infty_{\mathrm{loc}}L^3_{\mathrm{loc}}$ version on compact terminal windows. Summing over the finite class gives $\Phi_{\mathcal{C}}$. The final alternative is the local state-space criterion: a compound state with the detection threshold is retained as active, whereas a component with no compact-cylinder detection threshold is perturbative by the local residual theorem. □

Lemma 9.4 (Innermost-frame rescaling of strict outer states)**.** *Suppose all active local states have the same physical center x_* and split into distinct scale classes. If λ_1^n is the smallest active scale, then every outer scale $\lambda_j^n \gg \lambda_1^n$ satisfies*

$$\mu_j^n := \lambda_1^n/\lambda_j^n \to 0,$$

and in the innermost frame the j-th outer state has the local form

$$W_j^n(y) = \mu_j^n \widetilde{W}_{j,n}\left(\frac{x_*^n - x_j^n}{\lambda_j^n} + \mu_j^n y\right),$$

where $\widetilde{W}_{j,n}$ is the representative of the j-th state in its own outer frame.

Proof. The j-th local representative in physical coordinates is

$$u_j^n(x) = (\lambda_j^n)^{-1}\widetilde{W}_{j,n}\left(\frac{x - x_j^n}{\lambda_j^n}\right)$$

on the compact part of its outer terminal frame. Pull this back into the innermost frame $x = x_*^n + \lambda_1^n y$, with the critical rescaling $W_j^n(y) = \lambda_1^n u_j^n(x_*^n + \lambda_1^n y)$. Substituting,

$$W_j^n(y) = \frac{\lambda_1^n}{\lambda_j^n}\widetilde{W}_{j,n}\left(\frac{x_*^n + \lambda_1^n y - x_j^n}{\lambda_j^n}\right) = \mu_j^n \widetilde{W}_{j,n}\left(\frac{x_*^n - x_j^n}{\lambda_j^n} + \mu_j^n y\right),$$

which is the displayed form. The convergence $\mu_j^n \to 0$ is the hypothesis $\lambda_j^n \gg \lambda_1^n$. □

Lemma 9.5 (Local L^3-vanishing of strict outer states in the innermost frame)**.** *Under the hypotheses of Theorem 9.4, the outer states satisfy the following retained-frame alternative. If*

$$\xi_j^n := \frac{x_*^n - x_j^n}{\lambda_j^n}$$

stays in a compact subset of the outer frame and $\widetilde{W}_{j,n} \to W_j$ strongly in L^3_{loc}, then, for every $R < \infty$,

$$\|W_j^n\|_{L^3(B_R)} \to 0 \quad (n \to \infty).$$

If ξ_j^n leaves every compact subset of the outer frame, the contribution is assigned to the escaping-offset/exterior row of the local state-space decomposition rather than to a retained strict outer state.

Proof. Assume first that ξ_j^n stays in a fixed compact set K. Choose a compact K' containing $K + \mu_j^n B_R$ for all large n. By the formula in Theorem 9.4,

$$\|W_j^n\|_{L^3(B_R)}^3 = \int_{\xi_j^n + \mu_j^n B_R} |\widetilde{W}_{j,n}(z)|^3 \, dz.$$

The strong $L^3(K')$ convergence reduces this to the same integral with W_j. Since $|W_j|^3 \in L^1(K')$ and the sets $\xi_j^n + \mu_j^n B_R$ have measure tending to zero, absolute continuity of the integral gives convergence to zero. If ξ_j^n escapes every compact set, then the outer state is not represented on a fixed compact outer cylinder; this is exactly the escaping-offset/exterior boundary face in the local state-space compactification. □

Theorem 9.6 (Perturbative estimates for strict outer states)**.** *In the same setting as Theorem 9.4, the strict outer states are perturbative in the innermost retained frame:*

(i) *finite-offset regular outer states, namely those for which $(x_*^n - x_j^n)/\lambda_j^n$ is bounded, decompose on each fixed inner ball into a constant drift plus an $O((\mu_j^n)^2)$ shear remainder;*
(ii) *the constant drifts produced by item (i) are absorbed into the translation-modulation coefficient of the innermost frame;*
(iii) *the remaining pressure, cutoff, and nonlinear interactions generated by the outer states are perturbative in the compact scale-collapse topology in which the local perturbation criterion Theorem 9.10 and the pressure-tail decoupling Theorem 9.12 apply.*

The escaping-offset alternative $(x_*^n - x_j^n)/\lambda_j^n \to \infty$ is treated by Theorem 9.5.

Proof. In the escaping-offset case, Theorem 9.5 gives local L^3-vanishing on every innermost compact cylinder, and the local pressure-tail lemma gives the pressure and residual convergence.

It remains to consider bounded offsets. If the outer state is not regular at the observed offset, then some smaller offset cylinder has CKN quantity above the threshold of Theorem 9.34. That cylinder is a positive local state-space label at a larger same physical point scale, so the branch is assigned to the same-point cascade/secondary-core alternative before entering the retained innermost class. Thus, in the retained case, every bounded offset cylinder for the outer state is CKN-subcritical. The CKN criterion gives uniform C^1 bounds on a smaller offset cylinder, after subtracting the pressure mean.

On that retained regular cylinder, Taylor expansion at the offset gives a spatially constant leading term plus a shear remainder of size $O(\mu_j^n)$ in C^1 on every fixed inner ball. The Navier–Stokes scaling contributes the additional factor μ_j^n in the velocity amplitude, so the nonconstant velocity remainder is $O((\mu_j^n)^2)$ locally. The spatially constant term is absorbed into the translation modulation coefficient. The remaining velocity, pressure, cutoff, and modulation terms vanish in the compact-window topologies by Theorems 9.10 and 9.12. This is exactly the asserted perturbative reduction. □

Lemma 9.7 (Separated physical centers are locally invisible)**.** *Let $p \neq q$ be two active physical concentration points,*

$$x_p^* \neq x_q^*, \qquad d_{pq} := |x_p^* - x_q^*| > 0.$$

In the p-frame define the q-state contribution

$$W_{q\to p}^n(y) := \lambda_p^n (\lambda_q^n)^{-1} \phi^q \left(\frac{x_p^n + \lambda_p^n y - x_q^n}{\lambda_q^n} \right).$$

If $\phi^q \in L^3(\mathbb{R}^3)$, then for every $R < \infty$,

$$\|W_{q\to p}^n\|_{L^3(B_R)} \to 0.$$

If the corresponding representative pressure $\Pi^q \in L^{3/2}(\mathbb{R}^3)$, then the pure pressure contribution

$$P_{q\to p}^n(y) = (\lambda_p^n)^2 (\lambda_q^n)^{-2} \Pi^q \left(\frac{x_p^n + \lambda_p^n y - x_q^n}{\lambda_q^n} \right)$$

satisfies

$$\|P_{q\to p}^n\|_{L^{3/2}(B_R)} \to 0.$$

Proof. By the change of variables $x = x_p^n + \lambda_p^n y$,

$$\|W_{q\to p}^n\|_{L^3(B_R)}^3 = \int_{B(x_p^n, \lambda_p^n R)} (\lambda_q^n)^{-3} \left| \phi^q \left(\frac{x - x_q^n}{\lambda_q^n} \right) \right|^3 dx.$$

Set $z = (x - x_q^n)/\lambda_q^n$. Then

$$\|W_{q\to p}^n\|_{L^3(B_R)}^3 = \int_{B(c_n, \rho_n)} |\phi^q(z)|^3 \, dz,$$

where

$$c_n = \frac{x_p^n - x_q^n}{\lambda_q^n}, \qquad \rho_n = \frac{\lambda_p^n R}{\lambda_q^n}.$$

Since $x_p^n \to x_p^*$, $x_q^n \to x_q^*$, and $\lambda_q^n \to 0$,

$$|c_n| \sim d_{pq}/\lambda_q^n \to \infty.$$

Moreover

$$\frac{\rho_n}{|c_n|} = \frac{\lambda_p^n R}{|x_p^n - x_q^n|} \to 0.$$

Thus the balls $B(c_n, \rho_n)$ escape to spatial infinity: for every $A < \infty$, they are contained in $\{|z| > A\}$ for all sufficiently large n. Since $\phi^q \in L^3$,

$$\int_{|z|>A} |\phi^q(z)|^3 \, dz \to 0 \qquad (A \to \infty),$$

and hence $\|W_{q\to p}^n\|_{L^3(B_R)} \to 0$.

For the pressure term the same change of variables gives

$$\|P_{q\to p}^n\|_{L^{3/2}(B_R)}^{3/2} = \int_{B(c_n, \rho_n)} |\Pi^q(z)|^{3/2} \, dz.$$

The same escaping-ball argument and $\Pi^q \in L^{3/2}$ imply the stated pressure convergence. □

Definition 9.8 (Multipoint decoupling conditions)**.** For each active point and selected frame, the required multipoint decoupling conditions are:

(i) local velocity decoupling in the selected frame;
(ii) pressure decoupling modulo time-dependent constants in $L^1_\tau H^{-1}_y$ on compact cylinders;
(iii) localization errors from spatial cutoffs vanish in $L^1_\tau H^{-1}_y$;
(iv) repaired-gauge modulation coefficients remain bounded after exterior ambient terms are absorbed;
(v) each active point has positive finite critical mass in its own frame;
(vi) after states centered at other physical points are treated as perturbative exterior terms, the selected frame is tested against Theorem 7.27: either the first failed retained-state input is routed by Theorem 7.28 to a named local exit, or all retained compact scale-collapse inputs hold.

Theorem 9.9 (Reduction of active multibubbles)**.** *Assume the Type II local terminal state data, the active local mass floor, the same-point nonlinear decoupling conditions, the multipoint decoupling conditions of Theorem 9.8. Then no active multibubble Type II configuration can occur.*

Proof. By Theorem 9.2, there are only finitely many active local states on each selected terminal slice. Partition them by physical concentration point x^*_α, and within each physical point partition them by comparable scale class.

If a scale class contains several comparable local states, Theorem 9.3 combines them into one compound state. If the compound state has no detecting compact cylinder, it is perturbative by Theorem 9.31; if it is active, it is a single active local state for that frame.

If a physical point contains strict same-point scale separation, Theorem 9.6 and the same-point nonlinear decoupling conditions reduce the innermost active scale to a single repaired-gauge solution with perturbative outer interactions. The compact scale-collapse routing theorem Theorem 7.30 then rules out the corresponding retained same-point cascade; if one of its retained hypotheses fails, Theorem 7.28 assigns the branch to a named local alternative and it is not a retained active multibubble configuration.

It remains to consider separated physical points. Choose any active point x^*_α and place the selected frame at its active scale and center. By Theorem 9.7, every state centered at a different physical point is locally L^3-invisible in this frame. By the multipoint decoupling conditions, its pressure, cutoff, and modulation interactions are also perturbative in the compact scale-collapse topology. Therefore the selected frame contains a single active repaired-gauge solution whose scale tends to zero. If any retained compact scale-collapse input fails, the multipoint decoupling condition and Theorem 7.28 route the branch to a named local exit, so it is not a retained active multibubble configuration. If no retained input fails, the selected frame is a retained compact scale-collapse state in the sense of Theorem 7.27. Applying Theorem 7.30 excludes that selected active point. Since the chosen active point was arbitrary, no active multibubble configuration can occur. □

9.3. **Local perturbation and pressure decoupling.**

Lemma 9.10 (Local perturbation criterion)**.** *Let $I \subset \mathbb{R}$ be compact and $Q_R = B_R \times I$. Let U_n be a repaired-gauge solution on Q_{2R} satisfying*

$$\|U_n\|_{L^\infty_\tau L^3_y(B_{2R})} + \|U_n\|_{L^2_\tau H^1_y(B_{2R})} \le C_R.$$

Let S_n be a divergence-free perturbation on Q_{2R} with

$$S_n \to 0 \quad \text{in } L^\infty_\tau L^3_y(B_{2R}),$$

$$U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n \to 0 \quad \text{in } L^1_\tau L^2_y(B_R),$$

$$\mathcal{R}_n := \partial_\tau S_n - \nu\Delta S_n + a_n(S_n + y\cdot\nabla S_n) + b_n\cdot\nabla S_n \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R),$$

where $|a_n| + |b_n| \le M_{ab}$*, and suppose the pressure cross-term satisfies*

$$\nabla \mathcal{P}_{\mathrm{cross},n} \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R).$$

Then replacing $U_n + S_n$ *by* U_n *changes the repaired-gauge Navier–Stokes equation on* Q_R *by an error tending to zero in*

$$L^1_\tau H^{-1}_y(B_R).$$

Proof. The linear part is precisely $\mathcal{R}_n$, which tends to zero by hypothesis. The nonlinear difference is

$$(U_n + S_n)\cdot\nabla(U_n + S_n) - U_n\cdot\nabla U_n = \operatorname{div} F_n, \qquad F_n := U_n\otimes S_n + S_n\otimes U_n + S_n\otimes S_n,$$

using $\nabla\cdot U_n = \nabla\cdot S_n = 0$ distributionally. By the assumed $L^1_\tau L^2_y$ stress convergence, this divergence tends to zero in

$$L^1_\tau H^{-1}_y(B_R),$$

because for $\varphi \in H^1_0(B_R;\mathbb{R}^3)$,

$$\left|\int_{B_R}(U_n\otimes S_n + S_n\otimes U_n + S_n\otimes S_n) : \nabla\varphi\,dy\right| \le \|F_n\|_{L^2(B_R)} \times \|\nabla\varphi\|_{L^2(B_R)}.$$

The pressure contribution vanishes by the assumed cross-pressure convergence. Therefore the total perturbative error tends to zero in

$$L^1_\tau H^{-1}_y(B_R).$$

□

Lemma 9.11 (Static same-point outer profiles vanish locally)**.** *Let*

$$W_n(y) = \mu_n\phi(\xi_n + \mu_n y), \qquad \mu_n \to 0, \qquad \phi \in L^3(\mathbb{R}^3).$$

Then, for every $R < \infty$*,*

$$\|W_n\|_{L^3(B_R)} \to 0.$$

Proof. Changing variables $z = \xi_n + \mu_n y$,

$$\|W_n\|^3_{L^3(B_R)} = \int_{B(\xi_n,\mu_n R)} |\phi(z)|^3\,dz.$$

The measure of $B(\xi_n,\mu_n R)$ tends to zero. Since $|\phi|^3 \in L^1(\mathbb{R}^3)$, absolute continuity of the integral gives

$$\int_{B(\xi_n,\mu_n R)} |\phi(z)|^3\,dz \to 0,$$

uniformly in ξ_n. Hence $\|W_n\|_{L^3(B_R)} \to 0$. □

Lemma 9.12 (Pressure decoupling from kernel tails)**.** *Let* Γ_{ij} *be the Calderon–Zygmund kernel for*

$$P = \mathcal{R}_i\mathcal{R}_j(F_{ij}).$$

Assume

$$F_n \to 0 \quad \text{in } L^1_\tau L^2_y(B_A) \quad \text{for every fixed } A,$$

and

$$\sup_n \|F_n\|_{L^\infty_\tau L^{3/2}_y(\mathbb{R}^3)} < \infty.$$

Then the pressure generated by F_n *satisfies, for every* $R < \infty$*,*

$$P_n - c_{n,R} \to 0 \quad \text{in } L^1_\tau L^2_y(B_R)$$

for suitable constants $c_{n,R}(\tau)$, and consequently

$$\nabla P_n \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R).$$

Proof. Fix $A > 2R$ and write

$$F_n^{\mathrm{loc}} = F_n \mathbf{1}_{B_A}, \qquad F_n^{\mathrm{far}} = F_n \mathbf{1}_{\mathbb{R}^3 \setminus B_A}.$$

For the local part, Calderon–Zygmund boundedness gives

$$\|P_n^{\mathrm{loc}}\|_{L^1_\tau L^2_y(B_A)} \le C_A \|F_n\|_{L^1_\tau L^2_y(B_A)} \to 0.$$

For the far part define

$$c_{n,R}(\tau) = \int_{|z|>A} \Gamma_{ij}(-z) F_{n,ij}(z,\tau)\, dz,$$

first for compactly supported approximations and then by passage to the $L^{3/2}$ limit. For $y \in B_R$ and $|z| > A > 2R$,

$$|\Gamma_{ij}(y-z) - \Gamma_{ij}(-z)| \le C_R |z|^{-4}.$$

Hence

$$|P_n^{\mathrm{far}}(y,\tau) - c_{n,R}(\tau)| \le C_R \int_{|z|>A} |z|^{-4} |F_n(z,\tau)|\, dz.$$

By Holder's inequality with exponents $3/2$ and 3,

$$\int_{|z|>A} |z|^{-4}|F_n|\, dz \le \|F_n(\tau)\|_{L^{3/2}} \left(\int_{|z|>A} |z|^{-12}\, dz \right)^{1/3} \le C A^{-3} \|F_n(\tau)\|_{L^{3/2}}.$$

Therefore

$$\|P_n^{\mathrm{far}} - c_{n,R}\|_{L^1_\tau L^\infty_y(B_R)} \le C_{R,I} A^{-3} \sup_n \|F_n\|_{L^\infty_\tau L^{3/2}_y}.$$

Since B_R has finite measure, this also controls the far part in $L^1_\tau L^2_y(B_R)$. Taking $n \to \infty$ in the local term and then $A \to \infty$ in the far term gives

$$P_n - c_{n,R} \to 0 \quad \text{in } L^1_\tau L^2_y(B_R).$$

For $\varphi \in H^1_0(B_R; \mathbb{R}^3)$,

$$|\langle \nabla(P_n - c_{n,R}), \varphi\rangle| \le \left| \int_{B_R} (P_n - c_{n,R}) \nabla \cdot \varphi\, dy \right| \le \|P_n - c_{n,R}\|_{L^2(B_R)} \|\nabla \varphi\|_{L^2(B_R)}.$$

Thus $\nabla P_n \to 0$ in $L^1_\tau H^{-1}_y(B_R)$. ☐

Definition 9.13 (Terminal nonlinear local-state decoupling)**.** On every compact terminal-frame cylinder $B_R \times I$, let U_n be the retained active local-state evolution and let S_n be the nonretained local residual. Terminal nonlinear local-state decoupling means:

(i) $\nabla \cdot S_n = 0$ distributionally;

(ii) $S_n \to 0$ in $L^\infty_\tau L^3_y(B_{2R})$;

(iii)
$$U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n \to 0 \quad \text{in } L^1_\tau L^2_y(B_R);$$

(iv)
$$\partial_\tau S_n - \nu \Delta S_n + a_n(S_n + y \cdot \nabla S_n) + b_n \cdot \nabla S_n \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R);$$

(v) pressure sources involving at least one factor of S_n satisfy Theorem 9.12, or directly satisfy
$$\nabla \mathcal{P}_{\mathrm{cross},n} \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R);$$

(vi) cutoff errors introduced in isolating the selected frame vanish in $L^1_\tau H^{-1}_y(B_R)$;

(vii) the effective modulation coefficients remain bounded;
(viii) after subtracting S_n, the retained solution has positive finite critical mass, repaired-gauge representation, pressure reconstruction, Caccioppoli regularity, compactness, tightness, and the compact scale-collapse limiting inputs.

Theorem 9.14 (Same-point and multipoint decoupling from terminal decoupling)**.** *If terminal nonlinear local-state decoupling holds in the sense of Theorem 9.13, then both same-point scale-decoupling and separated-point frame-decoupling hold.*

Proof. For same-point active local states, group comparable scales by Theorem 9.3. For strict scale separation choose the innermost active scale as the selected terminal frame. By Theorem 9.11, every outer scale that remains in the selected frame is locally invisible in velocity, unless it was already grouped as comparable or routed to an adjacent state. Terminal nonlinear local-state decoupling supplies the nonlinear-stress, linear-residual, pressure-source, localization, modulation, and compact scale-collapse admissibility conditions on every compact cylinder. Theorem 9.10 then shows that the outer components and residual alter the selected equation only by an error tending to zero in $L^1_\tau H^{-1}_y(B_R)$. Thus the same-point configuration reduces to one retained compact scale-collapse solution with perturbative interactions.

For separated physical points, choose an active physical point x_p^* and its selected frame. For every local state centered at $x_q^* \neq x_p^*$, Theorem 9.7 gives local velocity convergence to zero in the p-frame. If a pure profile pressure is represented in L^2, the same escaping-ball calculation gives local L^2 pressure convergence and therefore H^{-1} convergence of the gradient. With only $L^{3/2}$-pressure, the mixed pressure sources are controlled by Theorem 9.12 through the L^2-strength hypotheses included in terminal nonlinear local-state decoupling. The same decoupling statement supplies localization-error convergence, bounded modulation, and compact scale-collapse admissibility. Applying Theorem 9.10 shows that separated physical-point states change the selected equation only by an error tending to zero in $L^1_\tau H^{-1}_{\mathrm{loc}}$. This proves separated-point decoupling. □

9.4. **Terminal local-state decoupling.**

Definition 9.15 (Terminal active local frame)**.** Partition active local states by physical point $x_j^n \to x_\alpha^*$. At a fixed physical point, group comparable scales. For a comparable-scale class $\mathcal{C}$, a representative frame $(x_{\mathcal{C}}^n, \lambda_{\mathcal{C}}^n)$ has compound local representative

$$\Phi_{\mathcal{C}}(y) = \sum_{j\in\mathcal{C}} \rho_j^{-1} W_j\left(\frac{y - z_j}{\rho_j}\right),$$

where W_j is the local representative of the j-th state in its own frame, transported as in Theorem 9.3. The class $\mathcal{C}$ is terminal at x_α^* if no active class at the same physical point has strictly smaller scale:

$$\lambda_{\mathcal{D}}^n/\lambda_{\mathcal{C}}^n \not\to 0 \quad \text{for every active class } \mathcal{D}.$$

The terminal frame is

$$y = \frac{x - x_{\mathcal{C}}^n}{\lambda_{\mathcal{C}}^n}, \qquad \tau = \int^t (\lambda_{\mathcal{C}}(s))^{-2}\, ds,$$

and

$$V_n(y,\tau) = \lambda_{\mathcal{C}}^n u(x_{\mathcal{C}}^n + \lambda_{\mathcal{C}}^n y, t_n + (\lambda_{\mathcal{C}}^n)^2\tau).$$

The retained solution U_n is the repaired-gauge nonlinear evolution of $\mathcal{C}$ in this frame, and $S_n := V_n - U_n$ is the nonretained remainder field.

Theorem 9.16 (Terminal windowed state-space completeness)**.** *For a sequence f_n and a frame (x_n, λ_n), define*

$$\mathfrak{m}_R(f_n; x_n, \lambda_n) := \limsup_{n\to\infty} \|\lambda_n f_n(x_n + \lambda_n \cdot)\|_{L^3(B_R)}$$

and

$$\mathfrak{M}_{R,I}(f_n; x_n, \lambda_n) := \limsup_{n\to\infty} \sup_{\tau\in I} \left\|\lambda_n f_n(x_n + \lambda_n \cdot, t_n + \lambda_n^2 \tau)\right\|_{L^3(B_R)}.$$

Terminal windowed state-space completeness means that if $\mathfrak{M}_{R,I}(f_n; x_n, \lambda_n) > 0$ for a sequence left in the remainder, then after passing to a subsequence the local state-space stratification assigns that residual mass either to a named adjacent local alternative or to the retained comparable class. In the latter case the component must have been grouped into the retained terminal class before selecting the frame. Hence no positive nonretained residual mass remains on a retained terminal branch.

Proof. This is the compact-window form of Theorem 10.88. If $\mathfrak{M}_{R,I}(f_n; x_n, \lambda_n) > 0$, then a subsequence and a smaller compact terminal cylinder carry positive local L^3 mass. The local compactness lemma Theorem 10.78 gives an occupation state, and Theorem 10.85 assigns that state to the first closed label in the local state-space decomposition. An adjacent label is handled by its local closure theorem and is not part of the retained terminal branch. A retained comparable label must already have been grouped into the selected terminal class by the terminal frame selection rule. Thus positive nonretained residual mass cannot persist on a retained terminal branch. □

Remark 9.17 (Status of Theorem 9.16). This theorem is deliberately local. It is a compact-window state-space exhaustion statement, not a terminal global profile theorem. Its proof uses the local compactness, closed-label, and residual-exhaustion lemmas in the terminal state-space discharge package; it does not use a whole-space profile decomposition or a heat-flow remainder estimate.

Lemma 9.18 (Terminal local velocity decoupling)**.** *Let $\mathcal{C}$ be a terminal active class and S_n be the nonretained remainder field in the $\mathcal{C}$-frame. Then for every compact cylinder $B_R \times I$,*

$$S_n \to 0 \quad \text{in } L^\infty_\tau L^3_y(B_R).$$

Proof. Assume otherwise. Then there exist $R < \infty$, a compact interval I, $\delta > 0$, a subsequence, and times $\tau_n \in I$ such that

$$\|S_n(\tau_n)\|_{L^3(B_R)} \geq \delta.$$

In physical variables, after the retained terminal class has been subtracted, the remainder carries at least δ critical L^3-mass inside a ball of radius $O(\lambda^n_{\mathcal{C}})$, centered at

$$x^n_{\mathcal{C}} + O(\lambda^n_{\mathcal{C}}),$$

at time

$$t_n + (\lambda^n_{\mathcal{C}})^2 \tau_n.$$

By terminal windowed state-space completeness, this nonvanishing mass has a first local state-space label. If the label is an adjacent alternative, the branch has left the retained terminal class and is handled by the corresponding local closure theorem, contradicting that it remains in S_n on the retained branch. If the label is the retained comparable class, then it was already grouped into $\mathcal{C}$ before the terminal frame was selected, so it cannot remain in S_n. This is again a contradiction.

Thus no such δ exists. The same argument covers same-point larger-scale and separated-point components: any nonzero mass seen in the terminal frame is either a named adjacent local state or part of the retained comparable class, contradicting the terminal partition. □

Lemma 9.19 (Uniform terminal H^1 bounds)**.** *Assume the repaired-gauge representation and the local Caccioppoli estimates hold on every compact terminal cylinder $B_{2R} \times I$. Then*

$$\sup_n \|V_n\|_{L^\infty_\tau L^3_y(B_{2R})} + \sup_n \|V_n\|_{L^2_\tau H^1_y(B_{2R})} \le C_R,$$

$$\sup_n \|U_n\|_{L^\infty_\tau L^3_y(B_{2R})} + \sup_n \|U_n\|_{L^2_\tau H^1_y(B_{2R})} \le C_R,$$

and, since $S_n = V_n - U_n$,

$$\sup_n \|S_n\|_{L^\infty_\tau L^3_y(B_{2R})} + \sup_n \|S_n\|_{L^2_\tau H^1_y(B_{2R})} \le C_R.$$

In particular, by Sobolev embedding on B_{2R},

$$U_n, S_n \quad \text{are bounded in} \quad L^2_\tau L^6_y(B_{2R}).$$

Proof. The first estimate is the repaired-gauge Caccioppoli estimate for V_n on the compact terminal cylinder. The retained solution U_n is itself a repaired-gauge Navier–Stokes solution with positive finite critical mass and bounded modulation on the same compact cylinder, so the same estimate applies to U_n. Since $S_n = V_n - U_n$, the displayed bound for S_n follows by the triangle inequality in the stated norms. The $L^2_\tau L^6_y$ bound follows from the Sobolev embedding $H^1(B_{2R}) \hookrightarrow L^6(B_{2R})$. □

Lemma 9.20 (Nonlinear stress decoupling in terminal frames)**.** *Let*

$$F_n := U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n.$$

Then, for every compact $B_R \times I$,

$$F_n \to 0 \quad \text{in } L^1_\tau L^2_y(B_R).$$

Proof. By Theorem 9.18,

$$\|S_n\|_{L^\infty_\tau L^3_y(B_R)} \to 0.$$

Using Holder's inequality in space and time,

$$\|U_n \otimes S_n\|_{L^1_\tau L^2_y(B_R)} \le \|S_n\|_{L^\infty_\tau L^3_y(B_R)} \|U_n\|_{L^1_\tau L^6_y(B_R)}$$

and

$$\|U_n\|_{L^1_\tau L^6_y(B_R)} \le |I|^{1/2} \|U_n\|_{L^2_\tau L^6_y(B_R)} \le C_{R,I}.$$

Thus $U_n \otimes S_n \to 0$ in $L^1_\tau L^2_y(B_R)$, and the same estimate gives $S_n \otimes U_n \to 0$. Finally,

$$\|S_n \otimes S_n\|_{L^1_\tau L^2_y(B_R)} \le \|S_n\|_{L^\infty_\tau L^3_y(B_R)} \|S_n\|_{L^1_\tau L^6_y(B_R)} \to 0,$$

because $\|S_n\|_{L^1_\tau L^6_y(B_R)} \le C_{R,I}$. Summing the three terms proves the local convergence. □

Lemma 9.21 (Quantitative nested-ball harmonic tail)**.** *Fix radii $0 < r < R < R_1 < R_2$ and $1 < p < \infty$. Let $\theta \in C^\infty_c(B_{R_2} \setminus \overline{B_{R_1}})$, extend θF by zero to $\mathbb{R}^3$, and set*

$$P^{\mathrm{ann}} = \mathcal{R}_i \mathcal{R}_j(\theta F_{ij}), \qquad H^{\mathrm{ann}} = P^{\mathrm{ann}} - (P^{\mathrm{ann}})_{B_{R_1}} \quad \text{on } B_{R_1}.$$

Then H^{ann} is harmonic and mean-zero on B_{R_1}. Moreover there is a constant $C_p(r, R, R_1, R_2, \theta) > 0$, depending only on p, the nested balls and the cutoff, such that

(i) Interior derivative bound. *For every harmonic mean-zero H on B_{R_1} and every $R < R_1$,*

$$\sup_{B_R} \big(|H| + R_1 |\nabla H|\big) \le C_p(R, R_1)\, \|H\|_{L^p(B_{R_1})}.$$

(ii) Annular harmonic-source bound. *The harmonic part generated by the source in the fixed annulus satisfies*

$$\|H^{\mathrm{ann}}\|_{L^p(B_R)} \le C_p(r, R, R_1, R_2, \theta)\, \|F\|_{L^p(B_{R_2} \setminus B_{R_1})}.$$

(iii) Quantitative converse for annular tails. *If* $\|H^{\mathrm{ann}}\|_{L^p(B_r)} \geq \eta > 0$*, then*

$$\|F\|_{L^p(B_{R_2}\setminus B_{R_1})} \geq C_p(r, R, R_1, R_2, \theta)^{-1}\eta.$$

This lemma controls only the annular harmonic contribution generated by sources in the displayed fixed annulus. Harmonic pressure coming from outside B_{R_2}*, or from a failure of the pressure reconstruction itself, is not silently bounded by this lemma and is instead read as a pressure/exterior state-space alternative.*

Proof. Item (i) is the standard interior derivative estimate for harmonic functions: by the mean-value property and Cauchy estimates on $B_{(R+R_1)/2}$, $|H| + R_1|\nabla H|$ on B_R is controlled by the L^p mass of H on B_{R_1} with constant depending only on p and R/R_1.

For item (ii), the source θF is supported outside B_{R_1}, so $-\Delta P^{\mathrm{ann}} = 0$ on B_{R_1}. Hence H^{ann} is harmonic there. Calderón–Zygmund boundedness on $\mathbb{R}^3$ gives

$$\|P^{\mathrm{ann}}\|_{L^p(\mathbb{R}^3)} \leq C_p\|\theta F\|_{L^p(\mathbb{R}^3)} \leq C_p(\theta)\|F\|_{L^p(B_{R_2}\setminus B_{R_1})},$$

and subtracting the mean on B_{R_1} changes the $L^p(B_R)$ norm by a geometric constant.

For item (iii), this is the contrapositive of item (ii): if the annular source were arbitrarily small, then item (ii) would force $\|H^{\mathrm{ann}}\|_{L^p(B_R)}$, and hence $\|H^{\mathrm{ann}}\|_{L^p(B_r)}$, to be arbitrarily small. ☐

Lemma 9.22 (Harmonic cross pressure routes to pressure/exterior alternative)**.** *Fix nested balls* $B_R \Subset B_{R_1} \Subset B_{R_2}$*,* $1 < p < \infty$*, and* $1 \leq q < \infty$*. Let*

$$P_{\mathrm{cross},n} = P^{\mathrm{loc}}_{\mathrm{cross},n} + H_{n,R_1} + c_{n,R_1}(\tau)$$

be the local pressure reconstruction on B_{R_1}*, where*

$$-\Delta P^{\mathrm{loc}}_{\mathrm{cross},n} = \partial_i\partial_j\big(\zeta(F_n)_{ij}\big), \qquad \zeta \equiv 1 \text{ on } B_{R_1}, \quad \operatorname{supp}\zeta \Subset B_{R_2},$$

and H_{n,R_1} *is harmonic with zero spatial mean on* B_{R_1}*. Suppose*

$$F_n \to 0 \quad \text{in } L^q_\tau L^p_y(B_{R_2}).$$

If H_{n,R_1} *does not converge to zero in* $L^q_\tau L^p_y(B_R)$*, then, after passing to a subsequence, the branch enters the ordered pressure/exterior-leakage alternative. Consequently, on any retained terminal branch,* $H_{n,R_1} \to 0$ *in* $L^q_\tau L^p_y(B_R)$*.*

Proof. The local part tends to zero by Calderon–Zygmund estimates:

$$\|P^{\mathrm{loc}}_{\mathrm{cross},n}\|_{L^q_\tau L^p_y(B_{R_1})} \leq C\|F_n\|_{L^q_\tau L^p_y(B_{R_2})} \to 0.$$

Hence any non-vanishing pressure oscillation modulo functions of time is carried by the normalized harmonic part. If

$$\limsup_{n\to\infty} \|H_{n,R_1}\|_{L^q_\tau L^p_y(B_R)} > 0,$$

then choose a finite smooth partition of unity $\{\theta_\ell\}_\ell \subset C^\infty_c(B_{R_2} \setminus \overline{B_{R_1}})$ for the recorded annular part of the pressure atlas. For each ℓ, the corresponding annular harmonic contribution

$$H^{\mathrm{ann}}_{n,\ell} := \mathcal{R}_i\mathcal{R}_j(\theta_\ell(F_n)_{ij}) - \fint_{B_{R_1}} \mathcal{R}_i\mathcal{R}_j(\theta_\ell(F_n)_{ij})\,dy$$

vanishes in $L^q_\tau L^p_y(B_R)$ by Theorem 9.21, because

$$\|F_n\|_{L^q_\tau L^p_y(B_{R_2}\setminus B_{R_1})} \to 0.$$

Summing over the finite partition shows that the whole harmonic contribution generated by sources recorded in the fixed annulus $B_{R_2} \setminus B_{R_1}$ vanishes.

By construction of the local pressure atlas, after the local Calderon–Zygmund piece and the recorded annular pieces have been removed, the remaining mean-zero harmonic part has only two possible origins: either the pressure gauge/reconstruction fails on the selected window, or it is a far-field pressure tail generated outside the fixed ball B_{R_2}. The former is one of the ordered pressure rows. In the latter case, Theorem 9.38 routes any exterior carrier to an exterior or separated-core concentration alternative; if no such exterior carrier exists, Theorem 9.44 gives convergence of the far-field pressure modulo functions of time on B_R, contradicting the positive lower bound for H_{n,R_1}. Therefore every non-vanishing harmonic cross pressure is a named pressure/exterior-leakage alternative. A retained terminal branch is the complementary case in the ordered validation, so the harmonic part vanishes there. $\square$

Lemma 9.23 (Terminal pressure decoupling)**.** *Fix nested balls $B_R \Subset B_{R_1} \Subset B_{R_2}$ and let $\zeta \in C_c^\infty(B_{R_2})$ be identically one on B_{R_1}. Let*

$$P^{\mathrm{loc}}_{\mathrm{cross},n} := \mathcal{R}_i\mathcal{R}_j\big(\zeta(F_n)_{ij}\big)$$

be the local Calderon–Zygmund pressure generated by the cross stress. Suppose the terminal pressure reconstruction is used on B_{R_2}, so that on B_{R_1}

$$P_{\mathrm{cross},n} = P^{\mathrm{loc}}_{\mathrm{cross},n} + H_{n,R_1} + c_{n,R_1}(\tau),$$

where $H_{n,R_1}(\cdot,\tau)$ is harmonic and normalized by $⨍_{B_{R_1}} H_{n,R_1}(\cdot,\tau)\,dy = 0$. On a retained terminal branch,

$$P_{\mathrm{cross},n} - c_{n,R} \to 0 \quad \textit{in } L^1_\tau L^2_y(B_R),$$

and

$$\nabla P_{\mathrm{cross},n} \to 0 \quad \textit{in } L^1_\tau H^{-1}_y(B_R).$$

Proof. Theorem 9.20 gives

$$F_n \to 0 \quad \text{in } L^1_\tau L^2_y(B_{R_2})$$

on every compact terminal cylinder. Calderon–Zygmund boundedness gives

$$\|P^{\mathrm{loc}}_{\mathrm{cross},n}\|_{L^1_\tau L^2_y(B_{R_1})} \le C_{R_1,R_2}\|F_n\|_{L^1_\tau L^2_y(B_{R_2})} \to 0.$$

For the harmonic part, apply Theorem 9.22 with $(q,p)=(1,2)$. A non-vanishing harmonic cross pressure is a pressure/exterior-leakage alternative and therefore is not present on a retained terminal branch. Hence $H_{n,R_1} \to 0$ in $L^1_\tau L^2_y(B_R)$. Adding the local part proves $P_{\mathrm{cross},n} - c_{n,R} \to 0$ in $L^1_\tau L^2_y(B_R)$.

Finally, for $\varphi \in H^1_0(B_R;\mathbb{R}^3)$,

$$|\langle \nabla(P_{\mathrm{cross},n} - c_{n,R}), \varphi\rangle| \le \|P_{\mathrm{cross},n} - c_{n,R}\|_{L^2(B_R)}\|\nabla\varphi\|_{L^2(B_R)}.$$

Integrating in time gives $\nabla P_{\mathrm{cross},n} \to 0$ in $L^1_\tau H^{-1}_y(B_R)$. $\square$

Lemma 9.24 (Linear residual decoupling)**.** *Let $V_n = U_n + S_n$ solve*

$$\partial_\tau V_n + (V_n\cdot\nabla)V_n + \nabla P_n = \nu\Delta V_n + a_n(V_n + y\cdot\nabla V_n) + b_n\cdot\nabla V_n.$$

Suppose U_n satisfies, in the same repaired gauge,

$$\partial_\tau U_n + (U_n\cdot\nabla)U_n + \nabla P_{U,n} = \nu\Delta U_n + a_n(U_n + y\cdot\nabla U_n) + b_n\cdot\nabla U_n + e_n,$$

with

$$e_n \to 0 \quad \textit{in } L^1_\tau H^{-1}_y(B_R)$$

on compact terminal cylinders. Then

$$\partial_\tau S_n - \nu\Delta S_n - a_n(S_n + y\cdot\nabla S_n) - b_n\cdot\nabla S_n \to 0 \quad \textit{in } L^1_\tau H^{-1}_y(B_R).$$

Equivalently, the same conclusion holds with the opposite sign convention for the modulation terms.

Proof. Subtracting the U_n-equation from the V_n-equation gives

$$\partial_\tau S_n - \nu\Delta S_n = -\operatorname{div} F_n - \nabla P_{\mathrm{cross},n} - e_n + a_n(S_n + y\cdot\nabla S_n) + b_n\cdot\nabla S_n.$$

Equivalently,

$$\partial_\tau S_n - \nu\Delta S_n - a_n(S_n + y\cdot\nabla S_n) - b_n\cdot\nabla S_n = -\operatorname{div} F_n - \nabla P_{\mathrm{cross},n} - e_n.$$

By Theorem 9.20,

$$F_n \to 0 \quad \text{in } L^1_\tau L^2_y(B_R),$$

hence

$$\operatorname{div} F_n \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R).$$

By Theorem 9.23,

$$\nabla P_{\mathrm{cross},n} \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R).$$

Together with $e_n \to 0$, this proves the displayed residual convergence.

For the opposite sign convention, the two residuals differ by

$$2a_n(S_n + y\cdot\nabla S_n) + 2b_n\cdot\nabla S_n.$$

By Theorem 9.18 and finite measure,

$$S_n \to 0 \quad \text{in } L^\infty_\tau L^2_y(B_R).$$

For $\varphi \in H^1_0(B_R)$,

$$|\langle y\cdot\nabla S_n, \varphi\rangle| = \left|\int_{B_R} S_n(3\varphi + y\cdot\nabla\varphi)\,dy\right| \le C_R\|S_n\|_{L^2(B_R)}\|\varphi\|_{H^1(B_R)},$$

and

$$|\langle b_n\cdot\nabla S_n, \varphi\rangle| \le |b_n|\,\|S_n\|_{L^2(B_R)}\|\nabla\varphi\|_{L^2(B_R)}.$$

The term S_n itself is controlled by the same L^2-convergence. Since a_n and b_n are bounded, the sign-convention difference tends to zero in $L^1_\tau H^{-1}_y(B_R)$. □

Lemma 9.25 (Terminal modulation compatibility)**.** *Same-point larger-scale profiles in a terminal frame have the form*

$$W^n_j(y,\tau) = \mu^n_j U^j(\theta^n_j(\tau), \xi^n_j + \mu^n_j y), \qquad \mu^n_j \to 0.$$

When the profile is regular at the observed offset, its leading term is a spatially constant ambient velocity. Absorbing this constant into the translation coefficient replaces b_n by $b_n - b_{\mathrm{amb},n}$, and the remaining terms vanish on compact terminal cylinders. If the offset escapes, the entire profile is locally invisible and no absorption is needed. In particular,

$$\sup_n \sup_{\tau\in I}\big(|a_n(\tau)| + |b_n(\tau)|\big) < \infty.$$

Proof. The local invisibility of the nonconstant remainder follows from Theorem 9.18 and the H^{-1} product estimates in Theorem 9.24. Separated physical-point profiles are locally invisible by Theorem 9.7; their pressures and cutoff commutators vanish by Theorem 9.23 and Theorem 9.24. Therefore only the spatially constant ambient velocity changes the translation coefficient, and after this absorption the effective modulation coefficients remain uniformly bounded. □

Lemma 9.26 (Terminal compact scale-collapse admissibility)**.** *The retained terminal solution U_n has positive finite critical mass, pressure reconstruction, bounded modulation, local windowed H^1 control, critical tightness on the selected windows, and the compactness input used by the compact scale-collapse argument.*

Proof. The terminal class is active, so after the perturbative residual and nonretained local states have been removed,

$$\inf_{\tau\in I}\|U_n(\tau)\|_{L^3(B_R)} \geq c_{\mathrm{act}} > 0$$

on the selected compact windows, after passing to the subsequences used in the local state-space selection. The upper bound follows from the retained local compact-cylinder bound:

$$\sup_{\tau\in I}\|U_n(\tau)\|_{L^3(B_R)} \leq C(M).$$

The terminal state-space completeness also gives the required tightness. If U_n were not uniformly L^3-tight on the selected windows, then there would be $\varepsilon > 0$, radii $R_k \to \infty$, and times τ_n such that

$$\int_{|y|>R_k} |U_n(y,\tau_n)|^3\,dy \geq \varepsilon.$$

Applying the local state-space stratification to this escaping sequence produces either an exterior-regular component, which is removed by the exterior regularity argument below, or an additional active local component at a different physical point or larger same-point scale. Such a component belongs to S_n and is handled by terminal decoupling. Hence no non-tight retained terminal solution remains.

The repaired-gauge representation gives pressure reconstruction and bounded modulation. The local Caccioppoli estimate gives local windowed H^1 control, and the Aubin–Lions compactness layer gives the compactness input on the selected windows. □

Theorem 9.27 (Terminal nonlinear local-state decoupling theorem)**.** *Assume local compact critical bounds, terminal windowed state-space completeness, local perturbative residual stability, the exterior annulus alternative, repaired-gauge representation, and local Caccioppoli regularity. Then terminal nonlinear local-state decoupling holds in the sense of Theorem 9.13.*

Proof. Let $\mathcal{C}$ be any terminal active compound class, let U_n be the retained repaired-gauge nonlinear solution in its terminal frame, and set

$$S_n = V_n - U_n.$$

Theorem 9.18 proves

$$S_n \to 0 \quad \text{in } L^\infty_\tau L^3_y(B_{2R})$$

on every compact terminal cylinder. Divergence-free compatibility follows from the divergence-free repaired-gauge equation and the local terminal decomposition.

Theorem 9.20 proves

$$U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n \to 0 \quad \text{in } L^1_\tau L^2_y(B_R).$$

The local compact critical bounds give the $L^\infty_\tau L^3_y$ control required in the pressure lemma below on the cylinders under consideration. Theorem 9.23 proves

$$\nabla P_{\mathrm{cross},n} \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R)$$

after subtracting spatially constant pressure terms. Theorem 9.24 proves

$$\partial_\tau S_n - \nu\Delta S_n + a_n(S_n + y\cdot\nabla S_n) + b_n\cdot\nabla S_n \to 0 \quad \text{in } L^1_\tau H^{-1}_y(B_R).$$

Theorem 9.25 gives modulation compatibility, and Theorem 9.26 gives compact scale-collapse admissibility of the retained terminal solution. These are the items in Theorem 9.13. □

Corollary 9.28 (Multibubble exclusion from terminal local decoupling)**.** *Under the Type II local terminal data, terminal windowed state-space completeness, local perturbative residual stability, the exterior annulus alternative, repaired-gauge representation, and local Caccioppoli regularity, no active multibubble Type II configuration occurs.*

Proof. Theorem 9.27 gives terminal nonlinear decoupling. Theorem 9.14 then supplies the same-point and separated-point decoupling hypotheses used in Theorem 9.9. The active local mass floor follows from the retained positive local critical annulus and local perturbative residual stability. Applying Theorem 9.9 gives the claimed exclusion of active multibubble configurations. □

9.5. **Small local residual components.** For a compact terminal cylinder $Q_{R,I} := B_R \times I$, define the local residual seminorm

$$\|W\|_{\mathcal{X}(Q_{R,I})} := \|W\|_{L^\infty(I;L^3(B_R))} + \|W\|_{L^2(I;H^1(B_R))}.$$

This is only a compact-window norm. It is not a whole-space heat-flow or global mild-solution norm.

Lemma 9.29 (Local residual product estimates)**.** *Let $Q_{2R,I} = B_{2R} \times I$ be compact. Suppose*

$$S_n \to 0 \quad \text{in } L^\infty(I; L^3(B_{2R}))$$

and

$$\sup_n \big(\|U_n\|_{\mathcal{X}(Q_{2R,I})} + \|S_n\|_{\mathcal{X}(Q_{2R,I})}\big) < \infty.$$

Then

$$U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n \to 0 \quad \text{in } L^1(I; L^2(B_R)).$$

If the pressures associated with these cross stresses are reconstructed by the local pressure decomposition, then their gradients converge to zero in

$$L^1(I; H^{-1}(B_R)).$$

Proof. Sobolev embedding on B_{2R} gives uniform

$$U_n, S_n \quad \text{in } L^2(I; L^6(B_{2R})).$$

Holder's inequality in space and time yields

$$\|U_n \otimes S_n\|_{L^1(I;L^2(B_R))} \le \|S_n\|_{L^\infty(I;L^3(B_R))} \|U_n\|_{L^1(I;L^6(B_R))} \to 0,$$

because

$$\|U_n\|_{L^1(I;L^6(B_R))} \le |I|^{1/2} \|U_n\|_{L^2(I;L^6(B_R))}.$$

The same estimate treats $S_n \otimes U_n$, and

$$\|S_n \otimes S_n\|_{L^1(I;L^2(B_R))} \le \|S_n\|_{L^\infty(I;L^3(B_R))} \|S_n\|_{L^1(I;L^6(B_R))} \to 0.$$

For the pressure statement, apply the local pressure reconstruction to the cross source. The preceding L^1L^2 convergence gives local convergence of the local pressure part by Calderon–Zygmund estimates. Any non-vanishing harmonic remainder is routed by Theorem 9.22 with $(q, p) = (1, 2)$ to the pressure/exterior alternative; on a retained branch this alternative is absent. Therefore the cross-pressure gradients vanish in $L^1H^{-1}(B_R)$. □

Theorem 9.30 (Small local residual components are perturbative)**.** *Let S_n be a nonretained terminal residual on every compact terminal cylinder $Q_{2R,I}$. Assume*

$$S_n \to 0 \quad \text{in } L^\infty(I; L^3(B_{2R}))$$

and that the local compact-window bounds, local pressure reconstruction, and bounded modulation estimates hold on $Q_{2R,I}$. Then S_n is perturbative in the terminal decoupling topology of Theorem 9.13.

Proof. Theorem 9.29 gives the vanishing nonlinear-stress and cross-pressure terms. The repaired-gauge equation for $V_n = U_n + S_n$ and the equation for the retained component U_n, subtracted on $Q_{R,I}$, give the linear residual identity of Theorem 9.24. The stress and pressure terms just proved vanish, and bounded modulation sends the remaining $a_n(S_n + y \cdot \nabla S_n) + b_n \cdot \nabla S_n$ terms to zero in $L^1(I; H^{-1}(B_R))$. Hence all perturbative clauses in Theorem 9.13 hold on $Q_{R,I}$. □

Corollary 9.31 (Small local components are not active)**.** *If a nonretained component has vanishing local $L^\infty L^3$ seminorm on every compact terminal cylinder and satisfies the local compact-window bounds, then it cannot define an active terminal state.*

Proof. By Theorem 9.30, such a component contributes only vanishing stress, pressure, modulation, and residual errors in the terminal decoupling topology. Active terminal states are precisely retained local state-space labels with positive nonperturbative local mass. Therefore a component that is perturbative on every compact terminal cylinder is not active. □

Theorem 9.32 (Positive local mass floor for active terminal components)**.** *There exists a local threshold $\varepsilon_{\mathrm{loc}} > 0$, depending only on the Caffarelli–Kohn–Nirenberg threshold and the finite covering constants of the selected compact cylinders, such that every retained active terminal component has a compact terminal cylinder $Q_{R,I}$ with*

$$\limsup_{n\to\infty} \|W_n\|_{L^\infty(I;L^3(B_R))} \geq \varepsilon_{\mathrm{loc}}.$$

Proof. Let $\varepsilon_{\mathrm{CKN}}$ be the threshold in Theorem 9.34. Fix a finite nested covering of each retained compact terminal cylinder by CKN cylinders whose smaller cylinders still cover the retained core. Choose $\eta_{\mathrm{det}} > 0$ to be $\varepsilon_{\mathrm{CKN}}$ divided by a sufficiently large constant depending only on this cover.

If every CKN cylinder in the cover had velocity-pressure quantity below η_{det}, then the CKN criterion would give a uniformly smooth bounded representative on the retained smaller cylinder. Such a bounded selected-window representative is not an active Type II terminal component; it is routed to the regular/bounded-window alternative. Hence an active retained component has a subcylinder on which the terminal CKN quantity is at least η_{det}.

If the pressure part of that CKN quantity carries at least half of η_{det} while the velocity part is arbitrarily small on all smaller subcylinders, then the local pressure reconstruction and harmonic-tail alternative, made explicit in Theorem 9.22 with $(q, p) = (3/2, 3/2)$, assign the branch to the pressure/exterior-leakage stratum. That stratum is adjacent and is not a retained active terminal component. Therefore on a retained active component the velocity part carries a fixed fraction of the detected CKN quantity on some smaller cylinder:

$$\rho^{-2} \int_{I_\rho} \int_{B_\rho} |W_n|^3 \, dy \, d\tau \geq c\, \eta_{\mathrm{det}}$$

along a subsequence. Since $|I_\rho| \simeq \rho^2$, there is a time in I_ρ with

$$\int_{B_\rho} |W_n(y,\tau)|^3 \, dy \geq c\, \eta_{\mathrm{det}}.$$

After rescaling this subcylinder to a fixed compact terminal cylinder, this is the asserted $L^\infty_\tau L^3_y$ lower bound with $\varepsilon_{\mathrm{loc}} := (c\, \eta_{\mathrm{det}})^{1/3}$.

If no such velocity lower bound existed on any compact terminal cylinder, then the component would have vanishing local $L^\infty L^3$ seminorm after subsequence extraction and would be perturbative by Theorem 9.31, contradicting retained activity. □

9.6. **Exterior regularity.**

Definition 9.33 (Exterior annuli and terminal CKN quantity)**.** Let (x_*, T^*) be the selected Type II core point. For $0 < r < R < \infty$, set

$$A_{r,R} := \overline{B_R(x_*)} \setminus B_r(x_*).$$

For $x_0 \in \mathbb{R}^3$ and $\rho > 0$, write

$$Q^*_\rho(x_0) := B_\rho(x_0) \times (T^* - \rho^2, T^*).$$

For a suitable weak solution (u, p), define the terminal local velocity-pressure quantity

$$\mathcal{C}_*(u, p; x_0, \rho) := \rho^{-2} \int_{T^*-\rho^2}^{T^*} \int_{B_\rho(x_0)} |u|^3 \, dx \, dt + \rho^{-2} \int_{T^*-\rho^2}^{T^*} \int_{B_\rho(x_0)} |p - (p)_{B_\rho(x_0)}(t)|^{3/2} \, dx \, dt.$$

The pressure is measured modulo functions of time.

Theorem 9.34 (Terminal Caffarelli–Kohn–Nirenberg regularity)**.** *There exists $\varepsilon_{\mathrm{CKN}} > 0$ with the following property. Let (u, p) be suitable on $Q^*_\rho(x_0)$. If*

$$\mathcal{C}_*(u, p; x_0, \rho) \le \varepsilon_{\mathrm{CKN}},$$

then for each integer $k \ge 0$ there are $0 < c_k < 1$ and $C_k < \infty$, independent of $(u, p), x_0, \rho$, such that u is smooth in

$$B_{c_k\rho}(x_0) \times (T^* - c_k\rho^2, T^*)$$

and, after subtracting a function of time from p,

$$\sup_{t \in (T^*-c_k\rho^2, T^*)} \Big(\rho^{k+1} \|\nabla^k u(t)\|_{L^\infty(B_{c_k\rho}(x_0))} + \rho^{k+2} \|\nabla^k p(t)\|_{L^\infty(B_{c_k\rho}(x_0))} \Big) \le C_k.$$

Proof. This is the standard endpoint form of the Caffarelli–Kohn–Nirenberg epsilon-regularity criterion [CKN82, Lin98], after translating the terminal time to 0 and rescaling $Q^*_\rho(x_0)$ to a unit parabolic cylinder. The pressure normalization by spatial averages is the usual normalization in the suitable weak-solution criterion. Interior parabolic bootstrapping gives the stated higher derivative bounds on smaller cylinders. □

Throughout the exterior regularity subsection we fix

$$\varepsilon_{\mathrm{ext}} := \frac{1}{2} \varepsilon_{\mathrm{CKN}}.$$

Definition 9.35 (No exterior concentration on a fixed annulus)**.** Let $0 < r < R < \infty$. A suitable Type II branch has no exterior concentration on $A_{r,R}$ if there exists $\rho_* > 0$ such that whenever

$$x_0 \in A_{r,R}, \qquad 0 < \rho \le \rho_*, \qquad Q^*_\rho(x_0) \subset (B_{2R}(x_*) \setminus \overline{B_{r/2}(x_*)}) \times (0, T^*),$$

one has

$$\mathcal{C}_*(u, p; x_0, \rho) \le \varepsilon_{\mathrm{ext}}.$$

Definition 9.36 (Exterior concentration branch)**.** An exterior concentration branch on $A_{r,R}$ is a sequence

$$x_n \in A_{r,R}, \qquad \rho_n \downarrow 0,$$

such that

$$Q^*_{\rho_n}(x_n) \subset (B_{2R}(x_*) \setminus \overline{B_{r/2}(x_*)}) \times (0, T^*)$$

and

$$\mathcal{C}_*(u, p; x_n, \rho_n) > \varepsilon_{\mathrm{ext}}.$$

The associated exterior rescalings are

$$U_n(y, s) := \rho_n u(x_n + \rho_n y, T^* + \rho_n^2 s), \qquad P_n(y, s) := \rho_n^2 p(x_n + \rho_n y, T^* + \rho_n^2 s),$$

for $s < 0$.

Lemma 9.37 (Exterior test dichotomy)**.** *Fix $0 < r < R < \infty$. Then either the branch has no exterior concentration on $A_{r,R}$, or it admits an exterior concentration branch on $A_{r,R}$.*

Proof. If the no-concentration condition fails, then for every m there are

$$x_m \in A_{r,R}, \qquad 0 < \rho_m \le m^{-1},$$

with $Q^*_{\rho_m}(x_m)$ contained in the enlarged annular cylinder and

$$\mathcal{C}_*(u, p; x_m, \rho_m) > \varepsilon_{\text{ext}}.$$

After discarding repetitions and passing to a subsequence, $\rho_m \to 0$. This is precisely an exterior concentration branch. The converse is immediate from the definitions. □

Theorem 9.38 (Exterior concentration stratification)**.** *Fix $0 < r < R < \infty$. A suitable Type II branch satisfies exactly one of the following alternatives on $A_{r,R}$:*

(i) *it has no exterior concentration on $A_{r,R}$;*
(ii) *it admits an exterior concentration branch on $A_{r,R}$.*

The second alternative is the complementary concentration branch. According to the relative position and scale of the exterior concentration sequence, it belongs to one of the exterior-profile, separated-core, multicore, noncompact exterior, or scale-collapse alternatives.

Proof. The dichotomy is Theorem 9.37. If an exterior concentration branch exists, then the annular terminal cylinders carry a positive Caffarelli–Kohn–Nirenberg quantity at scales $\rho_n \downarrow 0$ and centers a positive physical distance from the selected core. Let λ_n denote the selected core scale along the same terminal sequence. After passing to a subsequence, the relative scale-center configuration falls into one of the standard local concentration-compactness alternatives. If the exterior frame is orthogonal to the selected core frame, it gives an exterior profile. If only finitely many separated active centers remain at comparable terminal scales, it gives a separated-core or multicore configuration. If compactness fails in the exterior frame, one obtains the noncompact exterior alternative. If the exterior frame develops a further unresolved scale in its own renormalized variables, one obtains the scale-collapse alternative. These alternatives exhaust the subsequential configurations at positive physical distance from the selected core. Thus the failure of the annular regularity test is represented by a local concentration alternative rather than by an additional regularity hypothesis. □

Lemma 9.39 (Exterior CKN cover)**.** *Assume there is no exterior concentration on $A_{r,R}$. Fix an integer $k \ge 0$. If*

$$A_{r',R'} \Subset A_{r,R},$$

then there are points $x_j \in A_{r,R}$, radii $\rho_j > 0$, and a terminal time length $\delta > 0$ such that

$$A_{r',R'} \times (T^* - \delta, T^*) \subset \bigcup_j B_{c_k \rho_j}(x_j) \times (T^* - c_k \rho_j^2, T^*)$$

and

$$\mathcal{C}_*(u, p; x_j, \rho_j) \le \varepsilon_{\text{ext}}$$

for every j. Here c_k is the constant from Theorem 9.34.

Proof. Since $A_{r',R'} \Subset A_{r,R}$, there is $d > 0$ such that

$$B_{4d}(x) \subset B_{2R}(x_*) \setminus \overline{B_{r/2}(x_*)} \qquad \text{for all } x \in A_{r',R'}.$$

Let ρ_* be supplied by the no-concentration condition and set $\rho_0 := \min\{d, \rho_*\}$. By compactness, finitely many balls

$$B_{c_k\rho_0/2}(x_j), \qquad x_j \in A_{r,R},$$

cover $A_{r',R'}$. The terminal cylinders $Q^*_{\rho_0}(x_j)$ are contained in the enlarged annular cylinder, so the no-concentration condition gives the displayed smallness. Taking $\delta = c_k\rho_0^2/2$ gives the spacetime cover. □

Lemma 9.40 (Exterior pressure localization)**.** *Let $A_{r',R'} \Subset A_{r,R}$, and assume that u is bounded on a slightly larger annular terminal cylinder. Then on*

$$A_{r',R'} \times (T^* - \delta, T^*)$$

the pressure has a decomposition

$$p = p_{\mathrm{loc}} + p_{\mathrm{harm}} + c(t),$$

where $c(t)$ is a function of time, p_{harm} is harmonic in space on the smaller annulus, and for every $q < \infty$ and every integer $k \geq 0$

$$\|p_{\mathrm{loc}}\|_{L^\infty_t L^q_x(A_{r',R'})} + \|p_{\mathrm{harm}}\|_{L^\infty_t C^k_x(A_{r',R'})} \leq C_{r',R',q,k}.$$

Proof. Choose $\eta \in C^\infty_c(B_{2R}(x_*) \setminus \overline{B_{r/2}(x_*)})$ with $\eta \equiv 1$ on a slightly larger annulus containing $A_{r',R'}$. Define p_{loc} by

$$-\Delta p_{\mathrm{loc}} = \partial_i\partial_j(\eta u_i u_j)$$

using the Newtonian potential, with the usual spatial normalization. Since u is locally bounded on the support of η, Calderon–Zygmund estimates give $p_{\mathrm{loc}} \in L^\infty_t L^q_x$ for every $q < \infty$ on the terminal interval. The difference

$$p - p_{\mathrm{loc}}$$

is harmonic in space on the smaller annulus, modulo a function of time. The standard interior estimates for harmonic functions give the displayed C^k bounds after fixing the time-dependent additive constant. □

Theorem 9.41 (Exterior regularity on no-concentration annuli)**.** *Let u, p be a suitable weak solution on $\mathbb{R}^3 \times [0, T^*)$, and let (x_*, T^*) be the selected Type II core point. Fix $0 < r' < R' < \infty$. If there is no exterior concentration on some annulus $A_{r,R}$ with*

$$A_{r',R'} \Subset A_{r,R},$$

then for every integer $k \geq 0$ there are $\delta > 0$, a function $c(t)$, and $C_{r',R',k} < \infty$ such that

$$\sup_{t\in(T^*-\delta,T^*)} \Big(\|u(t)\|_{C^k(A_{r',R'})} + \|p(t) - c(t)\|_{C^k(A_{r',R'})}\Big) \leq C_{r',R',k}.$$

If the no-concentration hypothesis fails, the complementary exterior concentration alternative holds in the sense of Theorem 9.36.

Proof. The no-concentration condition and Theorem 9.39 give a finite cover of $A_{r',R'} \times (T^* - \delta, T^*)$ by terminal cylinders on which the Caffarelli–Kohn–Nirenberg quantity is below $\varepsilon_{\mathrm{CKN}}$. Applying Theorem 9.34 on each cylinder gives local C^k bounds for u and for p modulo functions of time on smaller cylinders. Since the cover is finite, taking the maximum over the cover gives the stated local bound.

The pressure normalizations on overlapping balls differ only by functions of time. Equivalently, one may use Theorem 9.40 after the $k = 0$ velocity bound has been obtained and then apply standard interior parabolic estimates on nested annuli. This gives a single time-dependent normalization $c(t)$ on $A_{r',R'}$. If the no-concentration hypothesis fails, the exterior test dichotomy gives an exterior concentration branch. □

Lemma 9.42 (Divergence-free core and exterior decomposition)**.** *Fix $0 < r_0 < R_0 < \infty$, and assume that there is no exterior concentration on an annulus containing*

$$\overline{B_{2r_0}(x_*)} \setminus B_{r_0}(x_*).$$

Let $\chi \in C_c^\infty(B_{2r_0}(x_*))$ satisfy

$$\chi \equiv 1 \quad \text{on } B_{r_0}(x_*).$$

Then there are divergence-free fields u_{core} and u_{ext} such that

$$u = u_{\text{core}} + u_{\text{ext}}, \qquad u_{\text{core}} = u \text{ on } B_{r_0}(x_*), \qquad u_{\text{ext}} = 0 \text{ on } B_{r_0}(x_*),$$

and all derivatives of u_{ext} are uniformly bounded on compact subannuli of the no-exterior-concentration region up to T^*.

Proof. Set

$$g = \nabla\chi \cdot u.$$

The support of g lies in the annulus

$$A_{r_0} = B_{2r_0}(x_*) \setminus B_{r_0}(x_*),$$

where u is smooth by Theorem 9.41. Since

$$\int_{A_{r_0}} g\,dx = \int_{\mathbb{R}^3} \nabla\chi \cdot u\,dx = -\int_{\mathbb{R}^3} \chi\,\nabla \cdot u\,dx = 0,$$

the Bogovskii operator on A_{r_0} gives a vector field

$$w = \mathcal{B}_{A_{r_0}} g$$

supported in A_{r_0}, smooth up to T^*, and satisfying

$$\nabla \cdot w = g.$$

Define

$$u_{\text{core}} := \chi u - w, \qquad u_{\text{ext}} := u - u_{\text{core}}.$$

Then

$$\nabla \cdot u_{\text{core}} = \nabla\chi \cdot u + \chi\nabla \cdot u - \nabla \cdot w = 0,$$

and therefore $\nabla \cdot u_{\text{ext}} = 0$. Since $\chi = 1$ and $w = 0$ on $B_{r_0}(x_*)$, $u_{\text{core}} = u$ and $u_{\text{ext}} = 0$ there. The uniform exterior bounds follow from Theorem 9.41 on compact subannuli and the standard boundedness of the Bogovskii construction on the fixed annulus. ∎

Lemma 9.43 (Exterior fields vanish in compact core frames)**.** *Let (x_n, λ_n) be a core frame with*

$$x_n \to x_*, \qquad \lambda_n \to 0.$$

For a compact frame ball B_R, define

$$V^n_{\text{ext}}(y,\tau) := \lambda_n u_{\text{ext}}(x_n + \lambda_n y, t_n + \lambda_n^2 \tau).$$

Then, for every compact $B_R \times I$,

$$V^n_{\text{ext}} \equiv 0$$

on $B_R \times I$ for all sufficiently large n. Consequently

$$V^n_{\text{ext}} \to 0 \quad \text{in every local } L^q_\tau W^{k,q}_y(B_R) \text{ norm.}$$

Proof. Since $x_n \to x_*$ and $\lambda_n \to 0$, for fixed R,

$$x_n + \lambda_n B_R \subset B_{r_0}(x_*)$$

for all sufficiently large n. On $B_{r_0}(x_*)$,

$$u_{\text{ext}} = 0.$$

Therefore $V^n_{\text{ext}} = 0$ on $B_R \times I$ for all large n, which implies convergence in every stated local norm. □

Lemma 9.44 (Exterior pressure vanishes modulo constants)**.** *Let p_{ext} be any pressure contribution generated by source terms supported in*

$$\{|x - x_*| \geq r_0\}.$$

In a core frame set

$$P^n_{\text{ext}}(y,\tau) := \lambda_n^2 p_{\text{ext}}(x_n + \lambda_n y, t_n + \lambda_n^2 \tau).$$

Then, for every compact $B_R \times I$, there are constants $c_{n,R}(\tau)$ such that

$$P^n_{\text{ext}} - c_{n,R} \to 0 \quad \textit{in } L^\infty_\tau C^1_y(B_R),$$

and hence

$$\nabla_y P^n_{\text{ext}} \to 0 \quad \textit{in } L^1_\tau H^{-1}_y(B_R).$$

Proof. The exterior pressure source is at positive physical distance from

$$x_n + \lambda_n B_R$$

for all sufficiently large n. Therefore p_{ext} is harmonic, smooth, and uniformly C^2-bounded on the physical neighborhood swept out by the compact frame cylinder. Take

$$c_{n,R}(\tau) := \lambda_n^2 p_{\text{ext}}(x_n, t_n + \lambda_n^2 \tau).$$

By the mean value theorem,

$$\|P^n_{\text{ext}} - c_{n,R}\|_{C^1_y(B_R)} \leq C_R \lambda_n^3 \sup |\nabla_x p_{\text{ext}}| + C_R \lambda_n^4 \sup |\nabla_x^2 p_{\text{ext}}|.$$

The right-hand side tends to zero because the physical derivatives are uniformly bounded and $\lambda_n \to 0$. This proves the asserted convergence. □

Lemma 9.45 (Positive-distance profiles vanish in compact core frames)**.** *Let (x_n, λ_n) be a core frame with $x_n \to x_*$ and $\lambda_n \to 0$. Let (z_n, μ_n) be a profile sequence satisfying*

$$\liminf_{n\to\infty} |z_n - x_*| \geq r_* > 0.$$

Assume moreover that either $|z_n - x_*| \to \infty$, or, after passing to a subsequence, z_n lies in a fixed annulus $A_{r_*/2,R_*}$ on which the no-exterior-concentration condition holds. Then this profile vanishes in every compact core frame:

$$\Lambda^{-1}_{\lambda_n,x_n} \Lambda_{\mu_n,z_n} \phi \to 0$$

locally in $L^3_y(B_R)$. In the escaping-center case its pressure contribution vanishes in the corresponding local $L^{3/2}$ pressure topology by critical profile orthogonality. In the fixed-annulus no-concentration case its pressure contribution vanishes modulo constants as in Theorem 9.44. Hence it is not part of the selected Type II core.

Proof. For fixed $R < \infty$, the physical frame ball

$$x_n + \lambda_n B_R$$

lies in $B_{r_*/4}(x_*)$ for all sufficiently large n, while the profile center z_n lies outside $B_{r_*/2}(x_*)$. Thus the profile is a positive physical distance from the compact core frame.

If $|z_n - x_*| \to \infty$, then the profile frame is orthogonal to the compact core frame. More explicitly, approximate the profile in $L^3(\mathbb{R}^3)$ by a compactly supported smooth field ϕ^ε. The rescaled compact part has support disjoint from $x_n + \lambda_n B_R$ for all large n, while the remainder has L^3-norm at most ε in every critical rescaling. Letting first $n \to \infty$ and then $\varepsilon \downarrow 0$ gives local $L^3_y(B_R)$ convergence to zero. The same truncation argument applied to the quadratic source, together with the local Calderon–Zygmund estimate for pressure modulo functions of time, gives convergence of the pressure contribution to zero in local $L^{3/2}$.

Otherwise, after passing to a subsequence, the centers lie in a fixed annulus on which the no-exterior-concentration condition holds. The exterior velocity part is covered by Theorem 9.43 after the core/exterior decomposition, and the exterior pressure part is covered by Theorem 9.44. This proves the claimed decoupling of positive-distance profiles from compact Type II core frames. ☐

Theorem 9.46 (Exterior regular components decouple from compact core frames)**.** *On the no-exterior-concentration stratum for the fixed annuli separating an exterior component from the selected Type II core point, that component is a regular exterior component and vanishes in compact Type II core frames, including its pressure contribution modulo constants.*

Proof. Apply the divergence-free decomposition of Theorem 9.42. The exterior component is identically zero in every compact core frame for large n by Theorem 9.43. Its pressure contribution is a locally scaled smooth harmonic field and vanishes modulo constants by Theorem 9.44. The cutoff and Bogovskii correction terms are supported in the fixed exterior annulus A_{r_0}, so they also vanish in compact core frames by the same argument.

Thus a regular exterior component decouples from the Type II core without changing the local repaired-gauge equation in any compact terminal frame, up to errors converging to zero in the perturbative topology used in Theorem 9.10 and Theorem 9.13. ☐

Corollary 9.47 (Exterior annulus alternative for the core decomposition)**.** *Fix a compact annulus $A_{r,R}$ around the selected core. Then exactly one of the following alternatives holds for the Type II branch on this annulus:*

(i) *the branch has no exterior concentration on $A_{r,R}$, and every compact subannulus of $A_{r,R}$ is covered by Theorem 9.41; consequently exterior regular contributions from this annulus vanish in compact core frames by Theorem 9.46;*

(ii) *the branch admits an exterior concentration branch on $A_{r,R}$, and therefore belongs to one of the adjacent exterior, separated-core, multicore, noncompact exterior, or scale-collapse alternatives described in Theorem 9.38.*

Proof. This is the exterior concentration stratification of Theorem 9.38, combined in the first alternative with the no-concentration regularity theorem and the exterior-removal theorem. ☐

10. Corrected Monotonicity and Scale-Drift Cost Criteria

This section records the remaining conditional analytic inputs needed for a localized cost-divergence exclusion argument on renormalized Type II branches. The statements are written entirely in the variables of the repaired gauge:

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0.$$

All pressure terms are understood modulo functions of time.

10.1. Hypotheses for a Conditional Cost Criterion.

Definition 10.1 (Conditional cost hypotheses)**.** A renormalized local Type II branch satisfies the conditional cost hypotheses if the following analytic facts hold.

(i) the branch lies in the local Type II concentration alternative;
(ii) the physical Navier–Stokes energy inequality gives bounded energy along the branch;
(iii) the branch is written in the repaired-gauge variables (V, P, a, b) above, including the scale, center, pressure normalization, and modulation data;
(iv) the localized cost is identified with

$$\tilde{\mathfrak{D}}_{R_0}(\tau) = \nu \int_{\mathbb{R}^3} |\nabla V(y,\tau)|^2 \phi_{R_0}(y)\, dy + a_+(\tau) \int_{\mathbb{R}^3} |V(y,\tau)|^2 \phi_{R_0}(y)\, dy,$$

or with an explicitly stated tail-comparable nonnegative replacement;
(v) a localized energy identity or suitable-solution local energy inequality is available for the cutoff used in the cost;
(vi) a corrected localized monotonicity inequality, in the sense of Theorem 10.6, is available.

Assumption 10.2 (Cost-divergence exclusion criterion)**.** For every renormalized local Type II branch satisfying Theorem 10.1, divergence of the corresponding nonnegative localized cost,

$$\int_{\tau_0}^{\infty} \mathcal{D}_{R_0}(\tau)\, d\tau = \infty,$$

excludes that branch from the local Type II class under consideration.

Lemma 10.3 (Concentration and energy)**.** *If a branch lies in the local Type II concentration alternative and satisfies the physical Navier–Stokes energy inequality, then it satisfies Theorem 10.1*(i) *and* (ii)*.*

Proof. The local Type II concentration alternative supplies the selected concentration branch. The physical energy inequality gives the bounded-energy part. These are precisely items (i) and (ii). □

Lemma 10.4 (Cost identification)**.** *Assume the localized cost is the Navier–Stokes cost $\tilde{\mathfrak{D}}_{R_0}$. Then divergence of*

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau$$

is the cost divergence in Theorem 10.2.

Proof. The active cost is assumed to be equal to $\tilde{\mathfrak{D}}_{R_0}$. Thus the cost appearing in the local Navier–Stokes estimates and the cost appearing in the conditional exclusion criterion are the same nonnegative functional. The divergence condition is therefore identical in the two formulations. □

Lemma 10.5 (Parabolic representation class)**.** *If the repaired-gauge representation theorem applies to a Type II branch, then the branch has the parabolic transport-diffusion structure required in Theorem 10.1.*

Proof. The repaired-gauge theorem supplies the renormalized orbit (V, P, a, b), the scale and center, the pressure reconstruction modulo functions of time, and the modulation coefficients. The displayed equation is a parabolic Navier–Stokes equation with transport, dilation, and translation drift terms. Consequently the branch has the parabolic transport-diffusion structure required by the conditional cost hypotheses. □

Definition 10.6 (Corrected localized monotonicity input)**.** A renormalized branch has corrected localized monotonicity if there are a localized renormalized energy $\mathcal{E}_{R_0}$, a nonnegative cost $\mathcal{D}_{R_0}$, a constant $c_0 > 0$, and a nonnegative remainder $\mathcal{R}_{R_0}$ such that, on the tail of the branch,

$$\frac{d}{d\tau}\mathcal{E}_{R_0}(\tau) + c_0\mathcal{D}_{R_0}(\tau) + \mathcal{R}_{R_0}(\tau) \le 0$$

in distributions. If the localized energy identity has an integrable error B_{R_0}, the corrected energy is

$$\mathcal{E}^{\mathrm{corr}}_{R_0}(\tau) = \mathcal{E}_{R_0}(\tau) + \int_\tau^\infty B_{R_0}(s)\,ds,$$

and the same inequality is required for $\mathcal{E}^{\mathrm{corr}}_{R_0}$. Multiplication of a nonnegative cost by $c_0 > 0$ does not change whether its tail integral diverges.

Lemma 10.7 (Energy and Caccioppoli bounds alone are not monotonicity)**.** *The global physical energy inequality and compact-cylinder Caccioppoli estimates do not, by themselves, imply the corrected localized monotonicity input of Theorem 10.6.*

Proof. The global physical energy inequality controls total physical energy and physical dissipation, but it does not identify a localized renormalized energy at the selected Type II scale whose derivative is monotone after pressure flux, cutoff flux, transport, and modulation terms are included. The Caccioppoli estimate gives local spacetime H^1 control from the local energy inequality. It is an a priori estimate, not a corrected monotonicity formula. Therefore an additional absorption or tail-correction argument is needed. □

Proposition 10.8 (Conditional cost-divergence exclusion)**.** *Let a renormalized Type II branch satisfy Theorem 10.1 and Theorem 10.2. If its nonnegative localized cost has divergent tail integral, then the branch is excluded from the local Type II class under consideration.*

Proof. Theorem 10.3 gives the local Type II concentration and bounded-energy inputs. Theorem 10.4 identifies the localized Navier–Stokes cost with the cost used in the exclusion criterion. Theorem 10.5 gives the parabolic transport-diffusion class. The corrected monotonicity input is an explicit hypothesis of Theorem 10.1. Once the cost diverges, Theorem 10.2 gives the stated exclusion. □

10.2. **Fixed-Cutoff Monotonicity.** Let $\phi = \phi_{R_0} \in C_c^\infty(\mathbb{R}^3)$, $0 \le \phi \le 1$, be the radial nonincreasing cutoff used in the localized cost. Define

$$\mathcal{E}_\phi(\tau) = \frac{1}{2}\int_{\mathbb{R}^3} |V(y,\tau)|^2\phi(y)\,dy,$$

$$A_\phi(\tau) = \int_{\mathbb{R}^3} |\nabla V(y,\tau)|^2\phi(y)\,dy, \qquad M_\phi(\tau) = \int_{\mathbb{R}^3} |V(y,\tau)|^2\phi(y)\,dy,$$

and

$$\tilde{\mathfrak{D}}_{R_0}(\tau) = \nu A_\phi(\tau) + a_+(\tau)M_\phi(\tau), \qquad a_+(\tau) = \max\{a(\tau), 0\}.$$

The localized energy identity used below is

$$\frac{d}{d\tau}\mathcal{E}_\phi = -\nu A_\phi + \frac{\nu}{2}\int |V|^2\Delta\phi + \frac{1}{2}\int |V|^2 V\cdot\nabla\phi + \int P\,V\cdot\nabla\phi$$

$$(13)\qquad -\frac{a}{2}M_\phi - \frac{a}{2}\int |V|^2 y\cdot\nabla\phi - \frac{1}{2}b\cdot\int |V|^2\nabla\phi.$$

Definition 10.9 (Fixed-cutoff error density)**.** Set $a_-(\tau)=\max\{-a(\tau),0\}$ and define

$$B_{R_0}(\tau) := \frac{\nu}{2}\left|\int |V|^2\Delta\phi\right| + \frac{1}{2}\left|\int |V|^2 V\cdot\nabla\phi\right| + \left|\int P\,V\cdot\nabla\phi\right| \\ + \frac{1}{2}|b(\tau)|\left|\int |V|^2\nabla\phi\right| + \frac{1}{2}a_-(\tau)M_\phi(\tau) + \frac{1}{2}a_+(\tau)\int |V|^2\big(-y\cdot\nabla\phi\big).$$

The fixed-cutoff finite-error condition is

$$\int_{\tau_0}^{\infty} B_{R_0}(\tau)\,d\tau < \infty.$$

Remark 10.10 (Pressure normalization). The pressure term in B_{R_0} is independent of the representative of the pressure modulo functions of time. Replacing P by $P+c(\tau)$ changes the pressure flux by

$$c(\tau)\int V\cdot\nabla\phi = -c(\tau)\int \phi\,\nabla\cdot V = 0.$$

Theorem 10.11 (Fixed-cutoff corrected monotonicity)**.** *Assume* (13) *holds on* (τ_0,∞) *and that* $B_{R_0}\in L^1(\tau_0,\infty)$. *Define*

$$\mathcal{E}_\phi^{\mathrm{corr}}(\tau) = \mathcal{E}_\phi(\tau) + \int_\tau^\infty B_{R_0}(s)\,ds.$$

Then, in distributions on (τ_0,∞),

$$\frac{d}{d\tau}\mathcal{E}_\phi^{\mathrm{corr}}(\tau) + \frac{1}{2}\tilde{\mathfrak{D}}_{R_0}(\tau) \le 0.$$

Proof. Add

$$\frac{1}{2}\tilde{\mathfrak{D}}_{R_0} = \frac{\nu}{2}A_\phi + \frac{1}{2}a_+M_\phi$$

to both sides of (13). The gradient term becomes

$$-\nu A_\phi + \frac{\nu}{2}A_\phi = -\frac{\nu}{2}A_\phi \le 0.$$

The core scale term satisfies

$$-\frac{a}{2}M_\phi + \frac{1}{2}a_+M_\phi = \frac{1}{2}a_-M_\phi \le B_{R_0}(\tau).$$

Since ϕ is radial nonincreasing, one has

$$y\cdot\nabla\phi(y)\le 0 \qquad \text{for all } y\in\mathbb{R}^3.$$

Hence

$$-\frac{a}{2}\int |V|^2 y\cdot\nabla\phi \le \frac{1}{2}a_+(\tau)\int |V|^2\big(-y\cdot\nabla\phi\big),$$

because the a_--part contributes a nonpositive term. Every remaining term in (13) is bounded above by its corresponding absolute-value contribution in B_{R_0}. Therefore

$$\frac{d}{d\tau}\mathcal{E}_\phi(\tau) + \frac{1}{2}\tilde{\mathfrak{D}}_{R_0}(\tau) \le B_{R_0}(\tau)$$

in distributions. Since $B_{R_0}\in L^1(\tau_0,\infty)$, the correction tail is finite for every $\tau\ge\tau_0$, absolutely continuous on compact subintervals, and

$$\frac{d}{d\tau}\int_\tau^\infty B_{R_0}(s)\,ds = -B_{R_0}(\tau).$$

Adding this identity gives the claimed inequality. ☐

Corollary 10.12 (Fixed-cutoff monotonicity input). *If the repaired-gauge representation, the cost identification, the localized energy identity, and the fixed-cutoff finite-error condition all hold, then Theorem 10.6 holds with $\mathcal{D}_{R_0} = \tilde{\mathfrak{D}}_{R_0}$ and $c_0 = 1/2$.*

Proof. The repaired-gauge representation supplies (V, P, a, b). The cost identification supplies $\tilde{\mathfrak{D}}_{R_0}$. The localized energy identity and finite-error condition allow Theorem 10.11 to be applied. □

Remark 10.13 (Finite-tail subconditions). The fixed-cutoff finite-error condition is equivalent to the conjunction of tail integrability for the viscous cutoff term, the convective flux, the pressure flux, the center-drift cutoff term, the negative-scale term, and the positive scale-shell term:

$$\left|\int |V|^2 \Delta\phi\right|, \qquad \left|\int |V|^2 V \cdot \nabla\phi\right|, \qquad \left|\int P\, V \cdot \nabla\phi\right|,$$
$$|b| \left|\int |V|^2 \nabla\phi\right|, \qquad a_- M_\phi, \qquad a_+ \int |V|^2 (-y \cdot \nabla\phi).$$

For radial nonincreasing cutoffs, only the positive part a_+ contributes to the scale-shell error. The negative part is favorable and does not belong to the residual finite-error burden. Uniform compact-window bounds do not by themselves give this global tail integrability on $[\tau_0, \infty)$.

Definition 10.14 (Fixed-cutoff cost hypotheses). A renormalized Type II branch satisfies the fixed-cutoff cost hypotheses if it has the local Type II concentration alternative, bounded physical energy, repaired-gauge representation, cost identification with $\tilde{\mathfrak{D}}_{R_0}$, divergent nonnegative localized cost, localized energy identity, and the finite-tail bound $\int_{\tau_0}^{\infty} B_{R_0} < \infty$.

Theorem 10.15 (Fixed-cutoff cost criterion). *Every branch satisfying the fixed-cutoff cost hypotheses is covered by Theorem 10.8.*

Proof. By Theorem 10.12, the finite-tail condition supplies the corrected monotonicity input. The remaining inputs are precisely the other items in Theorem 10.1, together with cost divergence. The local exclusion step is later proved on the canonical windowwise schedule in Theorem 10.42. □

Definition 10.16 (Compact cost-divergence replacement). A compact cost-divergence replacement is available when the compact Type II hypotheses of the preceding sections imply

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty,$$

and this divergence is the cost divergence used in Theorem 10.8. In particular, it may be supplied by the combination of classification of local alternatives, uniform critical tightness, local windowed H^1 control, and the cost identification above whenever those are the active hypotheses in the compact Type II theorem.

Theorem 10.17 (Compact cost-divergence criterion). *If a renormalized branch has all fixed-cutoff cost hypotheses except that cost divergence is supplied through the compact cost-divergence replacement of Theorem 10.16, then Theorem 10.8 applies.*

Proof. The compact cost-divergence replacement supplies the missing divergent cost. Substituting this into the fixed-cutoff hypotheses gives the hypotheses of Theorem 10.15. □

10.3. **Moving-Cutoff Monotonicity.** Let $\phi \in C_c^\infty(\mathbb{R}^3)$ be radial and nonincreasing, $0 \le \phi \le 1$, $\phi = 1$ on B_1, and $\phi = 0$ outside B_2. For $R(\tau) \ge R_0$, locally absolutely continuous and nondecreasing, set

$$\Phi(\tau, y) = \phi(y/R(\tau)), \qquad A_{R(\tau)} = \{R(\tau) \le |y| \le 2R(\tau)\},$$

$$\mathcal{E}_R(\tau) = \frac{1}{2}\int |V(y,\tau)|^2 \Phi(\tau,y)\,dy,$$

and

$$\tilde{\mathfrak{D}}_{R(\tau)}(\tau) = \nu \int |\nabla V|^2 \Phi + a_+(\tau)\int |V|^2\Phi.$$

Definition 10.18 (Moving-cutoff error density)**.** Define

$$\begin{aligned} B_{\mathrm{mov}}(\tau) := \frac{1}{2}\left|\int |V|^2 \partial_\tau \Phi\right| + \frac{\nu}{2}\left|\int |V|^2 \Delta\Phi\right| + \frac{1}{2}\left|\int |V|^2 V\cdot\nabla\Phi\right| + \left|\int P\,V\cdot\nabla\Phi\right| \\ + \frac{1}{2}|b(\tau)|\left|\int |V|^2\nabla\Phi\right| + \frac{1}{2}a_-(\tau)\int |V|^2\Phi + \frac{1}{2}a_+(\tau)\int |V|^2\big(-y\cdot\nabla\Phi\big). \end{aligned}$$

The moving finite-error condition is

$$\int_{\tau_0}^{\infty} B_{\mathrm{mov}}(\tau)\,d\tau < \infty.$$

Theorem 10.19 (Moving-cutoff corrected monotonicity)**.** *Assume the localized energy identity is valid with the time-dependent cutoff Φ, and assume $B_{\mathrm{mov}} \in L^1(\tau_0,\infty)$. Then*

$$\mathcal{E}_R^{\mathrm{corr}}(\tau) = \mathcal{E}_R(\tau) + \int_\tau^\infty B_{\mathrm{mov}}(s)\,ds$$

satisfies

$$\frac{d}{d\tau}\mathcal{E}_R^{\mathrm{corr}}(\tau) + \frac{1}{2}\tilde{\mathfrak{D}}_{R(\tau)}(\tau) \le 0$$

in distributions.

Proof. Use the localized energy identity with the time-dependent test function $\Phi(\tau,y)$. Compared with the fixed-cutoff identity, the only new term is

$$\frac{1}{2}\int |V|^2\partial_\tau\Phi,$$

which comes from differentiating the cutoff inside $\mathcal{E}_R$. All other terms are the fixed-cutoff terms with ϕ replaced by Φ. Add $\frac{1}{2}\tilde{\mathfrak{D}}_{R(\tau)}$ to both sides. The gradient term leaves

$$-\frac{\nu}{2}\int |\nabla V|^2\Phi \le 0,$$

and the core scale term contributes

$$\frac{1}{2}a_-\int |V|^2\Phi.$$

Since ϕ is radial nonincreasing, one has

$$y\cdot\nabla\Phi(\tau,y) \le 0 \qquad \text{for all } (\tau,y).$$

Therefore

$$-\frac{a}{2}\int |V|^2 y\cdot\nabla\Phi \le \frac{1}{2}a_+(\tau)\int |V|^2\big(-y\cdot\nabla\Phi\big),$$

because the a_--part is nonpositive. Each remaining term is bounded above by the corresponding absolute-value term in B_{mov}. Hence

$$\frac{d}{d\tau}\mathcal{E}_R(\tau) + \frac{1}{2}\tilde{\mathfrak{D}}_{R(\tau)}(\tau) \le B_{\mathrm{mov}}(\tau).$$

The moving finite-error condition makes the correction tail finite and absolutely continuous on compact subintervals. Differentiating that tail subtracts B_{mov}, giving the stated inequality. $\square$

Definition 10.20 (Summable moving-annulus errors)**.** The moving-annulus summability condition is the existence of a nondecreasing locally absolutely continuous $R(\tau) \to \infty$ such that each of the seven quantities

$$\left|\int |V|^2 \partial_\tau \Phi\right|, \quad \left|\int |V|^2 \Delta\Phi\right|, \quad \left|\int |V|^2 V \cdot \nabla\Phi\right|, \quad \left|\int P\, V \cdot \nabla\Phi\right|,$$
$$|b| \left|\int |V|^2 \nabla\Phi\right|, \quad a_- \int |V|^2 \Phi, \quad a_+ \int |V|^2 \big(-y \cdot \nabla\Phi\big)$$

is integrable on $[\tau_0, \infty)$.

Lemma 10.21 (Summable annuli imply moving finite error)**.** *If Theorem 10.20 holds, then* $B_{\mathrm{mov}} \in L^1(\tau_0, \infty)$.

Proof. The density B_{mov} is a finite positive linear combination of the seven quantities listed in Theorem 10.20. If each is integrable, then their linear combination is integrable. □

Definition 10.22 (Summable tightness schedule)**.** A summable tightness schedule is a nondecreasing sequence $R_n \to \infty$ such that, on unit windows $I_n = [\tau_0 + n, \tau_0 + n + 1]$,

$$\sum_{n=0}^{\infty} \sup_{\tau \in I_n} \|V(\tau)\|^3_{L^3(A_{R_n})} < \infty,$$

and the same schedule is compatible with the pressure-tail normalization used for P. This is stronger than uniform critical tightness, which permits large radii on each window but does not by itself give a monotone radius schedule with a summable series.

Definition 10.23 (Moving-cost compatibility)**.** The moving localized cost is compatible with the cost-divergence criterion if

$$\tilde{\mathfrak{D}}_{R(\tau)}(\tau) = \nu \int |\nabla V|^2 \phi_{R(\tau)} + a_+(\tau) \int |V|^2 \phi_{R(\tau)}$$

is the localized cost appearing in Theorem 10.2, or is compared to that cost by a positive constant on the tail so that cost divergence is unchanged.

Theorem 10.24 (Moving-cutoff replacement)**.** *Assume the repaired-gauge representation, localized energy identity, moving-cost compatibility, and summable moving-annulus errors. Then the moving cutoff supplies the corrected monotonicity input needed in Theorem 10.8.*

Proof. Theorem 10.21 gives $B_{\mathrm{mov}} \in L^1(\tau_0, \infty)$. Applying Theorem 10.19 gives corrected monotonicity with cost $\tilde{\mathfrak{D}}_{R(\tau)}$. The moving-cost compatibility assumption identifies this cost, or a tail-comparable version of it, with the localized cost in the conditional exclusion criterion. □

10.4. **Negative Scale Drift.** Set

$$M_R(\tau) = \int |V(y,\tau)|^2 \Phi(\tau, y)\, dy.$$

The negative-scale contribution in the moving cutoff argument is

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau.$$

Definition 10.25 (Negative scale-drift alternatives)**.** We say that the branch has finite negative drift if

$$\int_{\tau_0}^{\infty} a_-(\tau)\, d\tau < \infty.$$

It has bounded moving L^2 core mass if

$$\sup_{\tau \geq \tau_0} M_R(\tau) < \infty.$$

It has a moving L^2 core floor if there are $m_0 > 0$ and $\tau_1 \geq \tau_0$ such that

$$M_R(\tau) \geq m_0 \qquad \text{for a.e. } \tau \geq \tau_1.$$

The scale-collapse drift obstruction is the divergent alternative

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau = \infty.$$

Lemma 10.26 (Finite negative drift controls the scale term)**.** *If the branch has bounded moving L^2 core mass and finite negative drift, then*

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau < \infty.$$

Proof. Let $M_* = \sup_{\tau \geq \tau_0} M_R(\tau) < \infty$. Since $a_- \geq 0$,

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau \leq M_* \int_{\tau_0}^{\infty} a_-(\tau)\, d\tau < \infty.$$

□

Theorem 10.27 (Finite-or-infinite scale-negative alternative)**.** *For every renormalized orbit for which a_- and M_R are measurable and nonnegative, precisely one of the following alternatives holds:*

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau < \infty, \qquad \int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau = \infty.$$

Proof. The integral

$$I_R = \int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau$$

is an extended nonnegative number. Hence either $I_R < \infty$ or $I_R = \infty$, and the two cases are mutually exclusive. □

Lemma 10.28 (Logarithmic scale identity)**.** *For the repaired-gauge convention*

$$\tau_t = \lambda(t)^{-2}, \qquad a(\tau) = \lambda(t)\lambda_t(t),$$

one has

$$\int_{\tau_1}^{\tau_2} a(\tau)\, d\tau = \log \lambda(\tau_2) - \log \lambda(\tau_1)$$

whenever $\tau_0 \leq \tau_1 < \tau_2 < \infty$.

Proof. The chain rule gives

$$\frac{d}{d\tau} \log \lambda = \frac{\lambda_t}{\lambda} \frac{dt}{d\tau} = \frac{\lambda_t}{\lambda} \lambda^2 = \lambda \lambda_t = a.$$

Integrating over $[\tau_1, \tau_2]$ proves the identity. □

Theorem 10.29 (Collapse with core floor forces scale-drift obstruction)**.** *Assume*

$$\lambda(\tau) \to 0 \qquad \text{as } \tau \to \infty,$$

and assume the moving L^2 core floor of Theorem 10.25. Then

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau = \infty.$$

Proof. By Theorem 10.28,

$$\int_{\tau_1}^{\tau_2} a(\tau)\,d\tau = \log\lambda(\tau_2) - \log\lambda(\tau_1).$$

Since $\lambda(\tau) \to 0$, the right-hand side tends to $-\infty$ as $\tau_2 \to \infty$. If $\int_{\tau_1}^{\infty} a_-(\tau)\,d\tau < \infty$, then

$$\int_{\tau_1}^{\tau_2} a(\tau)\,d\tau = \int_{\tau_1}^{\tau_2} a_+(\tau)\,d\tau - \int_{\tau_1}^{\tau_2} a_-(\tau)\,d\tau \geq -\int_{\tau_1}^{\infty} a_-(\tau)\,d\tau > -\infty,$$

which contradicts the preceding limit. Thus $\int_{\tau_1}^{\infty} a_-(\tau)\,d\tau = \infty$.

By the moving L^2 core floor, $M_R(\tau) \geq m_0$ for a.e. $\tau \geq \tau_1$. Therefore

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\,d\tau \geq m_0 \int_{\tau_1}^{\infty} a_-(\tau)\,d\tau = \infty.$$

□

Corollary 10.30 (Scale-drift alternatives)**.** *In the moving-cutoff setup, the negative-scale term has the following exhaustive alternatives. If bounded moving L^2 core mass and finite negative drift hold, then the negative-scale term is integrable. If the weighted negative-drift integral is infinite, the branch lies in the scale-collapse drift obstruction. If genuine collapse and a moving L^2 core floor also hold, then the obstruction is forced.*

Proof. The first case is Theorem 10.26. The exhaustive split is Theorem 10.27. The final assertion is Theorem 10.29. □

10.5. **Scale-Drift Cost Criteria.**

Definition 10.31 (Scale-collapse drift cost)**.** For a renormalized branch define

$$\mathfrak{C}_{\mathrm{sc}}(\tau) = a_-(\tau) M_R(\tau), \qquad M_R(\tau) = \int |V(y,\tau)|^2 \phi_{R(\tau)}(y)\,dy,$$

and

$$\mathcal{C}_{\mathrm{sc}}[\tau_0,\infty) = \int_{\tau_0}^{\infty} \mathfrak{C}_{\mathrm{sc}}(\tau)\,d\tau.$$

Thus the scale-collapse drift obstruction is precisely $\mathcal{C}_{\mathrm{sc}}[\tau_0,\infty) = \infty$. The density is nonnegative and local in the renormalized core.

Definition 10.32 (Scale-collapse cost compatibility)**.** The scale-collapse drift cost is compatible with the cost-divergence criterion when:

(i) $\mathfrak{C}_{\mathrm{sc}}$ is measurable, nonnegative, and locally integrable on finite τ-intervals for renormalized branches;
(ii) divergence of $\mathcal{C}_{\mathrm{sc}}[\tau_0,\infty)$ is included among the hypotheses in Theorem 10.2;
(iii) the resulting conclusion is the same local Type II exclusion conclusion used for the fixed localized cost.

Theorem 10.33 (Scale-collapse drift cost criterion)**.** *If the scale-collapse drift obstruction holds and the cost $\mathfrak{C}_{\mathrm{sc}}$ is compatible with the cost-divergence criterion, then the divergent-cost hypothesis of Theorem 10.8 is satisfied for the scale-collapse branch.*

Proof. By Theorem 10.31, the obstruction is precisely

$$\mathcal{C}_{\mathrm{sc}}[\tau_0,\infty) = \infty.$$

By Theorem 10.32, this divergence is a nonnegative cost-divergence hypothesis with the same local conclusion. □

Corollary 10.34 (Scale-collapse exclusion by drift cost)**.** *Assume a renormalized local Type II branch has scale-collapse drift obstruction, compatible scale-collapse drift cost, and the remaining conditional cost hypotheses. If Theorem 10.2 holds, then the branch is excluded from the local Type II class under consideration.*

Proof. Theorem 10.33 supplies the divergent localized cost. Applying Theorem 10.8 gives the exclusion. □

Definition 10.35 (Absolute scale-drift cost)**.** Define

$$\mathfrak{C}_{\mathrm{abs}}(\tau) = \nu \int |\nabla V|^2 \phi_{R(\tau)} + |a(\tau)| M_R(\tau).$$

The absolute scale-drift cost is compatible with the cost-divergence criterion if divergence of $\int_{\tau_0}^{\infty} \mathfrak{C}_{\mathrm{abs}}(\tau)\, d\tau$ is included as a Type II cost-divergence condition with the same local conclusion.

Lemma 10.36 (Scale-collapse drift forces absolute-cost divergence)**.** *If the scale-collapse drift obstruction holds, then*

$$\int_{\tau_0}^{\infty} \mathfrak{C}_{\mathrm{abs}}(\tau)\, d\tau = \infty.$$

Proof. Since

$$\mathfrak{C}_{\mathrm{abs}}(\tau) = \nu \int |\nabla V|^2 \phi_{R(\tau)} + |a(\tau)| M_R(\tau) \geq a_-(\tau) M_R(\tau),$$

divergence of $\int a_- M_R$ forces divergence of $\int \mathfrak{C}_{\mathrm{abs}}$. □

Corollary 10.37 (Absolute scale-cost criterion)**.** *If the scale-collapse drift obstruction holds and the absolute scale-drift cost is compatible with the cost-divergence criterion, then the divergent-cost hypothesis of Theorem 10.8 is satisfied.*

Proof. Theorem 10.36 gives divergence of the absolute scale-drift cost, and compatibility makes this divergence a Type II cost-divergence condition. □

Theorem 10.38 (Moving cutoff with scale-collapse alternative)**.** *Assume the moving-cutoff setup and a renormalized local Type II branch.*

(i) *If the negative-scale term is integrable and the remaining moving-annulus errors are summable, then the moving-cutoff corrected monotonicity estimate is available.*
(ii) *If the scale-collapse drift obstruction holds and either the scale-collapse drift cost or the absolute scale-drift cost is compatible with the cost-divergence criterion, then the divergent-cost hypothesis is satisfied directly.*
(iii) *If, in addition, the remaining hypotheses of Theorem 10.1 and Theorem 10.2 hold, then the branch is excluded.*

Proof. The first branch is Theorem 10.24. The second branch is Theorem 10.33 or Theorem 10.37, depending on which compatible cost is used. The third branch is Theorem 10.8. □

10.6. **Cost-Divergence Exclusion on the Canonical Windowwise Schedule.** The conditional exclusion criterion of Theorem 10.2 is, given the corrected monotonicity input that is already part of Theorem 10.1, an arithmetic consequence of the nonnegativity of the localized renormalized energy together with the windowwise alternative argument of Theorems 8.149 and 8.154. We make this implication explicit here so that the assumption can be removed from the assembly hypotheses.

Lemma 10.39 (Nonnegativity of the corrected localized energy). *Let $\phi = \phi_{R_0} \in C_c^\infty(\mathbb{R}^3)$ be a radial nonnegative cutoff, and let $R(\tau)$ be either fixed at R_0 or the canonical windowwise interpolated radius of Theorem 8.139. Then $\mathcal{E}_\phi(\tau) = \frac{1}{2}\int |V(y,\tau)|^2\phi(y)\,dy \geq 0$ and $\mathcal{E}_R(\tau) = \frac{1}{2}\int |V(y,\tau)|^2\Phi(\tau,y)\,dy \geq 0$ for every τ, and the correction tails $\int_\tau^\infty B_{R_0}(s)\,ds$ and $\int_\tau^\infty B_{\mathrm{mov}}(s)\,ds$ of Theorems 10.9 and 10.18 are nonnegative. Consequently the corrected energies $\mathcal{E}_\phi^{\mathrm{corr}}$ and $\mathcal{E}_R^{\mathrm{corr}}$ of Theorems 10.11 and 10.19 are nonnegative.*

Proof. Both $\mathcal{E}_\phi$ and $\mathcal{E}_R$ are integrals of $|V|^2$ against a nonnegative weight, hence nonnegative. Inspecting the seven summands of B_{R_0} (Theorem 10.9) and the seven summands of B_{mov} (Theorem 10.18): five are absolute values of integrals and are therefore nonnegative; the remaining two are $\frac{1}{2}a_-(\tau)\int |V|^2\Phi$ and $\frac{1}{2}a_+(\tau)\int |V|^2(-y\cdot\nabla\Phi)$ (and likewise with Φ replaced by ϕ). In the first, $a_- \geq 0$ and the integrand $|V|^2\Phi \geq 0$, so the term is nonnegative. In the second, $a_+ \geq 0$, and $-y\cdot\nabla\Phi \geq 0$ because $\Phi(\tau,y) = \phi(y/R(\tau))$ with ϕ radial nonincreasing forces $\nabla\Phi(\tau,y)$ to point opposite to y; the term is nonnegative. Hence $B_{R_0}, B_{\mathrm{mov}} \geq 0$ pointwise, and their tail integrals are nonnegative. Adding a nonnegative tail to a nonnegative quantity preserves nonnegativity, so $\mathcal{E}_\phi^{\mathrm{corr}}$ and $\mathcal{E}_R^{\mathrm{corr}}$ are nonnegative. □

Lemma 10.40 (Corrected monotonicity is incompatible with cost divergence). *Let $\mathcal{E}^{\mathrm{corr}}\colon [\tau_0,\infty) \to [0,\infty)$ be locally integrable with finite initial value $\mathcal{E}^{\mathrm{corr}}(\tau_0) < \infty$, let $\mathcal{D}\colon [\tau_0,\infty) \to [0,\infty)$ be locally integrable, and let $c_0 > 0$. If*

$$\frac{d}{d\tau}\mathcal{E}^{\mathrm{corr}}(\tau) + c_0\mathcal{D}(\tau) \leq 0 \qquad \textit{in distributions on } (\tau_0,\infty),$$

then

$$\int_{\tau_0}^\infty \mathcal{D}(\tau)\,d\tau \leq \frac{1}{c_0}\,\mathcal{E}^{\mathrm{corr}}(\tau_0) < \infty.$$

In particular, the simultaneous validity of corrected monotonicity in this sense and of $\int_{\tau_0}^\infty \mathcal{D} = \infty$ is contradictory. The same conclusion holds if the inequality is strengthened to the form $(d/d\tau)\mathcal{E}^{\mathrm{corr}} + c_0\mathcal{D} + \mathcal{R} \leq 0$ with $\mathcal{R} \geq 0$, as in Theorem 10.6.

Proof. The distributional inequality $(\mathcal{E}^{\mathrm{corr}})' + c_0\mathcal{D} \leq 0$ on (τ_0,∞), with $\mathcal{D} \geq 0$ and $\mathcal{E}^{\mathrm{corr}} \geq 0$ locally integrable, implies that $\mathcal{E}^{\mathrm{corr}}$ admits a non-increasing right-continuous representative. Testing the inequality against a smooth approximation of $\mathbf{1}_{[\tau_0,\tau]}$ and passing to the limit using $\mathcal{D} \geq 0$ and $\mathcal{E}^{\mathrm{corr}} \geq 0$ (so the boundary terms have a definite sign) yields

$$c_0\int_{\tau_0}^\tau \mathcal{D}(\sigma)\,d\sigma \leq \mathcal{E}^{\mathrm{corr}}(\tau_0) - \mathcal{E}^{\mathrm{corr}}(\tau) \leq \mathcal{E}^{\mathrm{corr}}(\tau_0).$$

Letting $\tau \to \infty$ and using monotone convergence ($\mathcal{D} \geq 0$) yields the displayed bound. If $\int \mathcal{D} = \infty$ the left-hand side is infinite while the right-hand side is finite, a contradiction. □

Theorem 10.41 (Cost-divergence/corrected-monotonicity dichotomy on the canonical schedule). *Let ω be a renormalized local Type II branch in the pre-carrier validated compact-cost regime of Theorem 8.100. Equip ω with the canonical windowwise interpolated cutoff $\Phi(\tau,y) = \phi(y/R(\tau))$ of Theorem 8.139, and let $\widetilde{\mathfrak{D}}_{R(\tau)}$ be the moving localized cost of Theorem 10.23. Assume in addition the localized energy identity for the moving cutoff $\Phi(\tau,\cdot)$ of Theorem 10.1*(v) *(supplied, e.g., by Theorem 8.102 on the regime). If*

$$\int_{\tau_0}^\infty \widetilde{\mathfrak{D}}_{R(\tau)}(\tau)\,d\tau = \infty,$$

then the following sharpened dichotomy holds: at least one of the three components

$$J_1, \qquad J_6, \qquad J_7$$

fails to be in $L^1(\tau_1,\infty)$ *for every* $\tau_1 \ge \tau_0$. *More precisely, exactly one of the following two mutually exclusive alternatives holds:*

(D1) *at least one of* J_1, J_6, J_7 *fails to be in* $L^1(\tau_1,\infty)$ *for every* $\tau_1 \ge \tau_0$;
(D2) *every* J_k, $k \in \{1,\dots,7\}$, *belongs to* $L^1(\tau_0,\infty)$.

Moreover, case (D2) *is incompatible with cost divergence, so divergence of the cost forces case* (D1), *i.e. failure of one of* J_1, J_6, J_7.

Proof. The moving error density of Theorem 10.18 satisfies the identity

$$B_{\mathrm{mov}}(\tau) = \tfrac{1}{2}J_1(\tau) + \tfrac{\nu}{2}J_2(\tau) + \tfrac{1}{2}J_3(\tau) + J_4(\tau) + \tfrac{1}{2}J_5(\tau) + \tfrac{1}{2}J_6(\tau) + \tfrac{1}{2}J_7(\tau),$$

with J_k as in Theorem 8.113, because the seven nonnegative terms in B_{mov} coincide with $J_1,\dots,J_7$ up to the listed multiplicative constants, all of which are strictly positive (recall $\nu > 0$ is fixed throughout). In particular $B_{\mathrm{mov}} \in L^1(\tau_0,\infty)$ if and only if every $J_k \in L^1(\tau_0,\infty)$.

By Theorem 8.139, on the canonical windowwise schedule one has automatically

$$J_2,\ J_3,\ J_4,\ J_5 \in L^1(\tau_0,\infty).$$

Therefore the only J_k that can fail to be in L^1 on a tail are J_1, J_6, J_7. We obtain the following dichotomy: either at least one of J_1, J_6, J_7 fails to be in $L^1(\tau_1,\infty)$ for every $\tau_1 \ge \tau_0$ (case (D1)), or each of J_1, J_6, J_7 is integrable on $[\tau_0,\infty)$ and hence (combined with the automatic integrability of J_2, J_3, J_4, J_5) every J_k is integrable on $[\tau_0,\infty)$ (case (D2)).

In case (D2), Theorem 10.21 gives $B_{\mathrm{mov}} \in L^1(\tau_0,\infty)$, and the localized energy identity of Theorem 10.1(v) together with Theorem 10.19 produces

$$\frac{d}{d\tau}\mathcal{E}_R^{\mathrm{corr}}(\tau) + \tfrac{1}{2}\,\widetilde{\mathfrak{D}}_{R(\tau)}(\tau) \le 0 \qquad \text{in distributions on } (\tau_0,\infty),$$

with $c_0 = \frac{1}{2}$ in the sense of Theorem 10.6. We verify the finite initial value locally. By Theorem 8.109, after a harmless terminal restriction the initial time is a compact-cylinder local-energy time for the finite ball supporting $\Phi(\tau_0,\cdot)$. Hence

$$\mathcal{E}_R(\tau_0) = \tfrac{1}{2}\int |V(\tau_0)|^2\Phi(\tau_0,\cdot)\,dy < \infty.$$

Combined with the case-(D2) integrability $\|B_{\mathrm{mov}}\|_{L^1(\tau_0,\infty)} < \infty$,

$$\mathcal{E}_R^{\mathrm{corr}}(\tau_0) = \mathcal{E}_R(\tau_0) + \int_{\tau_0}^{\infty} B_{\mathrm{mov}}(s)\,ds < \infty.$$

By Theorem 10.39, $\mathcal{E}_R^{\mathrm{corr}} \ge 0$, so Theorem 10.40 yields

$$\int_{\tau_0}^{\infty} \widetilde{\mathfrak{D}}_{R(\tau)}(\tau)\,d\tau \le 2\,\mathcal{E}_R^{\mathrm{corr}}(\tau_0) < \infty,$$

contradicting the cost divergence. Hence case (D2) cannot occur, and case (D1) must hold; that is, at least one of J_1, J_6, J_7 fails to be in L^1 on every later tail. □

Theorem 10.42 (Cost-divergence exclusion on the canonical windowwise schedule)**.** *Fix the canonical windowwise moving family of Theorem 8.139 on the pre-carrier validated compact-cost regime of Theorem 8.100, and assume the localized energy identity for the moving cutoff (Theorem 8.102). Assume further that* (R1) *failure of* $J_6 \in L^1(\tau_1,\infty)$ *on every later tail* $[\tau_1,\infty)$ *places the branch into the negative scale-drift alternative* (vi) *of Theorem 8.133, in the sense of Theorem 8.156 (this is the same negative-drift alternative hypothesis already standing in that theorem, realized through Theorems 10.25 and 10.30); and* (R2) *the named adjacent strata of Theorem 8.105 together with the exterior/translation/positive-scale/negative-drift alternatives* (iii)–(vi) *of Theorem 8.133 are closed by their assigned local arguments in the ambient Type II* *class. Then for every renormalized local Type II branch ω in that regime satisfying Theorem 10.1, divergence of the localized cost*

$$\int_{\tau_0}^{\infty} \widetilde{\mathfrak{D}}_{R(\tau)}(\tau)\, d\tau = \infty$$

excludes ω from the local Type II class. Equivalently, the negative-drift alternative (R1) *together with the adjacent-stratum closures* (R2) *imply Theorem 10.2 as a theorem on the canonical windowwise schedule.*

Proof. Let ω be a candidate branch. By Theorem 10.41, case (D2) is impossible, so case (D1) occurs: at least one of J_1, J_6, J_7 fails to be in $L^1(\tau_1, \infty)$ for every $\tau_1 \geq \tau_0$.

Apply the alternative statement assigned to that index:

- if J_1 fails, the windowwise transition/leakage alternative of Theorem 8.149 (proved unconditionally on the regime via Theorem 8.148) places ω into the exterior/leakage alternative (iii) of Theorem 8.133, entering the named critical-tightness failure stratum of Theorem 8.105;
- if J_6 fails, hypothesis (R1) places ω into the negative scale-drift alternative (vi);
- if J_7 fails, the windowwise positive-scale alternative of Theorem 8.154 (proved unconditionally via the argument Theorem 8.151 and Theorem 5.40) places ω into the positive-scale alternative (v).

By hypothesis (R2), each of these named alternatives is closed by an already proved argument in the ambient Type II class. Therefore ω is excluded. This proves the conclusion of Theorem 10.2 as a theorem under the standing hypotheses (R1) and (R2). □

Corollary 10.43 (Cost-divergence exclusion holds under the same hypothesis set as adjacent closure)**.** *The hypotheses* (R1)–(R2) *of Theorem 10.42 on the canonical windowwise schedule coincide, after combining with the unconditional windowwise alternatives Theorems 8.149 and 8.154, with the standing hypothesis set of Theorem 8.156 (negative-drift alternative for J_6) plus the adjacent-stratum closure of Theorem 8.157. In particular, on any ambient Type II class on which the conclusion of Theorem 8.157 (compact-cost adjacent-stratum closure) is established, Theorem 10.2 is a theorem and may be removed from the hypothesis lists of Theorems 10.119 and 10.121. For the physical local Type II class considered in the final assembly, this adjacent closure is established by Theorem 8.162.*

Proof. The windowwise transition/leakage alternative Theorem 8.149 and windowwise positive-scale alternative Theorem 8.154 are unconditional on the regime. Combined with the negative-drift alternative for J_6 imported from Theorem 8.156, the alternative component (R1) of Theorem 10.42 is satisfied. The adjacent-stratum closure component (R2) is exactly the standing hypothesis closed under Theorem 8.157; in the physical local Type II class it is supplied by Theorem 8.162. Therefore Theorem 10.42 produces exclusion without any further hypothesis, and Theorem 10.2 is a theorem in that ambient class. □

Remark 10.44 (Status of the cost-divergence assumption)*.* The proof has three independent ingredients:

(i) the trivial arithmetic consequence Theorem 10.40 that corrected localized monotonicity together with finite initial corrected energy and $\mathcal{D} \geq 0$ forces $\int \mathcal{D} < \infty$;

(ii) the moving error-density identity $B_{\mathrm{mov}} = \sum_k c_k J_k$ of Theorems 8.113 and 10.18 (positive linear combination), combined with the canonical schedule integrability $J_2, J_3, J_4, J_5 \in L^1(\tau_0, \infty)$ of Theorem 8.139;

(iii) the windowwise alternative for J_1 and J_7 (Theorems 8.149 and 8.154, proved unconditionally on the regime) plus the negative-drift alternative for J_6 imported from Theorem 8.156.

Thus Theorem 10.2 no longer represents an independent analytic input. Its content is the conjunction of the trivial monotonicity-vs-divergence arithmetic and the windowwise alternative argument that is already used to close adjacent strata in Theorem 8.157.

10.7. **Absolutely Continuous Repaired-Gauge Charts and Renormalized Equations.** The following statements isolate the parts of the repaired-gauge representation that follow from an absolutely continuous concentration chart and an absolutely continuous repaired-gauge path. They do not prove extraction of the chart from arbitrary data and do not prove solvability of the gauge constraints.

Definition 10.45 (Absolutely continuous concentration chart)**.** Let u, p solve the Navier–Stokes equations on $\mathbb{R}^3 \times (t_0, T^*)$,

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \qquad \nabla \cdot u = 0.$$

An absolutely continuous concentration chart consists of absolutely continuous functions

$$x_c : (t_0, T^*) \to \mathbb{R}^3, \qquad \lambda : (t_0, T^*) \to (0, \infty),$$

and a strictly increasing absolutely continuous time ρ such that

$$\rho_t = \lambda^{-2} \quad \text{a.e.},$$

$$y = \frac{x - x_c(t)}{\lambda(t)}, \qquad u(x,t) = \lambda(t)^{-1} U(y, \rho(t)),$$

and

$$p(x,t) = \lambda(t)^{-2} Q(y, \rho(t)) + c(t)$$

for some scalar function $c(t)$. The chart is assumed to have the local integrability required for the distributional chain rule on compact sets in the y-variables. Define

$$A(\rho(t)) := \lambda(t)\lambda_t(t), \qquad B(\rho(t)) := \lambda(t) x_c'(t),$$

or equivalently

$$A = \partial_\rho \log \lambda, \qquad B = \lambda^{-1} \partial_\rho x_c,$$

where λ and x_c are regarded as functions of ρ.

Lemma 10.46 (Initial renormalized equation)**.** *Assume Definition 10.45. Then U, Q satisfy*

$$\partial_\rho U + (U \cdot \nabla)U + \nabla Q = \nu \Delta U + A(\rho)(U + y \cdot \nabla U) + B(\rho) \cdot \nabla U, \qquad \nabla \cdot U = 0$$

distributionally on compact y-sets.

Proof. For smooth functions the computation is pointwise. The distributional case is obtained by testing against compactly supported functions and using the assumed chart chain rule. The spatial derivatives are

$$\nabla_x u = \lambda^{-2} \nabla_y U, \qquad \Delta_x u = \lambda^{-3} \Delta_y U, \qquad (u \cdot \nabla_x) u = \lambda^{-3} (U \cdot \nabla_y) U.$$

Since

$$y_t = -\frac{x_c'}{\lambda} - \frac{\lambda_t}{\lambda} y, \qquad \rho_t = \lambda^{-2},$$

one has

$$\partial_t u = \lambda^{-3} \partial_\rho U - \lambda^{-2} \lambda_t (U + y \cdot \nabla U) - \lambda^{-2} x_c' \cdot \nabla U.$$

Also

$$\nabla_x p = \lambda^{-3} \nabla_y Q.$$

Substitution into the physical Navier–Stokes equation and multiplication by λ^3 gives

$$\partial_\rho U - \lambda \lambda_t (U + y \cdot \nabla U) - \lambda x_c' \cdot \nabla U + (U \cdot \nabla)U + \nabla Q = \nu \Delta U.$$

Moving the two chart-velocity terms to the right gives the displayed equation. Finally,

$$\nabla_x \cdot u = \lambda^{-2} \nabla_y \cdot U,$$

so $\nabla_y \cdot U = 0$. □

Definition 10.47 (Absolutely continuous repaired-gauge path)**.** An absolutely continuous repaired-gauge path for the initial chart consists, on every compact final time interval, of absolutely continuous functions

$$\mu(\tau) > 0, \qquad q(\tau) \in \mathbb{R}^3, \qquad \rho = \rho(\tau),$$

such that

$$\rho_\tau = \mu(\tau)^2 \quad \text{a.e.},$$
$$V(Y, \tau) = \mu(\tau) U(\mu(\tau)Y + q(\tau), \rho(\tau)),$$

and

$$P(Y, \tau) = \mu(\tau)^2 Q(\mu(\tau)Y + q(\tau), \rho(\tau)).$$

The scale and centering constraints defining the repaired gauge are assumed to hold along this path. This hypothesis assumes the path has already been selected; it does not prove solvability of the gauge constraints.

Lemma 10.48 (Physical chart after gauge repair)**.** *Assume Definitions 10.45 and 10.47. Let $t = t(\rho(\tau))$, and define*

$$\Lambda(\tau) := \lambda(t(\rho(\tau)))\mu(\tau), \qquad X(\tau) := x_c(t(\rho(\tau))) + \lambda(t(\rho(\tau)))q(\tau).$$

Then, on every compact final time interval, Λ and X are absolutely continuous, $\Lambda > 0$, and

$$\frac{d\tau}{dt} = \Lambda(\tau)^{-2}.$$

Moreover,

$$x = X(\tau) + \Lambda(\tau)Y$$

is the final physical chart and

$$u(x, t) = \Lambda(\tau)^{-1} V(Y, \tau), \qquad p(x, t) = \Lambda(\tau)^{-2} P(Y, \tau) + c(t).$$

Proof. The intermediate time ρ is strictly increasing because $\rho_t = \lambda^{-2} > 0$ a.e.; hence it has an absolutely continuous inverse on compact subintervals of its range. The compositions $\lambda(t(\rho(\tau)))$ and $x_c(t(\rho(\tau)))$ are therefore absolutely continuous on compact final windows. Products and sums with the absolutely continuous functions μ, q are absolutely continuous, so Λ and X are absolutely continuous.

The identity $\rho_\tau = \mu^2$ implies

$$\frac{d\tau}{d\rho} = \mu^{-2}.$$

Since $d\rho/dt = \lambda^{-2}$,

$$\frac{d\tau}{dt} = \frac{d\tau}{d\rho}\frac{d\rho}{dt} = \mu^{-2}\lambda^{-2} = (\lambda\mu)^{-2} = \Lambda^{-2}.$$

If $x = X + \Lambda Y$, then

$$x = x_c + \lambda q + \lambda\mu Y,$$

and hence the intermediate coordinate is

$$y = \frac{x - x_c}{\lambda} = q + \mu Y.$$

Therefore

$$u(x, t) = \lambda^{-1} U(q + \mu Y, \rho) = \lambda^{-1}\mu^{-1} V(Y, \tau) = \Lambda^{-1} V(Y, \tau),$$

and similarly

$$p(x,t) = \lambda^{-2}Q(q+\mu Y,\rho) + c(t) = \lambda^{-2}\mu^{-2}P(Y,\tau) + c(t) = \Lambda^{-2}P(Y,\tau) + c(t).$$

□

Lemma 10.49 (Modulation coefficients in repaired gauge)**.** *Define*

$$a(\tau) := \partial_\tau \log \Lambda(\tau), \qquad b(\tau) := \Lambda(\tau)^{-1}X_\tau(\tau),$$

where derivatives are understood a.e. on compact final windows. Under the hypotheses of Lemma 10.48,

$$a(\tau) = \mu(\tau)^2 A(\rho(\tau)) + \frac{\mu_\tau(\tau)}{\mu(\tau)}$$

and

$$b(\tau) = \mu(\tau)B(\rho(\tau)) + \mu(\tau)A(\rho(\tau))q(\tau) + \frac{q_\tau(\tau)}{\mu(\tau)}.$$

In particular $a, b \in L^1_{\mathrm{loc}}$ on every compact final time interval.

Proof. Since $\Lambda = \lambda(\rho(\tau))\mu(\tau)$,

$$\partial_\tau \log \Lambda = \rho_\tau \partial_\rho \log \lambda + \partial_\tau \log \mu = \mu^2 A + \frac{\mu_\tau}{\mu}.$$

This proves the formula for a. Next

$$X = x_c(\rho(\tau)) + \lambda(\rho(\tau))q(\tau).$$

Using

$$\partial_\rho x_c = \lambda B, \qquad \partial_\rho \lambda = \lambda A, \qquad \rho_\tau = \mu^2,$$

we get

$$X_\tau = \mu^2 \lambda B + \mu^2 \lambda A q + \lambda q_\tau.$$

Division by $\Lambda = \lambda\mu$ gives

$$\Lambda^{-1}X_\tau = \mu B + \mu A q + \frac{q_\tau}{\mu}.$$

Local integrability follows from absolute continuity and positivity of μ. On every compact final interval, μ is bounded above and below away from zero, q is bounded, and $\mu_\tau, q_\tau \in L^1$. The A-terms are locally integrable because

$$\int |A(\rho(\tau))|\mu(\tau)^2\, d\tau = \int |A(\rho)|\, d\rho,$$

and the extra factors μ, μ^{-1}, and q are bounded on the same compact window. Likewise, since μ is bounded away from zero,

$$\int |B(\rho(\tau))|\mu(\tau)\, d\tau \le C \int |B(\rho(\tau))|\mu(\tau)^2\, d\tau = C \int |B(\rho)|\, d\rho,$$

so the B-term in b is locally integrable. □

Lemma 10.50 (Renormalized equation in repaired gauge)**.** *The final variables V, P satisfy*

$$\partial_\tau V + (V\cdot\nabla)V + \nabla P = \nu\Delta V + a(\tau)(V + Y\cdot\nabla V) + b(\tau)\cdot\nabla V, \qquad \nabla\cdot V = 0$$

distributionally on compact Y-sets.

Proof. It suffices to compute in the smooth case and then pass to distributions by testing against compactly supported functions. Set

$$z = \mu Y + q.$$

Since $V(Y,\tau) = \mu U(z, \rho(\tau))$ and $\rho_\tau = \mu^2$, the spatial derivatives are

$$\nabla_Y V = \mu^2 \nabla_z U, \qquad \Delta_Y V = \mu^3 \Delta_z U,$$
$$(V \cdot \nabla_Y)V = \mu^3 (U \cdot \nabla_z)U, \qquad \nabla_Y P = \mu^3 \nabla_z Q.$$

Differentiation in τ gives

$$V_\tau = \mu_\tau U + \mu\mu^2 U_\rho + \mu(\mu_\tau Y + q_\tau) \cdot \nabla_z U.$$

Using the initial renormalized equation,

$$U_\rho = \nu \Delta_z U - (U \cdot \nabla_z)U - \nabla_z Q + A(U + z \cdot \nabla_z U) + B \cdot \nabla_z U,$$

we obtain

$$\begin{aligned} &V_\tau + (V \cdot \nabla_Y)V + \nabla_Y P - \nu \Delta_Y V \\ &\quad = \mu_\tau U + \mu^3 A(U + z \cdot \nabla_z U) + \mu^3 B \cdot \nabla_z U + \mu(\mu_\tau Y + q_\tau) \cdot \nabla_z U. \end{aligned}$$

Now

$$U = \mu^{-1} V, \qquad \nabla_z U = \mu^{-2} \nabla_Y V, \qquad z = \mu Y + q.$$

Therefore

$$\mu_\tau U + \mu^3 A U = \left(\frac{\mu_\tau}{\mu} + \mu^2 A \right) V,$$
$$\mu^3 A\, z \cdot \nabla_z U + \mu(\mu_\tau Y + q_\tau) \cdot \nabla_z U = \left(\mu^2 A + \frac{\mu_\tau}{\mu} \right) Y \cdot \nabla_Y V + \left(\mu A q + \frac{q_\tau}{\mu} \right) \cdot \nabla_Y V,$$
$$\mu^3 B \cdot \nabla_z U = \mu B \cdot \nabla_Y V.$$

Combining these identities, the coefficient multiplying

$$V + Y \cdot \nabla_Y V$$

is

$$\frac{\mu_\tau}{\mu} + \mu^2 A,$$

and the coefficient multiplying $\nabla_Y V$ is

$$\mu B + \mu A q + \frac{q_\tau}{\mu}.$$

By Lemma 10.49 these coefficients are a and b. Hence the displayed repaired-gauge equation holds in the distributional sense on compact Y-sets. Incompressibility follows from

$$\nabla_Y \cdot V = \mu^2 \nabla_z \cdot U = 0.$$

□

Lemma 10.51 (Pressure reconstruction under the charts)**.** *Assume the physical pressure satisfies*

$$-\Delta_x p = \partial_i \partial_j (u_i u_j)$$

in distributions. Then the intermediate pressure satisfies

$$-\Delta_y Q = \partial_i \partial_j (U_i U_j)$$

modulo functions of ρ, and the final pressure satisfies

$$-\Delta_Y P = \partial_i \partial_j (V_i V_j)$$

modulo functions of τ.

Proof. The additive function $c(t)$ in
$$p = \lambda^{-2} Q + c(t)$$
has zero spatial derivatives. Since
$$\Delta_x p = \lambda^{-4} \Delta_y Q, \qquad \partial_{x_i}\partial_{x_j}(u_i u_j) = \lambda^{-4}\partial_{y_i}\partial_{y_j}(U_i U_j),$$
the intermediate pressure equation follows. For the repaired variables,
$$P(Y) = \mu^2 Q(\mu Y + q), \qquad V(Y) = \mu U(\mu Y + q).$$
Therefore
$$-\Delta_Y P = \mu^4(-\Delta_z Q)(z) = \mu^4 \partial_{z_i}\partial_{z_j}(U_i U_j)(z) = \partial_{Y_i}\partial_{Y_j}(V_i V_j)(Y).$$
Adding functions of time to Q or P does not change the identity. □

Lemma 10.52 (Scaling of local L^r norms)**.** *For every $1 \le r < \infty$ for which the norms are finite,*
$$\|V(\tau)\|^r_{L^r_Y} = \Lambda(\tau)^{r-3} \|u(t(\tau))\|^r_{L^r_x}.$$
In particular,
$$\|V(\tau)\|_{L^3_Y} = \|u(t(\tau))\|_{L^3_x}.$$
Thus membership in $L^3(\mathbb{R}^3)$, finiteness, infiniteness, and positivity are invariant under the final chart.

Proof. Using
$$u(X + \Lambda Y, t) = \Lambda^{-1} V(Y, \tau), \qquad dx = \Lambda^3\, dY,$$
we obtain
$$\|u(t)\|^r_{L^r_x} = \int |\Lambda^{-1} V(Y,\tau)|^r \Lambda^3\, dY = \Lambda^{3-r} \|V(\tau)\|^r_{L^r_Y}.$$
Rearranging gives the stated formula. The case $r = 3$ is scale invariant. □

Lemma 10.53 (Regularity of scale, center, and modulation)**.** *If λ, x_c, μ, q are absolutely continuous and $\mu > 0$, then on compact final time intervals the final scale Λ, center X, and coefficients a, b are absolutely continuous or locally integrable as specified above: $\Lambda, X \in W^{1,1}_{\mathrm{loc}}$ and*
$$a, b \in L^1_{\mathrm{loc}}.$$

Proof. This follows from Lemmas 10.48 and 10.49. Positivity of μ and compactness of the time window give a positive lower bound for μ, so the divisions by μ in the formulas for a, b are legitimate. □

Theorem 10.54 (Consequences of absolutely continuous repaired-gauge charts)**.** *Assume an absolutely continuous concentration chart and an absolutely continuous repaired-gauge path in the sense of Definitions 10.45 and 10.47. Then the following analytic conclusions hold:*

(i) *the initial and repaired renormalized Navier–Stokes equations hold distributionally on compact renormalized sets;*
(ii) *the final physical chart satisfies*
$$u(x,t) = \Lambda(\tau)^{-1} V(Y,\tau), \qquad p(x,t) = \Lambda(\tau)^{-2} P(Y,\tau) + c(t), \qquad \frac{d\tau}{dt} = \Lambda^{-2};$$
(iii) *pressure reconstruction holds modulo functions of time;*
(iv) *the final modulation coefficients are the functions a, b in Lemma 10.49;*
(v) *the critical L^3 norm is invariant under the final chart;*
(vi) *the final scale, center, and modulation coefficients have the local absolute-continuity and integrability stated in Lemma 10.53.*

Proof. Lemma 10.46 gives the initial renormalized equation. Lemmas 10.48 and 10.50 give the final chart and the final repaired-gauge equation with the stated coefficients. Lemma 10.51 gives pressure reconstruction modulo functions of time. Lemma 10.52 gives critical norm invariance, and Lemma 10.53 gives the regularity inheritance. ☐

Corollary 10.55 (Remaining representation hypotheses)**.** *Once the absolutely continuous concentration chart and absolutely continuous repaired-gauge path are available, pressure reconstruction, modulation coefficient realization, local $W^{1,1}$ regularity of the chart, and critical L^3-invariance are analytic consequences rather than separate hypotheses. The remaining representation hypotheses are:*

(i) *extraction of an absolutely continuous concentration chart from the selected Type II branch;*
(ii) *solvability of the repaired scale and centering equations on the admissible profile class;*
(iii) *selection of the repaired scale and center as an absolutely continuous path satisfying $\rho_\tau = \mu^2$;*
(iv) *terminal admissibility of the final scale and time, including $\tau \to \infty$ and preservation of the Type II core scale.*

Proof. This is Theorem 10.54. The enumerated items are the complementary analytic hypotheses that must be available before that theorem can be applied: one must produce the initial concentration chart, solve the repaired gauge equations, select the resulting scale and center absolutely continuously, and verify that the resulting final chart is terminally admissible. ☐

10.8. Nontrivial Critical Core and Vanishing Critical L^3 Norm.

Definition 10.56 (Tail nontrivial critical core)**.** A repaired-gauge Type II branch (V, P, a, b) has a tail nontrivial critical core if there exist $R_* < \infty$, $\eta_* > 0$, and $\tau_* \geq \tau_0$ such that

$$\|V(\tau)\|_{L^3(B_{R_*})} \geq \eta_* \qquad \text{for all } \tau \geq \tau_*.$$

Equivalently, the selected terminal core remains nonzero in the critical topology on the renormalized tail.

Definition 10.57 (Subsequential nontrivial critical core)**.** A branch has subsequential nontriviality if there exist $R_* < \infty$, $\eta_* > 0$, and a sequence $\tau_n \to \infty$ such that

$$\|V(\tau_n)\|_{L^3(B_{R_*})} \geq \eta_* \qquad \text{for all } n.$$

Definition 10.58 (Core persistence)**.** Core persistence is the analytic implication that subsequential nontriviality of a selected terminal Type II core implies tail nontriviality:

$$\text{subsequential nontriviality} \Longrightarrow \text{tail nontriviality}.$$

Failure of this implication is a failure of persistence of the selected core, not a new Type II alternative.

Definition 10.59 (Vanishing critical L^3 alternative)**.** The vanishing critical L^3 alternative for a repaired-gauge branch is

$$\sup_{\tau \geq \tau_0} \|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty, \qquad \liminf_{\tau \to \infty} \|V(\tau)\|_{L^3(\mathbb{R}^3)} = 0.$$

Theorem 10.60 (Concentration extraction gives subsequential nontriviality)**.** *Assume a Type II branch has been obtained from a concentration extraction and placed in a repaired-gauge representation. Assume also that the extracted branch is selected as a terminal Type II core, rather than accounted for by scattering, exterior regularity, or loss of the extracted profile in the limiting procedure. Then the branch satisfies the subsequential nontriviality condition of Definition 10.57.*

Proof. The concentration extraction supplies a nonzero local occupation state on a compact terminal cylinder. The repaired-gauge representation transports that state to the renormalized trajectory $V(\tau)$ by Navier–Stokes critical symmetries and admissible chart maps. These symmetries preserve the local scale-critical L^3 quantity on corresponding compact cylinders.

Because the extracted state is nonzero in L^3_{loc} after choosing the terminal coordinate frame, there are $R_* < \infty$, $\eta_* > 0$, and the corresponding extraction times $\tau_n \to \infty$ such that

$$\|V(\tau_n)\|_{L^3(B_{R_*})} \geq \eta_*.$$

This is subsequential nontriviality. □

Remark 10.61 (Persistence is an additional analytic hypothesis)*.* Theorem 10.60 does not by itself rule out

$$\liminf_{\tau\to\infty} \|V(\tau)\|_{L^3(\mathbb{R}^3)} = 0.$$

That stronger conclusion requires the persistence hypothesis of Definition 10.58; a subsequence extraction theorem is not a uniform-in-time lower bound.

Theorem 10.62 (Vanishing critical L^3 norm excluded by persistent nontriviality)**.** *For a repaired-gauge Type II branch, core persistence together with subsequential nontriviality excludes the vanishing critical L^3 alternative of Definition 10.59. Equivalently, if the vanishing critical L^3 alternative occurs, then either subsequential nontriviality fails or core persistence fails.*

Proof. By core persistence and subsequential nontriviality, the branch has a tail nontrivial critical core. Hence there are $R_* < \infty$, $\eta_* > 0$, and τ_* such that

$$\|V(\tau)\|_{L^3(B_{R_*})} \geq \eta_* \qquad \text{for all } \tau \geq \tau_*.$$

Since $B_{R_*} \subset \mathbb{R}^3$, this gives the global lower bound

$$\|V(\tau)\|_{L^3(\mathbb{R}^3)} \geq \eta_* \qquad \text{for all } \tau \geq \tau_*.$$

Hence

$$\liminf_{\tau\to\infty} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \geq \eta_* > 0.$$

This contradicts the vanishing critical L^3 alternative, which requires the same liminf to be zero. Therefore a selected nontrivial core satisfying persistence cannot have vanishing critical L^3 norm. □

Corollary 10.63 (Critical L^3 alternatives after exclusion of vanishing norm)**.** *On selected Type II cores satisfying core persistence and subsequential nontriviality, the vanishing critical L^3 alternative is excluded from the ordered critical L^3 alternatives. The remaining possibilities are failure of membership in $L^3(\mathbb{R}^3)$ along the terminal tail, divergence of the critical L^3 norm, or a positive finite critical L^3 annulus.*

Proof. Apply Theorem 10.62 to exclude the vanishing critical L^3 alternative of Definition 10.59. The other alternatives are the remaining critical L^3 outcomes. □

Corollary 10.64 (Terminal boundedness after exclusion of vanishing norm)**.** *In the terminal boundedness alternatives of Theorem 10.111, if core persistence and subsequential nontriviality hold, then vanishing critical L^3 norm is not a terminal-sequence obstruction. The remaining obstructions are failure of sampling from the repaired-gauge trajectory, failure of membership in $L^3(\mathbb{R}^3)$ along the terminal tail, and divergence of the critical L^3 norm.*

Proof. Theorem 10.111 imports the critical L^3 alternatives. Corollary 10.63 excludes the vanishing critical L^3 alternative on selected nontrivial cores. The listed obstructions are the remaining terminal alternatives. □

Remark 10.65 (Remaining critical L^3 boundary)*.* The preceding results prove only the lower-bound half of critical normalization, modulo subsequential nontriviality and persistence. The remaining analytic hypotheses are persistence of the selected core, membership in $L^3(\mathbb{R}^3)$ along the terminal tail, classification of the infinite critical L^3-norm alternative, and an upper finite critical L^3 bound on the selected repaired-gauge branch. Once these are supplied, the positive finite critical annulus follows.

10.9. **Local Caccioppoli Estimate for Absolutely Continuous Suitable Branches.**

Definition 10.66 (Suitable repaired-gauge Type II branch)**.** A suitable repaired-gauge Type II branch is a repaired-gauge Type II branch for which:

(i) an absolutely continuous concentration chart and an absolutely continuous repaired-gauge path are available;
(ii) the conclusions of Theorem 10.54 hold, including pressure reconstruction and final modulation coefficients;
(iii) the physical branch (u, p) is a local suitable weak solution on the terminal interval:
$$u \in L_t^\infty L^2_{\mathrm{loc}} \cap L_t^2 H^1_{\mathrm{loc}}, \qquad p \in L^{3/2}_{\mathrm{loc}}, \qquad \nabla \cdot u = 0,$$
the Navier–Stokes equations hold distributionally, and the physical local energy inequality holds for every nonnegative smooth compactly supported test function.

Lemma 10.67 (Local energy inequality under absolutely continuous Type II charts)**.** *Let (u, p) be physically suitable on a compact physical cylinder, and let (V, P) be obtained from an absolutely continuous final chart*
$$u(x,t) = \Lambda(s)^{-1} V(Y,s), \qquad p(x,t) = \Lambda(s)^{-2} P(Y,s) + c(t), \qquad Y = \frac{x - X(s)}{\Lambda(s)},$$
with $s = \tau$, $dt = \Lambda(s)^2\,ds$, $\Lambda > 0$, and
$$X, \Lambda \in W^{1,1}_{\mathrm{loc}}.$$
Then (V, P) satisfies the renormalized local energy inequality on compact (Y, s)-cylinders:
$$\begin{aligned}
&\int_{\mathbb{R}^3} \frac{|V(s_2)|^2}{2}\phi(s_2)\,dY + \nu \int_{s_1}^{s_2}\int_{\mathbb{R}^3} |\nabla V|^2 \phi\,dY\,ds \\
&\le \int_{\mathbb{R}^3} \frac{|V(s_1)|^2}{2}\phi(s_1)\,dY + \int_{s_1}^{s_2}\int_{\mathbb{R}^3} \frac{|V|^2}{2}(\partial_s\phi + \nu\Delta\phi)\,dY\,ds \\
&\quad + \int_{s_1}^{s_2}\int_{\mathbb{R}^3} \left(\frac{|V|^2}{2} + P\right) V \cdot \nabla\phi\,dY\,ds \\
&\quad + \int_{s_1}^{s_2}\int_{\mathbb{R}^3} a(s)\frac{|V|^2}{2}(-\phi - Y\cdot\nabla\phi)\,dY\,ds \\
&\quad - \int_{s_1}^{s_2}\int_{\mathbb{R}^3} \frac{|V|^2}{2} b(s)\cdot\nabla\phi\,dY\,ds,
\end{aligned}$$
for every nonnegative $\phi \in C^\infty([s_1, s_2] \times \mathbb{R}^3)$ that is compactly supported in the spatial variable. Here a, b are the final modulation coefficients.

Proof. First suppose X, Λ are C^1. Use the physical test function
$$\psi(x,t) = \Lambda(s)^{-1}\phi\left(\frac{x - X(s)}{\Lambda(s)}, s\right), \qquad t = t(s),$$

where $dt = \Lambda^2 ds$. The spatial support of ϕ is compact, and Λ is bounded above and below on the selected compact time interval, so ψ is an admissible compactly supported test function in the physical local energy inequality.

Let
$$Y = \frac{x - X(s)}{\Lambda(s)}.$$
At fixed x,
$$\frac{dY}{ds} = -\frac{X_s}{\Lambda} - \frac{\Lambda_s}{\Lambda}Y.$$
Consequently,
$$\nabla_x \psi = \Lambda^{-2}\nabla_Y \phi, \qquad \Delta_x \psi = \Lambda^{-3}\Delta_Y \phi,$$
and, since $ds/dt = \Lambda^{-2}$,
$$\partial_t \psi = \Lambda^{-3}\left[\partial_s \phi - \frac{\Lambda_s}{\Lambda}(\phi + Y\cdot\nabla\phi) - \frac{X_s}{\Lambda}\cdot\nabla\phi\right].$$
Changing variables
$$x = X(s) + \Lambda(s)Y, \qquad dt = \Lambda^2 ds,$$
and using
$$u = \Lambda^{-1}V, \qquad p = \Lambda^{-2}P + c(t),$$
each term in the physical local energy inequality has the following renormalized form. The endpoint and dissipation terms transform as
$$\int \frac{|u(t_i)|^2}{2}\psi(t_i)\,dx = \int \frac{|V(s_i)|^2}{2}\phi(s_i)\,dY,$$
and
$$\nu\int_{t_1}^{t_2}\int |\nabla_x u|^2\psi\,dx\,dt = \nu\int_{s_1}^{s_2}\int |\nabla_Y V|^2\phi\,dY\,ds.$$
The test-function derivative term becomes
$$\begin{aligned}&\int_{t_1}^{t_2}\int \frac{|u|^2}{2}(\partial_t\psi + \nu\Delta_x\psi)\,dx\,dt\\ &\quad= \int_{s_1}^{s_2}\int \frac{|V|^2}{2}\left[\partial_s\phi + \nu\Delta_Y\phi - a(\phi + Y\cdot\nabla\phi) - b\cdot\nabla\phi\right]dY\,ds,\end{aligned}$$
where
$$a = \frac{\Lambda_s}{\Lambda}, \qquad b = \frac{X_s}{\Lambda}.$$
The transport and pressure term becomes
$$\begin{aligned}&\int_{t_1}^{t_2}\int \left(\frac{|u|^2}{2} + p\right)u\cdot\nabla_x\psi\,dx\,dt\\ &\quad= \int_{s_1}^{s_2}\int_{\mathbb{R}^3}\left(\frac{|V|^2}{2} + P\right)V\cdot\nabla_Y\phi\,dY\,ds + \int_{s_1}^{s_2}\Lambda(s)^2 c(t(s))\left(\int_{\mathbb{R}^3} V\cdot\nabla_Y\phi\,dY\right)ds.\end{aligned}$$
The pressure constant drops out because
$$\int_{\mathbb{R}^3} V\cdot\nabla_Y\phi\,dY = -\int_{\mathbb{R}^3}\phi\,\nabla_Y\cdot V\,dY = 0.$$
Combining the transformed terms gives the displayed renormalized local energy inequality.

For $X, \Lambda \in W^{1,1}_{\mathrm{loc}}$, the same identity follows at the level of test functions. On the selected compact time interval, Λ has positive lower and finite upper bounds. The pulled-back test

$$\psi(x,t) = \Lambda(s)^{-1}\phi((x - X(s))/\Lambda(s), s)$$

is compactly supported, nonnegative, bounded, has bounded spatial derivatives on the compact cylinder, and has time derivative in $L^1_t L^\infty_x$ because $X', \Lambda' \in L^1$ and ϕ is smooth compactly supported. The physical local energy inequality, initially stated for C^∞_c tests, extends to this class by standard mollification of the test function: the suitability classes give the local integrability needed for every term in the inequality, and the mollified tests keep support inside a slightly larger compact cylinder. Applying the extended inequality to ψ and changing variables gives the displayed renormalized inequality. The only derivatives of the chart that appear are the L^1-functions X', Λ', hence the modulation terms are locally integrable against local $|V|^2$. □

Lemma 10.68 (Pressure and modulation for suitable repaired-gauge branches)**.** *For a suitable repaired-gauge branch, the pressure identity*

$$-\Delta P = \partial_i \partial_j (V_i V_j)$$

holds modulo functions of τ, and the final coefficients a, b in the renormalized equation are measurable and locally integrable on compact renormalized windows.

Proof. This is the pressure and modulation part of Theorem 10.54. Pressure pullback comes from the physical pressure equation and the chart/gauge scaling. The final coefficients are forced by the absolutely continuous scale-translation gauge transform:

$$a = \mu^2 A + \frac{\mu_\tau}{\mu}, \qquad b = \mu B + \mu A q + \frac{q_\tau}{\mu}.$$

Since the chart and gauge parameters are absolutely continuous on compact windows, these coefficients are locally integrable. □

Theorem 10.69 (Caccioppoli estimate for suitable absolutely continuous repaired-gauge branches)**.** *Every suitable repaired-gauge Type II branch satisfies the local renormalized Caccioppoli estimate on each compact repaired-gauge cylinder. The constants may depend on the cylinder, the cutoff, and local upper and lower bounds for the chart. No local Caffarelli–Kohn–Nirenberg smallness, global windowed H^1, critical L^3-norm, tightness, bounded modulation, finite energy at infinity, or whole-space compactness conclusion is asserted.*

Proof. By the suitable repaired-gauge hypotheses and Theorem 10.54, the branch has an absolutely continuous final chart, an admissible absolutely continuous repaired-gauge path, pressure reconstruction, and final modulation coefficients. Lemma 10.67 pulls the physical local energy inequality to the repaired variables. Lemma 10.68 supplies the pressure and coefficient identities needed to interpret the renormalized inequality in the same PDE class used by the local windowed estimates.

Choose a standard compactly supported Caccioppoli cutoff

$$\phi(Y,\tau) = \eta(\tau)\zeta(Y)^2.$$

Let $I = (\tau_1, \tau_2)$, let $I' \Subset I$, and let $0 < r < R$. Take $0 \le \eta \le 1$, $\eta \in C^\infty_c(I)$, with $\eta \equiv 1$ on I', and take $0 \le \zeta \le 1$, $\zeta \in C^\infty_c(B_R)$, with $\zeta \equiv 1$ on B_r. Applying Lemma 10.67 on (τ_1, σ) and then taking the essential supremum over $\sigma \in I'$ controls the endpoint term. Applying the same inequality on the full interval I controls the dissipation term. Adding the two estimates gives

$$
\begin{aligned}
&\operatorname*{ess\,sup}_{\tau\in I'} \int_{B_r} \frac{|V(\tau)|^2}{2}\, dY + \nu \int_{I'}\int_{B_r} |\nabla V|^2\, dY\, d\tau \\
&\le C \int_I \int_{B_R} \frac{|V|^2}{2} \left(|\eta'|\zeta^2 + \nu\eta|\Delta(\zeta^2)|\right) dY\, d\tau \\
&\quad + C \int_I \int_{B_R} \left(\frac{|V|^2}{2} + |P|\right) |V|\, \eta |\nabla(\zeta^2)|\, dY\, d\tau \\
&\quad + C \int_I |a(\tau)| \int_{B_R} \frac{|V|^2}{2} \eta \left(\zeta^2 + |Y|\, |\nabla(\zeta^2)|\right) dY\, d\tau \\
&\quad + C \int_I |b(\tau)| \int_{B_R} \frac{|V|^2}{2} \eta |\nabla(\zeta^2)|\, dY\, d\tau.
\end{aligned}
$$

Here C is an absolute constant. Since the cutoff derivatives are bounded by constants depending only on I, I', r, R, this is the local Caccioppoli estimate on the compact renormalized cylinder.

No global estimate is used in this step. On the compact cylinder under consideration, the absolutely continuous chart has

$$0 < \Lambda_- \le \Lambda(\tau) \le \Lambda_+ < \infty.$$

Pulling back physical suitability therefore gives, for every compact spatial set K,

$$V \in L^\infty_\tau L^2_Y(K) \cap L^2_\tau H^1_Y(K), \qquad P \in L^{3/2}_{\tau,Y}(K).$$

Local Sobolev interpolation on the finite cylinder gives

$$V \in L^{10/3}_{\tau,Y}(K),$$

hence $V \in L^3_{\tau,Y}(K)$. Thus $PV \cdot \nabla\phi$ is integrable by Hölder's inequality. The representation identities give

$$a, b \in L^1_{\mathrm{loc}}(d\tau),$$

and

$$\tau \mapsto \int_K |V(Y,\tau)|^2\, dY$$

is locally essentially bounded; therefore the scale and translation modulation terms in the pulled-back local energy inequality are integrable on the same compact cylinder. This proves the local Caccioppoli estimate, but not a global windowed H^1 or critical L^3-norm bound. □

Corollary 10.70 (No separate Caccioppoli obstruction for suitable absolutely continuous branches)**.** *On suitable repaired-gauge branches, failure of local Caccioppoli regularity, failure of pressure reconstruction, and failure of local gauge regularity are not separate obstructions to the local H^1 analysis. The remaining hypotheses for the uniform windowed H^1 estimates are the separate bounded critical L^3 norm and modulation bounds required by those estimates.*

Proof. Theorem 10.69 gives local Caccioppoli regularity. Theorem 10.54 gives pressure reconstruction and the final modulation coefficients from the final chart and absolutely continuous gauge path. Thus the listed failures cannot occur separately on suitable repaired-gauge branches. The uniform windowed H^1 estimate may still require bounded critical norm and bounded modulation beyond the local Caccioppoli statement. □

10.10. **Caccioppoli Regularity Criterion.**

Definition 10.71 (Suitable physical hypotheses)**.** Let u, p be a local suitable weak solution of the three-dimensional Navier–Stokes equations on $\mathbb{R}^3 \times (0, T^*)$. Thus

$$u \in L_t^\infty L^2_{\mathrm{loc}} \cap L_t^2 H^1_{\mathrm{loc}}, \qquad p \in L^{3/2}_{\mathrm{loc}}, \qquad \nabla \cdot u = 0,$$

the equations hold distributionally, and for every nonnegative $\psi \in C_c^\infty(\mathbb{R}^3 \times (0, T^*))$ and a.e. $0 < t_1 < t_2 < T^*$,

$$\begin{aligned} &\int_{\mathbb{R}^3} \frac{|u(t_2)|^2}{2} \psi(t_2)\, dx + \nu \int_{t_1}^{t_2} \int_{\mathbb{R}^3} |\nabla u|^2 \psi \, dx\, dt \\ &\le \int_{\mathbb{R}^3} \frac{|u(t_1)|^2}{2} \psi(t_1)\, dx + \int_{t_1}^{t_2} \int_{\mathbb{R}^3} \frac{|u|^2}{2} (\partial_t \psi + \nu \Delta \psi)\, dx\, dt \\ &\quad + \int_{t_1}^{t_2} \int_{\mathbb{R}^3} \left(\frac{|u|^2}{2} + p \right) u \cdot \nabla \psi \, dx\, dt. \end{aligned}$$

Lemma 10.72 (Renormalized local energy inequality)**.** *Assume the suitable physical hypotheses of Theorem 10.71. Let*

$$u(x,t) = \lambda(t)^{-1} V\left(\frac{x - x_c(t)}{\lambda(t)}, \tau(t) \right), \qquad \frac{d\tau}{dt} = \lambda(t)^{-2},$$

with

$$P(y, \tau) = \lambda(t)^2 p(x_c(t) + \lambda(t) y, t),$$

where $\lambda > 0$, x_c, and τ are C^1 on compact windows. Assume that V, P satisfy the repaired-gauge equation

$$\partial_\tau V + (V \cdot \nabla) V + \nabla P = \nu \Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0.$$

Then, for every nonnegative $\phi \in C_c^\infty(\mathbb{R}^3 \times (\tau_0, \infty))$ and a.e. $\tau_1 < \tau_2$ in the corresponding renormalized time interval,

$$\begin{aligned} &\int_{\mathbb{R}^3} \frac{|V(\tau_2)|^2}{2} \phi(\tau_2)\, dy + \nu \int_{\tau_1}^{\tau_2} \int_{\mathbb{R}^3} |\nabla V|^2 \phi \, dy\, d\tau \\ &\le \int_{\mathbb{R}^3} \frac{|V(\tau_1)|^2}{2} \phi(\tau_1)\, dy + \int_{\tau_1}^{\tau_2} \int_{\mathbb{R}^3} \frac{|V|^2}{2} (\partial_\tau \phi + \nu \Delta \phi)\, dy\, d\tau \\ &\quad + \int_{\tau_1}^{\tau_2} \int_{\mathbb{R}^3} \left(\frac{|V|^2}{2} + P \right) V \cdot \nabla \phi \, dy\, d\tau \\ &\quad + \int_{\tau_1}^{\tau_2} \int_{\mathbb{R}^3} a(\tau) \frac{|V|^2}{2} \big(-\phi - y \cdot \nabla \phi \big)\, dy\, d\tau \\ &\quad - \int_{\tau_1}^{\tau_2} \int_{\mathbb{R}^3} \frac{|V|^2}{2} b(\tau) \cdot \nabla \phi \, dy\, d\tau. \end{aligned}$$

Proof. For smooth solutions, multiply the repaired-gauge equation by $V\phi$ and integrate by parts. Put $e = |V|^2/2$. Since $\nabla \cdot V = 0$,

$$V \cdot (V \cdot \nabla V) = V \cdot \nabla e = \nabla \cdot (eV), \qquad V \cdot \nabla P = \nabla \cdot (PV),$$

and

$$V \cdot (y \cdot \nabla V) = y \cdot \nabla e, \qquad V \cdot (b \cdot \nabla V) = b \cdot \nabla e.$$

The scale contribution is

$$\int a(2e + y \cdot \nabla e)\phi \, dy = \int ae(-\phi - y \cdot \nabla \phi)\, dy,$$

because

$$\int y \cdot \nabla e \, \phi \, dy = -\int e \, \nabla \cdot (y\phi) \, dy = -\int e(3\phi + y \cdot \nabla\phi) \, dy.$$

The translation contribution is

$$\int b \cdot \nabla e \, \phi \, dy = -\int e \, b \cdot \nabla\phi \, dy,$$

and the viscous term gives

$$\int \nu \Delta V \cdot V \phi \, dy = -\nu \int |\nabla V|^2 \phi \, dy + \nu \int e \Delta\phi \, dy.$$

Combining these identities gives the displayed equality in the smooth case.

For suitable weak solutions, apply the physical local energy inequality with

$$\psi(x,t) = \phi\left(\frac{x - x_c(t)}{\lambda(t)}, \tau(t)\right).$$

Use the change of variables

$$x = x_c(t) + \lambda(t)y, \qquad dt = \lambda(t)^2 \, d\tau.$$

The C^1 regularity of λ and x_c justifies the chain rule for the test function, and the Jacobian factors are those of the Navier–Stokes critical scaling. Passing to the variables y, τ gives the same formula with equality replaced by $\leq$. □

Lemma 10.73 (Caccioppoli estimate from the renormalized local energy inequality)**.** *Assume Theorem 10.72 on a compact renormalized cylinder $B_{2R} \times I$, bounded modulation on I, and local bounds*

$$V \in L^\infty(I; L^3(B_{2R})), \qquad P \in L^{3/2}(I \times B_{2R}).$$

Then the compact-cylinder renormalized Caccioppoli estimate used in Theorem 8.15 is justified.

Proof. Choose a standard nonnegative cutoff

$$\phi(y,\tau) = \eta(\tau)\zeta(y)^2,$$

where $\zeta \equiv 1$ on the inner ball and ζ is supported in the outer ball. Insert this test function in Theorem 10.72. The left-hand side contains

$$\nu \int |\nabla V|^2 \eta \zeta^2.$$

Every term on the right-hand side is finite under the stated hypotheses. The term

$$\int |V|^2 |\partial_\tau \phi + \Delta\phi|$$

is controlled by $V \in L^\infty(I; L^3(B_{2R}))$ and finite measure. The convective term

$$\int |V|^3 |\nabla\phi|$$

is controlled by the same local critical bound. For the pressure term, choose a time-dependent constant $c(\tau)$. Since

$$\int c(\tau) V \cdot \nabla\phi \, dy = -\int c(\tau)\phi \, \nabla \cdot V \, dy = 0,$$

Hölder's inequality controls

$$\int |P - c(\tau)| \, |V| \, |\nabla\phi|$$

by $P - c(\tau) \in L^{3/2}(I \times B_{2R})$ and $V \in L^\infty(I; L^3(B_{2R}))$. The modulation terms are bounded by

$$\|a\|_{L^\infty} + \|b\|_{L^\infty}$$

times the local L^2-mass, and the local L^2-mass is controlled on bounded balls by local L^3-mass. These estimates give the renormalized Caccioppoli estimate on compact cylinders. ☐

Proposition 10.74 (Caccioppoli regularity criterion)**.** *Assume the suitable physical hypotheses, the repaired-gauge representation, pressure reconstruction modulo functions of time, and C^1 scale and center functions on compact renormalized windows. Then the compact-cylinder Caccioppoli regularity hypothesis used in the local windowed H^1 estimate holds.*

Proof. The repaired-gauge representation writes the suitable weak solution in the variables V, P, a, b. Theorem 10.72 gives the renormalized local energy inequality, and Theorem 10.73 shows that this inequality justifies the renormalized Caccioppoli estimate on every compact cylinder. ☐

Corollary 10.75 (Caccioppoli estimates give windowed control under boundedness hypotheses)**.** *Assume the suitable physical hypotheses, the repaired-gauge representation, pressure reconstruction, a bounded critical L^3 norm on compact cylinders, and bounded modulation. Then local windowed H^1 control follows from Theorem 8.15. Thus failure of the Caccioppoli regularity hypothesis is not an independent rough-core alternative under these hypotheses.*

Proof. Theorem 10.74 supplies the Caccioppoli regularity hypothesis. The bounded critical norm, pressure reconstruction, and bounded modulation are the remaining hypotheses used in Theorem 8.15. Therefore the local windowed H^1 estimate holds. ☐

10.11. **Terminal Local State-Space Compactness.**

Definition 10.76 (Admissible retained terminal decomposition and completeness conclusion)**.** Let $Q_{2R,I} := B_{2R} \times I$ be a compact terminal-frame cylinder and let

$$V_n = U_n + S_n$$

be the local decomposition into the retained terminal component U_n and the nonretained residual S_n. The old critical-profile label is replaced by the following local notions.

(i) The decomposition is an admissible retained terminal decomposition if, on every $Q_{2R,I}$, the represented fields satisfy the repaired-gauge equation, local energy inequality, pressure reconstruction modulo functions of time, bounded modulation, and the compact-cylinder estimates

$$\sup_n \Big(\|V_n\|_{L^\infty(I;L^3(B_{2R}))} + \|V_n\|_{L^\infty(I;L^2(B_{2R}))} + \|V_n\|_{L^2(I;H^1(B_{2R}))} + \|P_n - c_{n,2R}\|_{L^{3/2}(Q_{2R,I})} \Big) < \infty,$$

and the same local velocity bounds hold for U_n. The time derivatives are bounded in $L^1(I; H^{-m}(B_{2R}))$ for some m large enough for the Aubin–Lions compactness argument.

(ii) A positive residual local label is assigned as follows. If, after passing to a subsequence,

$$\limsup_{n\to\infty} \|S_n\|_{L^\infty(I;L^3(B_R))} > 0,$$

then a selected subwindow and subcylinder carry an augmented local occupation state for the tuple $(V_n, P_n, U_n, S_n, a_n, b_n)$, together with the local residual concentration measures $|S_n|^3\, dy$ on selected time slices, whose residual coordinate or residual concentration measure is nonzero. The first closed label of this augmented state is read in the ordered local validation list of *Theorem* 8.86, with the compact local Type II row refined by *Theorem* 11.13.

(iii) The local terminal completeness conclusion is the statement that, if all named adjacent alternatives are closed by their local arguments and comparable retained classes have been grouped before selecting the terminal frame, then no nonretained residual with a positive residual local label remains on the retained terminal branch. Equivalently,

$$S_n \to 0 \quad \text{in } L^\infty(I; L^3(B_R))$$

on every compact terminal cylinder of the retained branch.

No global L^3 profile decomposition, global heat-flow norm, or global Kato estimate is part of this statement.

Local proof of terminal state-space completeness.

Lemma 10.77 (Interior-time selection on a compact terminal window)**.** *Let $I = [\tau_-, \tau_+]$ be a compact time-interval in the terminal-frame coordinates, and let $\tau_n \in I$ be a sequence of selected times for which $\|S_n(\tau_n)\|_{L^3(B_R)} \geq \delta > 0$. Then, after passing to a subsequence, exactly one of the following two admissible selections holds:*

(i) *there is a compact subinterval $J \Subset I$ such that $\tau_n \in J$ for all large n;*
(ii) *the selected times approach an endpoint of I, and the terminal window is replaced by the admissible recentered windows*

$$I_n^* := \tau_n + J_0, \qquad J_0 = [-\sigma, \sigma],$$

with fixed $\sigma > 0$, written in coordinates centered at τ_n. In these recentered coordinates the selected time is $0 \in J_0^\circ$ and

$$\|S_n(0)\|_{L^3(B_R)} \geq \delta.$$

No lower bound is transferred from one time to another.

Proof. After passing to a subsequence, $\tau_n \to \tau^* \in I$. If $\tau^* \in (\tau_-, \tau_+)$, choose $J \Subset I$ with $\tau^* \in J^\circ$. Then $\tau_n \in J$ for all large n, giving (i).

If $\tau^* \in \partial I$, we do not move the lower bound to a nearby time. Instead we use the admissibility of the terminal validation family under finite covariant time recentering: selected compact windows are local objects, and composing the terminal-frame chart with the finite translation $\tau \mapsto \tau - \tau_n$ produces another member of the same local validation family on every fixed model interval $J_0 = [-\sigma, \sigma]$. The compact-cylinder estimates are then the estimates of Theorem 10.76(i) on these recentered selected windows, with constants depending on the model cylinder and not on n. In the recentered coordinates the original selected time τ_n is exactly 0, so the lower bound remains $\|S_n(0)\|_{L^3(B_R)} \geq \delta$. This proves (ii) and uses no continuity in time of the L^3 norm. $\square$

Lemma 10.78 (Local compactness of terminal occupation states)**.** *Assume the estimates in Theorem 10.76(i) on $Q_{2R,I}$. Then every sequence of selected subwindows and subcylinders has a subsequential local occupation state. More precisely, after passing to a subsequence, V_n, U_n, and S_n converge weakly in $L^2(I; H^1(B_R))$, strongly in $L^2(I; L^q(B_R))$ for every $q < 6$, and their pressure gradients converge distributionally modulo functions of time.*

Proof. The velocity bounds and the time-derivative bound give compactness by the Aubin–Lions–Simon theorem applied to

$$H^1(B_R) \Subset L^2(B_R) \hookrightarrow H^{-m}(B_R).$$

Interpolation with the uniform $L^2(I; H^1(B_R))$ bound gives strong convergence in $L^2(I; L^q(B_R))$ for every $q < 6$. The pressure normalization bound gives weak compactness in $L^{3/2}(Q_{R,I})$; on overlaps the limits differ only by functions of time, so the gradients assemble distributionally. $\square$

Lemma 10.79 (Closed local labels for terminal residuals)**.** *In the augmented local topology for*

$$(V_n, P_n, U_n, S_n, a_n, b_n)$$

supplied by Theorem 10.78, the ordered validation rows in Theorem 8.86, with the compact local Type II row refined by Theorem 11.13, are closed under subsequential limits of selected compact terminal windows, after shrinking the tested cylinders inside the fixed compact cylinder when necessary.

Proof. The topology records the velocity variables weakly in L^2H^1, strongly in $L^2L^q_{\mathrm{loc}}$ for every $q < 6$, the selected-time residual measures $|S_n(\tau_n)|^3\,dy$ weakly-$*$ as finite Radon measures on compact balls, the pressures weakly in $L^{3/2}_{\mathrm{loc}}$ modulo functions of time, and the modulation, scale, center, cutoff, and cost variables in the compactified repaired-gauge coordinates used in the ordered validation procedure. The compactification adds boundary faces exactly for loss of representation, scale admissibility, bounded modulation, pressure reconstruction, local H^1 control, critical tightness, and cost compatibility.

Positive local velocity-pressure concentration is closed on smaller cylinders. For spacetime rows, the velocity part follows from strong local L^3 convergence after interpolating the strong L^2L^q convergence with the uniform $L^\infty L^3$ bound. For the pressure part, the weak $L^{3/2}$ bound alone gives only lower semicontinuity and would allow positive pressure mass to disappear in the weak limit. We therefore upgrade pressure convergence to strong local convergence on the retained branch as follows. After subtracting the reconstructed pressure of the limit branch, write the pressure difference on B_{R_1}, modulo functions of time, as

$$\widetilde{P}_n = P_n - P - c_n = \widetilde{P}_n^{\mathrm{loc}} + \widetilde{H}_n + c_{n,R_1}(\tau), \qquad \widetilde{P}_n^{\mathrm{loc}} = \mathcal{R}_i\mathcal{R}_j\big(\zeta((V_n\otimes V_n) - (V\otimes V))_{ij}\big),$$

with $\zeta \equiv 1$ on B_{R_1} and $\operatorname{supp}\zeta \Subset B_{R_2}$. Since $V_n \to V$ strongly in $L^2_\tau L^3_y(B_{R_2})$, while V_n and V are uniformly bounded in $L^\infty_\tau L^3_y(B_{R_2})$, interpolation gives strong convergence in $L^3_\tau L^3_y(B_{R_2})$. Hence

$$V_n \otimes V_n \to V \otimes V \quad \text{strongly in } L^{3/2}_\tau L^{3/2}_y(B_{R_2}).$$

Calderón–Zygmund continuity therefore gives strong convergence of $\widetilde{P}_n^{\mathrm{loc}}$ to zero in $L^{3/2}_\tau L^{3/2}_y(B_R)$. The harmonic remainder $\widetilde{H}_n$ is not controlled by a weak lower-semicontinuity argument: it is routed with $(q,p) = (3/2, 3/2)$ by the pressure/exterior alternative in Theorem 9.22. Thus on the retained branch the harmonic part vanishes locally modulo functions of time, while off that branch the nonzero harmonic part is a named pressure or exterior label. Consequently positive pressure mass cannot be hidden by weak convergence on the retained branch; pressure rows are strongly closed in the local $L^{3/2}_\tau L^{3/2}_y$ topology after shrinking cylinders. For selected-time rows, the velocity part is recorded by the weak-$*$ limits of the local measures $|S_n(\tau_n)|^3dy$, so a positive lower bound persists as mass of the limiting measure. Hence a lower CKN threshold cannot disappear after passing to a subsequence and shrinking the cylinder.

Multibubble and multicore splitting are finite collections of such positive CKN lower bounds with separated centers or separated scales; the separation relations are closed in the scale-center compactification. Same-point cascade labels are closed because they are the boundary faces $\lambda_i^n/\lambda_j^n \to 0$ or ∞ at a common physical center. Comparable same-point labels are closed by convergence of the relative scale-center parameters and are grouped by Theorem 9.3.

Exterior leakage and critical-tightness failure are closed because they are defined by a positive lower bound for local L^3 or CKN mass outside every retained compact set. If that mass returns to a fixed compact set along a subsequence, the state is relabelled by the corresponding compact concentration row; otherwise it remains on the exterior/tightness boundary face. Loss of local windowed H^1 control is the face at infinity for the nonnegative L^2H^1 seminorm on a compact cylinder.

Finite-cost scale collapse, negative-drift, and compact cost-divergence labels are determined by nonnegative localized cost functionals. Their sublevel or positive-carrier conditions are lower semicontinuous in the recorded occupation-state topology. The scale-rigid residual label is closed by the relative scale-center convergence and by the compact realization of the scale-rigid branch in Theorem 5.23. Representation, modulation, pressure, carrier, and cost-compatibility failures are not hidden analytic assumptions: they are the explicitly added boundary faces of the ordered validation compactification. Thus every first row of the ordered local validation is a closed diagnostic set. ☐

Lemma 10.80 (Residual concentration is visible in the augmented state)**.** *Let $V_n = U_n + S_n$ on a compact terminal cylinder, with the local $L^\infty L^3$ bounds of Theorem 10.76*(i)*. Suppose that, for selected times τ_n and a compact ball B_ρ,*

$$|S_n(\tau_n)|^3 dy\big|_{B_\rho} \overset{*}{\rightharpoonup} \mu_S, \qquad \mu_S(B_\rho) > 0.$$

Then, after shrinking B_ρ if necessary, either the full represented branch V_n carries a positive compact local concentration label, or the retained component U_n carries a positive comparable retained label in the same terminal frame, or a separated-center/separated-scale adjacent label has already occurred. In particular, residual L^3 concentration cannot be invisible in the augmented local state space by cancellation between V_n and U_n.

Proof. For every time slice,

$$|S_n|^3 = |V_n - U_n|^3 \le 4\big(|V_n|^3 + |U_n|^3\big).$$

Passing to weak-$*$ limits of the finite measures on a slightly smaller ball gives

$$\mu_S \le 4(\mu_V + \mu_U)$$

for subsequential limits of $|V_n(\tau_n)|^3 dy$ and $|U_n(\tau_n)|^3 dy$. Hence μ_V or μ_U has positive mass on some subball. If μ_V has positive mass, the full represented branch has a positive compact local concentration diagnostic and the ordered local state-space decomposition assigns it to the corresponding first label. If μ_U has positive mass, the mass belongs to the retained component already represented in the selected terminal frame. If it were not comparable to that frame, separated-center or separated-scale decoupling would place it in an adjacent row. If it is comparable but not already absorbed, then it is a retained class missed by the terminal grouping, contradicting Theorem 10.81 below.

This visibility argument does not assert a Brézis–Lieb identity for the moving pair $V_n = U_n + S_n$. Such an identity would require additional orthogonality hypotheses and is false for arbitrary moving decompositions. The local state-space proof uses only the domination inequality above and the maximality of the retained grouping: cancellation of V_n against U_n can make the represented measure small only by making the retained measure large, and a positive retained measure is itself a recorded comparable or adjacent local label. Hence positive residual L^3 mass cannot disappear from the augmented state-space diagnostics. ☐

Lemma 10.81 (Maximality of the selected retained frame)**.** *The terminal selection rule for the retained component U_n on a compact terminal cylinder $Q_{2R,I}$ produces a finite collection of comparable retained classes $\{U_n^\alpha\}_{\alpha \in A}$ such that:*

(i) Pairwise scale-center separation. *For $\alpha \ne \beta \in A$, the relative scale-center parameters $(\lambda_n^\alpha/\lambda_n^\beta, (x_n^\alpha - x_n^\beta)/\lambda_n^\beta)$ either tend to 0 or to ∞ (separated scales) or stay bounded with the centers escaping each other in scale-comparable units (separated centers).*

(ii) Maximality. *If a candidate retained class $(\tilde\lambda_n, \tilde x_n)$ at the same physical center and with scale parameter comparable to some λ_n^α survives the selection rule (i.e. carries a positive* *active local mass floor in the sense of Theorem 9.32), then it is already asymptotically equivalent to* $(\lambda_n^\alpha, x_n^\alpha)$ *and is absorbed into* U_n^α*.*

Proof. The terminal selection rule is local on the fixed compact cylinder $Q_{2R,I}$. It extracts only classes which carry the active local mass floor of Theorem 9.32 on a selected compact subcylinder. Classes whose witnesses leave every compact subcylinder, or form an infinite same-point scale chain, are not retained classes in this branch: they are read first as exterior, separated-scale, or cascade boundary faces in the ordered local validation. After those adjacent faces are removed, the remaining witnesses lie in a compact scale-center chart. A finite-overlap cover of that chart and the local bound $\sup_n \|V_n\|_{L^\infty(I;L^3(B_{2R}))} < \infty$ allow only finitely many disjoint active-floor witnesses. Hence the retained list on the compact terminal branch is finite.

For two retained classes in this finite list, either their relative scale-center parameters leave every compact subset of the relative scale-center chart, giving separated scales or separated centers, or those parameters remain in a compact subset. In the latter case the two concentrations are comparable in the same local observer coordinates and are grouped before the terminal frame is fixed. This proves the separation alternative in (i) for distinct retained classes.

For maximality (ii), suppose toward contradiction that $(\tilde\lambda_n, \tilde x_n)$ is comparable to $(\lambda_n^\alpha, x_n^\alpha)$ (i.e. both $\tilde\lambda_n/\lambda_n^\alpha \to c \in (0,\infty)$ and $|\tilde x_n - x_n^\alpha|/\lambda_n^\alpha \to 0$) but not asymptotically equivalent. Then the rescaled profile $\tilde\lambda_n V_n(x_n^\alpha + \tilde\lambda_n y, \cdot)$ carries an active local mass floor in a relative scale-center chart compactly comparable to the chart of U_n^α. Thus the two candidates are not separated by any ordered local boundary face. By Theorem 9.3, all same-point comparable concentrations in such a compact relative chart are assembled into one compound retained class before terminal selection is closed. The candidate $(\tilde\lambda_n, \tilde x_n)$ must therefore already be absorbed into U_n^α, contradicting the assumption that it is a surviving unabsorbed comparable class. □

Lemma 10.82 (Terminal local critical-mass failures are adjacent labels)**.** *Let a physical local Type II branch have passed covered entry and let a terminal active frame be selected from its local state-space decomposition. On each fixed compact terminal cylinder, exactly one of the following occurs after passing to a subsequence:*

(i) *the retained terminal component has positive finite local critical* L^3 *mass on the cylinder;*
(ii) *the lower active mass floor fails, so the candidate is not a retained terminal component;*
(iii) *the local* L^3 *upper bound fails on the compact cylinder, in which case the ordered local state space records an additional compact active label: a critical-mass adjacent label at the same retained frame, a comparable finite-mass retained class which must be grouped into the retained component, a multibubble/multicore label, a cascade label, a rough-core label, or an exterior/separated label.*

In the retained terminal branch only (i) *remains.*

Proof. The lower bound in (i) is the active local mass floor of Theorem 9.32. If this floor is absent, the candidate frame is not retained by the terminal selection rule, giving (ii).

Assume the lower floor is present and the finite local upper bound fails on a compact cylinder $B_R \times I$. Then there are times $\tau_n \in I$ and levels $M_n \to \infty$ with

$$\int_{B_R} |V_n(y,\tau_n)|^3 \, dy \geq M_n.$$

Cover B_R by finitely many smaller balls. On one ball, after passing to a subsequence and renormalizing at the corresponding local scale if necessary, the branch carries a further positive compact L^3 diagnostic. The ordered local state-space compactification records the first label of that diagnostic. If the diagnostic stays at the same comparable retained frame and the upper local L^3 bound still fails, the first label is the critical-mass adjacent face itself; by the retained-state definition this is not a retained terminal branch. If the diagnostic is comparable and finite-mass, then Theorem 10.81 says it was already grouped into the retained component. If it is not comparable, it is by definition one of the adjacent compact active labels: multibubble/multicore splitting, cascade, rough core, or exterior/separated concentration. These are precisely the labels in (iii).

On a retained terminal branch all comparable labels have already been grouped, and all non-comparable labels are adjacent local alternatives routed by the state-space closure package. Hence only the positive finite local critical mass alternative remains. This is a compact-cylinder statement; it does not require $V(\tau) \in L^3(\mathbb{R}^3)$ or a whole-space critical norm bound. □

Theorem 10.83 (Adjacent terminal labels are locally routed)**.** *For every admissible retained terminal decomposition, every first label in the ordered augmented local state space is either the retained comparable class or one of the already proved local alternatives. The adjacent labels are closed and routed as follows: representation, modulation, pressure, critical-mass, carrier, cost-compatibility, exterior tightness, local H^1-loss, multibubble, multicore, cascade, finite-cost scale-collapse, and scale-rigid residual labels are assigned to their named local alternatives and do not remain on the retained terminal branch.*

Proof. Theorem 10.79 proves closedness of the rows in the augmented local topology. Representation and modulation failures are the boundary faces of the repaired-gauge compactification and are exactly the representation alternatives. Pressure rows are routed by Theorem 9.22 and by the pressure reconstruction stability Theorem 3.12. Critical-mass rows are routed by Theorem 10.82: a lower-mass failure is not retained, and an upper local L^3-failure produces a comparable or adjacent active label. Exterior tightness and exterior leakage are routed by Theorem 9.38. Local H^1-loss is the rough-core row in Theorem 11.14. Multibubble, multicore, cascade, and gauge-degenerate rows are the alternatives of Theorem 11.13 and are eliminated or assigned by Theorem 11.14. Finite-cost scale-collapse is routed by Theorem 11.14 through the scale-cost theorem, and scale-rigid residual labels are routed by Theorem 5.40. Carrier and compact-cost compatibility rows are closed by the branchwise ambient adjacent wrapper Theorem 8.162 and the cost-divergence discharge Theorem 10.42. Thus every adjacent first label is already a named local alternative. The only first label not routed away is the retained comparable class, which is grouped into the selected terminal component before the terminal frame is fixed. □

Theorem 10.84 (Residual local mass gives an admissible augmented state)**.** *Assume the local compact-window estimates of Theorem 10.76*(i)*. If*

$$\limsup_{n\to\infty} \|S_n\|_{L^\infty(I;L^3(B_R))} > 0,$$

then, after passing to a subsequence, there are selected times, still denoted τ_n in the selected coordinates, a compact selected or recentered subcylinder $Q_{\rho,J}$, and an augmented local occupation state

$$\mathfrak{s} = \lim_{n\to\infty} (V_n, P_n, U_n, S_n, a_n, b_n)\big|_{Q_{\rho,J}}$$

whose residual coordinate or selected-time residual L^3 concentration measure is nonzero. The Navier–Stokes admissibility conditions in the ordered local validation are imposed on the full represented branch (V_n, P_n, a_n, b_n), not on S_n as an independent solution. Consequently the first nonretained residual row is either a named adjacent local alternative or the retained comparable local class.

Proof. Choose $\delta > 0$, times $\tau_n \in I$, and a subsequence such that

$$\|S_n(\tau_n)\|_{L^3(B_R)} \geq \delta.$$

By Theorem 10.77, after passing to a further subsequence, either the original selected times lie in a fixed interior subinterval $J \Subset I$, or we replace the terminal window by the admissible recentered windows supplied by that lemma and write the selected time as $0 \in J_0^\circ$. In both cases we keep the same selected time rather than transferring the lower bound to a different time; below J denotes the fixed model interval in the selected coordinates and τ_n denotes the selected interior time in those coordinates. Cover B_R by finitely many smaller balls compactly contained in B_R. One ball, after passing to a subsequence, carries residual L^3 mass bounded below by a fixed multiple of δ. The compact subinterval J supplied by Theorem 10.77 contains the selected times with positive distance from the boundary of the model window; we use this J without further time movement. The estimates in Theorem 10.76(i) and Theorem 10.78 give a subsequential limit of the augmented tuple on $Q_{\rho,J}$. In addition, the measures

$$|S_n(\tau_n)|^3\,dy\big|_{B_\rho}$$

have uniformly bounded total mass on compact balls and therefore converge weakly-$*$, after passing to a subsequence, to a finite nonzero Radon measure. The lower mass bound on the chosen ball passes to this measure. Thus the augmented state has a nonzero residual coordinate or, in the concentration case, a nonzero residual L^3 concentration measure. Theorem 10.80 shows that this residual diagnostic is visible either as a positive compact local label of the full represented branch, as a positive comparable retained label, or as an already named separated-center/separated-scale adjacent label. Hence it cannot be hidden by cancellation with U_n.

The PDE structure used by the state-space decomposition is carried by (V_n, P_n, a_n, b_n): it satisfies the repaired-gauge Navier–Stokes equation, the local energy inequality, pressure reconstruction modulo functions of time, bounded modulation, and the compact-cylinder bounds. The retained component U_n and residual coordinate S_n are bookkeeping variables in the augmented state space. Thus no step applies the Navier–Stokes decomposition theorem to S_n alone.

Evaluate the ordered local validation list of Theorem 8.86 on the augmented state. If a representation, pressure, modulation, exterior-tightness, local H^1, finite-cost, scale-rigid, or carrier/cost row fails first, that row is the named adjacent alternative. If the compact local Type II row is reached, apply Theorem 11.13 to the full represented branch (V_n, P_n) on the selected compact window. If the full branch produces a new compact concentration, splitting, cascade, scale-collapse, or scale-rigid state, this is again a named adjacent local alternative. The remaining possibility, using Theorem 10.80, is that the residual diagnostic is carried by the retained compact class. That class is comparable to the selected terminal frame; otherwise the separated-center or separated-scale decoupling lemmas would put it into one of the preceding adjacent rows. Hence the first nonretained residual row is a retained comparable class. ☐

Lemma 10.85 (Positive residual mass produces a local state-space label)**.** *Assume the local compact-window estimates of Theorem 10.76*(i)*. If*

$$\limsup_{n\to\infty} \|S_n\|_{L^\infty(I;L^3(B_R))} > 0,$$

then, after passing to a subsequence and selecting times and subcylinders, the residual produces one of the first closed labels listed in Theorem 10.76(ii)*.*

Proof. Theorem 10.84 constructs the augmented occupation state and verifies that the Navier–Stokes admissibility belongs to the full represented branch (V_n, P_n). The ordered first row of that state is closed by Theorem 10.79. Therefore, after passing to a subsequence, the positive residual mass has a first closed local label in the sense of Theorem 10.76(ii). ☐

Lemma 10.86 (Residual mass cannot remain in the retained terminal branch)**.** *Assume the local terminal frame has been chosen after grouping comparable retained classes. Then a nonretained* *residual satisfying the hypotheses of Theorem 10.85 cannot remain on the retained terminal branch.*

Proof. If the residual has positive local mass, then Theorem 10.85 assigns it to a first closed local label. By Theorem 10.83, an adjacent label is a named local alternative and is not part of the retained branch. If the label is the retained compact class, then the corresponding component is comparable to the selected retained terminal frame; by the terminal selection rule together with Theorem 10.81, comparable retained classes based at the same physical point were already grouped into U_n. It therefore cannot remain in S_n. Both cases contradict the assumption that positive residual mass persists in the nonretained component of the retained terminal branch. □

Lemma 10.87 (Subsequential exhaustion of local residual mass)**.** *Under the hypotheses of Theorem 10.86, every compact terminal cylinder satisfies*

$$S_n \to 0 \quad \textit{in } L^\infty(I; L^3(B_R)).$$

Proof. If the convergence failed, then a subsequence would have positive residual mass in the sense of Theorem 10.85. The preceding lemma shows that such mass cannot remain in the retained terminal branch after local state-space closure and grouping of comparable retained classes. This contradiction proves the convergence. □

Theorem 10.88 (Local terminal completeness from state-space stratification)**.** *Every admissible retained terminal decomposition in the sense of Theorem 10.76*(i) *satisfies the local terminal completeness conclusion of Theorem 10.76*(iii)*, after the terminal frame is selected by grouping comparable retained classes.*

Proof. Theorem 10.78 gives the local compactness needed to pass to occupation states. Theorem 10.79 identifies the closed local labels, and Theorem 10.83 routes every adjacent label to a named local alternative. Theorem 10.85 assigns any positive nonretained residual mass to one of those labels. Theorem 10.86 rules out persistence of such a label inside the retained branch once comparable retained classes have been grouped, and Theorem 10.81 certifies that the terminal selection rule actually performs that grouping maximally. Hence Theorem 10.87 gives $S_n \to 0$ on compact terminal cylinders. This is the local terminal completeness conclusion named in Theorem 10.76(iii). □

Remark 10.89 (Status of the terminal completeness statement). Theorem 10.88 is local. It uses only compact-cylinder estimates, Aubin–Lions compactness, pressure reconstruction modulo functions of time, and the local state-space stratification. It does not use a global critical profile decomposition, a global heat-flow smallness norm, or a global Kato perturbation theorem.

Definition 10.90 (Local compact terminal windows)**.** The local compact terminal-window hypothesis is the assertion that every terminal active sequence used in the terminal analysis is represented, on each compact terminal-frame cylinder $Q_{2R,I}$, by fields satisfying the local compact-cylinder bounds in Theorem 10.76(i). Equivalently, the analysis is carried out on the local state variables

$$(V_n, U_n, S_n, P_n, a_n, b_n)\big|_{Q_{2R,I}},$$

not on an arbitrary global sequence of initial data.

Theorem 10.91 (Local perturbative residual stability)**.** *Let $Q_{2R,I}$ be a compact terminal cylinder. Assume*

$$S_n \to 0 \quad \textit{in } L^\infty(I; L^3(B_{2R})),$$

and assume U_n and S_n are uniformly bounded in

$$L^2(I; H^1(B_{2R})) \cap L^\infty(I; L^3(B_{2R})).$$

Then all velocity interactions involving at least one factor S_n vanish on the smaller cylinder:

$$U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n \to 0 \quad \text{in } L^1(I; L^2(B_R)).$$

If the corresponding cross pressures are reconstructed by the local Calderon–Zygmund decomposition, then their gradients vanish in

$$L^1(I; H^{-1}(B_R)).$$

Proof. By Sobolev on B_{2R}, the L^2H^1 bounds give uniform $L^2(I; L^6(B_{2R}))$ bounds. Holder's inequality gives

$$\|U_n \otimes S_n\|_{L^1(I;L^2(B_R))} \le \|S_n\|_{L^\infty(I;L^3(B_R))} \|U_n\|_{L^1(I;L^6(B_R))} \to 0,$$

and the same estimate treats $S_n \otimes U_n$. The term $S_n \otimes S_n$ is identical, using the uniform L^1L^6 bound for S_n. The pressure statement follows from the local pressure-kernel estimate applied to the cross stress just shown to vanish in L^1L^2, with the pressure normalized modulo functions of time. Concretely, the local Calderón–Zygmund piece of the cross pressure vanishes in $L^1_\tau L^2_y(B_R)$ by Calderón–Zygmund continuity applied to the vanishing cross stress, and the harmonic remainder is routed by Theorem 9.22 with $(q, p) = (1, 2)$ (using the quantitative nested-ball harmonic-tail estimate of Theorem 9.21): on the retained terminal branch any non-vanishing harmonic remainder triggers a named pressure/exterior-leakage alternative, which is excluded. The two contributions sum to zero in $L^1_\tau H^{-1}_y(B_R)$. □

Definition 10.92 (Local perturbative stability)**.** Local perturbative stability means that every residual component which is small in $L^\infty(I; L^3(B_R))$ on compact terminal cylinders contributes only vanishing velocity-stress, pressure, and linear-residual terms in the local topologies used in Theorem 9.13.

Corollary 10.93 (Local residual stability gives perturbative stability)**.** *Theorem 10.91 implies the local perturbative stability condition of Theorem 10.92.*

Proof. This is exactly the conclusion of Theorem 10.91 on each compact terminal cylinder, combined with the local pressure and residual estimates already included in terminal nonlinear decoupling. □

Discharge of the retained compact-test closure.

Lemma 10.94 (Critical-tightness failure routes to an exterior alternative)**.** *Assume the retained compact single-core stratum: the branch V has passed the ordered local tests of Theorem 8.86 up to and including the local state-space decomposition Theorem 11.13, with the single-core alternative* (i) *retained. If the critical-tightness condition of Theorem 12.30 fails, i.e. there exist $\varepsilon_0 > 0$ and sequences $R_n \uparrow \infty$ and $\tau_n \ge \tau_0$ such that*

$$\int_{|y|>R_n} |V(y, \tau_n)|^3 \, dy \ge \varepsilon_0,$$

then, after passing to a subsequence, the branch admits an exterior concentration sequence on every annulus $A_{r,R}$ with r sufficiently large. By Theorem 9.38 the branch then belongs to one of the exterior-profile, separated-core, multicore, noncompact-exterior, or scale-collapse alternatives. All of these are non-retained alternatives in Theorem 11.13(ii),(iv) *and are excluded from the retained single-core stratum. Equivalently, on the retained stratum critical tightness holds automatically.*

Proof. Failure of critical tightness gives $\varepsilon_0 > 0$, $R_n \uparrow \infty$, and $\tau_n \ge \tau_0$ with

$$\int_{|y|>R_n} |V(y, \tau_n)|^3 \, dy \ge \varepsilon_0 \qquad \text{for all } n.$$

The integrand $|V|^3\,dy$ is scale-invariant under the parabolic rescaling defining V from u, so this lower bound is a positive critical-mass-floor that, in the original physical coordinates centred at (x_*, T^*), records positive L^3 mass on physical annuli $\{|x - x_*| > R_n e^{-\tau_n}\}$ at terminal times $t_n := T^* - e^{-2\tau_n} \to T^*$. Two cases occur after passing to a subsequence.

Case A: bounded distance from the core. Some $R^* < \infty$ and some $r_0 > 0$ admit infinitely many n with

$$\int_{r_0<|y|<R^*} |V(y,\tau_n)|^3\,dy \geq \varepsilon_0/2.$$

Suppose, towards contradiction, that there is no exterior concentration on the slightly larger annulus $A_{r_0/2,2R^*}$ in the physical-coordinate sense of Theorem 9.36. By Theorem 9.41 ($k = 0$), there is then a uniform C^0 bound for u on A_{r_0,R^*} on a terminal time interval, hence by integration the critical L^3 mass on the corresponding renormalized annulus is $O(e^{-3\tau})$, contradicting the positive lower bound $\varepsilon_0/2$ for $\tau_n \to \infty$. Therefore Theorem 9.37 forces the exterior concentration alternative on $A_{r_0/2,2R^*}$.

Case B: escape to infinity in the renormalized coordinate. For every $R < \infty$, $\int_{r_0<|y|<R} |V(y,\tau_n)|^3\,dy < \varepsilon_0/2$ infinitely often. Then a subsequence puts at least $\varepsilon_0/2$ of its mass on shells $\{|y| > R_k\}$ with $R_k \uparrow \infty$, producing a sequence of L^3-concentration centres x_n with $|x_n|/1 \to \infty$ in the renormalized frame. This is the noncompact-exterior alternative of Theorem 9.38, which is one of the listed exterior concentration alternatives.

In either case the dichotomy of Theorem 9.38 selects the exterior concentration branch. By the second clause of that theorem the branch is exterior profile, separated core, multicore, noncompact exterior, or scale collapse. Each is one of the alternatives (ii)–(iv) of Theorem 11.13 and disqualifies the branch from the retained single-core alternative (i). This contradicts the retained-stratum hypothesis, so on the retained stratum critical tightness holds. □

Lemma 10.95 (Local windowed H^1-loss routes to the rough-core alternative)**.** *Assume the retained compact single-core stratum, with the standing hypotheses of Theorem 8.15: a smooth repaired-gauge representation, the local compact-cylinder critical bound on the outer cylinder B_{2R}, bounded modulation $M_{ab} < \infty$, and the pressure normalization of Theorem 8.14. These standing hypotheses are part of the retained-stratum input package supplied by the ordered tests in Theorem 8.86*(i),(ii),(iv),(vii)*. Then, for every $R > 0$, the local windowed H^1 bound*

$$\sup_{T\geq\tau_0+1} \int_T^{T+1} \int_{B_R} |\nabla V|^2\,dy\,d\tau \leq C(R, \nu, M_{3,R}, M_{ab})$$

holds, so the local windowed H^1-control condition of Theorem 12.30 is automatic on the retained stratum. Equivalently, failure of local windowed H^1 control is incompatible with the retained-stratum input package and is routed to the rough-core alternative (iii) *of Theorem 11.13, which by Theorem 11.7 forces failure of compact CKN control on enclosing cylinders and therefore returns to the multibubble/cascade alternative* (ii)*, already excluded from the retained branch.*

Proof. The estimate is the conclusion of Theorem 8.15, applied with the standing inputs of the retained stratum: the local critical bound $M_{3,R}$ is the outcome of the positive finite critical L^3 annulus test Theorem 8.86(iv); the bounded modulation M_{ab} is supplied by Theorem 8.82; the pressure normalization is Theorem 8.14; and the smooth repaired-gauge representation is the output of Theorem 8.86(ii). The smoothness hypothesis of Theorem 8.15 is the standard one used throughout the local validation: the estimate is first proved for spatial-temporal mollifications $V^\varepsilon, P^\varepsilon$ and passes to the suitable weak limit by lower semicontinuity of the displayed gradient norm. All four standing inputs are present on the retained stratum by hypothesis. Hence the windowed H^1 bound holds and the corresponding test of Theorem 12.30 passes.

If, contrapositively, the windowed H^1 bound fails for some $R > 0$ along a terminal sequence, then by Theorem 8.15 at least one of the standing inputs must fail along that sequence (since the conjunction of the standing inputs implies the bound). By the ordered tests Theorem 8.86 this earlier-test failure routes the branch out of the retained stratum into the corresponding representation, repaired-gauge, critical-mass, or modulation alternative. Independently, in the local state-space terminology of Theorem 11.13 the same failure is recorded as the rough-core alternative (iii), and by Theorem 11.7 that alternative is equivalent to failure of compact CKN control on enclosing cylinders, hence to active multibubble or cascade behaviour, alternative (ii) of Theorem 11.13. Both routings exit the retained single-core stratum. Therefore on the retained stratum H^1-loss is impossible. □

Theorem 10.96 (Retained compact-test closure)**.** *On the retained compact single-core stratum, both compact tests of Theorem 12.30 pass: critical tightness holds and the local windowed H^1 control holds. Equivalently, the entries* (v),(vi) *of Theorem 8.86 are not genuine new assumptions on the retained stratum; their failure is routed to one of the named non-retained alternatives.*

Proof. Critical tightness holds by Theorem 10.94: failure routes the branch to the exterior-stratification alternatives, all non-retained. Local windowed H^1 control holds by Theorem 10.95: failure routes the branch to the rough-core alternative, which by Theorem 11.7 is absorbed into the multibubble/cascade alternative, also non-retained. Hence both compact tests are theorems, not assumptions, on the retained stratum. □

Corollary 10.97 (Discharge of the retained compact-test closure assumption)**.** *The retained compact-test closure obligation listed as item 3 of the proof-obligation tracker is discharged on every retained compact single-core branch produced by the ordered local validation procedure of Theorem 8.86. No additional global tightness or global windowed H^1 hypothesis is required.*

Proof. Theorem 10.96 reduces both compact tests to local consequences of inputs already present on the retained stratum or to named non-retained alternatives. No global tightness or H^1 bound is invoked. □

10.12. **Discharge of the Type II analytic-data exhaustiveness assumption.** The exhaustion hypothesis of Theorem 12.4 is the entry condition for every admissible local Type II branch. It is the item formerly listed as item 1 of the proof-obligation tracker. We discharge it by an ordered local extraction package: each component of Theorem 12.2 is either supplied by an existing local lemma on the admissible class $\mathcal{U}_{\mathrm{II}}^{NS}$, or its first failure is routed to a named alternative that is class-incompatible or already covered by the local state-space stratification. No global critical profile theorem and no global L^3 bound are invoked.

Lemma 10.98 (Concentration extraction routes)**.** *Let $\omega = (u, p, T^*) \in \mathcal{U}_{\mathrm{II}}^{NS}$. Exactly one of the following holds:*

(a) *the local concentration dichotomy Theorem 3.5 produces $x_0 \in \Sigma(T^*)$, $\eta > 0$, and $r_n \downarrow 0$ with $C((x_0, T^*), r_n) + D((x_0, T^*), r_n) \geq \eta$ for all n, and the concentration point lies in the local Type II alternative;*

(b) *the singular set $\Sigma(T^*)$ is empty;*

(c) *$\Sigma(T^*) \neq \emptyset$, and every concentration point of $\Sigma(T^*)$ lies in the local Type I alternative.*

Cases (b) *and* (c) *are class-incompatible with Theorem 12.1. Hence on $\mathcal{U}_{\mathrm{II}}^{NS}$ only case* (a) *occurs, and the first ordered failure of Theorem 12.3 ("concentration extraction fails") does not arise on the admissible class.*

Proof. If $\Sigma(T^*) = \emptyset$, case (b) holds. If $\Sigma(T^*) \neq \emptyset$, apply the local concentration dichotomy at singular points. If some singular point is not in the local Type I alternative, choose that point as

x_0; the Caffarelli–Kohn–Nirenberg failure at x_0 supplies $\eta > 0$ and $r_n \downarrow 0$ with

$$C((x_0, T^*), r_n) + D((x_0, T^*), r_n) \geq \eta,$$

and case (a) holds. If no such point exists, every concentration point lies in the local Type I alternative and case (c) holds. The three cases are therefore disjoint and exhaustive.

In case (b) there is no terminal singular concentration point, so the branch is the local continuation/no-terminal-concentration alternative rather than a local Type II branch. This is excluded by clause (iv) of Theorem 12.1; no global continuation estimate is used. In case (c) every concentration point lies in the local Type I alternative of Theorem 3.5, which is exactly the local Type I scale-invariant criterion. This contradicts clause (iii) of Theorem 12.1. In case (a) the dichotomy supplies the positive CKN concentration scales and the exclusion of the Type I alternative at (x_0, T^*); this is the content of the first analytic datum of Theorem 12.2. □

Lemma 10.99 (Supercritical scale alternative routes). *Let $\omega \in \mathcal{U}_{\mathrm{II}}^{NS}$ and let (x_0, T^*) be a concentration point produced by Theorem 10.98* (a). *Then the supercritical scale alternative of Theorem 12.2* (ii) *holds at (x_0, T^*).*

Proof. The dichotomy of Theorem 3.5 at (x_0, T^*) is exclusive: either a local Type I scale-invariant bound holds, or no such local Type I bound holds. The first option contradicts clause (iii) of Theorem 12.1, which excludes branches in the local Type I class. Hence the second option holds at every concentration point produced by the previous lemma; this is exactly the supercritical Type II scale alternative. The second ordered failure of Theorem 12.3 therefore does not arise on $\mathcal{U}_{\mathrm{II}}^{NS}$. □

Lemma 10.100 (Local profile data routes). *Let $\omega \in \mathcal{U}_{\mathrm{II}}^{NS}$ and let (x_0, T^*) be a concentration point in the local Type II alternative as produced by Theorems 10.98 and 10.99. Run the local scale-translation selection of Theorem 3.6 at the positive concentration scales and test whether it produces a nondegenerate compact single core. Then exactly one of the following holds:*

(a) *the localized scale and center are nondegenerate and no represented selected-window subsequence has a nonzero bounded selected-window limit on any compact cylinder; Theorem 3.7 supplies represented variables (V_n, P_n, a_n, b_n) on compact cylinders satisfying the represented Navier–Stokes equation* (1)*; the represented selected windows form a local Type II sequence in the sense of Theorem 3.2, providing the local profile data of Theorem 12.2* (iii)*;*

(b) *the localized nondegeneracy fails, and the branch falls into the multibubble, cascade, or gauge-degenerate alternative of Theorem 5.6* (ii)–(v)*;*

(c) *nondegeneracy holds but the represented selected windows admit a nonzero bounded selected-window limit on some compact cylinder; the branch then enters the bounded-selected-window alternative of Theorem 3.1 and exits Type II by Theorem 5.25.*

In cases (b) *and* (c) *the branch is non-retained: case* (b) *is treated by the multibubble/cascade machinery of Section 5, and case* (c) *is the bounded selected-window exit of the scale-rigid stratum. Neither case is a compact single-core branch.*

Proof. The local scale-translation selection at (x_0, T^*) attempts to extract a single compact core at the positive CKN concentration scales $r_n \downarrow 0$. Either the localized scale and center are nondegenerate (cases (a) and (c)), or they are degenerate (case (b)); under nondegeneracy, either no represented selected-window subsequence has a nonzero bounded selected-window limit (case (a)), or some such subsequence does (case (c)). The three cases are exhaustive.

In case (a), Theorem 3.6 applies and rewrites the shrinking cylinders $Q_{r_n}(x_0, T^*)$ as fixed selected windows. The represented sequence has positive local CKN concentration on a fixed compact cylinder; combined with the standing absence of nonzero bounded selected-window limits, it is a local Type II sequence in the sense of Theorem 3.2. Theorem 3.7 then supplies the represented variables (V_n, P_n, a_n, b_n) on compact cylinders satisfying (1); this is the local profile datum of Theorem 12.2 (iii).

In case (b), failure of localized nondegeneracy is, by the second sentence of Theorem 3.7, the multibubble, cascade, or gauge-degenerate alternative. The corresponding stratum is named in Theorem 5.6 (ii)–(v) and is non-retained: it is handled by the multibubble/cascade machinery and never enters the retained compact single-core stratum produced by Theorem 8.86.

In case (c), the represented selected-window sequence has a nonzero bounded selected-window limit on some compact cylinder, placing the branch in Theorem 3.1; Theorem 5.25 then routes the branch out of Type II via the scale-rigid bounded-window exit, which is also non-retained.

Hence the third ordered failure of Theorem 12.3 ("local profile data are insufficient") is routed to the multibubble/cascade/gauge-degenerate non-retained alternative or to the scale-rigid bounded-window exit; on the retained compact single-core stratum it does not arise. □

Theorem 10.101 (Type II analytic-data routing on the admissible class)**.** *The Type II analytic data of Theorem 12.2 are exhaustive by ordered local routing on $\mathcal{U}_{\mathrm{II}}^{NS}$: every $\omega \in \mathcal{U}_{\mathrm{II}}^{NS}$ either has the concentration profile, the supercritical scale alternative, and the local profile data, or its first applicable failure in Theorem 12.3* (i)–(iv) *is matched to a named alternative that is class-incompatible with Theorem 12.1 or to a non-retained state-space branch already covered by the local stratification. In particular, after those named exits are removed, every retained admissible branch has the three Type II analytic data.*

Proof. Process the three analytic data of Theorem 12.2 in the order specified by Theorem 12.3.

Datum (i): concentration extraction. Theorem 10.98 shows that on $\mathcal{U}_{\mathrm{II}}^{NS}$ the only realised case is the dichotomy case (a) producing (x_0, T^*) with positive CKN concentration in the local Type II alternative; the other two cases are class-incompatible.

Datum (ii): supercritical scale alternative. Theorem 10.99 then forces the supercritical Type II scale alternative at (x_0, T^*); the Type I alternative is class-incompatible by clause (iii) of Theorem 12.1.

Datum (iii): local profile data. Theorem 10.100 either supplies the local profile data via the local scale-translation selection and Theorem 3.7, or routes the branch to the multibubble/cascade/gauge-degenerate non-retained alternative Theorem 5.6 (ii)–(v), or to the bounded selected-window exit Theorem 5.25.

Residual datum (iv). The fourth ordered failure of Theorem 12.3, "the admissible Type II branch lies outside the concentration, scale, and profile alternatives under consideration", is the catch-all for a branch that is not one of the local Navier–Stokes Type II alternatives after the first three tests have been applied. Clause (iv) of Theorem 12.1 excludes continuation, Type I, boundary, and non-Navier–Stokes-domain alternatives. Any remaining local branch outside the retained profile alternatives is one of the named non-retained state-space exits produced in the preceding steps.

In every case the branch either has all three analytic data or its first ordered failure is recorded as a class-incompatible alternative or as a non-retained state-space branch already covered by the local stratification. This is the ordered local-routing exhaustiveness statement associated with Theorem 12.5. Removing the named exits leaves only the first alternative of that corollary, namely the full Type II analytic data. □

Corollary 10.102 (Discharge of the Type II analytic-data exhaustiveness assumption)**.** *The ordered-routing exhaustion hypothesis used by Theorem 12.4 is now a theorem on $\mathcal{U}_{\mathrm{II}}^{NS}$: item 1 of the proof-obligation tracker is discharged by ordered local extraction. On retained admissible branches the three Type II analytic data hold; non-retained failures are assigned to named local* *alternatives before the downstream compact analysis is invoked. No global critical profile theorem and no global L^3 bound are invoked.*

Proof. Theorem 10.101 establishes ordered local routing of all failures of the Type II analytic data on $\mathcal{U}_{\mathrm{II}}^{NS}$. By Theorem 12.5, after the named routed exits are removed, the remaining retained branch is in the first alternative and therefore has the Type II analytic data required by Theorem 12.4. Each step uses only the local concentration dichotomy Theorem 3.5, the local scale-translation selection Theorem 3.6, the local representation input Theorem 3.7, the local Type II definition Theorem 3.2, the admissibility clauses of Theorem 12.1, and the multibubble/cascade stratum Theorem 5.6, together with the bounded-window exit Theorem 5.25. □

Definition 10.103 (Terminal local compactness input hypotheses)**.** The terminal local compactness *input* hypotheses consist of the local compact terminal-window hypothesis Theorem 10.90 and local perturbative stability Theorem 10.92, together with item (i) of Theorem 10.76 (the admissibility estimates on the represented branch). These hypotheses do *not* include the local terminal completeness conclusion (iii) of Theorem 10.76.

Definition 10.104 (Terminal local compactness package)**.** The terminal local compactness package is the input hypotheses Theorem 10.103 together with the local terminal completeness theorem Theorem 10.88. This is a record of what has been proved and is not an additional assumption.

Theorem 10.105 (Terminal critical local compactness)**.** *Assume the terminal local compactness input hypotheses of Theorem 10.103. Assume also that the ordered adjacent local labels are routed as in Theorem 10.83 and that comparable retained classes are grouped before the terminal frame is fixed. Then, on every retained terminal branch and every compact terminal cylinder, all nonretained residual components vanish in the local perturbative topology of Theorem 9.13.*

Proof. The local compact terminal-window hypothesis gives the compact-cylinder estimates. Applying Theorem 10.88 under the stated routing and grouping gives

$$S_n \to 0 \quad \text{in } L^\infty(I; L^3(B_R))$$

on each compact terminal cylinder. Local perturbative stability then converts this velocity convergence into the nonlinear-stress, pressure, and residual convergences required in terminal nonlinear decoupling. □

Corollary 10.106 (Terminal completeness from local compactness hypotheses)**.** *The terminal local completeness statement in the final argument may be replaced by the terminal local compactness package of Theorem 10.104.*

Proof. Theorem 10.105 gives the asserted terminal local completeness. □

Remark 10.107 (Boundary of the terminal compactness hypothesis)*.* The preceding compactness conclusion applies only to terminal windows on which the repaired-gauge representation, pressure reconstruction, local Caccioppoli control, bounded modulation, and local state-space labels are available. If one of these inputs is absent, the branch is assigned to the corresponding representation, pressure, rough-core, modulation, or state-space alternative; it is not treated by importing a global profile theorem.

Definition 10.108 (Terminal sequences sampled from the renormalized orbit)**.** A terminal active sequence is sampled from the renormalized orbit if it is of the form

$$u_{0,n} = \Lambda_{\lambda_n, x_n} V(\tau_n), \qquad \tau_n \to \infty, \qquad \lambda_n > 0, \qquad x_n \in \mathbb{R}^3,$$

or by the equivalent inverse critical change of variables, where

$$(\Lambda_{\lambda,x_0}f)(x) = \lambda^{-1} f\left(\frac{x-x_0}{\lambda}\right).$$

Lemma 10.109 (Critical symmetry preserves the L^3 norm)**.** *For every $f \in L^3(\mathbb{R}^3)$, $\lambda > 0$, and $x_0 \in \mathbb{R}^3$,*

$$\|\Lambda_{\lambda,x_0}f\|_{L^3} = \|f\|_{L^3}.$$

Proof. By the change of variables $x = x_0 + \lambda y$,

$$\|\Lambda_{\lambda,x_0}f\|_{L^3}^3 = \int_{\mathbb{R}^3} \lambda^{-3} \left| f\left(\frac{x-x_0}{\lambda}\right)\right|^3 dx = \int_{\mathbb{R}^3} |f(y)|^3\, dy.$$

Taking cube roots gives the claim. □

Theorem 10.110 (Bounded terminal sequences from the critical annulus)**.** *Assume the renormalized orbit satisfies the local compact-window bounds*

$$0 < \inf_{\tau\in I} \|V(\tau)\|_{L^3(B_R)} \le \sup_{\tau\in I} \|V(\tau)\|_{L^3(B_{2R})} < \infty,$$

on the terminal interval under consideration, together with the local Caccioppoli and pressure bounds on $Q_{2R,I}$. If every terminal active window is sampled from the renormalized orbit, then every terminal active window satisfies the local compact terminal-window hypothesis.

Proof. The local L^3 upper bound is one component of the compact-window hypothesis. Local $L^\infty L^2$, $L^2 H^1$, and pressure bounds are supplied by the local Caccioppoli estimate and pressure reconstruction on $Q_{2R,I}$. The repaired-gauge equation gives the required time-derivative bound in H^{-m}. The lower bound records that the retained window is nontrivial. □

Theorem 10.111 (Terminal compactness alternatives)**.** *For a candidate terminal active sequence in the Type II analysis, the following alternatives exhaust the possible local compactness outcomes, after passing to a subsequence when necessary:*

(i) *the sequence is not sampled from the renormalized orbit;*
(ii) *the repaired-gauge representation, pressure reconstruction, or local Caccioppoli estimate is unavailable on the selected compact cylinder;*
(iii) *the local L^3 upper bound fails along a terminal sampling subsequence;*
(iv) *the retained local L^3 lower bound vanishes along a terminal sampling subsequence;*
(v) *after excluding the preceding alternatives, the terminal active sequence satisfies the local compact terminal-window hypothesis.*

Proof. If the terminal sequence is not sampled from the renormalized orbit, then (i) holds. If one of the local PDE inputs is missing, then (ii) holds and the branch is assigned to the corresponding local alternative. If the local critical upper bound fails, then (iii) holds after passing to a subsequence. If the retained lower bound vanishes, then (iv) holds. Otherwise the local hypotheses of Theorem 10.110 holds, and (v) follows. □

Corollary 10.112 (Local compactness from orbit sampling)**.** *In the presence of terminal orbit sampling, local repaired-gauge estimates, and a positive finite local terminal L^3 annulus, the local compact terminal-window hypothesis in Theorem 10.90 follows from Theorem 10.110.*

Proof. The theorem gives the required compact-cylinder bounds for every terminal active window.
□

Definition 10.113 (Terminal coordinate-frame construction)**.** The terminal coordinate-frame construction means that every terminal active frame used in the analysis is obtained from the repaired-gauge orbit as follows. Choose sampling times $\tau_n \to \infty$ and critical symmetry parameters (λ_n, x_n); form the local terminal windows by applying the critical coordinate-frame map to V on compact time intervals around τ_n; group comparable retained local classes based at the same physical point before selecting the terminal active frame. No external data are introduced.

Theorem 10.114 (Terminal coordinate-frame construction from local state selection)**.** *Assume terminal active frames are selected by first partitioning local active states by physical concentration point, then grouping comparable retained states at the same point into compound local states, and finally choosing a minimal-scale retained class at that point. Then the terminal coordinate-frame construction of Theorem 10.113 holds for the selected terminal frames.*

Proof. By the stated selection rule, each terminal active frame is attached to a compound active class after all comparable retained states at the same physical point have been grouped. The scale and center of the terminal frame are the scale and center of this compound class. The terminal data are then obtained by applying the Navier–Stokes critical change of variables to the local renormalized branch along the chosen sampling sequence $\tau_n \to \infty$. Thus the construction uses only the repaired-gauge orbit and the Navier–Stokes critical symmetries, and it satisfies all clauses of Theorem 10.113. □

Theorem 10.115 (Terminal coordinate frames are sampled from the orbit)**.** *Assume the terminal coordinate-frame construction. Then every terminal active sequence belongs to the class described in Theorem 10.108.*

Proof. By definition of the terminal coordinate-frame construction, each terminal active frame is obtained by choosing times τ_n, selecting scale and center parameters (λ_n, x_n), and applying the Navier–Stokes critical change of variables to the renormalized solution $V(\tau_n)$. Hence the associated terminal data have the form

$$u_{0,n} = \Lambda_{\lambda_n, x_n} V(\tau_n),$$

or the equivalent inverse formulation, which is the orbit-sampling condition. □

Theorem 10.116 (Repaired-gauge representation gives terminal orbit sampling)**.** *Assume a repaired-gauge representation of the Type II branch and the terminal coordinate-frame construction. Then every terminal active sequence is sampled from the renormalized orbit.*

Proof. The repaired-gauge representation supplies the renormalized orbit $V(\tau)$ and the critical coordinate maps between physical variables and renormalized variables. The terminal coordinate-frame construction supplies the terminal sampling times and scale-center parameters. Combining these hypotheses gives

$$u_{0,n} = \Lambda_{\lambda_n, x_n} V(\tau_n),$$

which is the assertion of Theorem 10.108. □

Corollary 10.117 (Failure of terminal orbit sampling)**.** *If a sequence reaches the terminal local-state stage but is not sampled from the renormalized orbit, then either the repaired-gauge representation is not available for that sequence or the terminal coordinate-frame construction has not been applied as specified.*

Proof. This is the contrapositive of Theorem 10.116. □

Corollary 10.118 (Terminal compactness from explicit PDE hypotheses)**.** *Assume the repaired-gauge representation, the terminal coordinate-frame construction, a positive finite local terminal L^3 annulus for V, local Caccioppoli and pressure reconstruction estimates on compact terminal* *cylinders, bounded modulation, and local perturbative residual stability. Then the terminal local completeness statement in the final argument holds.*

Proof. Theorem 10.116 gives terminal orbit sampling. Together with the positive finite local L^3 annulus and the local PDE estimates, Theorem 10.110 gives local compact terminal windows. Local perturbative residual stability is supplied by Theorem 10.93. The local adjacent labels are routed by Theorem 10.83, and Theorem 10.88 gives the terminal local completeness conclusion. Therefore the terminal local compactness package of Theorem 10.104 hold, and Theorem 10.106 gives terminal local completeness. □

10.13. **Terminal Conditional Assembly.**

Definition 10.119 (Final conditional hypotheses)**.** The final conditional hypotheses for this section are:

(i) every renormalized local Type II branch enters one of the local alternatives used in Theorem 11.13;
(ii) compact single-core branches are excluded by Theorem 11.5;
(iii) multibubble, cascade, gauge-degenerate, and scale-rigid terminal branches are excluded under the hypotheses of Theorem 11.6;
(iv) finite-cost scale-collapse branches are excluded by Theorem 11.8, while scale-collapse branches with divergent negative scale-drift cost are covered by Theorem 10.33 or Theorem 10.37 and then excluded by the cost-divergence theorem Theorem 10.42;
(v) failure of uniform critical tightness is excluded, equivalently uniform critical tightness holds for every renormalized branch in the class;
(vi) failure of local windowed H^1 control is excluded, equivalently the required local windowed H^1 control holds for every renormalized branch in the class;

Theorem 10.120 (Final conditional Type II exclusion)**.** *Under the final conditional hypotheses of Theorem 10.119, no renormalized local Type II branch in the stated class remains.*

Proof. Let a renormalized local Type II branch be given. By item (i), it lies in one of the alternatives in Theorem 11.13. Items (ii) and (iii) exclude the compact single-core, multibubble, cascade, gauge-degenerate, and scale-rigid alternatives. Item (iv) excludes finite-cost scale collapse and also excludes the divergent negative-drift scale-collapse branch whenever the corresponding cost is one of the costs covered by the cost-divergence theorem Theorem 10.42. Items (v) and (vi) exclude failure of uniform critical tightness and failure of local windowed H^1 control. These cases exhaust the alternatives, so no branch remains. □

Definition 10.121 (Expanded terminal hypotheses)**.** The following analytic hypotheses imply the final conditional hypotheses:

(i) the local alternative decomposition for renormalized Type II branches;
(ii) the compact single-core, multibubble/cascade, finite-cost scale-collapse, and scale-rigidity exclusions from the preceding sections;
(iii) a critical-tightness estimate giving uniform critical tightness;
(iv) an exhaustive terminal compactness partition and a local finite critical-mass statement for retained active terminal strata;
(v) local perturbative residual stability and terminal state-space completeness on compact windows;
(vi) removal of physically exterior regular components;
(vii) the repaired-gauge representation theorem;
(viii) the local Caccioppoli/windowed regularity input;
(ix) the local compact scale-collapse routing theorem Theorem 7.30.

Lemma 10.122 (Terminal hypotheses imply critical tightness)**.** *The expanded terminal hypotheses imply uniform critical tightness.*

Proof. The repaired-gauge representation turns each renormalized orbit into the terminal windows used in the local state-space analysis. The terminal compactness partition, finite critical-mass statement, local perturbative stability, and terminal state-space completeness give the required terminal completeness. Exterior regular components are removed by hypothesis. Together with the multibubble and scale-collapse exclusions, this leaves only possible loss of critical tightness; the critical-tightness estimate excludes that loss. □

Lemma 10.123 (Terminal hypotheses imply windowed H^1 control)**.** *The expanded terminal hypotheses imply local windowed H^1 control.*

Proof. The windowed H^1 estimate requires a repaired-gauge branch, local pressure and modulation control, compact-cylinder critical bounds, and the Caccioppoli/windowed regularity input. These are among the expanded terminal hypotheses. Hence the required local windowed H^1 control holds, and the failure of local windowed H^1 control is unavailable. □

Lemma 10.124 (Expanded terminal hypotheses imply final hypotheses)**.** *The expanded terminal hypotheses imply the final conditional hypotheses of Theorem 10.119.*

Proof. Items (i) and (ii) of Theorem 10.121 give the local alternative decomposition and the compact, multibubble, scale-collapse, and scale-rigidity exclusions. The cost-divergence criterion is supplied separately by the cost-divergence theorem Theorem 10.42 and its status corollary Theorem 10.43. By Theorem 10.122 they also give critical tightness, and by Theorem 10.123 they give the local windowed H^1 control. These are the hypotheses listed in Theorem 10.119. □

Theorem 10.125 (Expanded terminal Type II exclusion)**.** *If the expanded terminal hypotheses hold, then no renormalized local Type II branch in the stated class remains.*

Proof. Theorem 10.124 gives the final conditional hypotheses. Applying Theorem 10.120 gives the exclusion. □

Remark 10.126 (Branchwise reading). Compact single-core branches are removed by the compact criterion; retained scale-collapse branches are removed by the local compact scale-collapse routing theorem Theorem 7.30, or by Theorem 10.33 when the scale-collapse drift cost is compatible with the cost-divergence theorem; multibubble and non-tight cases are removed by the terminal profile and critical-tightness inputs; failures of local windowed H^1 control are removed by the local Caccioppoli estimate; and every divergent compatible localized cost on the canonical windowwise schedule is handled by Theorem 10.42.

Theorem 10.127 (Expanded terminal hypotheses hold branchwise)**.** *Let ω be a physical local Type II branch after Theorem 11.21. Then either ω exits through a named local alternative already eliminated by the local state-space packages, or the nine items of Theorem 10.121 hold on a retained terminal tail.*

Proof. The proof is an ordered verification of the nine terminal inputs.

Local decomposition and structural exclusions. Covered entry supplies a positive local Type II concentration sequence or an earlier named exit. In the retained case Theorem 11.13 gives the local alternative decomposition. The compact single-core, multibubble/cascade, finite-cost scale-collapse, retained compact scale-collapse, and scale-rigid alternatives are eliminated by Theorems 5.40, 7.30, 11.5, 11.6 and 11.8.

Critical tightness and compact terminal completeness. On the retained compact stratum, critical tightness and local windowed H^1-control are theorems by Theorem 10.96. If either test fails before the retained stratum is reached, the failure is an exterior, rough-core, or multibubble/cascade state-space exit, routed by Theorems 9.38, 11.7 and 11.14. The finite critical-mass input used here is local on compact terminal windows: Theorem 10.82 proves that failure of positive finite local L^3 mass is either non-retention or an adjacent active label. The proof does not require the whole-space critical norm $\|V(\tau)\|_{L^3(\mathbb{R}^3)}$. With this local finite-mass input, the terminal compactness partition is supplied by Theorems 10.105 and 10.118; their adjacent labels are routed by Theorem 10.83, whose carrier and cost rows now use the ambient adjacent wrapper Theorem 8.162.

Residual stability, representation, exterior removal, and local regularity. Terminal state-space completeness is Theorem 9.16, proved locally from the discharged critical profile decomposition Theorem 10.88. Local perturbative residual stability is Theorem 10.93. Physically exterior regular components are removed by Theorem 9.38. The repaired-gauge representation is the ordered consequence of Theorems 12.13 and 12.20 together with the absolutely continuous chart consequences Theorem 10.54; if the chart or gauge row fails, that failure is the representation alternative and not a retained terminal branch. The local Caccioppoli/windowed regularity input is supplied by Theorems 10.69 and 10.70 on suitable repaired-gauge branches, with retained windowed control supplied by Theorem 10.96. Finally, compact scale-collapse routing is exactly Theorem 7.30.

Thus each item in Theorem 10.121 is either verified on the retained terminal tail or its first failure is one of the named local alternatives already closed above. The verification uses only local compact windows, ordered state-space labels, suitable local energy, and compact terminal estimates; it does not import a global profile theorem, global Kato smallness, or a whole-space L^3 bound. □

Corollary 10.128 (Expanded terminal wrapper discharged)**.** *The expanded terminal hypotheses are no longer an active assumption for the physical no-Type-II theorem: every physical local Type II branch either satisfies them on a retained terminal tail or exits through a named local alternative closed by the local packages.*

Proof. This is Theorem 10.127. □

11. Assembly of the Local Type II Exclusion

11.1. **Local Entry.** Let (u,p) be a suitable weak solution of the three-dimensional Navier–Stokes equations

$$\partial_t u + (u\cdot\nabla)u + \nabla p = \nu\Delta u, \qquad \nabla\cdot u = 0,$$

on $\mathbb{R}^3\times(T-\delta,T)$. For $z_0=(x_0,T)$ and $r>0$, set

$$Q_r(z_0) = B_r(x_0)\times(T-r^2,T),$$
$$C(z_0,r) = r^{-2}\int_{Q_r(z_0)} |u|^3\,dx\,dt,$$

and

$$D(z_0,r) = r^{-2}\int_{T-r^2}^{T}\int_{B_r(x_0)} |p(x,t)-(p)_{B_r(x_0)}(t)|^{3/2}\,dx\,dt.$$

The pressure is always taken modulo functions of time, and the spatial mean normalization is the one used in the Caffarelli–Kohn–Nirenberg criterion.

Definition 11.1 (Singular set and local Type II point)**.** Let $\Sigma(T)$ be the set of points x_0 for which u is not locally bounded in any backward cylinder $Q_r(x_0,T)$. A point $x_0\in\Sigma(T)$ is in the local Type II alternative if it carries a positive local Caffarelli–Kohn–Nirenberg concentration sequence and no local Type I scale-invariant bound holds at (x_0,T).

Definition 11.2 (Positive local Type II concentration sequence)**.** A sequence $r_n\downarrow 0$ is a positive local Type II concentration sequence at (x_0,T) if there exists $\eta>0$ such that

$$C((x_0,T),r_n) + D((x_0,T),r_n) \ge \eta \qquad \text{for all } n,$$

and (x_0,T) is in the local Type II alternative. The associated parabolic rescalings are

$$u_n(y,s) = r_n u(x_0+r_n y, T+r_n^2 s), \qquad p_n(y,s) = r_n^2 p(x_0+r_n y, T+r_n^2 s).$$

Theorem 11.3 (Local concentration and Type I/Type II dichotomy)**.** *If $\Sigma(T)\neq\emptyset$, then for every $x_0\in\Sigma(T)$ there are $\eta>0$ and $r_n\downarrow 0$ such that*

$$C((x_0,T),r_n) + D((x_0,T),r_n) \ge \eta \qquad \textit{for all } n.$$

At the same point exactly one of the following alternatives holds: there is a local Type I scale-invariant bound, or (x_0,T) is in the local Type II alternative.

Proof. This is the pointwise CKN concentration dichotomy of Theorem 3.5, restated in the notation used in the final assembly. □

11.2. **Local Theorems Used in the Assembly.** The local results needed for the final assembly are restated in the following form.

Theorem 11.4 (Repaired-gauge representation)**.** *Let a positive local Type II concentration sequence contain a nondegenerate selected core. Then, after passing to selected windows, there are local scale-center variables $\lambda(\tau)>0$, $x_c(\tau)$, renormalized fields*

$$V(y,\tau) = \lambda(\tau)u(x_c(\tau)+\lambda(\tau)y, t(\tau)), \qquad P(y,\tau) = \lambda(\tau)^2 p(x_c(\tau)+\lambda(\tau)y, t(\tau)) + \pi(\tau),$$

and modulation coefficients a,b such that

$$\partial_\tau V + (V\cdot\nabla)V + \nabla P = \nu\Delta V + a(\tau)(V+y\cdot\nabla V) + b(\tau)\cdot\nabla V, \qquad \nabla\cdot V = 0.$$

On each compact renormalized window one has the local suitable energy inequality, local pressure reconstruction modulo functions of time, stability under subsequences, and the compact-window bounds proved in Section 4. Failure of the nondegenerate scale-center selection is part of the multibubble, cascade, or scale-rigidity alternatives.

Proof. Theorem 4.35 is precisely the compact-window representation theorem. □

Theorem 11.5 (Compact single-core exclusion)**.** *No compact single-core vanishing-dissipation branch can be a local Type II sequence. More explicitly, let a represented selected-window sequence have:*

(i) *positive local CKN concentration on a fixed compact cylinder;*
(ii) *uniformly bounded compact-cylinder suitable estimates $A + E + C + D$;*
(iii) *nondegenerate local scale-center representation;*
(iv) *bounded modulation coefficients;*
(v) *local pressure reconstruction and pressure stability modulo spatial means;*
(vi) *vanishing localized dissipation on some retained compact cylinder.*

Then the sequence is incompatible with the local Type II condition.

Proof. Theorem 3.18 gives the stated exclusion. □

Theorem 11.6 (Multibubble and cascade exclusion)**.** *In the local terminal class where terminal admissibility is supplied by Theorem 5.41 and scale-rigidity is discharged by Theorem 5.40, no local multibubble, cascade, or gauge-degenerate Type II branch can occur in the class produced by the local active-core extraction. Equivalently, every failure of the single nondegenerate compact-core alternative is excluded by the local multibubble/cascade analysis.*

Proof. Theorem 5.42, together with its removal corollary Theorem 5.43, gives the stated exclusion. □

Theorem 11.7 (Rough-core reduction)**.** *On any compact renormalized window satisfying the local energy inequality and bounded modulation hypotheses, finite outer velocity-pressure CKN control $\mathcal{C} + \mathcal{D}$ implies finite inner windowed H^1 control. Hence failure of local windowed H^1 control on a retained inner cylinder forces failure of the compact-cylinder $\mathcal{C} + \mathcal{D}$ bounds on every relevant enclosing cylinder.*

Proof. Theorem 6.7 gives the stated reduction. □

Theorem 11.8 (Finite-cost scale-collapse exclusion)**.** *No renormalized branch with finite scale-collapsing cost, and hence no branch with finite absolute scale cost, can have genuine scale collapse together with a nonvanishing selected core.*

Proof. Theorems 7.7 and 7.9 give the stated exclusion. □

Theorem 11.9 (Scale-collapse decomposition of alternatives)**.** *Let a represented Type II branch have genuine scale collapse and a nonvanishing selected core. After passing to selected terminal windows, exactly one of the alternatives isolated in Section 7 occurs:*

(i) *finite scale-collapsing cost;*
(ii) *finite absolute scale cost;*
(iii) *thin-drift alternative;*
(iv) *autonomous-modulation alternative;*
(v) *compactness alternative;*
(vi) *a nonzero autonomous reduced limit in the scale-rigid state.*

In the local Type II class of alternatives used by the multibubble/cascade and scale-collapse analyses, alternatives (iii)–(vi) *do not constitute additional compact single-core mechanisms: the thin-drift, modulation, and compactness alternatives lie outside the retained compact scale-collapse branch, and the remaining retained scale-collapse state is removed by Theorem 7.30.*

Proof. The assertion follows by combining the scale-collapse decomposition of alternatives with the local compact scale-collapse routing theorem Theorem 7.30. □

Definition 11.10 (Virial-dissipative terminal state)**.** Let $G_\nu(y) = e^{-|y|^2/(4\nu)}$. A scale-rigid terminal state is called velocity-virial dissipative if its selected terminal representative (V, P) satisfies, after the usual pressure normalization, the autonomous self-similar equation

$$\partial_\tau V + (V\cdot\nabla)V + \nabla P = \nu\Delta V - \frac{1}{2}(V + y\cdot\nabla V), \qquad \nabla\cdot V = 0,$$

and the Gaussian identity below is justified by cutoff approximation on the selected renormalized windows. It is called swirl-virial dissipative if, in addition, the representative is axisymmetric, $\Gamma = RV_\theta$ is the scale-invariant circulation, and the corresponding weighted circulation identity below is justified.

Theorem 11.11 (Virial exclusion for dissipative terminal states)**.** *Let a scale-rigid terminal state satisfy the hypotheses of Theorem 11.10 and be bounded for all negative renormalized times in the sense needed for the Gaussian integrals below.*

(a) *Suppose the velocity virial defect is absorbable: for some $\kappa > 0$ and a.e. τ,*

$$-\frac{1}{2\nu}\int_{\mathbb{R}^3}|V|^2(V\cdot y)G_\nu\,dy - \frac{1}{\nu}\int_{\mathbb{R}^3}P(V\cdot y)G_\nu\,dy \le (1-\kappa)\int_{\mathbb{R}^3}|V|^2G_\nu\,dy + 2\nu(1-\kappa)\int_{\mathbb{R}^3}|\nabla V|^2G_\nu\,dy.$$

Then

$$\int_{\mathbb{R}^3}|V(y,\tau)|^2G_\nu(y)\,dy = 0 \qquad \text{for every } \tau.$$

Consequently no retained terminal core with positive Gaussian velocity mass can belong to this branch.

(b) *Suppose the terminal state is axisymmetric and the circulation defect is absorbable: for some $\kappa > 0$ and a.e. τ,*

$$-\frac{1}{2\nu}\int_{\mathbb{R}^3}\Gamma^2(V\cdot y)G_\nu\,dy \le (1-\kappa)\int_{\mathbb{R}^3}\Gamma^2G_\nu\,dy + 2\nu(1-\kappa)\int_{\mathbb{R}^3}|\nabla\Gamma|^2G_\nu\,dy.$$

Then

$$\int_{\mathbb{R}^3}\Gamma(y,\tau)^2G_\nu(y)\,dy = 0 \qquad \text{for every } \tau.$$

Thus every retained axisymmetric terminal core whose remaining obstruction is a positive Gaussian circulation mass is excluded by the weighted circulation identity.

Proof. For the velocity statement set

$$\mathcal{E}(\tau) = \int_{\mathbb{R}^3}|V(y,\tau)|^2G_\nu(y)\,dy.$$

The Gaussian virial identity gives

$$\partial_\tau\mathcal{E} = -2\nu\int_{\mathbb{R}^3}|\nabla V|^2G_\nu\,dy - \int_{\mathbb{R}^3}|V|^2G_\nu\,dy - \frac{1}{2\nu}\int_{\mathbb{R}^3}|V|^2(V\cdot y)G_\nu\,dy - \frac{1}{\nu}\int_{\mathbb{R}^3}P(V\cdot y)G_\nu\,dy.$$

The absorbability hypothesis therefore implies

$$\partial_\tau\mathcal{E} + \kappa\mathcal{E} + 2\nu\kappa\int_{\mathbb{R}^3}|\nabla V|^2G_\nu\,dy \le 0.$$

In particular $\partial_\tau\mathcal{E} + \kappa\mathcal{E} \le 0$. Integrating from $s < \tau$ to τ gives

$$\mathcal{E}(\tau) \le e^{-\kappa(\tau-s)}\mathcal{E}(s).$$

The ancient boundedness of the terminal state makes $\sup_{s<\tau}\mathcal{E}(s) < \infty$. Sending $s \to -\infty$ yields $\mathcal{E}(\tau) = 0$.

For the axisymmetric statement set

$$\mathcal{G}(\tau) = \int_{\mathbb{R}^3} \Gamma(y,\tau)^2 G_\nu(y)\, dy.$$

The weighted circulation identity is

$$\partial_\tau \mathcal{G} + \mathcal{G} + 2\nu \int_{\mathbb{R}^3} |\nabla\Gamma|^2 G_\nu \, dy = -\frac{1}{2\nu}\int_{\mathbb{R}^3} \Gamma^2 (V\cdot y) G_\nu \, dy.$$

The assumed absorption gives

$$\partial_\tau \mathcal{G} + \kappa\mathcal{G} + 2\nu\kappa \int_{\mathbb{R}^3} |\nabla\Gamma|^2 G_\nu \, dy \le 0.$$

The same backward Gronwall argument gives $\mathcal{G}(\tau) = 0$ for every τ. □

Remark 11.12 (Controlled swirl alone). If the axisymmetric representative satisfies $\Gamma \in L^\infty$, the same weighted circulation identity gives the a priori estimate

$$\mathcal{G}(\tau)^{1/2} \lesssim \|V\|_{L^\infty}\|\Gamma\|_{L^\infty}$$

for bounded ancient terminal states. This estimate enters the local decomposition of alternatives but does not by itself exclude a terminal state. The exclusion requires the stated absorption, vanishing conclusion, or an equivalent condition which makes the virial inequality dissipative.

11.3. **Assembly of Alternatives.**

Theorem 11.13 (Local Type II decomposition into local alternatives)**.** *Every positive local Type II concentration sequence in Theorem 11.2, after passing to subsequences and selected windows, enters one of the following alternatives:*

(i) *a compact single-core vanishing-dissipation branch;*
(ii) *a multibubble, cascade, or gauge-degenerate branch;*
(iii) *loss of local windowed H^1 control on a retained compact cylinder;*
(iv) *a finite-cost scale-collapse branch with nonvanishing selected core;*
(v) *a retained compact scale-collapse or scale-rigid terminal state.*

In the last alternative, any terminal subbranch satisfying Theorem 11.11 is eliminated before the remaining scale-rigid state is passed to the local scale-rigidity and compact scale-collapse routing theorems.

Proof. This decomposition is obtained by combining Theorems 11.4, 11.6, 11.7 and 11.9. The representation theorem gives the nondegenerate single-core chart whenever a unique retained core is selected. Failure of that selection, splitting of positive CKN mass, compactness failure, or scale cascade is the multibubble/cascade side. On a represented retained core, rough-core loss is equivalent by Theorem 11.7 to failure of compact CKN control on an enclosing cylinder, hence returns to the multibubble/cascade side. Genuine scale collapse is divided by Theorem 11.9 into finite-cost scale collapse or one of the retained compact scale-collapse/scale-rigid terminal states. These cases exhaust the local alternatives. □

Proposition 11.14 (Elimination of local Type II states)**.** *No positive local Type II concentration sequence can lie in any of the alternatives* (i)–(v) *of Theorem 11.13.*

Proof. Alternative (i) is excluded by the compact single-core criterion, Theorem 11.5.

Alternative (ii) is excluded by the local multibubble and cascade theorem, Theorem 11.6.

Alternative (iii) is not an independent terminal state. By Theorem 11.7, rough-core loss of local H^1 control forces failure of the compact-cylinder $\mathcal{C} + \mathcal{D}$ bounds on enclosing compact cylinders. Such failure yields an active compact-cylinder concentration branch and therefore belongs to the multibubble/cascade alternative, already excluded by Theorem 11.6.

Alternative (iv) is excluded by the finite-cost scale-collapse theorem, Theorem 11.8.

Alternative (v) is the retained compact scale-collapse or scale-rigid terminal state. If this terminal state is velocity-virial dissipative, Theorem 11.11 forces its Gaussian velocity mass to vanish, which contradicts retention of a positive terminal core. If it is axisymmetric and swirl-virial dissipative with positive retained circulation mass, the same theorem forces $\Gamma \equiv 0$, contradicting that positive circulation mass. The remaining retained compact scale-collapse states are excluded by Theorem 7.30. The remaining scale-rigid terminal states are excluded by Theorem 5.40: the state either has a nonzero bounded selected-window limit, which is incompatible with the local Type II condition, or it reduces to the compact single-core branch, which is excluded by Theorem 11.5. Hence it cannot be a local Type II branch. □

11.4. **Physical Entry into the Covered Local Type II Path.**

Definition 11.15 (Physical local Type II singular germ)**.** Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T-\delta, T)$. A tuple (u, p, x_0, T) is a physical local Type II singular germ if:

(i) $x_0 \in \Sigma(T)$;
(ii) the local Type I scale-invariant bound of Theorem 3.5 fails at (x_0, T);
(iii) the germ is an interior Navier–Stokes germ, meaning that all sufficiently small backward cylinders $Q_r(x_0, T)$ are contained in the Navier–Stokes domain under consideration.

Definition 11.16 (Covered local Type II entry)**.** A physical local Type II singular germ has covered local Type II entry if one of the following two alternatives occurs:

(i) one of its Caffarelli–Kohn–Nirenberg concentration sequences, after passing to a subsequence and applying the selected-window reductions used in this paper, enters one of the alternatives in Theorem 11.13;
(ii) before the selected-window state decomposition is reached, the ordered local tests assign the germ to a class-incompatible alternative or to a named non-retained local state-space exit.

Lemma 11.17 (Physical Type II germs are admissible local branches)**.** *Let (u, p, x_0, T) be a physical local Type II singular germ. After translating space and time and restricting to a smaller terminal interval if necessary, the terminal germ defines an element of $\mathcal{U}_{\mathrm{II}}^{NS}$ in the sense of Theorem 12.1.*

Proof. Choose $0 < \delta' < \delta$ so that the cylinders under consideration remain interior. Set

$$\tilde{u}(y, s) = u(y + x_0, T - \delta' + s), \qquad \tilde{p}(y, s) = p(y + x_0, T - \delta' + s), \qquad 0 < s < \delta',$$

and $T^* = \delta'$. This space-time translation and restriction preserve the local Navier–Stokes equations, suitability, and the pressure normalization modulo functions of time. The terminal time T^* is finite by construction. Since $x_0 \in \Sigma(T)$, the terminal germ is not a local continuation branch. By the definition of physical local Type II germ, the local Type I scale-invariant bound fails at (x_0, T), so the germ is not in the local Type I class. The interior-domain clause excludes boundary and non-Navier–Stokes-domain alternatives. These are exactly the admissibility clauses in Theorem 12.1. □

Lemma 11.18 (Physical Type II germs carry local Type II concentration sequences)**.** *Let (u, p, x_0, T) be a physical local Type II singular germ. Then there are $\eta > 0$ and $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \qquad \text{for all } n,$$

and r_n is a positive local Type II concentration sequence in the sense of Theorem 11.2.

Proof. Since $x_0 \in \Sigma(T)$, Theorem 3.5 gives $\eta > 0$ and $r_n \downarrow 0$ such that the displayed CKN lower bound holds. The same dichotomy has exactly two pointwise alternatives: the local Type I scale-invariant bound holds, or the point lies in the local Type II alternative. The first alternative is excluded by Theorem 11.15. Hence (x_0, T) lies in the local Type II alternative, and the displayed concentration sequence is exactly a positive local Type II concentration sequence as defined in Theorem 11.2. □

Lemma 11.19 (No extra outside-class row for physical Type II germs)**.** *Let (u, p, x_0, T) be a physical local Type II singular germ and let ω be the admissible terminal branch supplied by Theorem 11.17. Any failure of the C1 analytic data for ω is either class-incompatible or one of the named non-retained local exits in Theorem 10.101. In particular, there is no additional outside-covered-class alternative for such a physical Type II germ.*

Proof. By Theorem 11.17, $\omega \in \mathcal{U}_{\mathrm{II}}^{NS}$. The ordered C1 discharge Theorem 10.101 applies on this class. Its conclusion is precisely that the first failure of Theorem 12.3 is assigned either to a class-incompatible alternative excluded by Theorem 12.1, or to a named non-retained state-space branch already covered by the local stratification. Thus a physical Type II germ has no residual "outside the covered class" row after the ordered local tests are run. □

Theorem 11.20 (Covered-class entry for physical Type II germs)**.** *Every physical local Type II singular germ has covered local Type II entry in the sense of Theorem 11.16. More explicitly, after passing to a CKN concentration sequence and to the selected windows used in this paper, the germ enters one of the alternatives in Theorem 11.13, unless it has already exited through one of the class-incompatible or named non-retained local alternatives of Theorem 10.101.*

Proof. Let (u, p, x_0, T) be a physical local Type II singular germ. By Theorem 11.17, the germ defines an admissible terminal branch. Run the ordered C1 analytic-data tests on that branch. If a first failure occurs, then Theorem 11.19 routes it to a class-incompatible alternative or to a named non-retained local state-space exit. This is covered entry in the sense of Theorem 11.16 (ii).

It remains to consider the retained case, when no C1 failure occurs. By Theorem 11.18, the germ carries a positive local Type II concentration sequence. The decomposition theorem Theorem 11.13 is formulated exactly for such sequences: after passing to subsequences and selected windows, the sequence enters one of the five alternatives listed there. This is covered entry in the sense of Theorem 11.16 (i). Thus every case is a covered local state-space outcome. The proof uses only pointwise CKN concentration, local first-failure routing, and selected-window state-space decomposition; it does not use a global $L^3(\mathbb{R}^3)$ bound, a global profile decomposition, global Kato smallness, or vanishing localized dissipation beyond what is produced inside the selected local branch. □

Corollary 11.21 (Discharge of covered-class entry)**.** *The covered-class entry obligation for the no-Type-II theorem is discharged: every physical local Type II singular germ either enters the local state-decomposition path of Theorem 11.13, or exits earlier through a class-incompatible or named non-retained local alternative.*

Proof. This is Theorem 11.20. □

Remark 11.22 (Scope of the covered-entry discharge)*.* Theorem 11.21 proves only the entry statement: a physical local Type II germ is not lost before the local state-space analysis begins. It does not by itself supply the ambient adjacent-stratum closure hypotheses or the expanded terminal hypotheses used in the final assembly. Nor does it assert that every named non-retained exit has already been eliminated; it asserts that any such exit is recorded in the local state-space taxonomy rather than being an untracked outside-class branch. The closure of those named exits is supplied either by the retired local packages cited in the tracker or by the ambient/terminal wrappers Theorems 8.162 and 10.128. The entry argument is also independent of Theorem 11.24; it uses the pointwise CKN dichotomy, the local Type I/Type II split, and the ordered C1 routing theorem, so the discharge is not circular.

Theorem 11.23 (Unconditional local no-Type-II theorem)**.** *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$. Then there is no physical local Type II singular germ (u, p, x_0, T) in the sense of Theorem 11.15. Equivalently, no singular point at time T can lie in the local Type II alternative after the local Type I/Type II separation.*

Proof. Assume for contradiction that (u, p, x_0, T) is a physical local Type II singular germ. By Theorem 11.21, the germ either enters the local state-decomposition path or exits earlier through a class-incompatible or named non-retained local alternative.

A class-incompatible exit contradicts the definition of physical local Type II germ and the admissibility theorem Theorem 11.17. A named non-retained exit is closed by the ordered local packages and, where carrier or cost rows are involved, by the ambient adjacent wrapper Theorem 8.162. If the branch reaches the retained terminal path, the expanded terminal wrapper Theorem 10.128 supplies the expanded terminal hypotheses of Theorem 10.121, and Theorem 10.125 excludes the branch.

Equivalently, once the branch has entered the state-decomposition path, Theorem 11.13 lists the five possible local Type II states and Theorem 11.14 eliminates all five. The previous paragraph records why no physical branch can avoid that list by an untracked terminal or adjacent-row failure. Hence every possible outcome of the assumed physical Type II germ is impossible, a contradiction. □

11.5. **Assembly Theorem.**

Theorem 11.24 (Local Type II exclusion)**.** *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$. Then no point of $\Sigma(T)$ lies in the local Type II alternative.*

Proof. Theorem 11.23 excludes physical local Type II singular germs. If a point $x_0 \in \Sigma(T)$ lay in the local Type II alternative, then it would satisfy the three clauses of Theorem 11.15: singularity gives $x_0 \in \Sigma(T)$, the local Type I bound fails by definition of the Type II alternative, and the theorem is stated for interior cylinders of $\mathbb{R}^3 \times (T - \delta, T)$. This contradicts Theorem 11.23. □

Corollary 11.25 (Final local dichotomy after Type I separation)**.** *Every singular point at time T belongs to the local Type I alternative. Equivalently, once the Type I case is covered by the Type I part of the theory, the local Type II class of alternatives contains no remaining singular branch.*

Proof. Let $x_0 \in \Sigma(T)$. By Theorem 11.3, x_0 carries a positive CKN concentration sequence and lies either in the local Type I alternative or in the local Type II alternative. Theorem 11.24 excludes the latter. □

12. PDE Form of the Type II Classification Inputs

This section records, in PDE notation, the Type II classification inputs used in the local Type II analysis. The statements are conditional classification statements for admissible Type II branches. They do not assert the existence of such branches and do not add any regularity theorem beyond the analytic hypotheses explicitly stated below.

12.1. **Admissible Type II branches and exhaustion.**

Definition 12.1 (Admissible local Type II class)**.** Let $\mathcal{U}_{\mathrm{II}}^{NS}$ be the class of admissible terminal branches (u, p, T^*) such that:

(i) (u, p) is a three-dimensional incompressible Navier–Stokes solution on a terminal interval $[0, T^*)$ in the analytic class under consideration;
(ii) $T^* < \infty$ is the terminal time of the branch;
(iii) the branch is not in the local Type I class associated with the scale-invariant Type I criterion;
(iv) the branch is a Type II branch rather than a continuation, Type I, boundary, or non-Navier–Stokes-domain alternative.

Membership in $\mathcal{U}_{\mathrm{II}}^{NS}$ specifies the class under analysis; no nonemptiness assertion is made.

Definition 12.2 (Type II analytic data)**.** A branch $\omega = (u, p, T^*) \in \mathcal{U}_{\mathrm{II}}^{NS}$ has the Type II analytic data if the following three analytic assertions hold:

(i) a concentration profile or blow-up germ modulo the Navier–Stokes symmetry group;
(ii) a supercritical scale alternative, meaning that the branch enters the Type II scale alternative rather than the Type I scale alternative;
(iii) local profile data in the Navier–Stokes profile class used for the subsequent compactness and representation arguments.

The analytic data are *strictly exhaustive* if every admissible Type II branch has these three properties. They are *exhaustive by ordered local routing* if every branch which fails one of the three properties is assigned, at its first failed C1 test, to one of the ordered failures in Theorem 12.3 and that failure is then routed to a class-incompatible alternative or to a named non-retained local state-space exit. On a retained subbranch, ordered local-routing exhaustion reduces to strict exhaustion.

Definition 12.3 (Ordered exhaustion failures)**.** For an admissible Type II branch, the ordered failures of the Type II analytic data are:

(i) concentration extraction fails;
(ii) concentration extraction succeeds but the supercritical scale alternative does not hold;
(iii) concentration extraction and the scale alternative hold but the local profile data are insufficient for the subsequent compactness and representation arguments;
(iv) the admissible Type II branch lies outside the concentration, scale, and profile alternatives under consideration.

The first applicable failure in this order is the exhaustion alternative.

Theorem 12.4 (Type II branch exhaustion)**.** *Assume the Type II analytic data of Theorem 12.2 are exhaustive by ordered local routing on $\mathcal{U}_{\mathrm{II}}^{NS}$, and that each ordered failure is assigned to a class-incompatible alternative or to a named non-retained local state-space exit. Then every $\omega \in \mathcal{U}_{\mathrm{II}}^{NS}$ either has the concentration profile, the supercritical scale alternative, and the local profile data, or is already in one of those named exits. In particular, every retained admissible Type II branch has the three analytic data.*

Proof. Fix $\omega = (u, p, T^*) \in \mathcal{U}_{\mathrm{II}}^{NS}$. By ordered local routing, exactly one of two possibilities occurs. If no ordered failure occurs, then the concentration-extraction, scale, and profile components all hold, so ω has the three Type II analytic data. If an ordered failure occurs, the first such failure is, by hypothesis, assigned to a class-incompatible alternative or to a named non-retained local state-space exit. Therefore a retained admissible branch cannot be in the failure case, and must have the three analytic data. □

Corollary 12.5 (Ordered exhaustion classification)**.** *For each $\omega \in \mathcal{U}_{\mathrm{II}}^{NS}$, exactly one ordered alternative occurs: the Type II analytic data hold, or one of the four failures in Theorem 12.3 occurs. The data are strictly exhaustive on $\mathcal{U}_{\mathrm{II}}^{NS}$ if and only if the first alternative holds for every $\omega \in \mathcal{U}_{\mathrm{II}}^{NS}$. They are exhaustive by ordered local routing if every failure alternative is assigned to a class-incompatible alternative or to a named non-retained local state-space exit; on the retained subbranch this is equivalent to the first alternative.*

Proof. Test the four analytic requirements in the stated order. If concentration extraction, the scale alternative, and the profile condition all hold, the branch has the Type II analytic data. If a test fails, the first failed test gives the corresponding ordered failure. The alternatives are disjoint by the first failure convention and exhaustive by construction. The final equivalence is the definition of strict exhaustion. The ordered-routing statement follows by recording the target assigned to each first failure. Once class-incompatible and named non-retained exits have been removed, a retained branch cannot lie in any failure alternative, so it lies in the data alternative. □

Theorem 12.6 (Entry into the Type II alternatives)**.** *Assume the Type II analytic data are exhaustive by ordered local routing. Then each admissible Type II branch enters the subsequent analysis in exactly one of the following ways:*

(i) *it reaches the compact Type II cost-divergence hypotheses after the representation, critical-mass, tightness, and windowed-H^1 tests;*
(ii) *it falls into an ordered representation failure, critical-mass failure, loss of critical tightness, or loss of local windowed H^1 control.*
(iii) *it has already exited through a routed C1 concentration, scale, profile, or outside-class failure.*

Proof. By Theorem 12.4, an admissible Type II branch either has the concentration, scale, and profile data or has already exited through a routed C1 failure. In the latter case item (iii) holds. In the former case the representation step either gives a repaired-gauge representation or one of the ordered representation failures in Theorem 12.12. If a repaired-gauge orbit exists, the critical L^3 test gives either the positive finite annulus or one of the ordered critical-mass alternatives in Theorem 12.22. If the positive finite annulus holds, the remaining compact cost-divergence inputs are critical tightness and local windowed H^1 control. Their failures are exactly the two local failures in Theorem 12.30. If both hold, the compact Type II cost-divergence criterion applies. □

12.2. **Repaired-gauge representation from Type II analytic data.**

Definition 12.7 (Raw Type II chart)**.** A raw Type II chart consists of

$$(u, p, T^*, x_c, \lambda, \tau, V, P, \mathcal{I}), \qquad \mathcal{I} = (t_0, T^*),$$

where u, p solve

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \qquad \nabla \cdot u = 0,$$

$\lambda(t) > 0$, $x_c(t) \in \mathbb{R}^3$, $x_c, \lambda \in AC_{\mathrm{loc}}$, $\lambda(t) \to 0$ as $t \uparrow T^*$, and

$$\frac{d\tau}{dt} = \lambda(t)^{-2}, \qquad \tau(t) \to \infty \quad \text{as } t \uparrow T^*.$$

The renormalized variables are

$$y = \frac{x - x_c(t)}{\lambda(t)}, \qquad u(x,t) = \lambda(t)^{-1} V(y, \tau(t)), \qquad p(x,t) = \lambda(t)^{-2} P(y, \tau(t)),$$

with enough regularity to justify the distributional chain rule on compact y-sets.

Lemma 12.8 (Raw renormalized equation)**.** *For a raw Type II chart, define*

$$a_{\mathrm{raw}}(\tau(t)) = \lambda(t)\lambda_t(t), \qquad b_{\mathrm{raw}}(\tau(t)) = \lambda(t) x_c'(t).$$

Then V, P satisfy, distributionally on compact y-sets,

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a_{\mathrm{raw}}(\tau)(V + y \cdot \nabla V) + b_{\mathrm{raw}}(\tau) \cdot \nabla V, \qquad \nabla \cdot V = 0.$$

Proof. Write

$$u(x,t) = \lambda(t)^{-1} V(y, \tau(t)), \qquad y = \frac{x - x_c(t)}{\lambda(t)}, \qquad \tau_t = \lambda^{-2}.$$

Then

$$\nabla_x u = \lambda^{-2} \nabla_y V, \qquad \Delta_x u = \lambda^{-3} \Delta_y V, \qquad (u \cdot \nabla_x) u = \lambda^{-3} (V \cdot \nabla_y) V.$$

Since

$$y_t = -\frac{x_c'(t)}{\lambda(t)} - \frac{\lambda_t(t)}{\lambda(t)} y,$$

one has

$$\partial_t u = \lambda^{-3} \partial_\tau V - \lambda_t \lambda^{-2} (V + y \cdot \nabla V) - \lambda^{-2} x_c'(t) \cdot \nabla V.$$

Moreover $\nabla_x p = \lambda^{-3} \nabla_y P$. Substitution in Navier–Stokes and multiplication by λ^3 give

$$\partial_\tau V - \lambda \lambda_t (V + y \cdot \nabla V) - \lambda x_c'(t) \cdot \nabla V + (V \cdot \nabla) V + \nabla P = \nu \Delta V.$$

Moving the modulation terms to the right-hand side gives the equation with $a_{\mathrm{raw}} = \lambda \lambda_t$ and $b_{\mathrm{raw}} = \lambda x_c'$. Finally, $\nabla_x \cdot u = \lambda^{-2} \nabla_y \cdot V$, so incompressibility of u implies $\nabla_y \cdot V = 0$. $\square$

Definition 12.9 (Repaired-gauge realization)**.** Let $0 < p < 3$, $\Theta_0 > 0$, and let ψ_R be a smooth compactly supported centering cutoff. A raw Type II chart admits a repaired-gauge realization if, after an admissible time-dependent scale and translation correction, the profile satisfies

$$G_{\mathrm{sc}}(V) = \int_{\mathbb{R}^3} |y|^{-p} |V(y)|^3 \, dy - \Theta_0 = 0,$$

and

$$G_j(V) = \int_{\mathbb{R}^3} y_j |V(y)|^2 \psi_R(y) \, dy = 0, \qquad j = 1, 2, 3,$$

for a.e. renormalized time. The corrected profile belongs to the admissible gauge class $\mathcal{A}_p$, has the regularity needed to differentiate these functionals, and all time-dependent symmetry corrections have been absorbed into final modulation coefficients a, b.

Definition 12.10 (Repaired-gauge representation)**.** A Type II branch has a repaired-gauge representation if it has:

(i) a raw Type II chart;
(ii) a repaired-gauge realization;
(iii) pressure reconstruction modulo functions of time;
(iv) final modulation coefficients a, b appearing in the repaired-gauge equation.

The representation consists of the tuple

$$(V, P, a, b, G_{\mathrm{sc}}, G_1, G_2, G_3, \tau_0).$$

Theorem 12.11 (Representation implication)**.** *Assume a Type II profile branch has a raw Type II chart, a repaired-gauge realization, pressure reconstruction modulo functions of time, and final modulation coefficients. Then the branch has a repaired-gauge representation in the sense of Theorem 12.10. In particular, (V, P, a, b) satisfies the PDE class required by the compact Type II cost-divergence criterion.*

Proof. The raw chart gives V, P and, by Theorem 12.8, the renormalized Navier–Stokes equation with raw modulation coefficients. The repaired-gauge realization places the profile on the scale and centering gauge surface and absorbs the time-dependent symmetry correction into updated coefficients. The pressure reconstruction identifies the pressure associated with V, modulo the usual functions of time. The modulation data identify the final coefficients a, b. These are the components of Theorem 12.10. □

Definition 12.12 (Ordered representation failures)**.** If a repaired-gauge representation is not obtained, the ordered failures are:

(i) no raw orbit is supplied by the concentration and profile data;
(ii) the raw orbit exists but the repaired gauge cannot be realized;
(iii) pressure reconstruction modulo functions of time is unavailable;
(iv) the final modulation coefficients are unavailable.

The first applicable failure is the representation alternative.

Corollary 12.13 (Ordered representation classification)**.** *Every Type II profile branch in the admissible class has exactly one ordered alternative: a repaired-gauge representation, or one of the four failures in Theorem 12.12.*

Proof. Verify the four representation inputs in order. If all are present, Theorem 12.11 gives the repaired-gauge representation. If not, the first missing input gives the corresponding ordered failure. This is exhaustive and disjoint by first-failure ordering. □

12.3. **Minimal chart and gauge realization.**

Definition 12.14 (Minimal representation data)**.** The minimal data for a Type II branch satisfying the analytic data consist of:

(i) an absolutely continuous concentration chart
$$(u, p, T^*, x_c, \lambda, \rho, U, Q, \mathcal{I}),$$
where $\mathcal{I} = (t_0, T^*)$, $\lambda(t) > 0$, $\lambda(t) \to 0$, $\rho_t = \lambda^{-2}$, and
$$u(x,t) = \lambda(t)^{-1} U(y, \rho(t)), \qquad p(x,t) = \lambda(t)^{-2} Q(y, \rho(t)) + c(t), \qquad y = \frac{x - x_c(t)}{\lambda(t)};$$

(ii) an absolutely continuous repaired-gauge correction consisting of $\rho = \rho(\tau)$, $\mu(\tau) > 0$, and $q(\tau) \in \mathbb{R}^3$, with
$$\rho_\tau = \mu(\tau)^2,$$
$$V(Y,\tau) = \mu(\tau) U(\mu(\tau)Y + q(\tau), \rho(\tau)), \qquad P(Y,\tau) = \mu(\tau)^2 Q(\mu(\tau)Y + q(\tau), \rho(\tau)),$$
and
$$G_{\mathrm{sc}}(V(\tau)) = 0, \qquad G_j(V(\tau)) = 0, \quad j = 1, 2, 3.$$
The condition $\rho_\tau = \mu^2$ preserves the viscosity coefficient after the time-dependent scale correction.

Lemma 12.15 (Chart data imply the raw orbit)**.** *Under the chart data in Theorem 12.14, U, Q satisfy*

$$\partial_\rho U + (U \cdot \nabla)U + \nabla Q = \nu\Delta U + A(\rho)(U + y \cdot \nabla U) + B(\rho) \cdot \nabla U, \qquad \nabla \cdot U = 0,$$

where

$$A(\rho(t)) = \lambda(t)\lambda_t(t), \qquad B(\rho(t)) = \lambda(t)x_c'(t).$$

Proof. The chart data are the raw orbit data with raw time ρ. The additive pressure function $c(t)$ has zero spatial gradient. The chain-rule computation is the one in Theorem 12.8, with τ replaced by ρ. Thus spatial derivatives scale as $\nabla_x u = \lambda^{-2}\nabla_y U$, $\Delta_x u = \lambda^{-3}\Delta_y U$, the convection term scales as $\lambda^{-3}(U \cdot \nabla)U$, and $\rho_t = \lambda^{-2}$. The terms generated by λ_t and x_c' give $A = \lambda\lambda_t$ and $B = \lambda x_c'$. Incompressibility pulls back as $\nabla_x \cdot u = \lambda^{-2}\nabla_y \cdot U$. □

Lemma 12.16 (Pressure pullback)**.** *Under the chart data,*

$$-\Delta Q = \partial_i\partial_j(U_iU_j)$$

in distributions, modulo functions of ρ. After any admissible repaired-gauge correction, the transformed pressure satisfies

$$-\Delta P = \partial_i\partial_j(V_iV_j)$$

in distributions, modulo functions of τ.

Proof. Taking divergence of the physical Navier–Stokes equation gives

$$-\Delta p = \partial_i\partial_j(u_iu_j),$$

with pressure determined up to a function of time. Pulling this identity back through $p = \lambda^{-2}Q + c(t)$, $u = \lambda^{-1}U$, and $y = (x - x_c)/\lambda$ gives the pressure equation for Q. If

$$V(Y) = \mu U(\mu Y + q), \qquad P(Y) = \mu^2 Q(\mu Y + q),$$

then

$$-\Delta_Y P = \mu^4(-\Delta Q)(\mu Y + q) = \mu^4\partial_i\partial_j(U_iU_j)(\mu Y + q) = \partial_i\partial_j(V_iV_j).$$

Adding a function of ρ or τ to the pressure does not affect the identity. □

Lemma 12.17 (Forced modulation coefficients)**.** *Assume the minimal data of Theorem 12.14. Then (V, P) satisfies*

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu\Delta V + a(\tau)(V + Y \cdot \nabla V) + b(\tau) \cdot \nabla V,$$

where

$$a(\tau) = \mu(\tau)^2 A(\rho(\tau)) + \frac{\mu_\tau(\tau)}{\mu(\tau)}$$

and

$$b(\tau) = \mu(\tau)B(\rho(\tau)) + \mu(\tau)A(\rho(\tau))q(\tau) + \frac{q_\tau(\tau)}{\mu(\tau)}.$$

Proof. Let

$$z = \mu(\tau)Y + q(\tau), \qquad V(Y,\tau) = \mu(\tau)U(z, \rho(\tau)), \qquad \rho_\tau = \mu^2.$$

The raw equation in (z, ρ) is

$$U_\rho + (U \cdot \nabla_z)U + \nabla_z Q = \nu\Delta_z U + A(U + z \cdot \nabla_z U) + B \cdot \nabla_z U.$$

The identities

$$\nabla_Y V = \mu^2\nabla_z U, \qquad \Delta_Y V = \mu^3\Delta_z U, \qquad (V \cdot \nabla_Y)V = \mu^3(U \cdot \nabla_z)U, \qquad \nabla_Y P = \mu^3\nabla_z Q$$

together with differentiation in τ give

$$V_\tau + (V\cdot\nabla_Y)V + \nabla_Y P - \nu\Delta_Y V = \left(\frac{\mu_\tau}{\mu} + \mu^2 A\right)(V + Y\cdot\nabla_Y V) + \left(\mu B + \mu A q + \frac{q_\tau}{\mu}\right)\cdot\nabla_Y V.$$

This is the displayed equation. Since μ, q, ρ are absolutely continuous and A, B are the raw chart coefficients, a, b are measurable and locally integrable on every finite renormalized window. $\square$

Theorem 12.18 (Minimal representation theorem)**.** *If a Type II branch satisfying the analytic data has the minimal data of Theorem 12.14, then it has a repaired-gauge representation in the sense of Theorem 12.10.*

Proof. Theorem 12.15 gives the raw orbit. The gauge correction places the profile on the repaired scale and centering surface, hence gives the repaired-gauge realization. Theorem 12.16 gives pressure reconstruction modulo functions of time. Theorem 12.17 gives the final modulation coefficients. These are the four inputs in Theorem 12.10. $\square$

Definition 12.19 (Minimal representation failures)**.** Outside the admissible repaired-gauge representation class, the ordered failures are:

(i) no absolutely continuous concentration chart is available;
(ii) the chart exists but the repaired scale and centering gauges cannot be solved admissibly and absolutely continuously.

Pressure-pullback and modulation-coefficient failures are not independent alternatives when the chart and the absolutely continuous gauge solve exist.

Corollary 12.20 (Ordered minimal representation classification)**.** *Every Type II branch satisfying the analytic data has exactly one ordered alternative: a repaired-gauge representation, failure of the raw chart, or failure of the repaired-gauge solve.*

Proof. If the chart is absent, the first failure occurs. If the chart exists but the admissible absolutely continuous gauge solve is absent, the second failure occurs. If both are present, Theorem 12.18 gives the repaired-gauge representation. The alternatives are disjoint by their order and exhaustive for the representation row. $\square$

12.4. **Critical L^3 annulus.**

Definition 12.21 (Critical mass and positive finite annulus)**.** For a repaired-gauge renormalized Type II branch (V, P, a, b) on $[\tau_0, \infty)$, define

$$N_3(\tau) = \|V(\tau)\|_{L^3(\mathbb{R}^3)}.$$

The orbit has positive finite critical mass if there are constants $0 < \eta \le M < \infty$ such that

$$\eta \le N_3(\tau) \le M, \qquad \tau \ge \tau_0.$$

This condition replaces the exact normalization $\|V(\tau)\|_{L^3} = 1$; no amplitude rescaling is performed.

Definition 12.22 (Critical-mass alternatives)**.** For a repaired-gauge orbit, the ordered critical-mass alternatives are:

(i) N_3 is not a well-defined extended measurable critical norm on the renormalized branch;
(ii) $N_3(\tau) = \infty$ for some τ, or $\sup_{\tau\ge\tau_0} N_3(\tau) = \infty$;
(iii) $\sup_{\tau\ge\tau_0} N_3(\tau) < \infty$ and $\liminf_{\tau\to\infty} N_3(\tau) = 0$;
(iv) after increasing the initial time at most once, the positive finite critical annulus of Theorem 12.21 holds.

The first applicable case is the corresponding alternative.

Theorem 12.23 (Nonzero-mass compact Type II cost divergence)**.** *Let (V, P, a, b) be a repaired-gauge renormalized Type II branch. Assume:*

(i) *the positive finite critical annulus holds:*

$$0 < \eta \le \|V(\tau)\|_{L^3(\mathbb{R}^3)} \le M < \infty;$$

(ii) *uniform critical tightness holds:*

$$\forall \varepsilon > 0 \ \exists R_\varepsilon : \quad \sup_{\tau \ge \tau_0} \int_{|y| > R_\varepsilon} |V(y, \tau)|^3 \, dy < \varepsilon;$$

(iii) *local windowed H^1 control holds:*

$$\forall m \ge 1 : \quad \sup_{n \in \mathbb{N}} \int_{\tau_0 + n}^{\tau_0 + n + 1} \|V(\tau)\|_{H^1(B_m)}^2 \, d\tau < \infty.$$

Then finite total localized renormalization cost is impossible:

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau) \, d\tau = \infty.$$

Proof. The proof is the compact Type II cost-divergence proof with the exact normalization $\|V(\tau)\|_{L^3} = 1$ replaced by the lower bound $N_3(\tau) \ge \eta$. Assume for contradiction that the total localized renormalization cost is finite. The unit-window selection and compactness argument gives times $\sigma_j \to \infty$ such that, after passing to a subsequence,

$$V(\sigma_j) \to V_\infty \quad \text{strongly in } L^3_{\text{loc}},$$

the selected sequence has vanishing local dissipation on the exhaustion used by the compactness argument, and consequently

$$\nabla V_\infty = 0$$

distributionally on every fixed ball. These conclusions use only tightness, local windowed H^1 control, and finite total cost.

The low-dissipation limit is spatially constant on each fixed ball. The constants are compatible as the balls increase, so $V_\infty \equiv c$ on $\mathbb{R}^3$. The upper critical-mass bound gives, for every R,

$$|c|^3 |B_R| = \lim_{j \to \infty} \int_{B_R} |V(y, \sigma_j)|^3 \, dy \le M^3.$$

Letting $R \to \infty$ forces $c = 0$. Thus

$$V(\sigma_j) \to 0 \quad \text{strongly in } L^3_{\text{loc}}.$$

Choose R so that the critical L^3-tail outside B_R is $< \eta^3/4$ uniformly in τ. Strong convergence on B_R gives

$$\int_{B_R} |V(y, \sigma_j)|^3 \, dy < \eta^3/4$$

for j large. Therefore

$$\|V(\sigma_j)\|_{L^3(\mathbb{R}^3)}^3 < \eta^3/2,$$

contradicting $N_3(\sigma_j) \ge \eta$. Hence the total localized renormalization cost is infinite. □

Theorem 12.24 (Critical L^3-annulus classification)**.** *For a repaired-gauge Type II branch, exactly one of the ordered alternatives in Theorem 12.22 occurs. In the fourth case the compact Type II cost-divergence criterion remains valid with the positive finite critical annulus in place of exact unit L^3-normalization.*

Proof. If N_3 is not a well-defined measurable extended norm, the first alternative holds. Otherwise N_3 is an extended nonnegative measurable function. If it is infinite at some time, or has infinite supremum on the tail, the second alternative holds. If the supremum is finite but $\liminf_{\tau\to\infty} N_3(\tau) = 0$, the third alternative holds. In the remaining case,

$$0 < \liminf_{\tau\to\infty} N_3(\tau) \quad \text{and} \quad \sup_{\tau\geq\tau_0} N_3(\tau) < \infty.$$

After increasing the initial time, there exist $0 < \eta \leq M < \infty$ such that

$$\eta \leq N_3(\tau) \leq M$$

for all remaining τ. Relabeling this tail initial time as τ_0 gives the fourth alternative. The cost-divergence statement is Theorem 12.23. □

12.5. Localized cost and the Type II scale-exclusion condition.

Definition 12.25 (Localized Type II cost)**.** For a repaired-gauge Type II branch, let

$$\tilde{\mathfrak{D}}_{R_0}(\tau) = \nu \int_{\mathbb{R}^3} |\nabla V(y,\tau)|^2 \phi_{R_0}(y)\, dy + a_+(\tau) \int_{\mathbb{R}^3} |V(y,\tau)|^2 \phi_{R_0}(y)\, dy,$$

where $a_+ = \max(a, 0)$, ϕ_{R_0} is the compact Type II cutoff, and $\nu > 0$. A Type II scale-exclusion cost is a measurable nonnegative function $\mathfrak{C}_{\mathrm{II}}$ that is locally integrable on finite τ-intervals and whose divergence on $[\tau_0, \infty)$ gives the Type II scale-exclusion condition.

Definition 12.26 (Cost comparison)**.** A cost comparison for a repaired-gauge Type II branch consists of $R_0, \phi_{R_0}, \mathfrak{C}_{\mathrm{II}}, c_0, \tau_1$, with $c_0 > 0$ and $\tau_1 \geq \tau_0$, such that

$$\mathfrak{C}_{\mathrm{II}}(\tau) \geq c_0\, \tilde{\mathfrak{D}}_{R_0}(\tau) \qquad \text{for a.e. } \tau \geq \tau_1.$$

The canonical Navier–Stokes choice is

$$\mathfrak{C}_{\mathrm{II}}(\tau) = \tilde{\mathfrak{D}}_{R_0}(\tau), \qquad c_0 = 1, \qquad \tau_1 = \tau_0.$$

Lemma 12.27 (Identity-cost comparison)**.** *If $\tilde{\mathfrak{D}}_{R_0}$ is measurable, nonnegative, and locally integrable on finite τ-intervals, then the canonical choice $\mathfrak{C}_{\mathrm{II}} = \tilde{\mathfrak{D}}_{R_0}$ defines a valid Type II scale-exclusion cost and satisfies the cost comparison with $c_0 = 1$ and $\tau_1 = \tau_0$.*

Proof. The assumed measurability, nonnegativity, and local finite-interval integrability give the cost properties in Theorem 12.25. The identity

$$\mathfrak{C}_{\mathrm{II}} = \tilde{\mathfrak{D}}_{R_0}$$

gives the comparison in Theorem 12.26 with $c_0 = 1$ and $\tau_1 = \tau_0$. □

Theorem 12.28 (Cost comparison theorem)**.** *Let (V, P, a, b) be a repaired-gauge Type II branch. Assume:*

(i) *$\tilde{\mathfrak{D}}_{R_0}$ is measurable, nonnegative, and locally integrable on finite τ-intervals;*
(ii)
$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty;$$
(iii) *either the canonical cost $\mathfrak{C}_{\mathrm{II}} = \tilde{\mathfrak{D}}_{R_0}$ is used, or a cost comparison as in Theorem 12.26 is available.*

Then the Type II scale-exclusion condition associated with $\mathfrak{C}_{\mathrm{II}}$ holds.

Proof. In the canonical case, Theorem 12.27 supplies the cost comparison. In the general case, the comparison is assumed. Hence there are $c_0 > 0$ and $\tau_1 \geq \tau_0$ such that

$$\mathfrak{C}_{\mathrm{II}}(\tau) \geq c_0 \tilde{\mathfrak{D}}_{R_0}(\tau)$$

for a.e. $\tau \geq \tau_1$. Since $\tilde{\mathfrak{D}}_{R_0} \geq 0$ and is locally integrable on finite intervals, divergence on $[\tau_0, \infty)$ implies

$$\int_{\tau_1}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty.$$

Therefore

$$\int_{\tau_1}^{\infty} \mathfrak{C}_{\mathrm{II}}(\tau)\, d\tau \geq c_0 \int_{\tau_1}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty.$$

Since $\mathfrak{C}_{\mathrm{II}}$ is nonnegative and locally integrable on finite intervals,

$$\int_{\tau_0}^{\infty} \mathfrak{C}_{\mathrm{II}}(\tau)\, d\tau = \int_{\tau_0}^{\tau_1} \mathfrak{C}_{\mathrm{II}}(\tau)\, d\tau + \int_{\tau_1}^{\infty} \mathfrak{C}_{\mathrm{II}}(\tau)\, d\tau = \infty.$$

By definition of the Type II scale-exclusion cost, this divergence is the Type II scale-exclusion condition. □

Corollary 12.29 (Compact cost divergence implies scale exclusion)**.** *Assume the hypotheses of Theorem 12.23 and use the canonical Navier–Stokes cost $\mathfrak{C}_{\mathrm{II}} = \tilde{\mathfrak{D}}_{R_0}$. Then the Type II scale-exclusion condition holds. If, in addition, the supercritical scale alternative is present and the branch remains in the retained compact cost-divergence stratum after the ordered validation tests of Theorem 8.86, then the Type II exclusion conclusion holds.*

Proof. Theorem 12.23 gives

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty.$$

By Theorem 12.27, the canonical cost satisfies the cost comparison. Theorem 12.28 converts the infinite PDE cost into the Type II scale-exclusion condition, hence the validated compact cost-divergence conclusion. Under the retained-stratum validation hypothesis in the statement, Theorem 8.95 applies and gives the Type II exclusion conclusion. Thus no separate Navier–Stokes cost-exclusion hypothesis is used. □

12.6. **Two-alternative finite-cost classification.**

Definition 12.30 (Remaining compact cost-divergence tests)**.** For a repaired-gauge Type II orbit, define the critical-tightness condition by

$$\forall \varepsilon > 0\ \exists R_\varepsilon : \quad \sup_{\tau \geq \tau_0} \int_{|y|>R_\varepsilon} |V(y,\tau)|^3\, dy < \varepsilon.$$

Failure of critical tightness is the condition

$$\exists \varepsilon_0 > 0\ \forall R > 0\ \exists \tau_R \geq \tau_0 : \quad \int_{|y|>R} |V(y,\tau_R)|^3\, dy \geq \varepsilon_0.$$

Define local windowed H^1 control by

$$\forall m \geq 1 : \quad \sup_{n\in\mathbb{N}} \int_{\tau_0+n}^{\tau_0+n+1} \|V(\tau)\|_{H^1(B_m)}^2\, d\tau < \infty.$$

Failure of local windowed H^1 control is the condition

$$\exists m \geq 1 : \quad \sup_{n\in\mathbb{N}} \int_{\tau_0+n}^{\tau_0+n+1} \|V(\tau)\|_{H^1(B_m)}^2\, d\tau = \infty.$$

Definition 12.31 (Compact Type II data)**.** A Type II branch has compact Type II data if it has:

(i) the Type II analytic data of Theorem 12.2;
(ii) the repaired-gauge representation of Theorem 12.10;
(iii) the positive finite critical annulus of Theorem 12.21;
(iv) the locally integrable nonnegative cost and cost comparison of Theorems 12.25 and 12.26.

This condition does not include critical tightness or windowed H^1 control; those are the two remaining compact cost-divergence tests.

Definition 12.32 (Routed pre-compact-data exits)**.** A routed pre-compact-data exit is the first failed ordered test before the compact cost-divergence tests are reached. Thus it is one of the routed C1 exhaustion failures in Theorem 12.3, one of the ordered representation failures in Theorem 12.12 or Theorem 12.19, one of the critical-mass alternatives Theorem 12.22 (i)–(iii), or a failure of local cost well-posedness or cost comparison before Theorems 12.25 and 12.26 is available. The first-failure convention makes these exits disjoint from the compact Type II data of Theorem 12.31.

Lemma 12.33 (Pre-compact-data routing)**.** *Every admissible Type II branch has exactly one of the following ordered outcomes:*

(i) *it has compact Type II data in the sense of Theorem 12.31;*
(ii) *it has a routed pre-compact-data exit in the sense of Theorem 12.32.*

Proof. Fix an admissible branch. The ordered C1 routing hypothesis of Theorem 12.4 is supplied by Theorem 10.101. Applying Theorem 12.4, if the concentration, scale, and local profile data are not all present, the first failed C1 test is a routed pre-compact-data exit. Otherwise the first component of Theorem 12.31 holds.

Next run the ordered representation classification Theorem 12.13, equivalently the minimal representation classification Theorem 12.20 when the minimal chart package is used. A first representation, repaired-gauge, pressure, or modulation failure is a routed pre-compact-data exit. If no such failure occurs, the branch has the repaired-gauge representation of Theorem 12.10.

On that repaired-gauge branch, apply the critical L^3-annulus classification Theorem 12.24. Alternatives (i)–(iii) of Theorem 12.22 are routed pre-compact-data exits. In the remaining case the positive finite critical annulus Theorem 12.21 holds on a terminal tail.

Finally test whether the localized cost is well posed and compatible with the Navier–Stokes identity cost. Availability of the local cost and comparison is the fourth component of Theorem 12.31. If it is absent, the first absence is the local cost-well-posedness or cost-comparison failure listed in Theorem 12.32; if it is present, all four components of compact Type II data hold. The first-failure convention makes the two outcomes disjoint and exhaustive. □

Lemma 12.34 (Compact positive case is excluded)**.** *Assume an admissible Type II branch has compact Type II data and satisfies both compact cost-divergence tests in Theorem 12.30. If the branch remains in the retained compact cost-divergence stratum after the ordered validation tests of Theorem 8.86, then the branch satisfies the Type II exclusion conclusion.*

Proof. The compact Type II data give the supercritical scale alternative, a repaired-gauge orbit (V, P, a, b), the positive finite critical annulus, and the cost comparison. Together with critical tightness and local windowed H^1 control, the hypotheses of Theorem 12.23 hold. Thus

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau)\, d\tau = \infty.$$

Theorem 12.28 converts this divergence into the Type II scale-exclusion condition, hence the validated compact cost-divergence conclusion. Since the branch remains in the retained compact cost-divergence stratum, Theorem 8.95 gives the Type II exclusion conclusion. □

Theorem 12.35 (Compact exclusion or two-alternative classification)**.** *Assume adjacent-stratum closure for the compact-cost regime in the sense of Theorem 8.96. Then each admissible Type II branch has exactly one ordered alternative:*

(i) *a routed pre-compact-data exit;*
(ii) *the Type II exclusion conclusion, either on the retained compact cost-divergence stratum or on a treated adjacent stratum;*
(iii) *failure of critical tightness;*
(iv) *failure of local windowed H^1 control after critical tightness has been verified.*

Proof. Fix an admissible Type II branch. By Theorem 12.33, either the branch has a routed pre-compact-data exit or it has compact Type II data. In the first case item (i) holds. In the second case only the two compact cost-divergence tests remain.

Evaluate critical tightness. If tightness fails, the failure-of-tightness alternative holds. If tightness holds, evaluate local windowed H^1 control. If the windowed H^1 condition fails, the windowed-control alternative holds. If it holds, Theorem 8.90 places the branch either in the retained compact cost-divergence stratum or in a named adjacent exit stratum. In the retained case, Theorem 12.34 gives the Type II exclusion conclusion. In an adjacent exit stratum, adjacent-stratum closure gives the same Type II exclusion conclusion. The alternatives are exhaustive and disjoint by this ordered evaluation. □

Definition 12.36 (Finite-cost branch not excluded by compact cost divergence)**.** An admissible Type II branch is finite-cost if

$$\int_{\tau_0}^{\infty} \mathfrak{C}_{\mathrm{II}}(\tau)\, d\tau < \infty.$$

In the canonical Navier–Stokes cost this is equivalent to finite localized PDE cost, because $\mathfrak{C}_{\mathrm{II}} = \tilde{\mathfrak{D}}_{R_0}$. It is not excluded by compact cost divergence if the compact Type II exclusion conclusion does not hold for that branch.

Corollary 12.37 (Two-alternative finite-cost classification)**.** *Under the hypotheses of Theorem 12.35, every finite-cost admissible Type II branch that is not in a routed pre-compact-data exit and is not excluded by the retained compact cost-divergence argument or by the treated adjacent-stratum alternatives is either non-tight or fails local windowed H^1 control.*

Proof. Apply Theorem 12.35. Since the branch is not in a routed pre-compact-data exit, item (i) is unavailable. Since it is not excluded by retained compact cost-divergence or by the treated adjacent-stratum alternatives, item (ii) is unavailable. Therefore the ordered alternative is failure of critical tightness or failure of local windowed H^1 control. The finite-cost assumption records that the resulting branch remains finite-cost. □

Remark 12.38 (Content of the classification inputs). The preceding statements put the Type II classification inputs in PDE form. Exhaustion failure, missing repaired-gauge representation, ambiguous critical L^3 normalization, and cost-comparison failure are all replaced by ordered alternatives. Once those alternatives are excluded or assumed absent, the only finite-cost Type II alternatives not excluded by the compact cost-divergence criterion are failure of critical tightness and failure of local windowed H^1 control.

13. Conclusion

The paper gives a local branchwise exclusion mechanism for Type II Navier–Stokes concentration. Beginning with a positive Caffarelli–Kohn–Nirenberg concentration sequence, the argument passes to repaired-gauge variables around a retained core and keeps the analysis inside local PDE estimates: suitable energy inequalities, pressure reconstruction, compactness on fixed cylinders, Caccioppoli estimates, profile decoupling, and localized scale-cost identities.

The compact single-core mechanism is the central obstruction. A retained core with positive local CKN density and vanishing localized dissipation has only two compact limits after passing to selected windows. The zero limit contradicts the persistence of the velocity-pressure concentration by strong convergence and pressure stability, while a nonzero spatially constant limit gives a bounded selected-window profile and hence exits the Type II regime. The rest of the proof shows that the remaining alternatives to this compact mechanism are also accounted for: rough cores are reduced to compact-cylinder control, active multibubbles and cascades are separated by the profile analysis, and finite-cost scale collapse is incompatible with a nonvanishing retained core.

The scale-collapse analysis is intentionally explicit about its limits. Finite scale-collapsing cost, and hence finite absolute scale cost in the covered setting, are excluded within the renormalized setting. Thick autonomous windows produce nontrivial autonomous limits only after the stated compactness and modulation hypotheses are available. The retained compact autonomous branch is not closed by silently invoking a global Liouville theorem: it is routed through Theorem 7.30 into the scale-rigid discharge. If a retained input fails, the failure is one of the named thin-drift, modulation, compactness, exterior, rough-core, or multicore local exits.

Thus, under the analytic inputs and local routing hypotheses made explicit in the text, every physical local Type II concentration branch is eliminated. The final local dichotomy is that a singular point must belong to the local Type I alternative once the Type II branch has been removed. The contribution is an unconditional no-Type-II theorem within the local state-space framework developed here; it is not an unconditional resolution of the Navier–Stokes regularity problem, since the Type I side is a separate problem.

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